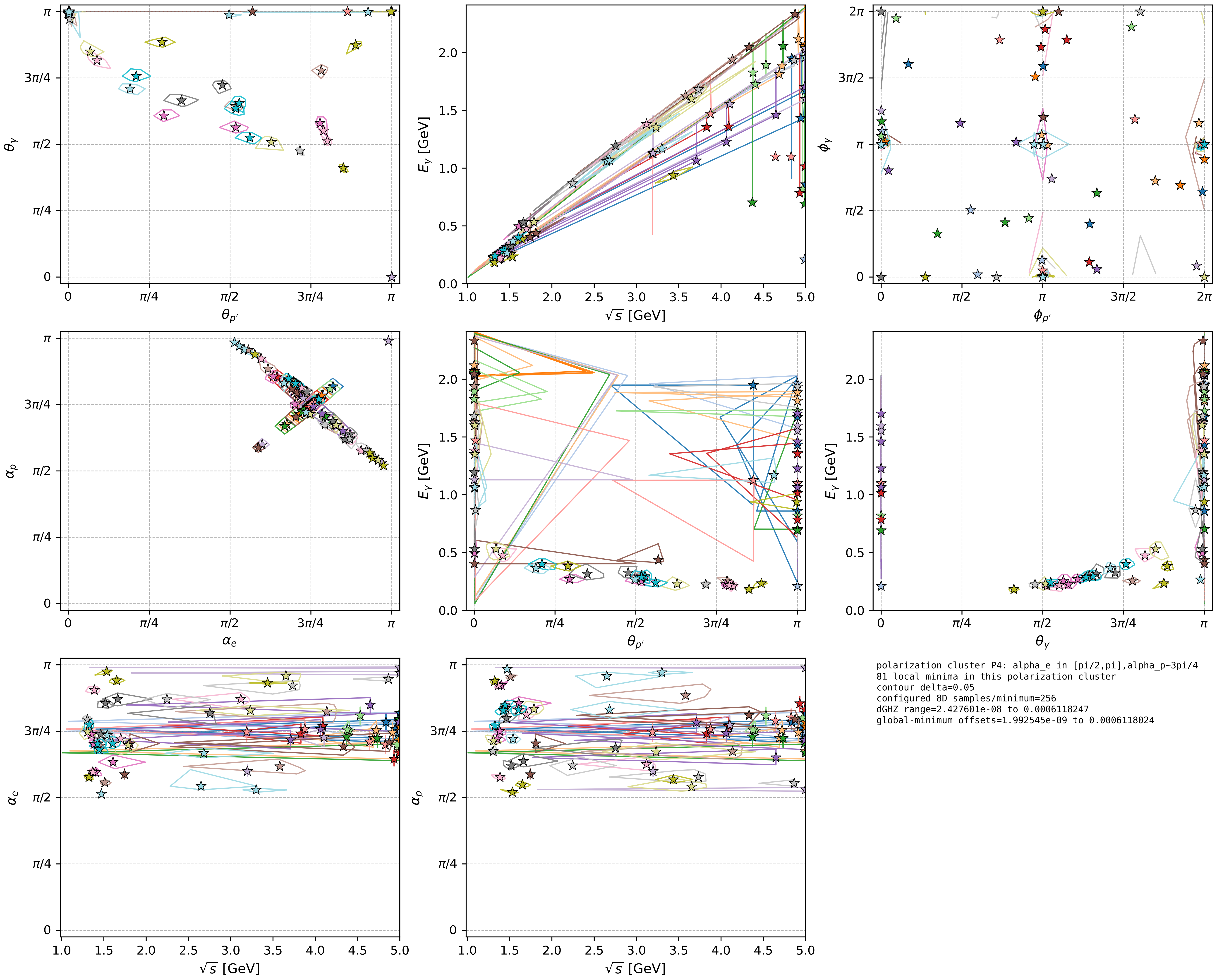
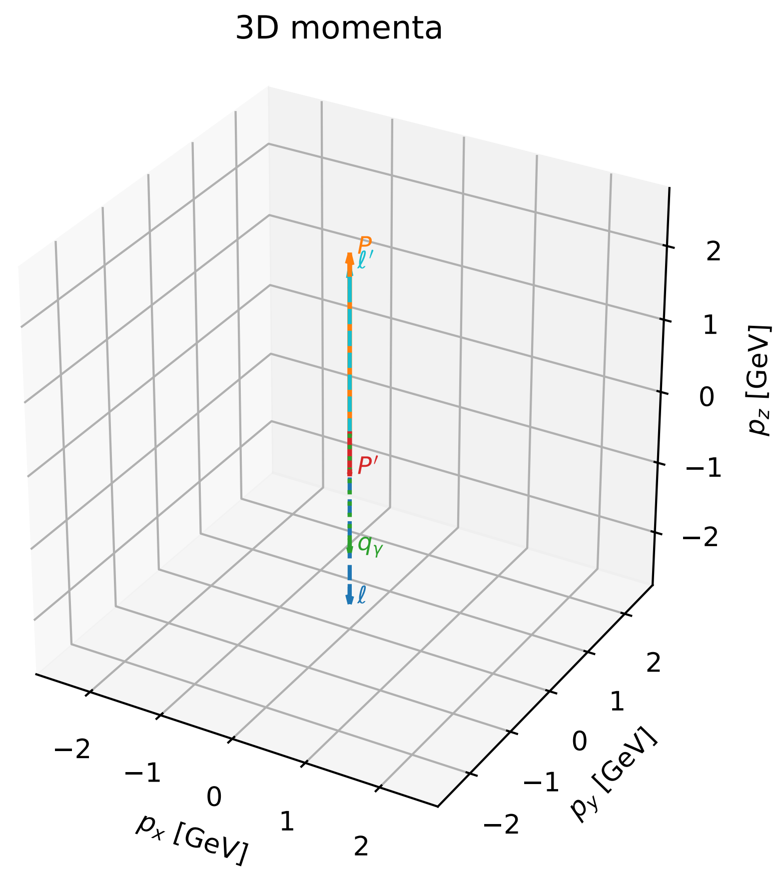
electron: polarization cluster P4; all local-minimum configurations and pairwise projections of their 8D contours



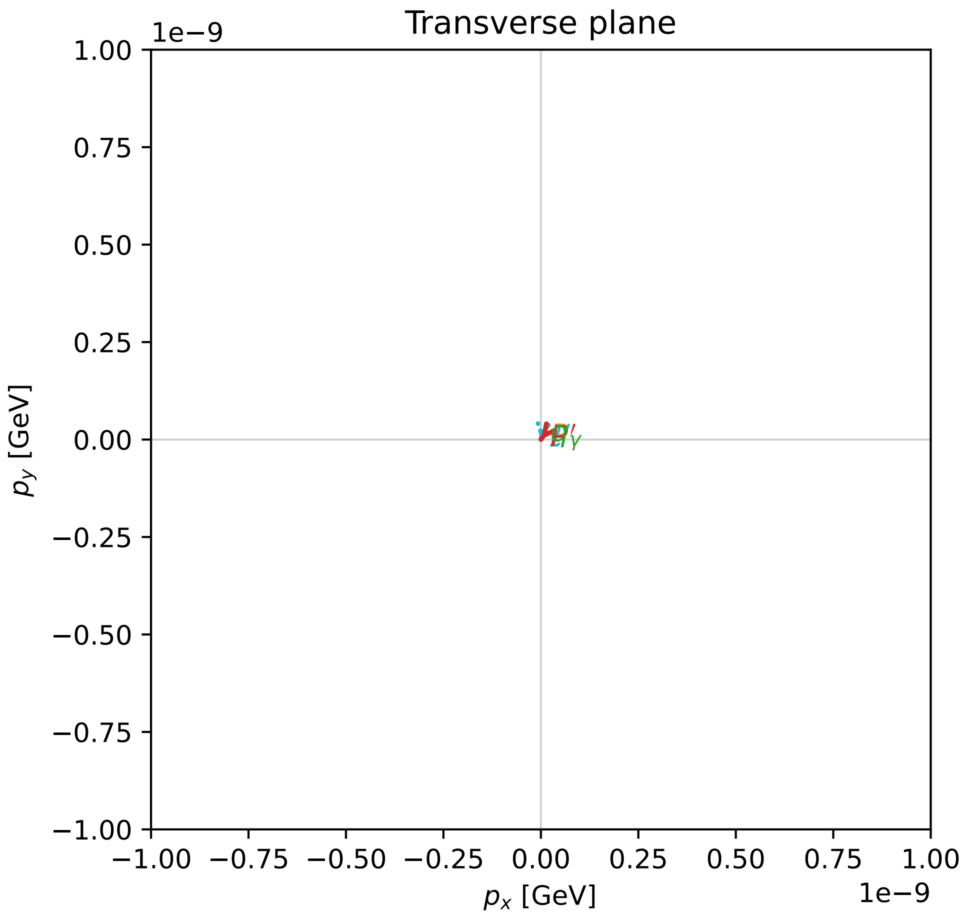
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

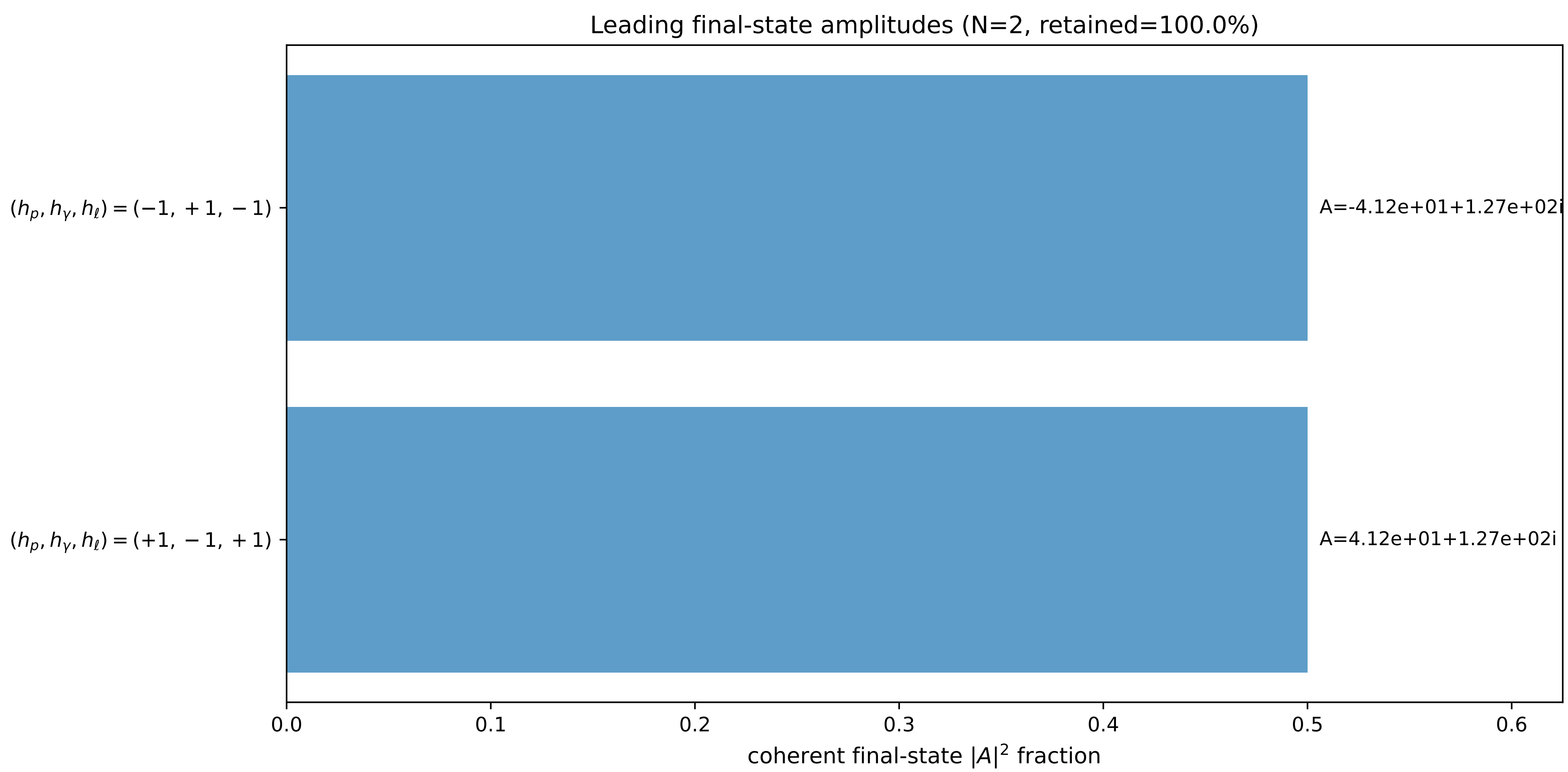


dghz_mixing_angles_polarization_04_local_minimum_8 d_{\rm GHZ}=2.4276e-08
region: polarization_cluster_04_local_minimum_8
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.332037 rad
incoming lepton: -0.68982|+> +0.723981|->
proton mixing angle: alpha_p=2.3320298 rad
incoming proton: -0.689815|+> +0.723986|->

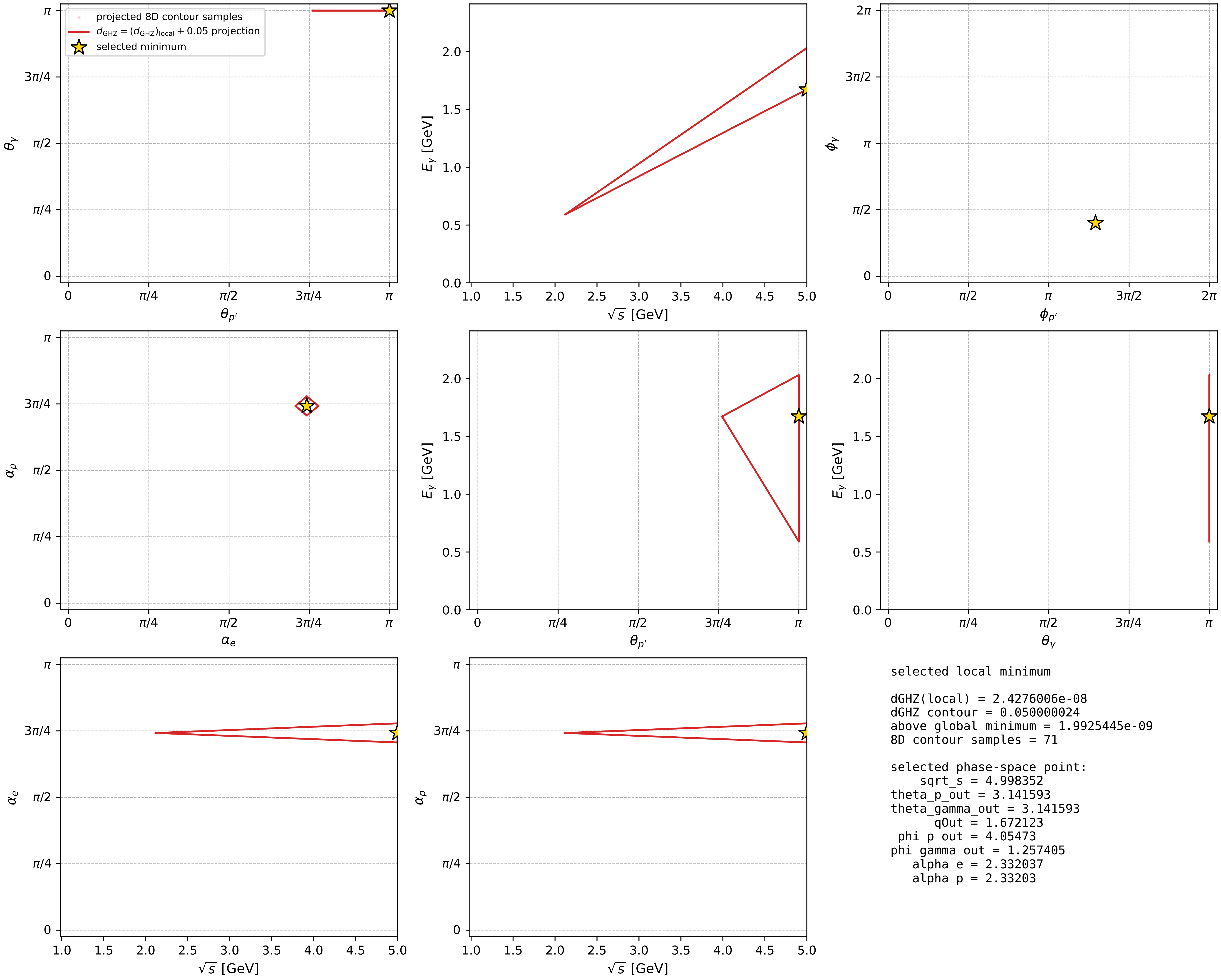
s=24.9835, $\sqrt{s}$=4.99835, $|\vec{P}|$=2.41116, $|\vec{P}'|$=0.561095
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=4.05473, ϕ_γ=1.2574
E_γ=1.67212, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.56305
Q^2=21.5386, x_B=0.923711, t=-6.60174, W^2=2.6587, y=0.967382

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4112 p_x= 0.0000 p_y= 0.0000 p_z= -2.4112
P E= 2.5872 p_x= 0.0000 p_y= 0.0000 p_z= 2.4112
ℓ' E= 2.2332 p_x= -0.0000 p_y= -0.0000 p_z= 2.2332
P' E= 1.0930 p_x= -0.0000 p_y= -0.0000 p_z= -0.5611
q_γ E= 1.6721 p_x= 0.0000 p_y= 0.0000 p_z= -1.6721

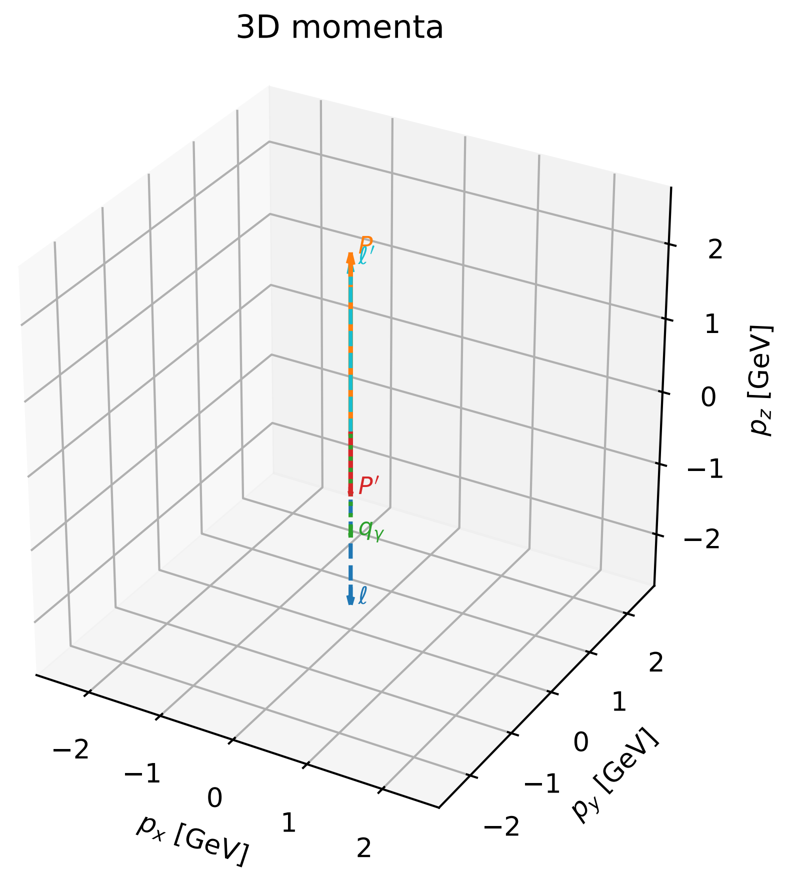




electron: polarization cluster P4, local minimum 8; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



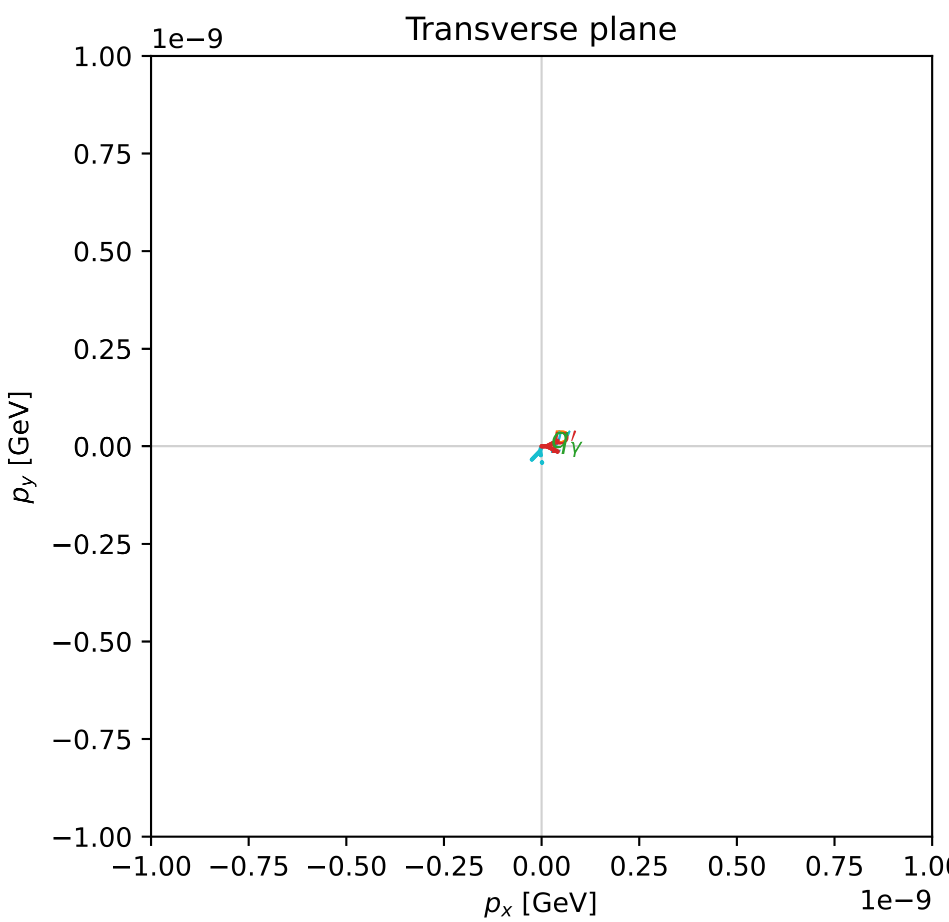
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

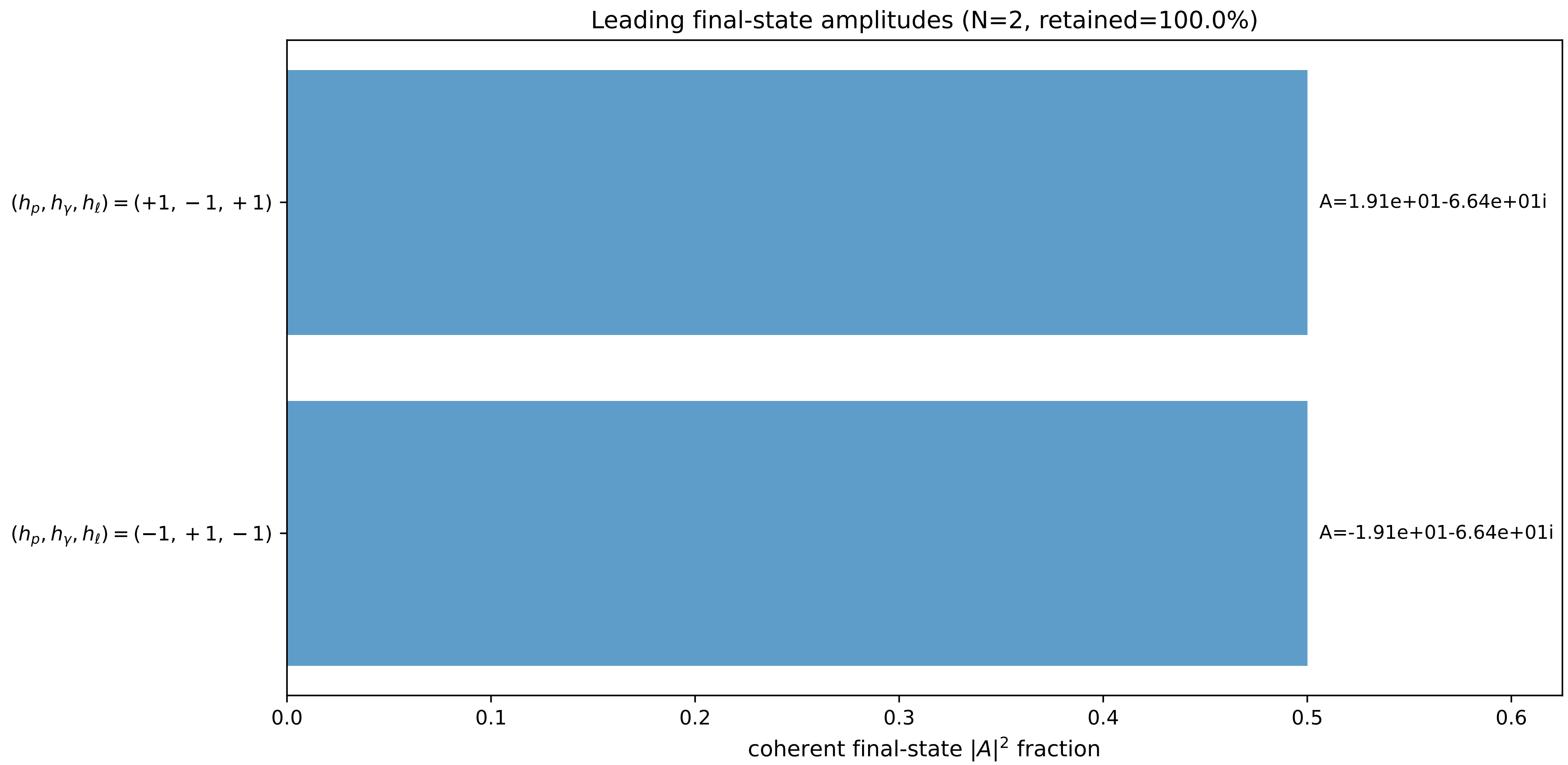


dghz_mixing_angles_polarization_04_local_minimum_9 d_{\rm GHZ}=2.43084e-08
region: polarization_cluster_04_local_minimum_9
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.380064 rad
incoming lepton: -0.723782|+> +0.690029|->
proton mixing angle: alpha_p=2.3800831 rad
incoming proton: -0.723795|+> +0.690015|->

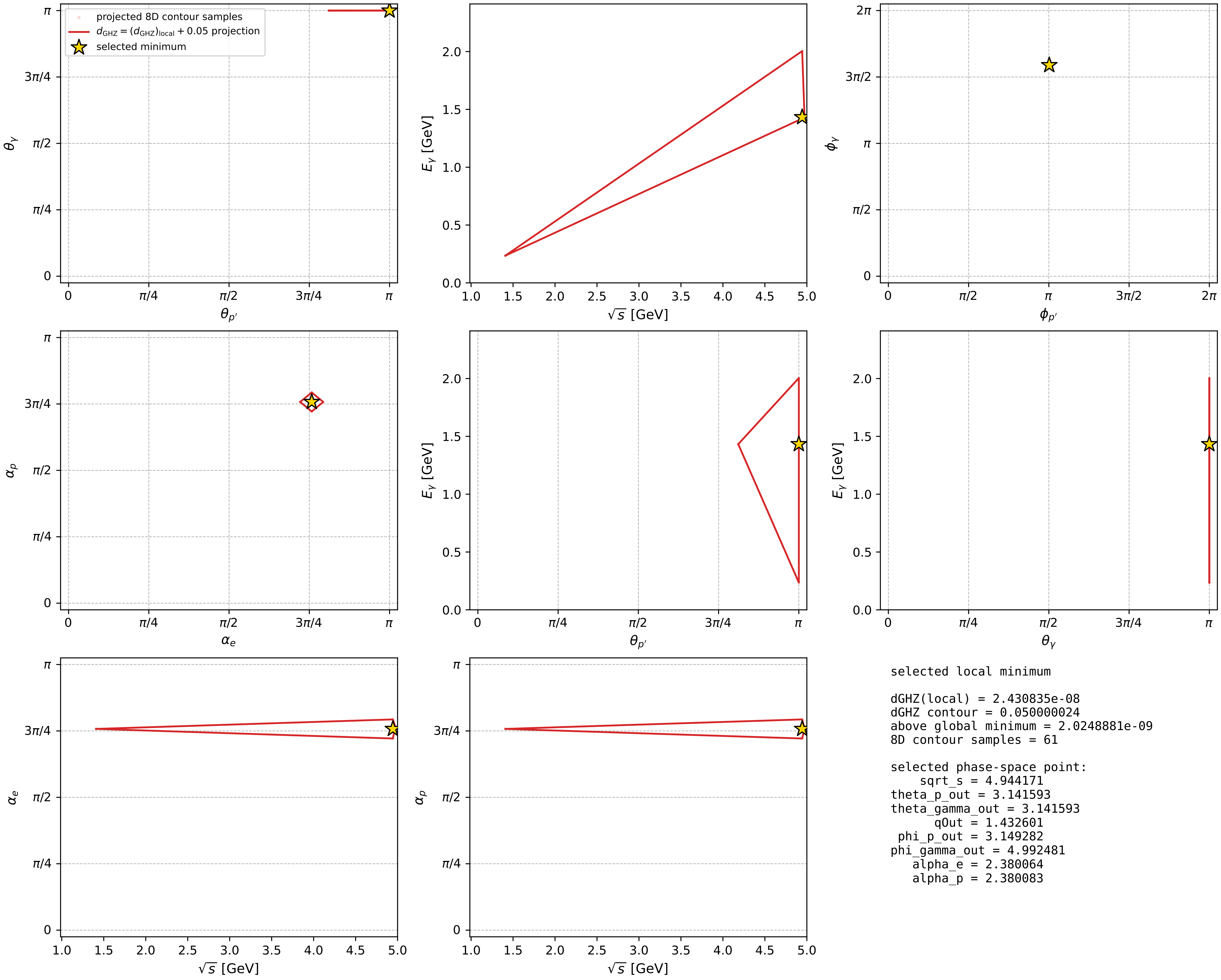
s=24.4448, $\sqrt{s}$=4.94417, $|\vec{P}|$=2.38311, $|\vec{P}'|$=0.827879
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=3.14928, ϕ_γ=4.99248
E_γ=1.4326, $\theta_{\ell'}$=0, $\phi_{\ell'}$=1.26818
Q^2=21.5479, x_B=0.946724, t=-8.5944, W^2=2.09243, y=0.965859

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3831 p_x= 0.0000 p_y= 0.0000 p_z= -2.3831
P E= 2.5611 p_x= 0.0000 p_y= 0.0000 p_z= 2.3831
ℓ' E= 2.2605 p_x= 0.0000 p_y= 0.0000 p_z= 2.2605
P' E= 1.2511 p_x= -0.0000 p_y= -0.0000 p_z= -0.8279
q_γ E= 1.4326 p_x= 0.0000 p_y= -0.0000 p_z= -1.4326

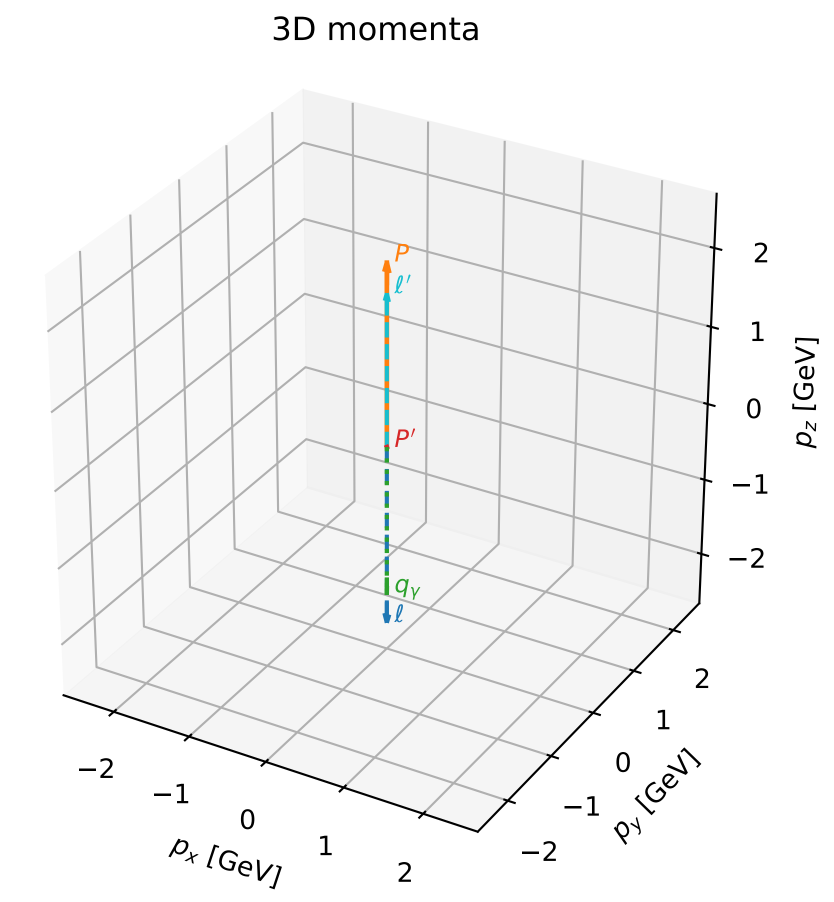




electron: polarization cluster P4, local minimum 9; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



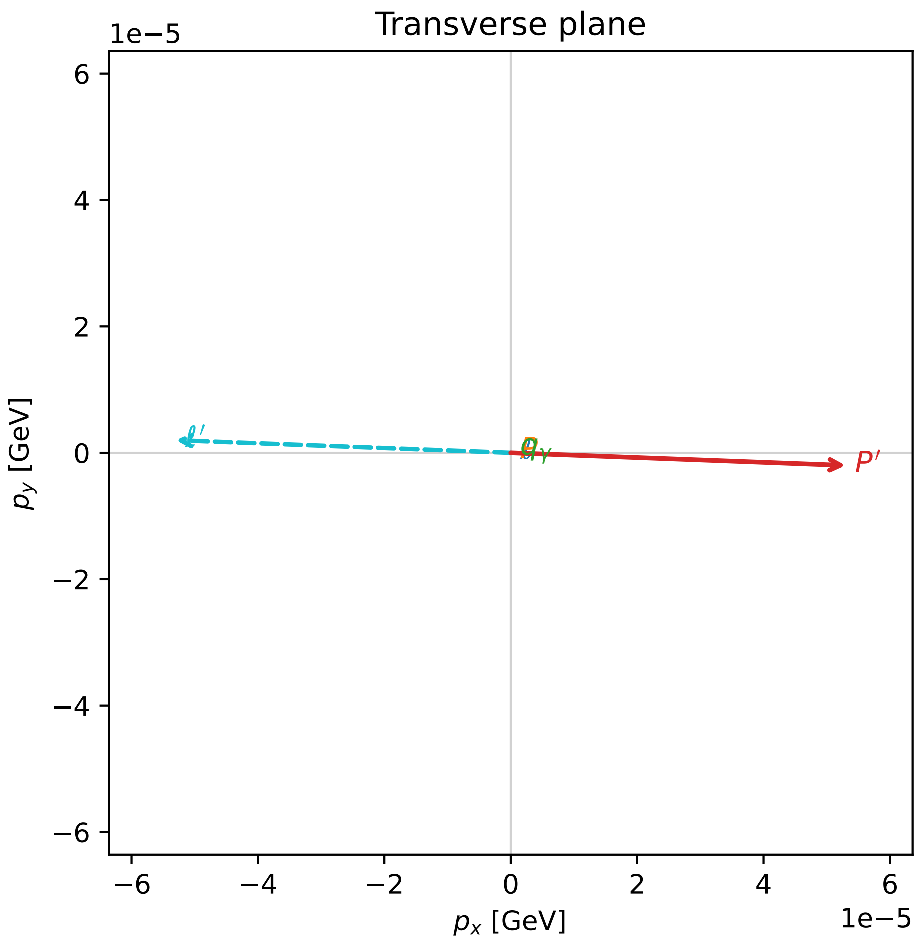
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

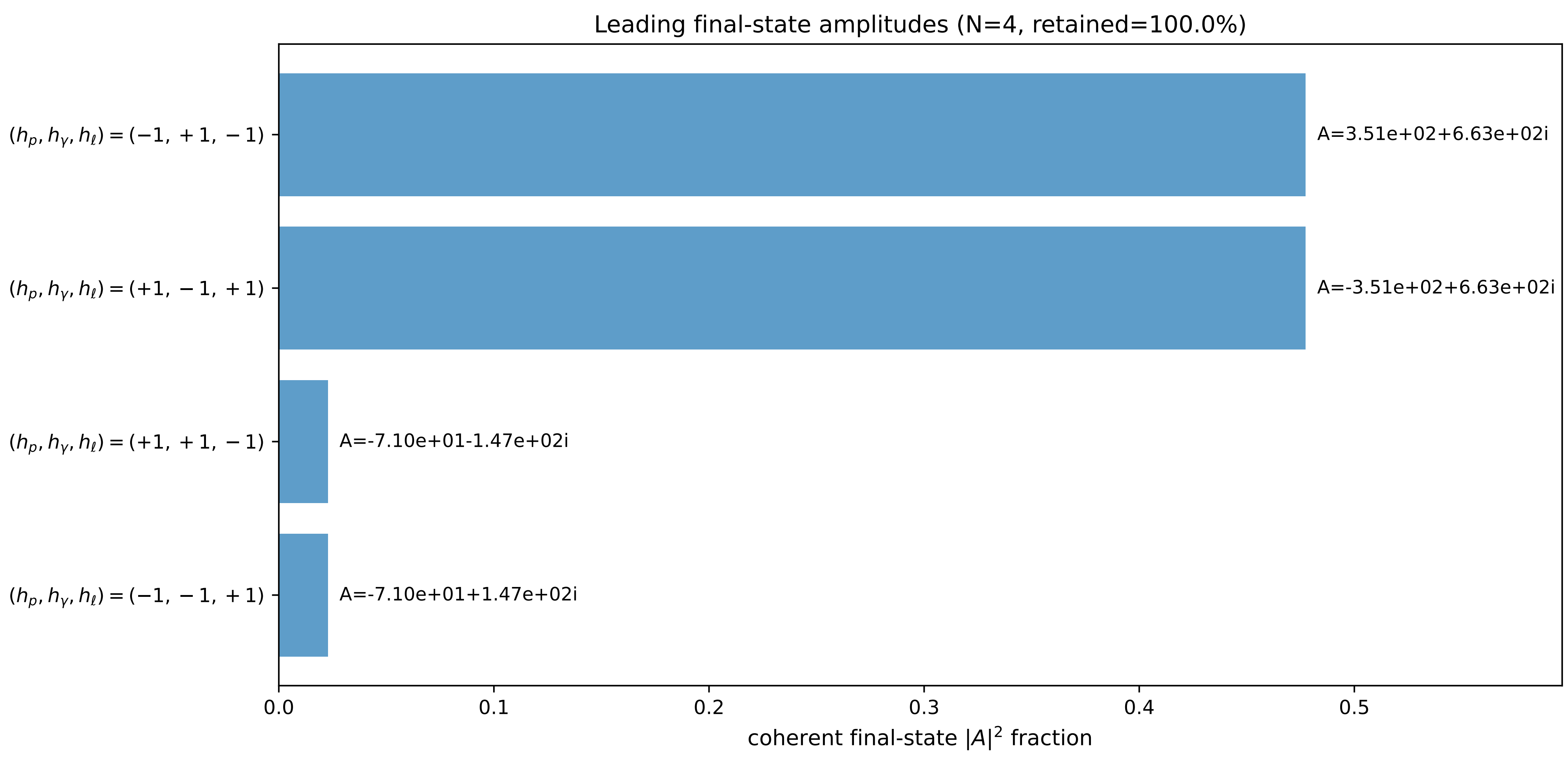


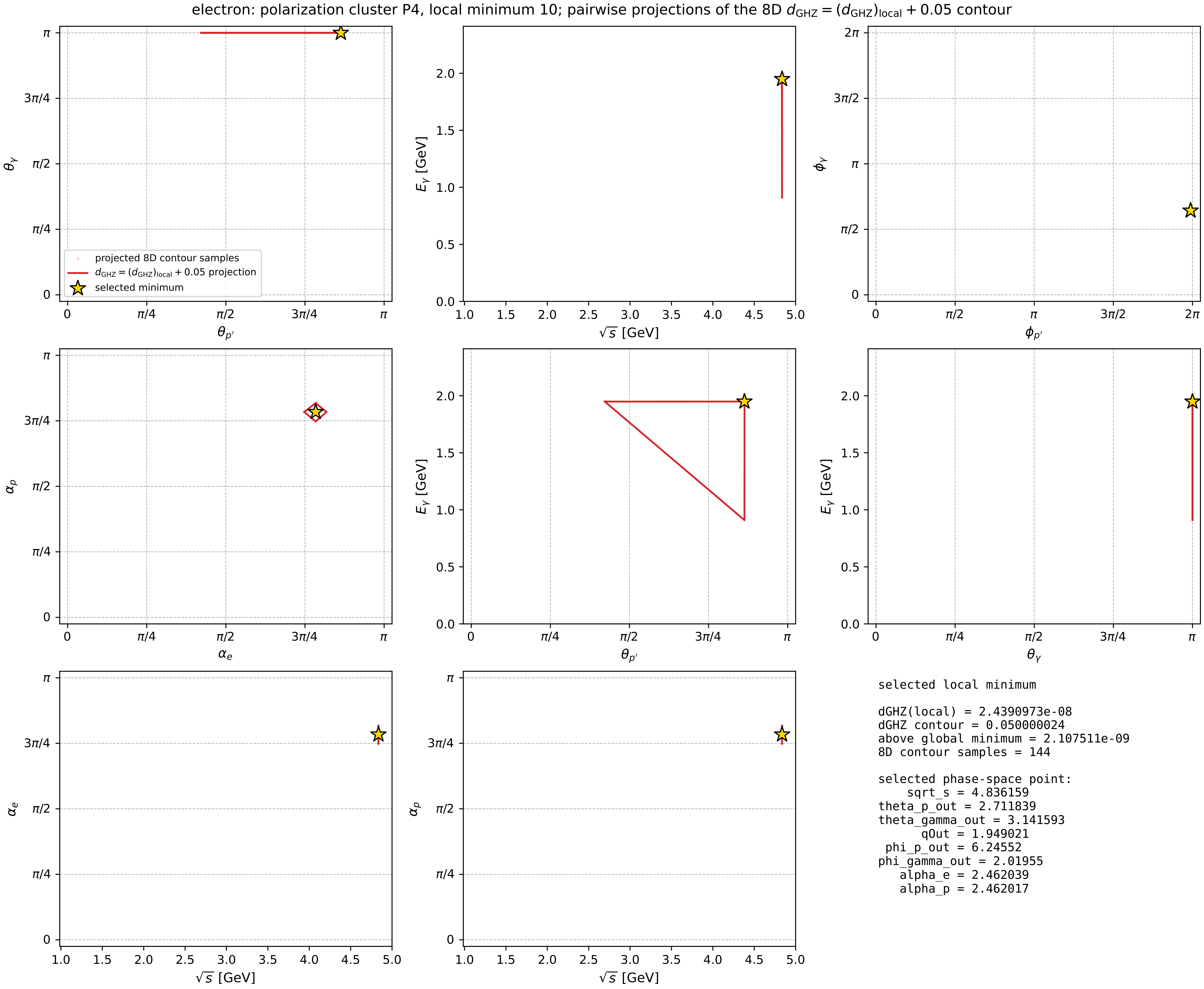

```
dghz_mixing_angles_polarization_04_local_minimum_10  d_{\rm GHZ}=2.4391e-08
region: polarization_cluster_04_local_minimum_10
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4620394 rad
incoming lepton: -0.777854|+> +0.628446|->
proton mixing angle: alpha_p=2.4620173 rad
incoming proton: -0.77784|+> +0.628463|->
```

s=23.3884, $\sqrt{s}$=4.83616, $|\vec{P}|$=2.32711, $|\vec{P}'|$=0.000127265
$\theta_{p'}$=2.71184, θ_γ=3.14159, $\phi_{p'}$=6.24552, ϕ_γ=2.01955
E_γ=1.94902, $\theta_{\ell'}$=2.72042e-05, $\phi_{\ell'}$=3.10393
Q^2=18.1435, x_B=0.832293, t=-2.94782, W^2=4.53576, y=0.968491

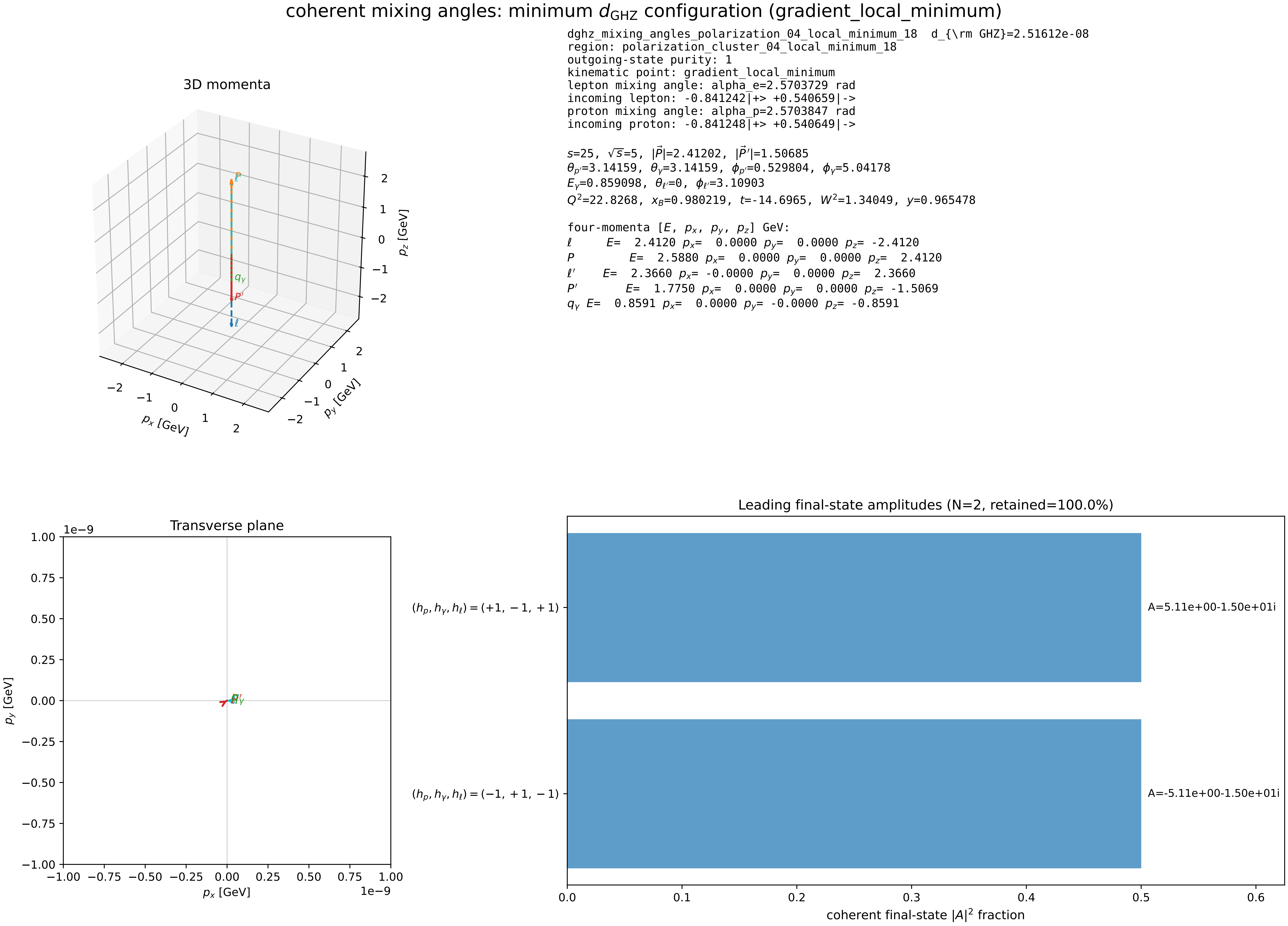
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3271 p_x= 0.0000 p_y= 0.0000 p_z= -2.3271
P E= 2.5090 p_x= 0.0000 p_y= 0.0000 p_z= 2.3271
ℓ' E= 1.9491 p_x= -0.0001 p_y= 0.0000 p_z= 1.9491
P' E= 0.9380 p_x= 0.0001 p_y= -0.0000 p_z= -0.0001
q_γ E= 1.9490 p_x= -0.0000 p_y= 0.0000 p_z= -1.9490

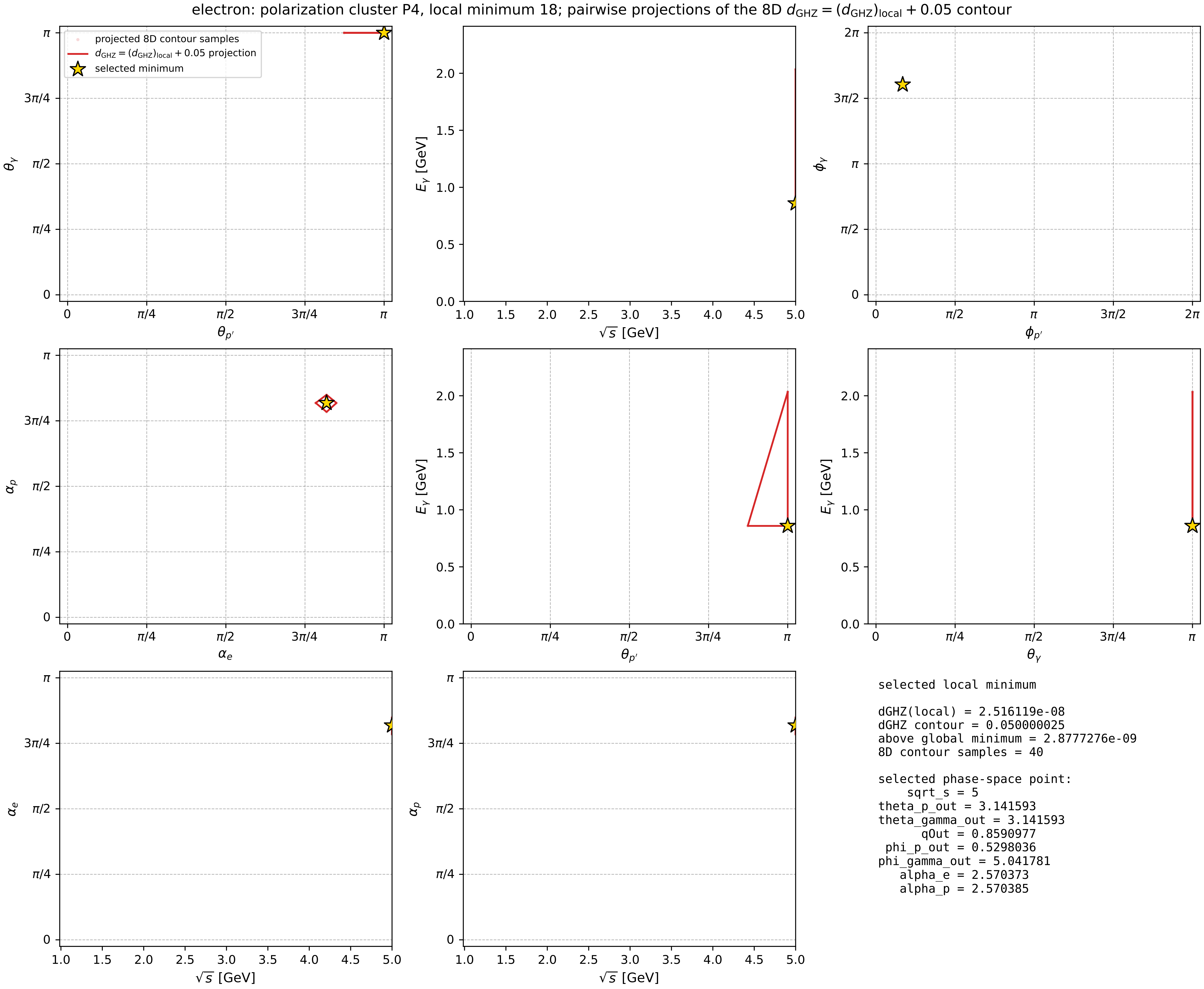




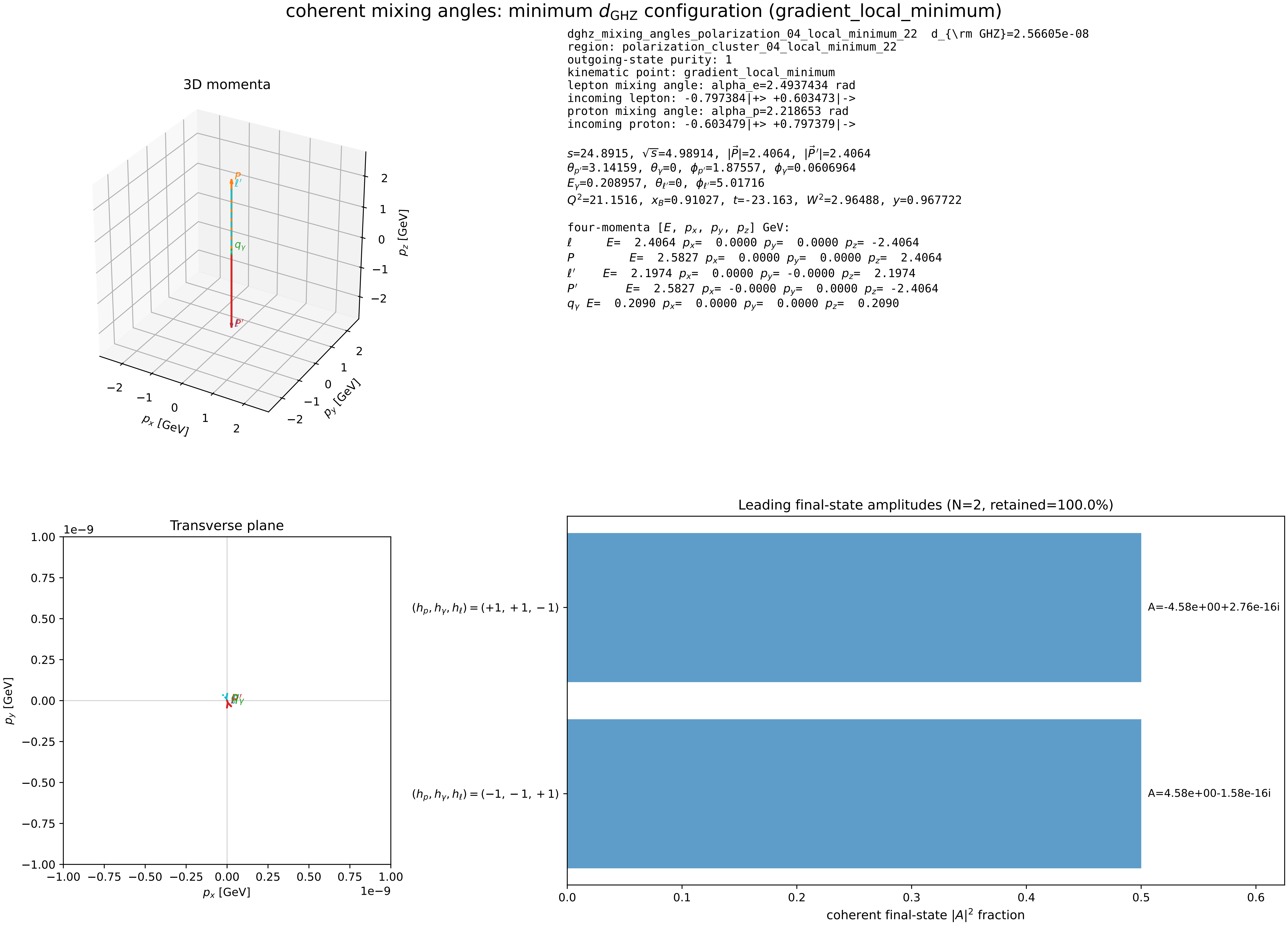


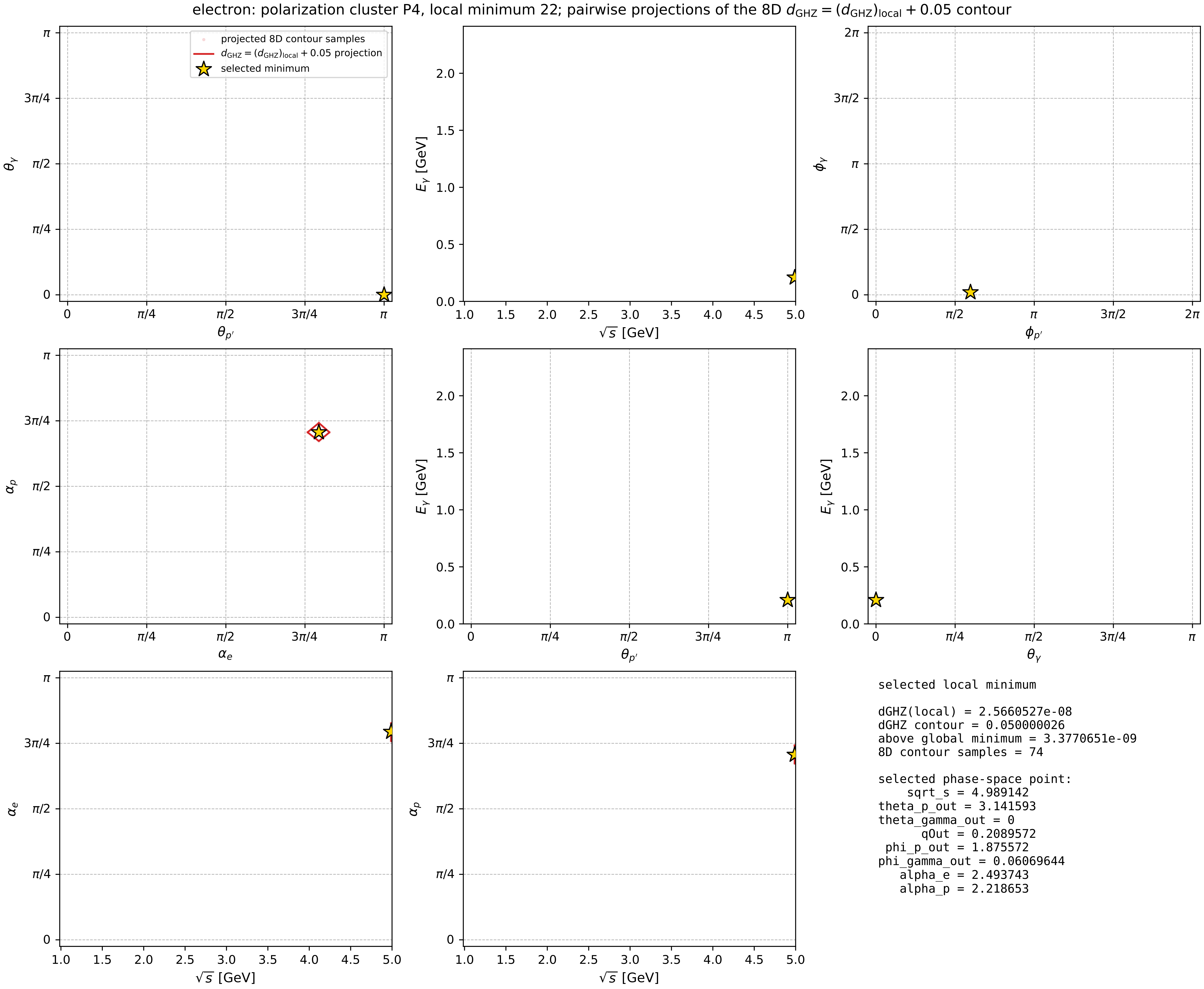
electron: polarization cluster P4, local minimum 10; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 2.4390973e-08
dGHZ contour = 0.050000024
above global minimum = 2.107511e-09
8D contour samples = 144
selected phase-space point:
sqrt_s = 4.836159
theta_p_out = 2.711839
theta_gamma_out = 3.141593
qOut = 1.949021
phi_p_out = 6.24552
phi_gamma_out = 2.01955
alpha_e = 2.462039
alpha_p = 2.462017


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_18 d_{\rm GHZ}=2.51612e-08
region: polarization_cluster_04_local_minimum_18
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.5703729 rad
incoming lepton: -0.841242|+> +0.540659|->
proton mixing angle: alpha_p=2.5703847 rad
incoming proton: -0.841248|+> +0.540649|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=1.50685
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=0.529804, ϕ_γ=5.04178
E_γ=0.859098, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.10903
Q^2=22.8268, x_B=0.980219, t=-14.6965, W^2=1.34049, y=0.965478
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.3660 p_x= -0.0000 p_y= 0.0000 p_z= 2.3660
P' E= 1.7750 p_x= 0.0000 p_y= 0.0000 p_z= -1.5069
q_γ E= 0.8591 p_x= 0.0000 p_y= -0.0000 p_z= -0.8591
Transverse plane
1e−9
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=5.11e+00-1.50e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=-5.11e+00-1.50e+01i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P4, local minimum 18; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 2.516119e-08
dGHZ contour = 0.050000025
above global minimum = 2.8777276e-09
8D contour samples = 40
selected phase-space point:
sqrt_s = 5
theta_p_out = 3.141593
theta_gamma_out = 3.141593
qOut = 0.8590977
phi_p_out = 0.5298036
phi_gamma_out = 5.041781
alpha_e = 2.570373
alpha_p = 2.570385


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_22 d_{\rm GHZ}=2.56605e-08
region: polarization_cluster_04_local_minimum_22
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4937434 rad
incoming lepton: -0.797384|+> +0.603473|->
proton mixing angle: alpha_p=2.218653 rad
incoming proton: -0.603479|+> +0.797379|->
s=24.8915, $\sqrt{s}$=4.98914, $|\vec{P}|$=2.4064, $|\vec{P}'|$=2.4064
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=1.87557, ϕ_γ=0.0606964
E_γ=0.208957, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.01716
Q^2=21.1516, x_B=0.91027, t=-23.163, W^2=2.96488, y=0.967722
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4064 p_x= 0.0000 p_y= 0.0000 p_z= -2.4064
P E= 2.5827 p_x= 0.0000 p_y= 0.0000 p_z= 2.4064
ℓ' E= 2.1974 p_x= 0.0000 p_y= -0.0000 p_z= 2.1974
P' E= 2.5827 p_x= -0.0000 p_y= 0.0000 p_z= -2.4064
q_γ E= 0.2090 p_x= 0.0000 p_y= 0.0000 p_z= 0.2090
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e−9
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-4.58e+00+2.76e-16i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=4.58e+00-1.58e-16i
coherent final-state $|A|^2$ fraction



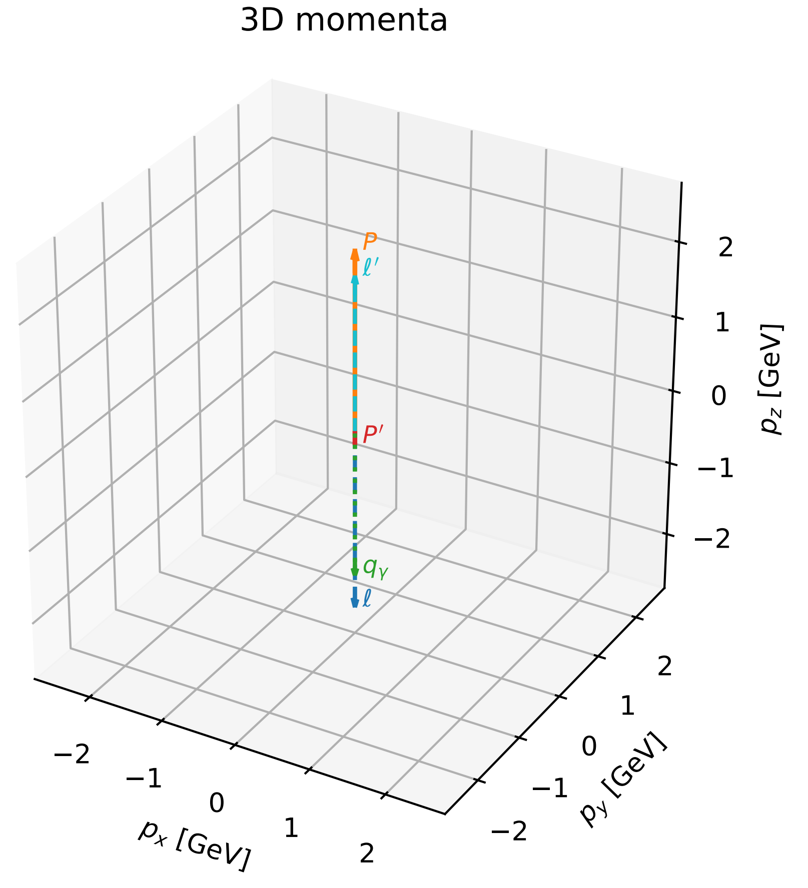
electron: polarization cluster P4, local minimum 22; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 2.5660527e-08
dGHZ contour = 0.050000026
above global minimum = 3.3770651e-09
8D contour samples = 74
selected phase-space point:
sqrt_s = 4.989142
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 0.2089572
phi_p_out = 1.875572
phi_gamma_out = 0.06069644
alpha_e = 2.493743
alpha_p = 2.218653

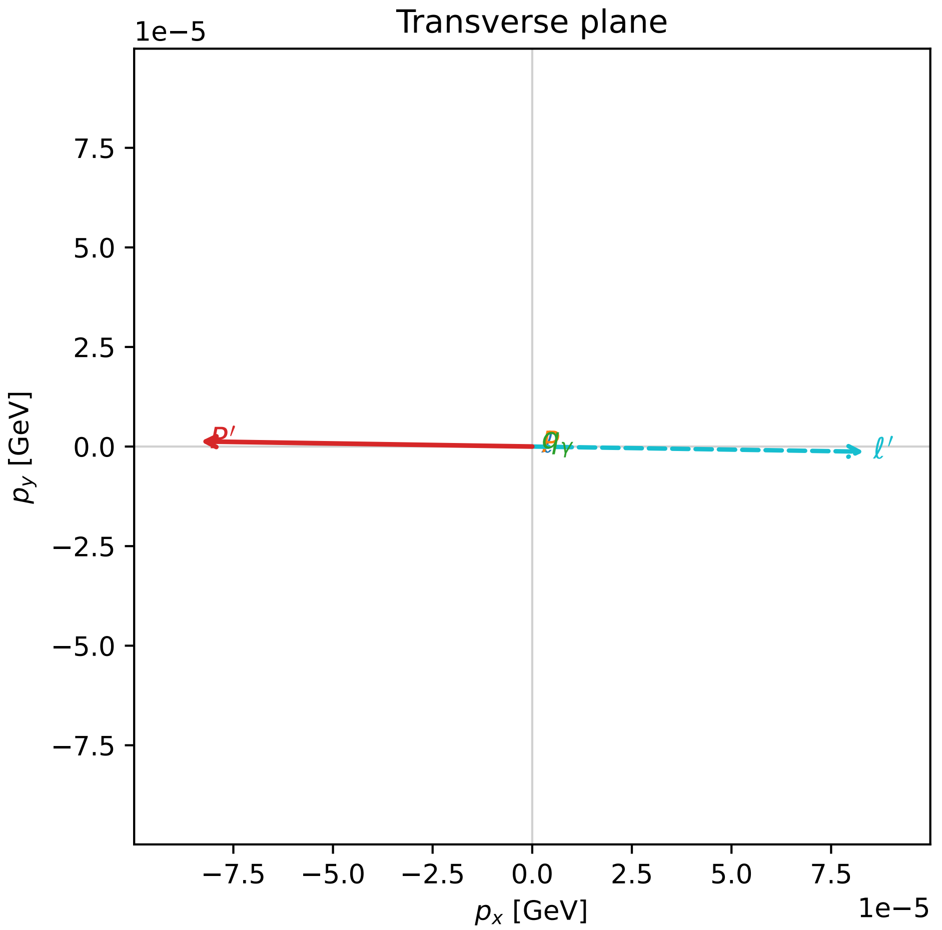
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

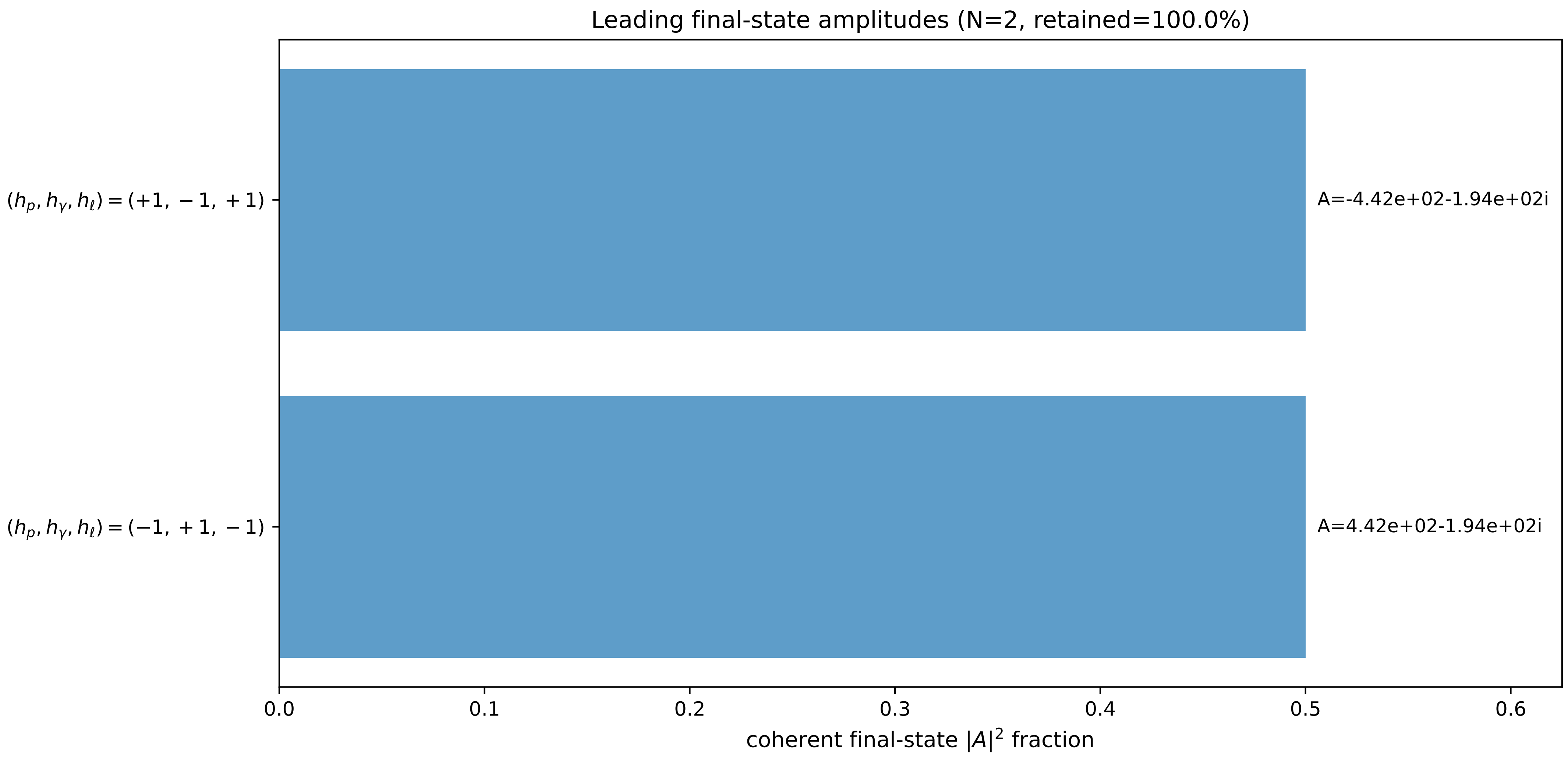
dghz_mixing_angles_polarization_04_local_minimum_29 d_{\rm GHZ}=2.60368e-08
region: polarization_cluster_04_local_minimum_29
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3616845 rad
incoming lepton: -0.710978|+> +0.703214|->
proton mixing angle: alpha_p=2.3617041 rad
incoming proton: -0.710992|+> +0.7032|->

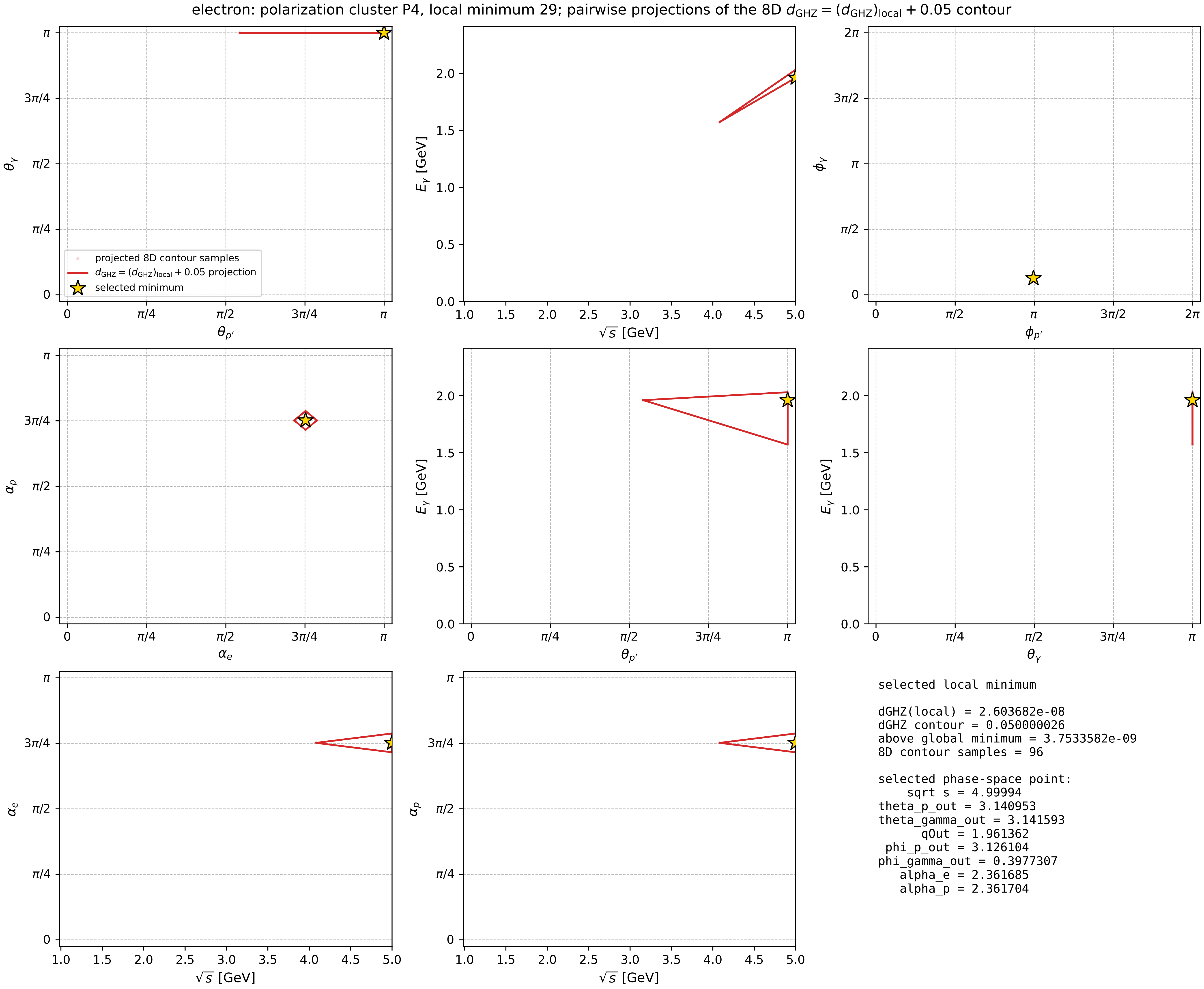
s=24.9994, $\sqrt{s}$=4.99994, $|\vec{P}|$=2.41198, $|\vec{P}'|$=0.13022
$\theta_{p'}$=3.14095, θ_{γ}=3.14159, $\phi_{p'}$=3.1261, ϕ_{γ}=0.397731
E_{γ}=1.96136, $\theta_{\ell'}$=3.98045e-05, $\phi_{\ell'}$=6.2677
Q^2=20.1795, x_B=0.86298, t=-3.77005, W^2=4.08383, y=0.969481

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0916 p_x= 0.0001 p_y= -0.0000 p_z= 2.0916
P' E= 0.9470 p_x= -0.0001 p_y= 0.0000 p_z= -0.1302
q_{γ} E= 1.9614 p_x= 0.0000 p_y= 0.0000 p_z= -1.9614

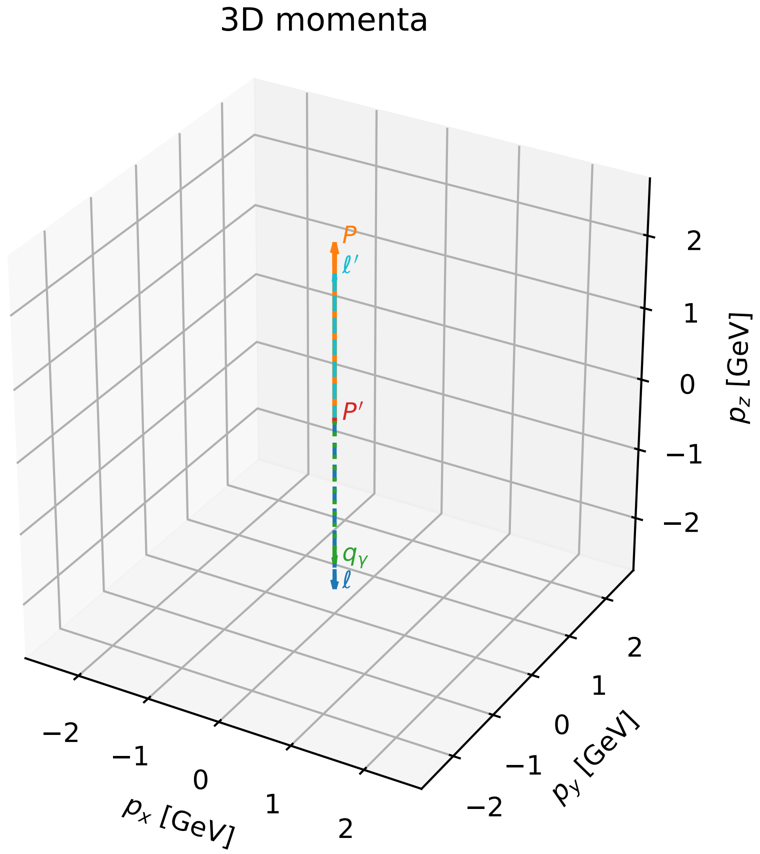







electron: polarization cluster P4, local minimum 29; pairwise projections of the 8D $d_{GHZ} = (d_{GHZ})_{local} + 0.05$ contour
projected 8D contour samples
$d_{GHZ} = (d_{GHZ})_{local} + 0.05$ projection
selected minimum
θ_{γ}
$\theta_{p'}$
E_{γ} [GeV]
$\sqrt{s}$ [GeV]
ϕ_{γ}
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 2.603682e-08
dGHZ contour = 0.050000026
above global minimum = 3.7533582e-09
8D contour samples = 96
selected phase-space point:
sqrt_s = 4.99994
theta_p_out = 3.140953
theta_gamma_out = 3.141593
qOut = 1.961362
phi_p_out = 3.126104
phi_gamma_out = 0.3977307
alpha_e = 2.361685
alpha_p = 2.361704

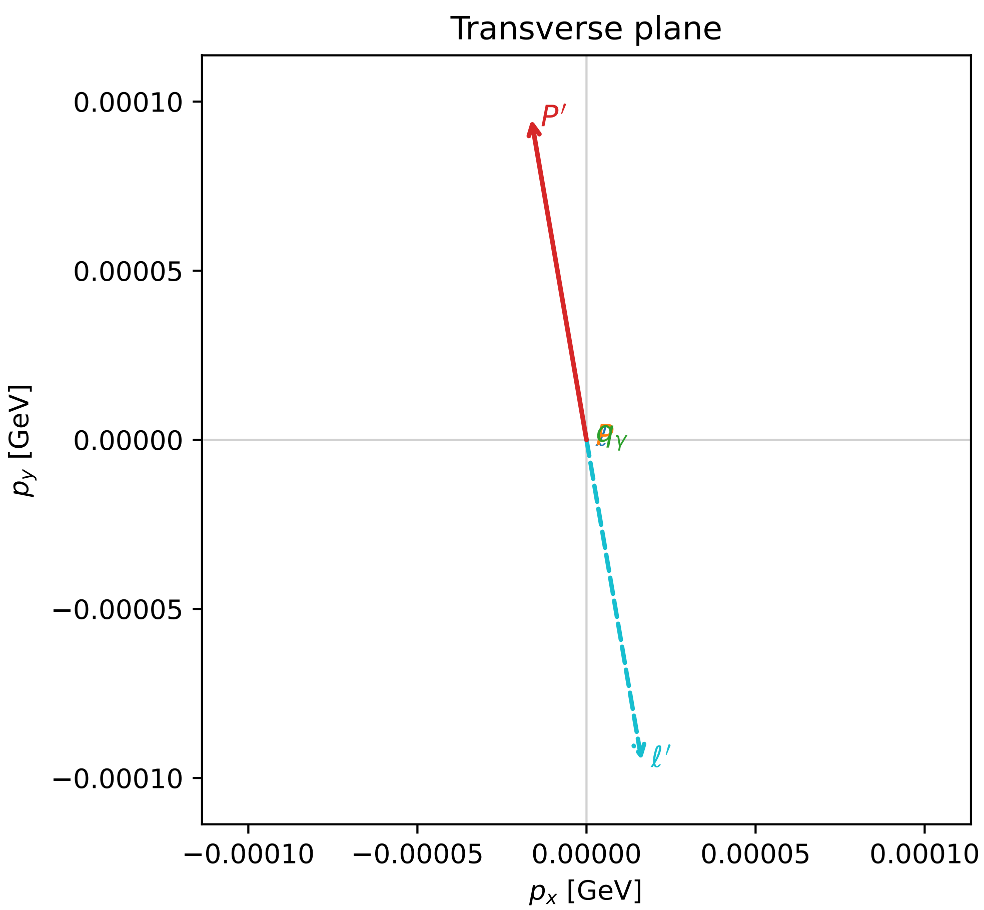
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

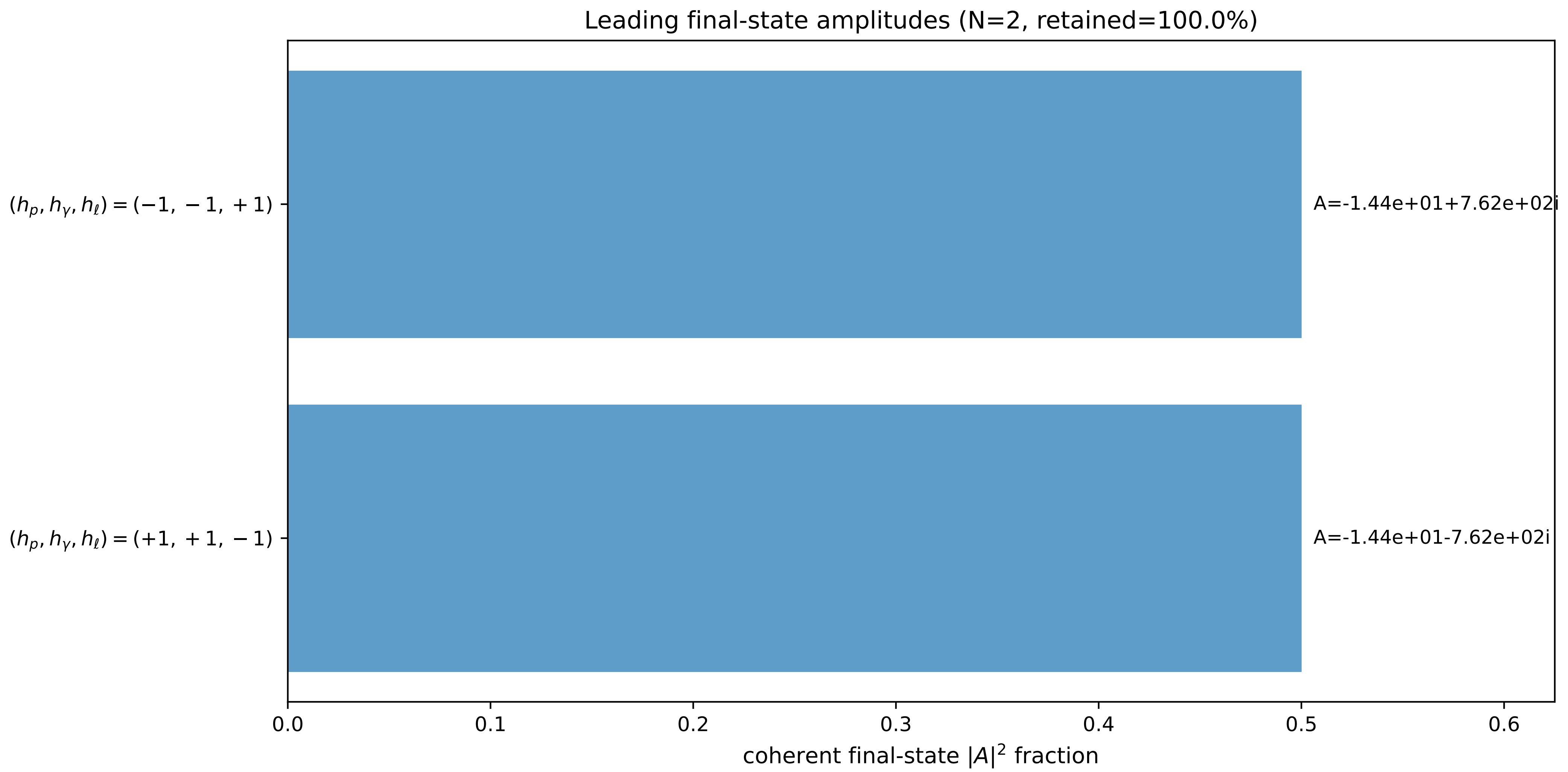


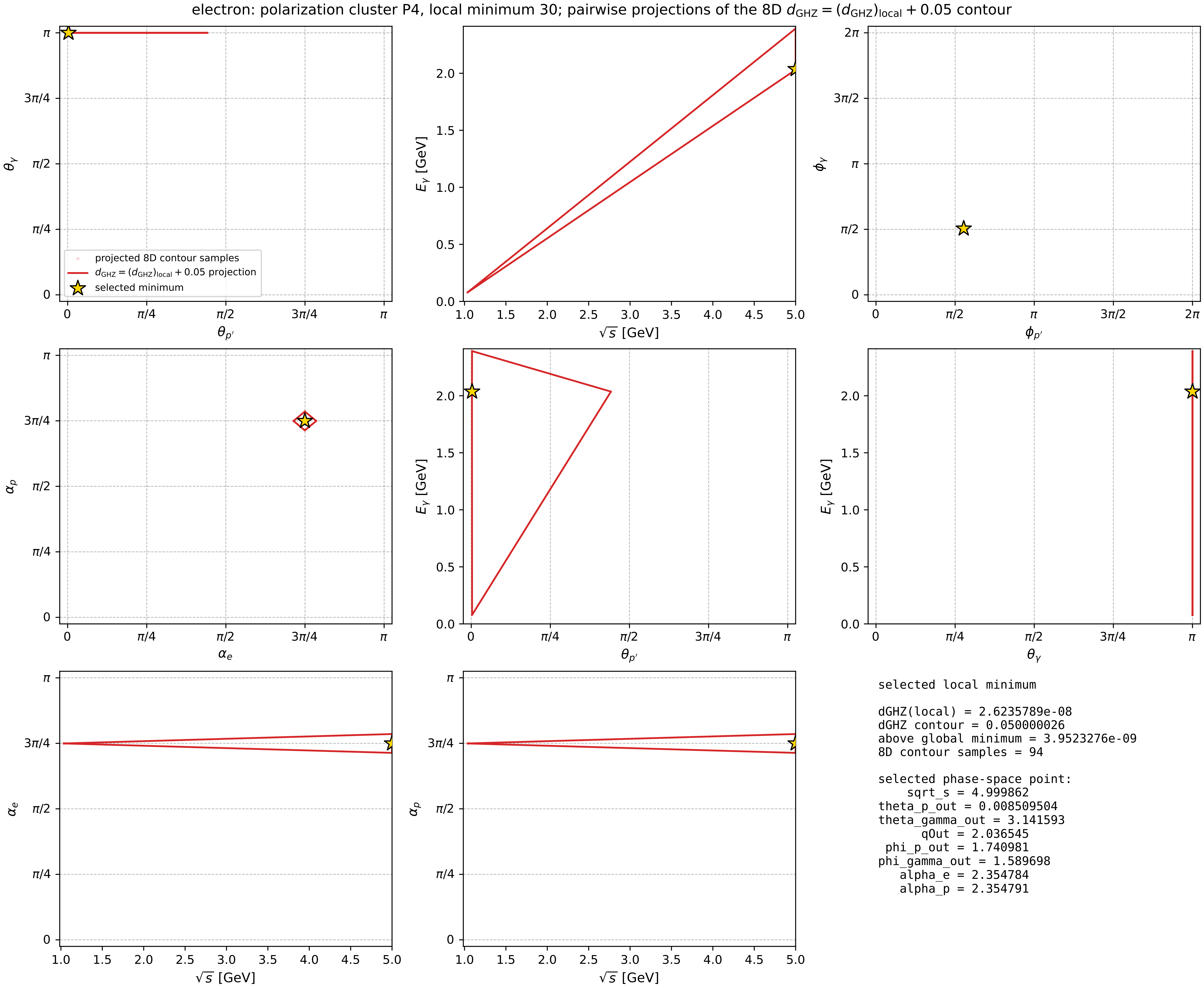
dghz_mixing_angles_polarization_04_local_minimum_30 d_{\rm GHZ}=2.62358e-08
region: polarization_cluster_04_local_minimum_30
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3547844 rad
incoming lepton: -0.706109|+> +0.708103|->
proton mixing angle: alpha_p=2.3547913 rad
incoming proton: -0.706114|+> +0.708098|->

s=24.9986, $\sqrt{s}$=4.99986, $|\vec{P}|$=2.41194, $|\vec{P}'|$=0.0112972
$\theta_{p'}$=0.0085095, θ_γ=3.14159, $\phi_{p'}$=1.74098, ϕ_γ=1.5897
E_γ=2.03655, $\theta_{\ell'}$=4.7467e-05, $\phi_{\ell'}$=4.88257
Q^2=19.5391, x_B=0.834792, t=-3.0411, W^2=4.74669, y=0.970447

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4119 p_x= 0.0000 p_y= 0.0000 p_z= -2.4119
P E= 2.5879 p_x= 0.0000 p_y= 0.0000 p_z= 2.4119
ℓ' E= 2.0252 p_x= 0.0000 p_y= -0.0001 p_z= 2.0252
P' E= 0.9381 p_x= -0.0000 p_y= 0.0001 p_z= 0.0113
q_γ E= 2.0365 p_x= -0.0000 p_y= 0.0000 p_z= -2.0365

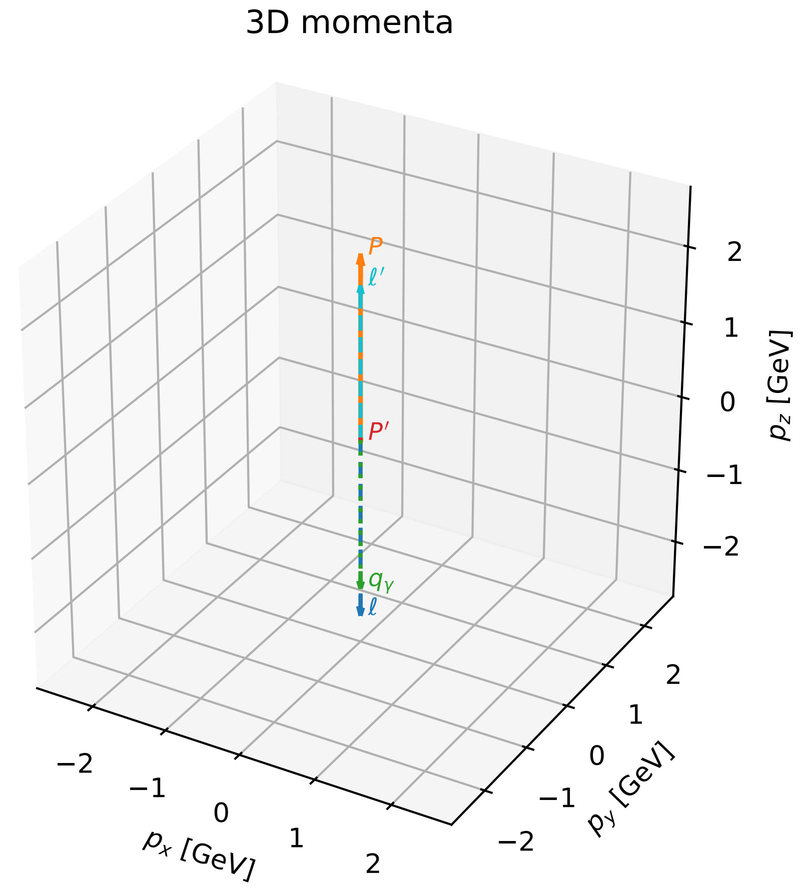






electron: polarization cluster P4, local minimum 30; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 2.6235789e-08
dGHZ contour = 0.050000026
above global minimum = 3.9523276e-09
8D contour samples = 94
selected phase-space point:
sqrt_s = 4.999862
theta_p_out = 0.008509504
theta_gamma_out = 3.141593
qOut = 2.036545
phi_p_out = 1.740981
phi_gamma_out = 1.589698
alpha_e = 2.354784
alpha_p = 2.354791

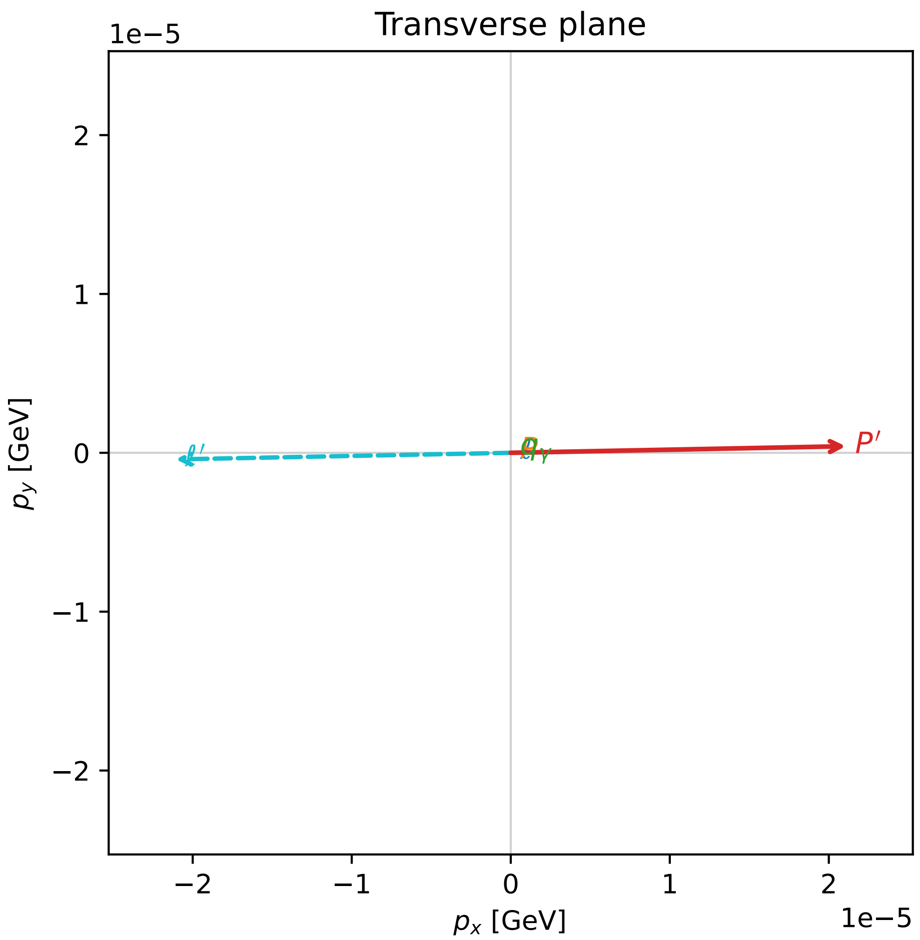
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

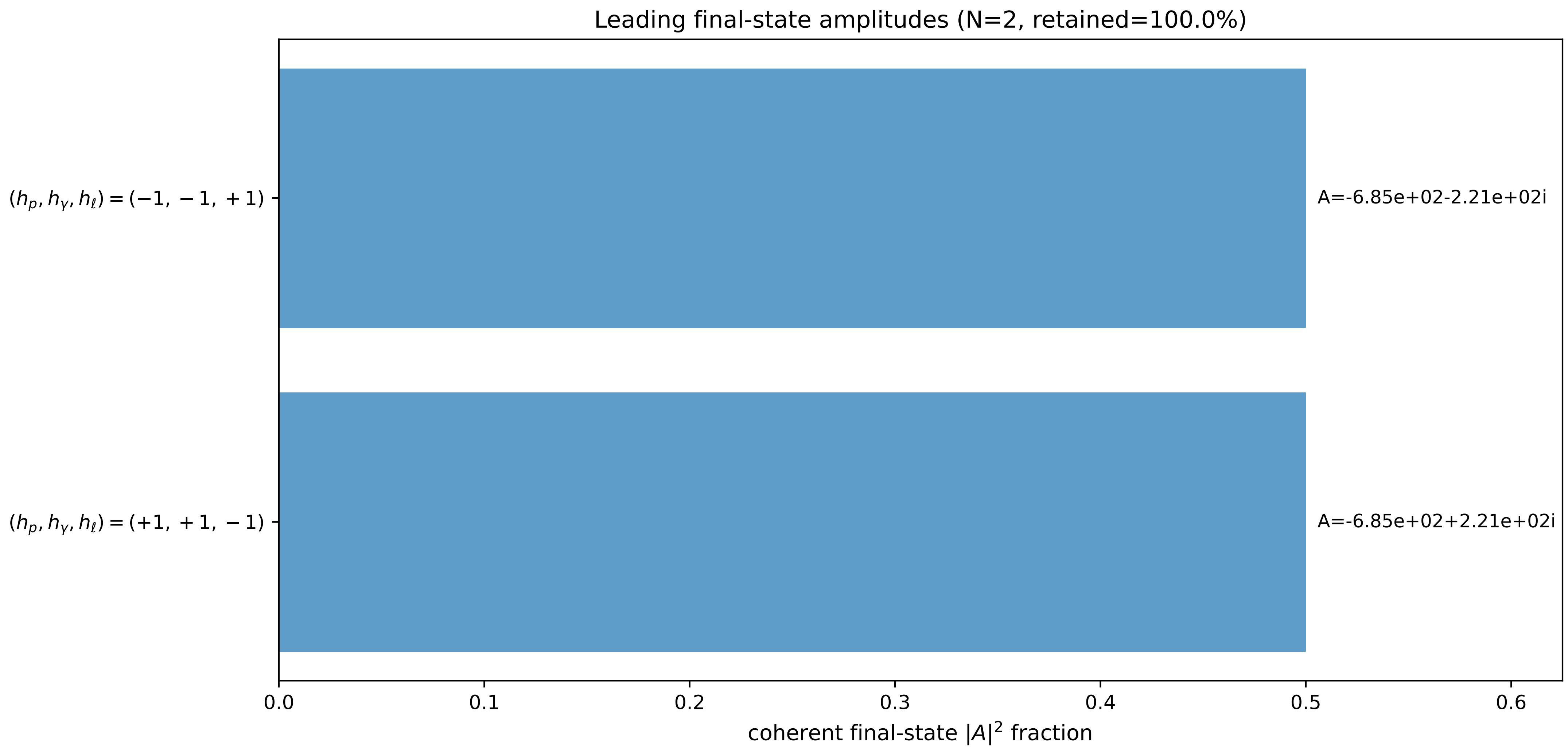


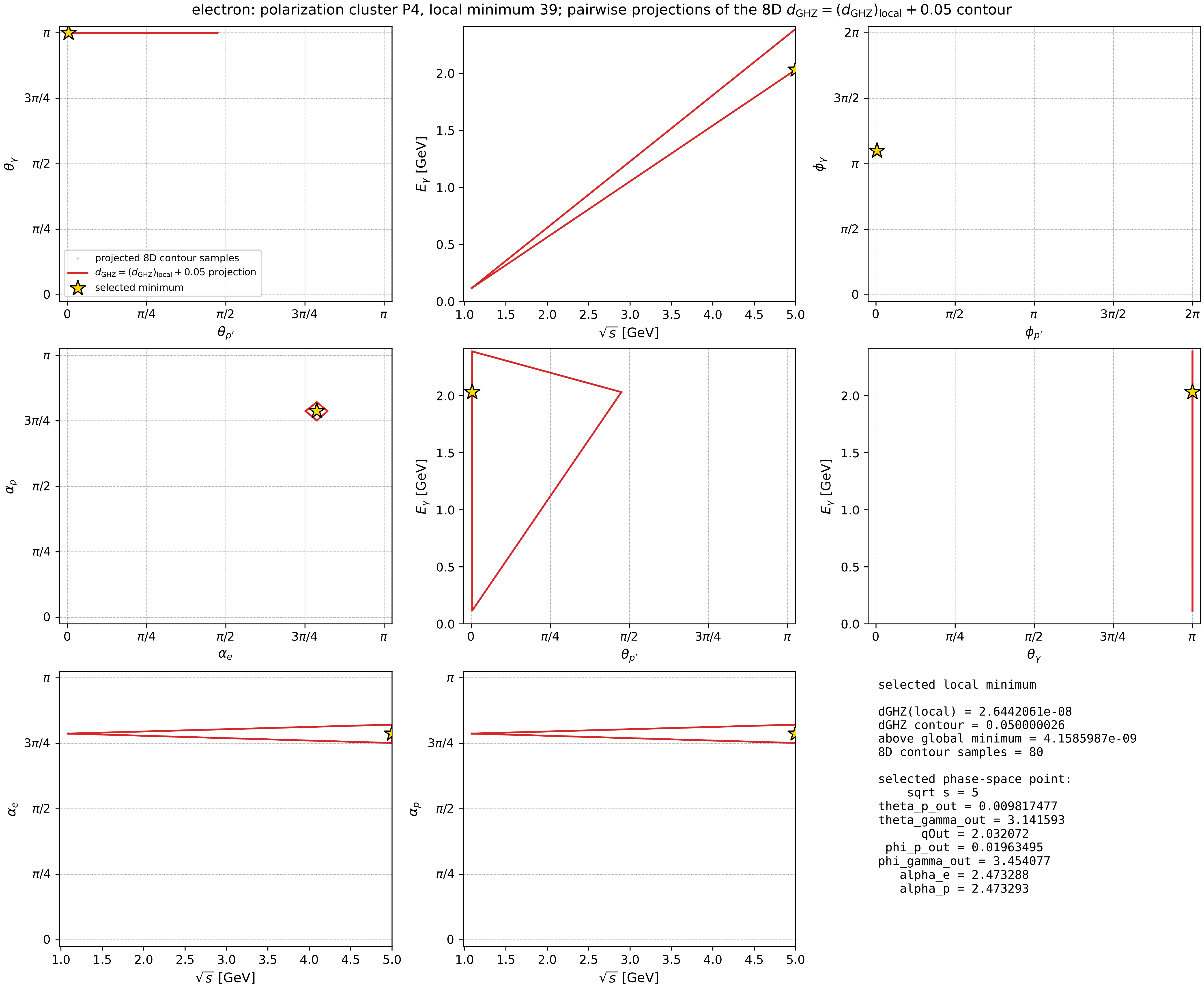
dghz_mixing_angles_polarization_04_local_minimum_39 d_{\rm GHZ}=2.64421e-08
region: polarization_cluster_04_local_minimum_39
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4732875 rad
incoming lepton: -0.784873|+> +0.619657|->
proton mixing angle: alpha_p=2.4732933 rad
incoming proton: -0.784877|+> +0.619652|->

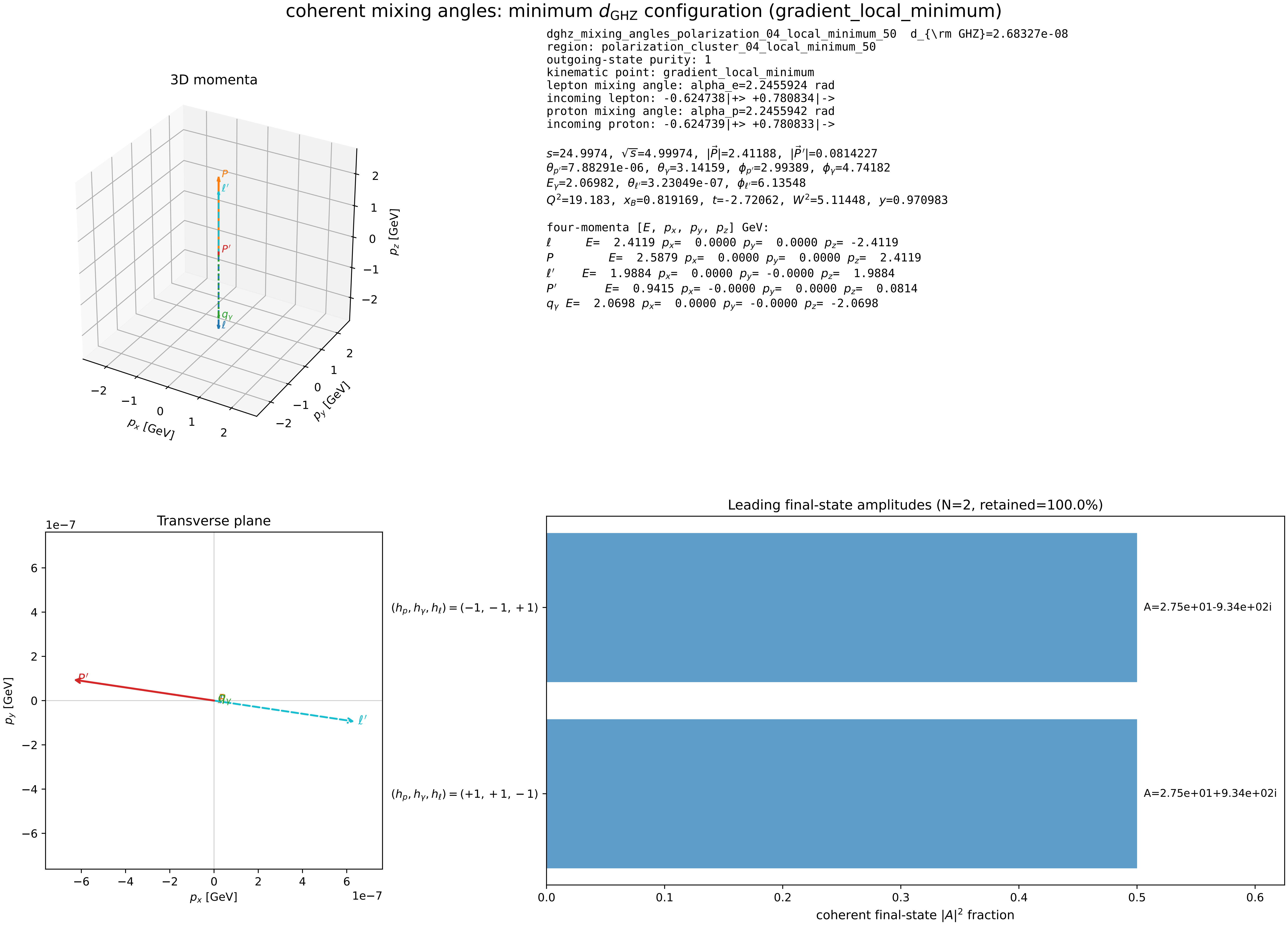
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.00214662
$\theta_{p'}$=0.00981748, θ_γ=3.14159, $\phi_{p'}$=0.019635, ϕ_γ=3.45408
E_γ=2.03207, $\theta_{\ell'}$=1.03817e-05, $\phi_{\ell'}$=3.16123
Q^2=19.5848, x_B=0.836754, t=-3.08503, W^2=4.70074, y=0.970381

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0299 p_x= -0.0000 p_y= -0.0000 p_z= 2.0299
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0021
q_γ E= 2.0321 p_x= -0.0000 p_y= -0.0000 p_z= -2.0321

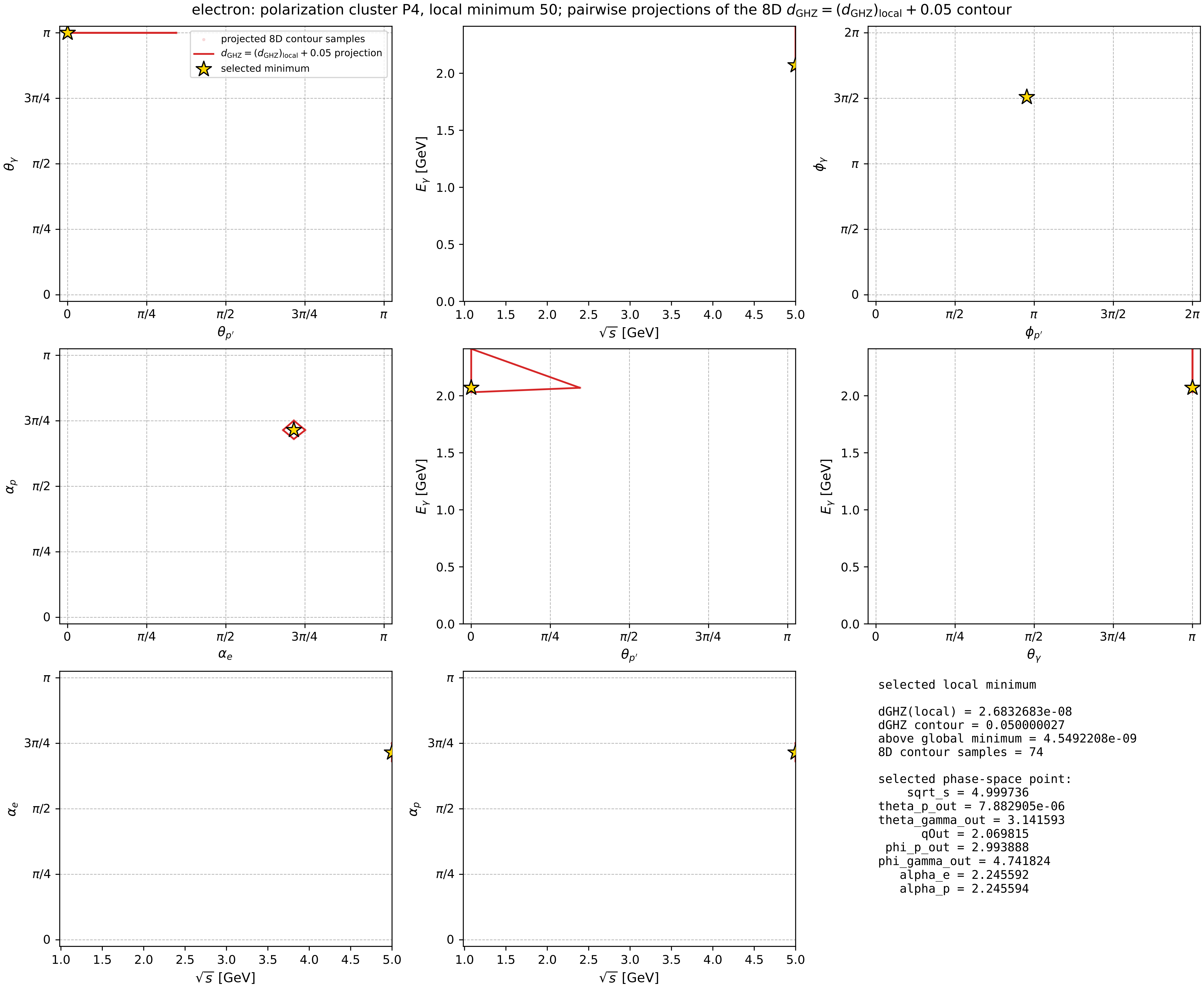




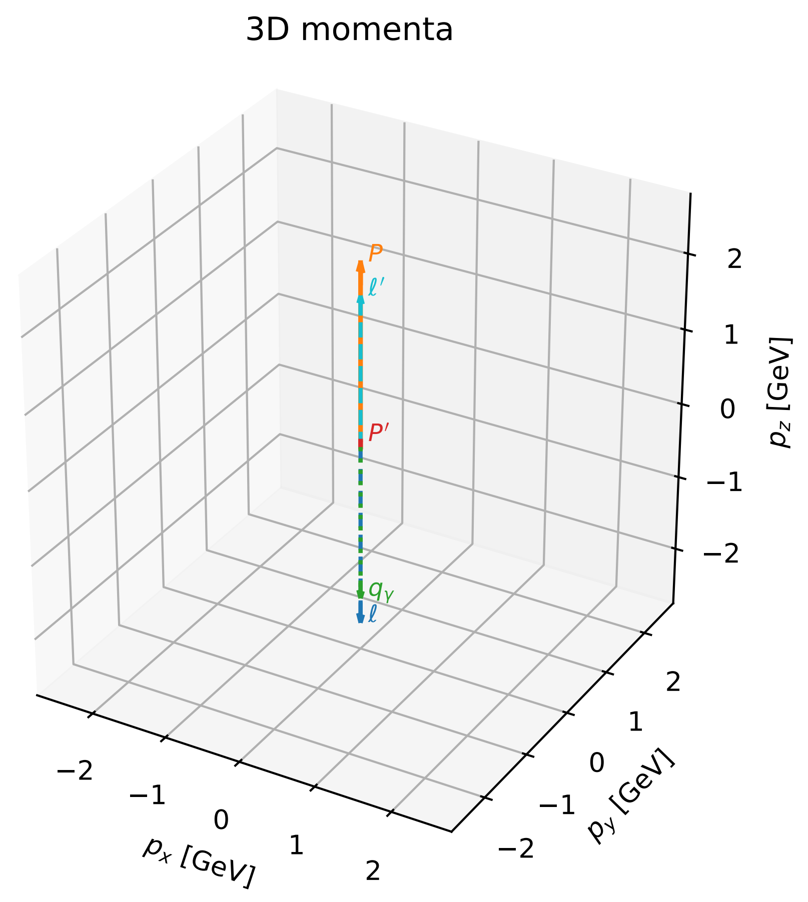

electron: polarization cluster P4, local minimum 39; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 2.6442061e-08
dGHZ contour = 0.050000026
above global minimum = 4.1585987e-09
8D contour samples = 80
selected phase-space point:
sqrt_s = 5
theta_p_out = 0.009817477
theta_gamma_out = 3.141593
qOut = 2.032072
phi_p_out = 0.01963495
phi_gamma_out = 3.454077
alpha_e = 2.473288
alpha_p = 2.473293



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_50 d_{\rm GHZ}=2.68327e-08
region: polarization_cluster_04_local_minimum_50
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2455924 rad
incoming lepton: -0.624738|+> +0.780834|->
proton mixing angle: alpha_p=2.2455942 rad
incoming proton: -0.624739|+> +0.780833|->
s=24.9974, $\sqrt{s}$=4.99974, $|\vec{P}|$=2.41188, $|\vec{P}'|$=0.0814227
$\theta_{p'}$=7.88291e-06, θ_γ=3.14159, $\phi_{p'}$=2.99389, ϕ_γ=4.74182
E_γ=2.06982, $\theta_{\ell'}$=3.23049e-07, $\phi_{\ell'}$=6.13548
Q^2=19.183, x_B=0.819169, t=-2.72062, W^2=5.11448, y=0.970983
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4119 p_x= 0.0000 p_y= 0.0000 p_z= -2.4119
P E= 2.5879 p_x= 0.0000 p_y= 0.0000 p_z= 2.4119
ℓ' E= 1.9884 p_x= 0.0000 p_y= -0.0000 p_z= 1.9884
P' E= 0.9415 p_x= -0.0000 p_y= 0.0000 p_z= 0.0814
q_γ E= 2.0698 p_x= 0.0000 p_y= -0.0000 p_z= -2.0698
Transverse plane
1e-7
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=2.75e+01-9.34e+02i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=2.75e+01+9.34e+02i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 50; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 2.6832683e-08
dGHZ contour = 0.050000027
above global minimum = 4.5492208e-09
8D contour samples = 74
selected phase-space point:
sqrt_s = 4.999736
theta_p_out = 7.882905e-06
theta_gamma_out = 3.141593
qOut = 2.069815
phi_p_out = 2.993888
phi_gamma_out = 4.741824
alpha_e = 2.245592
alpha_p = 2.245594

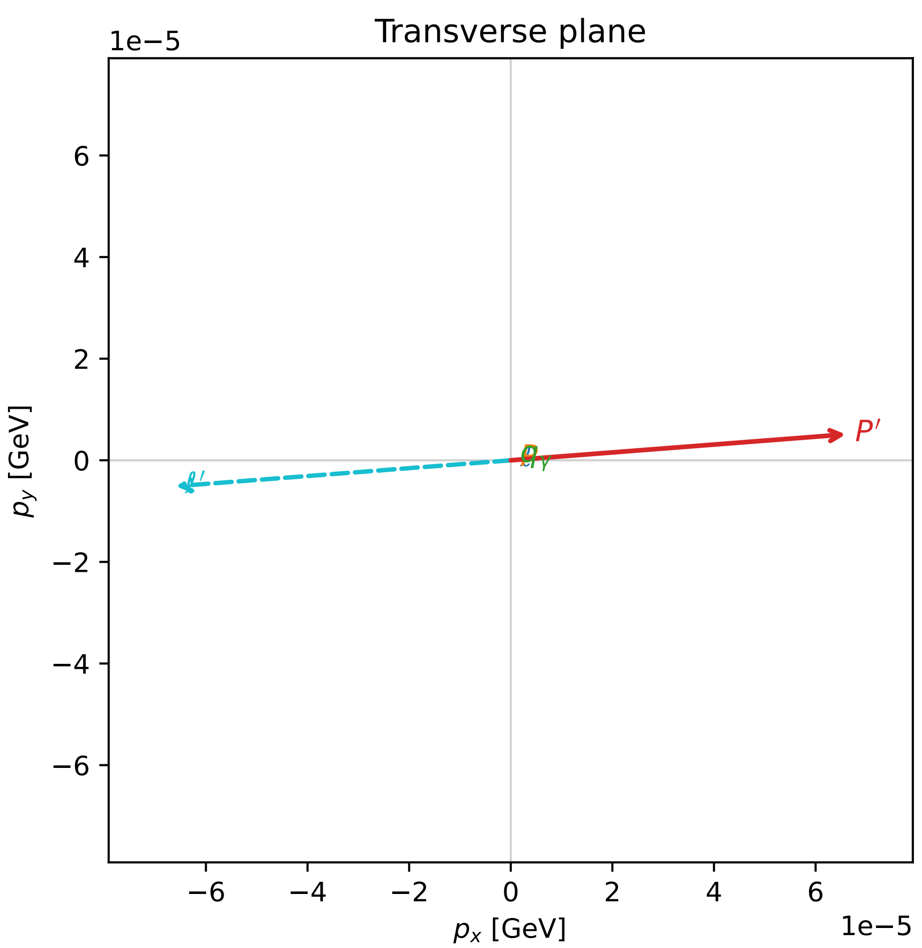
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

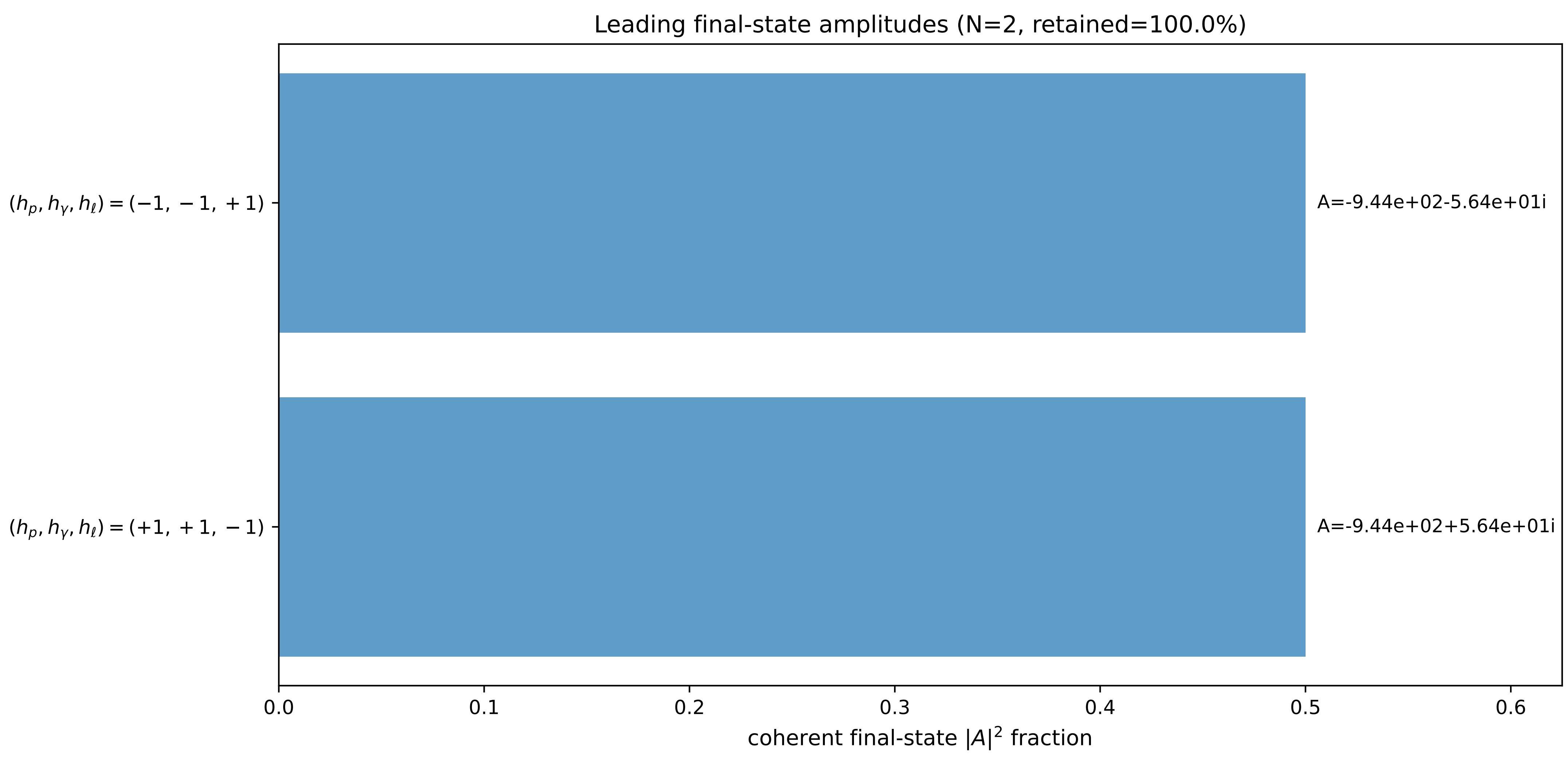


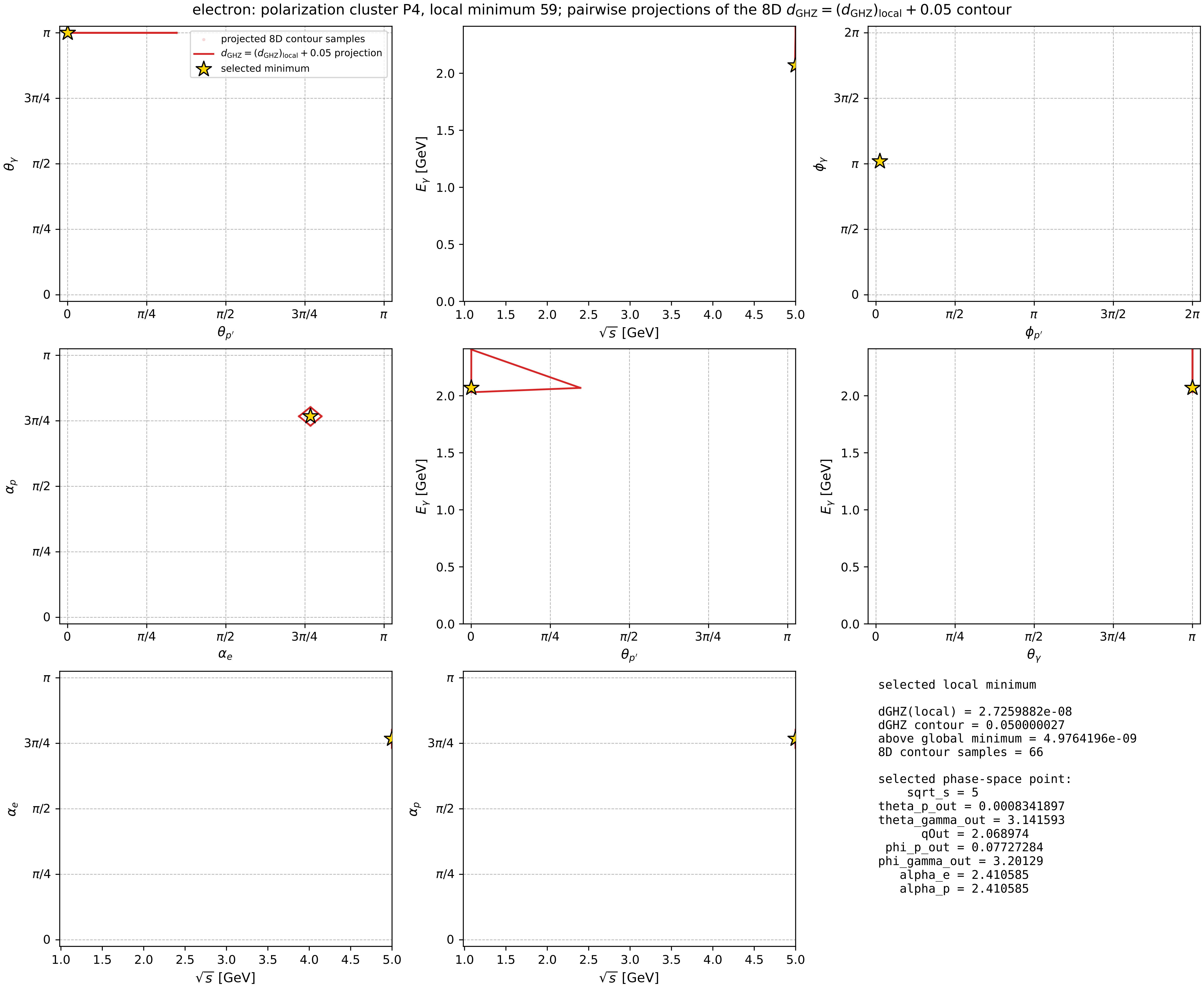
dghz_mixing_angles_polarization_04_local_minimum_59 d_{\rm GHZ}=2.72599e-08
region: polarization_cluster_04_local_minimum_59
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4105855 rad
incoming lepton: -0.744502|+> +0.66762|->
proton mixing angle: alpha_p=2.4105852 rad
incoming proton: -0.744502|+> +0.66762|->

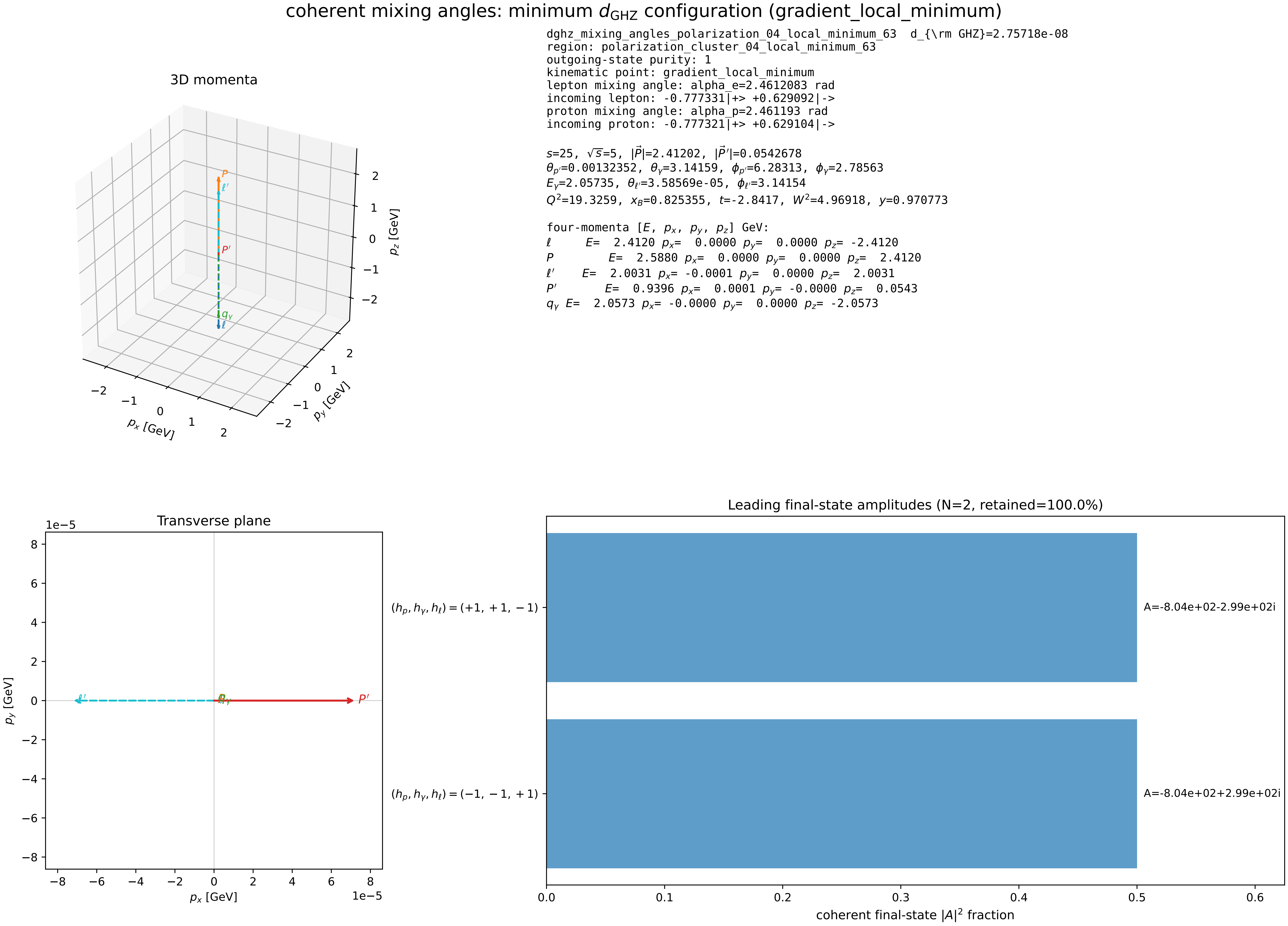
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0792932
$\theta_{p'}$=0.00083419, θ_γ=3.14159, $\phi_{p'}$=0.0772728, ϕ_γ=3.20129
E_γ=2.06897, $\theta_{\ell'}$=3.32443e-05, $\phi_{\ell'}$=3.21887
Q^2=19.1966, x_B=0.819668, t=-2.73017, W^2=5.10319, y=0.970969

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 1.9897 p_x= -0.0001 p_y= -0.0000 p_z= 1.9897
P' E= 0.9413 p_x= 0.0001 p_y= 0.0000 p_z= 0.0793
q_γ E= 2.0690 p_x= -0.0000 p_y= -0.0000 p_z= -2.0690

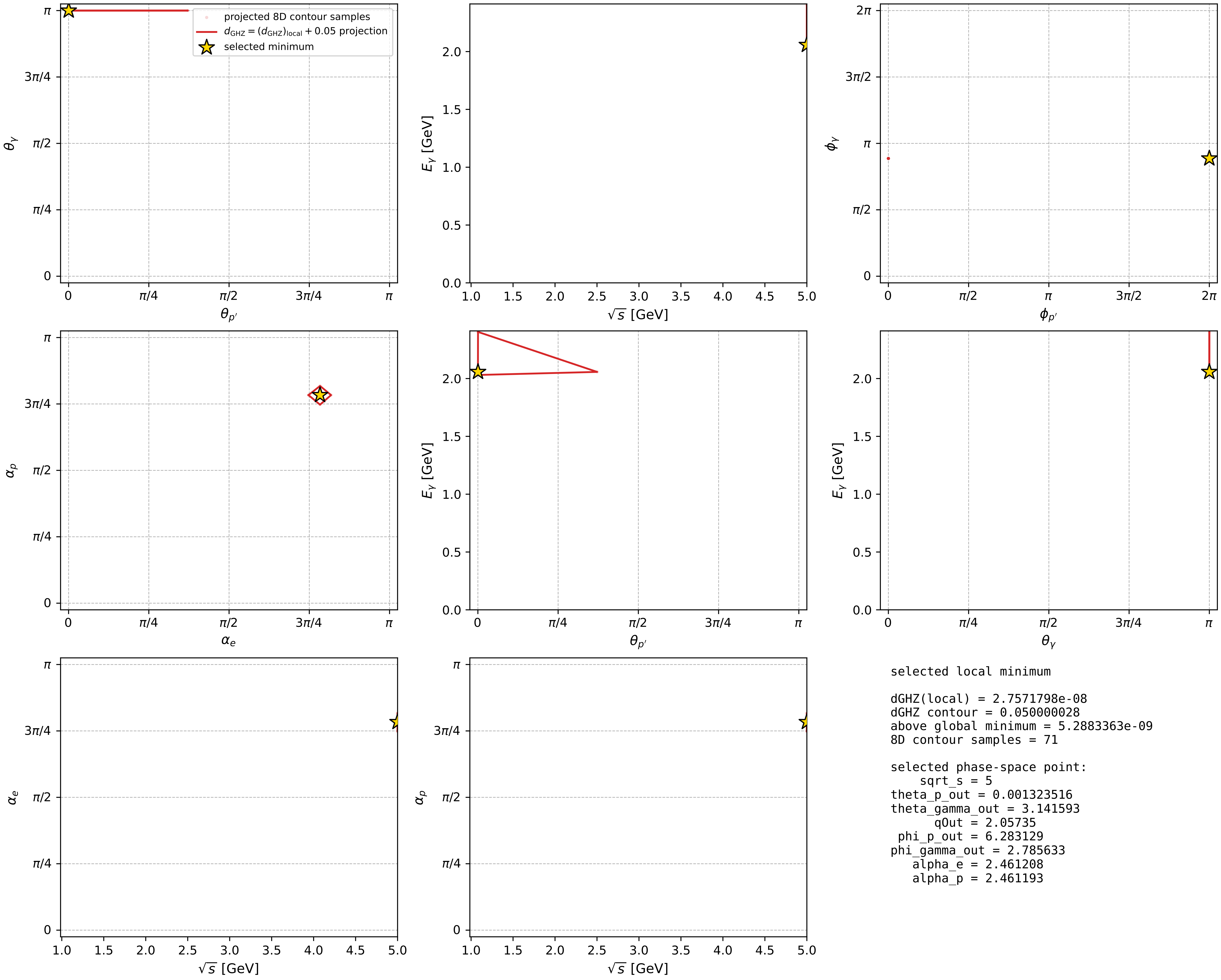




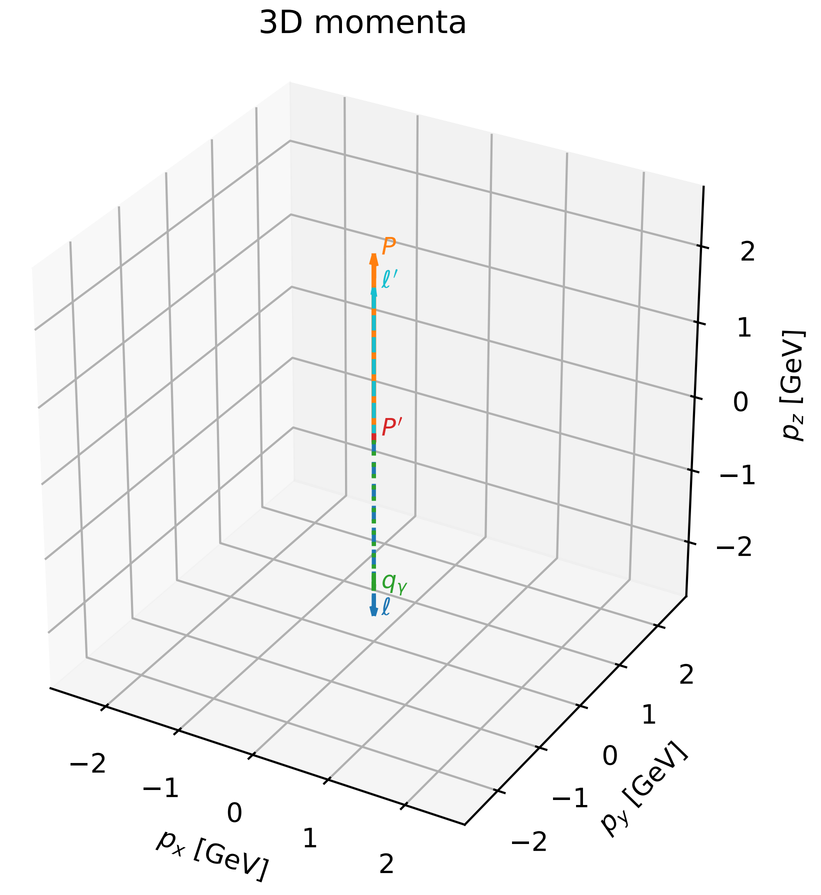

electron: polarization cluster P4, local minimum 59; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 2.7259882e-08
dGHZ contour = 0.050000027
above global minimum = 4.9764196e-09
8D contour samples = 66
selected phase-space point:
sqrt_s = 5
theta_p_out = 0.0008341897
theta_gamma_out = 3.141593
qOut = 2.068974
phi_p_out = 0.07727284
phi_gamma_out = 3.20129
alpha_e = 2.410585
alpha_p = 2.410585


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_63 d_{\rm GHZ}=2.75718e-08
region: polarization_cluster_04_local_minimum_63
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4612083 rad
incoming lepton: -0.777331|+> +0.629092|->
proton mixing angle: alpha_p=2.461193 rad
incoming proton: -0.777321|+> +0.629104|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0542678
$\theta_{p'}$=0.00132352, θ_γ=3.14159, $\phi_{p'}$=6.28313, ϕ_γ=2.78563
E_γ=2.05735, $\theta_{\ell'}$=3.58569e-05, $\phi_{\ell'}$=3.14154
Q^2=19.3259, x_B=0.825355, t=-2.8417, W^2=4.96918, y=0.970773
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0031 p_x= -0.0001 p_y= 0.0000 p_z= 2.0031
P' E= 0.9396 p_x= 0.0001 p_y= -0.0000 p_z= 0.0543
q_γ E= 2.0573 p_x= -0.0000 p_y= 0.0000 p_z= -2.0573
Transverse plane
1e−5
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-8.04e+02-2.99e+02i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-8.04e+02+2.99e+02i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P4, local minimum 63; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



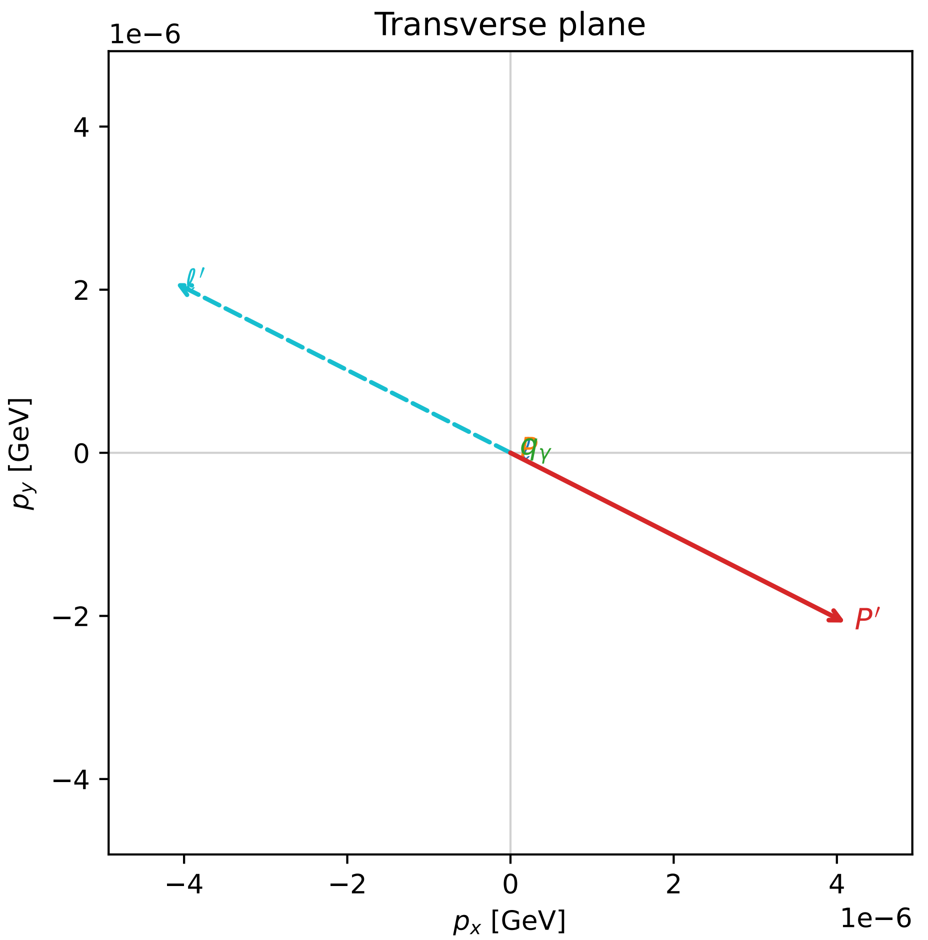
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

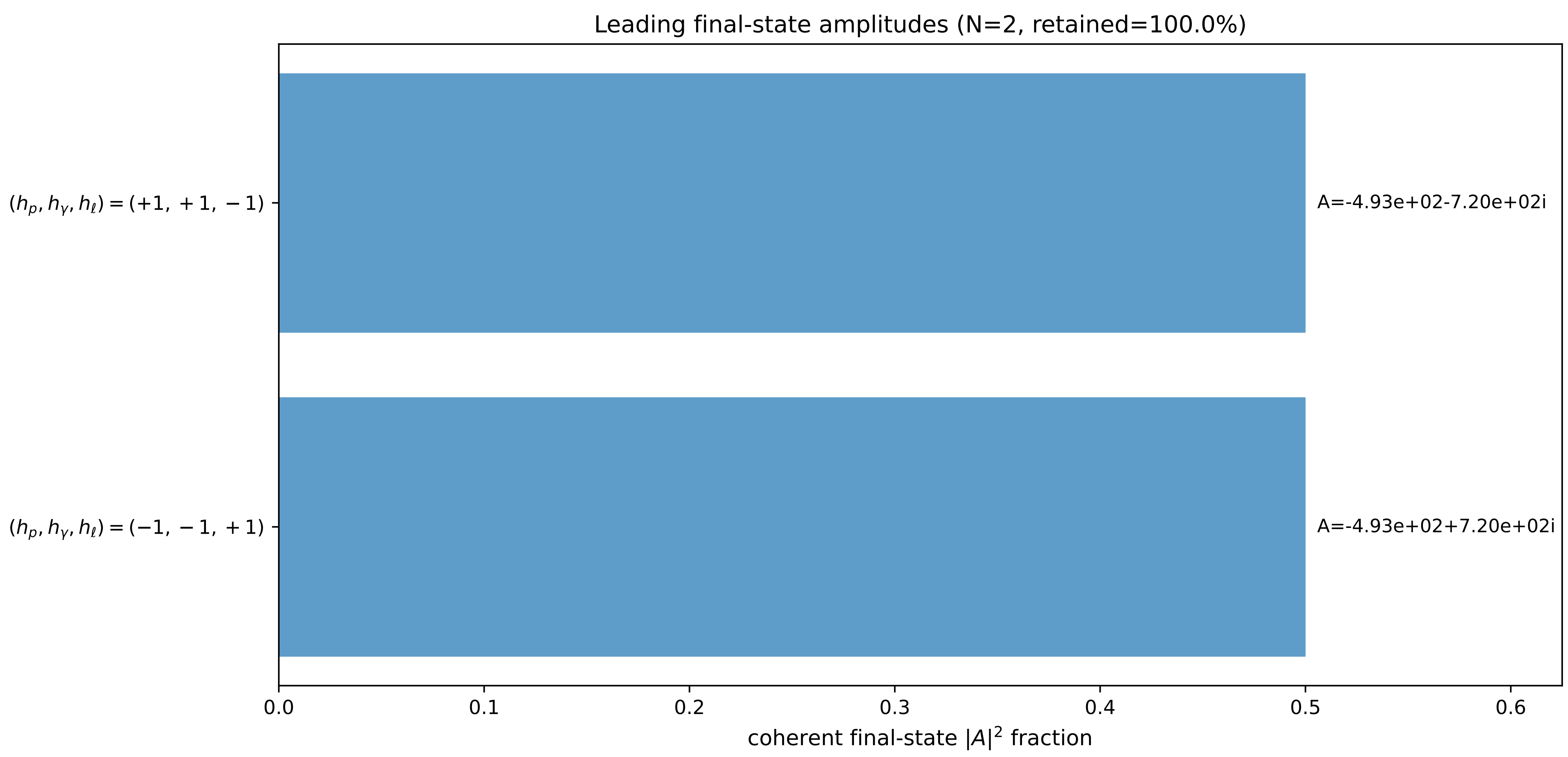


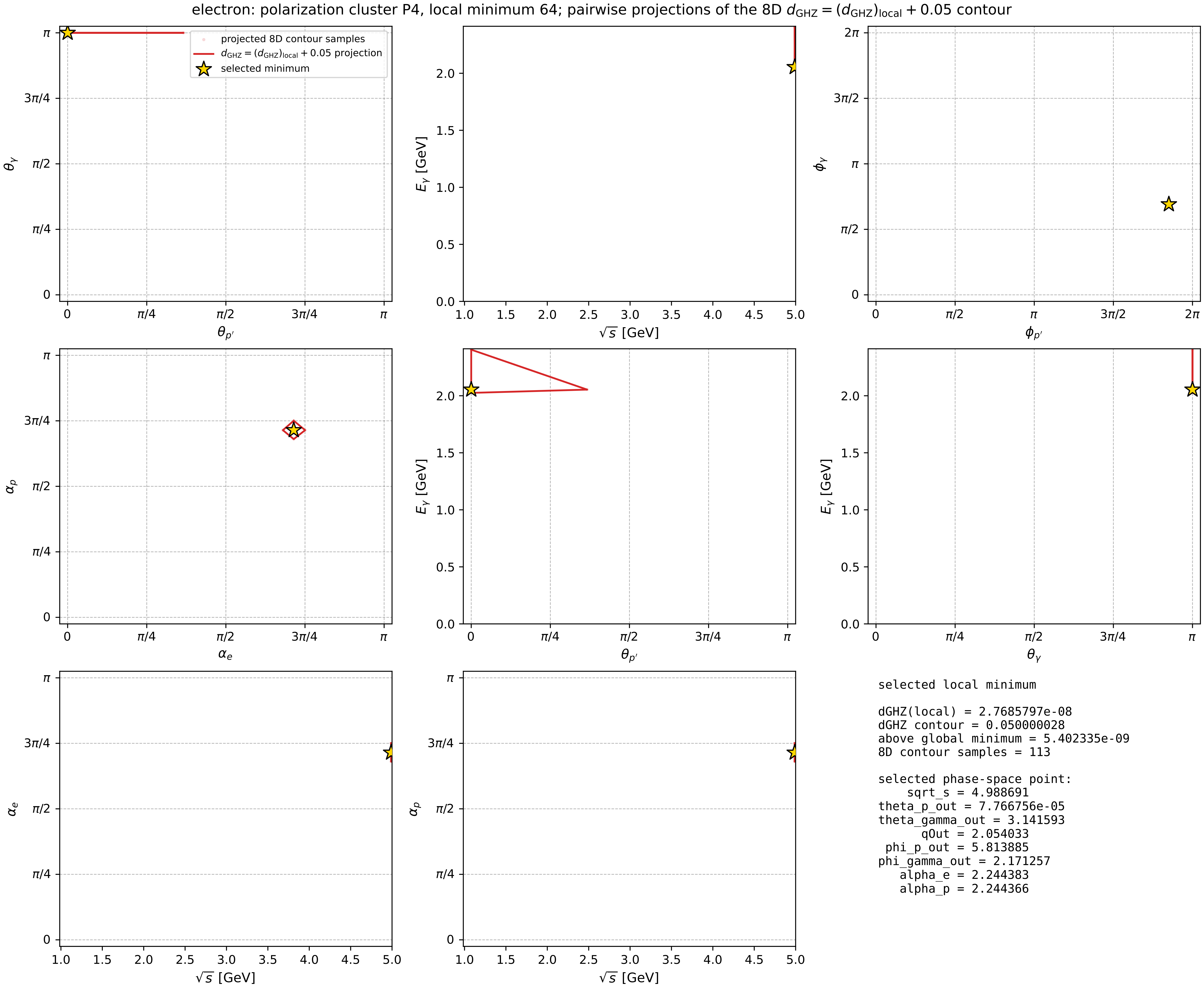
dghz_mixing_angles_polarization_04_local_minimum_64 d_{\rm GHZ}=2.76858e-08
region: polarization_cluster_04_local_minimum_64
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2443831 rad
incoming lepton: -0.623793|+> +0.781589|->
proton mixing angle: alpha_p=2.2443657 rad
incoming proton: -0.62378|+> +0.7816|->

s=24.887, $\sqrt{s}$=4.98869, $|\vec{P}|$=2.40616, $|\vec{P}'|$=0.0592441
$\theta_{p'}$=7.76676e-05, θ_γ=3.14159, $\phi_{p'}$=5.81389, ϕ_γ=2.17126
E_γ=2.05403, $\theta_{\ell'}$=2.30665e-06, $\phi_{\ell'}$=2.67229
Q^2=19.1991, x_B=0.823871, t=-2.80969, W^2=4.98427, y=0.970691

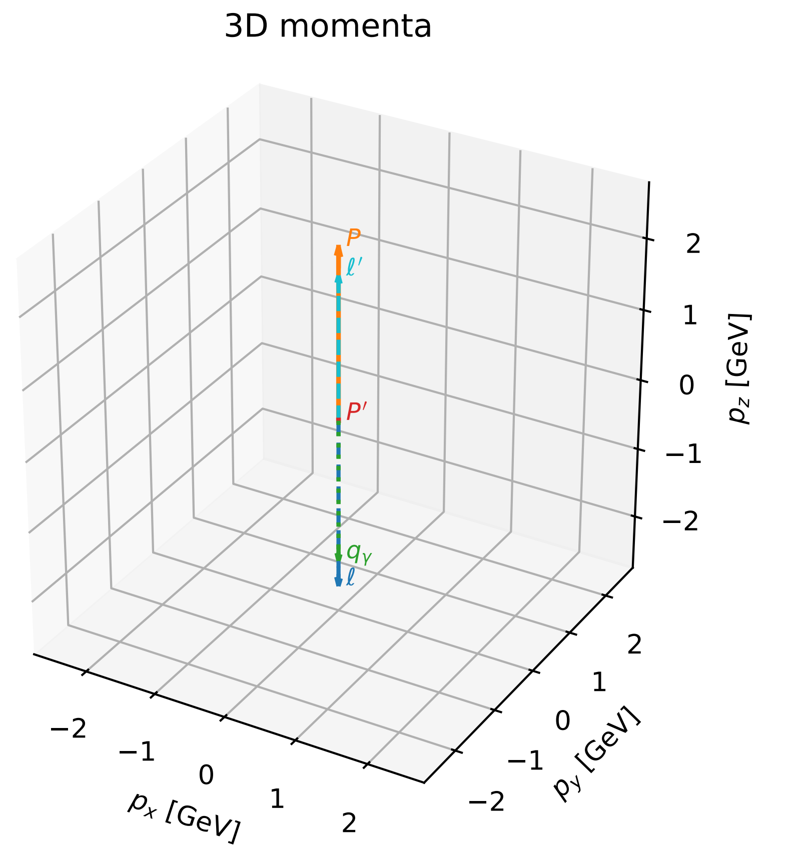
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4062 p_x= 0.0000 p_y= 0.0000 p_z= -2.4062
P E= 2.5825 p_x= 0.0000 p_y= 0.0000 p_z= 2.4062
ℓ' E= 1.9948 p_x= -0.0000 p_y= 0.0000 p_z= 1.9948
P' E= 0.9399 p_x= 0.0000 p_y= -0.0000 p_z= 0.0592
q_γ E= 2.0540 p_x= -0.0000 p_y= 0.0000 p_z= -2.0540






electron: polarization cluster P4, local minimum 64; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 2.7685797e-08
dGHZ contour = 0.050000028
above global minimum = 5.402335e-09
8D contour samples = 113
selected phase-space point:
sqrt_s = 4.988691
theta_p_out = 7.766756e-05
theta_gamma_out = 3.141593
qOut = 2.054033
phi_p_out = 5.813885
phi_gamma_out = 2.171257
alpha_e = 2.244383
alpha_p = 2.244366

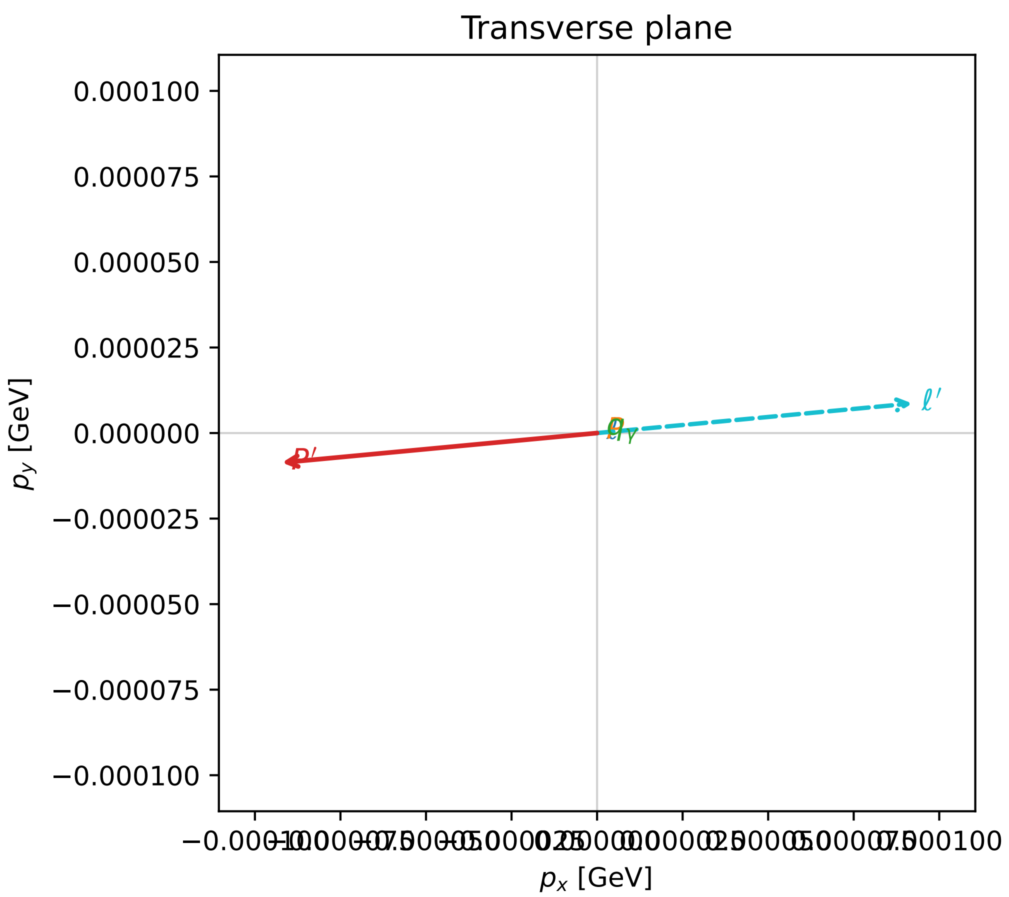
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

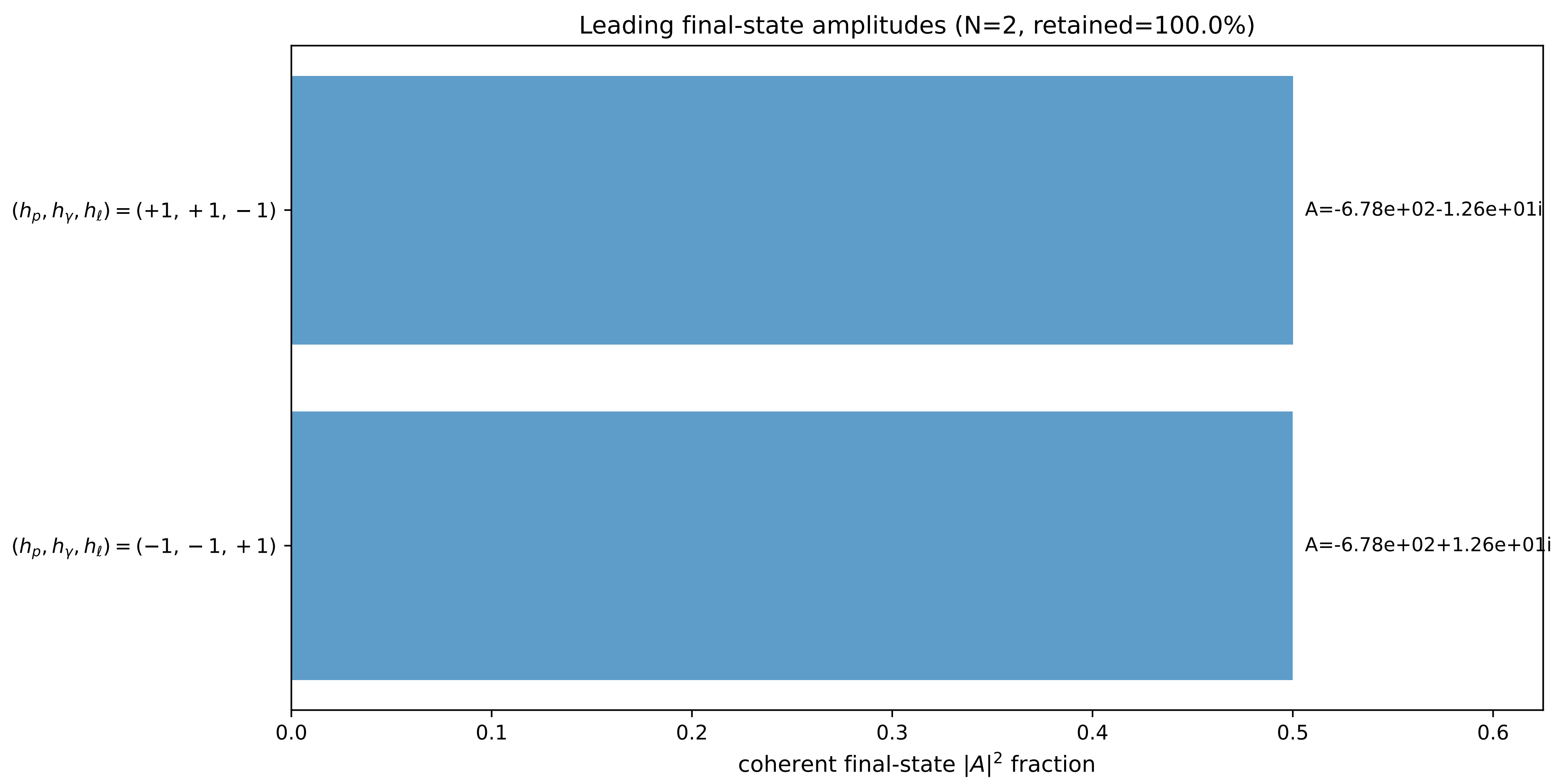


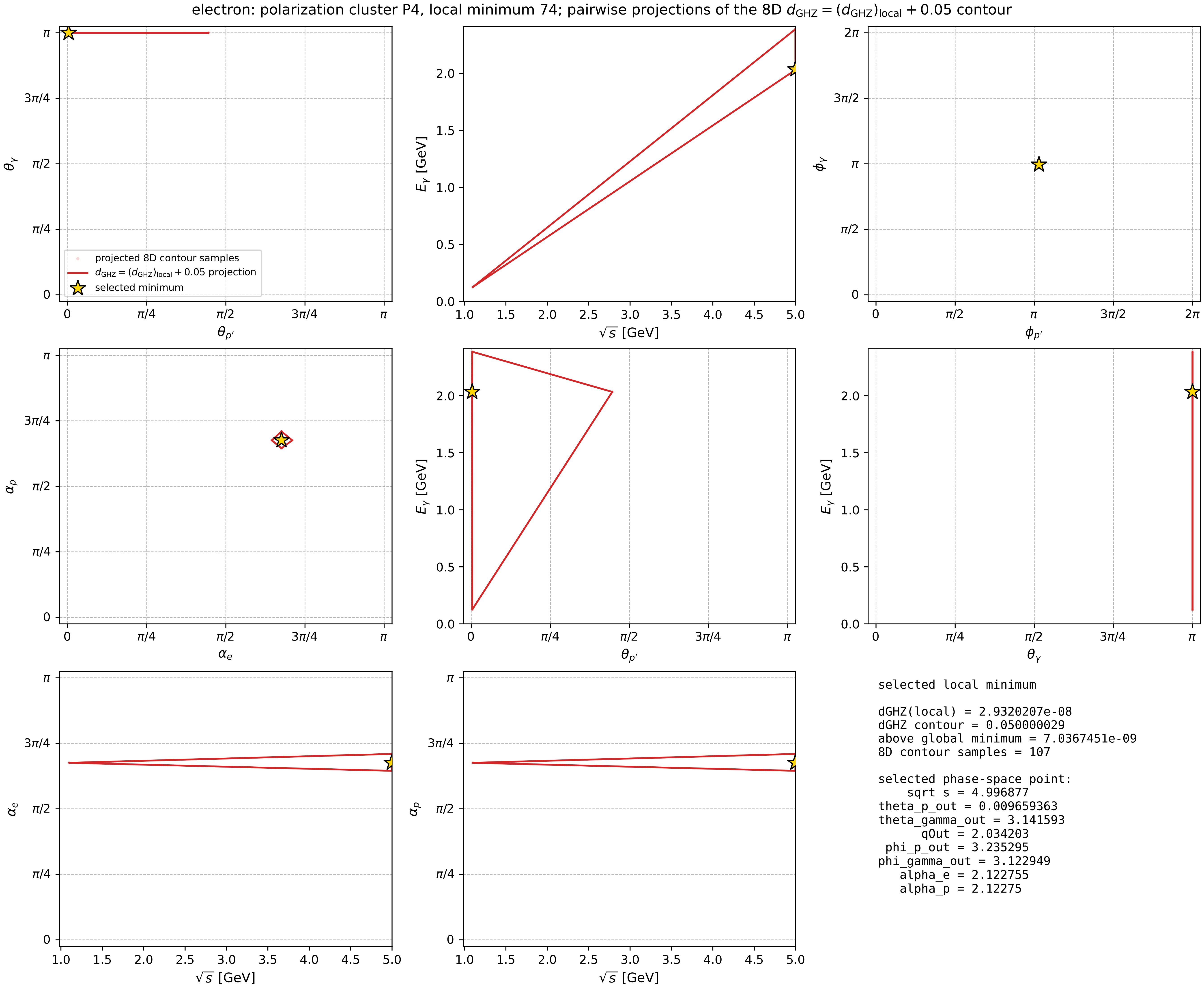
dghz_mixing_angles_polarization_04_local_minimum_74 d_{\rm GHZ}=2.93202e-08
region: polarization_cluster_04_local_minimum_74
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.1227555 rad
incoming lepton: -0.524356|+> +0.851499|->
proton mixing angle: alpha_p=2.12275 rad
incoming proton: -0.524352|+> +0.851502|->

s=24.9688, $\sqrt{s}$=4.99688, $|\vec{P}|$=2.4104, $|\vec{P}'|$=0.00957823
$\theta_{p'}$=0.00965936, θ_γ=3.14159, $\phi_{p'}$=3.2353, ϕ_γ=3.12295
E_γ=2.0342, $\theta_{\ell'}$=4.56964e-05, $\phi_{\ell'}$=0.0937025
Q^2=19.5206, x_B=0.835073, t=-3.04662, W^2=4.73517, y=0.970402

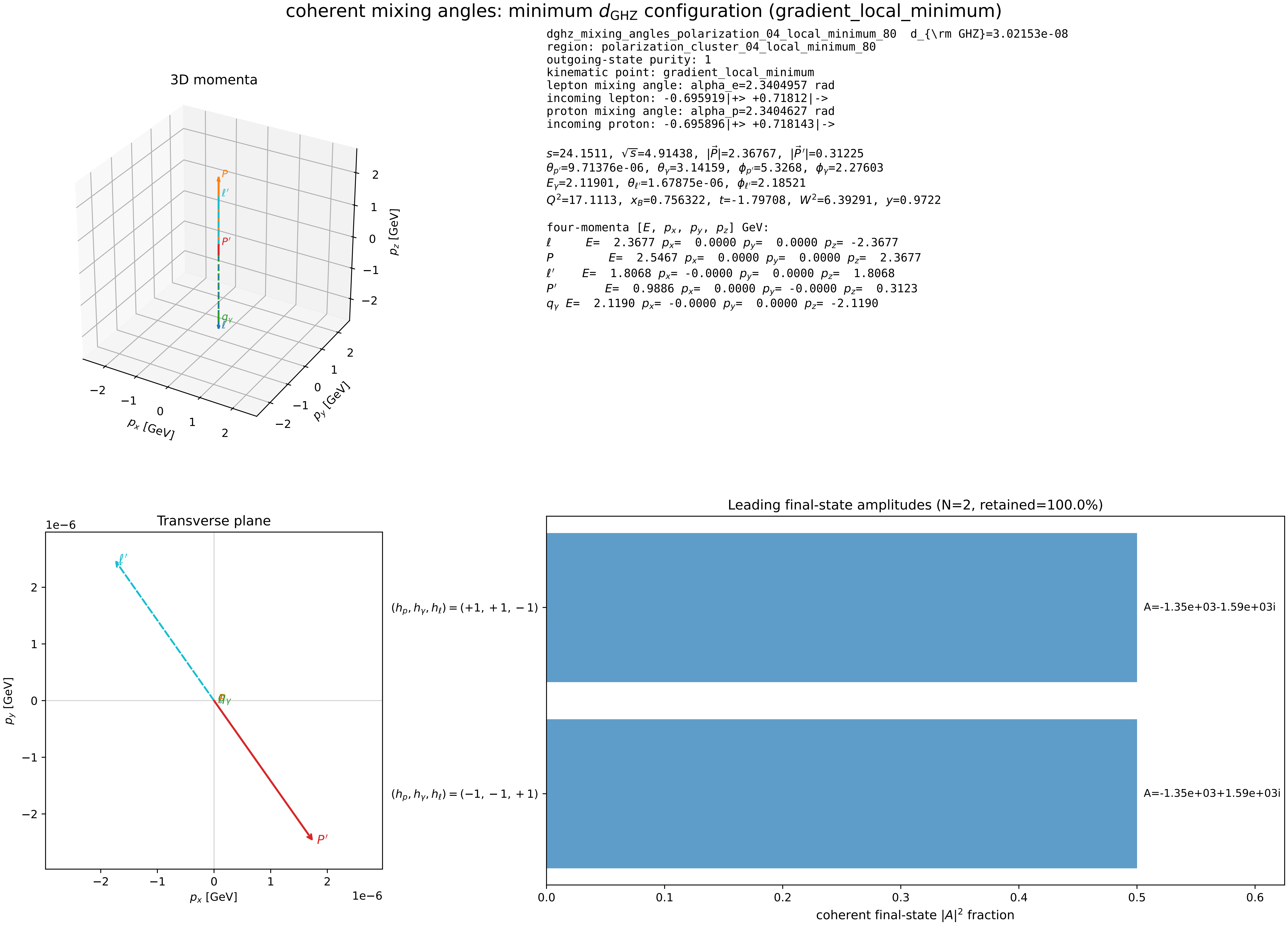
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4104 p_x= 0.0000 p_y= 0.0000 p_z= -2.4104
P E= 2.5865 p_x= 0.0000 p_y= 0.0000 p_z= 2.4104
ℓ' E= 2.0246 p_x= 0.0001 p_y= 0.0000 p_z= 2.0246
P' E= 0.9380 p_x= -0.0001 p_y= -0.0000 p_z= 0.0096
q_γ E= 2.0342 p_x= -0.0000 p_y= 0.0000 p_z= -2.0342



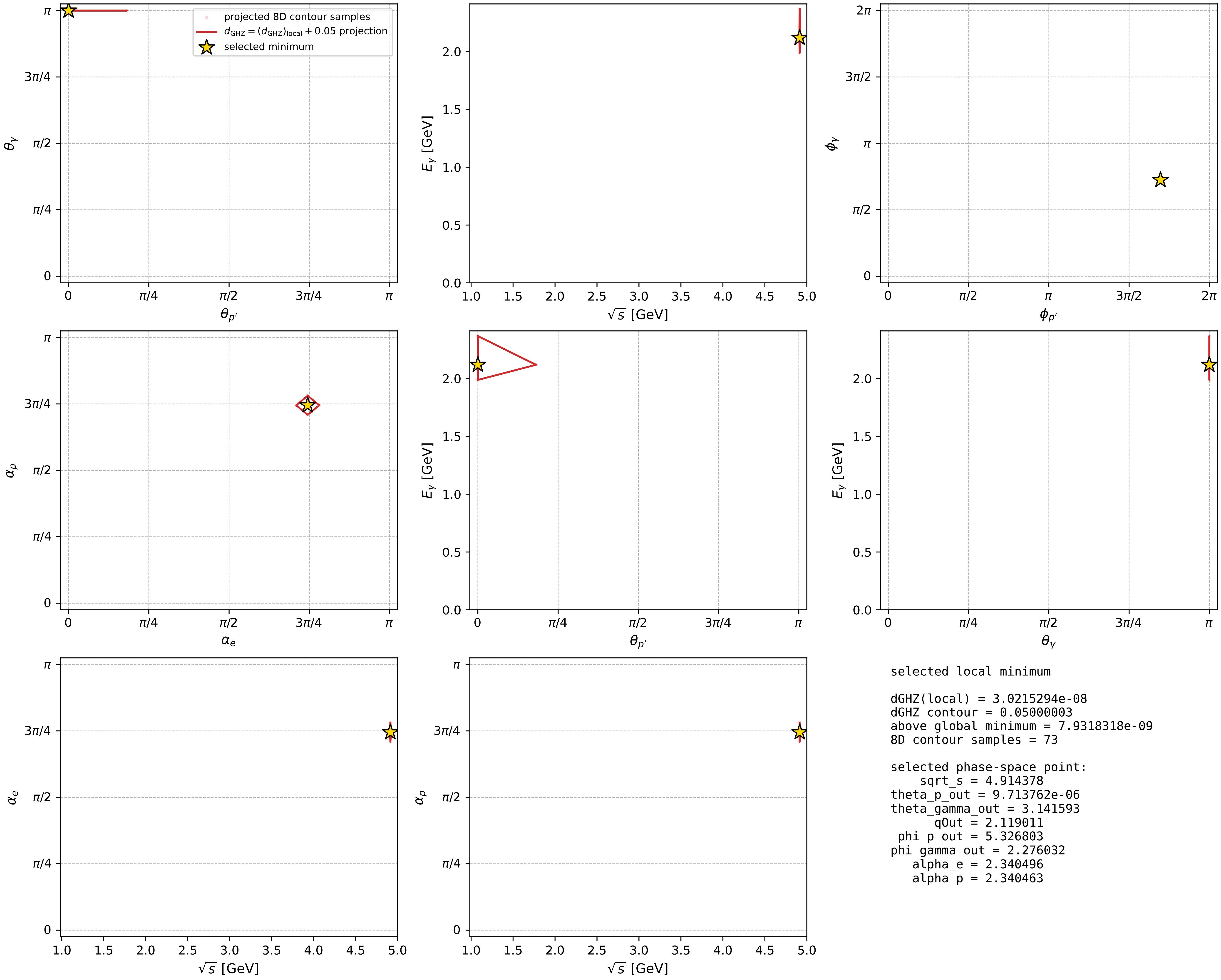


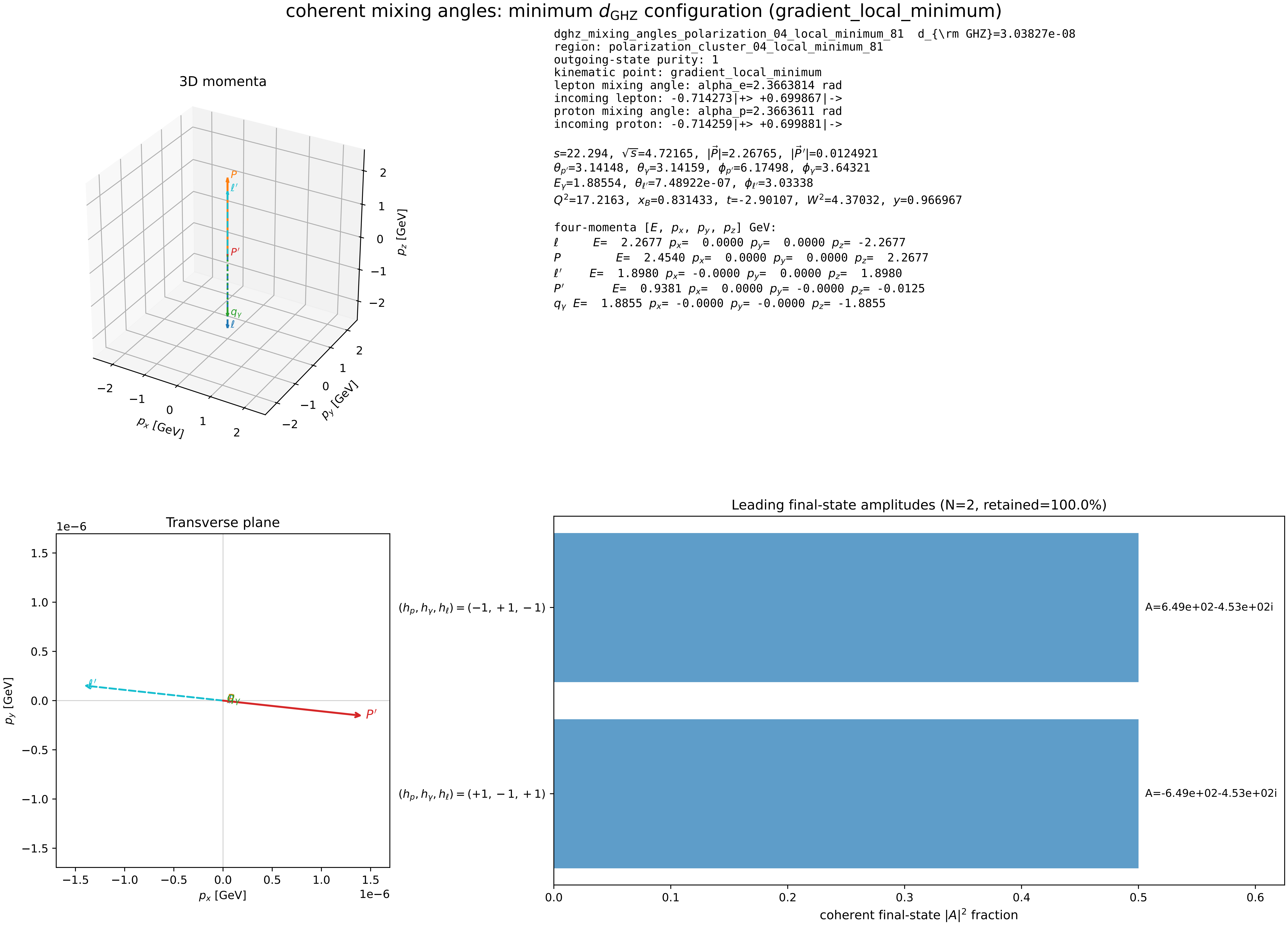


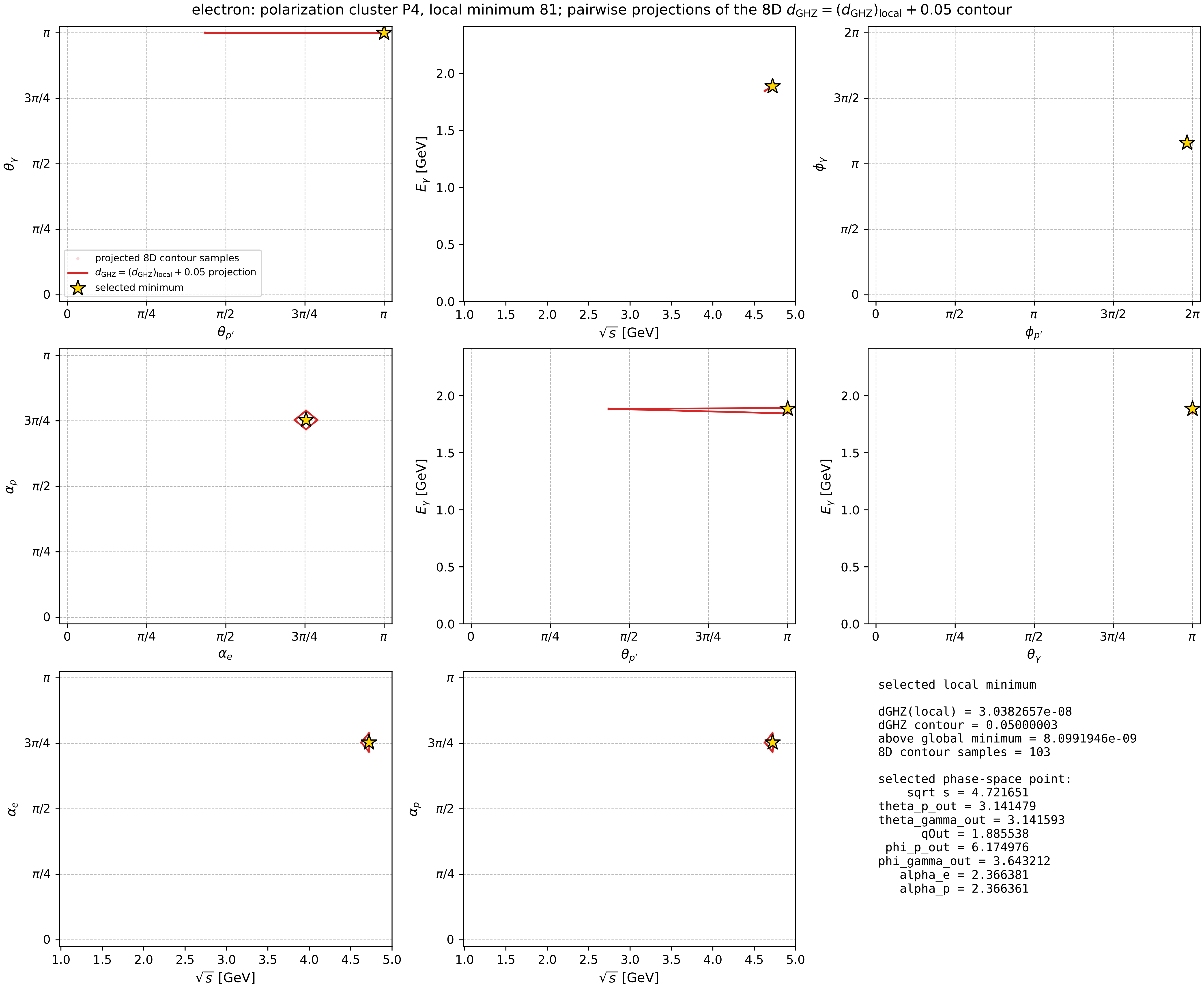
electron: polarization cluster P4, local minimum 74; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 2.9320207e-08
dGHZ contour = 0.050000029
above global minimum = 7.0367451e-09
8D contour samples = 107
selected phase-space point:
sqrt_s = 4.996877
theta_p_out = 0.009659363
theta_gamma_out = 3.141593
qOut = 2.034203
phi_p_out = 3.235295
phi_gamma_out = 3.122949
alpha_e = 2.122755
alpha_p = 2.12275


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_80 d_{\rm GHZ}=3.02153e-08
region: polarization_cluster_04_local_minimum_80
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3404957 rad
incoming lepton: -0.695919|+> +0.71812|->
proton mixing angle: alpha_p=2.3404627 rad
incoming proton: -0.695896|+> +0.718143|->
s=24.1511, $\sqrt{s}$=4.91438, $|\vec{P}|$=2.36767, $|\vec{P}'|$=0.31225
$\theta_{p'}$=9.71376e-06, θ_γ=3.14159, $\phi_{p'}$=5.3268, ϕ_γ=2.27603
E_γ=2.11901, $\theta_{\ell'}$=1.67875e-06, $\phi_{\ell'}$=2.18521
Q^2=17.1113, x_B=0.756322, t=-1.79708, W^2=6.39291, y=0.9722
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3677 p_x= 0.0000 p_y= 0.0000 p_z= -2.3677
P E= 2.5467 p_x= 0.0000 p_y= 0.0000 p_z= 2.3677
ℓ' E= 1.8068 p_x= -0.0000 p_y= 0.0000 p_z= 1.8068
P' E= 0.9886 p_x= 0.0000 p_y= -0.0000 p_z= 0.3123
q_γ E= 2.1190 p_x= -0.0000 p_y= 0.0000 p_z= -2.1190
Transverse plane
1e−6
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-1.35e+03-1.59e+03i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-1.35e+03+1.59e+03i
coherent final-state $|A|^2$ fraction

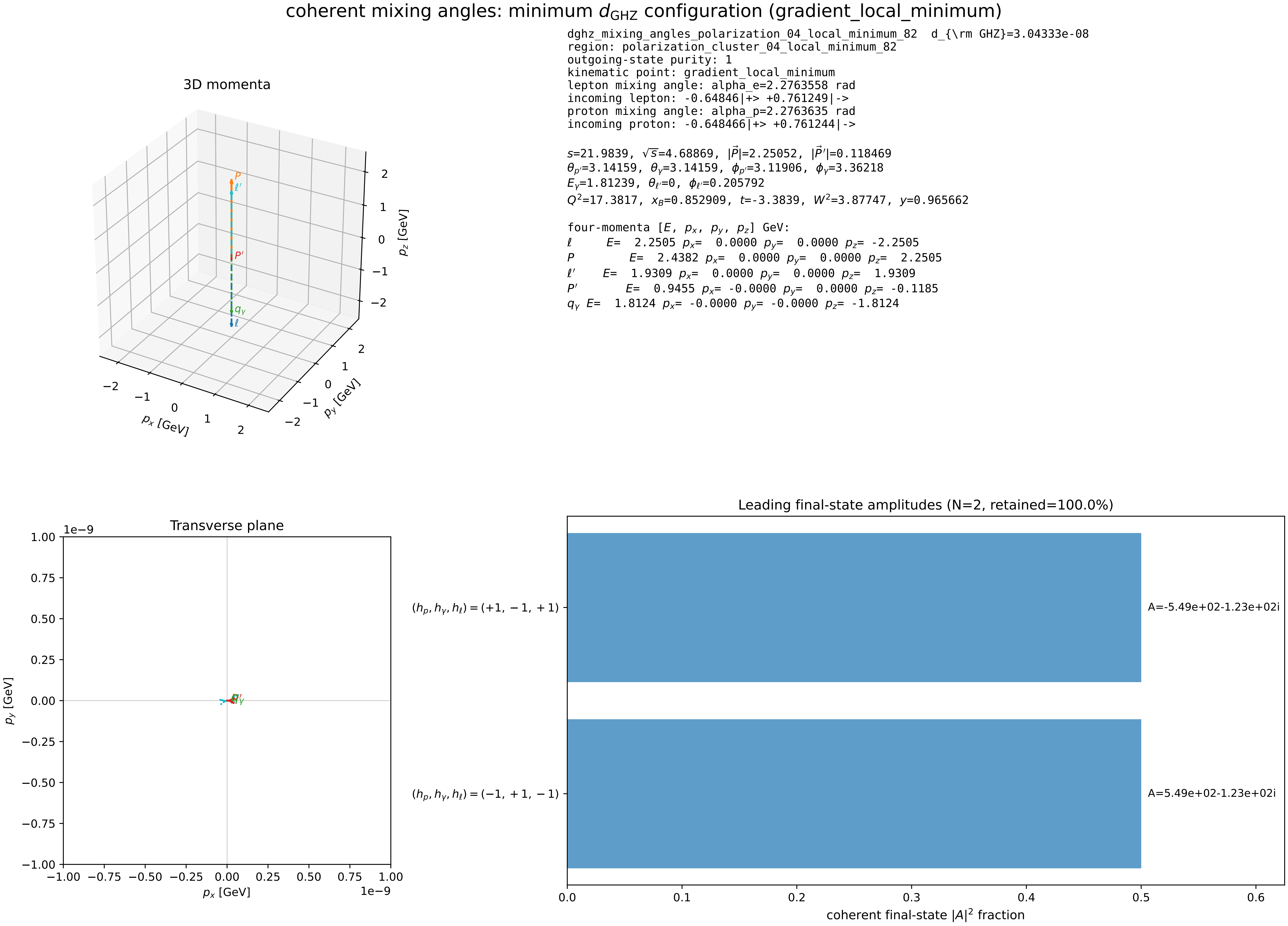
electron: polarization cluster P4, local minimum 80; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour

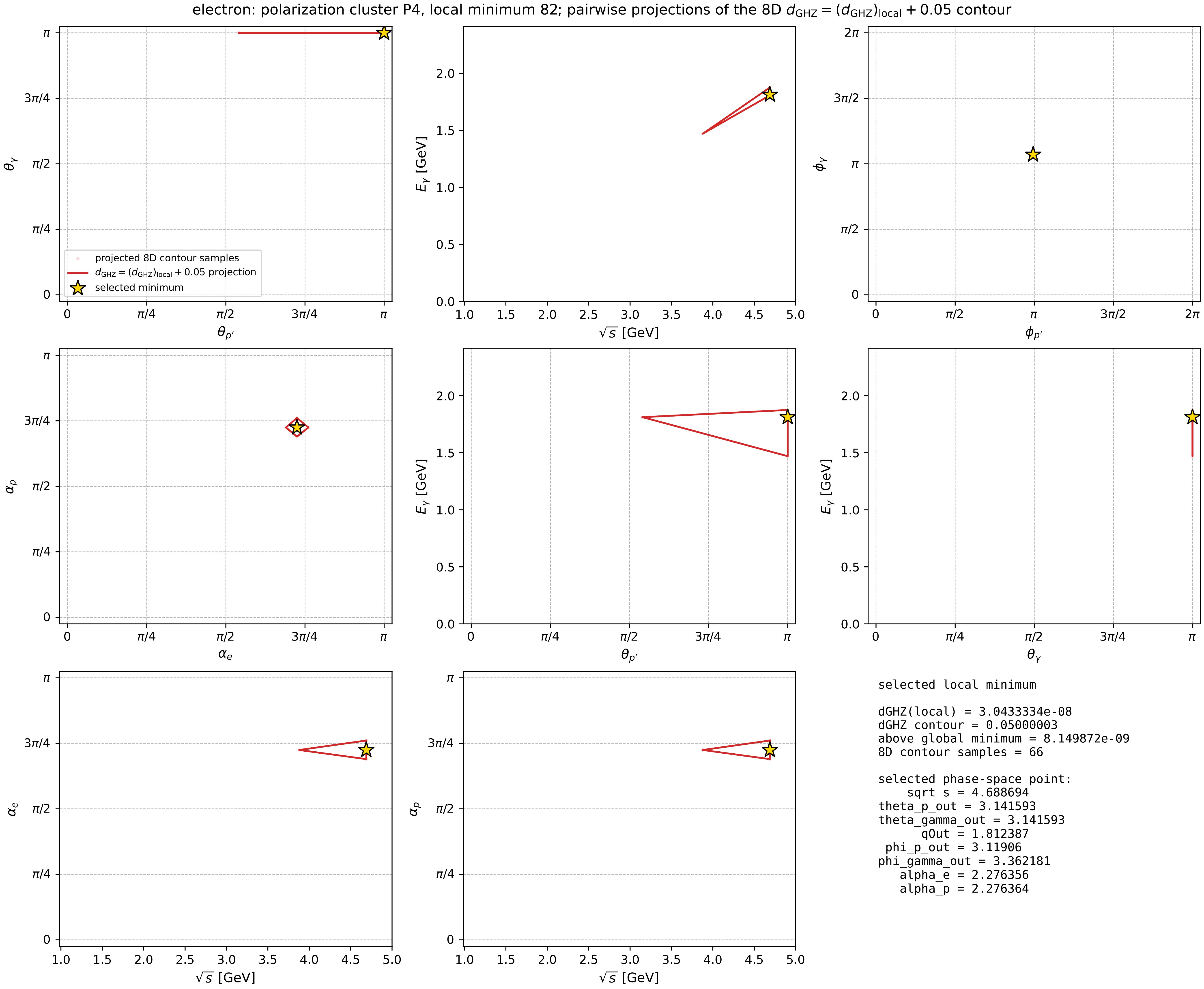



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_81 d_{\rm GHZ}=3.03827e-08
region: polarization_cluster_04_local_minimum_81
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3663814 rad
incoming lepton: -0.714273|+> +0.699867|->
proton mixing angle: alpha_p=2.3663611 rad
incoming proton: -0.714259|+> +0.699881|->
s=22.294, $\sqrt{s}$=4.72165, $|\vec{P}|$=2.26765, $|\vec{P}'|$=0.0124921
$\theta_{p'}$=3.14148, θ_γ=3.14159, $\phi_{p'}$=6.17498, ϕ_γ=3.64321
E_γ=1.88554, $\theta_{\ell'}$=7.48922e-07, $\phi_{\ell'}$=3.03338
Q^2=17.2163, x_B=0.831433, t=-2.90107, W^2=4.37032, y=0.966967
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2677 p_x= 0.0000 p_y= 0.0000 p_z= -2.2677
P E= 2.4540 p_x= 0.0000 p_y= 0.0000 p_z= 2.2677
ℓ' E= 1.8980 p_x= -0.0000 p_y= 0.0000 p_z= 1.8980
P' E= 0.9381 p_x= 0.0000 p_y= -0.0000 p_z= -0.0125
q_γ E= 1.8855 p_x= -0.0000 p_y= -0.0000 p_z= -1.8855
3D momenta
P
ℓ'
P'
q_γ
ℓ
2
1
0
−1
−2
p_z [GeV]
−2
−1
0
1
2
p_x [GeV]
2
1
0
−1
−2
p_y [GeV]
Transverse plane
1e−6
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
p_y [GeV]
ℓ'
q_γ
P'
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
p_x [GeV]
1e−6
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=6.49e+02-4.53e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=-6.49e+02-4.53e+02i
0.0
0.1
0.2
0.3
0.4
0.5
0.6
coherent final-state $|A|^2$ fraction



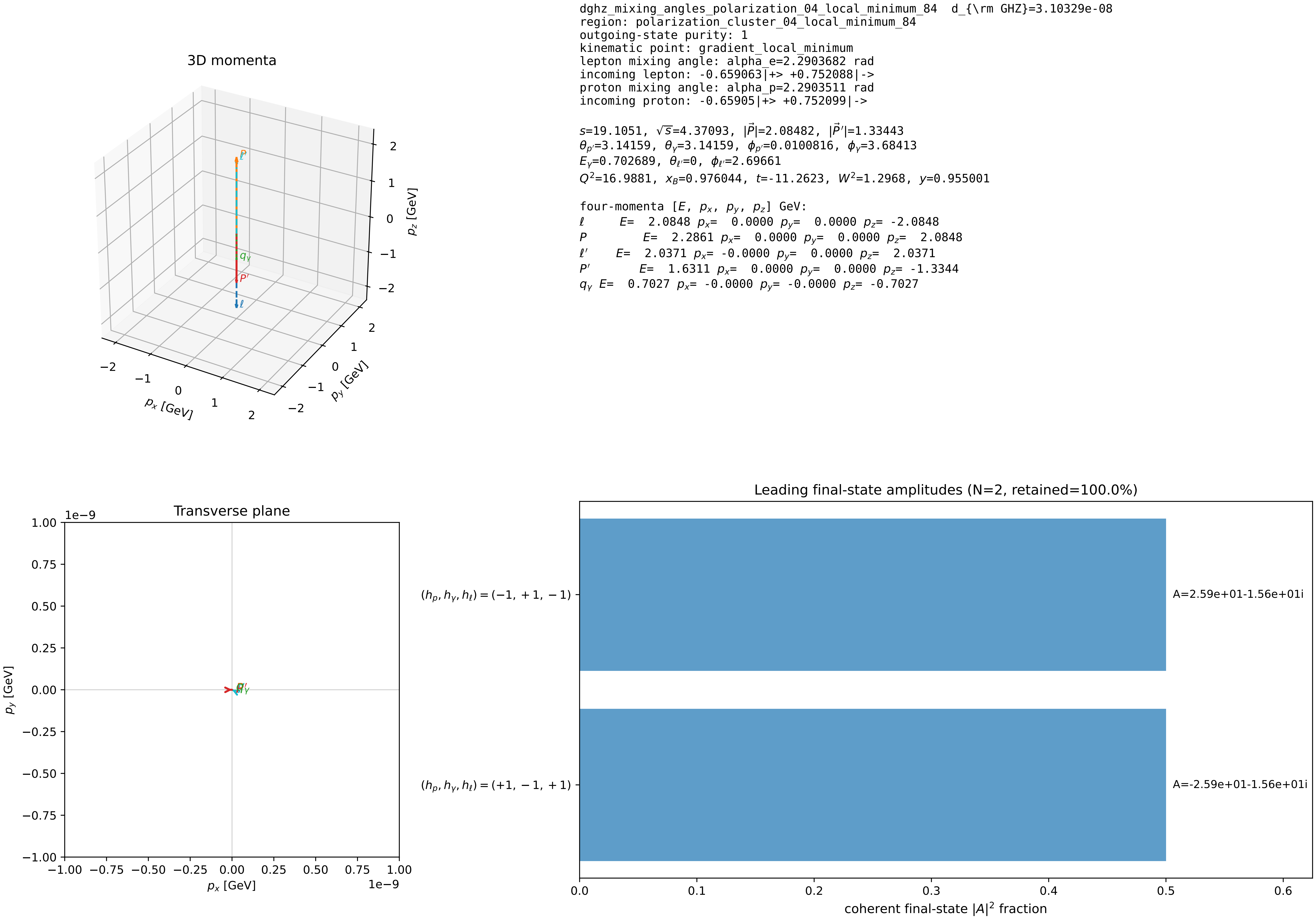
electron: polarization cluster P4, local minimum 81; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 3.0382657e-08
dGHZ contour = 0.05000003
above global minimum = 8.0991946e-09
8D contour samples = 103
selected phase-space point:
sqrt_s = 4.721651
theta_p_out = 3.141479
theta_gamma_out = 3.141593
qOut = 1.885538
phi_p_out = 6.174976
phi_gamma_out = 3.643212
alpha_e = 2.366381
alpha_p = 2.366361


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_82 d_{\rm GHZ}=3.04333e-08
region: polarization_cluster_04_local_minimum_82
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2763558 rad
incoming lepton: -0.64846|+> +0.761249|->
proton mixing angle: alpha_p=2.2763635 rad
incoming proton: -0.648466|+> +0.761244|->
s=21.9839, $\sqrt{s}$=4.68869, $|\vec{P}|$=2.25052, $|\vec{P}'|$=0.118469
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=3.11906, ϕ_γ=3.36218
E_γ=1.81239, $\theta_{\ell'}$=0, $\phi_{\ell'}$=0.205792
Q^2=17.3817, x_B=0.852909, t=-3.3839, W^2=3.87747, y=0.965662
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2505 p_x= 0.0000 p_y= 0.0000 p_z= -2.2505
P E= 2.4382 p_x= 0.0000 p_y= 0.0000 p_z= 2.2505
ℓ' E= 1.9309 p_x= 0.0000 p_y= 0.0000 p_z= 1.9309
P' E= 0.9455 p_x= -0.0000 p_y= 0.0000 p_z= -0.1185
q_γ E= 1.8124 p_x= -0.0000 p_y= -0.0000 p_z= -1.8124
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=-5.49e+02-1.23e+02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=5.49e+02-1.23e+02i
coherent final-state $|A|^2$ fraction

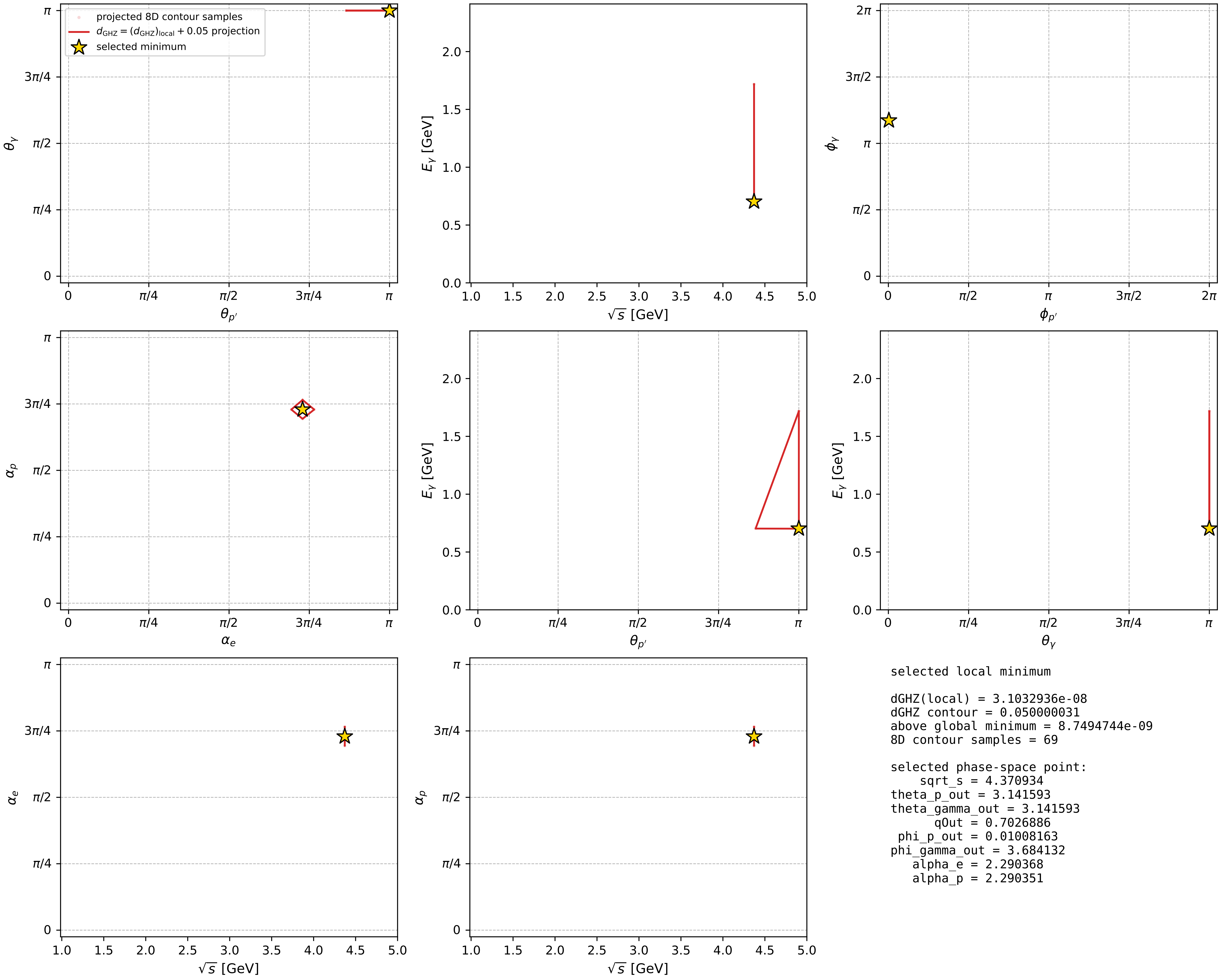


electron: polarization cluster P4, local minimum 82; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 3.0433334e-08
dGHZ contour = 0.05000003
above global minimum = 8.149872e-09
8D contour samples = 66
selected phase-space point:
sqrt_s = 4.688694
theta_p_out = 3.141593
theta_gamma_out = 3.141593
qOut = 1.812387
phi_p_out = 3.11906
phi_gamma_out = 3.362181
alpha_e = 2.276356
alpha_p = 2.276364

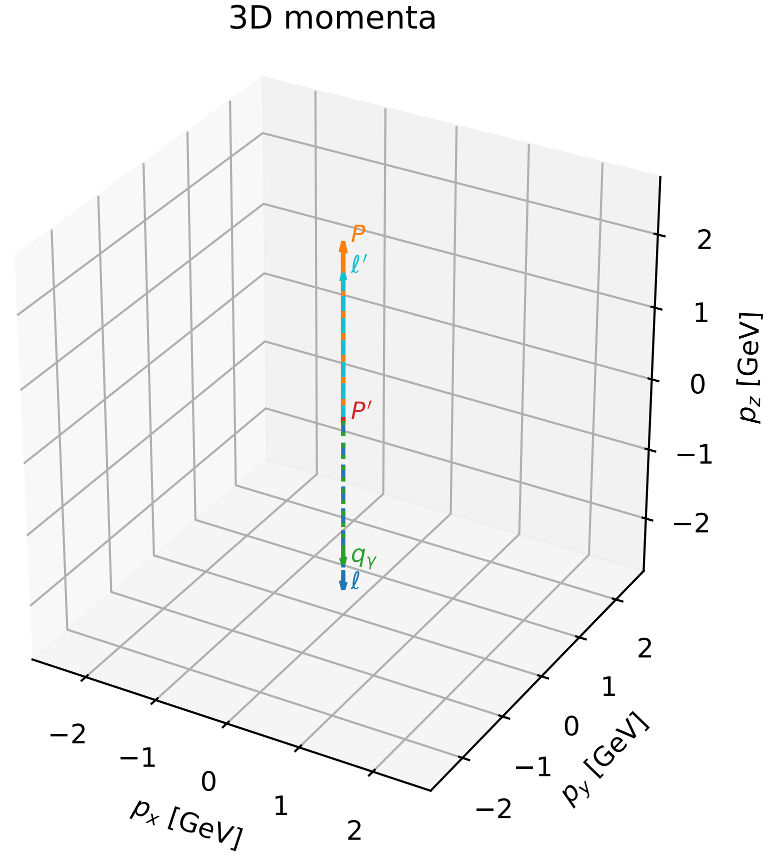
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



electron: polarization cluster P4, local minimum 84; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



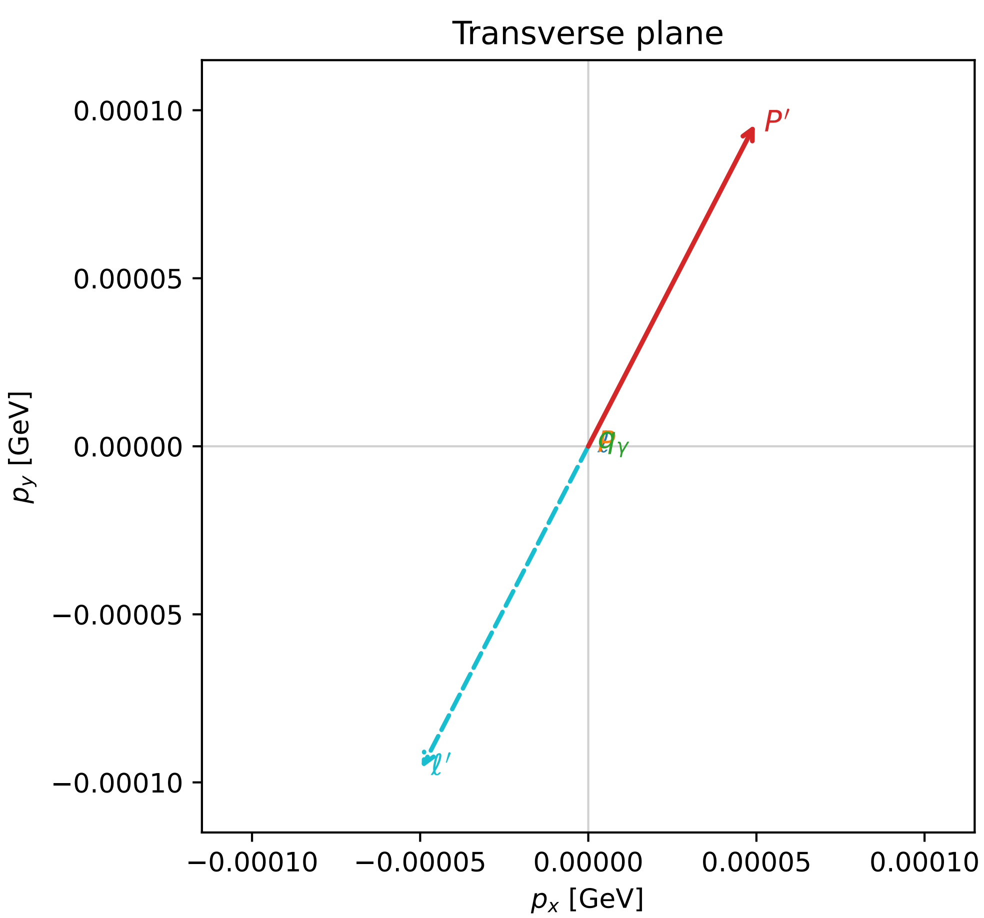
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

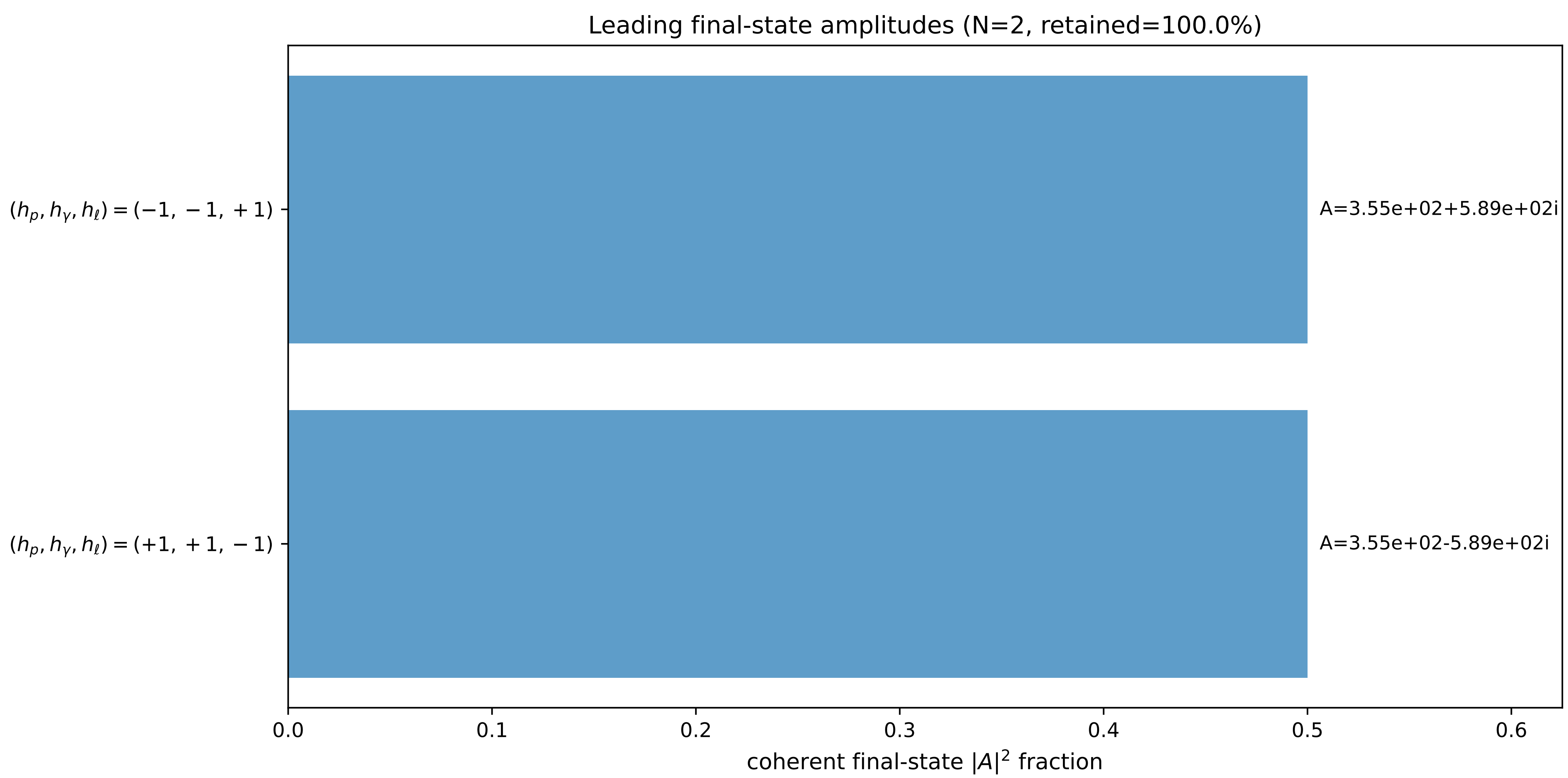


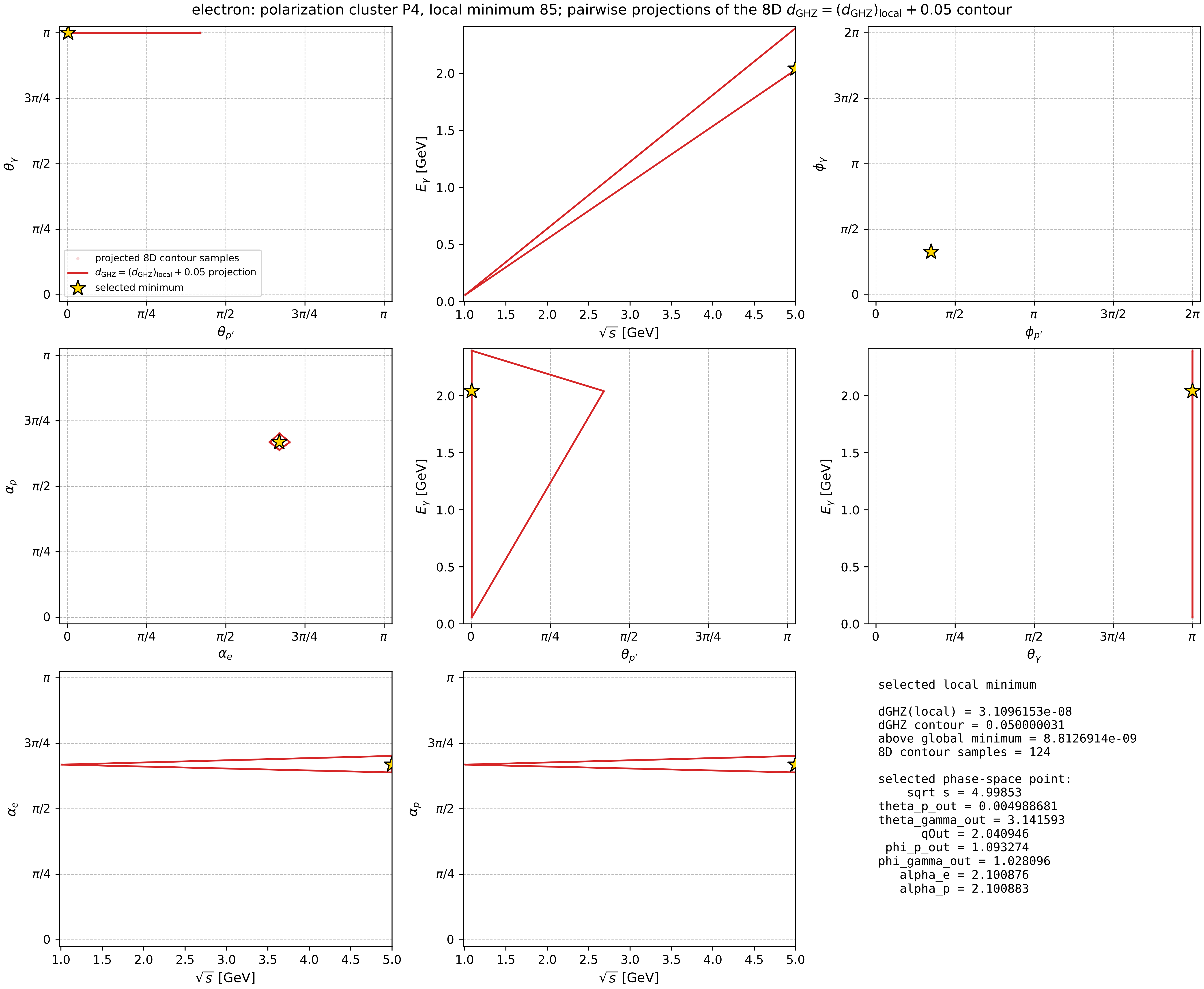
dghz_mixing_angles_polarization_04_local_minimum_85 d_{\rm GHZ}=3.10962e-08
region: polarization_cluster_04_local_minimum_85
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.1008763 rad
incoming lepton: -0.505602|+> +0.862767|->
proton mixing angle: alpha_p=2.1008833 rad
incoming proton: -0.505608|+> +0.862763|->

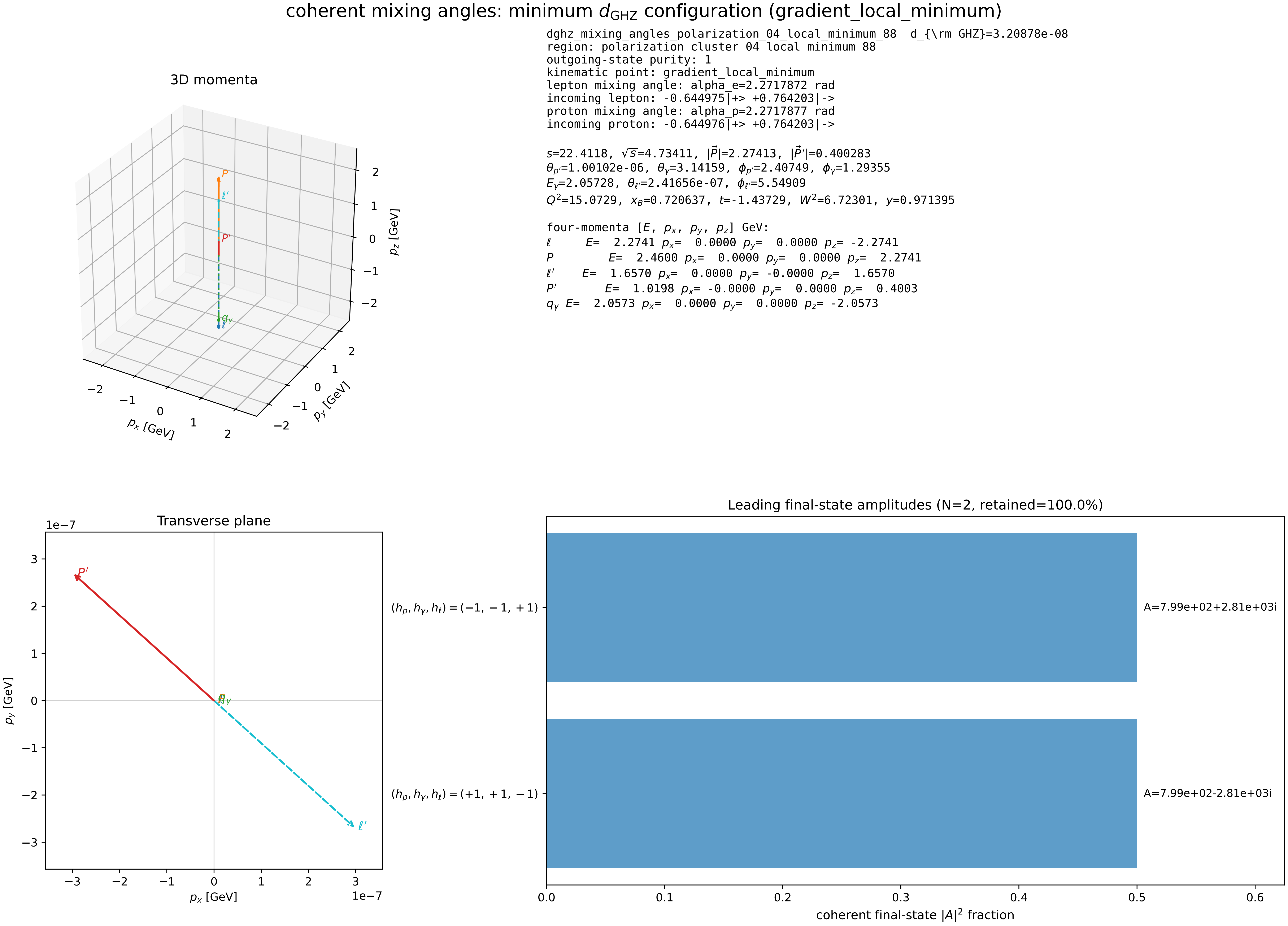
s=24.9853, $\sqrt{s}$=4.99853, $|\vec{P}|$=2.41125, $|\vec{P}'|$=0.0216116
$\theta_{p'}$=0.00498868, θ_γ=3.14159, $\phi_{p'}$=1.09327, ϕ_γ=1.0281
E_γ=2.04095, $\theta_{\ell'}$=5.33904e-05, $\phi_{\ell'}$=4.23487
Q^2=19.4765, x_B=0.832523, t=-2.99111, W^2=4.79789, y=0.970509

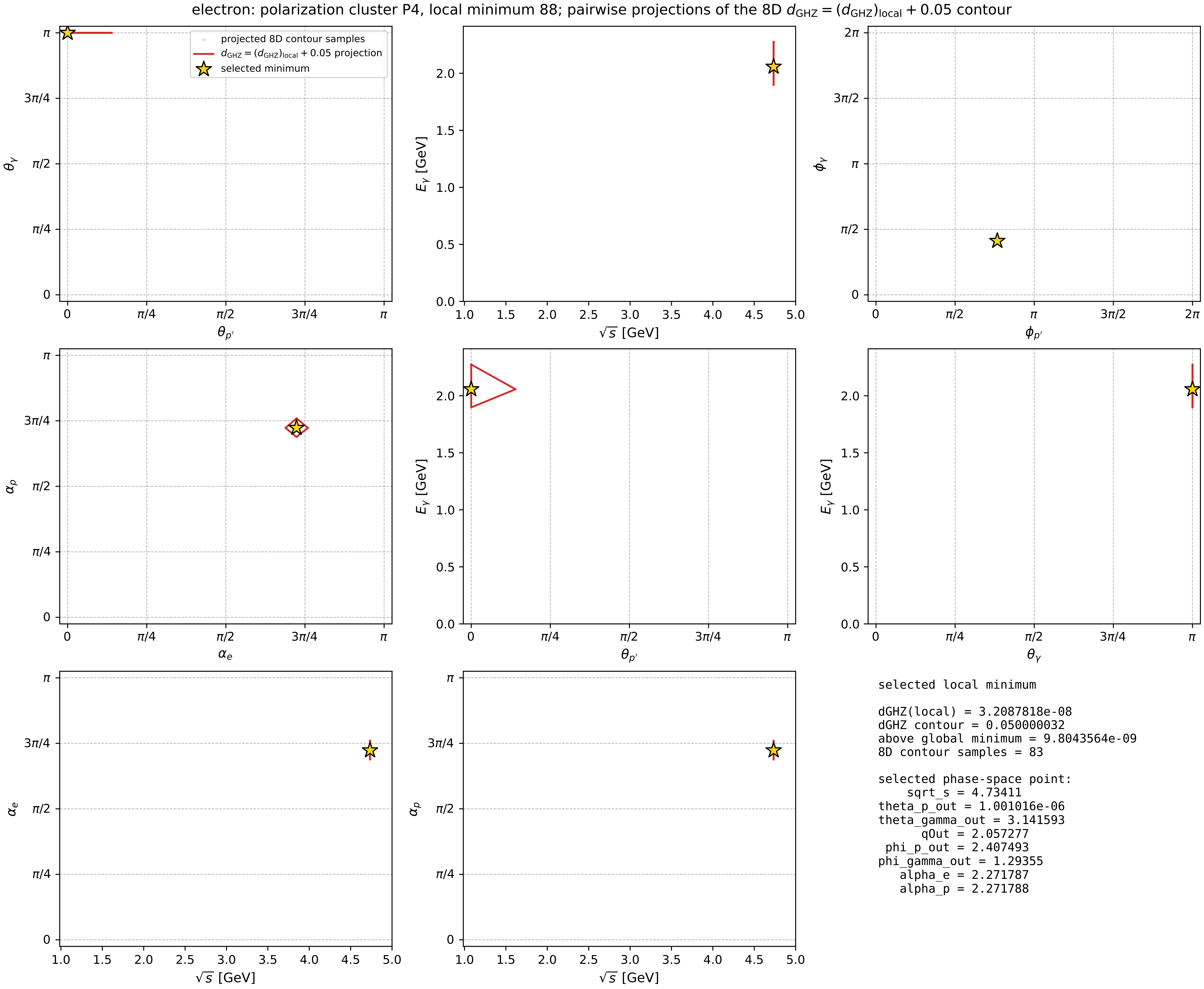
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4113 p_x= 0.0000 p_y= 0.0000 p_z= -2.4113
P E= 2.5873 p_x= 0.0000 p_y= 0.0000 p_z= 2.4113
ℓ' E= 2.0193 p_x= -0.0000 p_y= -0.0001 p_z= 2.0193
P' E= 0.9382 p_x= 0.0000 p_y= 0.0001 p_z= 0.0216
q_γ E= 2.0409 p_x= 0.0000 p_y= 0.0000 p_z= -2.0409

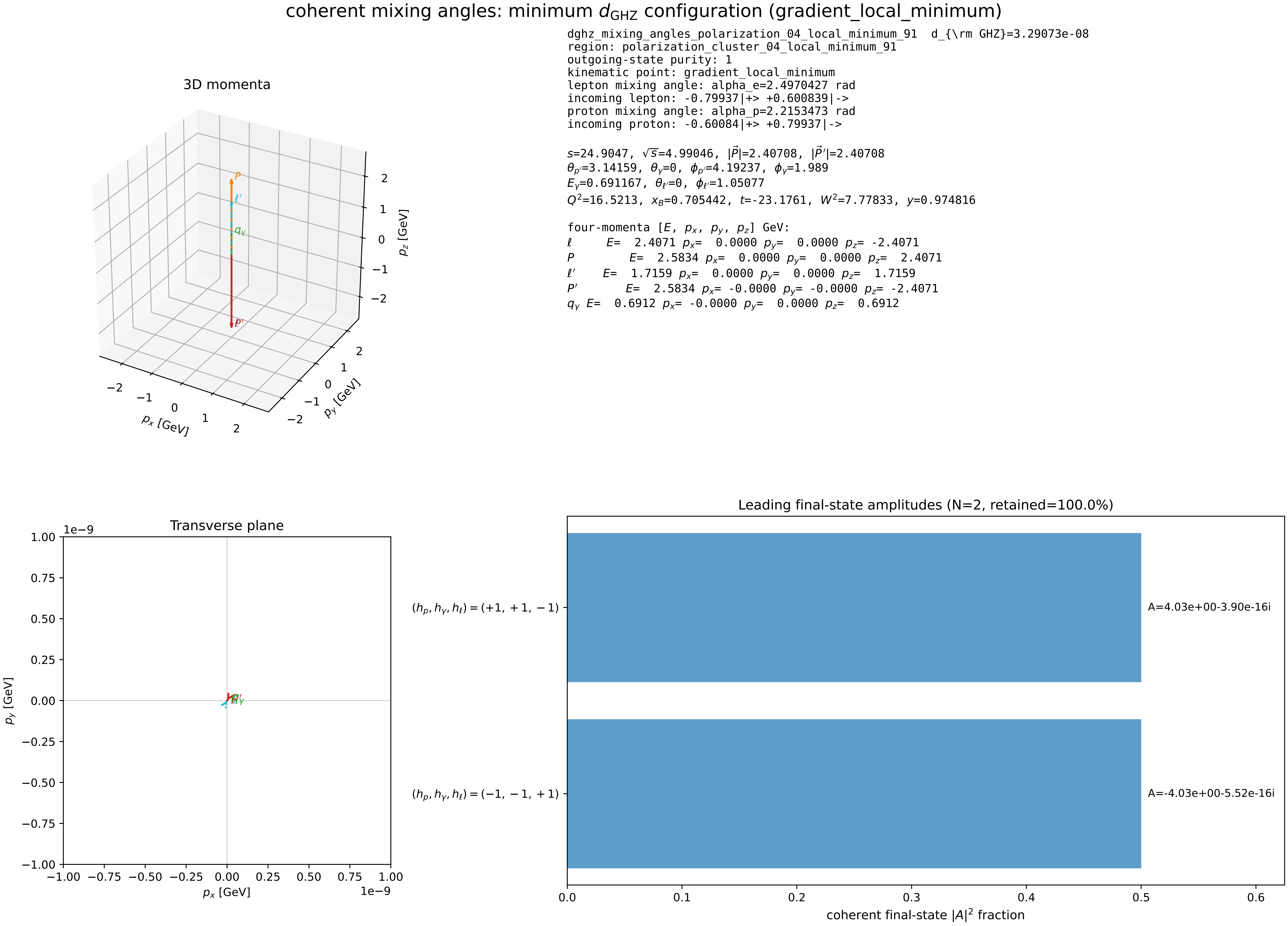


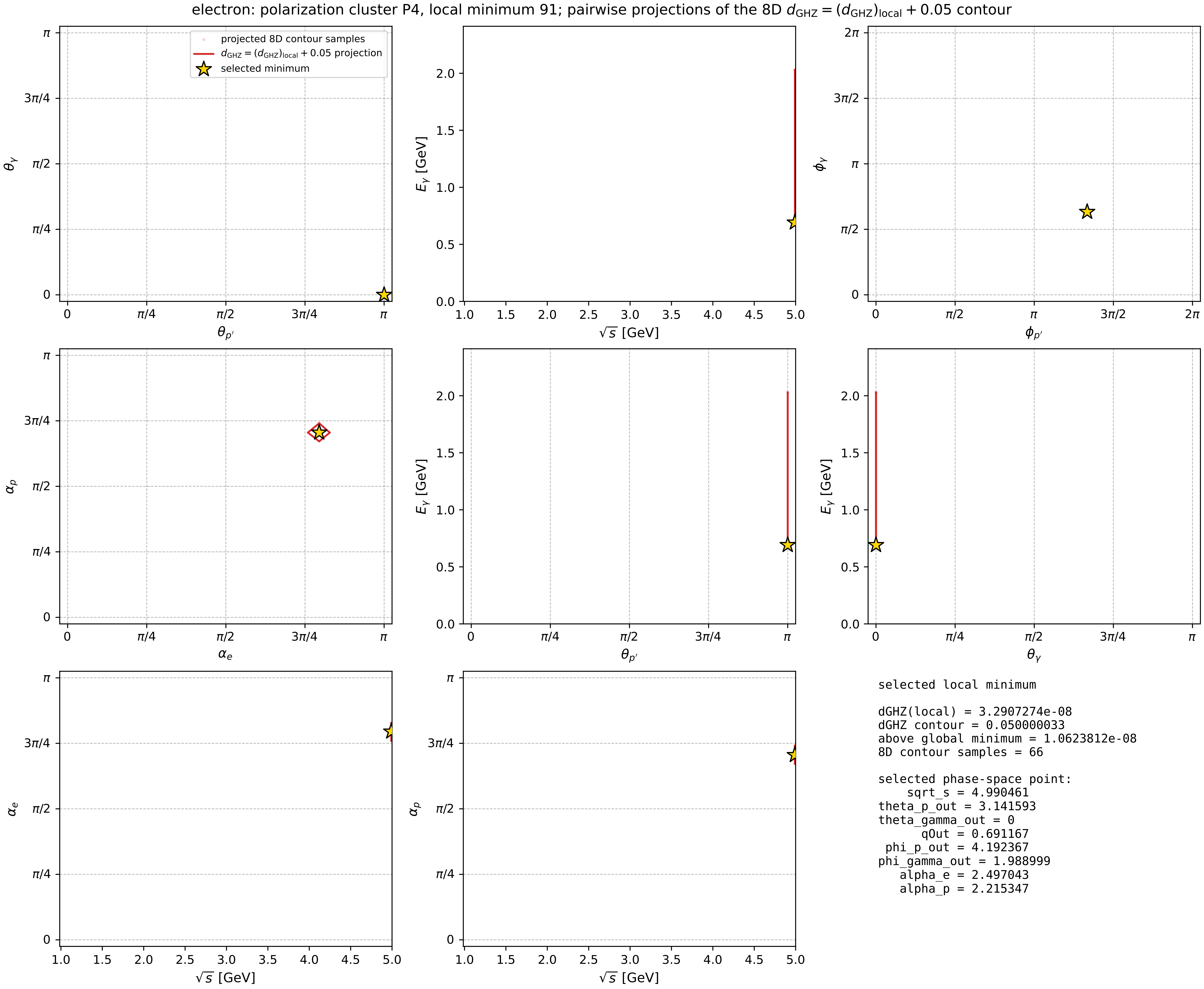


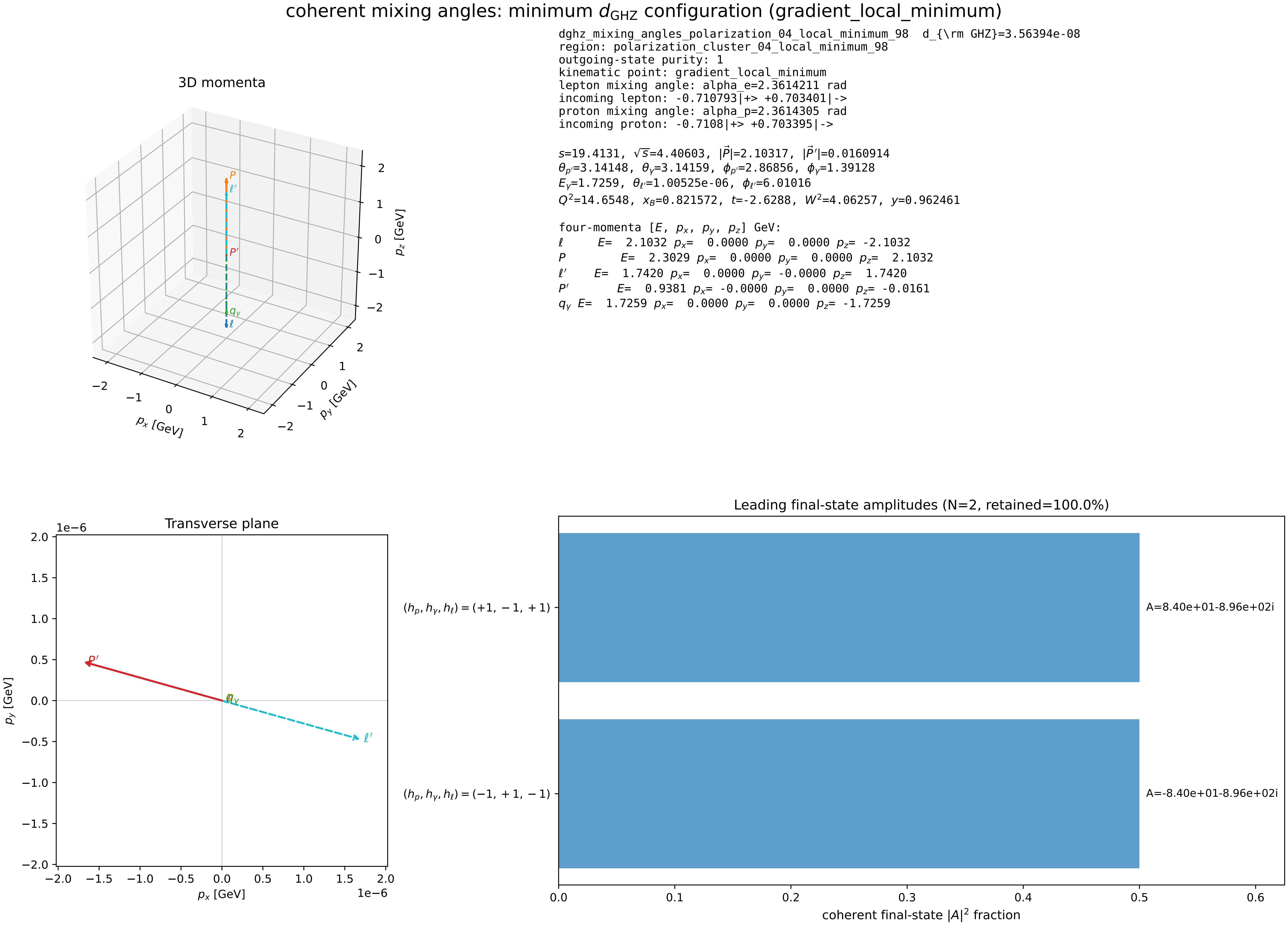

electron: polarization cluster P4, local minimum 85; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 3.1096153e-08
dGHZ contour = 0.050000031
above global minimum = 8.8126914e-09
8D contour samples = 124
selected phase-space point:
sqrt_s = 4.99853
theta_p_out = 0.004988681
theta_gamma_out = 3.141593
qOut = 2.040946
phi_p_out = 1.093274
phi_gamma_out = 1.028096
alpha_e = 2.100876
alpha_p = 2.100883

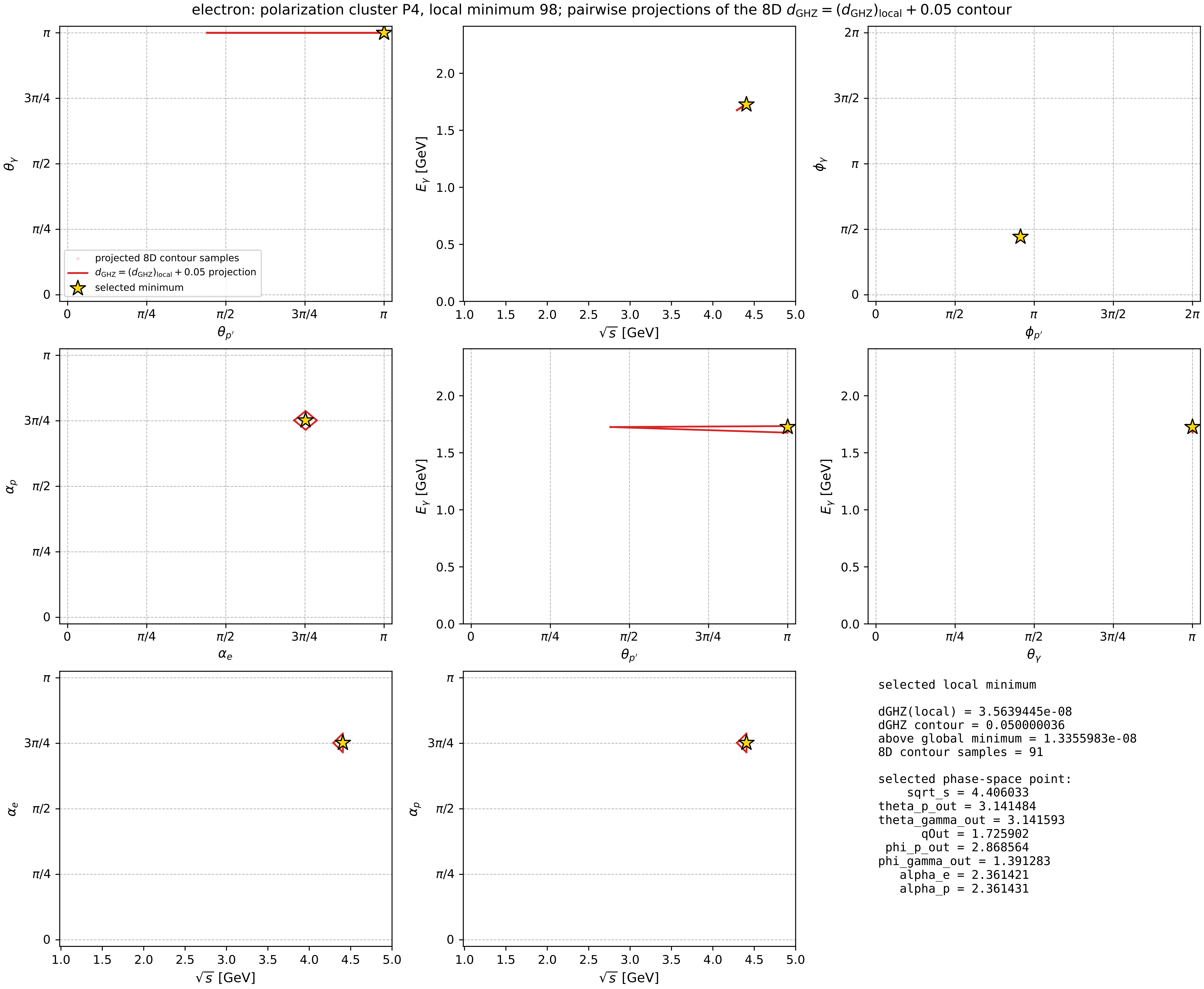

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
3D momenta
dghz_mixing_angles_polarization_04_local_minimum_88 d_{\rm GHZ}=3.20878e-08
region: polarization_cluster_04_local_minimum_88
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2717872 rad
incoming lepton: -0.644975|+> +0.764203|->
proton mixing angle: alpha_p=2.2717877 rad
incoming proton: -0.644976|+> +0.764203|->
s=22.4118, $\sqrt{s}$=4.73411, $|\vec{P}|$=2.27413, $|\vec{P}'|$=0.400283
$\theta_{p'}$=1.00102e-06, θ_γ=3.14159, $\phi_{p'}$=2.40749, ϕ_γ=1.29355
E_γ=2.05728, $\theta_{\ell'}$=2.41656e-07, $\phi_{\ell'}$=5.54909
Q^2=15.0729, x_B=0.720637, t=-1.43729, W^2=6.72301, y=0.971395
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2741 p_x= 0.0000 p_y= 0.0000 p_z= -2.2741
P E= 2.4600 p_x= 0.0000 p_y= 0.0000 p_z= 2.2741
ℓ' E= 1.6570 p_x= 0.0000 p_y= -0.0000 p_z= 1.6570
P' E= 1.0198 p_x= -0.0000 p_y= 0.0000 p_z= 0.4003
q_γ E= 2.0573 p_x= 0.0000 p_y= 0.0000 p_z= -2.0573
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e−7
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=7.99e+02+2.81e+03i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=7.99e+02-2.81e+03i
coherent final-state $|A|^2$ fraction

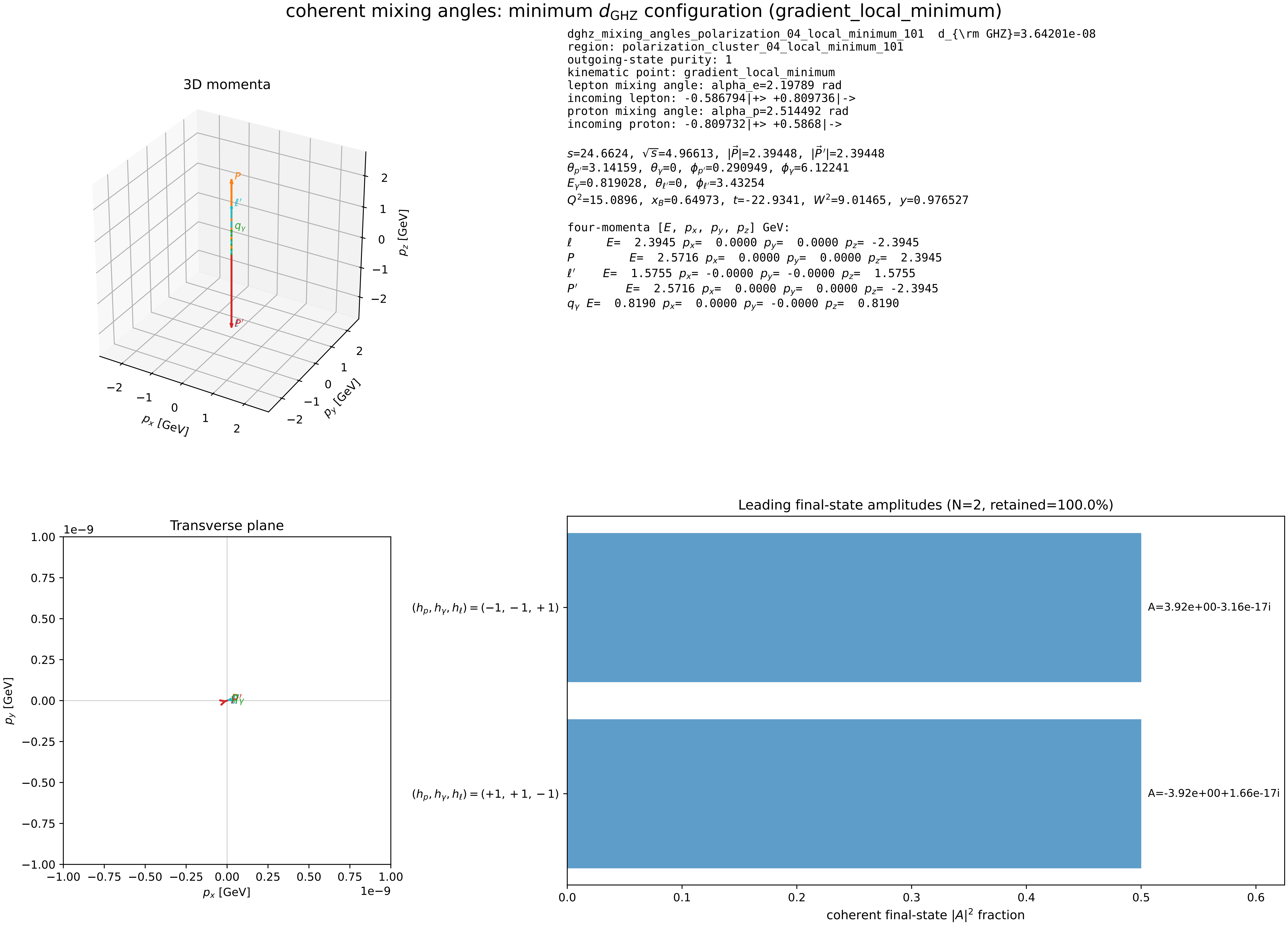

electron: polarization cluster P4, local minimum 88; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 3.2087818e-08
dGHZ contour = 0.050000032
above global minimum = 9.8043564e-09
8D contour samples = 83
selected phase-space point:
sqrt_s = 4.73411
theta_p_out = 1.001016e-06
theta_gamma_out = 3.141593
qOut = 2.057277
phi_p_out = 2.407493
phi_gamma_out = 1.29355
alpha_e = 2.271787
alpha_p = 2.271788

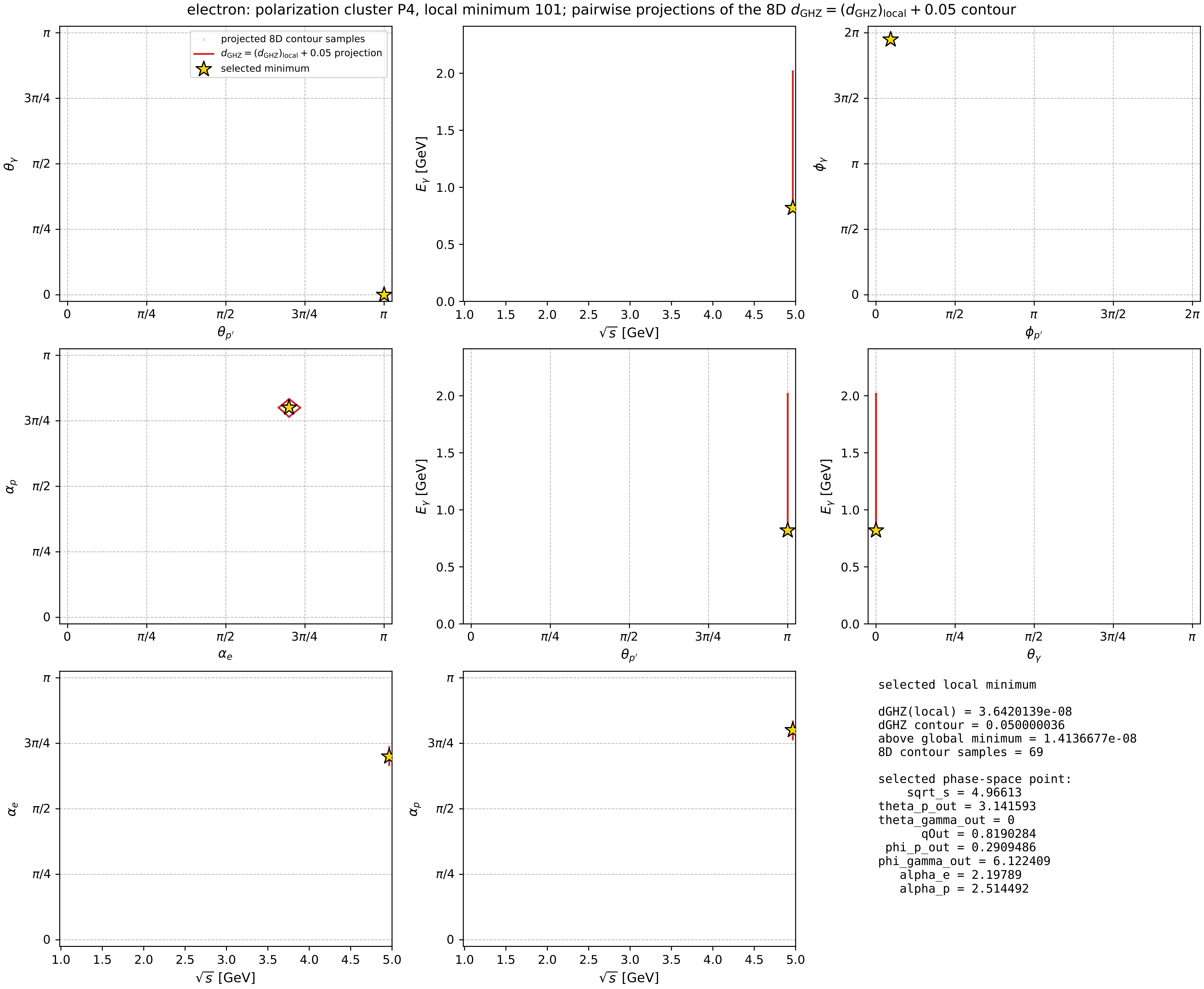

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
P
ℓ'
q_γ
P'
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_91 d_{\rm GHZ}=3.29073e-08
region: polarization_cluster_04_local_minimum_91
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4970427 rad
incoming lepton: -0.79937|+> +0.600839|->
proton mixing angle: alpha_p=2.2153473 rad
incoming proton: -0.60084|+> +0.79937|->
s=24.9047, $\sqrt{s}$=4.99046, $|\vec{P}|$=2.40708, $|\vec{P}'|$=2.40708
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=4.19237, ϕ_γ=1.989
E_γ=0.691167, $\theta_{\ell'}$=0, $\phi_{\ell'}$=1.05077
Q^2=16.5213, x_B=0.705442, t=-23.1761, W^2=7.77833, y=0.974816
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4071 p_x= 0.0000 p_y= 0.0000 p_z= -2.4071
P E= 2.5834 p_x= 0.0000 p_y= 0.0000 p_z= 2.4071
ℓ' E= 1.7159 p_x= 0.0000 p_y= 0.0000 p_z= 1.7159
P' E= 2.5834 p_x= -0.0000 p_y= -0.0000 p_z= -2.4071
q_γ E= 0.6912 p_x= -0.0000 p_y= 0.0000 p_z= 0.6912
Transverse plane
1e−9
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=4.03e+00-3.90e-16i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-4.03e+00-5.52e-16i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 91; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 3.2907274e-08
dGHZ contour = 0.050000033
above global minimum = 1.0623812e-08
8D contour samples = 66
selected phase-space point:
sqrt_s = 4.990461
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 0.691167
phi_p_out = 4.192367
phi_gamma_out = 1.988999
alpha_e = 2.497043
alpha_p = 2.215347

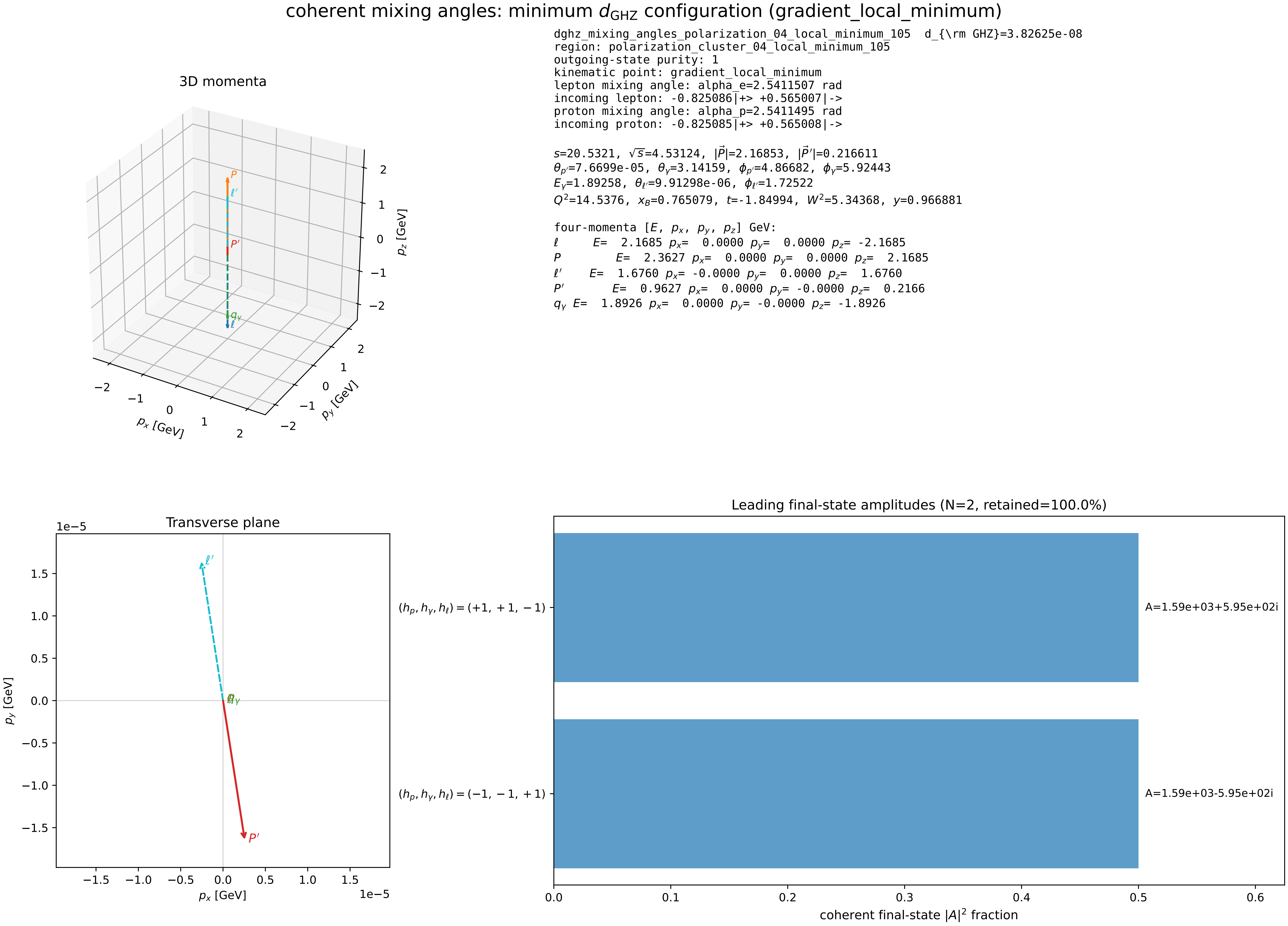

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
P
ℓ'
P'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_98 d_{\rm GHZ}=3.56394e-08
region: polarization_cluster_04_local_minimum_98
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3614211 rad
incoming lepton: -0.710793|+> +0.703401|->
proton mixing angle: alpha_p=2.3614305 rad
incoming proton: -0.7108|+> +0.703395|->
s=19.4131, $\sqrt{s}$=4.40603, $|\vec{P}|$=2.10317, $|\vec{P}'|$=0.0160914
$\theta_{p'}$=3.14148, θ_γ=3.14159, $\phi_{p'}$=2.86856, ϕ_γ=1.39128
E_γ=1.7259, $\theta_{\ell'}$=1.00525e-06, $\phi_{\ell'}$=6.01016
Q^2=14.6548, x_B=0.821572, t=-2.6288, W^2=4.06257, y=0.962461
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.1032 p_x= 0.0000 p_y= 0.0000 p_z= -2.1032
P E= 2.3029 p_x= 0.0000 p_y= 0.0000 p_z= 2.1032
ℓ' E= 1.7420 p_x= 0.0000 p_y= -0.0000 p_z= 1.7420
P' E= 0.9381 p_x= -0.0000 p_y= 0.0000 p_z= -0.0161
q_γ E= 1.7259 p_x= 0.0000 p_y= 0.0000 p_z= -1.7259
Transverse plane
1e−6
p_y [GeV]
p_x [GeV]
P'
ℓ'
q_γ
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=8.40e+01-8.96e+02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=-8.40e+01-8.96e+02i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 98; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 3.5639445e-08
dGHZ contour = 0.050000036
above global minimum = 1.3355983e-08
8D contour samples = 91
selected phase-space point:
sqrt_s = 4.406033
theta_p_out = 3.141484
theta_gamma_out = 3.141593
qOut = 1.725902
phi_p_out = 2.868564
phi_gamma_out = 1.391283
alpha_e = 2.361421
alpha_p = 2.361431

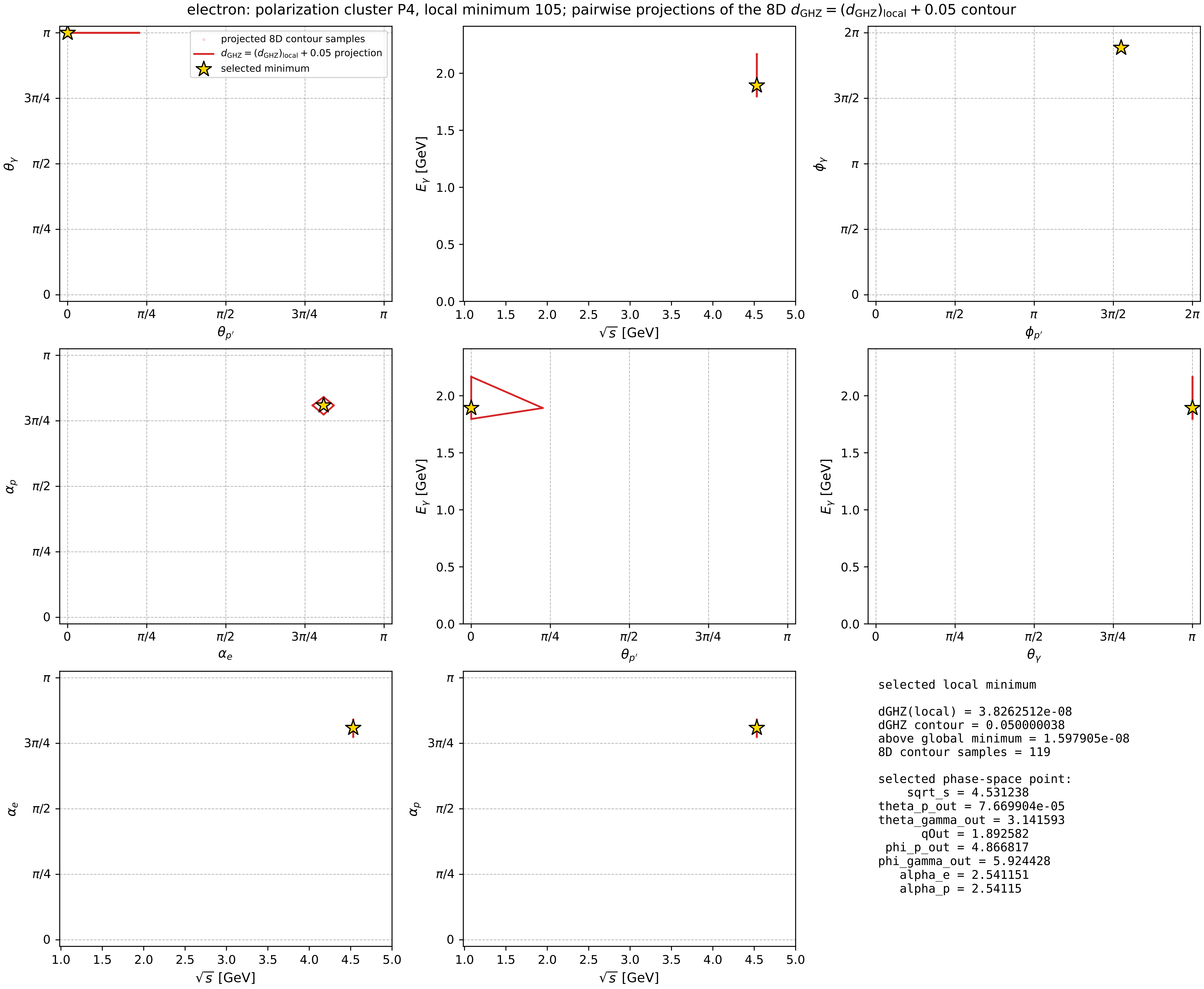

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_101 d_{\rm GHZ}=3.64201e-08
region: polarization_cluster_04_local_minimum_101
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.19789 rad
incoming lepton: -0.586794|+> +0.809736|->
proton mixing angle: alpha_p=2.514492 rad
incoming proton: -0.809732|+> +0.5868|->
s=24.6624, $\sqrt{s}$=4.96613, $|\vec{P}|$=2.39448, $|\vec{P}'|$=2.39448
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=0.290949, ϕ_γ=6.12241
E_γ=0.819028, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.43254
Q^2=15.0896, x_B=0.64973, t=-22.9341, W^2=9.01465, y=0.976527
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3945 p_x= 0.0000 p_y= 0.0000 p_z= -2.3945
P E= 2.5716 p_x= 0.0000 p_y= 0.0000 p_z= 2.3945
ℓ' E= 1.5755 p_x= -0.0000 p_y= -0.0000 p_z= 1.5755
P' E= 2.5716 p_x= 0.0000 p_y= 0.0000 p_z= -2.3945
q_γ E= 0.8190 p_x= 0.0000 p_y= -0.0000 p_z= 0.8190
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
1e−9
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=3.92e+00-3.16e-17i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-3.92e+00+1.66e-17i
coherent final-state $|A|^2$ fraction



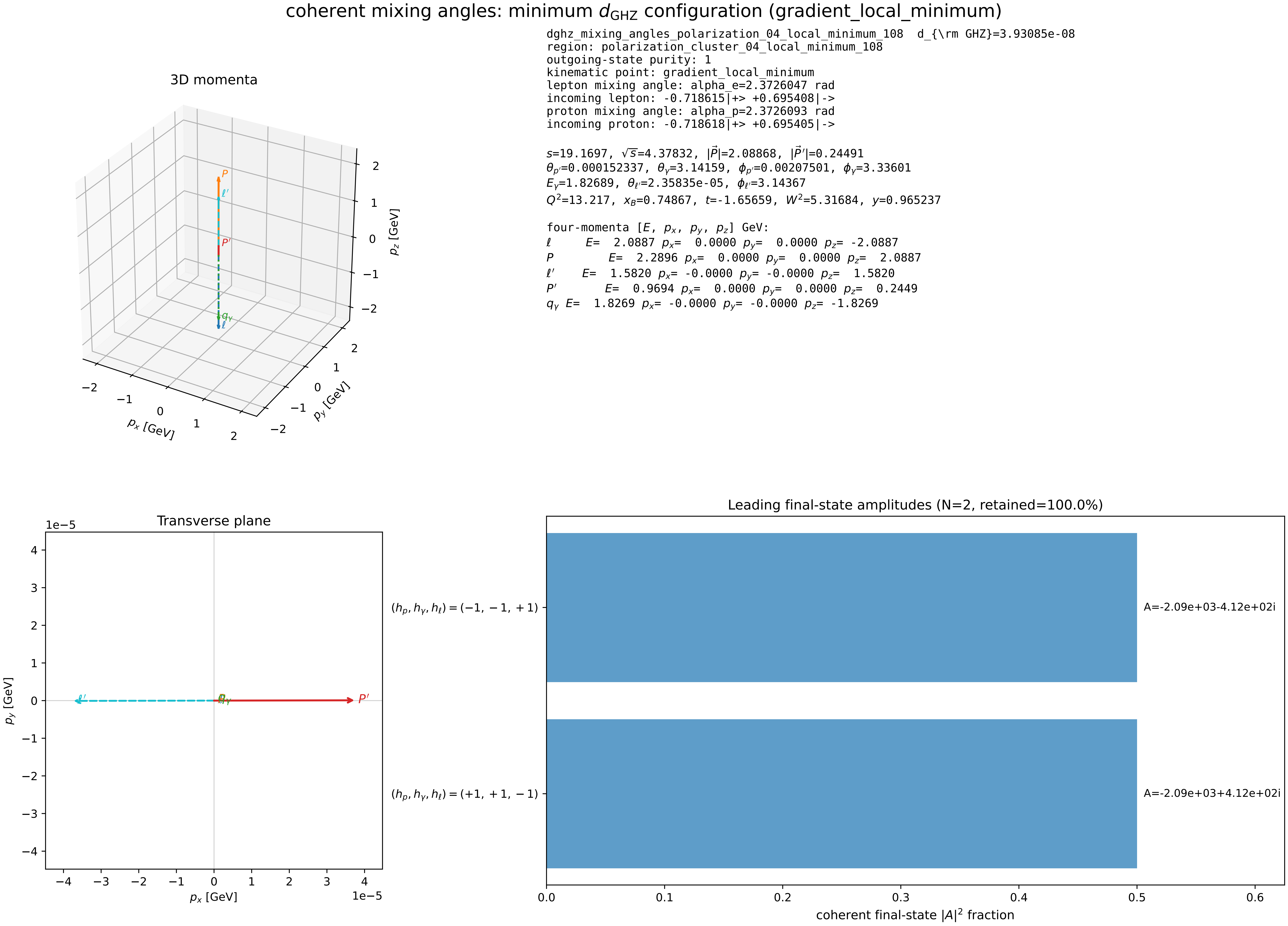
electron: polarization cluster P4, local minimum 101; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 3.6420139e-08
dGHZ contour = 0.050000036
above global minimum = 1.4136677e-08
8D contour samples = 69
selected phase-space point:
sqrt_s = 4.96613
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 0.8190284
phi_p_out = 0.2909486
phi_gamma_out = 6.122409
alpha_e = 2.19789
alpha_p = 2.514492

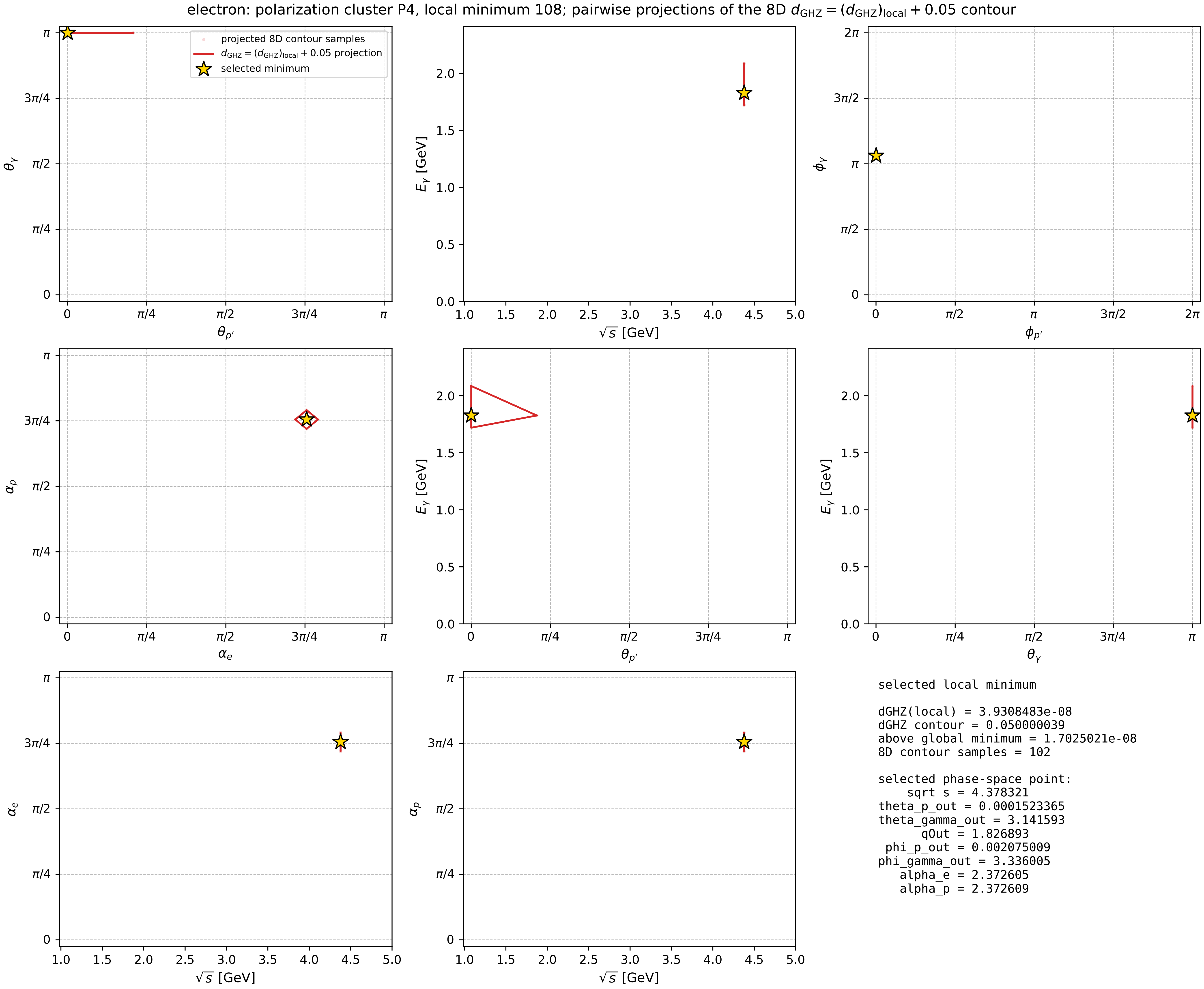


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
dghz_mixing_angles_polarization_04_local_minimum_105 d_{\rm GHZ}=3.82625e-08
region: polarization_cluster_04_local_minimum_105
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.5411507 rad
incoming lepton: -0.825086|+> +0.565007|->
proton mixing angle: alpha_p=2.5411495 rad
incoming proton: -0.825085|+> +0.565008|->
s=20.5321, $\sqrt{s}$=4.53124, $|\vec{P}|$=2.16853, $|\vec{P}'|$=0.216611
$\theta_{p'}$=7.6699e-05, θ_γ=3.14159, $\phi_{p'}$=4.86682, ϕ_γ=5.92443
E_γ=1.89258, $\theta_{\ell'}$=9.91298e-06, $\phi_{\ell'}$=1.72522
Q^2=14.5376, x_B=0.765079, t=-1.84994, W^2=5.34368, y=0.966881
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.1685 p_x= 0.0000 p_y= 0.0000 p_z= -2.1685
P E= 2.3627 p_x= 0.0000 p_y= 0.0000 p_z= 2.1685
ℓ' E= 1.6760 p_x= -0.0000 p_y= 0.0000 p_z= 1.6760
P' E= 0.9627 p_x= 0.0000 p_y= -0.0000 p_z= 0.2166
q_γ E= 1.8926 p_x= 0.0000 p_y= -0.0000 p_z= -1.8926
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e-5
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=1.59e+03+5.95e+02i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=1.59e+03-5.95e+02i
coherent final-state $|A|^2$ fraction



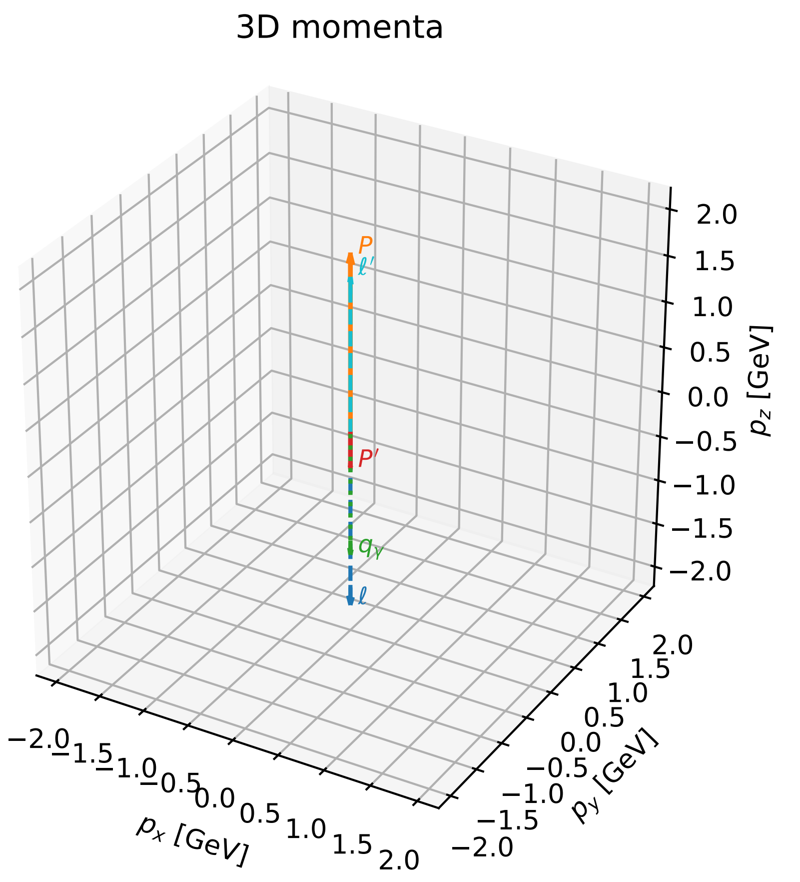
electron: polarization cluster P4, local minimum 105; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 3.8262512e-08
dGHZ contour = 0.050000038
above global minimum = 1.597905e-08
8D contour samples = 119
selected phase-space point:
sqrt_s = 4.531238
theta_p_out = 7.669904e-05
theta_gamma_out = 3.141593
qOut = 1.892582
phi_p_out = 4.866817
phi_gamma_out = 5.924428
alpha_e = 2.541151
alpha_p = 2.54115


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
dghz_mixing_angles_polarization_04_local_minimum_108 d_{\rm GHZ}=3.93085e-08
region: polarization_cluster_04_local_minimum_108
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3726047 rad
incoming lepton: -0.718615|+> +0.695408|->
proton mixing angle: alpha_p=2.3726093 rad
incoming proton: -0.718618|+> +0.695405|->
s=19.1697, $\sqrt{s}$=4.37832, $|\vec{P}|$=2.08868, $|\vec{P}'|$=0.24491
$\theta_{p'}$=0.000152337, θ_γ=3.14159, $\phi_{p'}$=0.00207501, ϕ_γ=3.33601
E_γ=1.82689, $\theta_{\ell'}$=2.35835e-05, $\phi_{\ell'}$=3.14367
Q^2=13.217, x_B=0.74867, t=-1.65659, W^2=5.31684, y=0.965237
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.0887 p_x= 0.0000 p_y= 0.0000 p_z= -2.0887
P E= 2.2896 p_x= 0.0000 p_y= 0.0000 p_z= 2.0887
ℓ' E= 1.5820 p_x= -0.0000 p_y= -0.0000 p_z= 1.5820
P' E= 0.9694 p_x= 0.0000 p_y= 0.0000 p_z= 0.2449
q_γ E= 1.8269 p_x= -0.0000 p_y= -0.0000 p_z= -1.8269
P
ℓ'
P'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e−5
ℓ'
P'
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-2.09e+03-4.12e+02i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-2.09e+03+4.12e+02i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P4, local minimum 108; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 3.9308483e-08
dGHZ contour = 0.050000039
above global minimum = 1.7025021e-08
8D contour samples = 102
selected phase-space point:
sqrt_s = 4.378321
theta_p_out = 0.0001523365
theta_gamma_out = 3.141593
qOut = 1.826893
phi_p_out = 0.002075009
phi_gamma_out = 3.336005
alpha_e = 2.372605
alpha_p = 2.372609

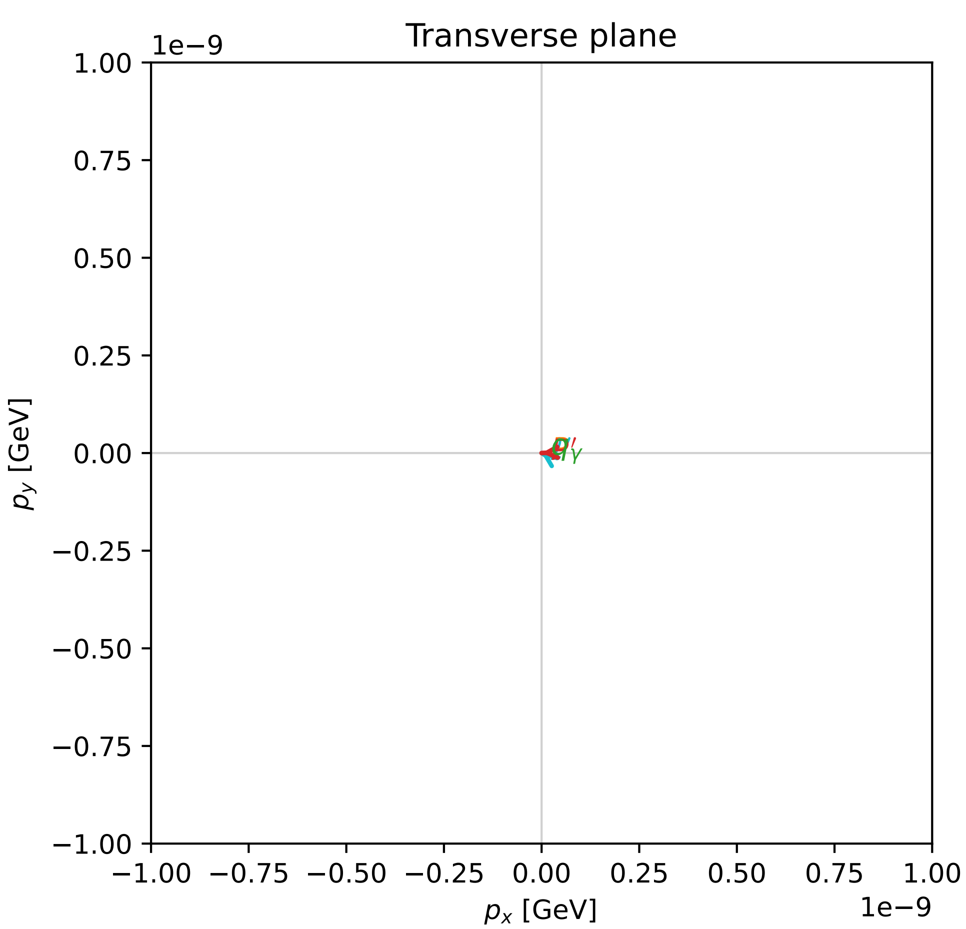
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

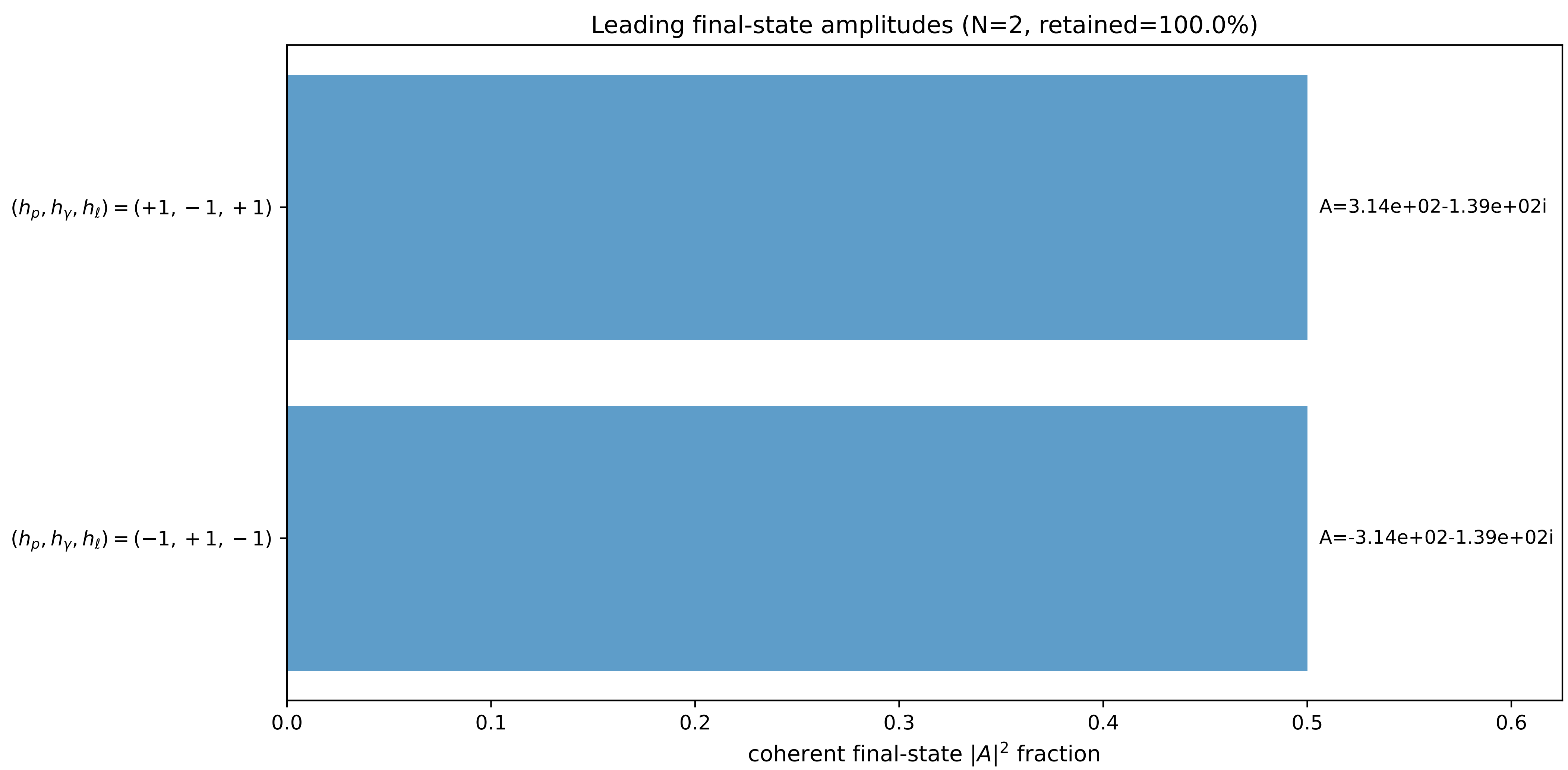


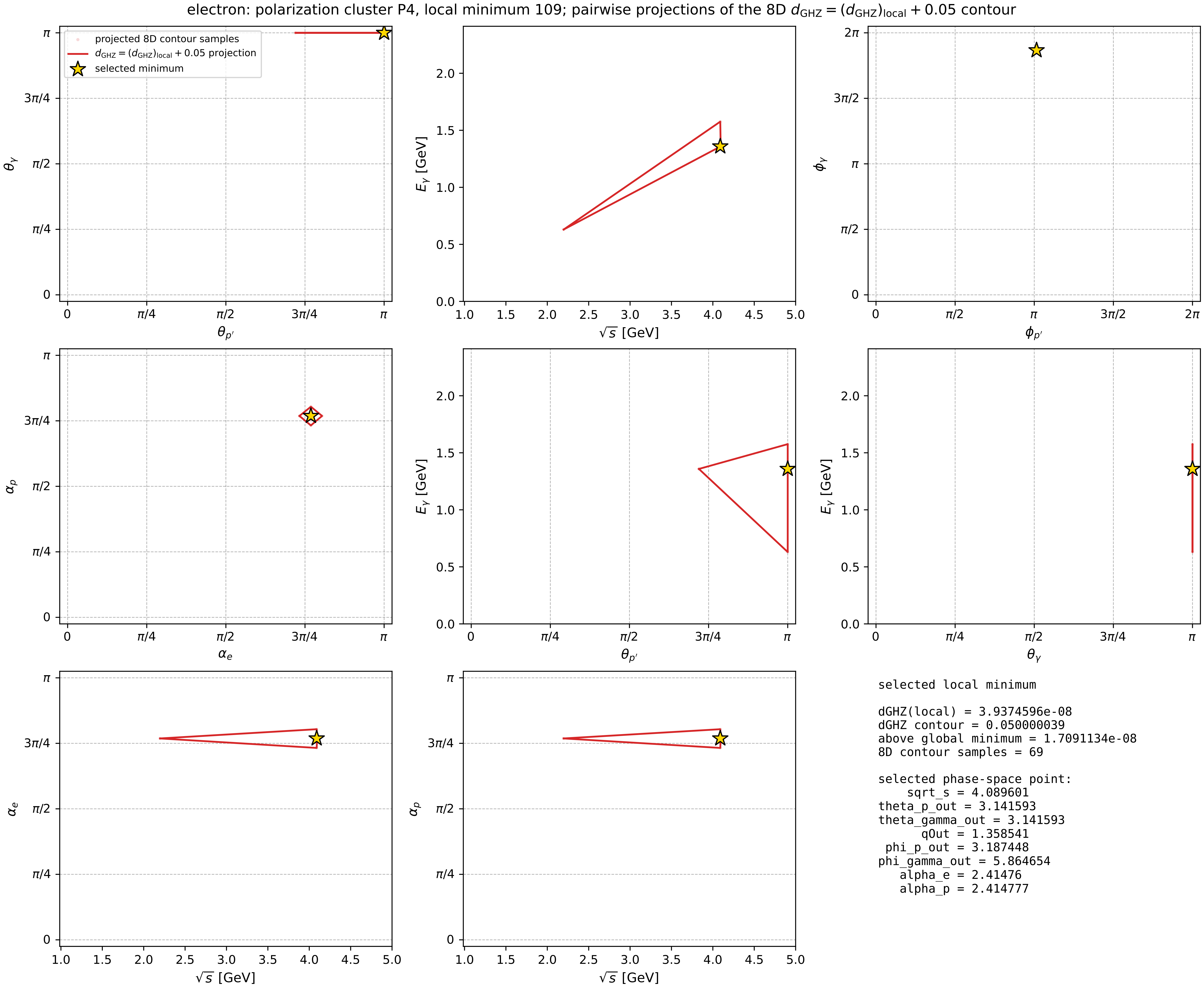
dghz_mixing_angles_polarization_04_local_minimum_109 d_{\rm GHZ}=3.93746e-08
region: polarization_cluster_04_local_minimum_109
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4147598 rad
incoming lepton: -0.747283|+> +0.664506|->
proton mixing angle: alpha_p=2.4147771 rad
incoming proton: -0.747294|+> +0.664493|->

s=16.7248, $\sqrt{s}$=4.0896, $|\vec{P}|$=1.93723, $|\vec{P}'|$=0.365737
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=3.18745, ϕ_γ=5.86465
E_γ=1.35854, $\theta_{\ell'}$=0, $\phi_{\ell'}$=2.56558
Q^2=13.3613, x_B=0.884674, t=-3.99128, W^2=2.62161, y=0.953176

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.9372 p_x= 0.0000 p_y= 0.0000 p_z= -1.9372
P E= 2.1524 p_x= 0.0000 p_y= 0.0000 p_z= 1.9372
ℓ' E= 1.7243 p_x= -0.0000 p_y= 0.0000 p_z= 1.7243
P' E= 1.0068 p_x= -0.0000 p_y= -0.0000 p_z= -0.3657
q_γ E= 1.3585 p_x= 0.0000 p_y= -0.0000 p_z= -1.3585

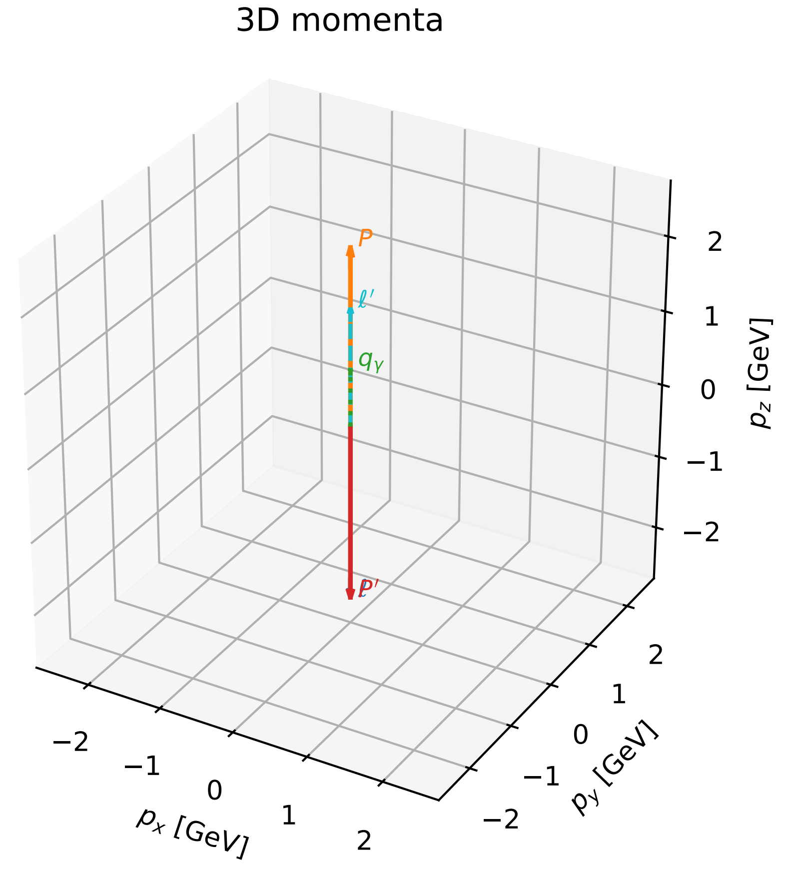






electron: polarization cluster P4, local minimum 109; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_{γ}
$\theta_{p'}$
E_{γ} [GeV]
$\sqrt{s}$ [GeV]
ϕ_{γ}
$\phi_{p'}$
α_p
α_e
θ_{γ}
selected local minimum
dGHZ(local) = 3.9374596e-08
dGHZ contour = 0.050000039
above global minimum = 1.7091134e-08
8D contour samples = 69
selected phase-space point:
sqrt_s = 4.089601
theta_p_out = 3.141593
theta_gamma_out = 3.141593
qOut = 1.358541
phi_p_out = 3.187448
phi_gamma_out = 5.864654
alpha_e = 2.41476
alpha_p = 2.414777

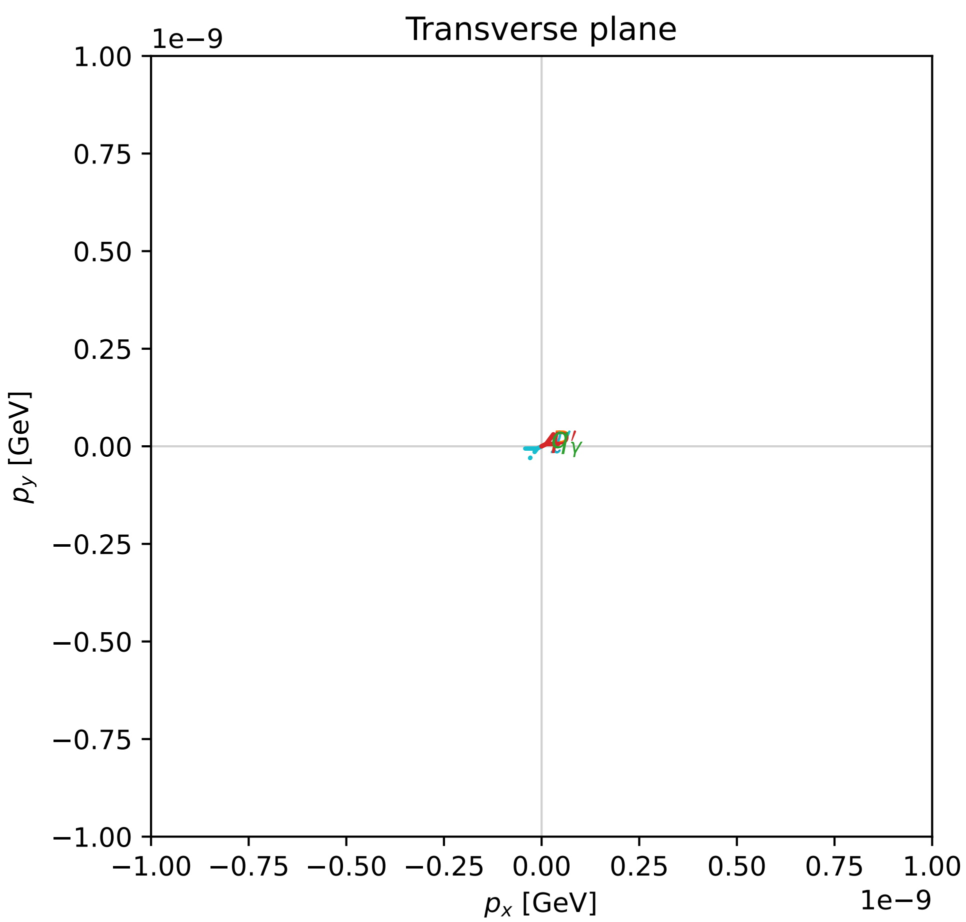
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

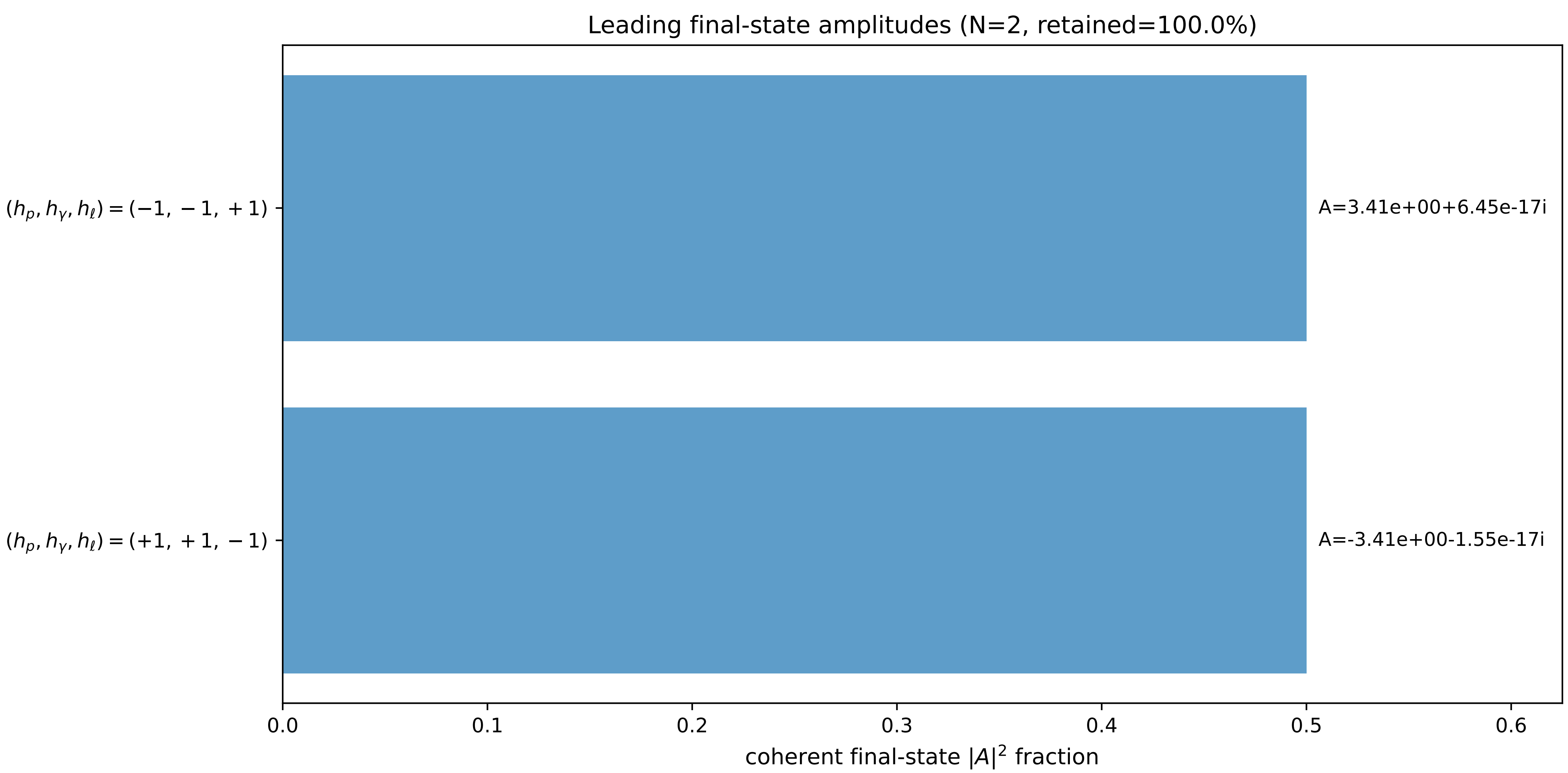


dghz_mixing_angles_polarization_04_local_minimum_117 d_{\rm GHZ}=4.36167e-08
region: polarization_cluster_04_local_minimum_117
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.0273398 rad
incoming lepton: -0.440848|+> +0.897582|->
proton mixing angle: alpha_p=2.6850445 rad
incoming proton: -0.89758|+> +0.440852|->

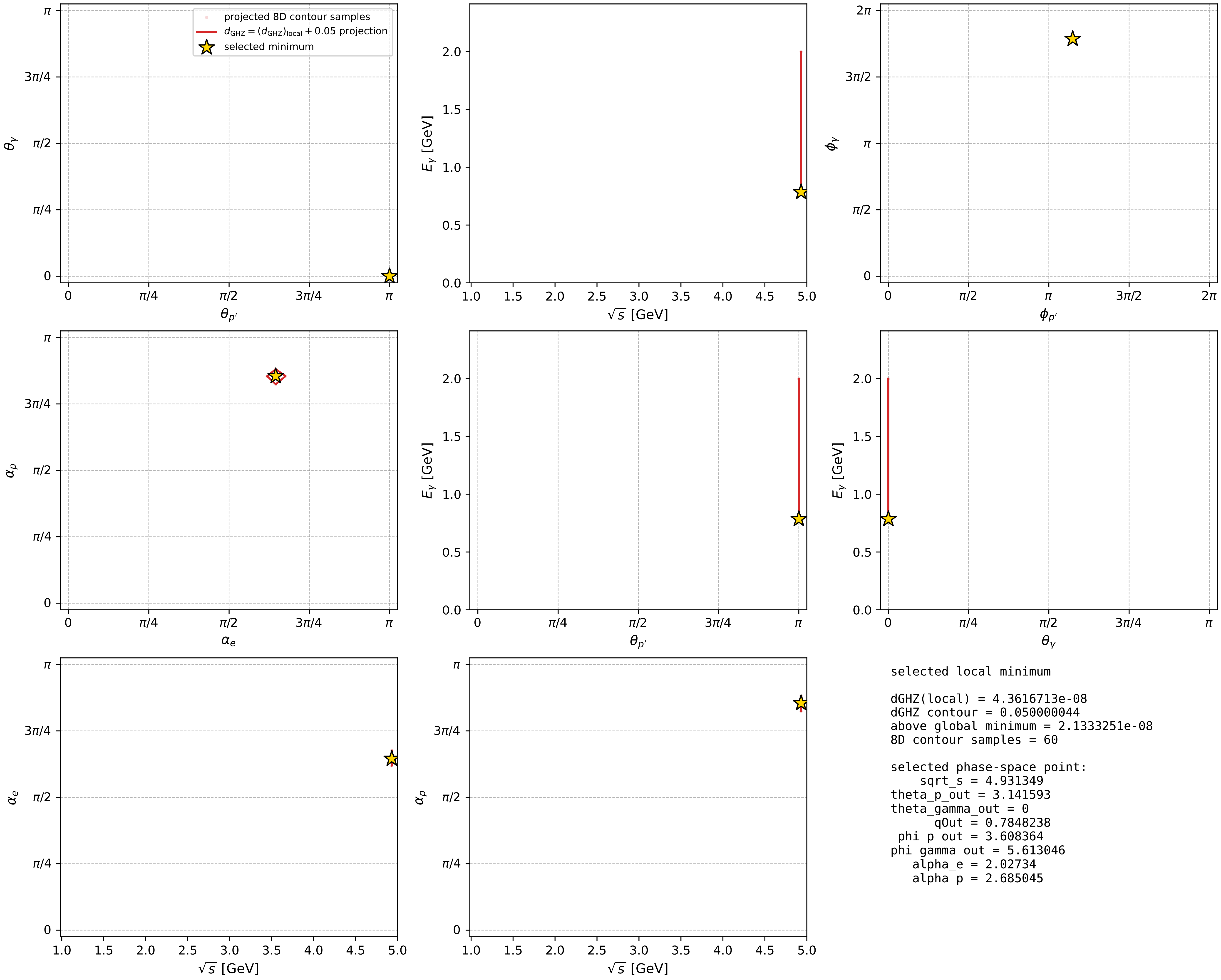
s=24.3182, $\sqrt{s}$=4.93135, $|\vec{P}|$=2.37646, $|\vec{P}'|$=2.37646
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=3.60836, ϕ_γ=5.61305
E_γ=0.784824, $\theta_{\ell'}$=0, $\phi_{\ell'}$=0.466771
Q^2=15.1299, x_B=0.66155, t=-22.5903, W^2=8.62032, y=0.975768

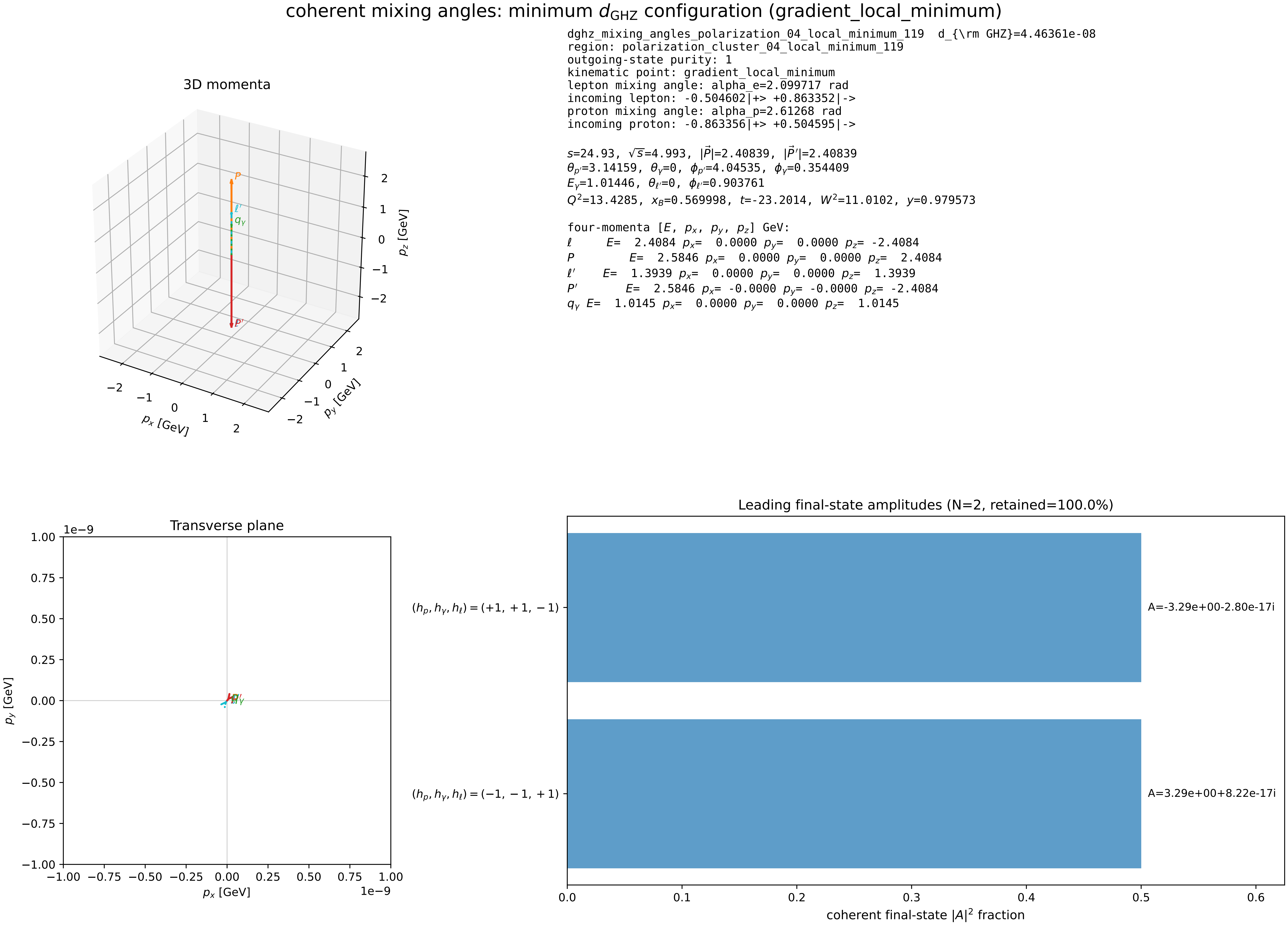
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3765 p_x= 0.0000 p_y= 0.0000 p_z= -2.3765
P E= 2.5549 p_x= 0.0000 p_y= 0.0000 p_z= 2.3765
ℓ' E= 1.5916 p_x= 0.0000 p_y= 0.0000 p_z= 1.5916
P' E= 2.5549 p_x= -0.0000 p_y= -0.0000 p_z= -2.3765
q_γ E= 0.7848 p_x= 0.0000 p_y= -0.0000 p_z= 0.7848





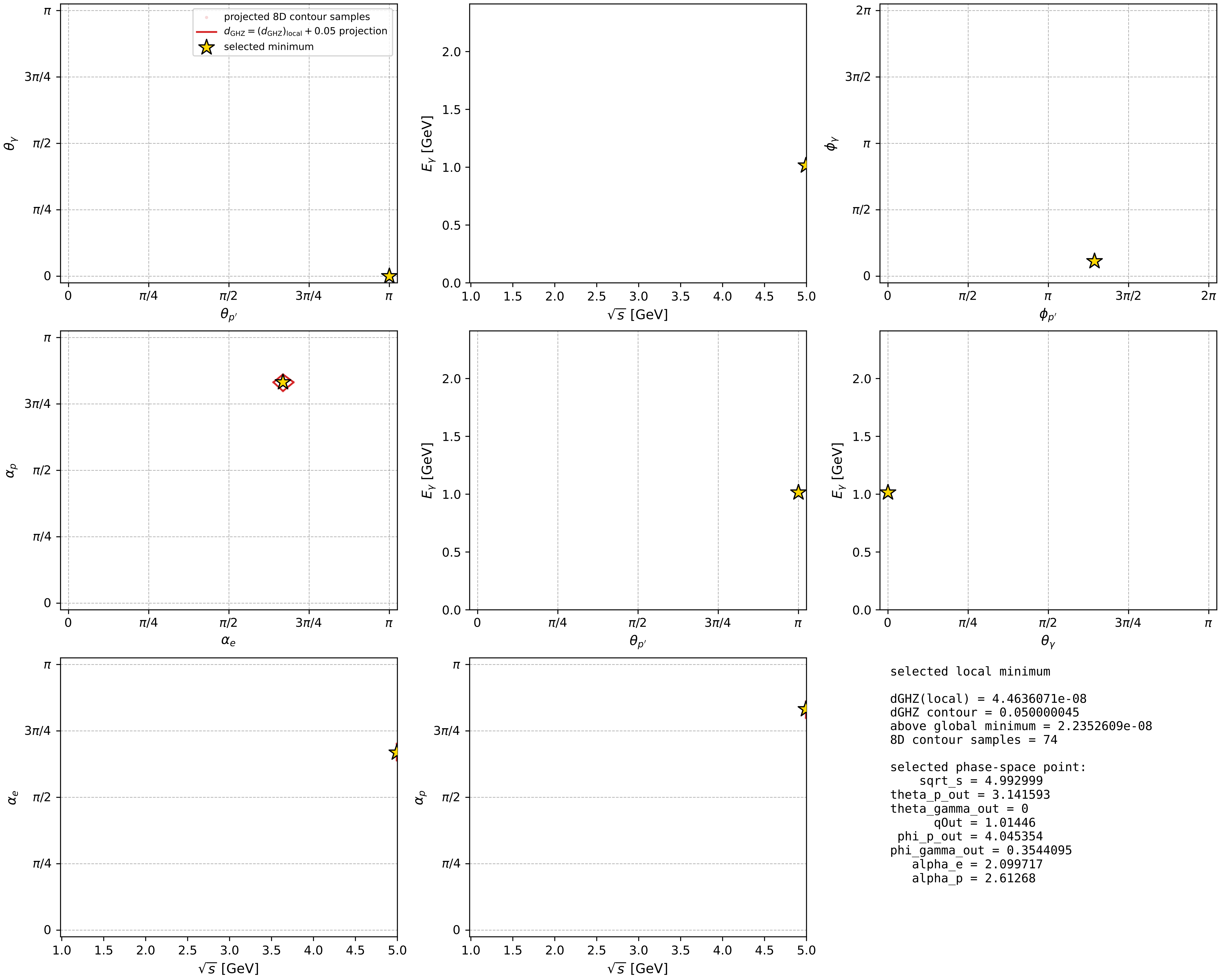
electron: polarization cluster P4, local minimum 117; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



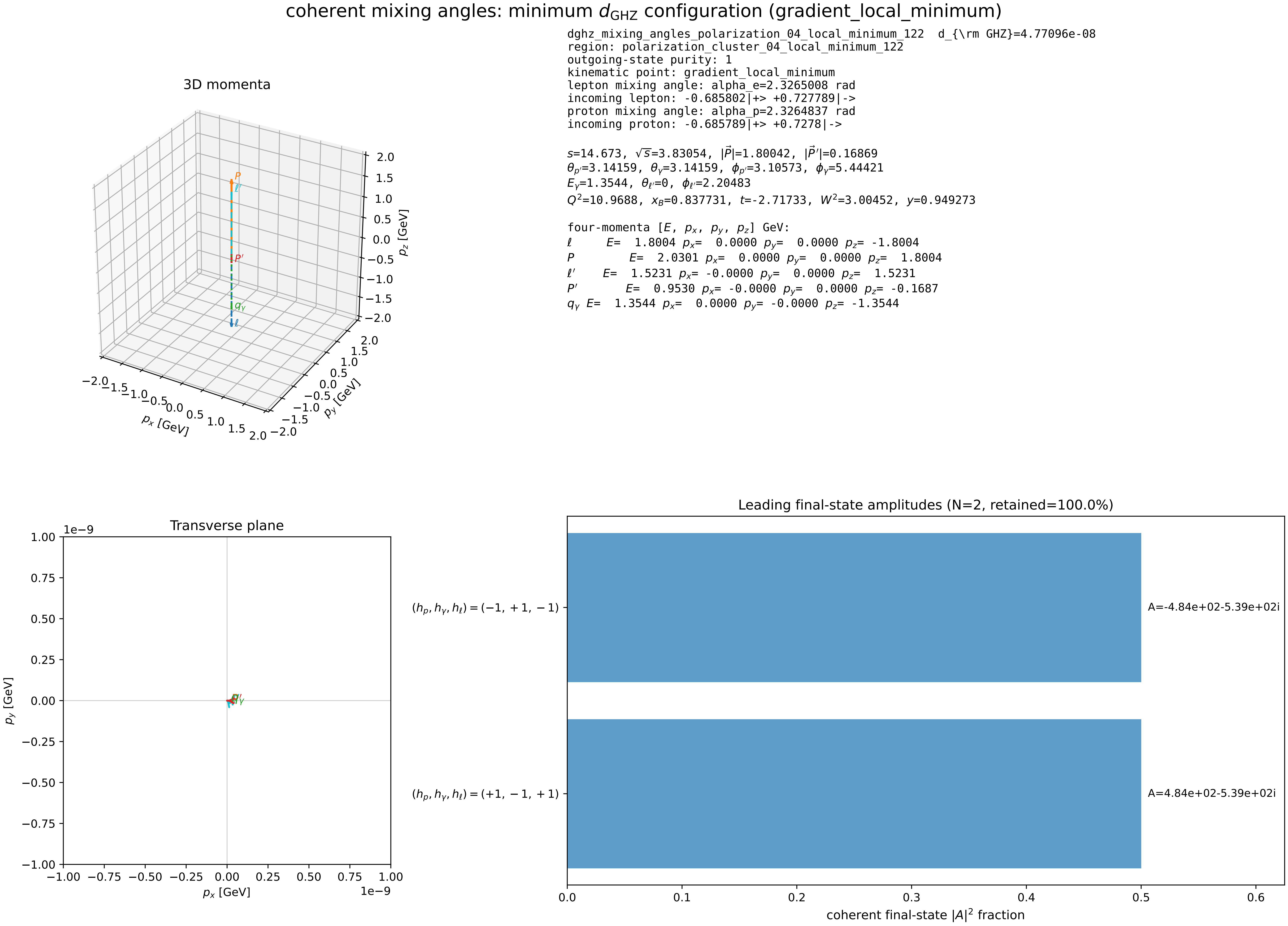


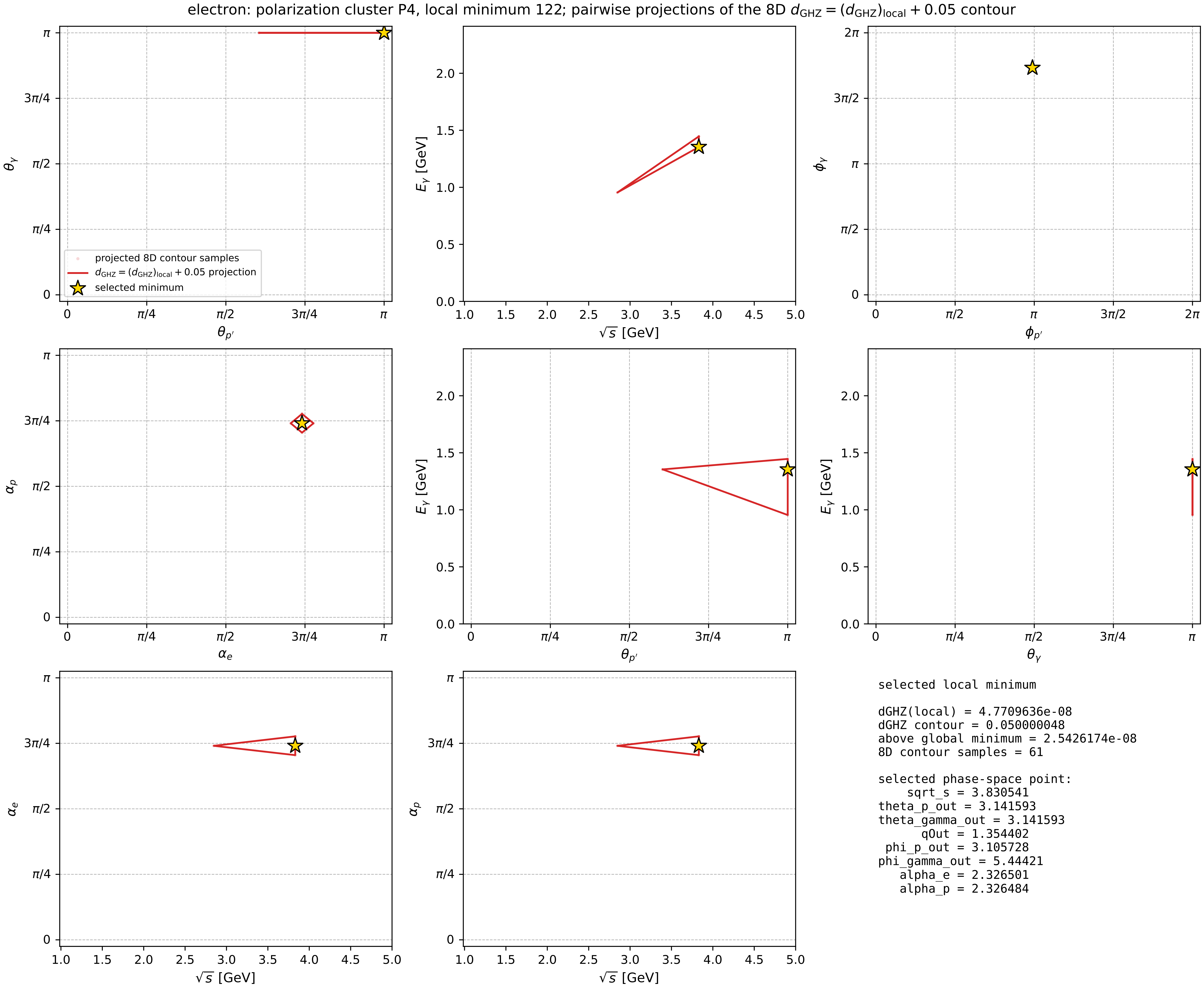
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
dghz_mixing_angles_polarization_04_local_minimum_119 d_{\rm GHZ}=4.46361e-08
region: polarization_cluster_04_local_minimum_119
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.099717 rad
incoming lepton: -0.504602|+> +0.863352|->
proton mixing angle: alpha_p=2.61268 rad
incoming proton: -0.863356|+> +0.504595|->
s=24.93, $\sqrt{s}$=4.993, $|\vec{P}|$=2.40839, $|\vec{P}'|$=2.40839
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=4.04535, ϕ_γ=0.354409
E_γ=1.01446, $\theta_{\ell'}$=0, $\phi_{\ell'}$=0.903761
Q^2=13.4285, x_B=0.569998, t=-23.2014, W^2=11.0102, y=0.979573
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4084 p_x= 0.0000 p_y= 0.0000 p_z= -2.4084
P E= 2.5846 p_x= 0.0000 p_y= 0.0000 p_z= 2.4084
ℓ' E= 1.3939 p_x= 0.0000 p_y= 0.0000 p_z= 1.3939
P' E= 2.5846 p_x= -0.0000 p_y= -0.0000 p_z= -2.4084
q_γ E= 1.0145 p_x= 0.0000 p_y= 0.0000 p_z= 1.0145
P
ℓ'
q_γ
P'
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e−9
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-3.29e+00-2.80e-17i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=3.29e+00+8.22e-17i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P4, local minimum 119; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour

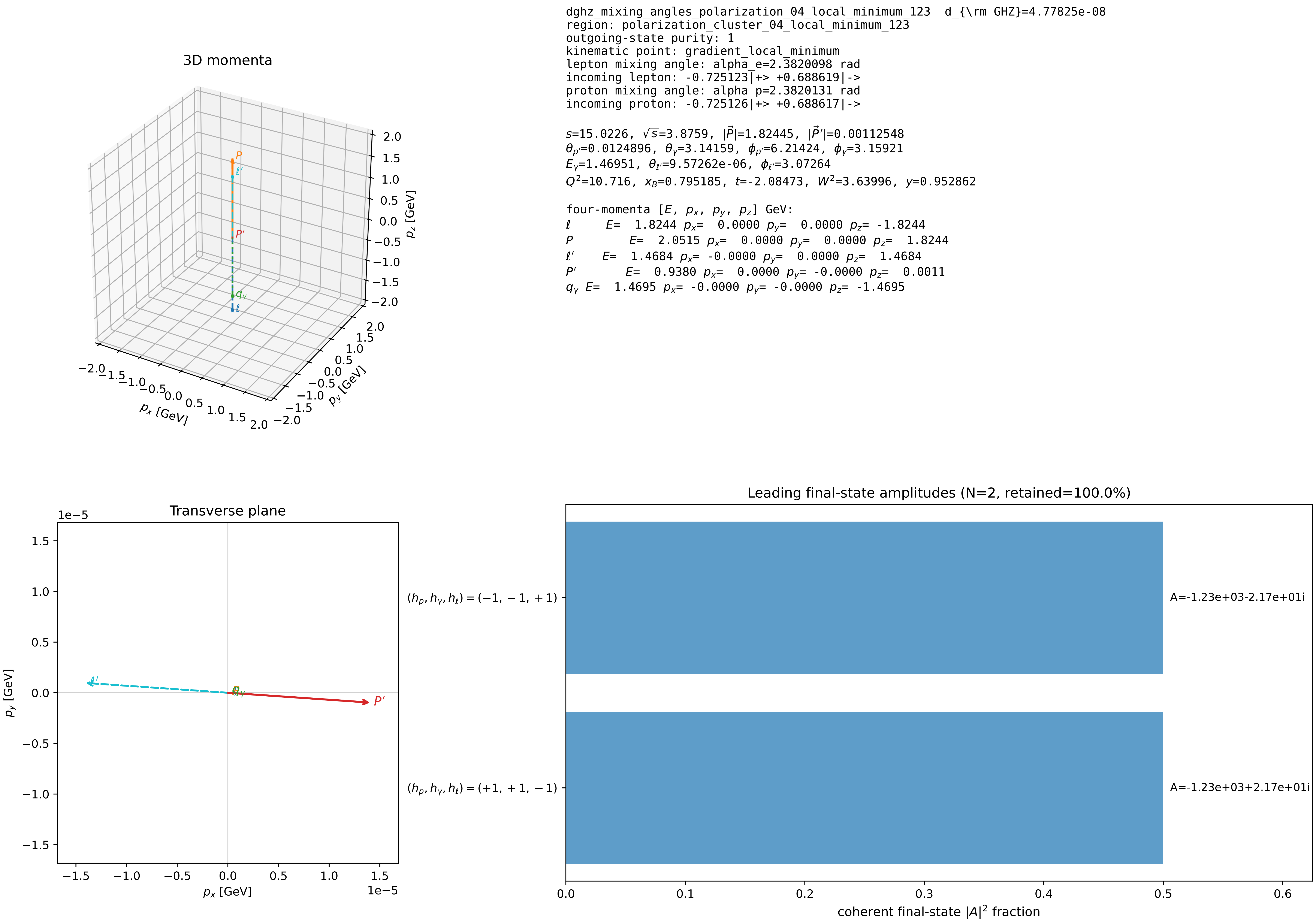



```
selected local minimum

dGHZ(local) = 4.4636071e-08
dGHZ contour = 0.050000045
above global minimum = 2.2352609e-08
8D contour samples = 74

selected phase-space point:
    sqrt_s = 4.992999
theta_p_out = 3.141593
theta_gamma_out = 0
      qOut = 1.01446
 phi_p_out = 4.045354
phi_gamma_out = 0.3544095
   alpha_e = 2.099717
   alpha_p = 2.61268
```

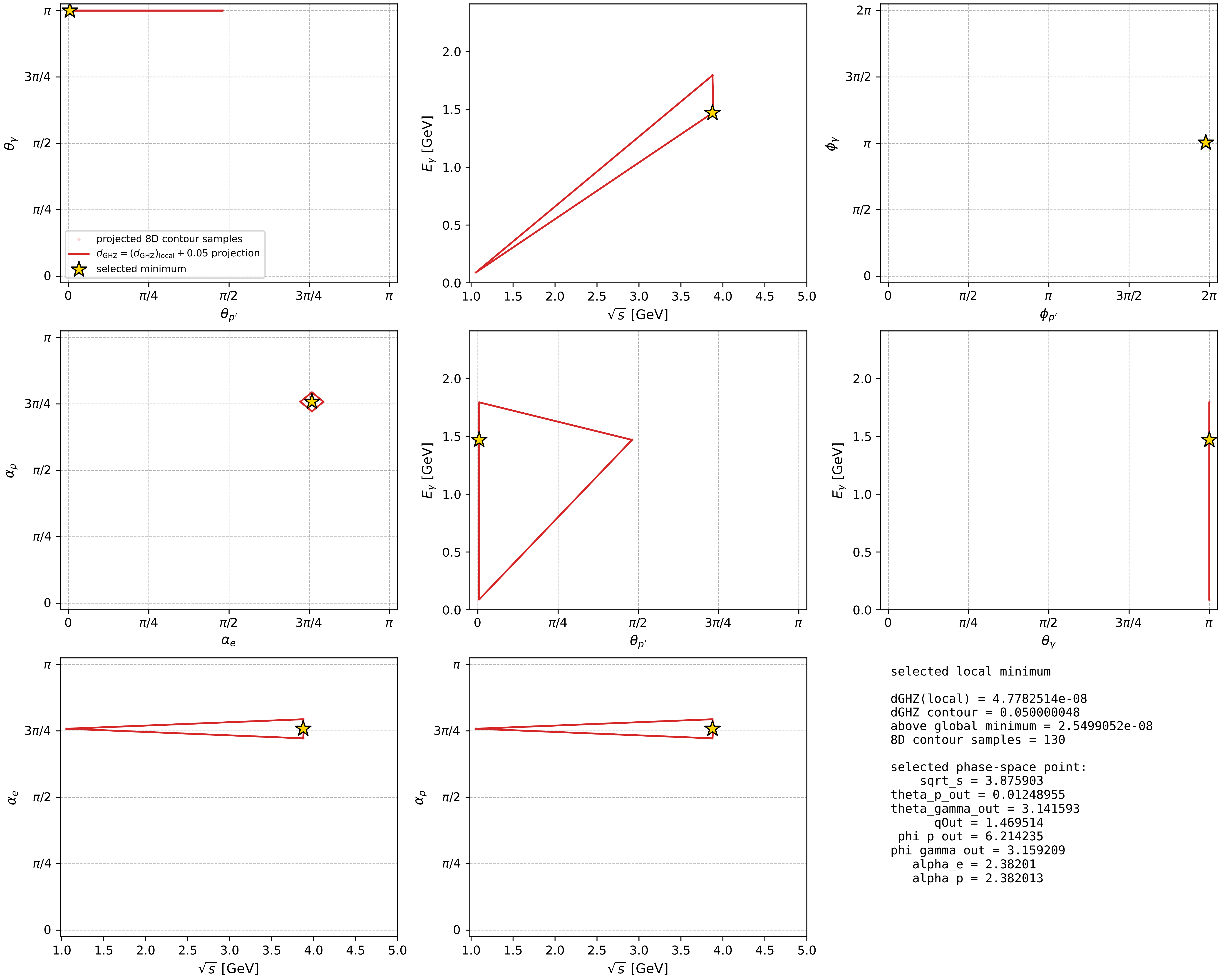

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_122 d_{\rm GHZ}=4.77096e-08
region: polarization_cluster_04_local_minimum_122
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3265008 rad
incoming lepton: -0.685802|+> +0.727789|->
proton mixing angle: alpha_p=2.3264837 rad
incoming proton: -0.685789|+> +0.7278|->
s=14.673, $\sqrt{s}$=3.83054, $|\vec{P}|$=1.80042, $|\vec{P}'|$=0.16869
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=3.10573, ϕ_γ=5.44421
E_γ=1.3544, $\theta_{\ell'}$=0, $\phi_{\ell'}$=2.20483
Q^2=10.9688, x_B=0.837731, t=-2.71733, W^2=3.00452, y=0.949273
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.8004 p_x= 0.0000 p_y= 0.0000 p_z= -1.8004
P E= 2.0301 p_x= 0.0000 p_y= 0.0000 p_z= 1.8004
ℓ' E= 1.5231 p_x= -0.0000 p_y= 0.0000 p_z= 1.5231
P' E= 0.9530 p_x= -0.0000 p_y= 0.0000 p_z= -0.1687
q_γ E= 1.3544 p_x= 0.0000 p_y= -0.0000 p_z= -1.3544
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e−9
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=-4.84e+02-5.39e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=4.84e+02-5.39e+02i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 122; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
$\theta_{p'}$
θ_γ
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 4.7709636e-08
dGHZ contour = 0.050000048
above global minimum = 2.5426174e-08
8D contour samples = 61
selected phase-space point:
sqrt_s = 3.830541
theta_p_out = 3.141593
theta_gamma_out = 3.141593
qOut = 1.354402
phi_p_out = 3.105728
phi_gamma_out = 5.44421
alpha_e = 2.326501
alpha_p = 2.326484

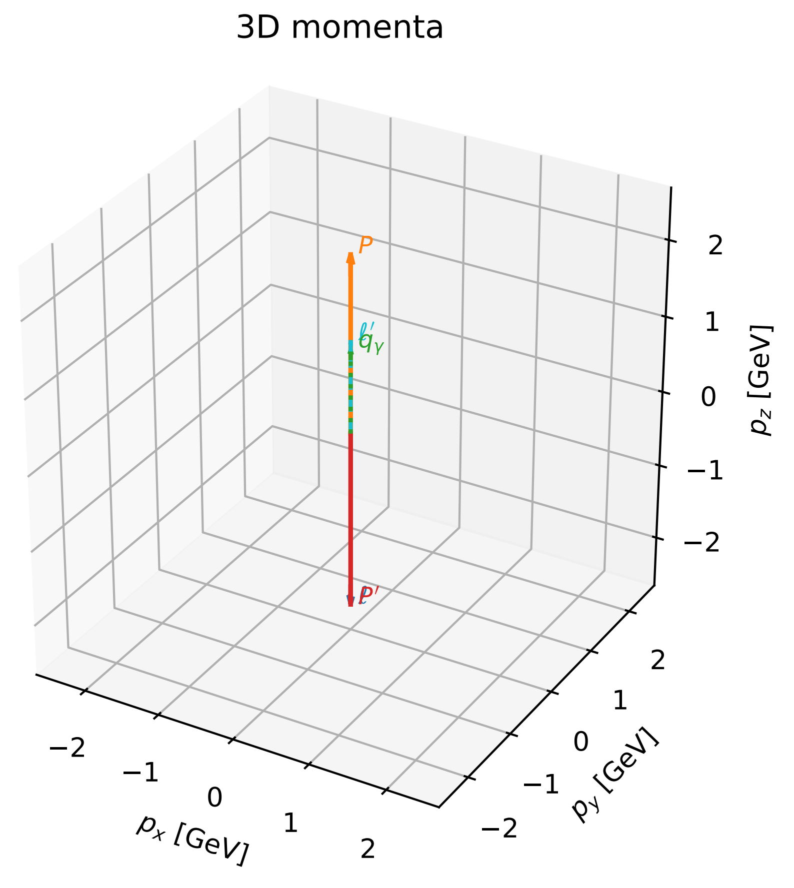
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)



electron: polarization cluster P4, local minimum 123; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



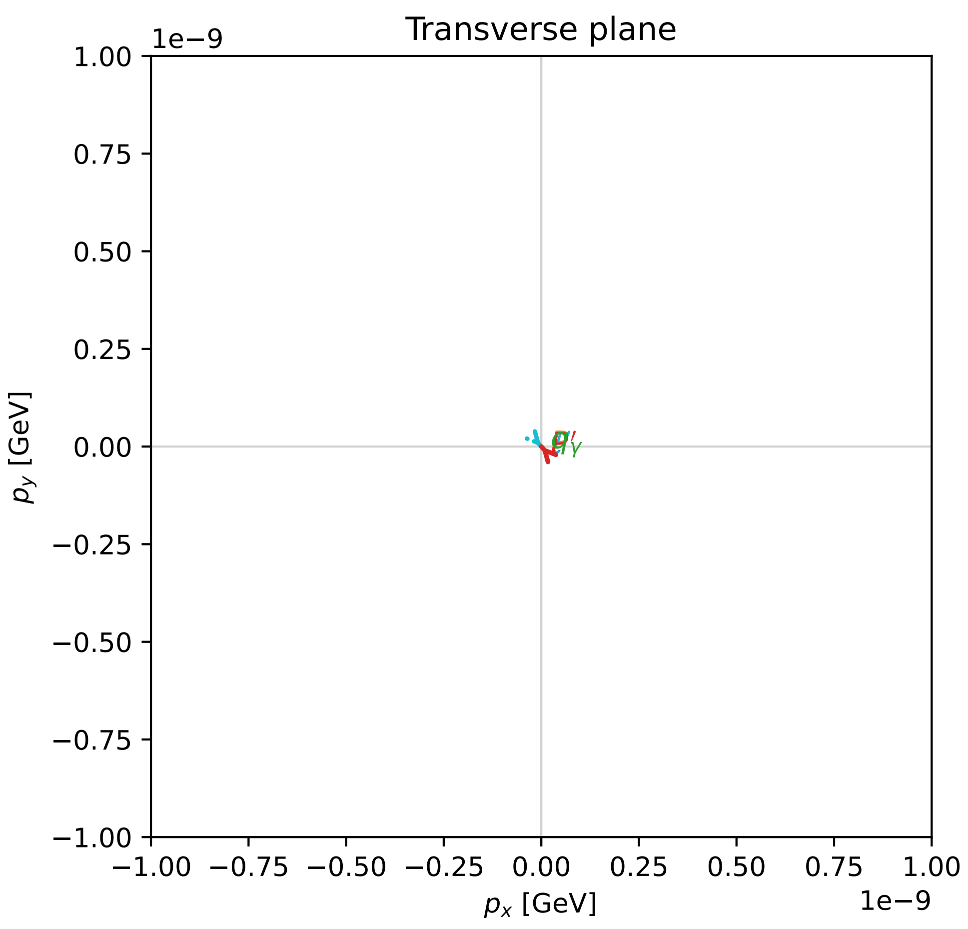
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

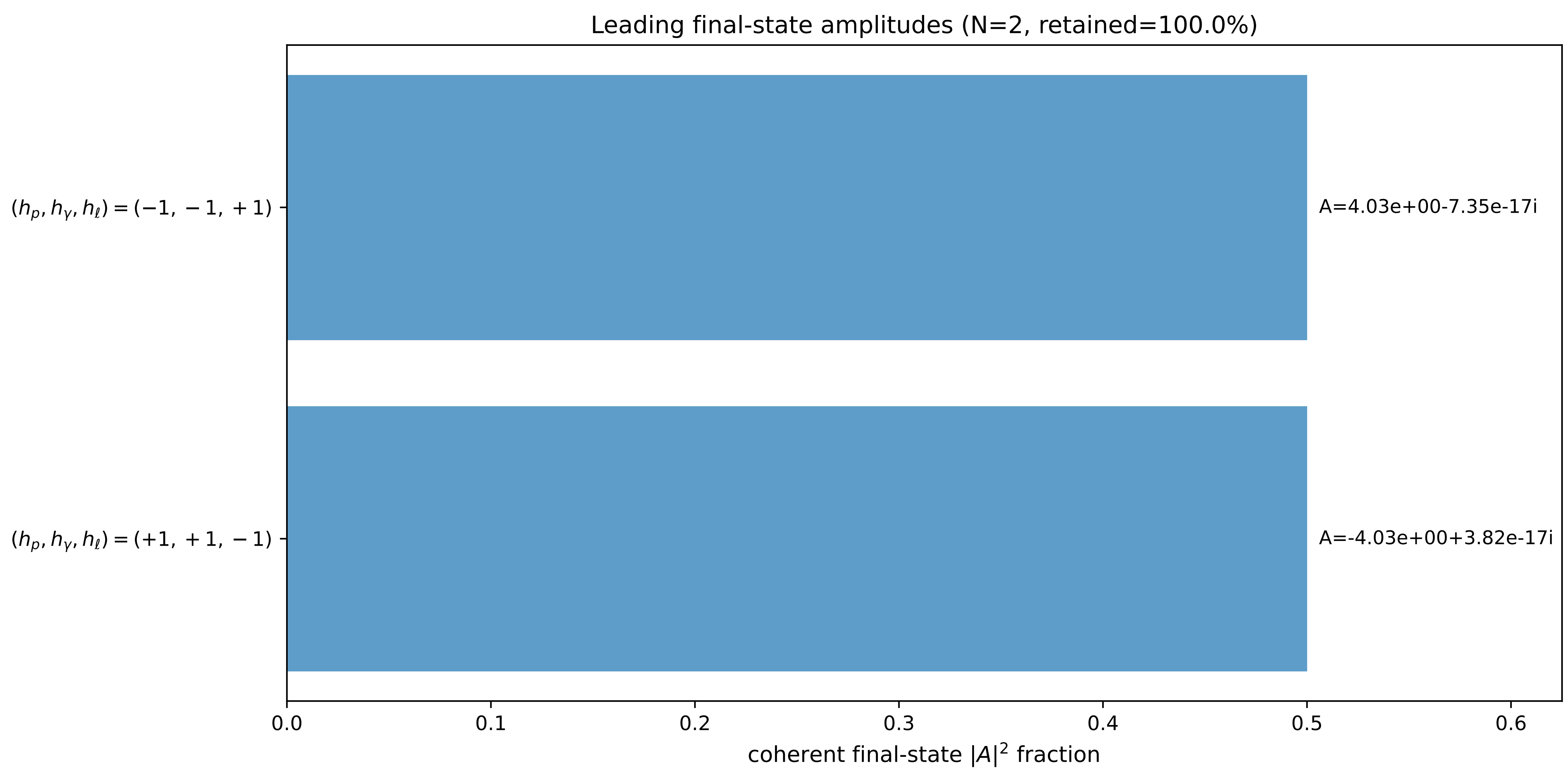


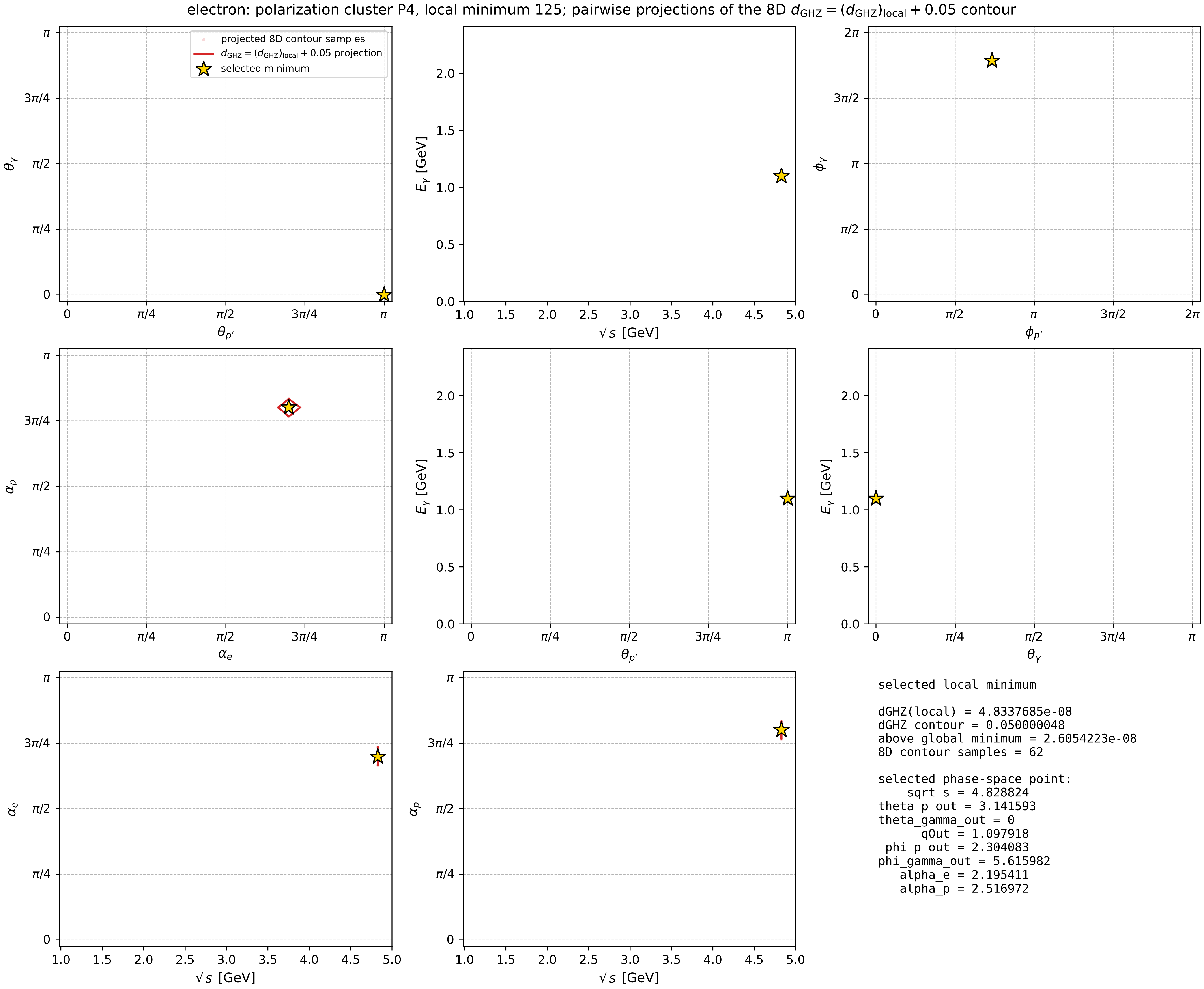
dghz_mixing_angles_polarization_04_local_minimum_125 d_{\rm GHZ}=4.83377e-08
region: polarization_cluster_04_local_minimum_125
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.1954113 rad
incoming lepton: -0.584785|+> +0.811188|->
proton mixing angle: alpha_p=2.5169716 rad
incoming proton: -0.811185|+> +0.58479|->

s=23.3175, $\sqrt{s}$=4.82882, $|\vec{P}|$=2.32331, $|\vec{P}'|$=2.32331
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=2.30408, ϕ_γ=5.61598
E_γ=1.09792, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.44568
Q^2=11.3878, x_B=0.517838, t=-21.591, W^2=11.4831, y=0.980098

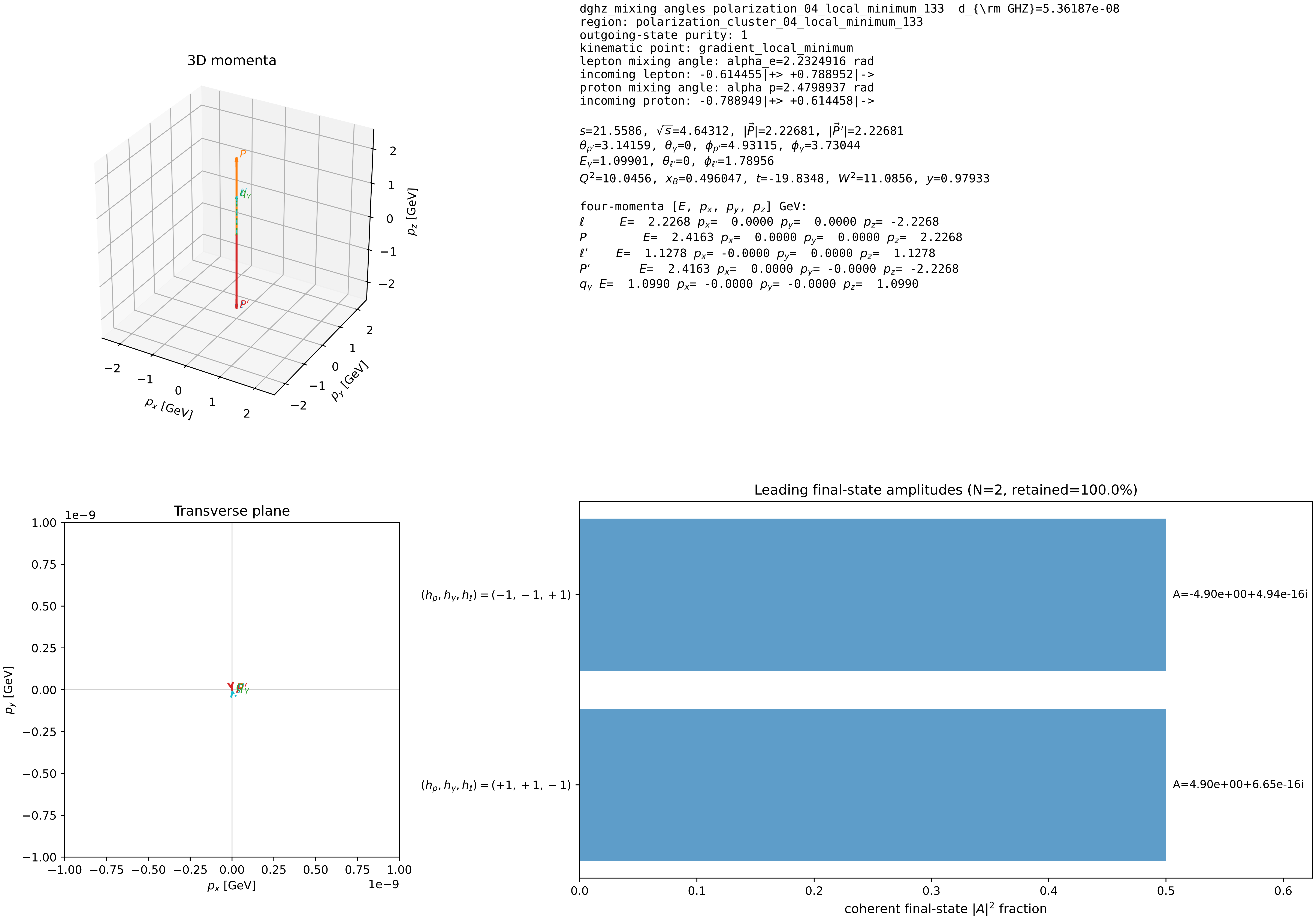
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3233 p_x= 0.0000 p_y= 0.0000 p_z= -2.3233
P E= 2.5055 p_x= 0.0000 p_y= 0.0000 p_z= 2.3233
ℓ' E= 1.2254 p_x= 0.0000 p_y= -0.0000 p_z= 1.2254
P' E= 2.5055 p_x= -0.0000 p_y= 0.0000 p_z= -2.3233
q_γ E= 1.0979 p_x= 0.0000 p_y= -0.0000 p_z= 1.0979

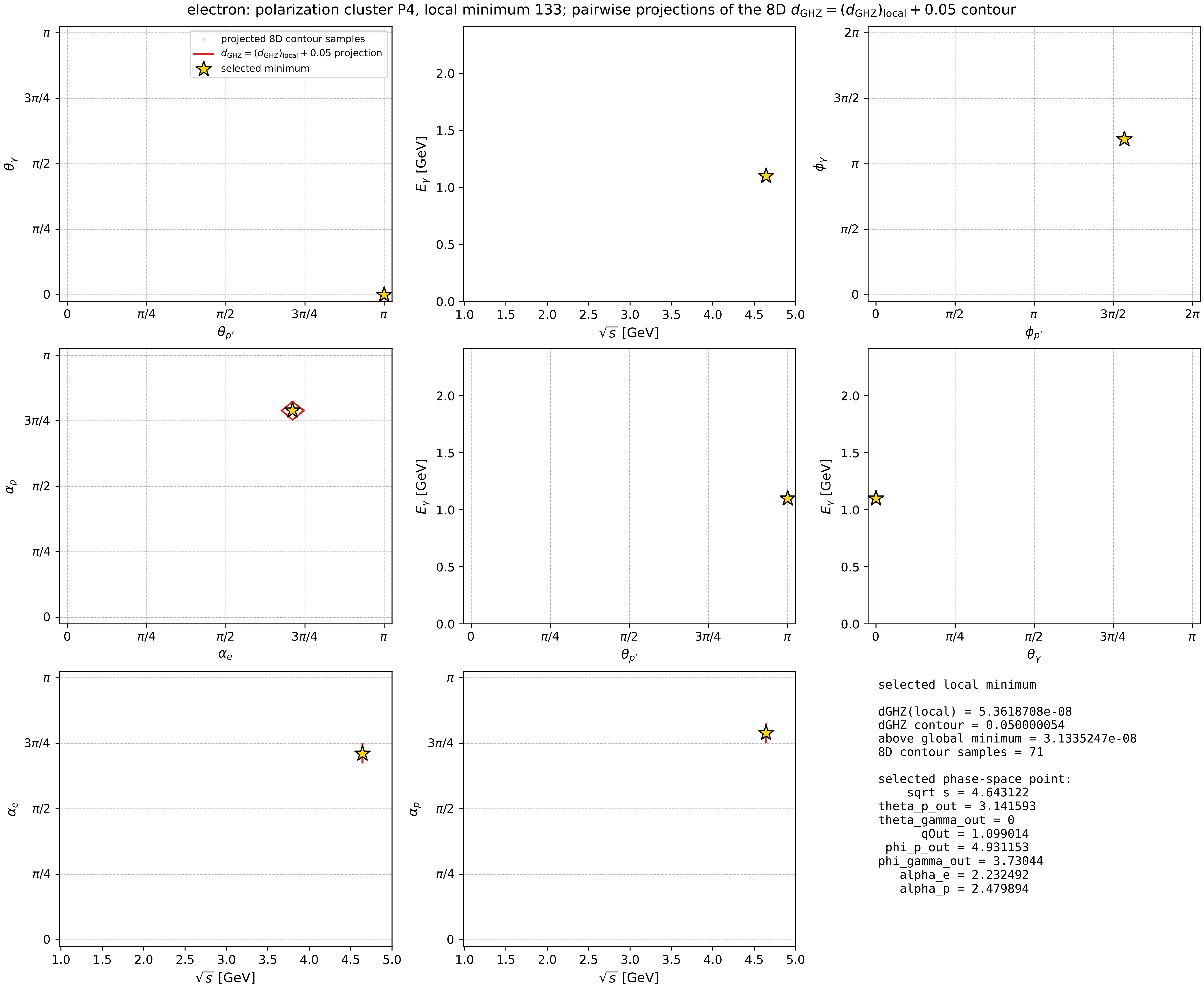




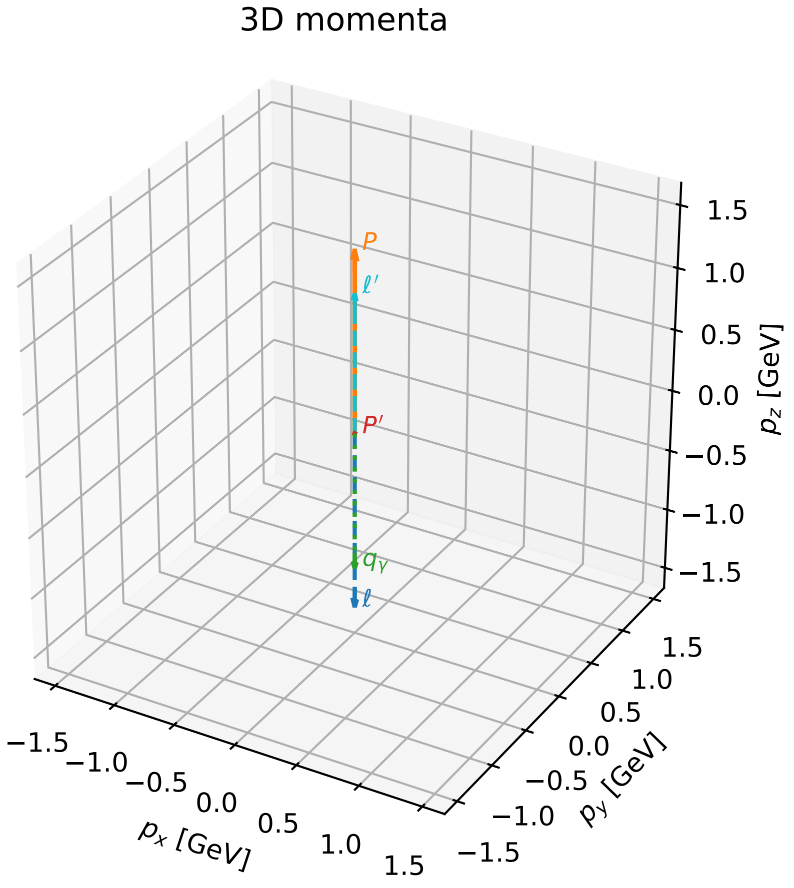

electron: polarization cluster P4, local minimum 125; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_{γ}
$\theta_{p'}$
E_{γ} [GeV]
$\sqrt{s}$ [GeV]
ϕ_{γ}
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 4.8337685e-08
dGHZ contour = 0.050000048
above global minimum = 2.6054223e-08
8D contour samples = 62
selected phase-space point:
sqrt_s = 4.828824
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 1.097918
phi_p_out = 2.304083
phi_gamma_out = 5.615982
alpha_e = 2.195411
alpha_p = 2.516972

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)




electron: polarization cluster P4, local minimum 133; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 5.3618708e-08
dGHZ contour = 0.050000054
above global minimum = 3.1335247e-08
8D contour samples = 71
selected phase-space point:
sqrt_s = 4.643122
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 1.099014
phi_p_out = 4.931153
phi_gamma_out = 3.73044
alpha_e = 2.232492
alpha_p = 2.479894

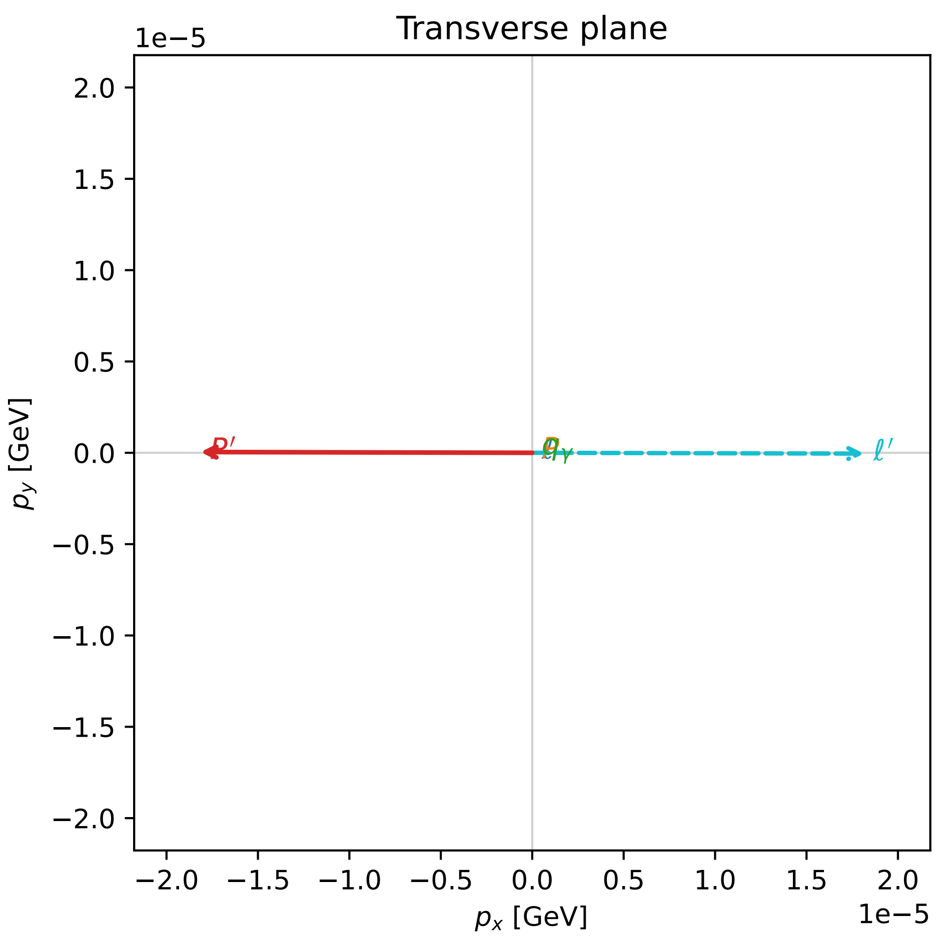
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

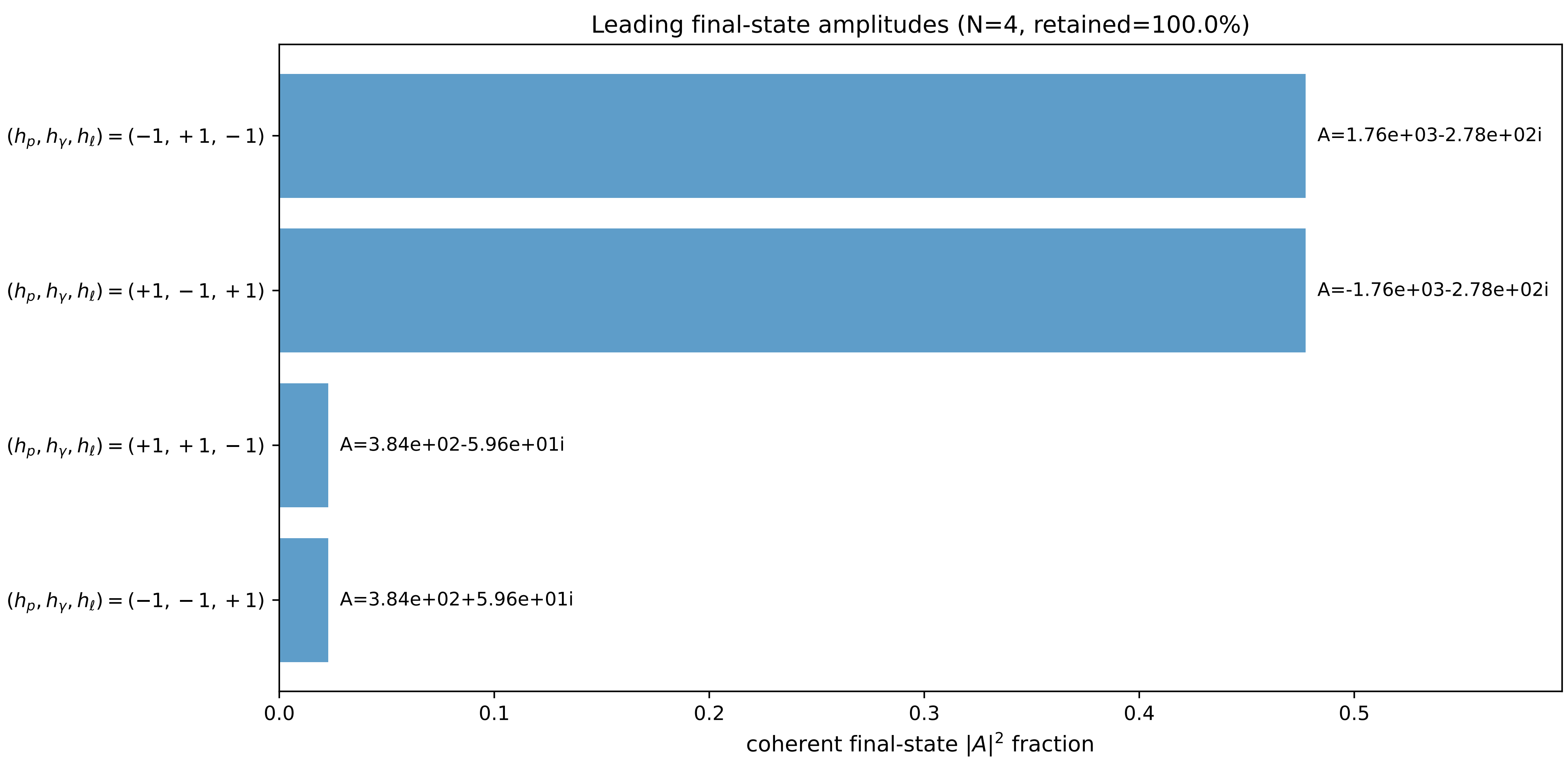


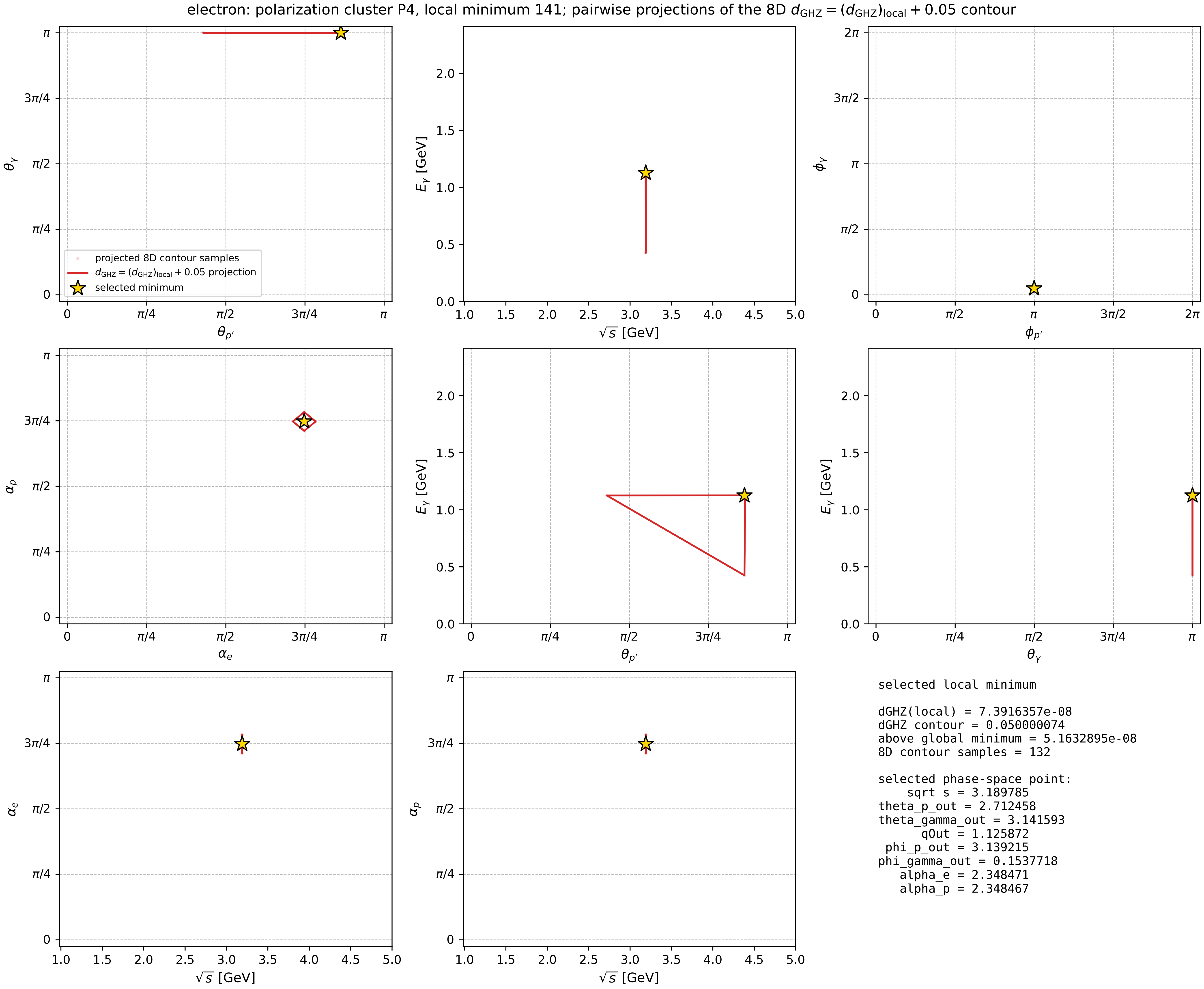
dghz_mixing_angles_polarization_04_local_minimum_141 d_{\rm GHZ}=7.39164e-08
region: polarization_cluster_04_local_minimum_141
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3484711 rad
incoming lepton: -0.701624|+> +0.712547|->
proton mixing angle: alpha_p=2.3484668 rad
incoming proton: -0.701621|+> +0.71255|->

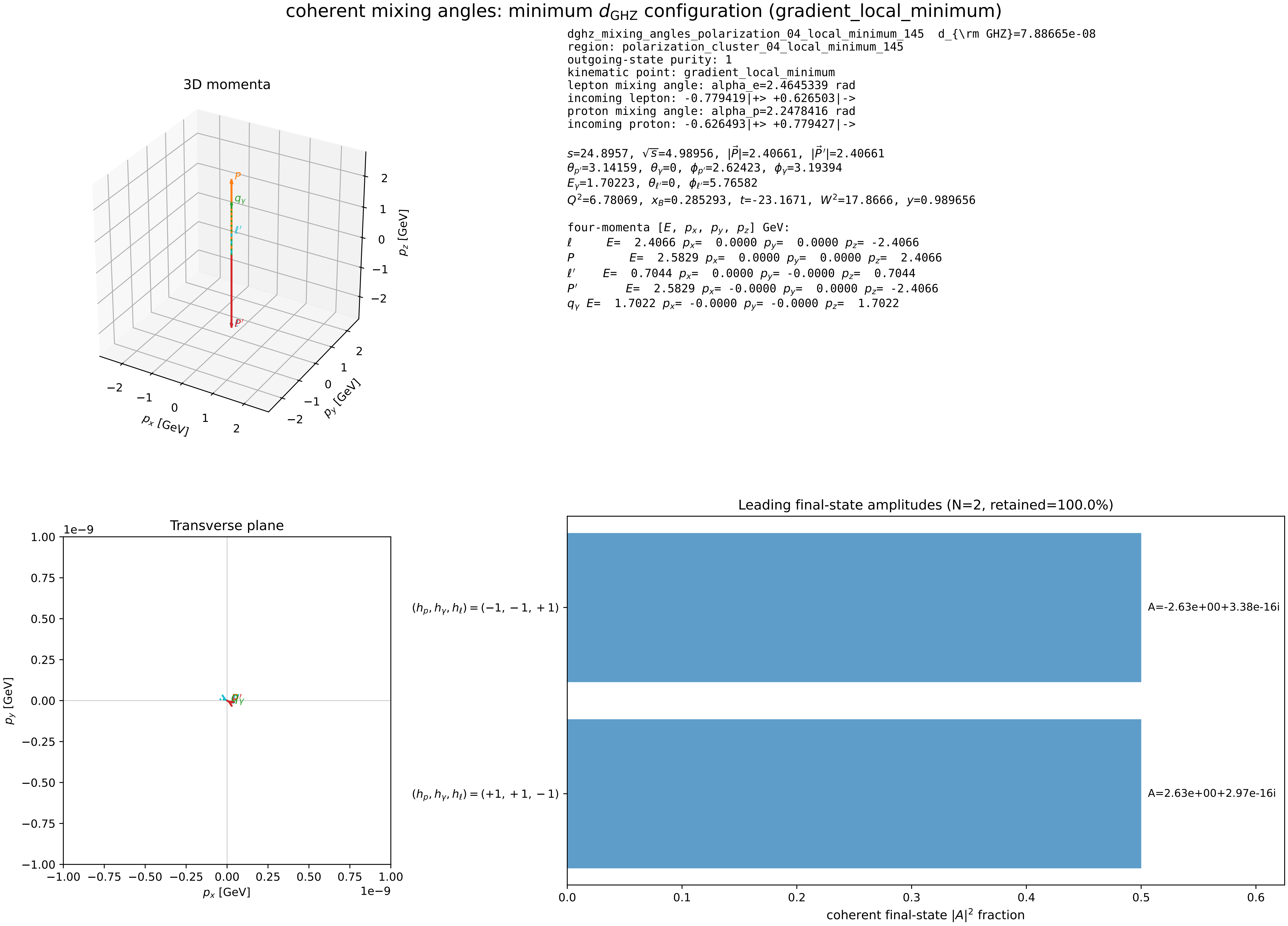
s=10.1747, $\sqrt{s}$=3.18978, $|\vec{P}|$=1.45698, $|\vec{P}'|$=4.3601e-05
$\theta_{p'}$=2.71246, θ_γ=3.14159, $\phi_{p'}$=3.13922, ϕ_γ=0.153772
E_γ=1.12587, $\theta_{\ell'}$=1.61129e-05, $\phi_{\ell'}$=6.28081
Q^2=6.56171, x_B=0.756501, t=-1.49118, W^2=2.99189, y=0.933176

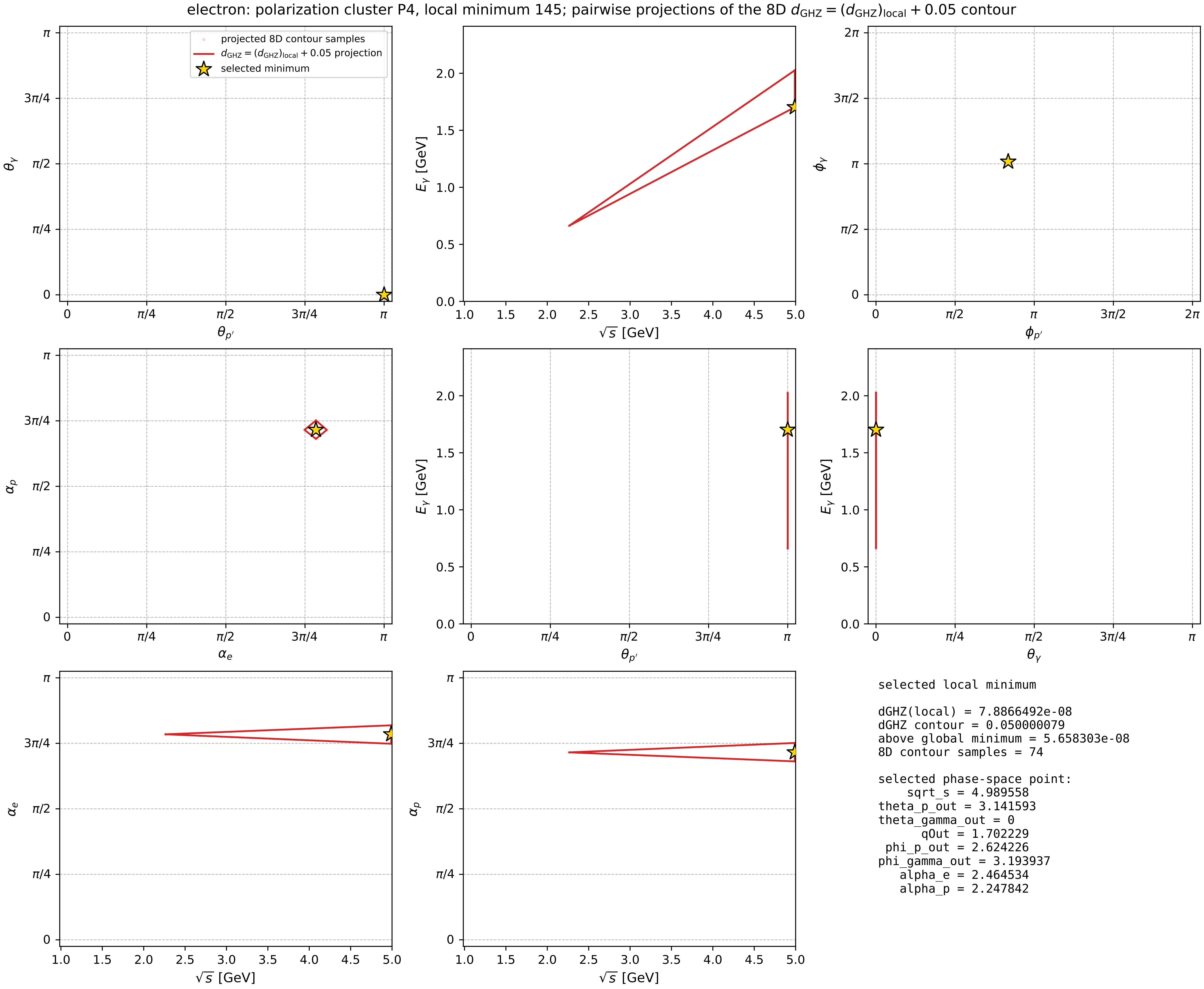
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.4570 p_x= 0.0000 p_y= 0.0000 p_z= -1.4570
P E= 1.7328 p_x= 0.0000 p_y= 0.0000 p_z= 1.4570
ℓ' E= 1.1259 p_x= 0.0000 p_y= -0.0000 p_z= 1.1259
P' E= 0.9380 p_x= -0.0000 p_y= 0.0000 p_z= -0.0000
q_γ E= 1.1259 p_x= 0.0000 p_y= 0.0000 p_z= -1.1259



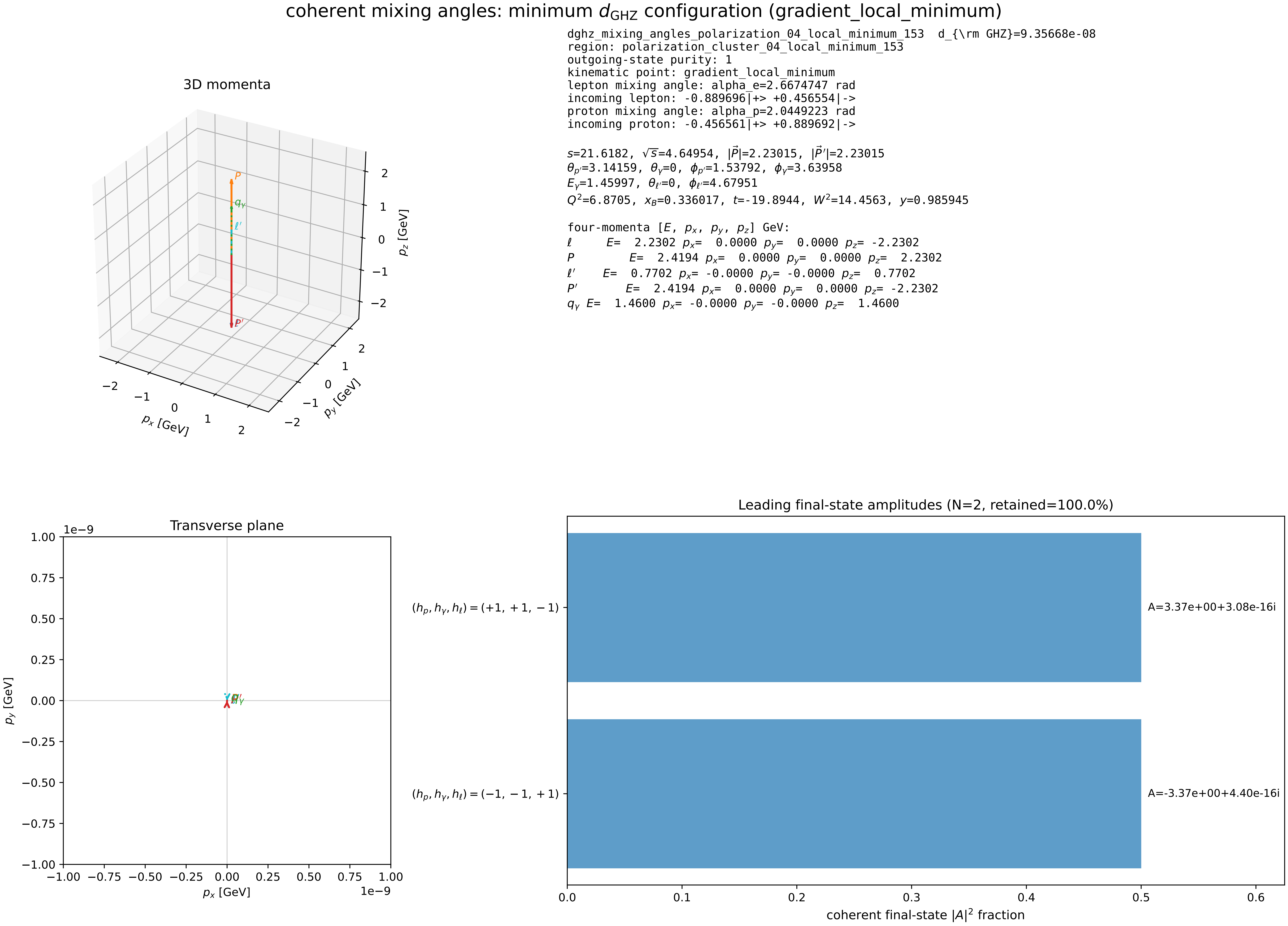


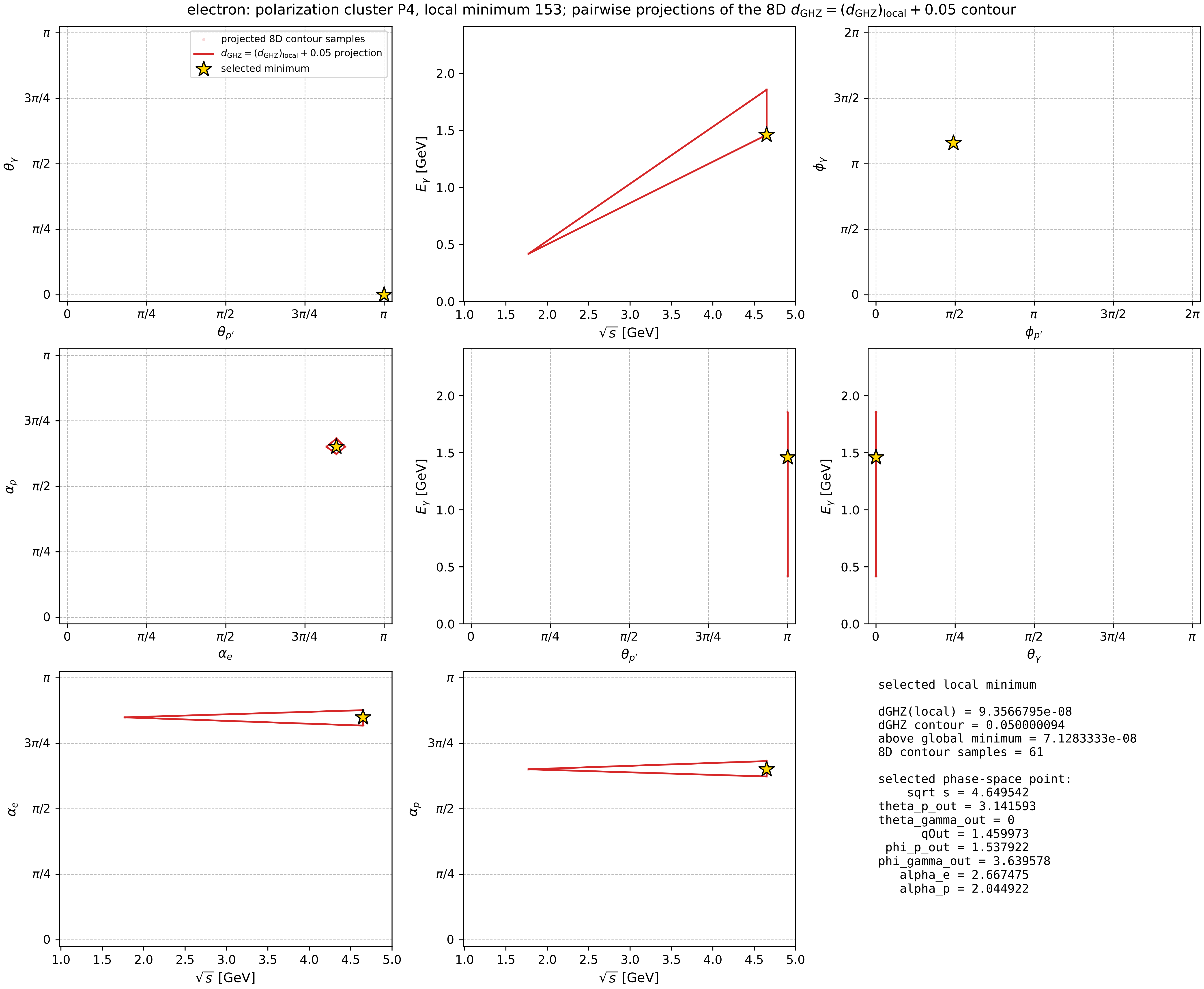

electron: polarization cluster P4, local minimum 141; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 7.3916357e-08
dGHZ contour = 0.050000074
above global minimum = 5.1632895e-08
8D contour samples = 132
selected phase-space point:
sqrt_s = 3.189785
theta_p_out = 2.712458
theta_gamma_out = 3.141593
qOut = 1.125872
phi_p_out = 3.139215
phi_gamma_out = 0.1537718
alpha_e = 2.348471
alpha_p = 2.348467


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
q_γ
ℓ'
P'
dghz_mixing_angles_polarization_04_local_minimum_145 d_{\rm GHZ}=7.88665e-08
region: polarization_cluster_04_local_minimum_145
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4645339 rad
incoming lepton: -0.779419|+> +0.626503|->
proton mixing angle: alpha_p=2.2478416 rad
incoming proton: -0.626493|+> +0.779427|->
s=24.8957, $\sqrt{s}$=4.98956, $|\vec{P}|$=2.40661, $|\vec{P}'|$=2.40661
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=2.62423, ϕ_γ=3.19394
E_γ=1.70223, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.76582
Q^2=6.78069, x_B=0.285293, t=-23.1671, W^2=17.8666, y=0.989656
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4066 p_x= 0.0000 p_y= 0.0000 p_z= -2.4066
P E= 2.5829 p_x= 0.0000 p_y= 0.0000 p_z= 2.4066
ℓ' E= 0.7044 p_x= 0.0000 p_y= -0.0000 p_z= 0.7044
P' E= 2.5829 p_x= -0.0000 p_y= 0.0000 p_z= -2.4066
q_γ E= 1.7022 p_x= -0.0000 p_y= -0.0000 p_z= 1.7022
Transverse plane
1e−9
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-2.63e+00+3.38e-16i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=2.63e+00+2.97e-16i
coherent final-state $|A|^2$ fraction



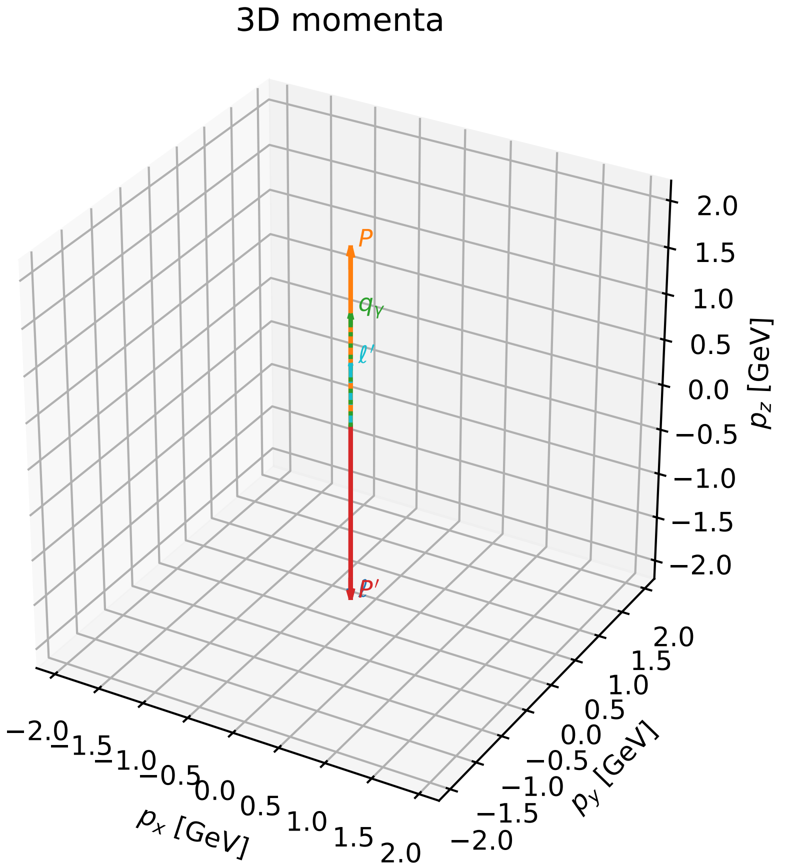
electron: polarization cluster P4, local minimum 145; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 7.8866492e-08
dGHZ contour = 0.050000079
above global minimum = 5.658303e-08
8D contour samples = 74
selected phase-space point:
sqrt_s = 4.989558
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 1.702229
phi_p_out = 2.624226
phi_gamma_out = 3.193937
alpha_e = 2.464534
alpha_p = 2.247842


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_153 d_{\rm GHZ}=9.35668e-08
region: polarization_cluster_04_local_minimum_153
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.6674747 rad
incoming lepton: -0.889696|+> +0.456554|->
proton mixing angle: alpha_p=2.0449223 rad
incoming proton: -0.456561|+> +0.889692|->
s=21.6182, $\sqrt{s}$=4.64954, $|\vec{P}|$=2.23015, $|\vec{P}'|$=2.23015
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=1.53792, ϕ_γ=3.63958
E_γ=1.45997, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.67951
Q^2=6.8705, x_B=0.336017, t=-19.8944, W^2=14.4563, y=0.985945
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2302 p_x= 0.0000 p_y= 0.0000 p_z= -2.2302
P E= 2.4194 p_x= 0.0000 p_y= 0.0000 p_z= 2.2302
ℓ' E= 0.7702 p_x= -0.0000 p_y= -0.0000 p_z= 0.7702
P' E= 2.4194 p_x= 0.0000 p_y= 0.0000 p_z= -2.2302
q_γ E= 1.4600 p_x= -0.0000 p_y= -0.0000 p_z= 1.4600
3D momenta
P
q_γ
ℓ'
P'
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e−9
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=3.37e+00+3.08e-16i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-3.37e+00+4.40e-16i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P4, local minimum 153; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_{γ}
$\theta_{p'}$
E_{γ} [GeV]
$\sqrt{s}$ [GeV]
ϕ_{γ}
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 9.3566795e-08
dGHZ contour = 0.050000094
above global minimum = 7.1283333e-08
8D contour samples = 61
selected phase-space point:
sqrt_s = 4.649542
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 1.459973
phi_p_out = 1.537922
phi_gamma_out = 3.639578
alpha_e = 2.667475
alpha_p = 2.044922

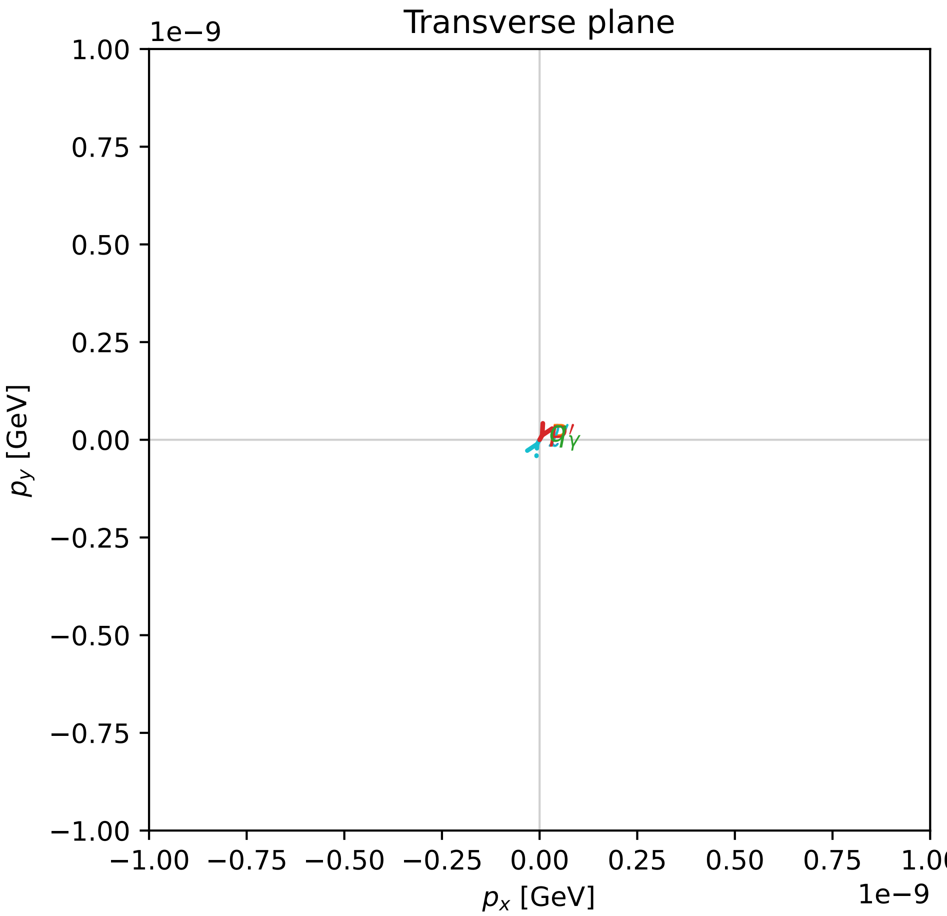
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

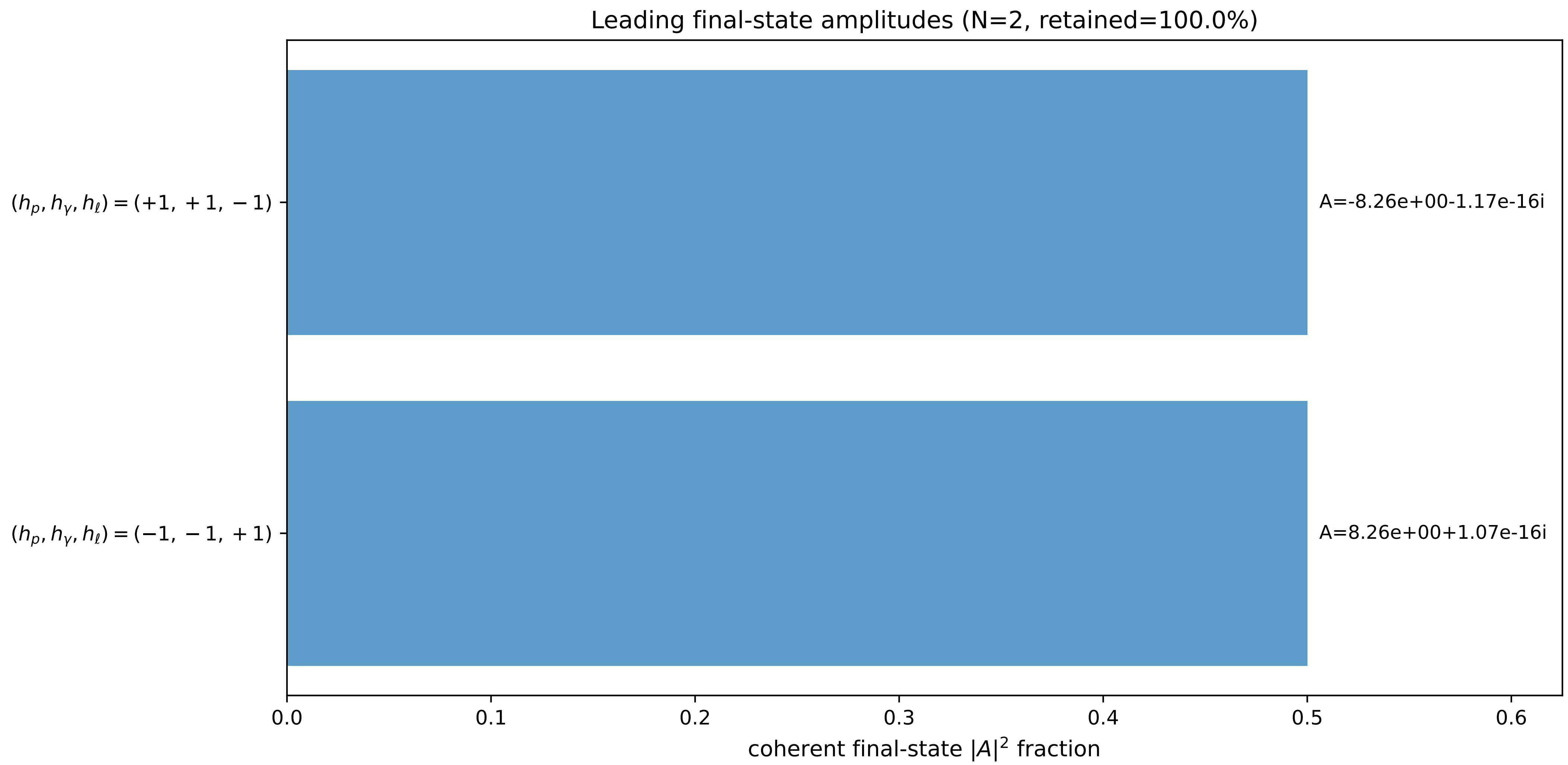


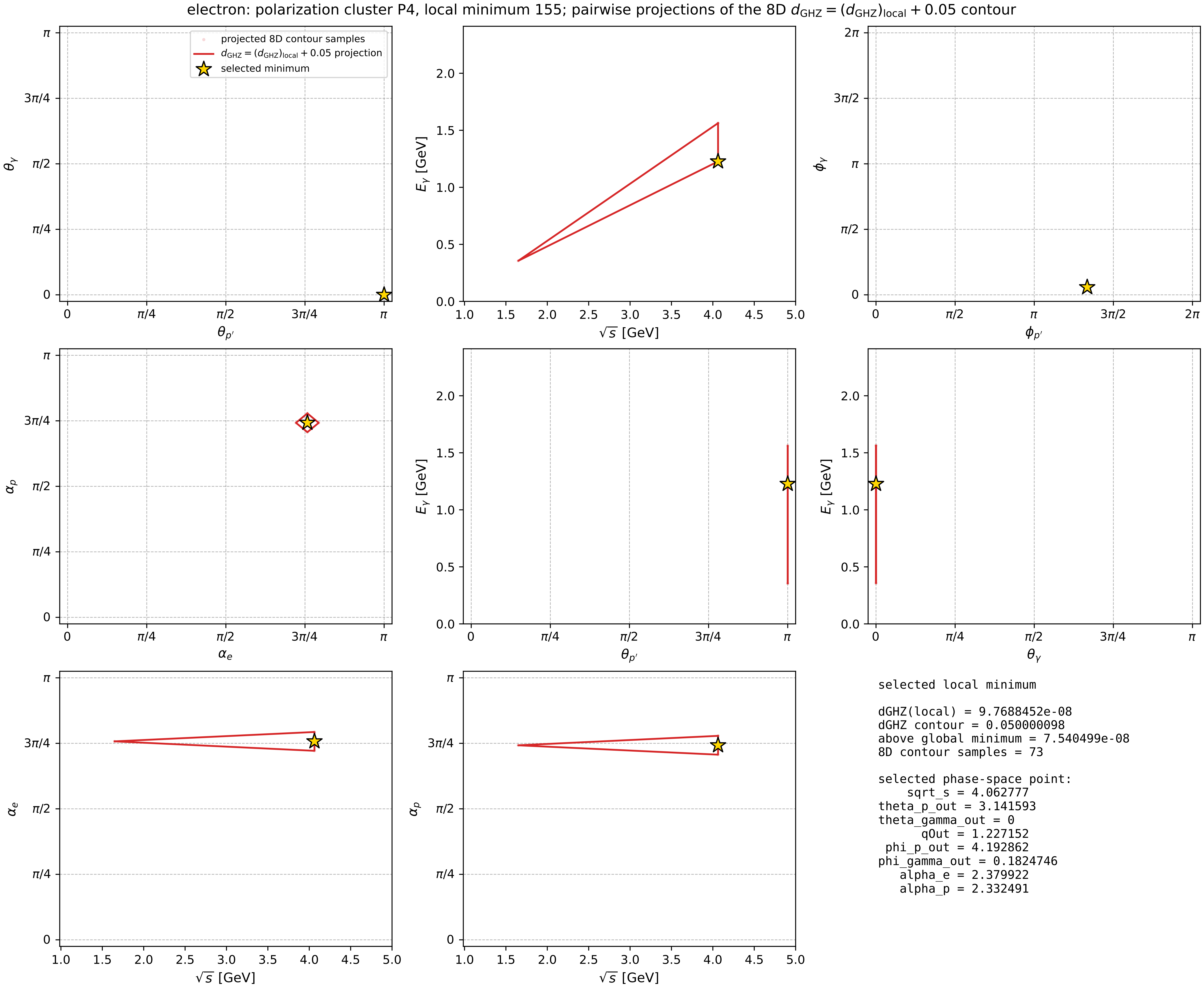

```
dghz_mixing_angles_polarization_04_local_minimum_155  d_{\rm GHZ}=9.76885e-08
region: polarization_cluster_04_local_minimum_155
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.379922 rad
incoming lepton: -0.723684|+> +0.690131|->
proton mixing angle: alpha_p=2.3324913 rad
incoming proton: -0.690149|+> +0.723667|->
```

s=16.5062, $\sqrt{s}$=4.06278, $|\vec{P}|$=1.92311, $|\vec{P}'|$=1.92311
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=4.19286, ϕ_γ=0.182475
E_γ=1.22715, $\theta_{\ell'}$=0, $\phi_{\ell'}$=1.05127
Q^2=5.35358, x_B=0.34934, t=-14.7934, W^2=10.8511, y=0.98071

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.9231 p_x= 0.0000 p_y= 0.0000 p_z= -1.9231
P E= 2.1397 p_x= 0.0000 p_y= 0.0000 p_z= 1.9231
ℓ' E= 0.6960 p_x= 0.0000 p_y= 0.0000 p_z= 0.6960
P' E= 2.1397 p_x= -0.0000 p_y= -0.0000 p_z= -1.9231
q_γ E= 1.2272 p_x= 0.0000 p_y= 0.0000 p_z= 1.2272

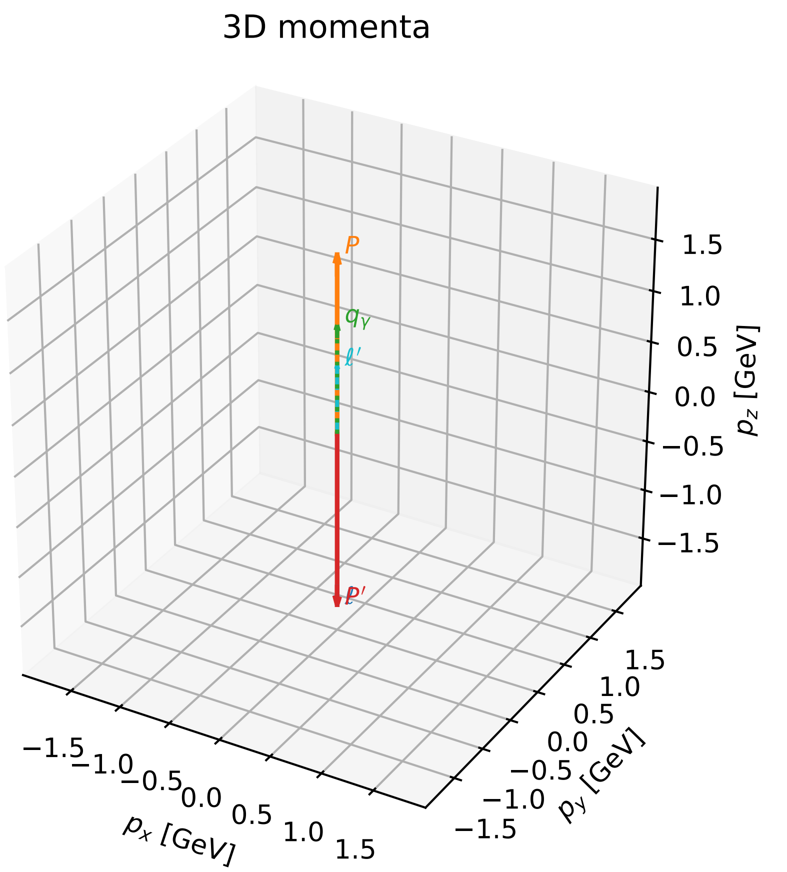






electron: polarization cluster P4, local minimum 155; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
$\theta_{p'}$
θ_γ
selected local minimum
dGHZ(local) = 9.7688452e-08
dGHZ contour = 0.050000098
above global minimum = 7.540499e-08
8D contour samples = 73
selected phase-space point:
sqrt_s = 4.062777
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 1.227152
phi_p_out = 4.192862
phi_gamma_out = 0.1824746
alpha_e = 2.379922
alpha_p = 2.332491

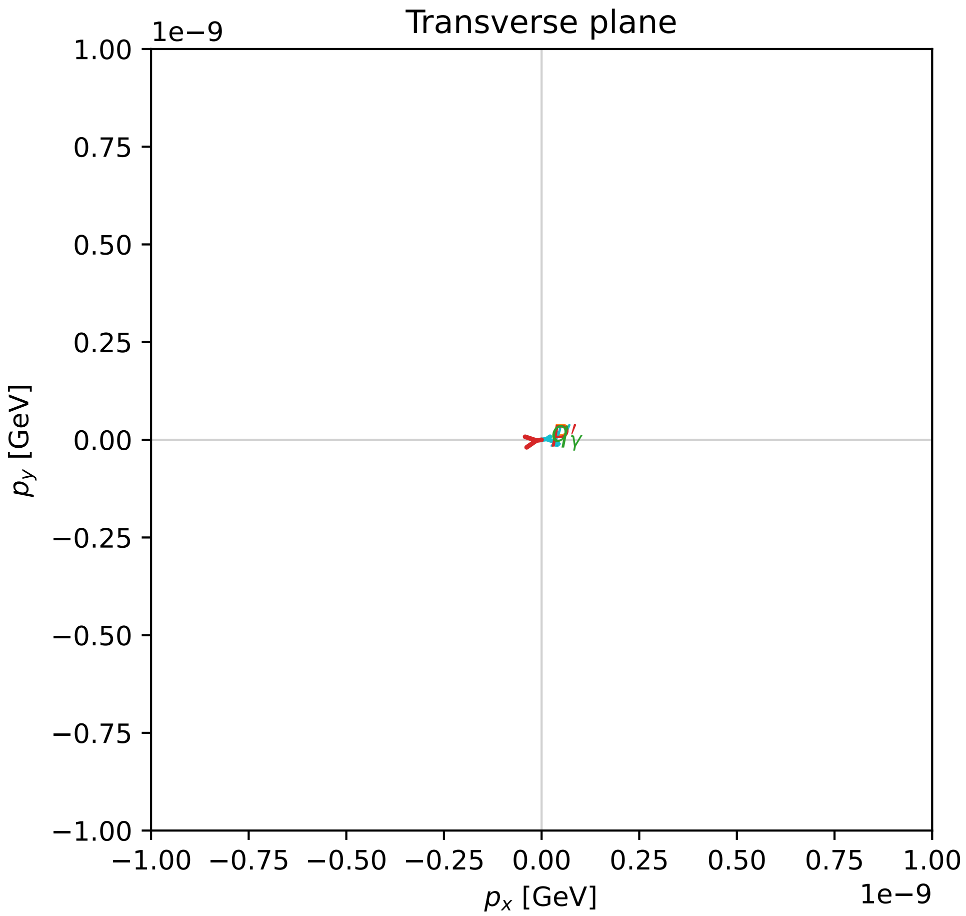
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

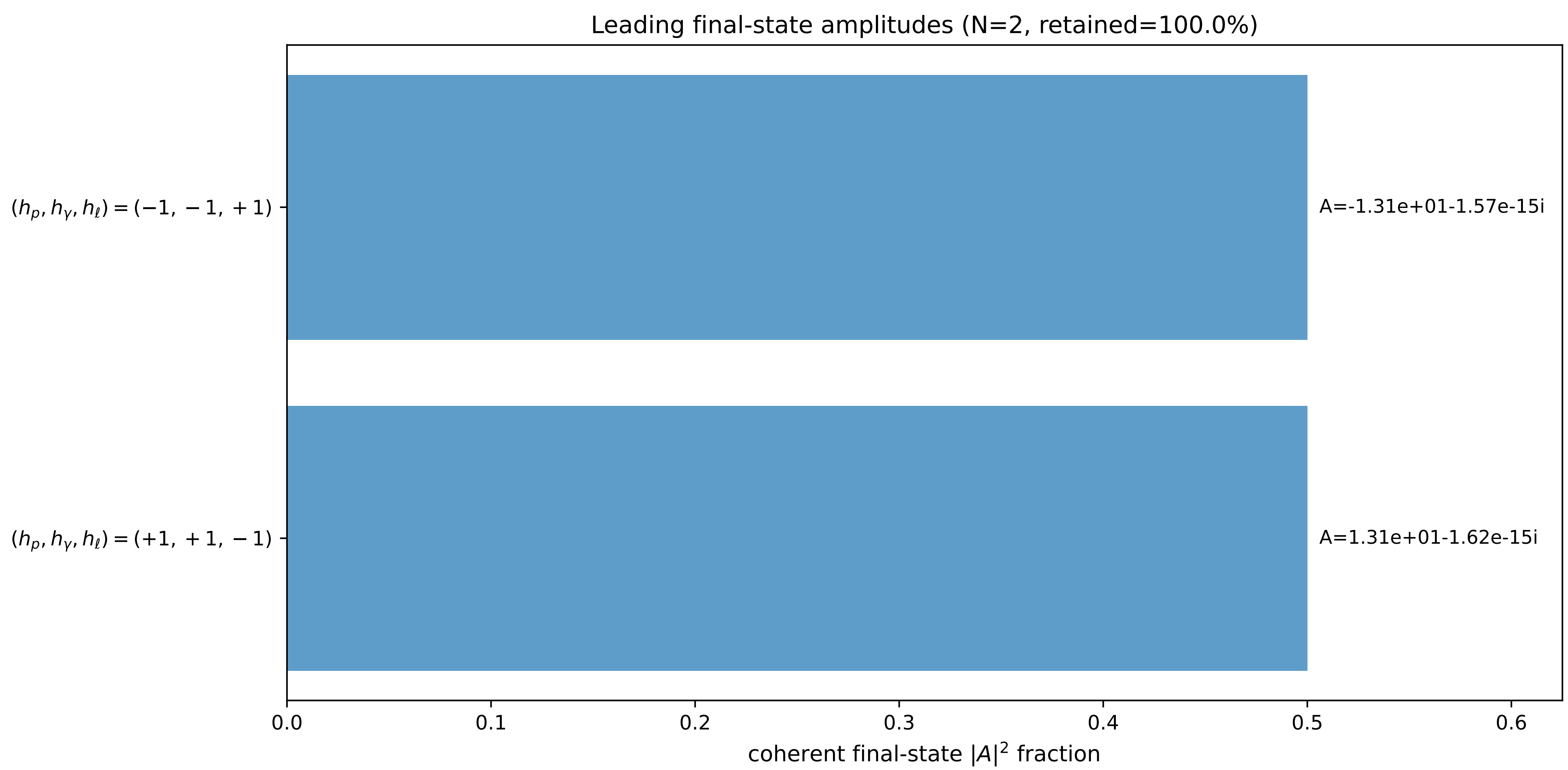


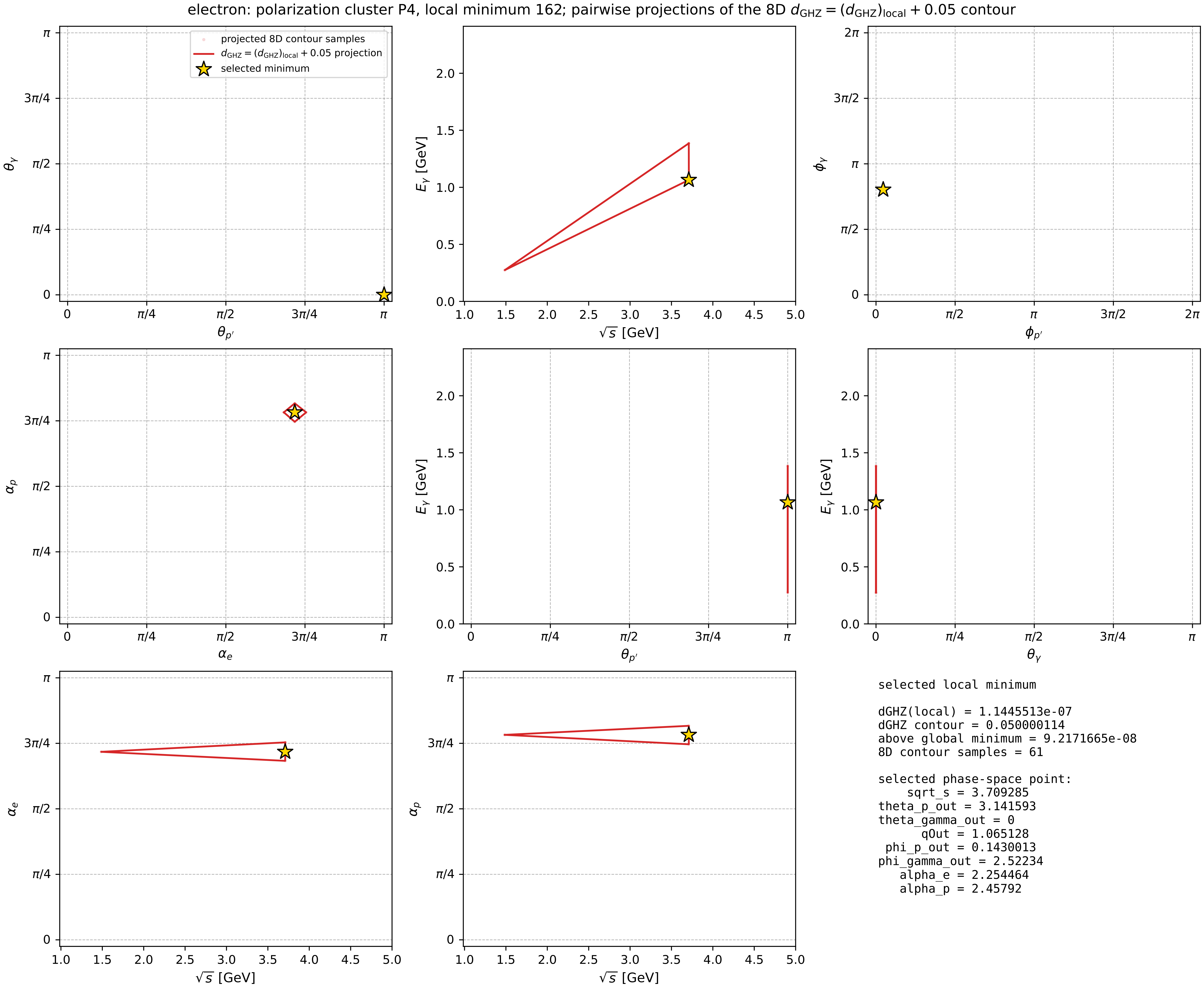

```
dghz_mixing_angles_polarization_04_local_minimum_162  d_{\rm GHZ}=1.14455e-07
region: polarization_cluster_04_local_minimum_162
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.254464 rad
incoming lepton: -0.631641|+> +0.775261|->
proton mixing angle: alpha_p=2.4579197 rad
incoming proton: -0.775258|+> +0.631645|->
```

s=13.7588, $\sqrt{s}$=3.70928, $|\vec{P}|$=1.73604, $|\vec{P}'|$=1.73604
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=0.143001, ϕ_γ=2.52234
E_γ=1.06513, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.28459
Q^2=4.65894, x_B=0.370915, t=-12.0554, W^2=8.78157, y=0.975287

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.7360 p_x= 0.0000 p_y= 0.0000 p_z= -1.7360
P E= 1.9732 p_x= 0.0000 p_y= 0.0000 p_z= 1.7360
ℓ' E= 0.6709 p_x= -0.0000 p_y= -0.0000 p_z= 0.6709
P' E= 1.9732 p_x= 0.0000 p_y= 0.0000 p_z= -1.7360
q_γ E= 1.0651 p_x= -0.0000 p_y= 0.0000 p_z= 1.0651

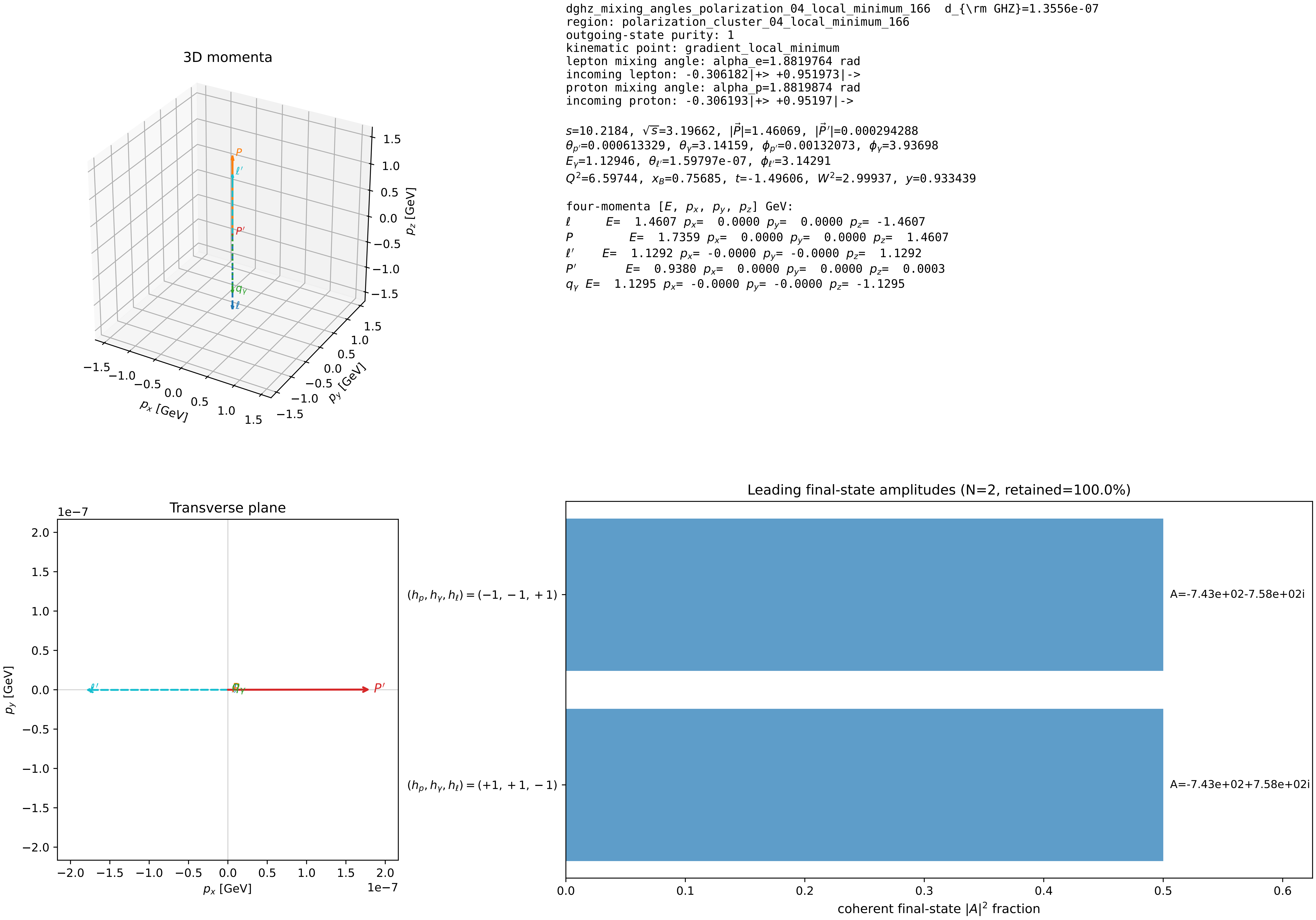


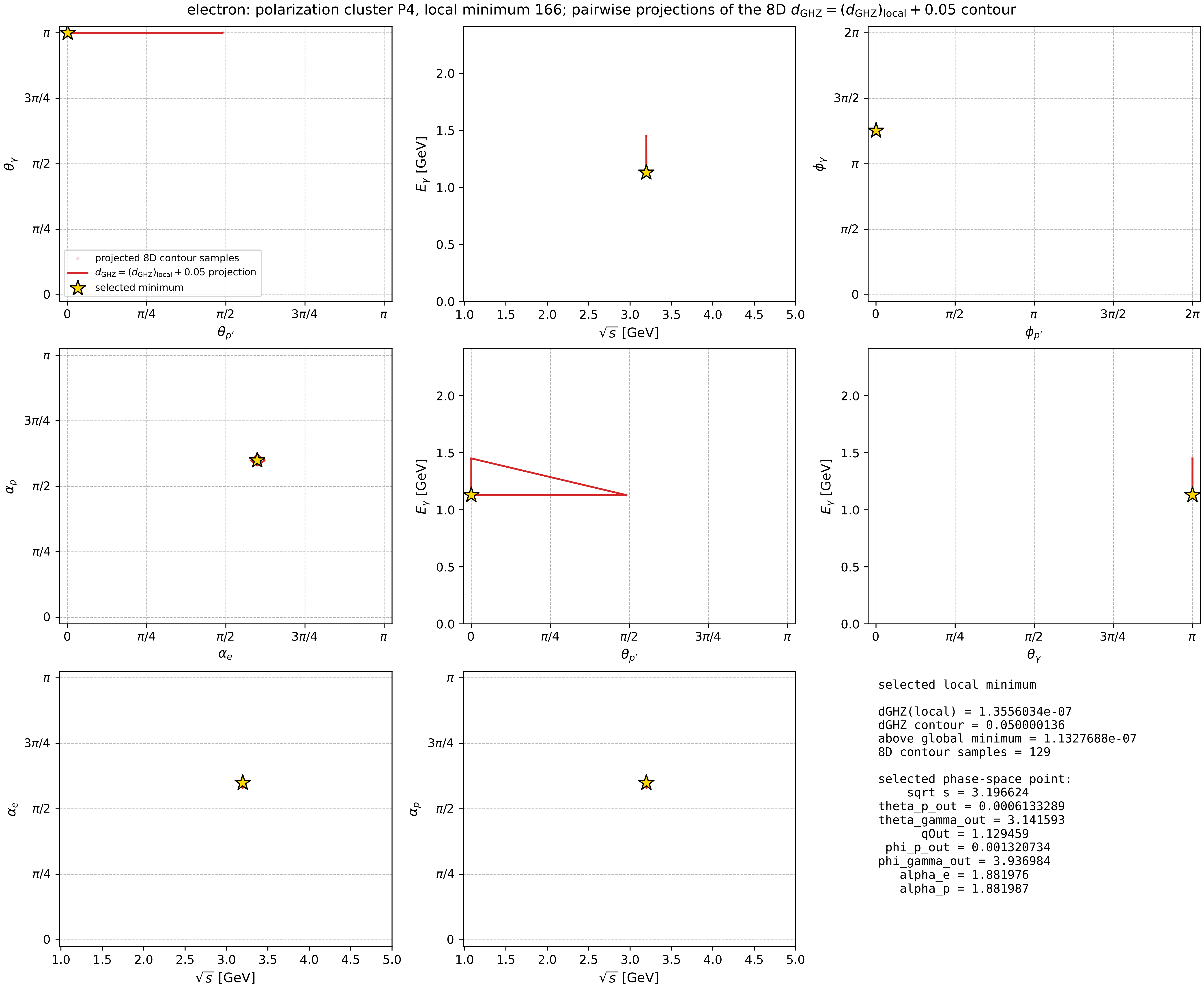


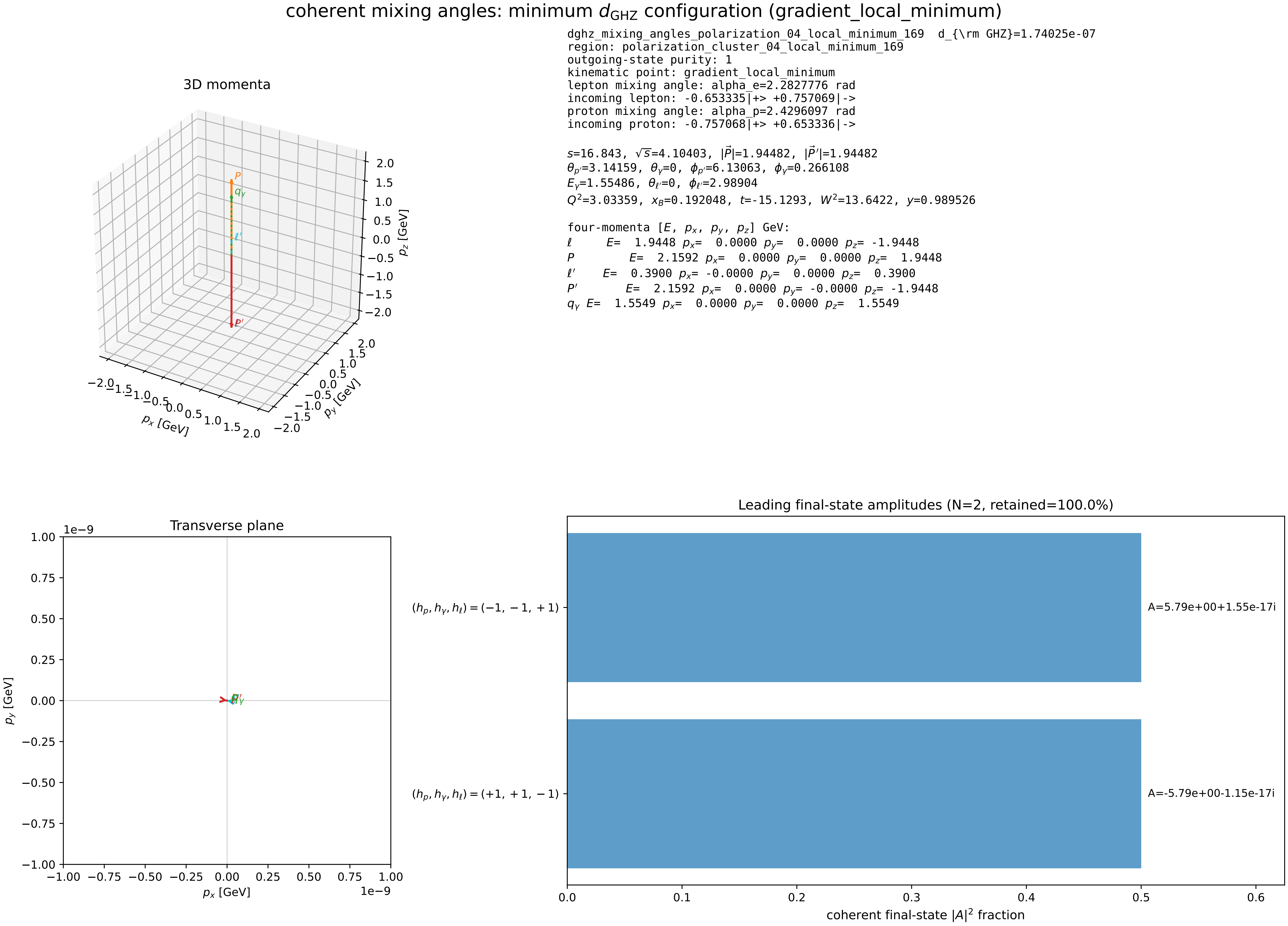


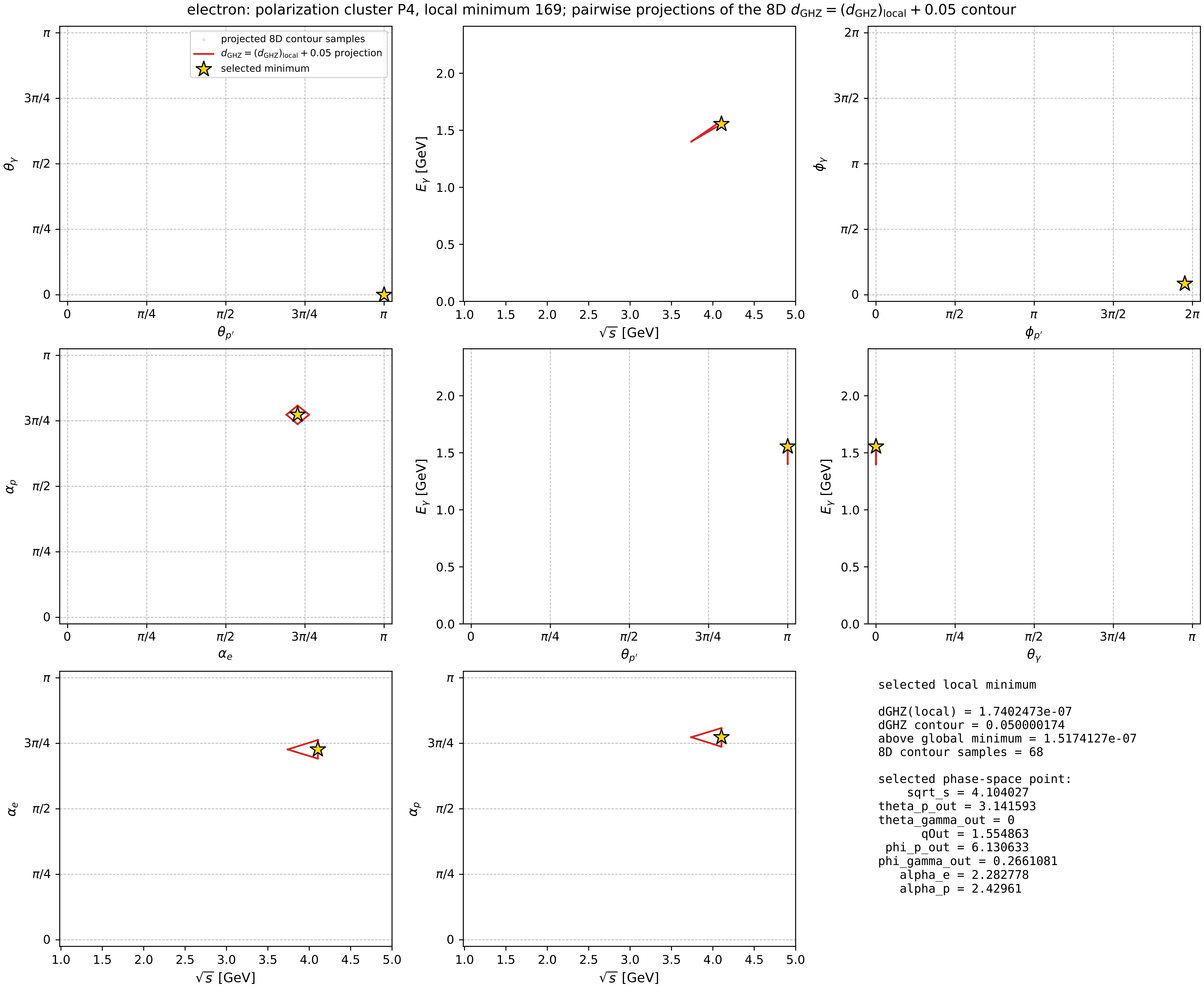
electron: polarization cluster P4, local minimum 162; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 1.1445513e-07
dGHZ contour = 0.050000114
above global minimum = 9.2171665e-08
8D contour samples = 61
selected phase-space point:
sqrt_s = 3.709285
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 1.065128
phi_p_out = 0.1430013
phi_gamma_out = 2.52234
alpha_e = 2.254464
alpha_p = 2.45792

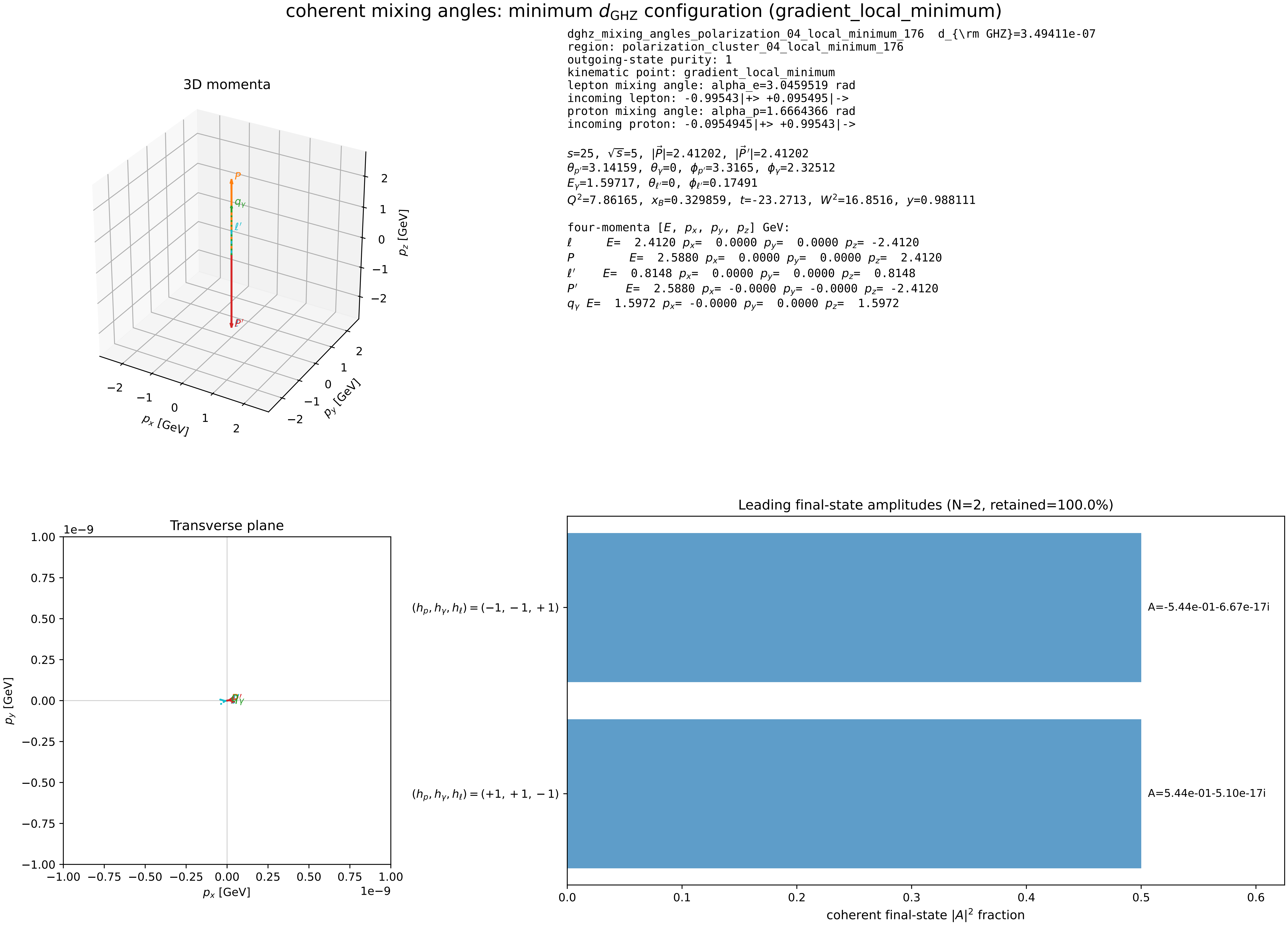
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

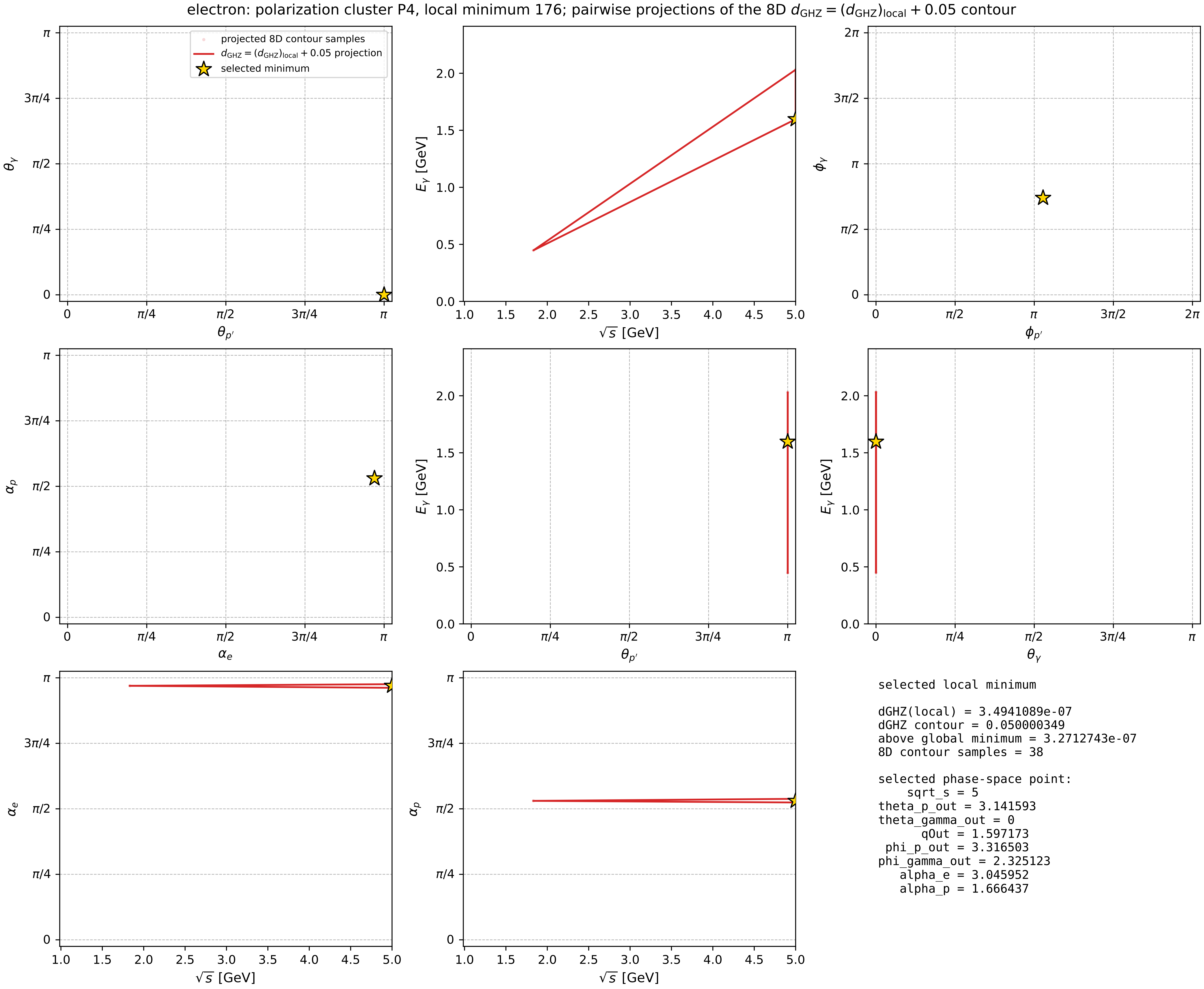


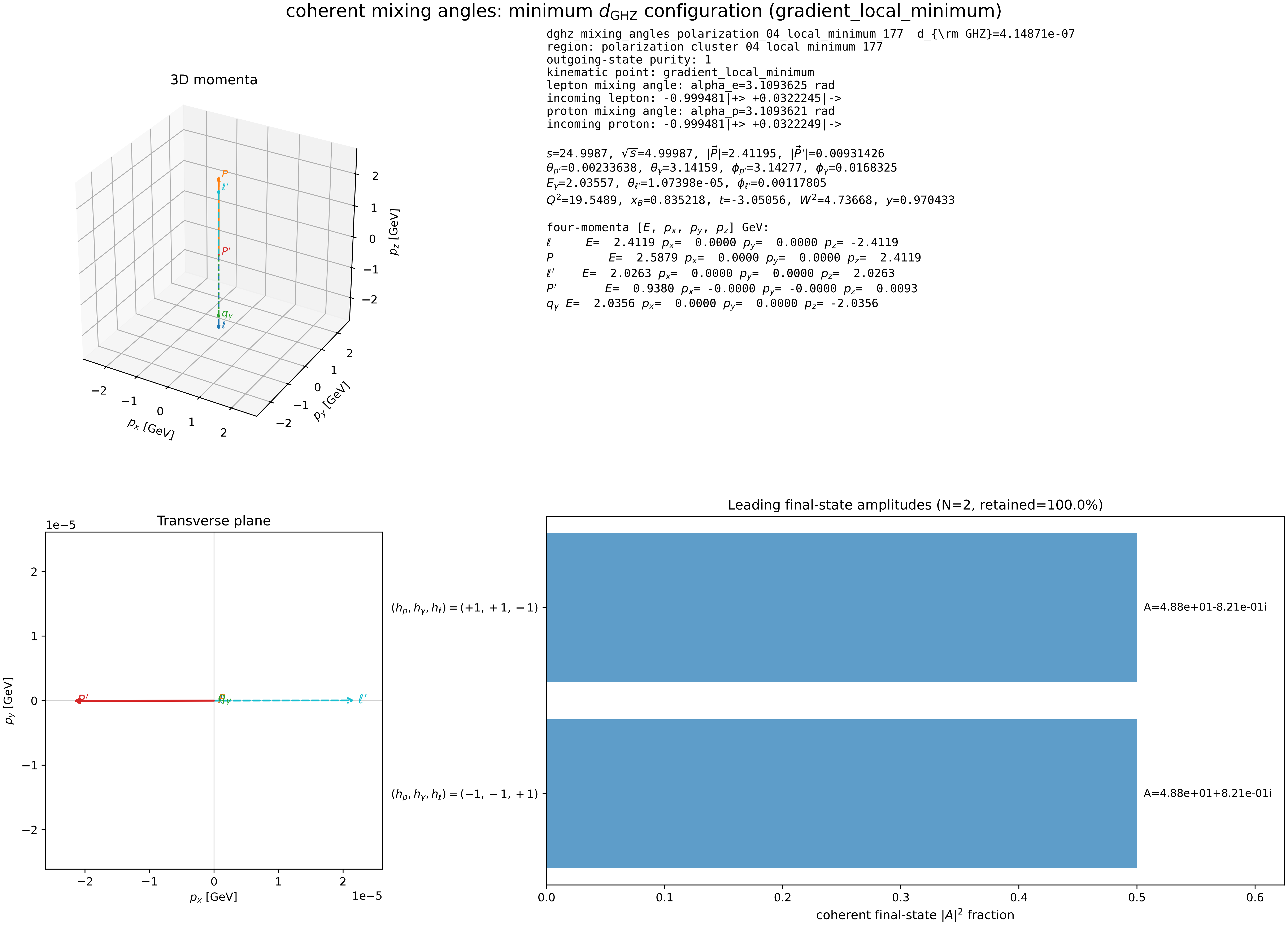

electron: polarization cluster P4, local minimum 166; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 1.3556034e-07
dGHZ contour = 0.050000136
above global minimum = 1.1327688e-07
8D contour samples = 129
selected phase-space point:
sqrt_s = 3.196624
theta_p_out = 0.0006133289
theta_gamma_out = 3.141593
qOut = 1.129459
phi_p_out = 0.001320734
phi_gamma_out = 3.936984
alpha_e = 1.881976
alpha_p = 1.881987

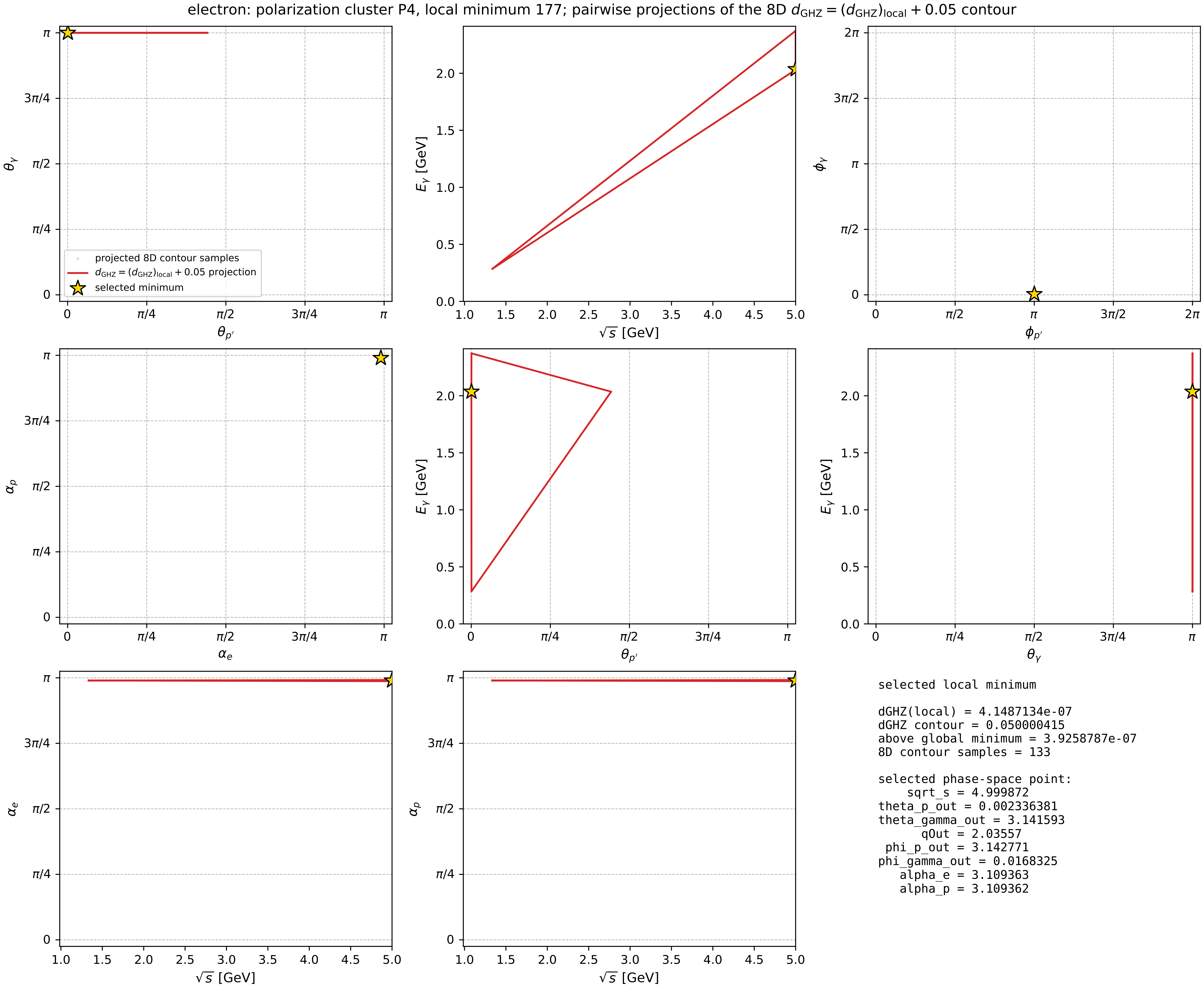

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_169 d_{\rm GHZ}=1.74025e-07
region: polarization_cluster_04_local_minimum_169
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2827776 rad
incoming lepton: -0.653335|+> +0.757069|->
proton mixing angle: alpha_p=2.4296097 rad
incoming proton: -0.757068|+> +0.653336|->
s=16.843, $\sqrt{s}$=4.10403, $|\vec{P}|$=1.94482, $|\vec{P}'|$=1.94482
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=6.13063, ϕ_γ=0.266108
E_γ=1.55486, $\theta_{\ell'}$=0, $\phi_{\ell'}$=2.98904
Q^2=3.03359, x_B=0.192048, t=-15.1293, W^2=13.6422, y=0.989526
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.9448 p_x= 0.0000 p_y= 0.0000 p_z= -1.9448
P E= 2.1592 p_x= 0.0000 p_y= 0.0000 p_z= 1.9448
ℓ' E= 0.3900 p_x= -0.0000 p_y= 0.0000 p_z= 0.3900
P' E= 2.1592 p_x= 0.0000 p_y= -0.0000 p_z= -1.9448
q_γ E= 1.5549 p_x= 0.0000 p_y= 0.0000 p_z= 1.5549
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=5.79e+00+1.55e-17i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-5.79e+00-1.15e-17i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 169; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
θ_γ
selected local minimum
dGHZ(local) = 1.7402473e-07
dGHZ contour = 0.050000174
above global minimum = 1.5174127e-07
8D contour samples = 68
selected phase-space point:
sqrt_s = 4.104027
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 1.554863
phi_p_out = 6.130633
phi_gamma_out = 0.2661081
alpha_e = 2.282778
alpha_p = 2.42961

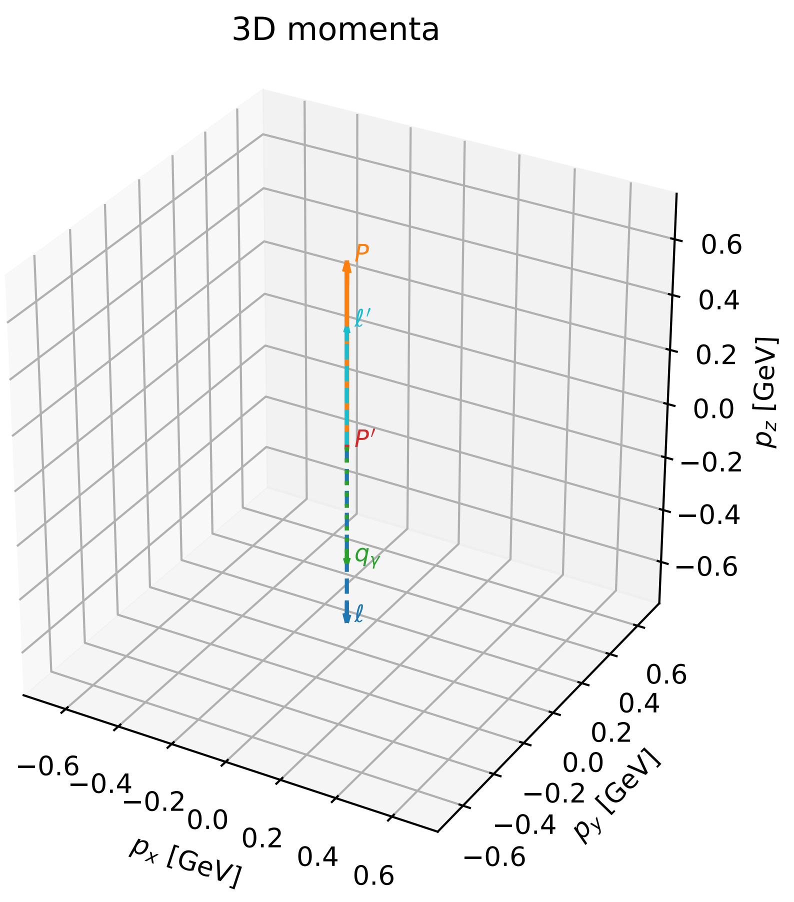

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_176 d_{\rm GHZ}=3.49411e-07
region: polarization_cluster_04_local_minimum_176
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=3.0459519 rad
incoming lepton: -0.99543|+> +0.095495|->
proton mixing angle: alpha_p=1.6664366 rad
incoming proton: -0.0954945|+> +0.99543|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.41202
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=3.3165, ϕ_γ=2.32512
E_γ=1.59717, $\theta_{\ell'}$=0, $\phi_{\ell'}$=0.17491
Q^2=7.86165, x_B=0.329859, t=-23.2713, W^2=16.8516, y=0.988111
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 0.8148 p_x= 0.0000 p_y= 0.0000 p_z= 0.8148
P' E= 2.5880 p_x= -0.0000 p_y= -0.0000 p_z= -2.4120
q_γ E= 1.5972 p_x= -0.0000 p_y= 0.0000 p_z= 1.5972
3D momenta
P
q_γ
ℓ'
P'
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e−9
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-5.44e-01-6.67e-17i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=5.44e-01-5.10e-17i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 176; pairwise projections of the 8D $d_{GHZ} = (d_{GHZ})_{local} + 0.05$ contour
projected 8D contour samples
$d_{GHZ} = (d_{GHZ})_{local} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 3.4941089e-07
dGHZ contour = 0.050000349
above global minimum = 3.2712743e-07
8D contour samples = 38
selected phase-space point:
sqrt_s = 5
theta_p_out = 3.141593
theta_gamma_out = 0
qOut = 1.597173
phi_p_out = 3.316503
phi_gamma_out = 2.325123
alpha_e = 3.045952
alpha_p = 1.666437


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
dghz_mixing_angles_polarization_04_local_minimum_177 d_{\rm GHZ}=4.14871e-07
region: polarization_cluster_04_local_minimum_177
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=3.1093625 rad
incoming lepton: -0.999481|+> +0.0322245|->
proton mixing angle: alpha_p=3.1093621 rad
incoming proton: -0.999481|+> +0.0322249|->
s=24.9987, $\sqrt{s}$=4.99987, $|\vec{P}|$=2.41195, $|\vec{P}'|$=0.00931426
$\theta_{p'}$=0.00233638, θ_γ=3.14159, $\phi_{p'}$=3.14277, ϕ_γ=0.0168325
E_γ=2.03557, $\theta_{\ell'}$=1.07398e-05, $\phi_{\ell'}$=0.00117805
Q^2=19.5489, x_B=0.835218, t=-3.05056, W^2=4.73668, y=0.970433
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4119 p_x= 0.0000 p_y= 0.0000 p_z= -2.4119
P E= 2.5879 p_x= 0.0000 p_y= 0.0000 p_z= 2.4119
ℓ' E= 2.0263 p_x= 0.0000 p_y= 0.0000 p_z= 2.0263
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0093
q_γ E= 2.0356 p_x= 0.0000 p_y= 0.0000 p_z= -2.0356
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
1e−5
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=4.88e+01-8.21e-01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=4.88e+01+8.21e-01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 177; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 4.1487134e-07
dGHZ contour = 0.050000415
above global minimum = 3.9258787e-07
8D contour samples = 133
selected phase-space point:
sqrt_s = 4.999872
theta_p_out = 0.002336381
theta_gamma_out = 3.141593
qOut = 2.03557
phi_p_out = 3.142771
phi_gamma_out = 0.0168325
alpha_e = 3.109363
alpha_p = 3.109362

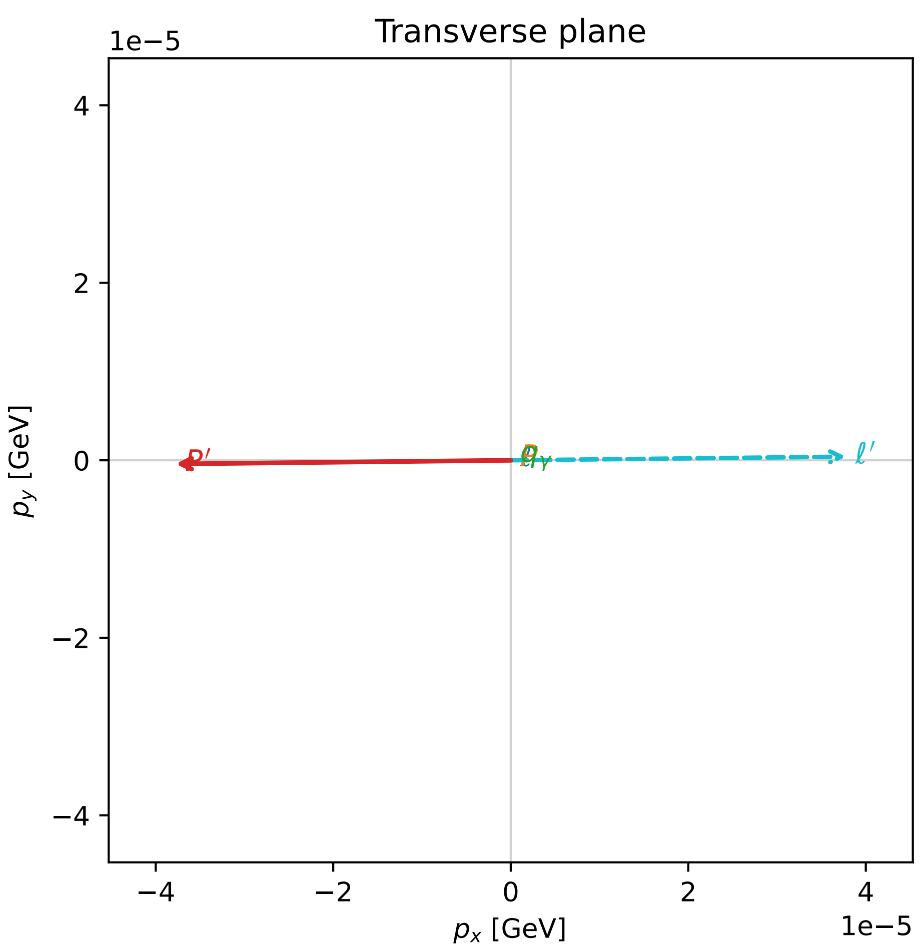
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

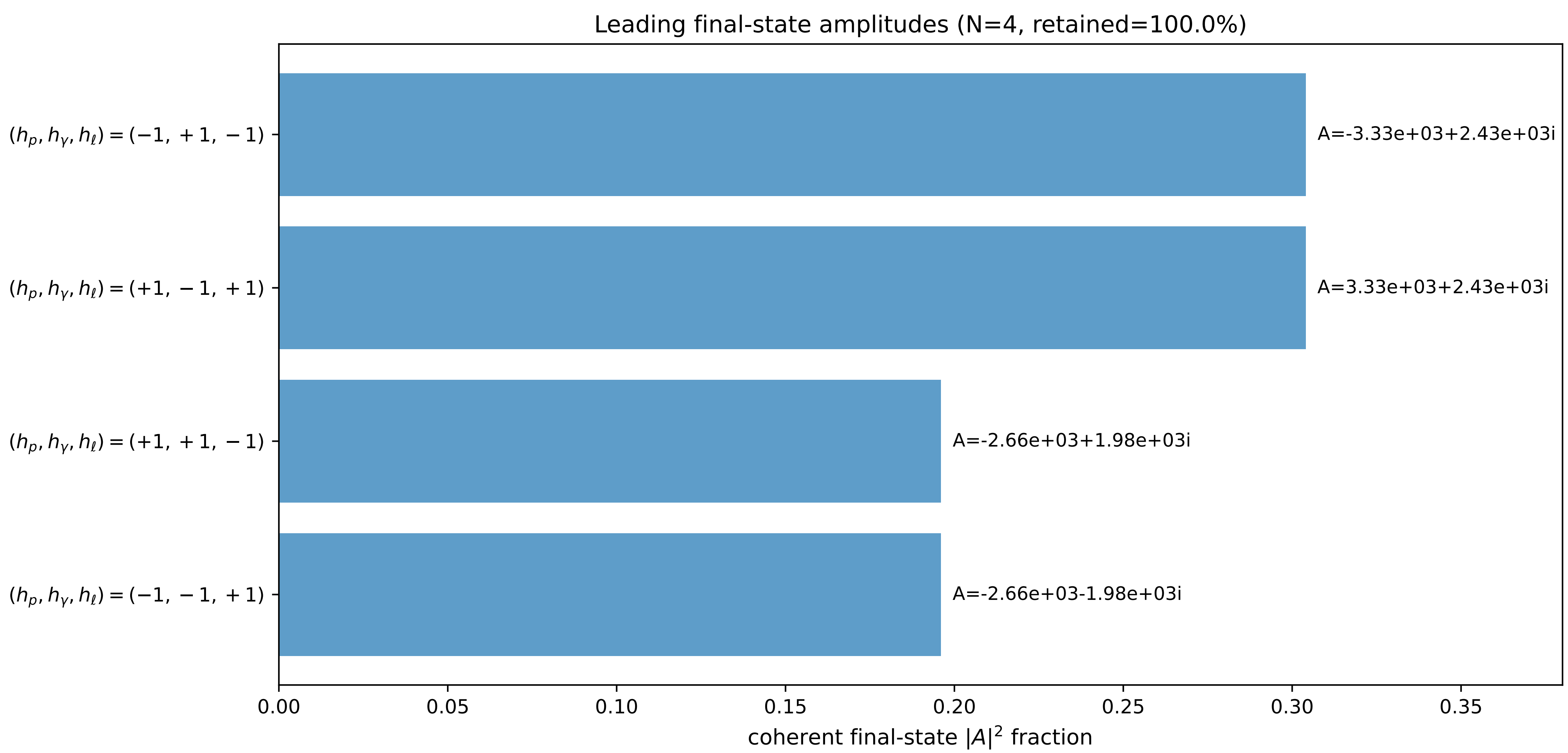


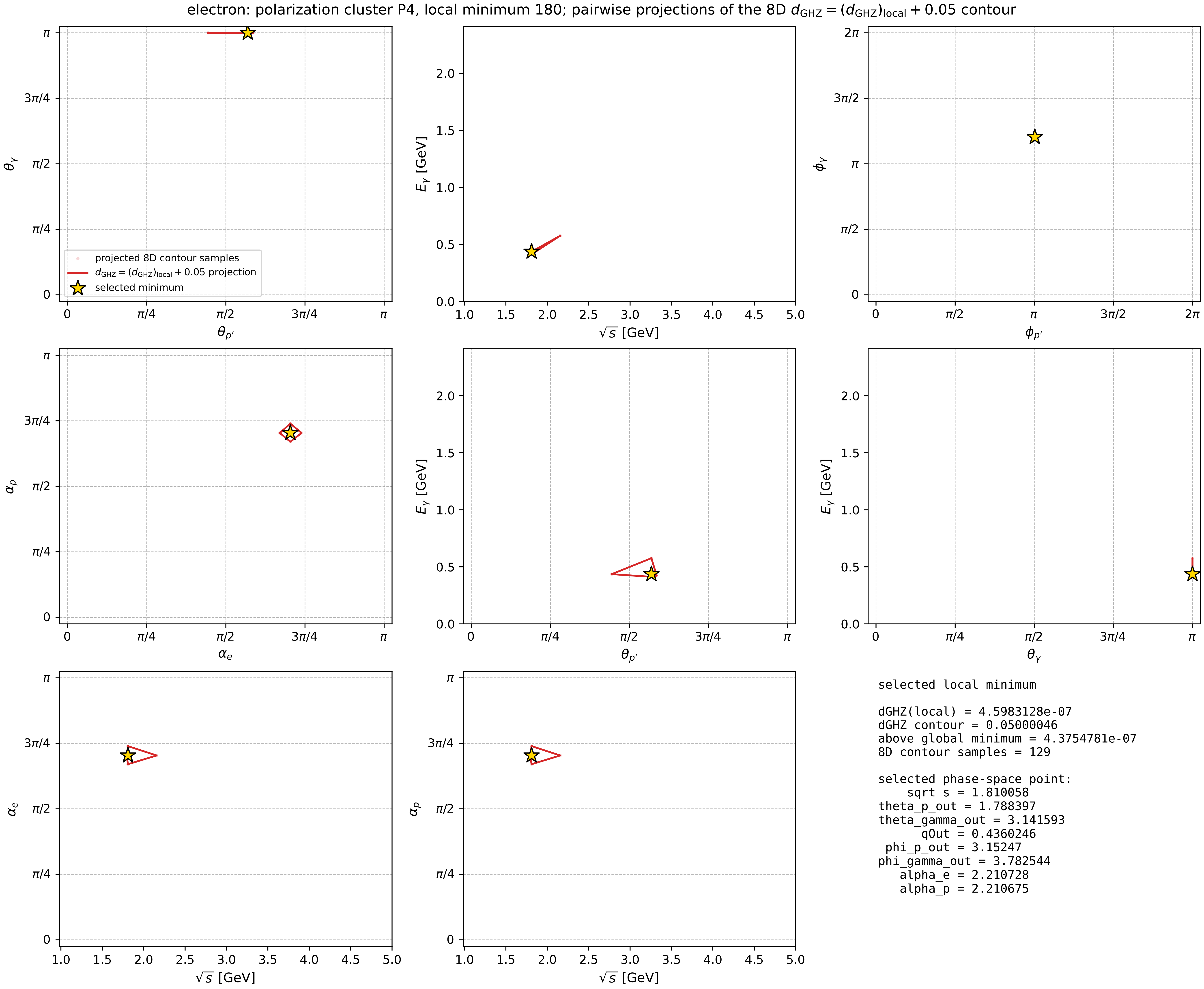
dghz_mixing_angles_polarization_04_local_minimum_180 d_{\rm GHZ}=4.59831e-07
region: polarization_cluster_04_local_minimum_180
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2107281 rad
incoming lepton: -0.597141|+> +0.802137|->
proton mixing angle: alpha_p=2.2106753 rad
incoming proton: -0.597098|+> +0.802168|->

s=3.27631, $\sqrt{s}$=1.81006, $|\vec{P}|$=0.661986, $|\vec{P}'|$=3.86629e-05
$\theta_{p'}$=1.7884, θ_γ=3.14159, $\phi_{p'}$=3.15247, ϕ_γ=3.78254
E_γ=0.436025, $\theta_{\ell'}$=8.65788e-05, $\phi_{\ell'}$=0.0108771
Q^2=1.15459, x_B=0.585324, t=-0.394106, W^2=1.69782, y=0.823115

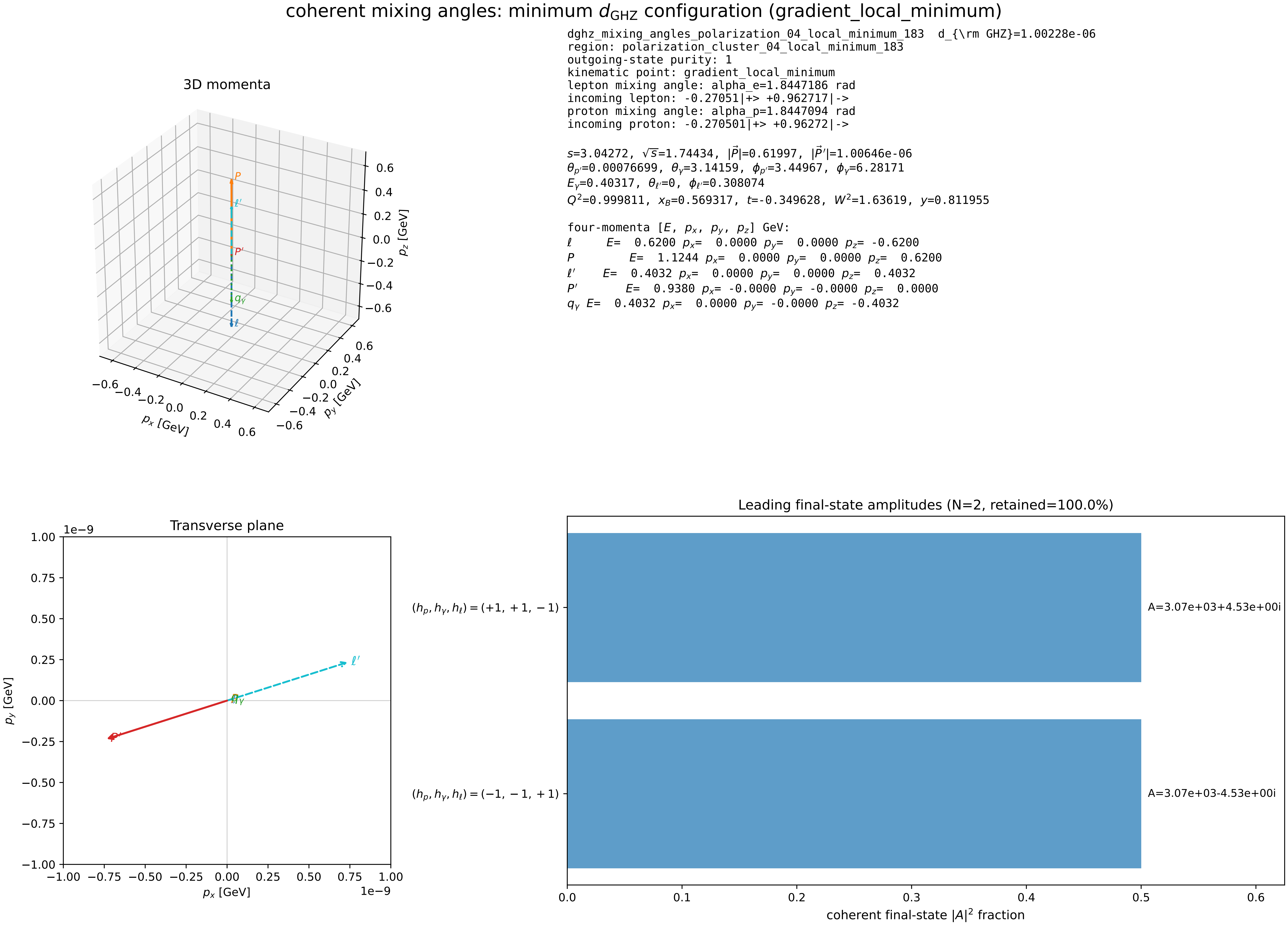
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.6620 p_x= 0.0000 p_y= 0.0000 p_z= -0.6620
P E= 1.1481 p_x= 0.0000 p_y= 0.0000 p_z= 0.6620
ℓ' E= 0.4360 p_x= 0.0000 p_y= 0.0000 p_z= 0.4360
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0000
q_γ E= 0.4360 p_x= -0.0000 p_y= -0.0000 p_z= -0.4360



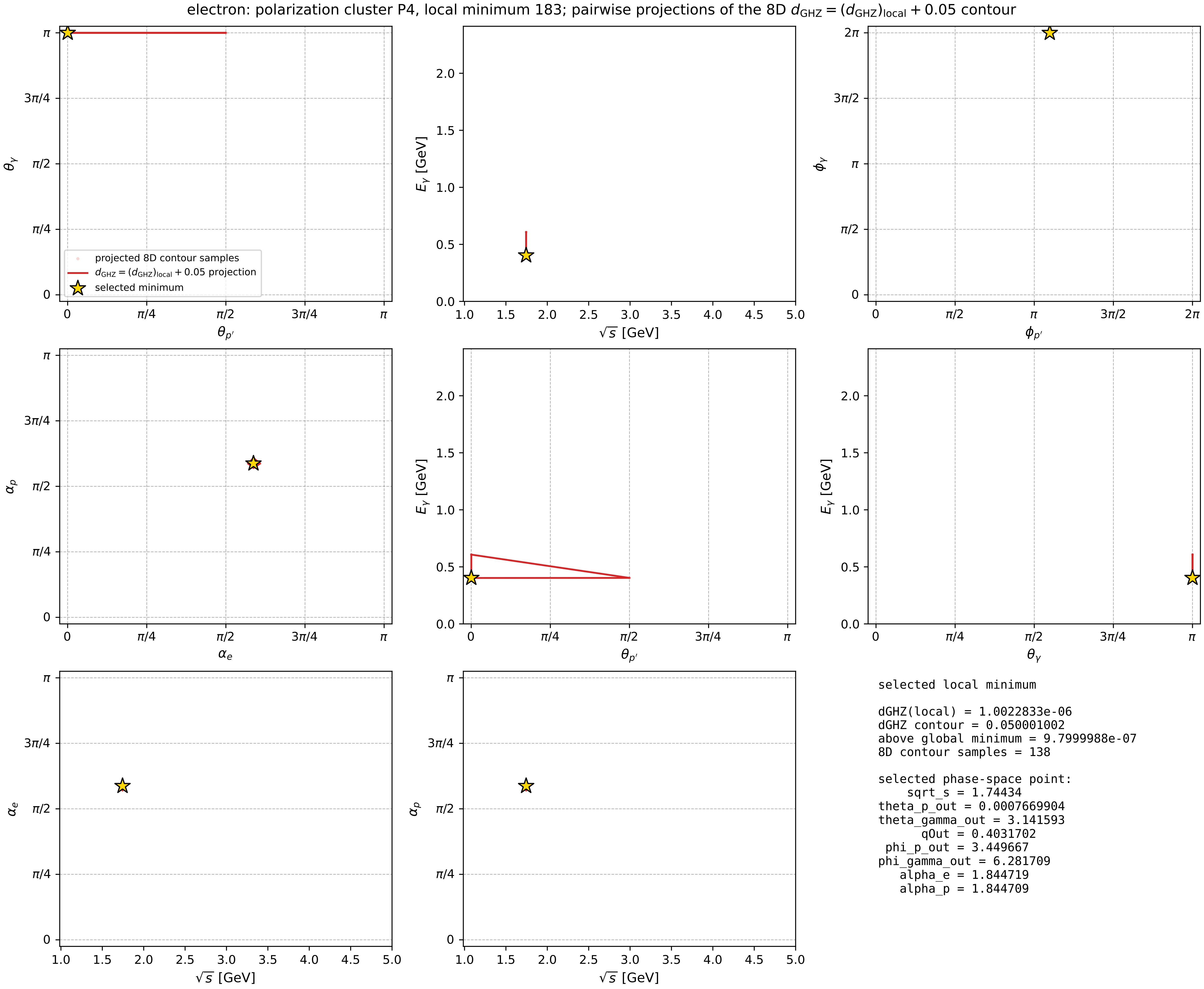




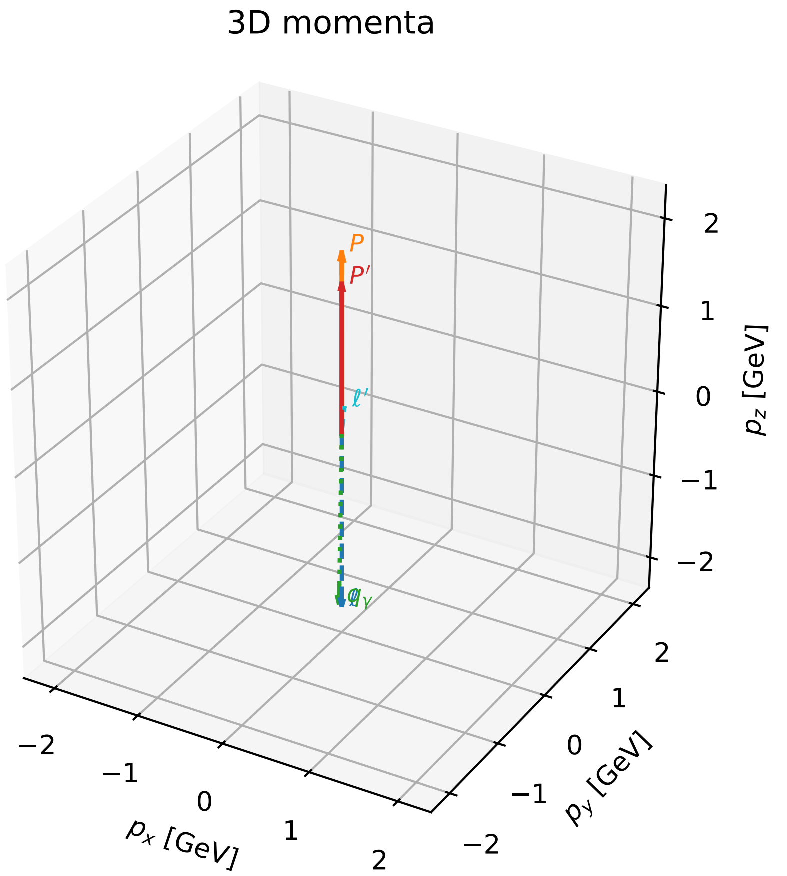
electron: polarization cluster P4, local minimum 180; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 4.5983128e-07
dGHZ contour = 0.05000046
above global minimum = 4.3754781e-07
8D contour samples = 129
selected phase-space point:
sqrt_s = 1.810058
theta_p_out = 1.788397
theta_gamma_out = 3.141593
qOut = 0.4360246
phi_p_out = 3.15247
phi_gamma_out = 3.782544
alpha_e = 2.210728
alpha_p = 2.210675



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
P
ℓ'
P'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_183 d_{\rm GHZ}=1.00228e-06
region: polarization_cluster_04_local_minimum_183
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.8447186 rad
incoming lepton: -0.27051|+> +0.962717|->
proton mixing angle: alpha_p=1.8447094 rad
incoming proton: -0.270501|+> +0.96272|->
s=3.04272, $\sqrt{s}$=1.74434, $|\vec{P}|$=0.61997, $|\vec{P}'|$=1.00646e-06
$\theta_{p'}$=0.00076699, θ_γ=3.14159, $\phi_{p'}$=3.44967, ϕ_γ=6.28171
E_γ=0.40317, $\theta_{\ell'}$=0, $\phi_{\ell'}$=0.308074
Q^2=0.999811, x_B=0.569317, t=-0.349628, W^2=1.63619, y=0.811955
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.6200 p_x= 0.0000 p_y= 0.0000 p_z= -0.6200
P E= 1.1244 p_x= 0.0000 p_y= 0.0000 p_z= 0.6200
ℓ' E= 0.4032 p_x= 0.0000 p_y= 0.0000 p_z= 0.4032
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0000
q_γ E= 0.4032 p_x= 0.0000 p_y= -0.0000 p_z= -0.4032
Transverse plane
1e−9
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=3.07e+03+4.53e+00i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=3.07e+03-4.53e+00i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 183; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 1.0022833e-06
dGHZ contour = 0.050001002
above global minimum = 9.7999988e-07
8D contour samples = 138
selected phase-space point:
sqrt_s = 1.74434
theta_p_out = 0.0007669904
theta_gamma_out = 3.141593
qOut = 0.4031702
phi_p_out = 3.449667
phi_gamma_out = 6.281709
alpha_e = 1.844719
alpha_p = 1.844709

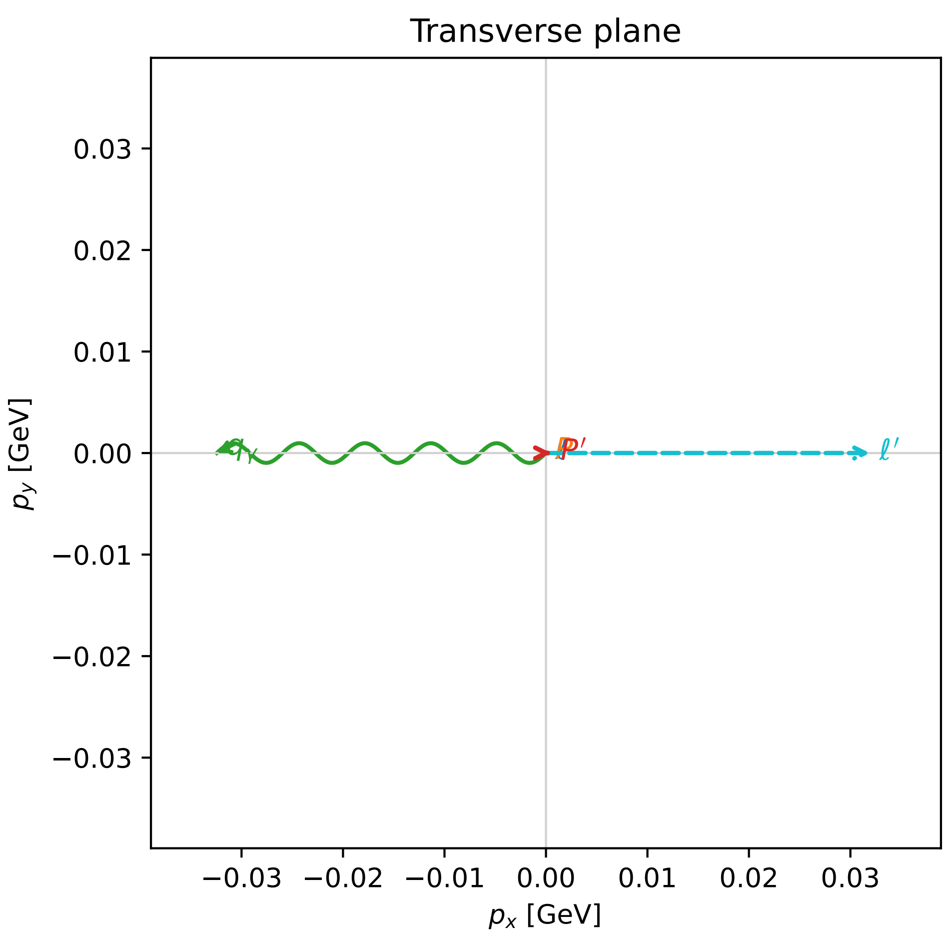
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

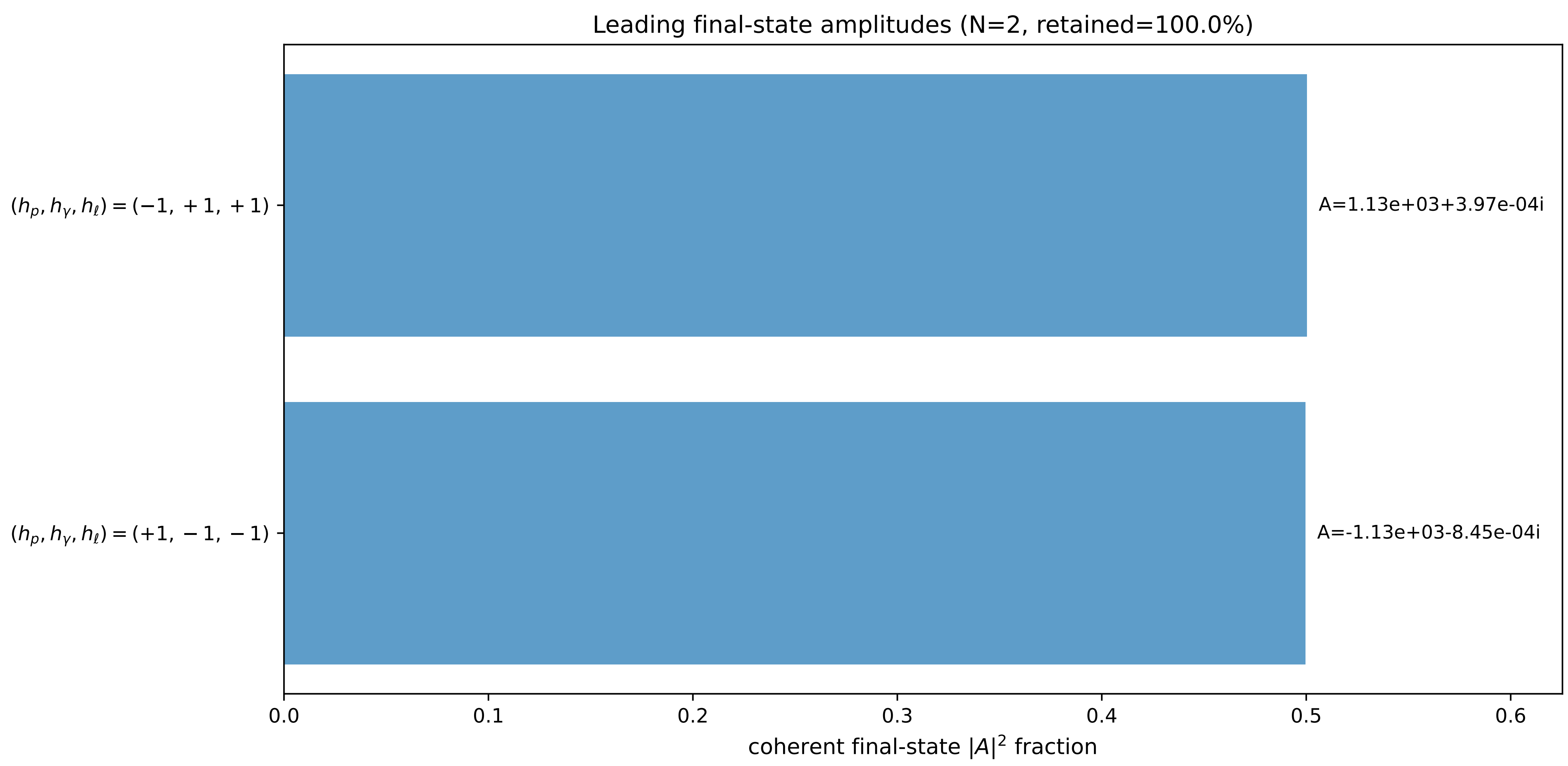


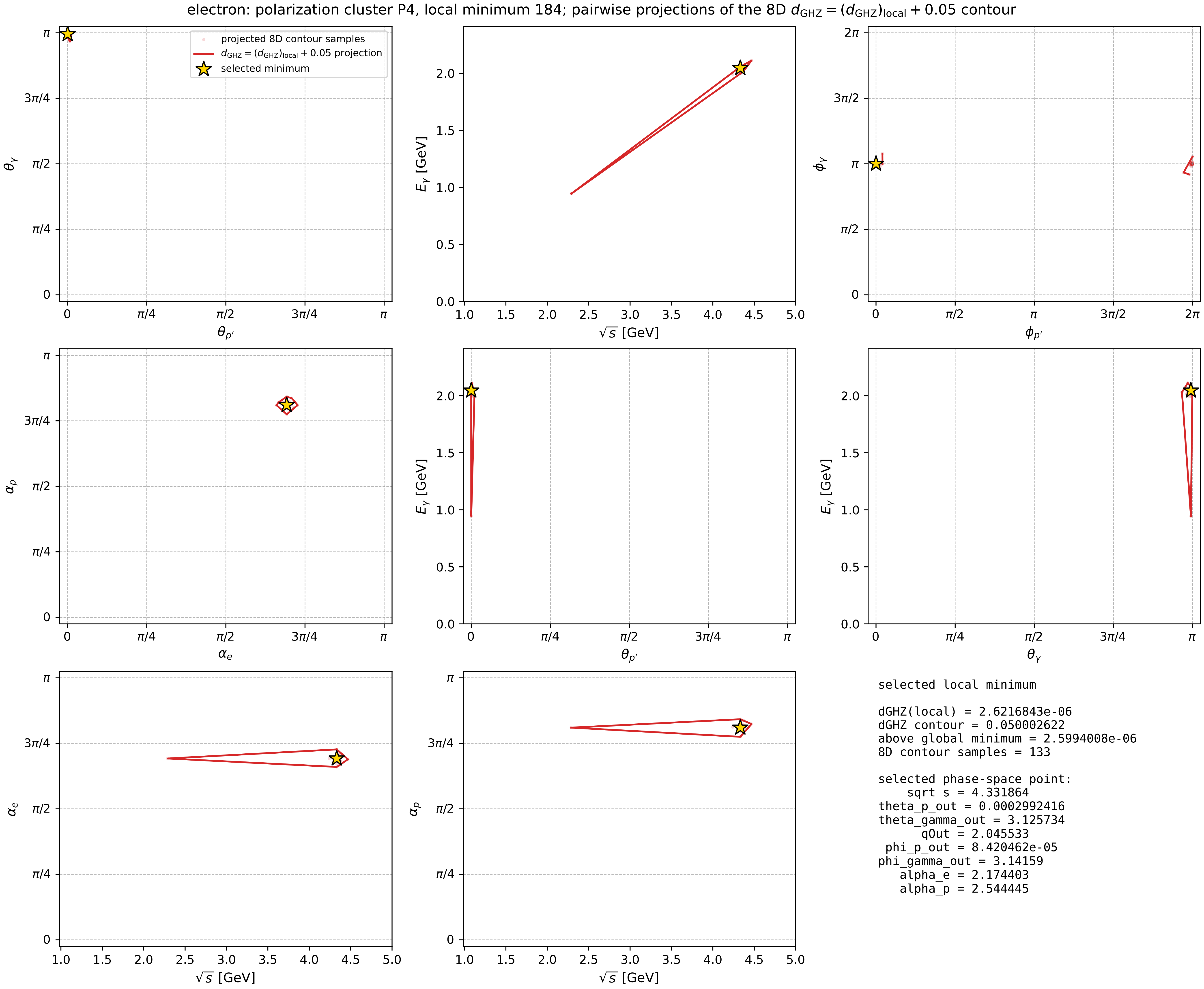
dghz_mixing_angles_polarization_04_local_minimum_184 d_{\rm GHZ}=2.62168e-06
region: polarization_cluster_04_local_minimum_184
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.1744032 rad
incoming lepton: -0.567616|+> +0.823294|->
proton mixing angle: alpha_p=2.5444445 rad
incoming proton: -0.826943|+> +0.562286|->

s=18.765, $\sqrt{s}$=4.33186, $|\vec{P}|$=2.06438, $|\vec{P}'|$=1.71704
$\theta_{p'}$=0.000299242, θ_γ=3.12573, $\phi_{p'}$=8.42046e-05, ϕ_γ=3.14159
E_γ=2.04553, $\theta_{\ell'}$=0.0969549, $\phi_{\ell'}$=6.28318
Q^2=2.71682, x_B=0.153105, t=-0.023959, W^2=15.9079, y=0.992152

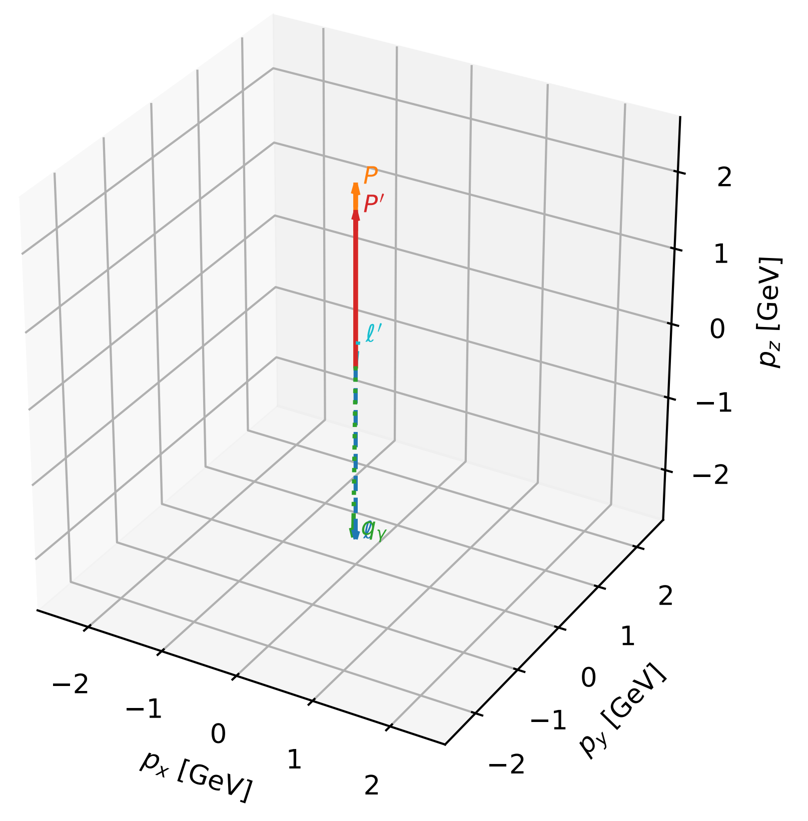
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.0644 p_x= 0.0000 p_y= 0.0000 p_z= -2.0644
P E= 2.2675 p_x= 0.0000 p_y= 0.0000 p_z= 2.0644
ℓ' E= 0.3298 p_x= 0.0319 p_y= -0.0000 p_z= 0.3282
P' E= 1.9565 p_x= 0.0005 p_y= 0.0000 p_z= 1.7170
q_γ E= 2.0455 p_x= -0.0324 p_y= 0.0000 p_z= -2.0453






electron: polarization cluster P4, local minimum 184; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 2.6216843e-06
dGHZ contour = 0.050002622
above global minimum = 2.5994008e-06
8D contour samples = 133
selected phase-space point:
sqrt_s = 4.331864
theta_p_out = 0.0002992416
theta_gamma_out = 3.125734
qOut = 2.045533
phi_p_out = 8.420462e-05
phi_gamma_out = 3.14159
alpha_e = 2.174403
alpha_p = 2.544445

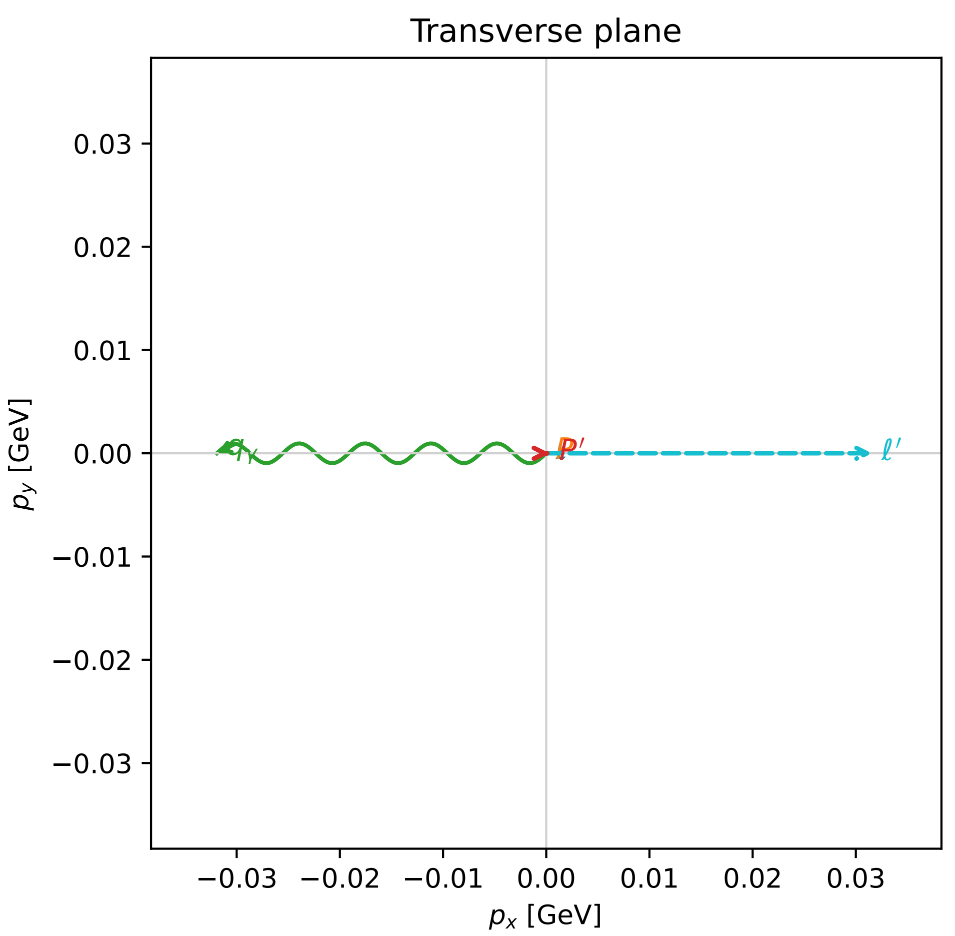
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

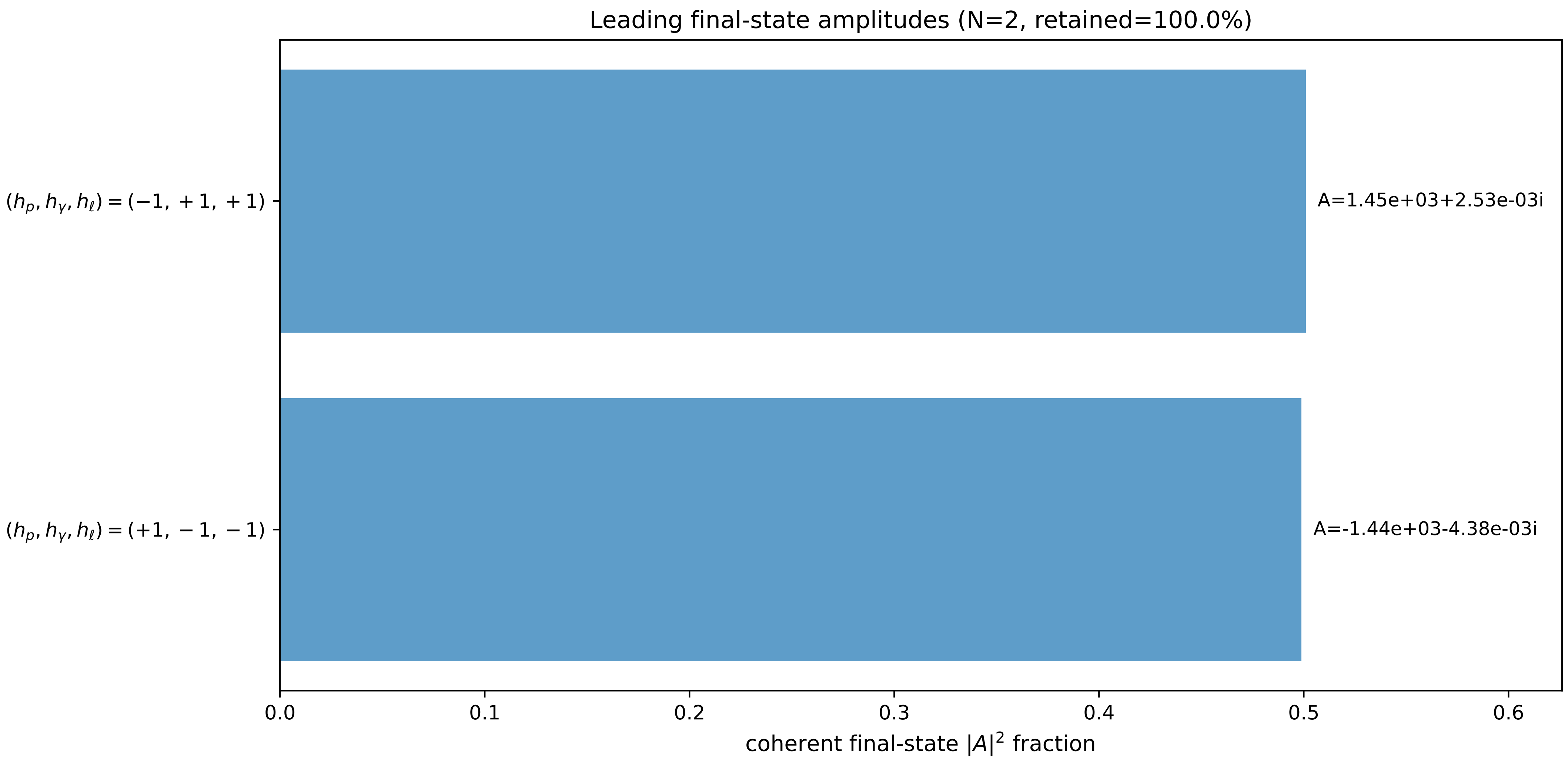


dghz_mixing_angles_polarization_04_local_minimum_186 d_{\rm GHZ}=7.14153e-06
region: polarization_cluster_04_local_minimum_186
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2229182 rad
incoming lepton: -0.606874|+> +0.794798|->
proton mixing angle: alpha_p=2.4957371 rad
incoming proton: -0.798585|+> +0.601882|->

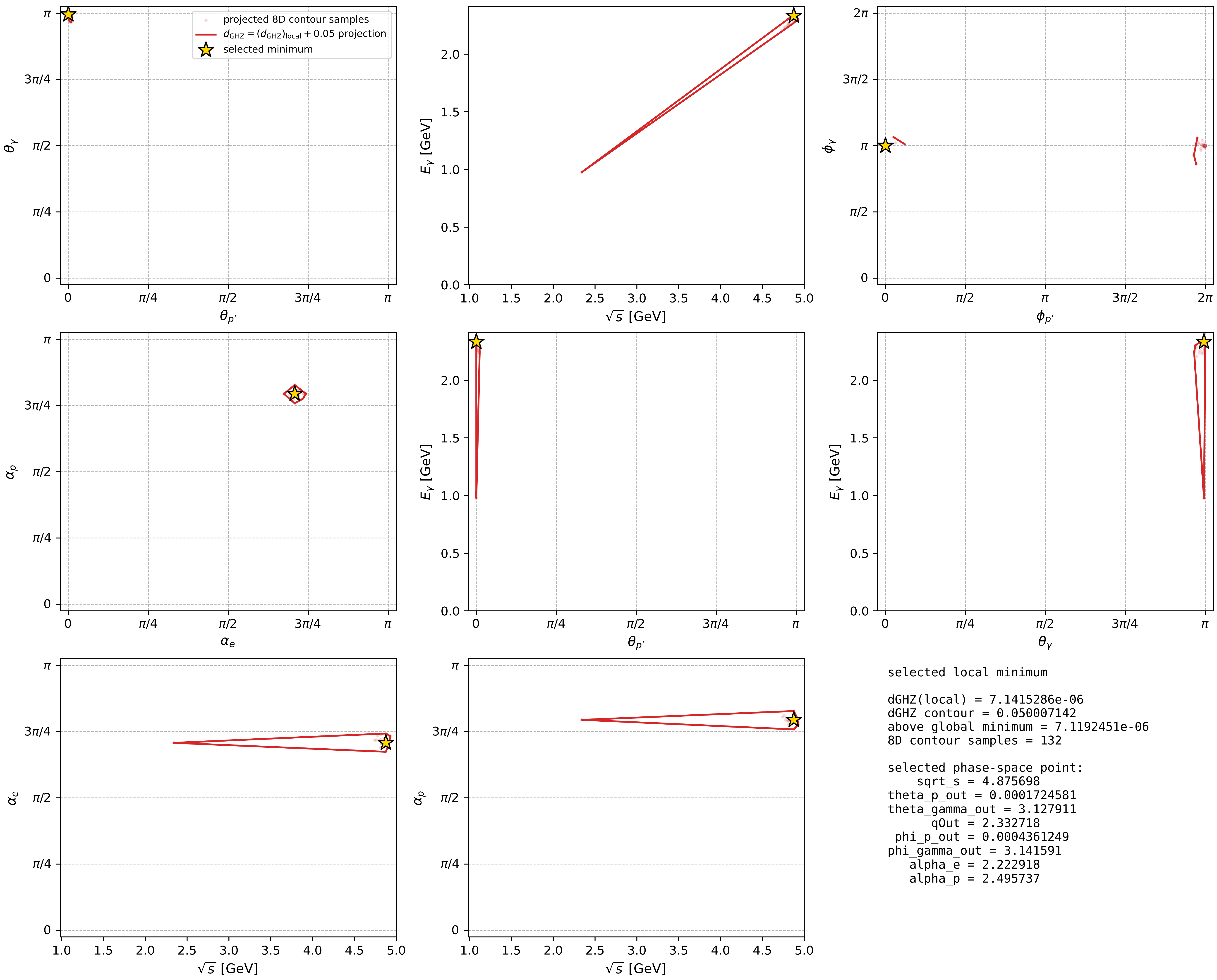
s=23.7724, $\sqrt{s}$=4.8757, $|\vec{P}|$=2.34762, $|\vec{P}'|$=2.00059
$\theta_{p'}$=0.000172458, θ_γ=3.12791, $\phi_{p'}$=0.000436125, ϕ_γ=3.14159
E_γ=2.33272, $\theta_{\ell'}$=0.0948293, $\phi_{\ell'}$=6.28318
Q^2=3.12386, x_B=0.137221, t=-0.0189857, W^2=20.5212, y=0.994436

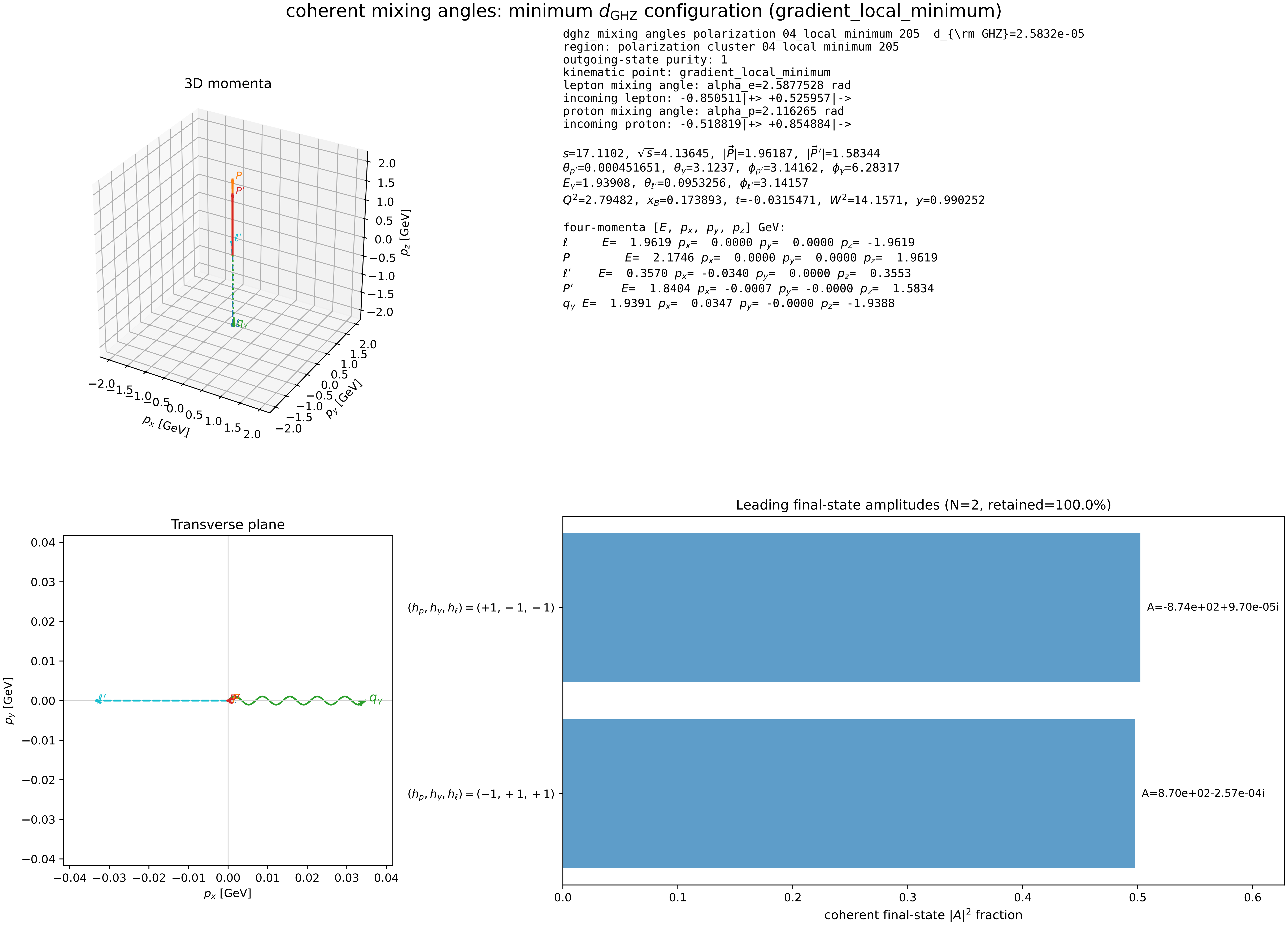
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3476 p_x= 0.0000 p_y= 0.0000 p_z= -2.3476
P E= 2.5281 p_x= 0.0000 p_y= 0.0000 p_z= 2.3476
ℓ' E= 0.3334 p_x= 0.0316 p_y= -0.0000 p_z= 0.3319
P' E= 2.2096 p_x= 0.0003 p_y= 0.0000 p_z= 2.0006
q_γ E= 2.3327 p_x= -0.0319 p_y= 0.0000 p_z= -2.3325





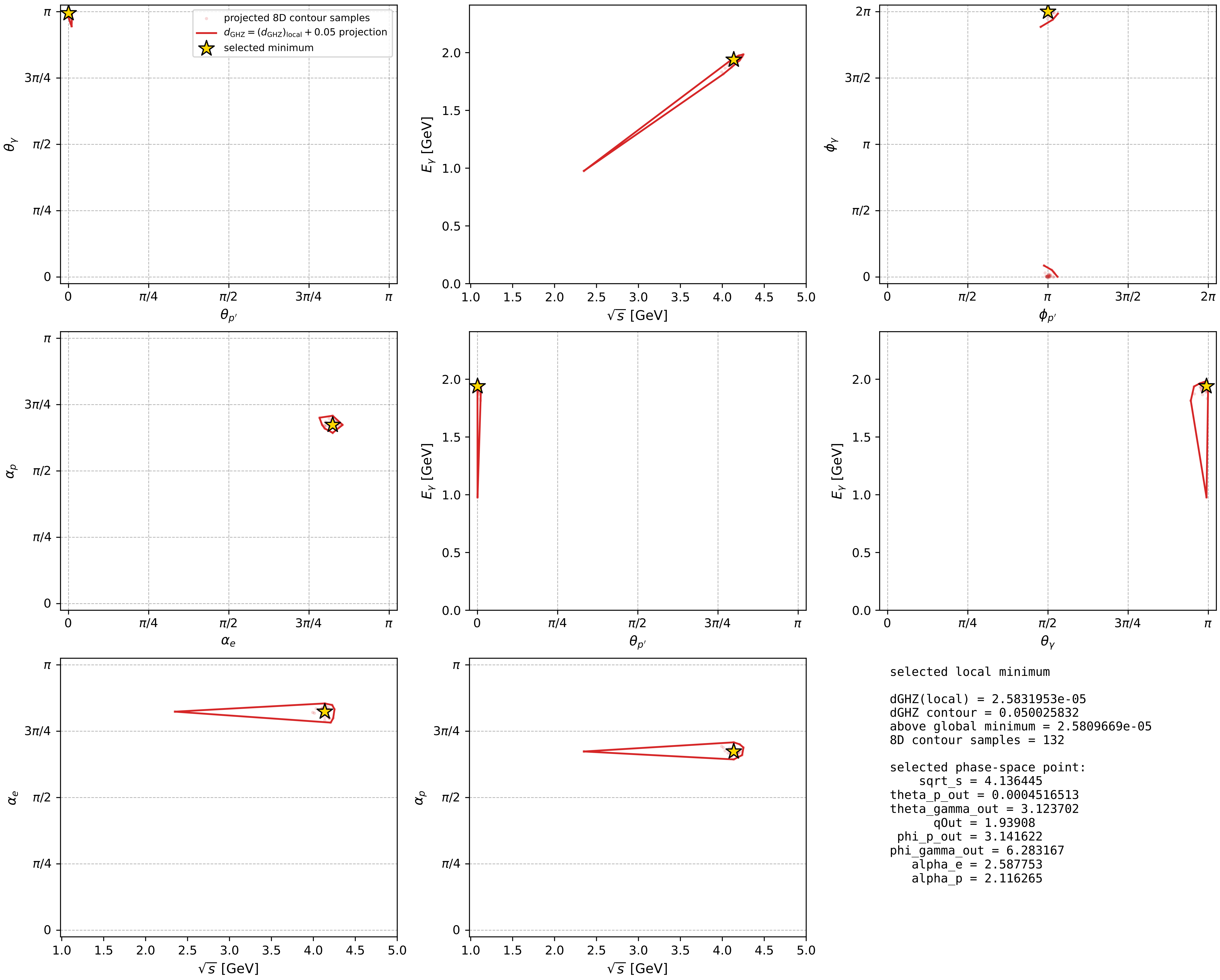
electron: polarization cluster P4, local minimum 186; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



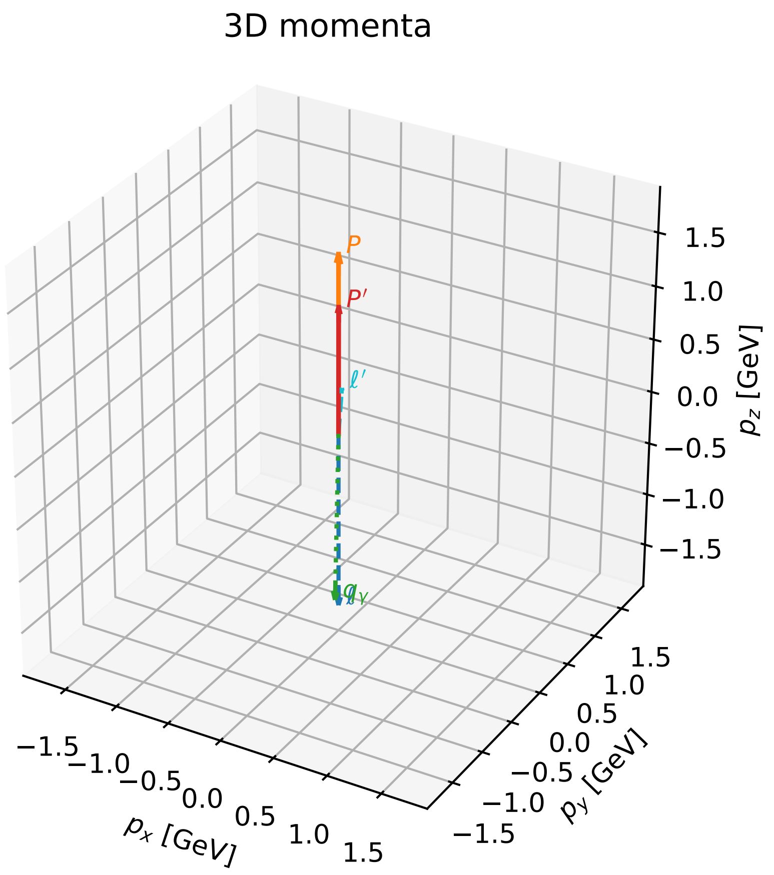


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_205 d_{\rm GHZ}=2.5832e-05
region: polarization_cluster_04_local_minimum_205
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.5877528 rad
incoming lepton: -0.850511|+> +0.525957|->
proton mixing angle: alpha_p=2.116265 rad
incoming proton: -0.518819|+> +0.854884|->
s=17.1102, $\sqrt{s}$=4.13645, $|\vec{P}|$=1.96187, $|\vec{P}'|$=1.58344
$\theta_{p'}$=0.000451651, θ_γ=3.1237, $\phi_{p'}$=3.14162, ϕ_γ=6.28317
E_γ=1.93908, $\theta_{\ell'}$=0.0953256, $\phi_{\ell'}$=3.14157
Q^2=2.79482, x_B=0.173893, t=-0.0315471, W^2=14.1571, y=0.990252
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.9619 p_x= 0.0000 p_y= 0.0000 p_z= -1.9619
P E= 2.1746 p_x= 0.0000 p_y= 0.0000 p_z= 1.9619
ℓ' E= 0.3570 p_x= -0.0340 p_y= 0.0000 p_z= 0.3553
P' E= 1.8404 p_x= -0.0007 p_y= -0.0000 p_z= 1.5834
q_γ E= 1.9391 p_x= 0.0347 p_y= -0.0000 p_z= -1.9388
3D momenta
P
P'
ℓ'
q_γ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-8.74e+02+9.70e-05i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=8.70e+02-2.57e-04i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P4, local minimum 205; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



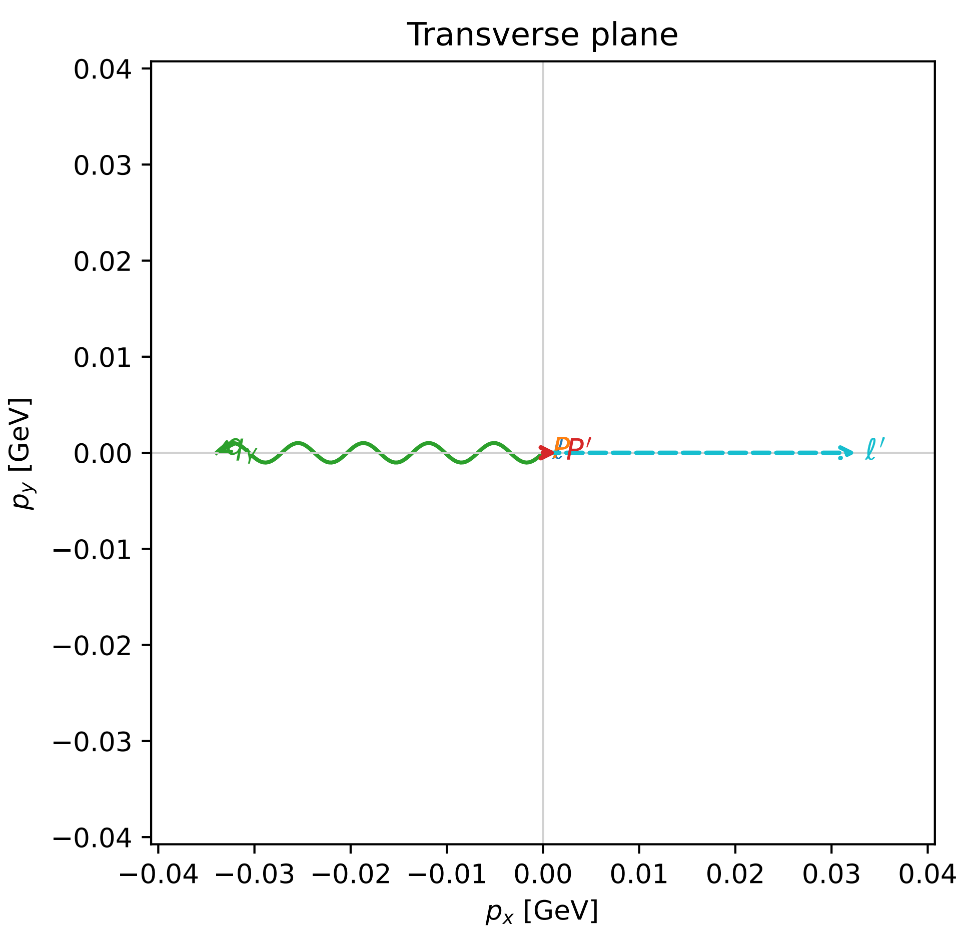
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

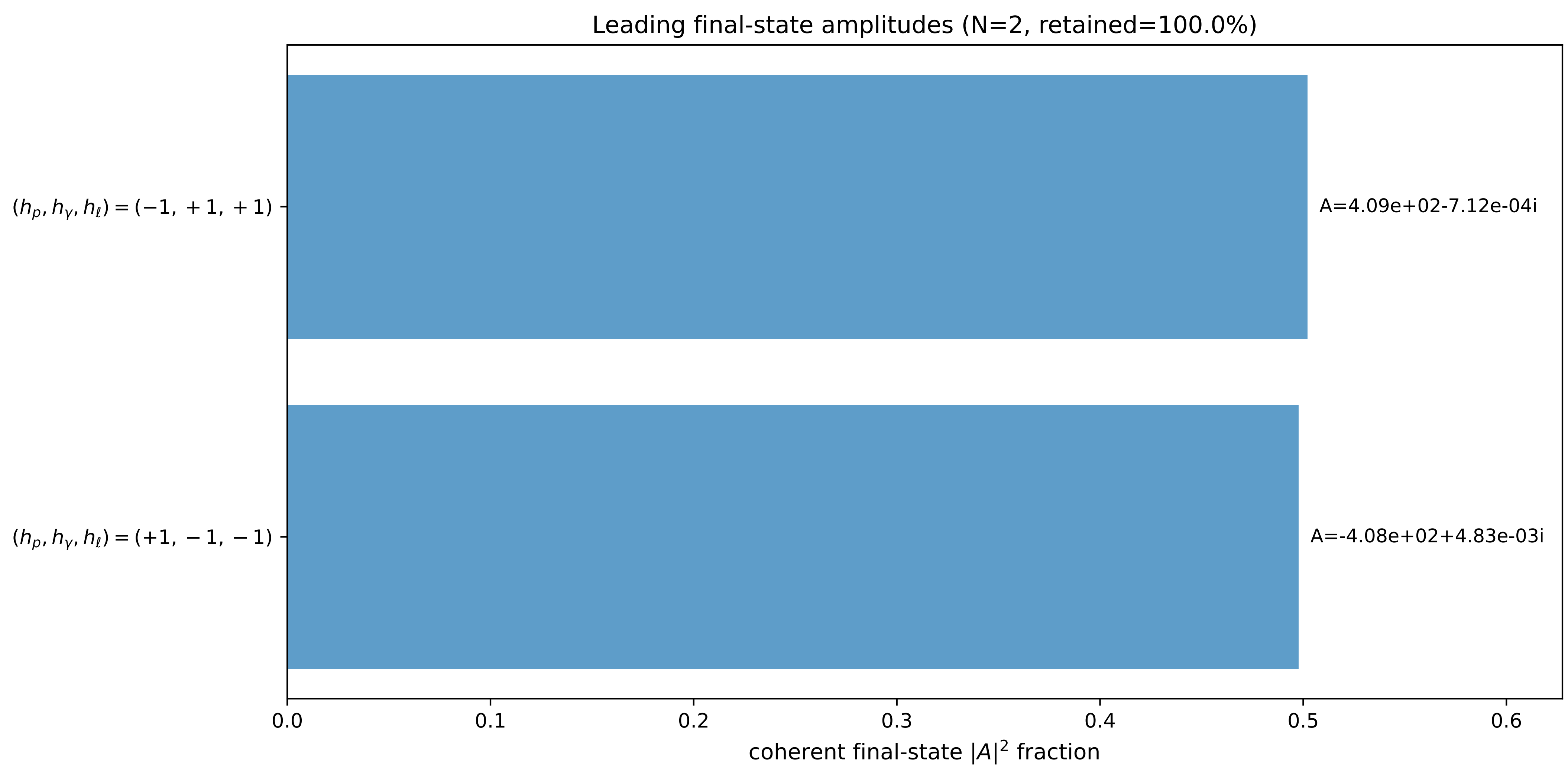


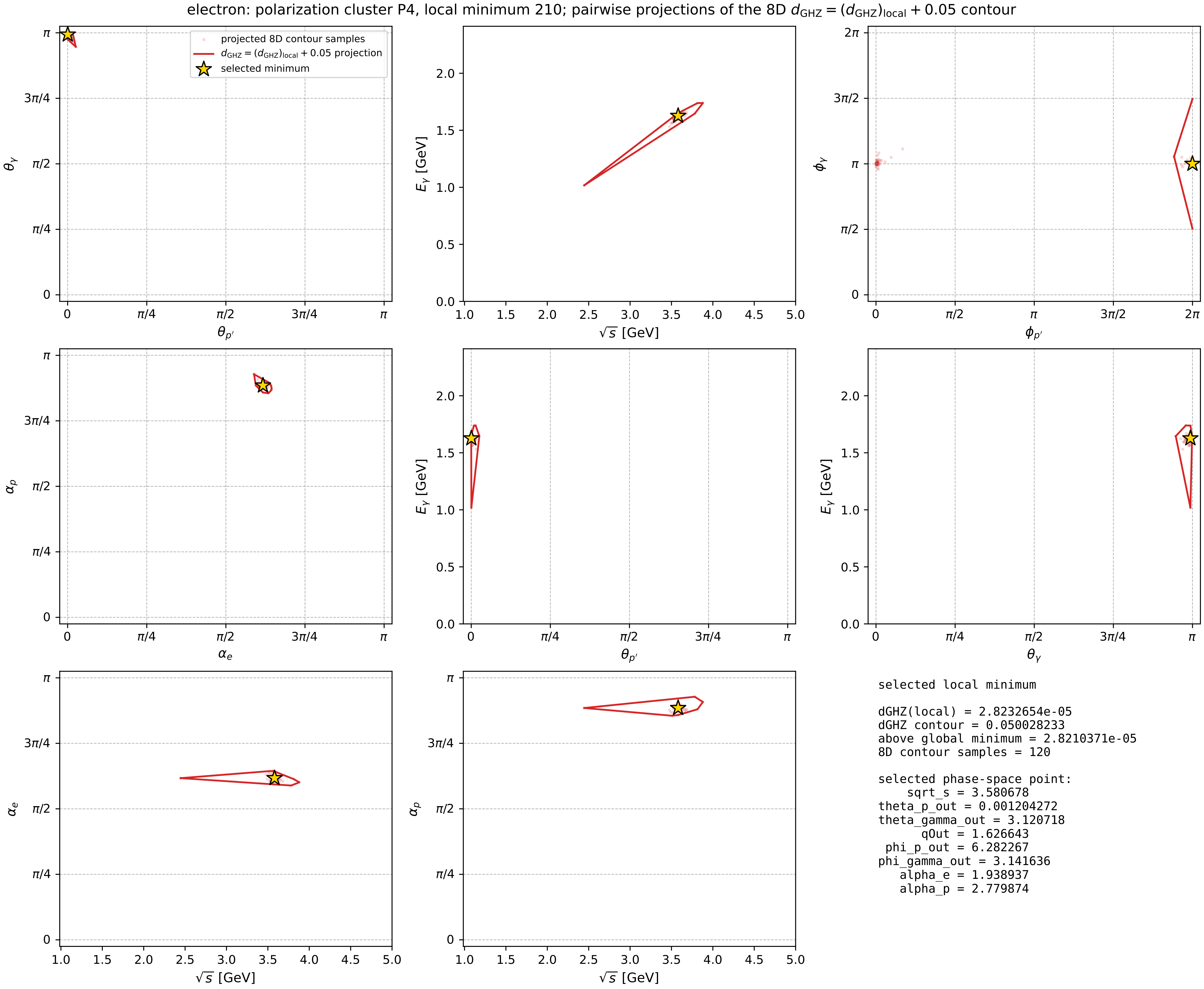
dghz_mixing_angles_polarization_04_local_minimum_210 d_{\rm GHZ}=2.82327e-05
region: polarization_cluster_04_local_minimum_210
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.938937 rad
incoming lepton: -0.359881|+> +0.932998|->
proton mixing angle: alpha_p=2.7798744 rad
incoming proton: -0.93529|+> +0.353882|->

s=12.8213, $\sqrt{s}$=3.58068, $|\vec{P}|$=1.66748, $|\vec{P}'|$=1.1839
$\theta_{p'}$=0.00120427, θ_γ=3.12072, $\phi_{p'}$=6.28227, ϕ_γ=3.14164
E_γ=1.62664, $\theta_{\ell'}$=0.0733953, $\phi_{\ell'}$=8.56722e-05
Q^2=2.95466, x_B=0.252116, t=-0.0716446, W^2=9.64462, y=0.981411

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.6675 p_x= 0.0000 p_y= 0.0000 p_z= -1.6675
P E= 1.9132 p_x= 0.0000 p_y= 0.0000 p_z= 1.6675
ℓ' E= 0.4436 p_x= 0.0325 p_y= 0.0000 p_z= 0.4424
P' E= 1.5105 p_x= 0.0014 p_y= -0.0000 p_z= 1.1839
q_γ E= 1.6266 p_x= -0.0340 p_y= -0.0000 p_z= -1.6263

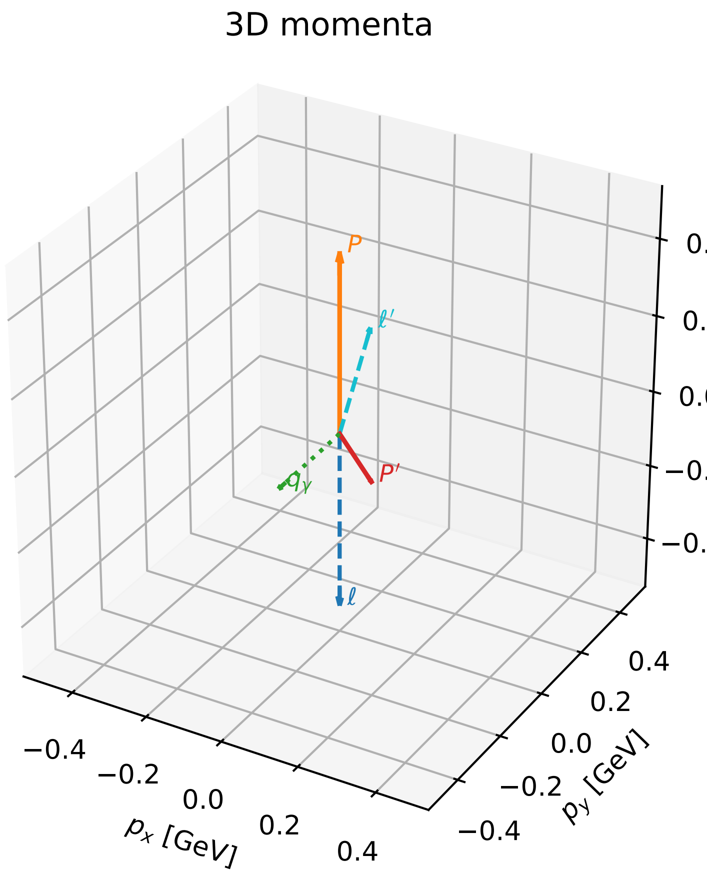






electron: polarization cluster P4, local minimum 210; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 2.8232654e-05
dGHZ contour = 0.050028233
above global minimum = 2.8210371e-05
8D contour samples = 120
selected phase-space point:
sqrt_s = 3.580678
theta_p_out = 0.001204272
theta_gamma_out = 3.120718
qOut = 1.626643
phi_p_out = 6.282267
phi_gamma_out = 3.141636
alpha_e = 1.938937
alpha_p = 2.779874

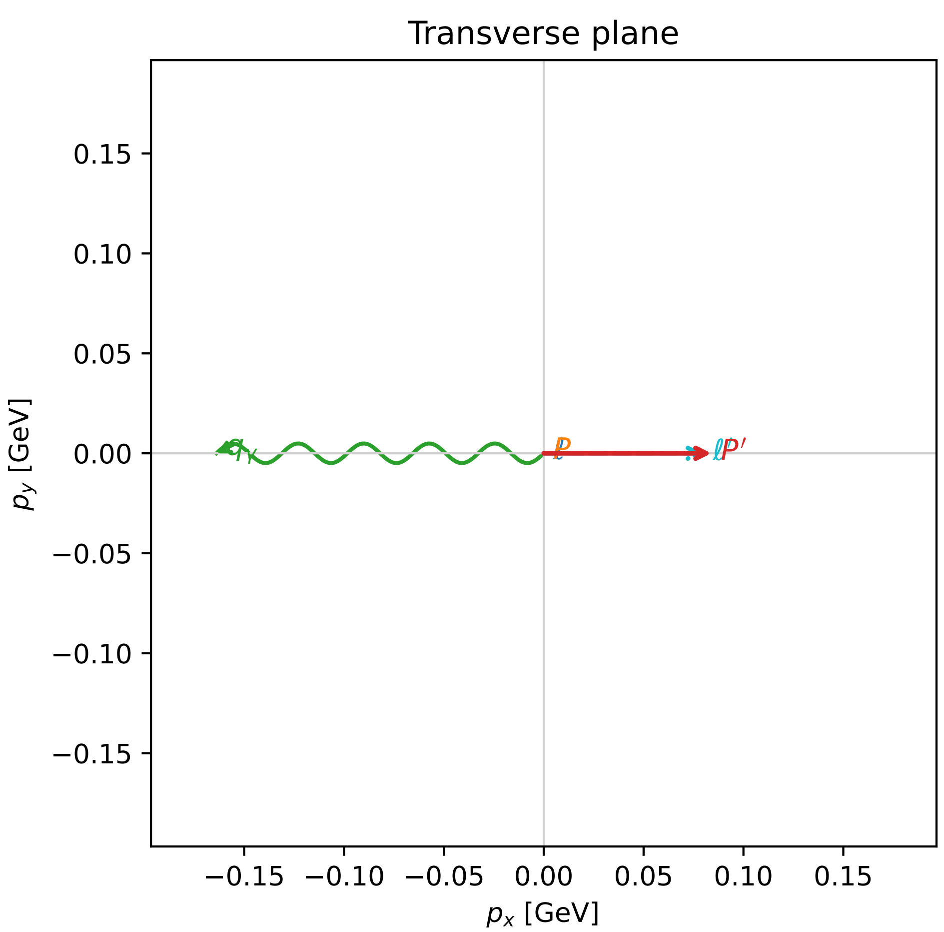
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

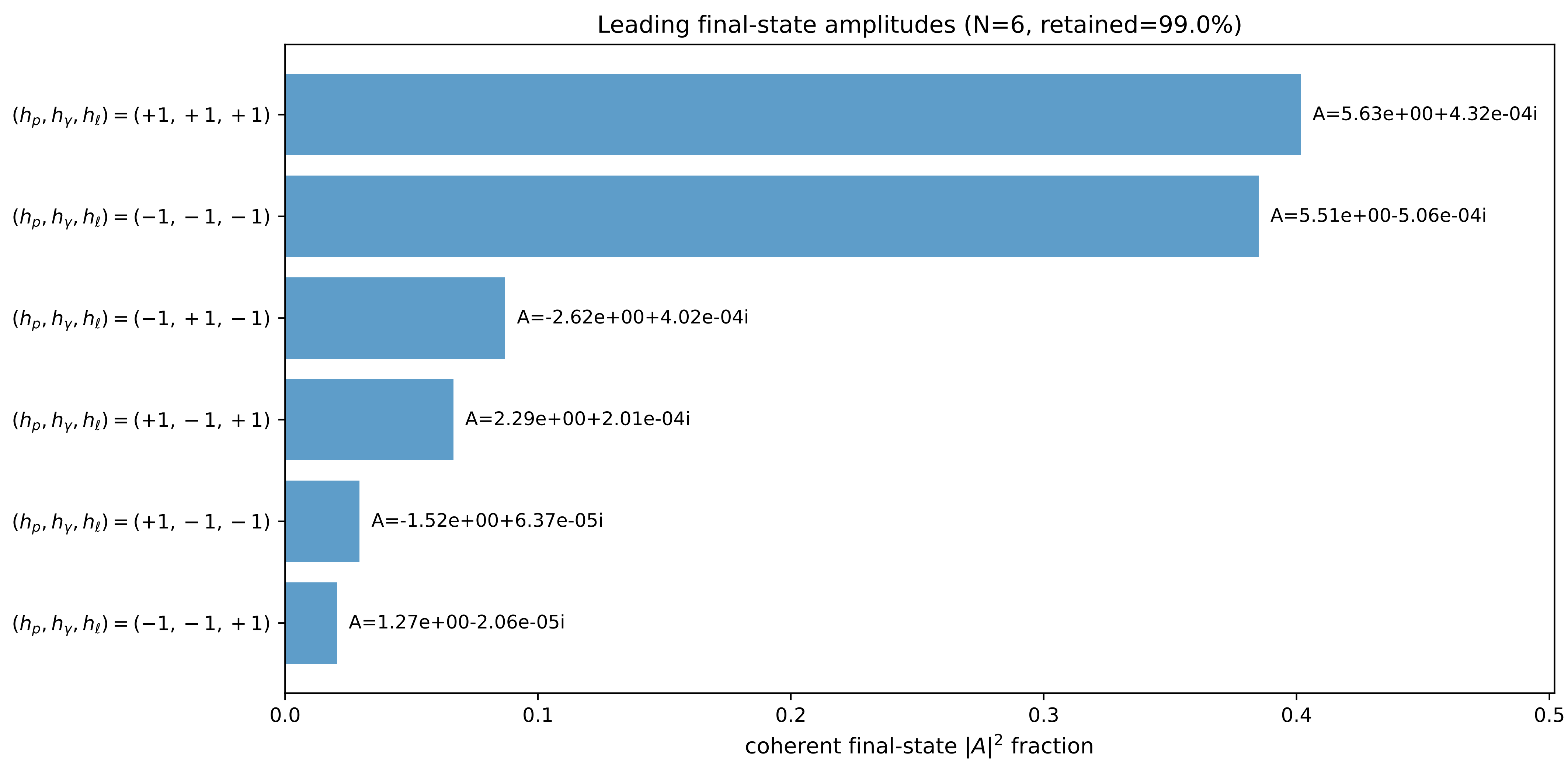


dghz_mixing_angles_polarization_04_local_minimum_213 d_{\rm GHZ}=3.38546e-05
region: polarization_cluster_04_local_minimum_213
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.7484301 rad
incoming lepton: -0.176701|+> +0.984265|->
proton mixing angle: alpha_p=3.0008821 rad
incoming proton: -0.990117|+> +0.140247|->

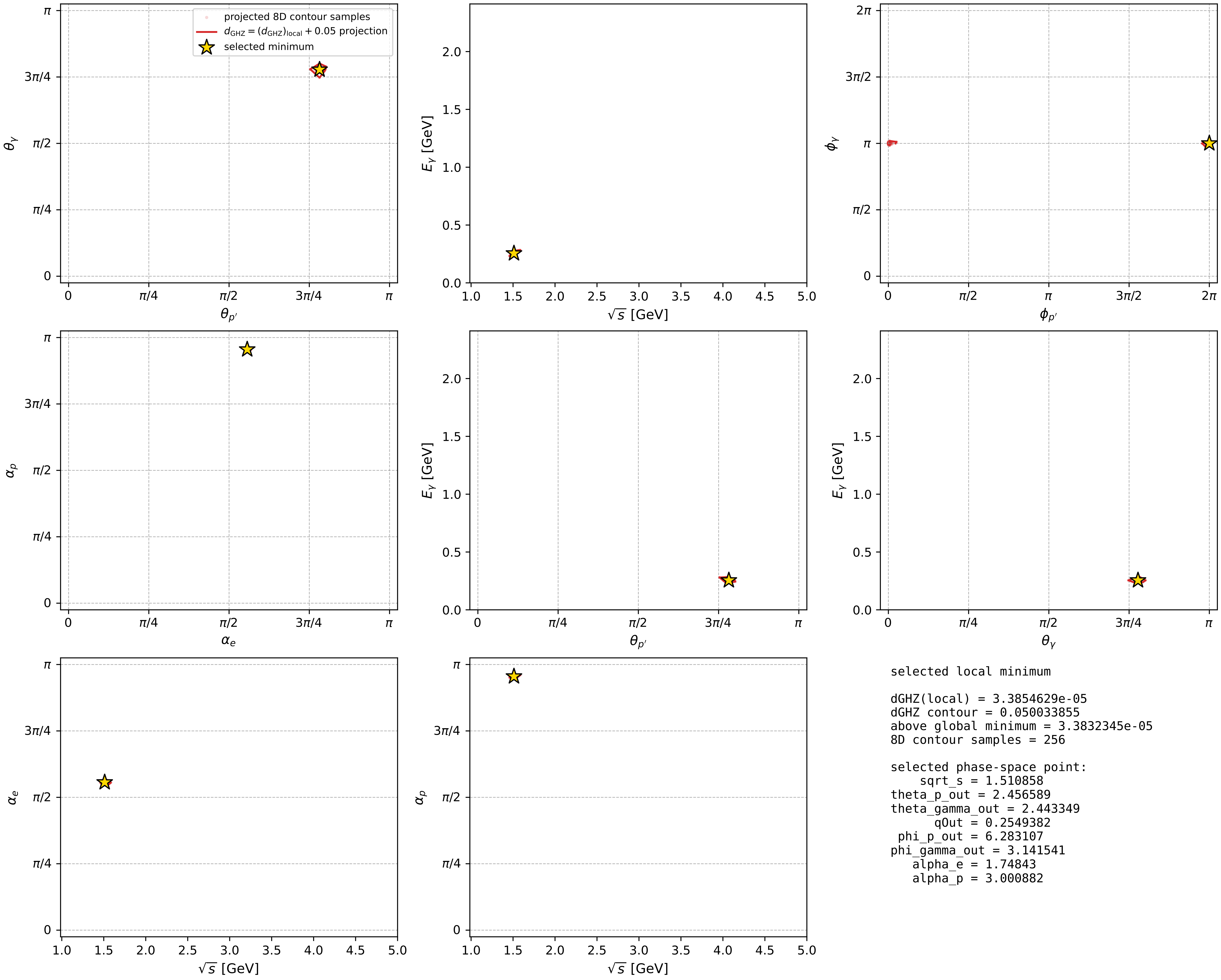
s=2.28269, $\sqrt{s}$=1.51086, $|\vec{P}|$=0.464255, $|\vec{P}'|$=0.132706
$\theta_{p'}$=2.45659, θ_γ=2.44335, $\phi_{p'}$=6.28311, ϕ_γ=3.14154
E_γ=0.254938, $\theta_{\ell'}$=0.262024, $\phi_{\ell'}$=6.28316
Q^2=0.563257, x_B=0.544911, t=-0.318714, W^2=1.35025, y=0.736835

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4643 p_x= 0.0000 p_y= 0.0000 p_z= -0.4643
P E= 1.0466 p_x= 0.0000 p_y= 0.0000 p_z= 0.4643
ℓ' E= 0.3086 p_x= 0.0799 p_y= -0.0000 p_z= 0.2980
P' E= 0.9473 p_x= 0.0840 p_y= -0.0000 p_z= -0.1028
q_γ E= 0.2549 p_x= -0.1639 p_y= 0.0000 p_z= -0.1953

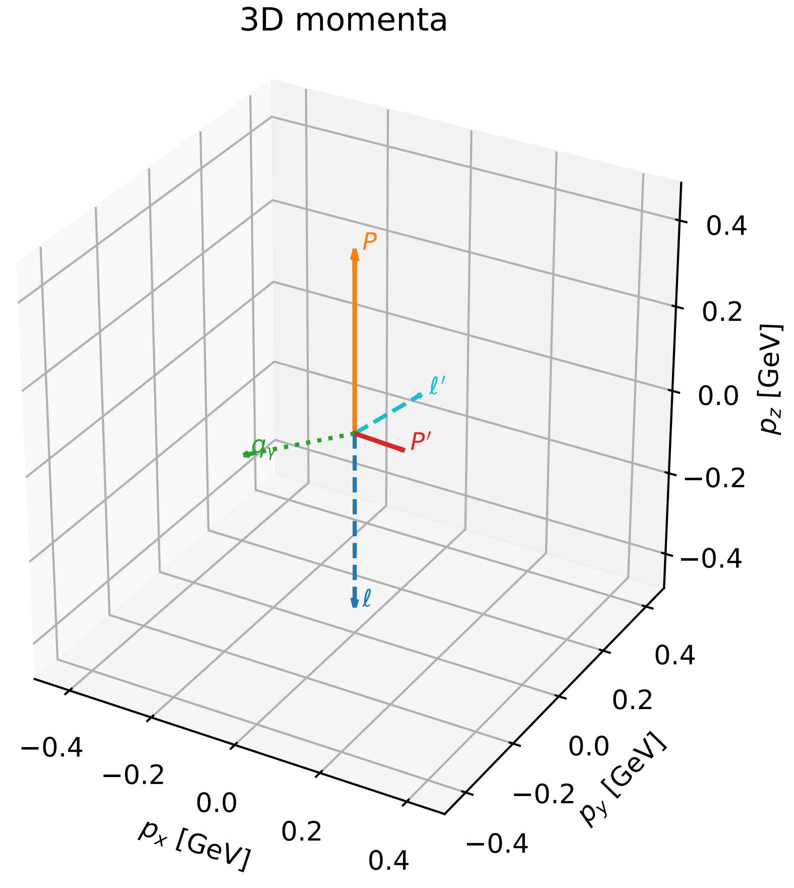




electron: polarization cluster P4, local minimum 213; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



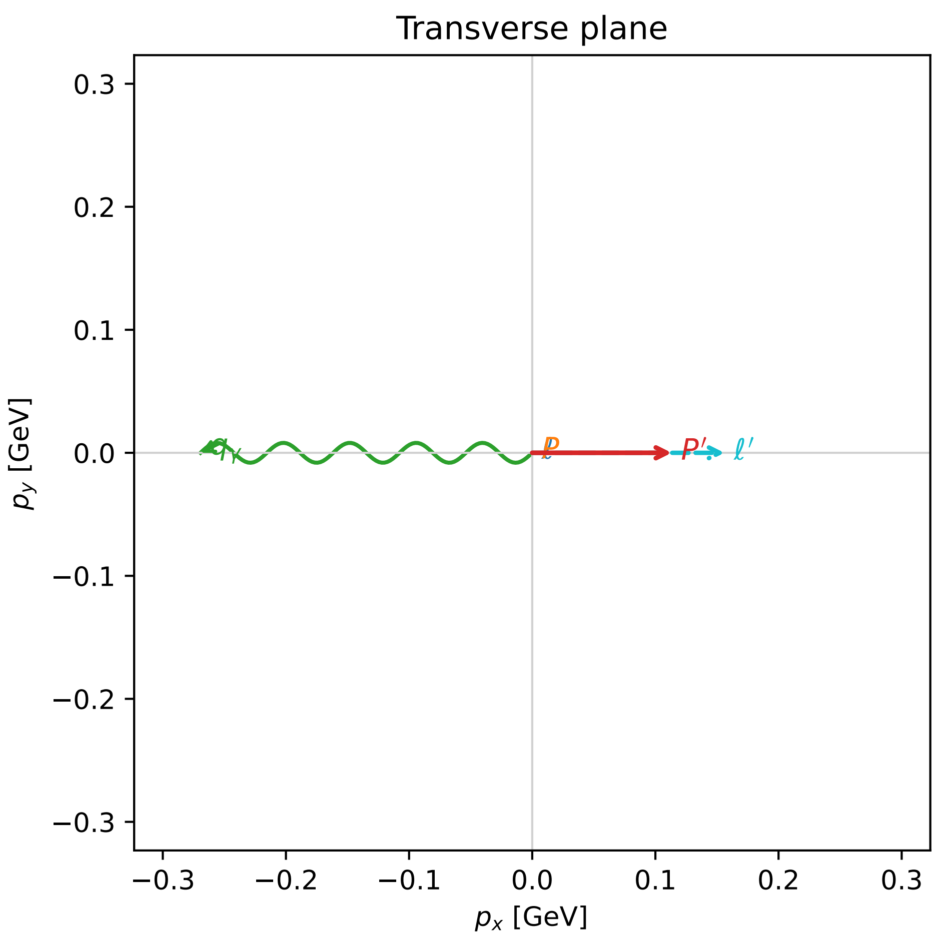
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

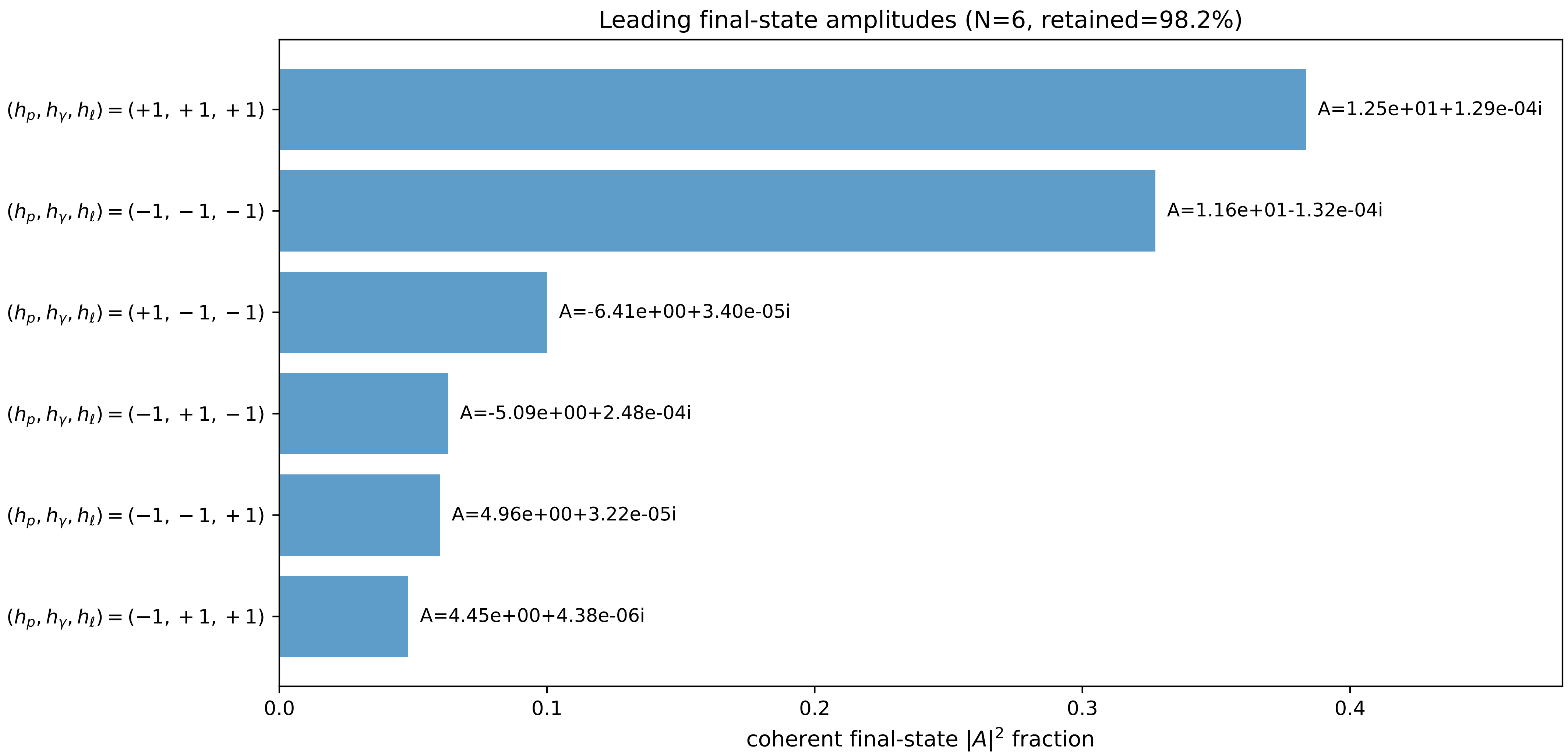


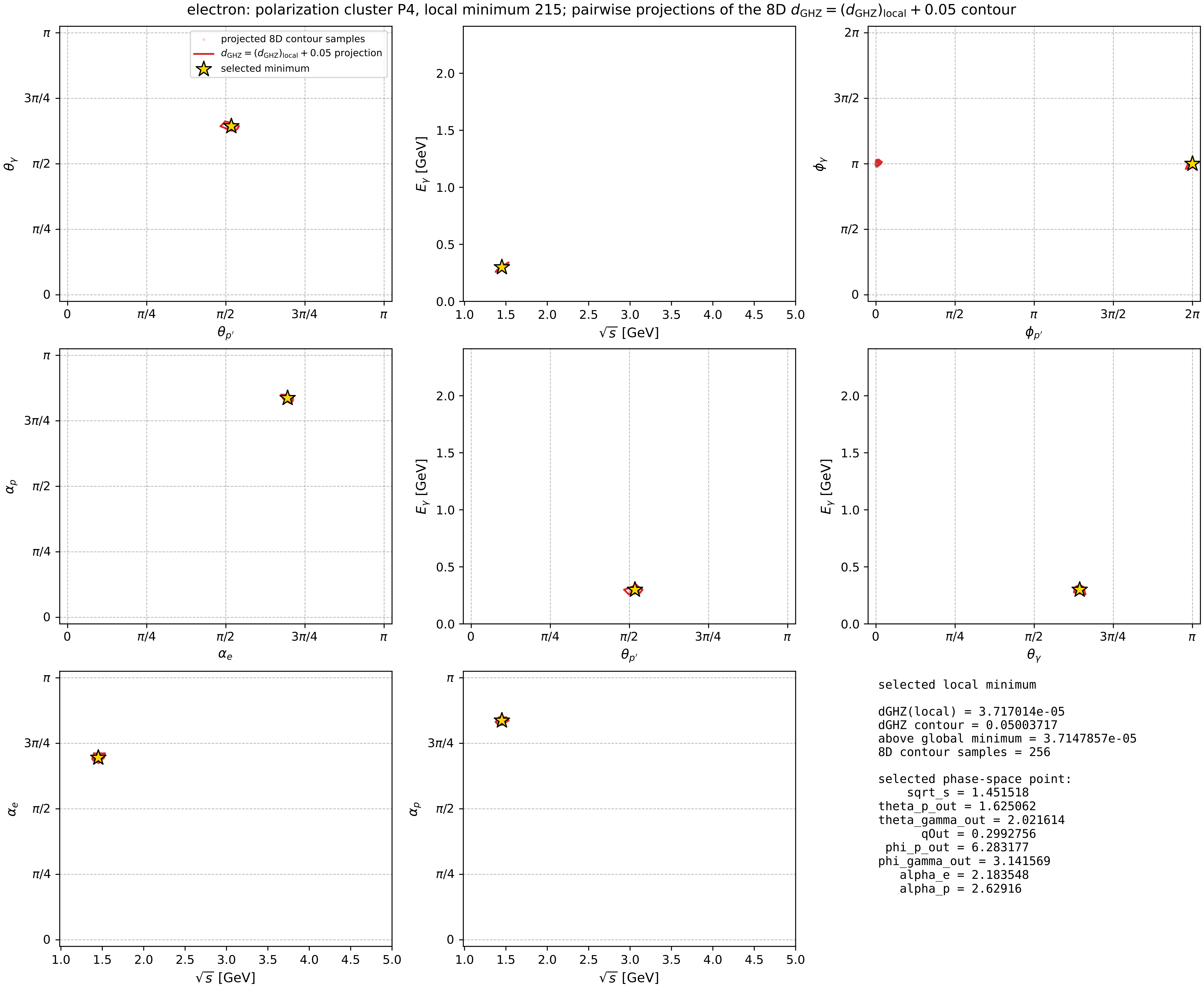
dghz_mixing_angles_polarization_04_local_minimum_215 d_{\rm GHZ}=3.71701e-05
region: polarization_cluster_04_local_minimum_215
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.1835482 rad
incoming lepton: -0.575121|+> +0.818068|->
proton mixing angle: alpha_p=2.6291605 rad
incoming proton: -0.871555|+> +0.490298|->

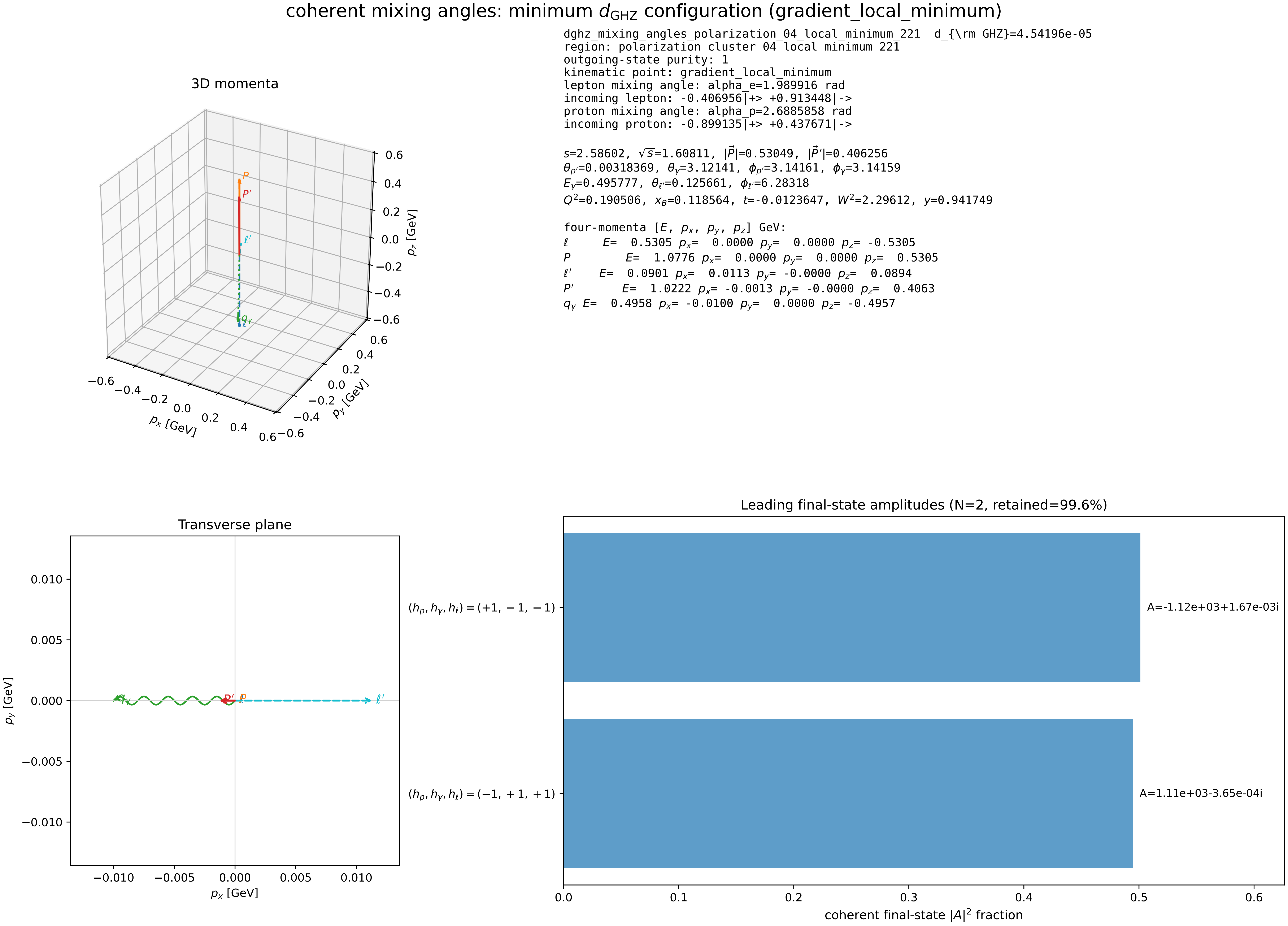
s=2.1069, $\sqrt{s}$=1.45152, $|\vec{P}|$=0.422682, $|\vec{P}'|$=0.113422
$\theta_{p'}$=1.62506, θ_γ=2.02161, $\phi_{p'}$=6.28318, ϕ_γ=3.14157
E_γ=0.299276, $\theta_{\ell'}$=0.852178, $\phi_{\ell'}$=6.28315
Q^2=0.290768, x_B=0.317533, t=-0.189668, W^2=1.50479, y=0.746264

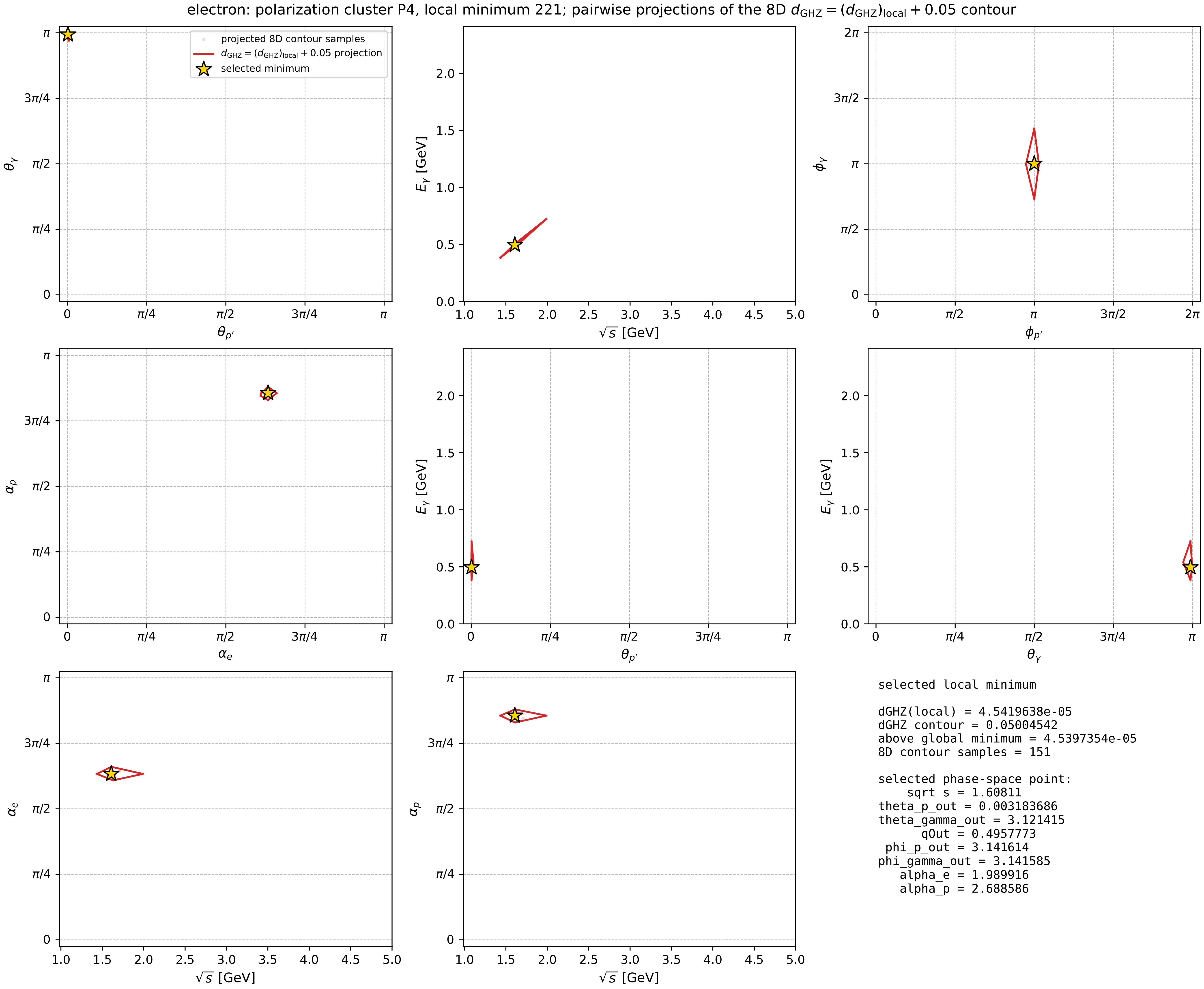
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4227 p_x= 0.0000 p_y= 0.0000 p_z= -0.4227
P E= 1.0288 p_x= 0.0000 p_y= 0.0000 p_z= 0.4227
ℓ' E= 0.2074 p_x= 0.1561 p_y= -0.0000 p_z= 0.1365
P' E= 0.9448 p_x= 0.1133 p_y= -0.0000 p_z= -0.0062
q_γ E= 0.2993 p_x= -0.2694 p_y= 0.0000 p_z= -0.1304





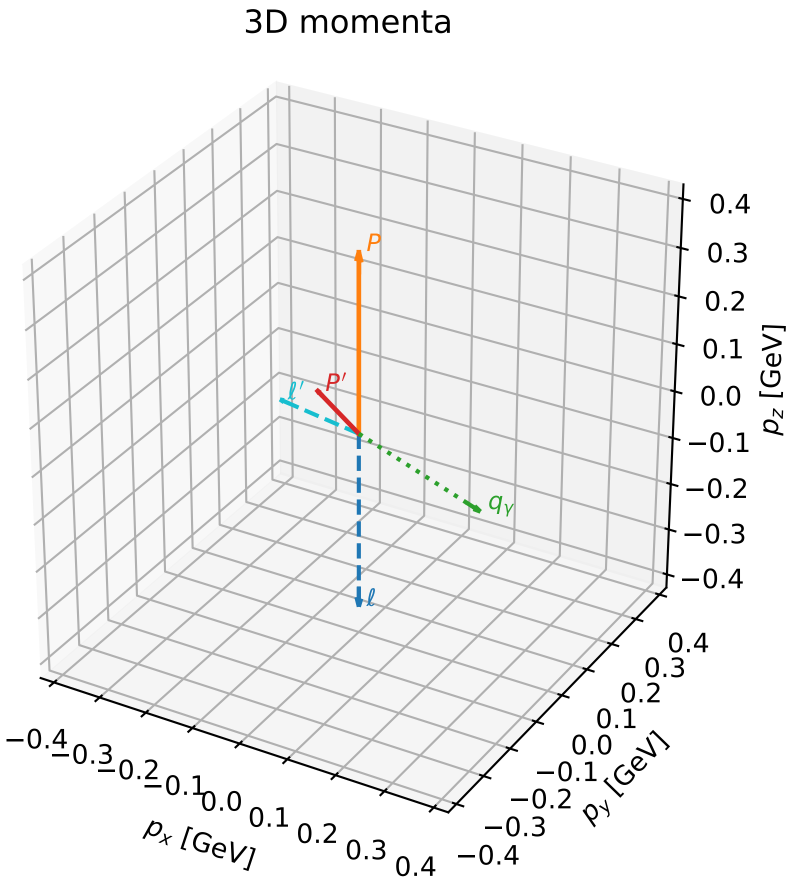

electron: polarization cluster P4, local minimum 215; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 3.717014e-05
dGHZ contour = 0.05003717
above global minimum = 3.7147857e-05
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.451518
theta_p_out = 1.625062
theta_gamma_out = 2.021614
qOut = 0.2992756
phi_p_out = 6.283177
phi_gamma_out = 3.141569
alpha_e = 2.183548
alpha_p = 2.62916


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
P
P'
ℓ'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_221 d_{\rm GHZ}=4.54196e-05
region: polarization_cluster_04_local_minimum_221
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.989916 rad
incoming lepton: -0.406956|+> +0.913448|->
proton mixing angle: alpha_p=2.6885858 rad
incoming proton: -0.899135|+> +0.437671|->
s=2.58602, $\sqrt{s}$=1.60811, $|\vec{P}|$=0.53049, $|\vec{P}'|$=0.406256
$\theta_{p'}$=0.00318369, θ_γ=3.12141, $\phi_{p'}$=3.14161, ϕ_γ=3.14159
E_γ=0.495777, $\theta_{\ell'}$=0.125661, $\phi_{\ell'}$=6.28318
Q^2=0.190506, x_B=0.118564, t=-0.0123647, W^2=2.29612, y=0.941749
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5305 p_x= 0.0000 p_y= 0.0000 p_z= -0.5305
P E= 1.0776 p_x= 0.0000 p_y= 0.0000 p_z= 0.5305
ℓ' E= 0.0901 p_x= 0.0113 p_y= -0.0000 p_z= 0.0894
P' E= 1.0222 p_x= -0.0013 p_y= -0.0000 p_z= 0.4063
q_γ E= 0.4958 p_x= -0.0100 p_y= 0.0000 p_z= -0.4957
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=99.6%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-1.12e+03+1.67e-03i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=1.11e+03-3.65e-04i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P4, local minimum 221; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 4.5419638e-05
dGHZ contour = 0.05004542
above global minimum = 4.5397354e-05
8D contour samples = 151
selected phase-space point:
sqrt_s = 1.60811
theta_p_out = 0.003183686
theta_gamma_out = 3.121415
qOut = 0.4957773
phi_p_out = 3.141614
phi_gamma_out = 3.141585
alpha_e = 1.989916
alpha_p = 2.688586

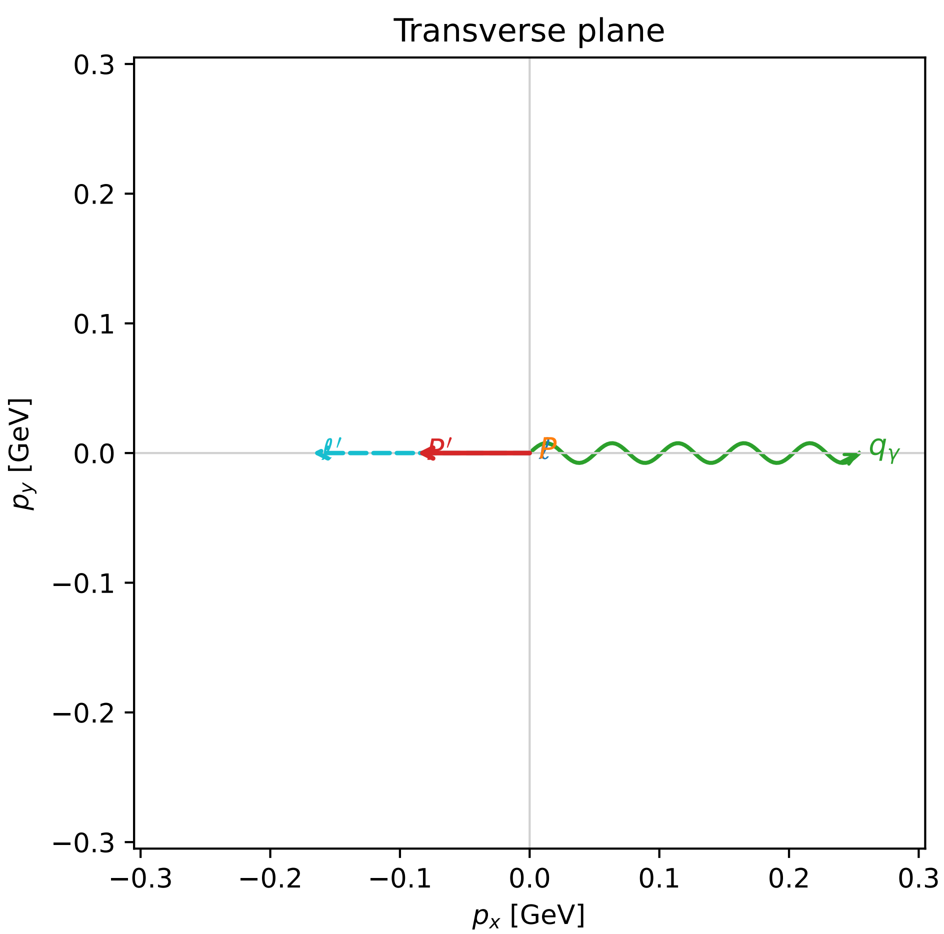
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

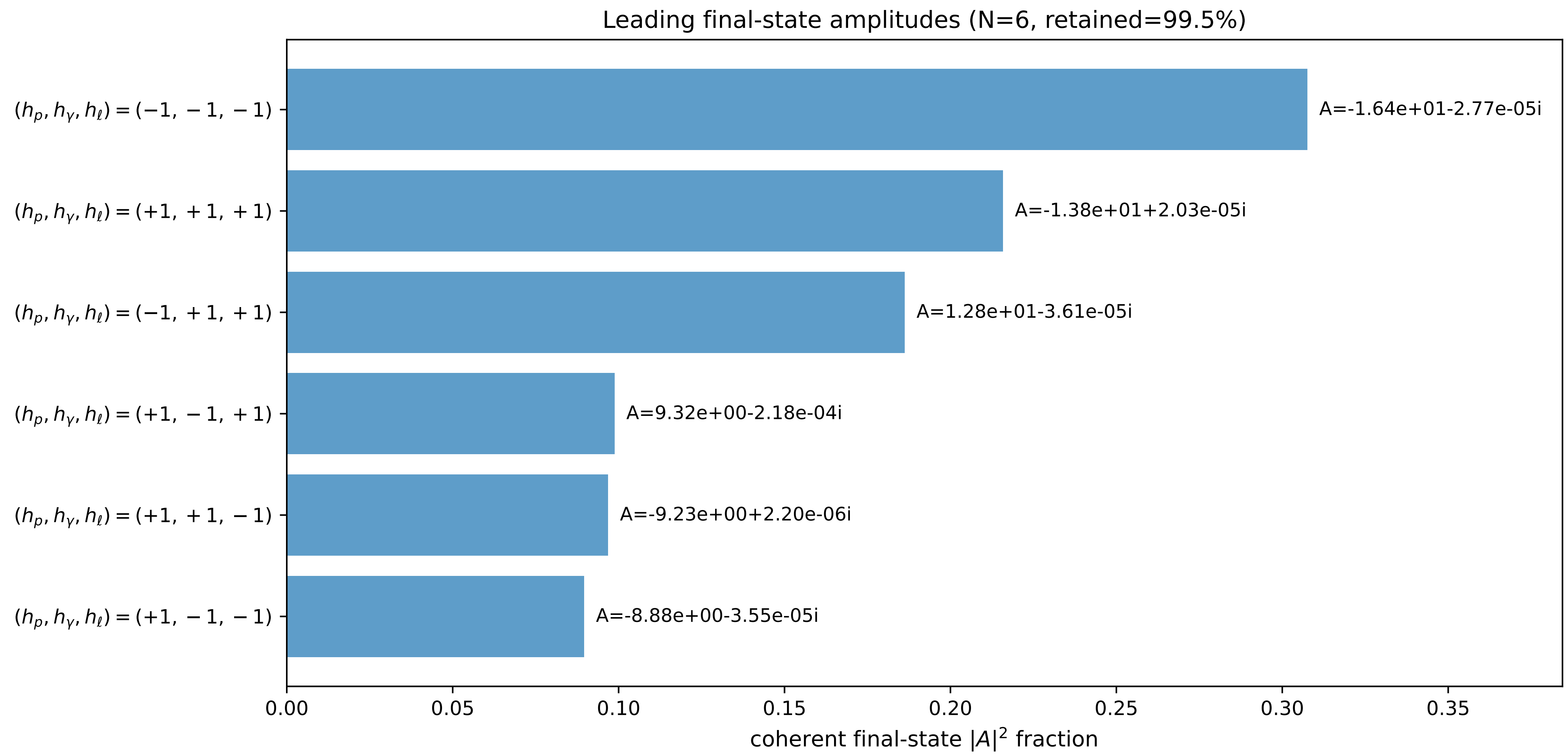


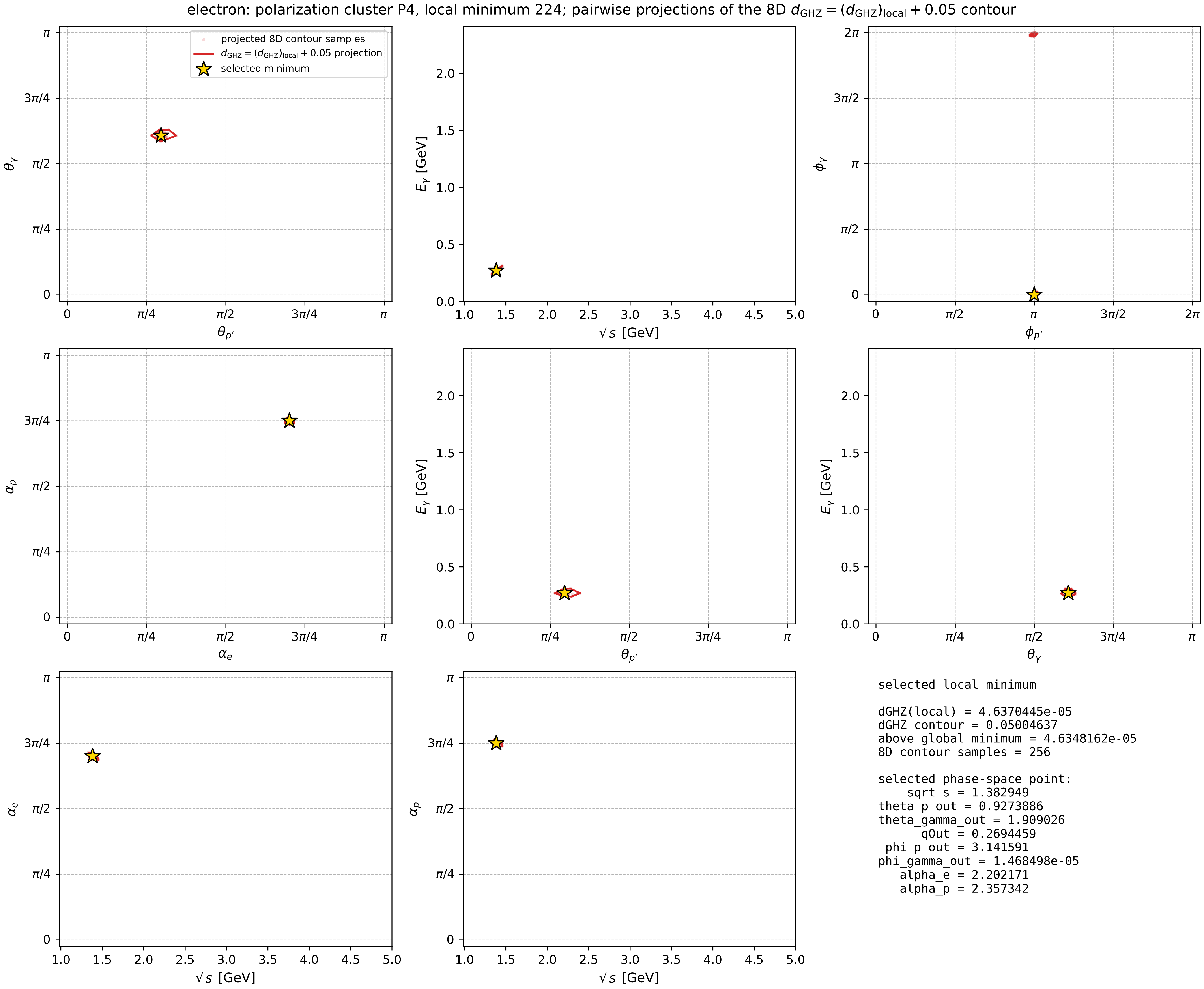
dghz_mixing_angles_polarization_04_local_minimum_224 d_{\rm GHZ}=4.63704e-05
region: polarization_cluster_04_local_minimum_224
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.202171 rad
incoming lepton: -0.590255|+> +0.807217|->
proton mixing angle: alpha_p=2.3573418 rad
incoming proton: -0.707918|+> +0.706295|->

s=1.91255, $\sqrt{s}$=1.38295, $|\vec{P}|$=0.37337, $|\vec{P}'|$=0.108346
$\theta_{p'}$=0.927389, θ_γ=1.90903, $\phi_{p'}$=3.14159, ϕ_γ=1.4685e-05
E_γ=0.269446, $\theta_{\ell'}$=1.4261, $\phi_{\ell'}$=3.14162
Q^2=0.144624, x_B=0.203939, t=-0.0983369, W^2=1.44437, y=0.686696

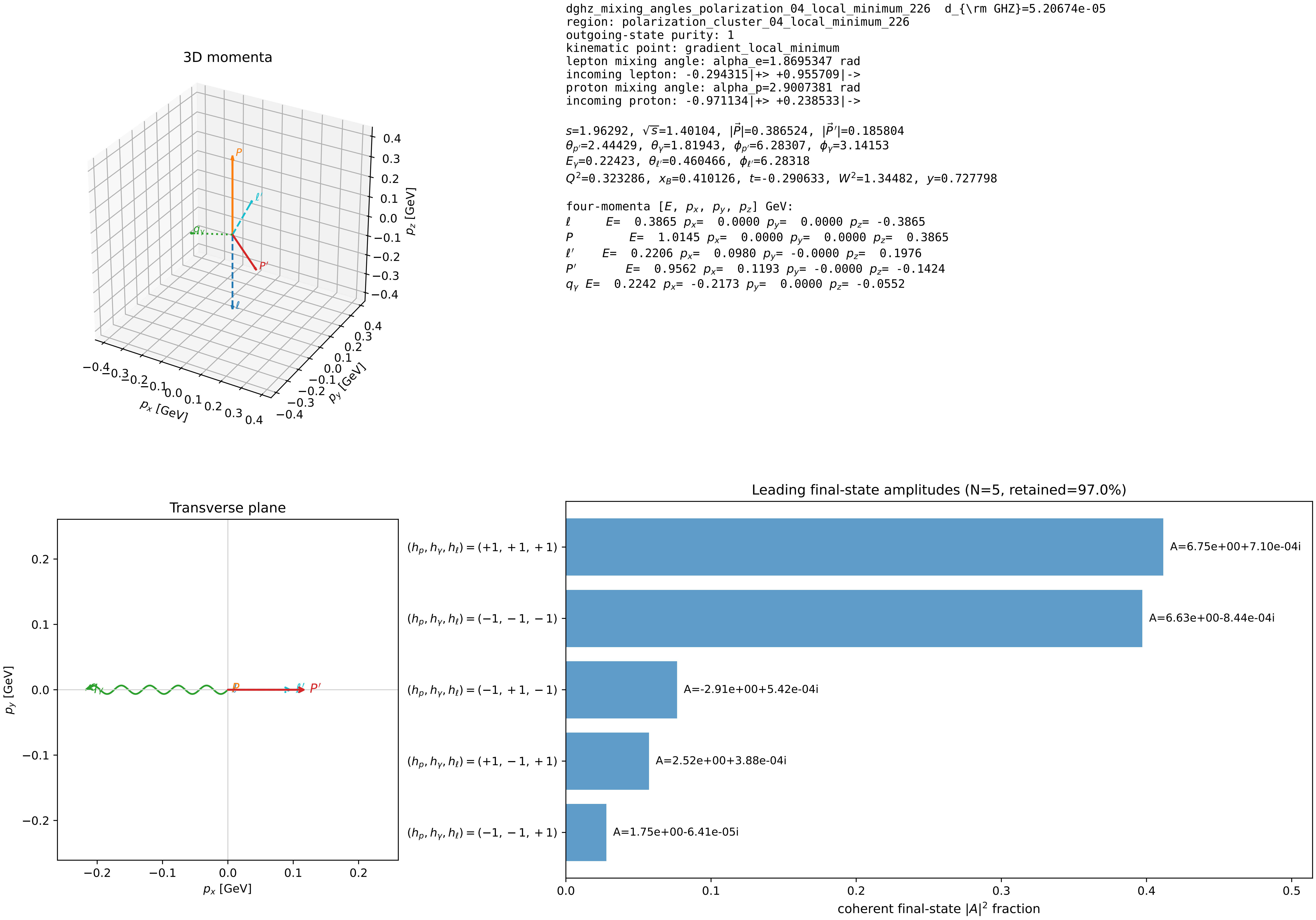
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3734 p_x= 0.0000 p_y= 0.0000 p_z= -0.3734
P E= 1.0096 p_x= 0.0000 p_y= 0.0000 p_z= 0.3734
ℓ' E= 0.1693 p_x= -0.1675 p_y= -0.0000 p_z= 0.0244
P' E= 0.9442 p_x= -0.0867 p_y= 0.0000 p_z= 0.0650
q_γ E= 0.2694 p_x= 0.2542 p_y= 0.0000 p_z= -0.0894



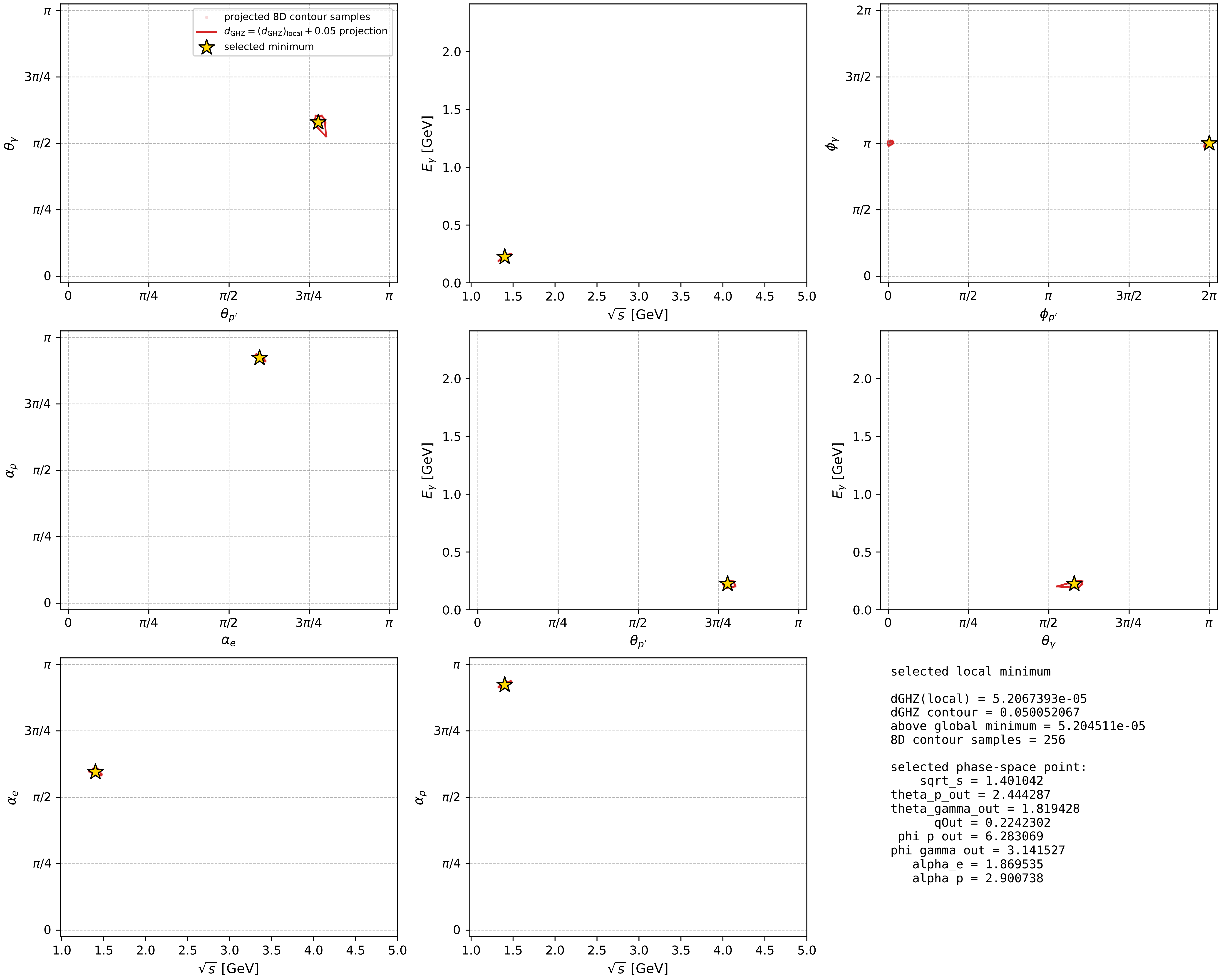



electron: polarization cluster P4, local minimum 224; pairwise projections of the 8D $d_{GHZ} = (d_{GHZ})_{local} + 0.05$ contour
projected 8D contour samples
$d_{GHZ} = (d_{GHZ})_{local} + 0.05$ projection
selected minimum
selected local minimum
dGHZ(local) = 4.6370445e-05
dGHZ contour = 0.05004637
above global minimum = 4.6348162e-05
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.382949
theta_p_out = 0.9273886
theta_gamma_out = 1.909026
qOut = 0.2694459
phi_p_out = 3.141591
phi_gamma_out = 1.468498e-05
alpha_e = 2.202171
alpha_p = 2.357342

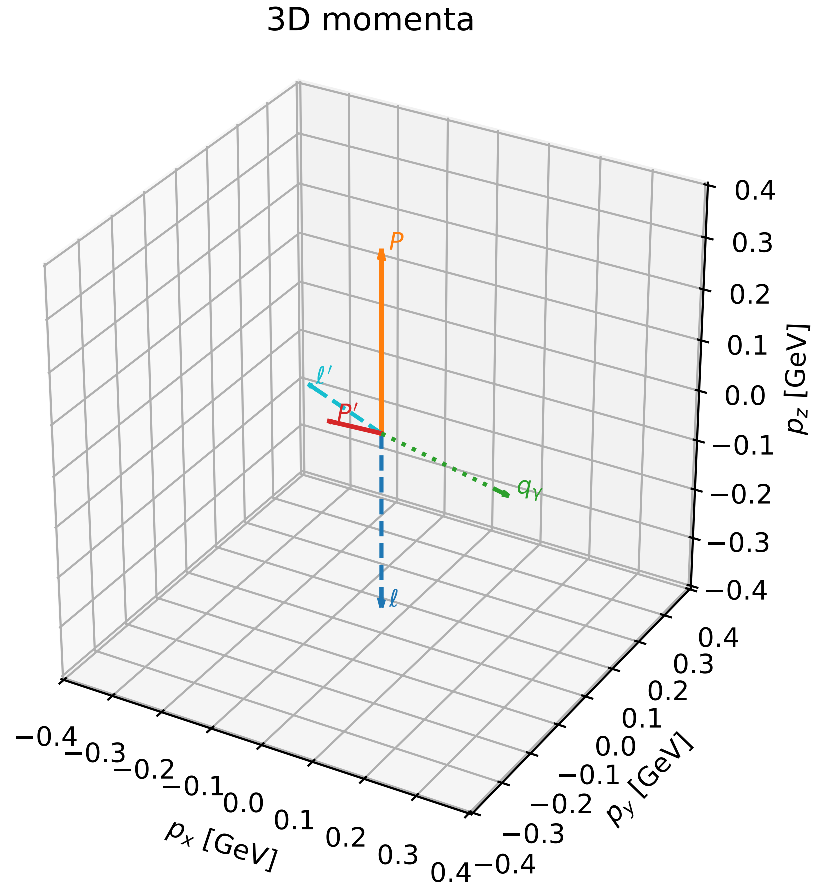
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)



electron: polarization cluster P4, local minimum 226; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



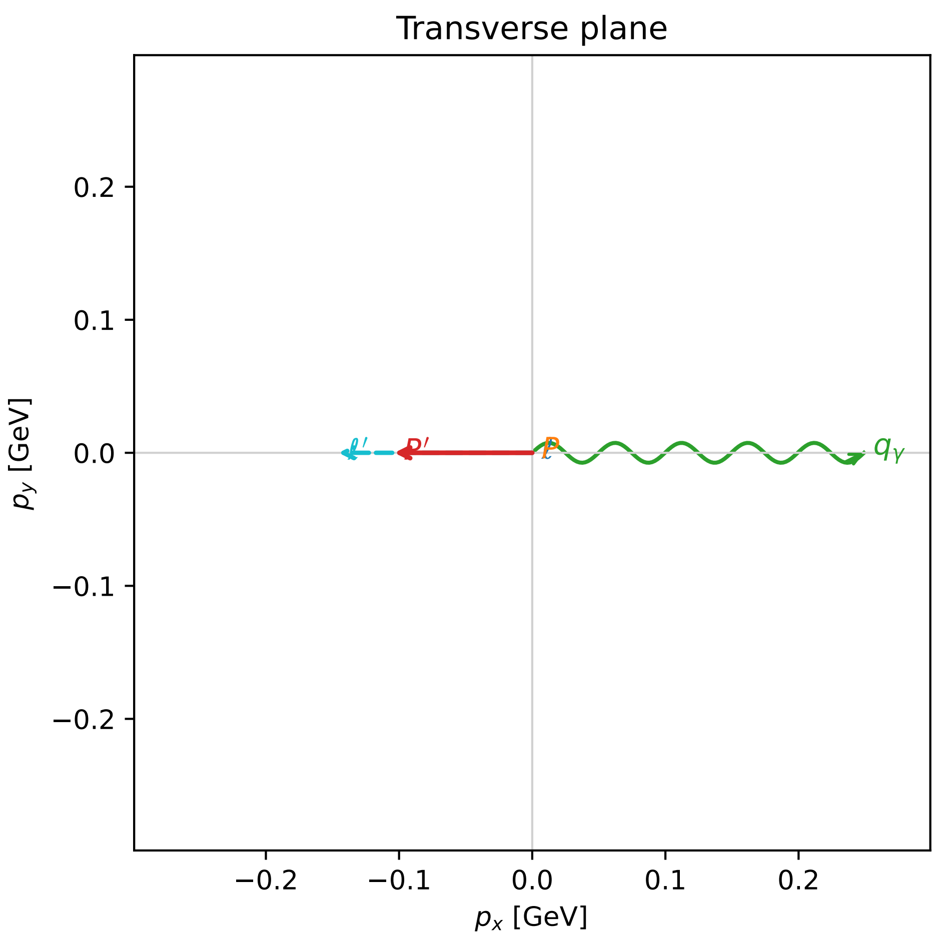
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

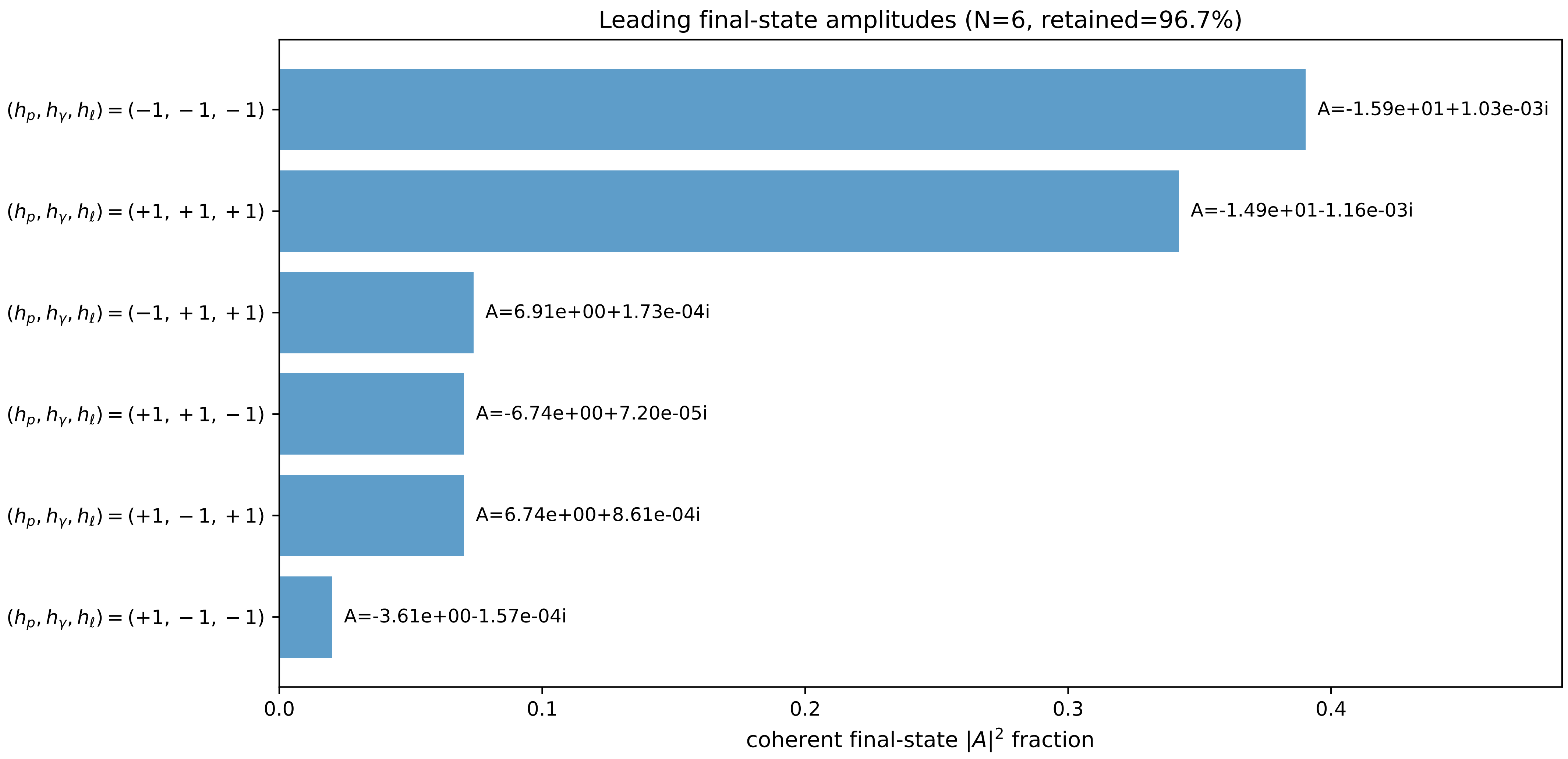


dghz_mixing_angles_polarization_04_local_minimum_227 d_{\rm GHZ}=5.40414e-05
region: polarization_cluster_04_local_minimum_227
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3201738 rad
incoming lepton: -0.681183|+> +0.732113|->
proton mixing angle: alpha_p=2.2653317 rad
incoming proton: -0.640029|+> +0.768351|->

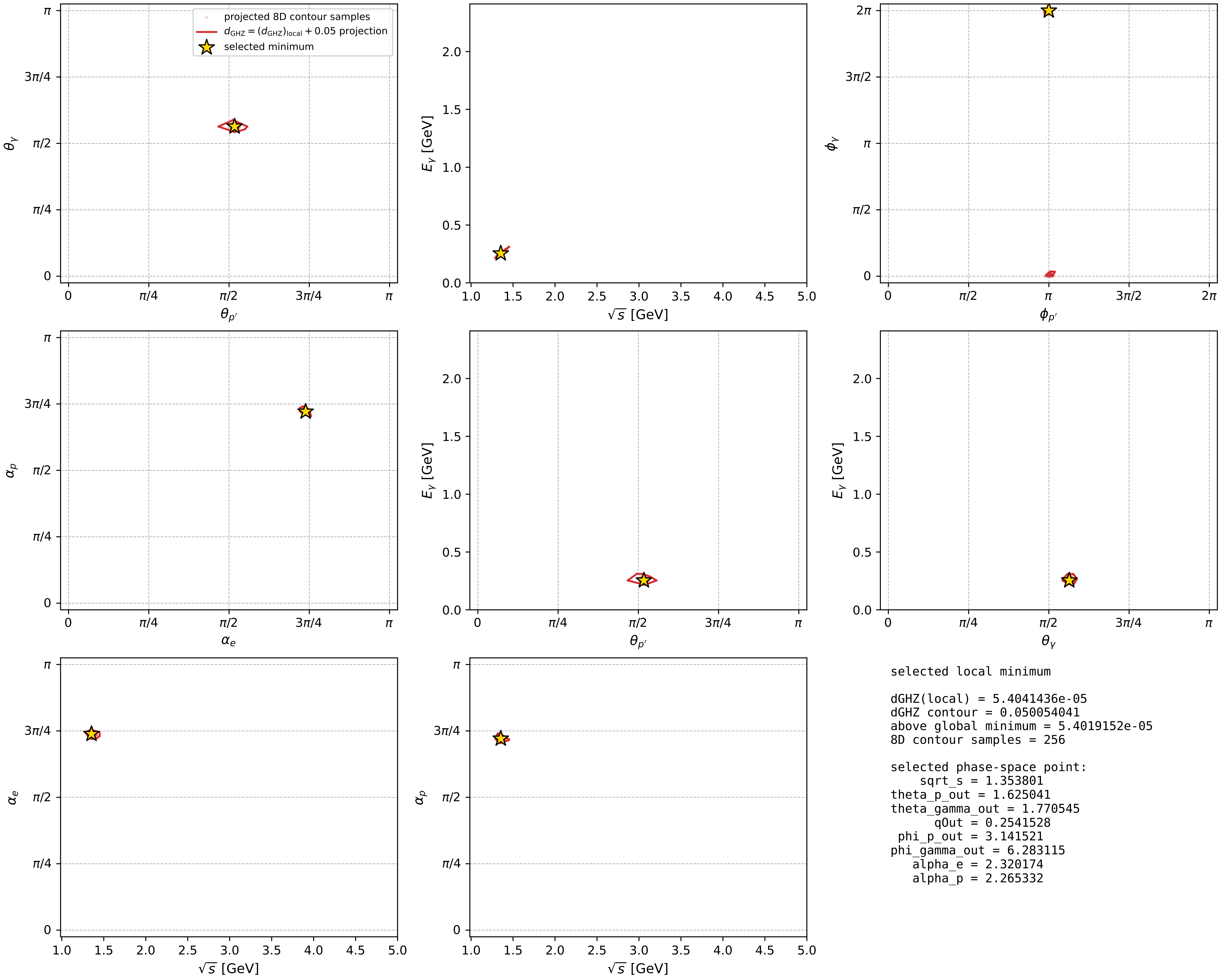
s=1.83278, $\sqrt{s}$=1.3538, $|\vec{P}|$=0.351947, $|\vec{P}'|$=0.103749
$\theta_{p'}$=1.62504, θ_γ=1.77055, $\phi_{p'}$=3.14152, ϕ_γ=6.28312
E_γ=0.254153, $\theta_{\ell'}$=1.20307, $\phi_{\ell'}$=3.14152
Q^2=0.149213, x_B=0.219446, t=-0.135211, W^2=1.41059, y=0.71354

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3519 p_x= 0.0000 p_y= 0.0000 p_z= -0.3519
P E= 1.0019 p_x= 0.0000 p_y= 0.0000 p_z= 0.3519
ℓ' E= 0.1559 p_x= -0.1455 p_y= 0.0000 p_z= 0.0561
P' E= 0.9437 p_x= -0.1036 p_y= 0.0000 p_z= -0.0056
q_γ E= 0.2542 p_x= 0.2491 p_y= -0.0000 p_z= -0.0504

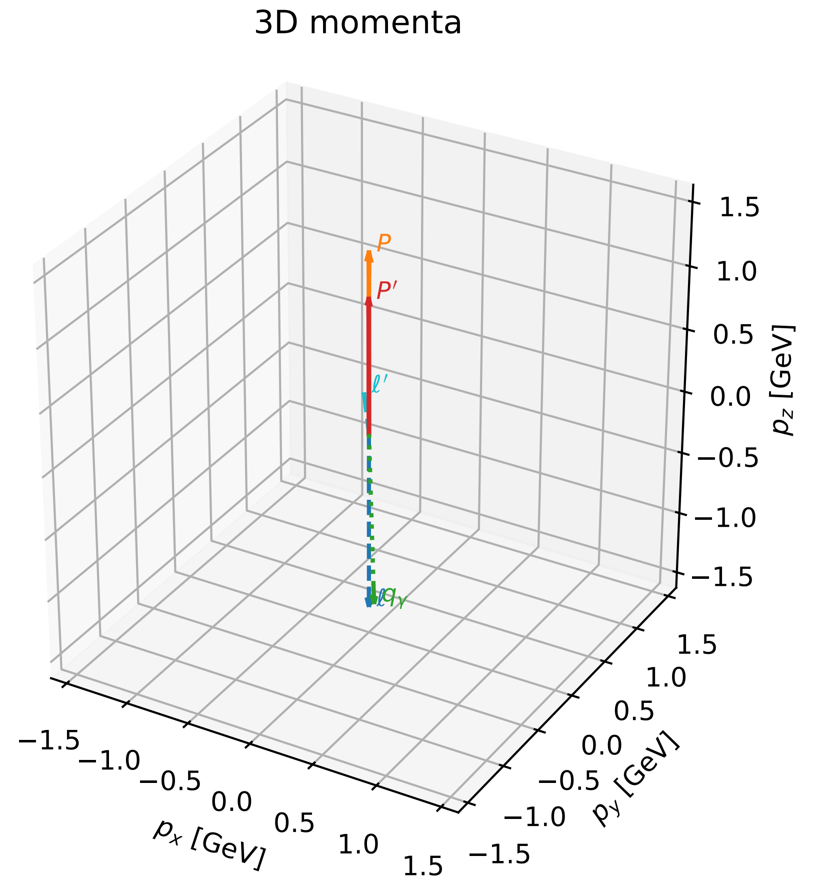




electron: polarization cluster P4, local minimum 227; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



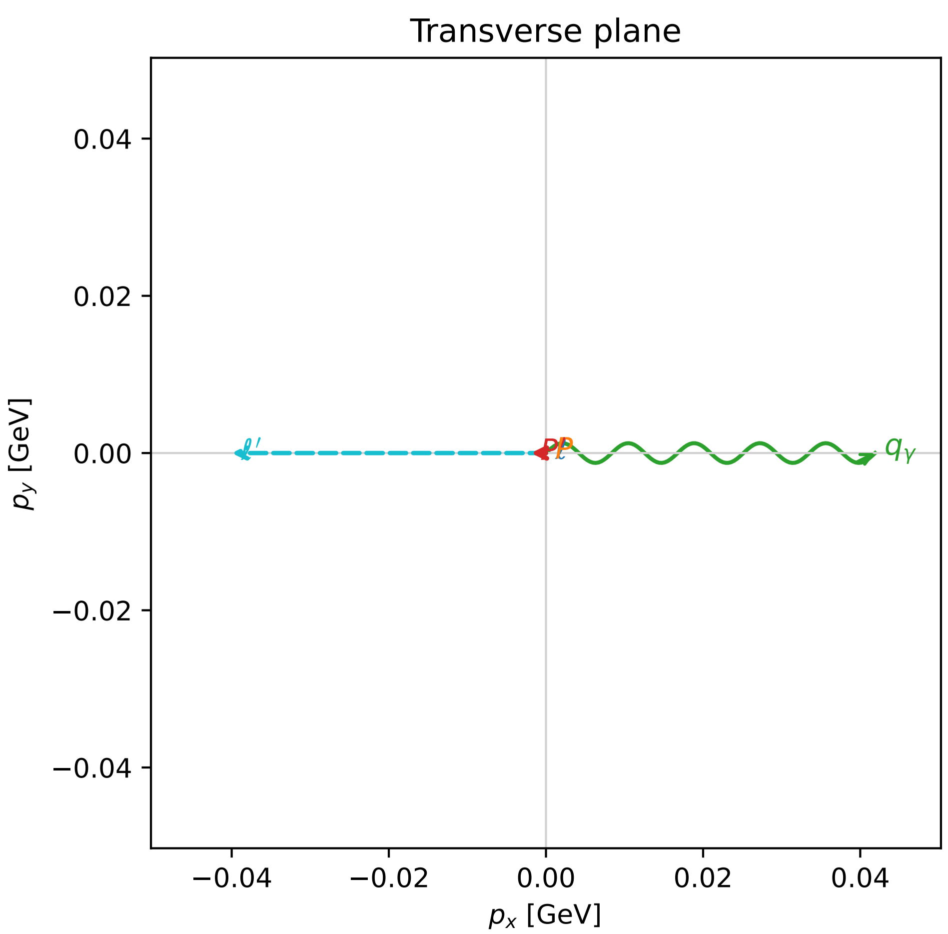
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

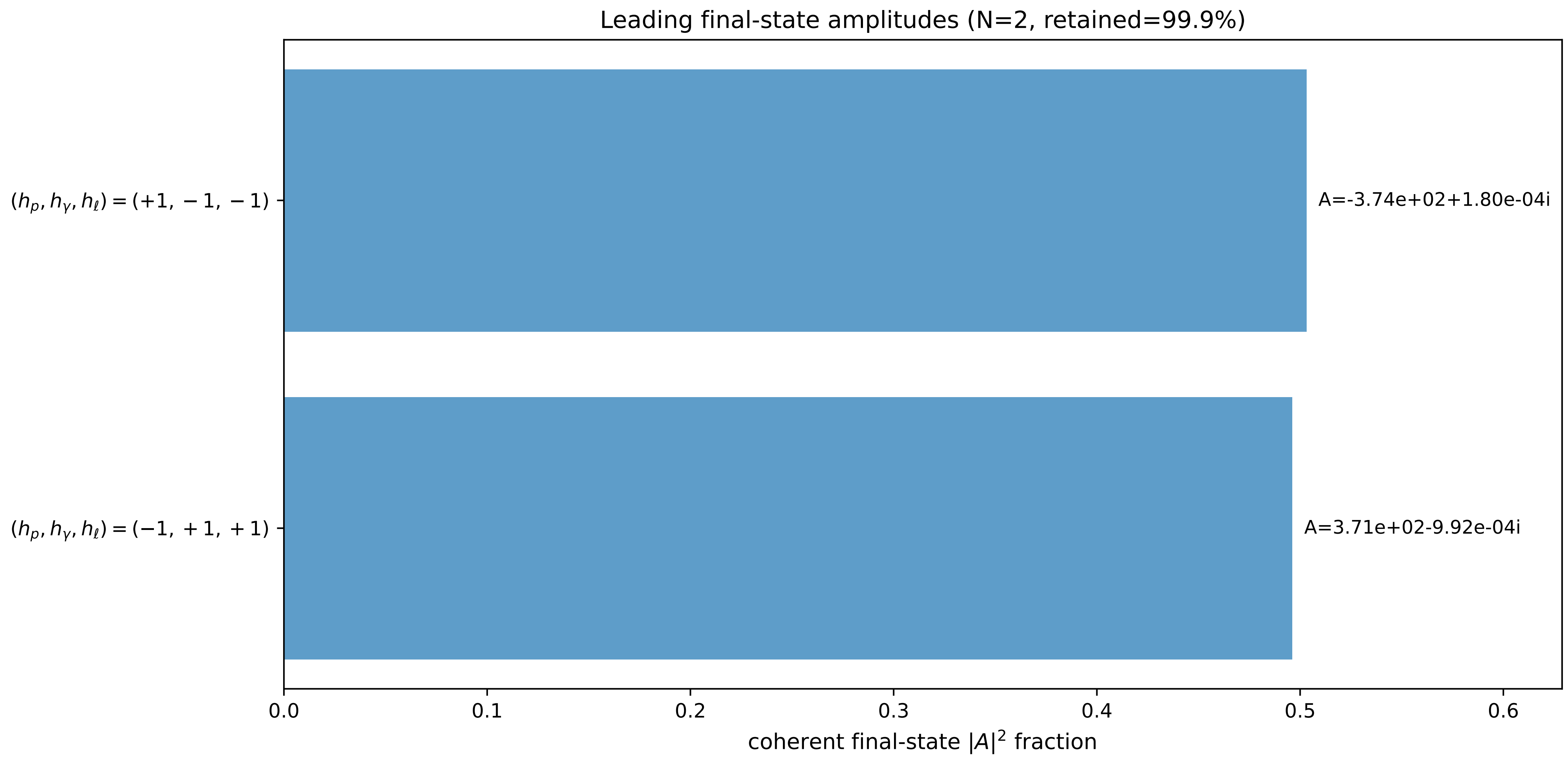


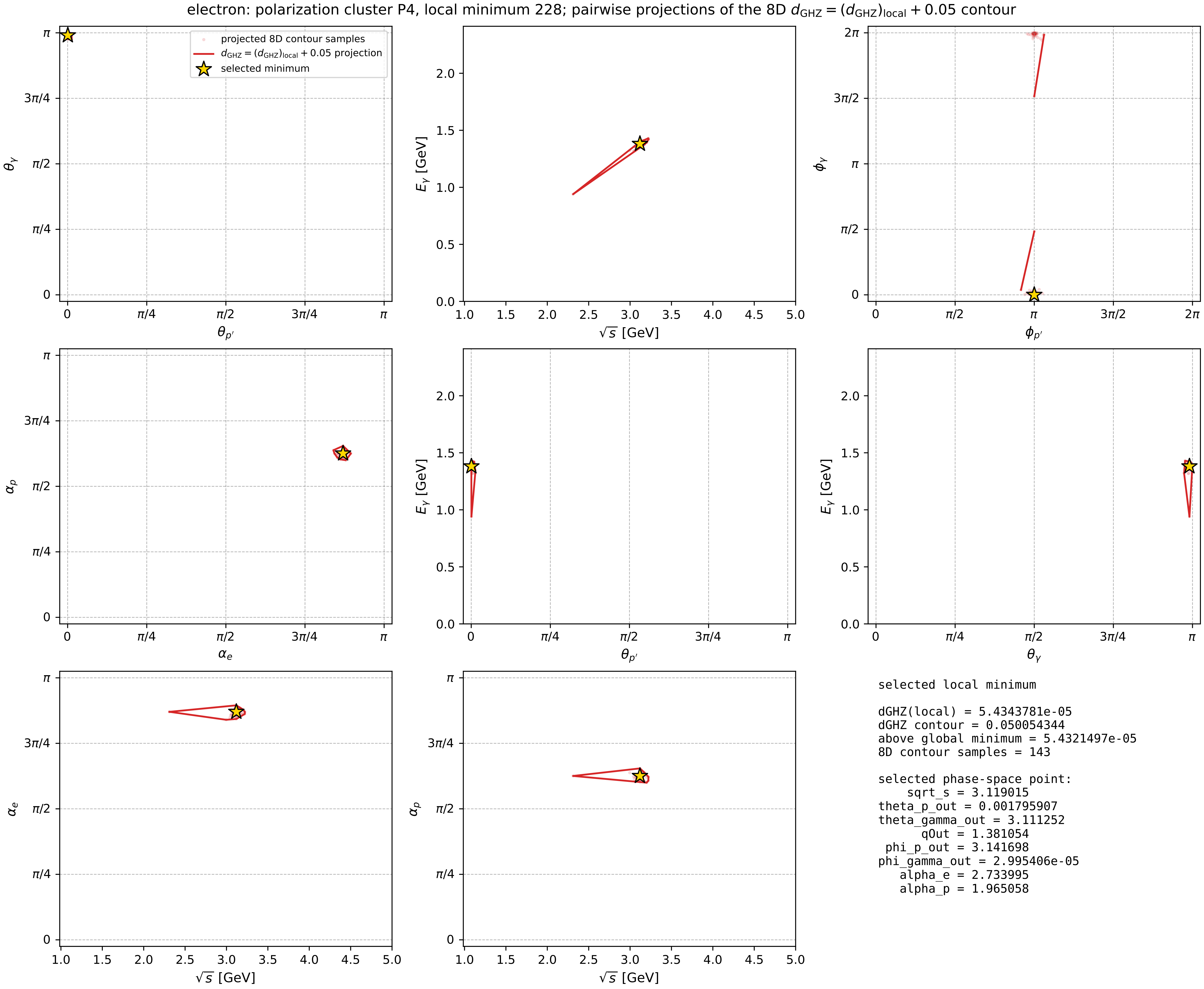
dghz_mixing_angles_polarization_04_local_minimum_228 d_{\rm GHZ}=5.43438e-05
region: polarization_cluster_04_local_minimum_228
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.7339948 rad
incoming lepton: -0.918076|+> +0.396405|->
proton mixing angle: alpha_p=1.9650585 rad
incoming proton: -0.384127|+> +0.92328|->

s=9.72826, $\sqrt{s}$=3.11902, $|\vec{P}|$=1.41846, $|\vec{P}'|$=1.06154
$\theta_{p'}$=0.00179591, θ_γ=3.11125, $\phi_{p'}$=3.1417, ϕ_γ=2.99541e-05
E_γ=1.38105, $\theta_{\ell'}$=0.124754, $\phi_{\ell'}$=3.14162
Q^2=1.81636, x_B=0.209741, t=-0.0467601, W^2=7.7235, y=0.978708

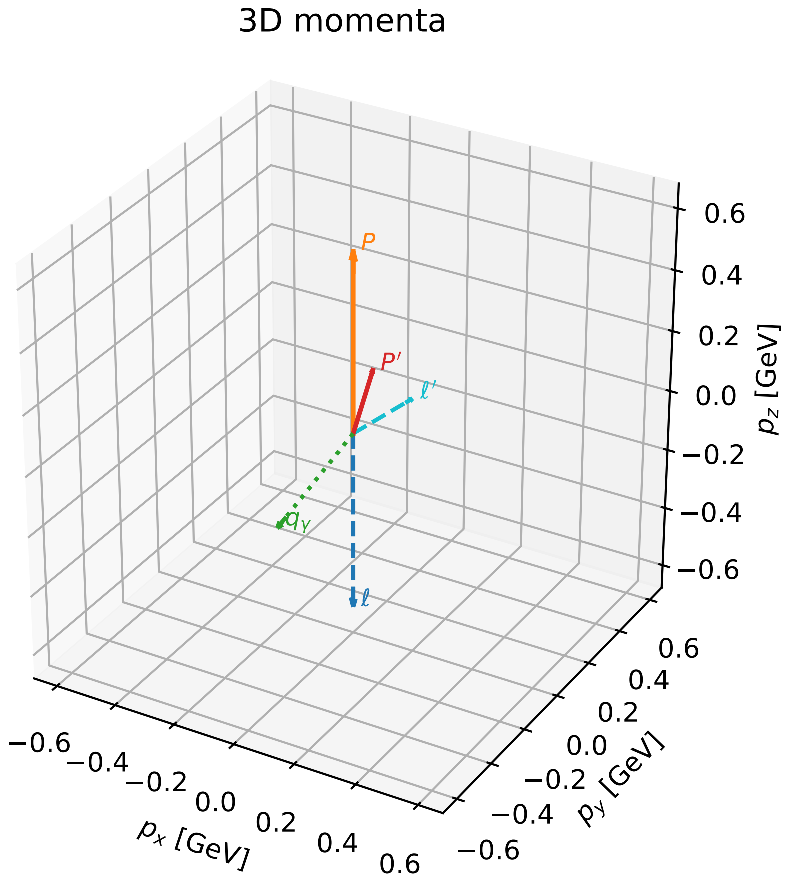
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.4185 p_x= 0.0000 p_y= 0.0000 p_z= -1.4185
P E= 1.7006 p_x= 0.0000 p_y= 0.0000 p_z= 1.4185
ℓ' E= 0.3214 p_x= -0.0400 p_y= -0.0000 p_z= 0.3189
P' E= 1.4166 p_x= -0.0019 p_y= -0.0000 p_z= 1.0615
q_γ E= 1.3811 p_x= 0.0419 p_y= 0.0000 p_z= -1.3804






electron: polarization cluster P4, local minimum 228; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 5.4343781e-05
dGHZ contour = 0.050054344
above global minimum = 5.4321497e-05
8D contour samples = 143
selected phase-space point:
sqrt_s = 3.119015
theta_p_out = 0.001795907
theta_gamma_out = 3.111252
qOut = 1.381054
phi_p_out = 3.141698
phi_gamma_out = 2.995406e-05
alpha_e = 2.733995
alpha_p = 1.965058

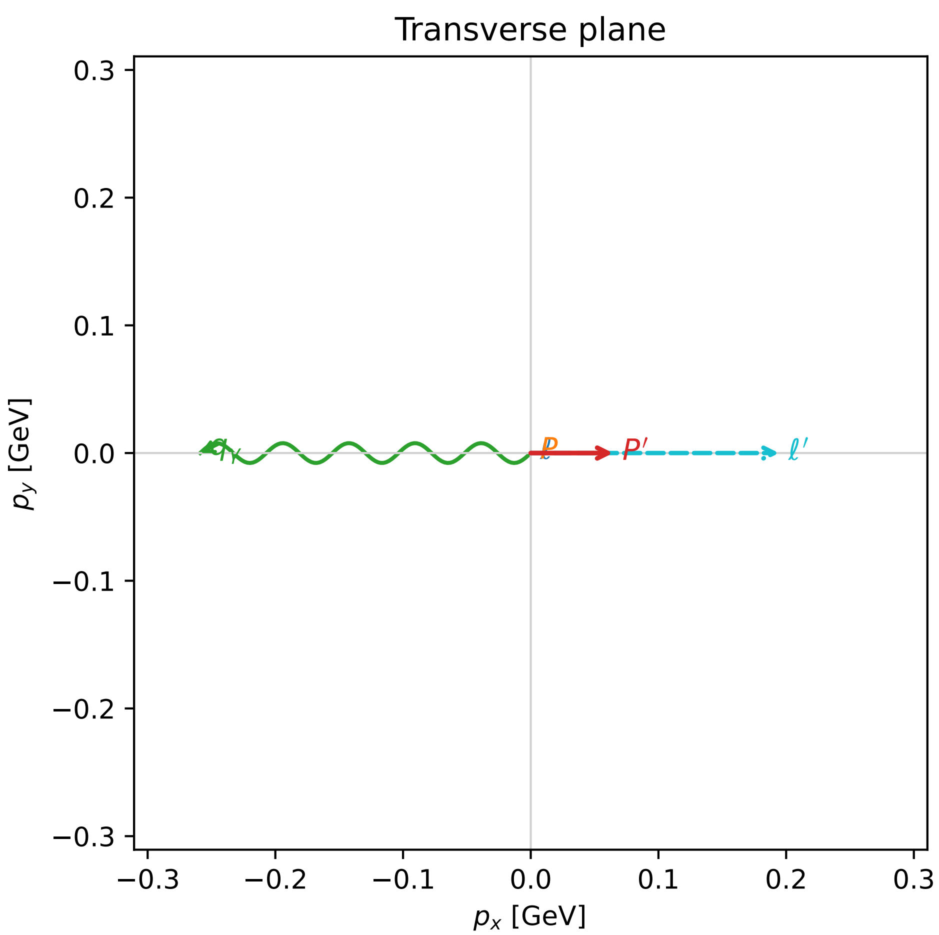
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

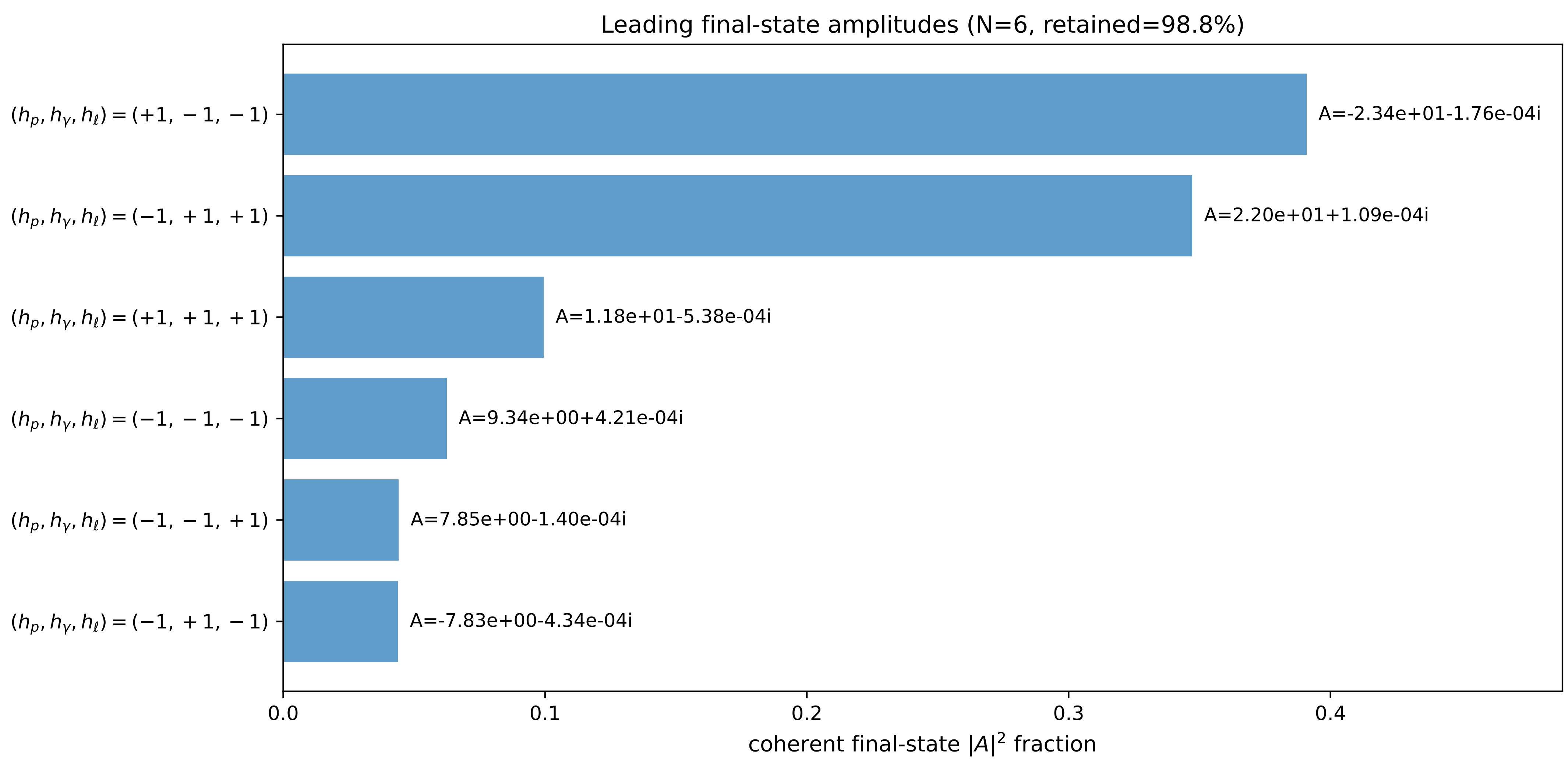


dghz_mixing_angles_polarization_04_local_minimum_230 d_{\rm GHZ}=5.72107e-05
region: polarization_cluster_04_local_minimum_230
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2668516 rad
incoming lepton: -0.641196|+> +0.767377|->
proton mixing angle: alpha_p=2.5530007 rad
incoming proton: -0.831723|+> +0.55519|->

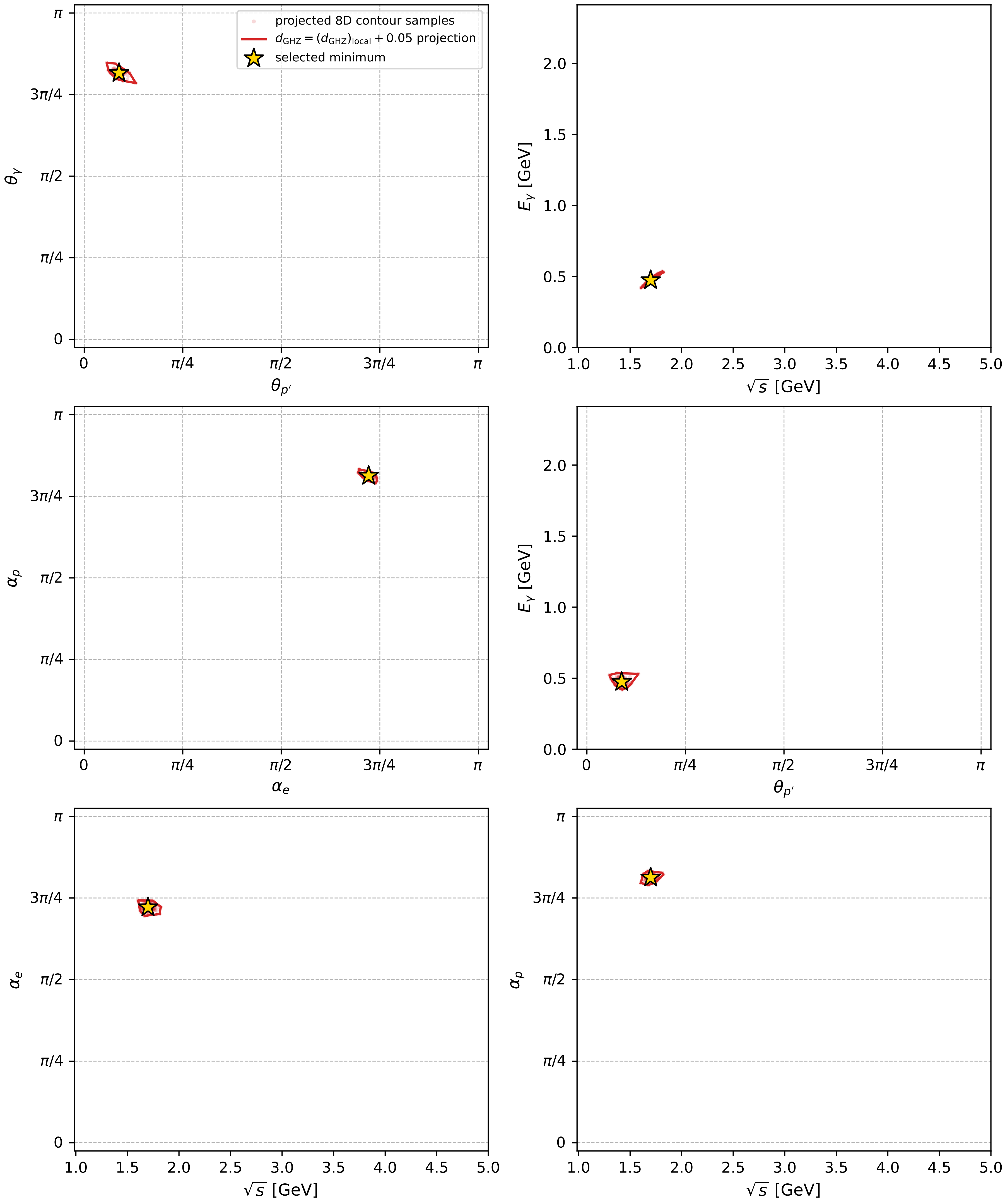
s=2.88835, $\sqrt{s}$=1.69952, $|\vec{P}|$=0.590906, $|\vec{P}'|$=0.235444
$\theta_{p'}$=0.277285, θ_γ=2.56371, $\phi_{p'}$=4.57598e-05, ϕ_γ=3.14163
E_γ=0.473865, $\theta_{\ell'}$=0.85088, $\phi_{\ell'}$=3.51569e-05
Q^2=0.507024, x_B=0.309784, t=-0.116956, W^2=2.00952, y=0.814885

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5909 p_x= 0.0000 p_y= 0.0000 p_z= -0.5909
P E= 1.1086 p_x= 0.0000 p_y= 0.0000 p_z= 0.5909
ℓ' E= 0.2586 p_x= 0.1944 p_y= 0.0000 p_z= 0.1705
P' E= 0.9671 p_x= 0.0645 p_y= 0.0000 p_z= 0.2265
q_γ E= 0.4739 p_x= -0.2588 p_y= -0.0000 p_z= -0.3969

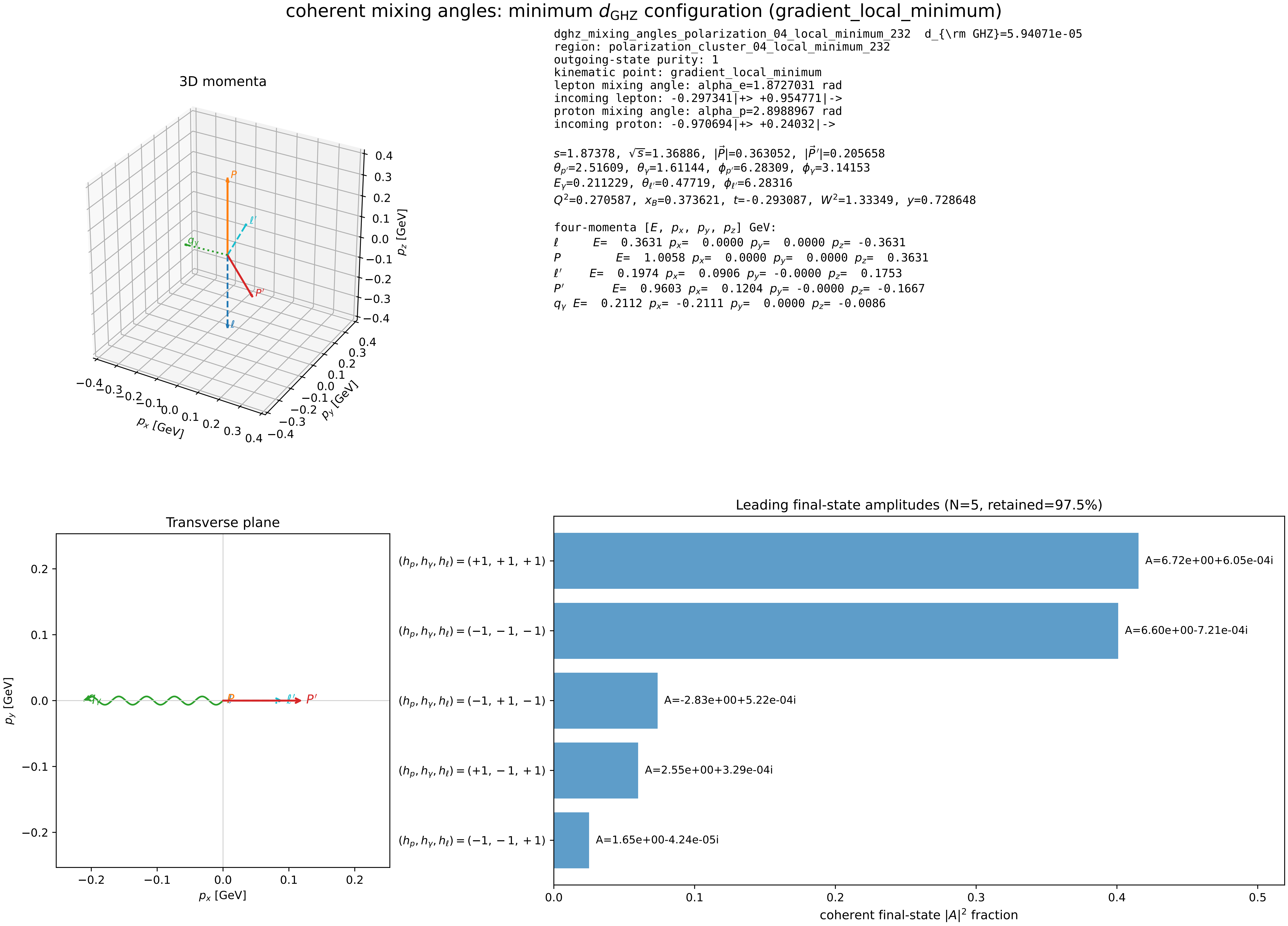




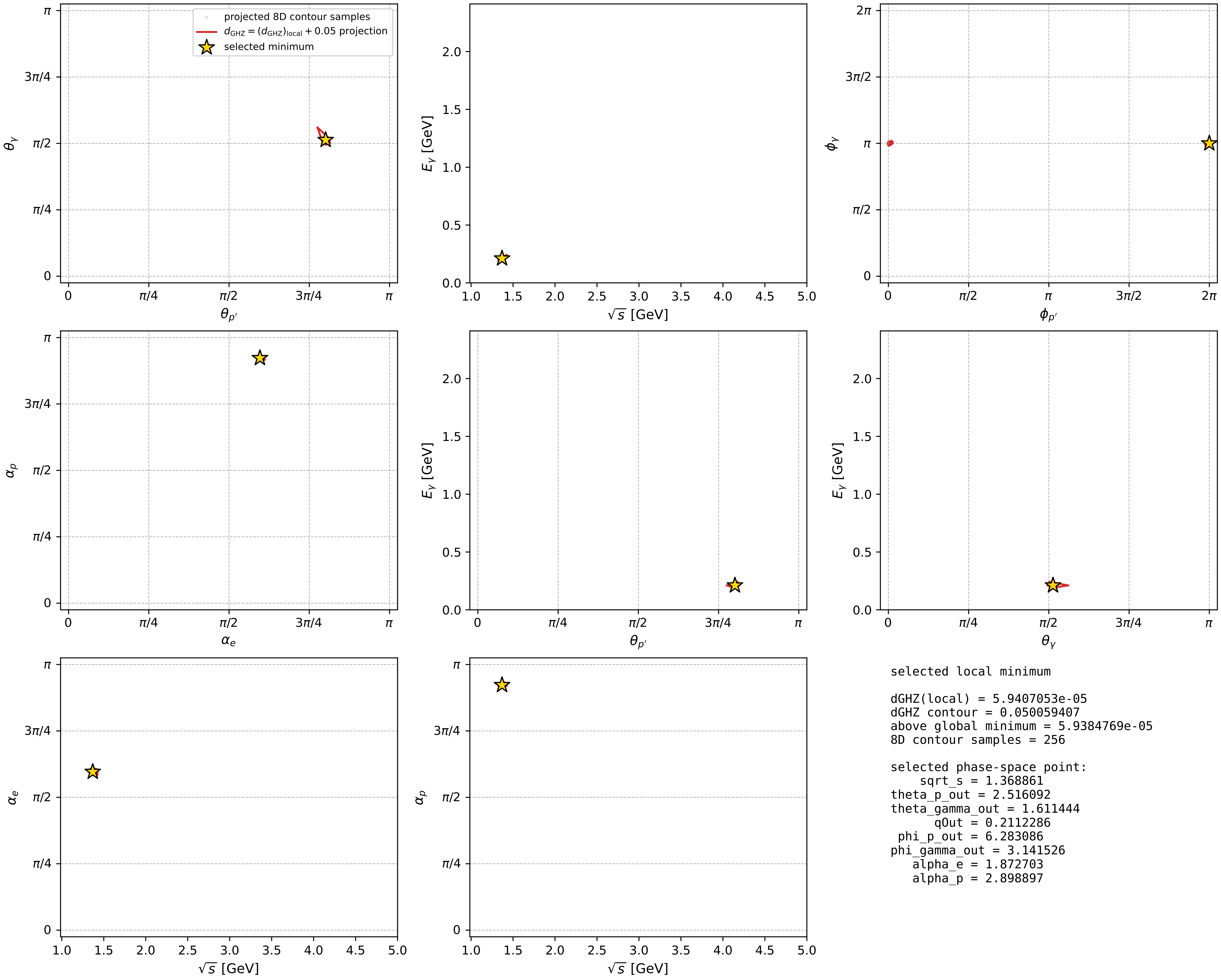
electron: polarization cluster P4, local minimum 230; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour

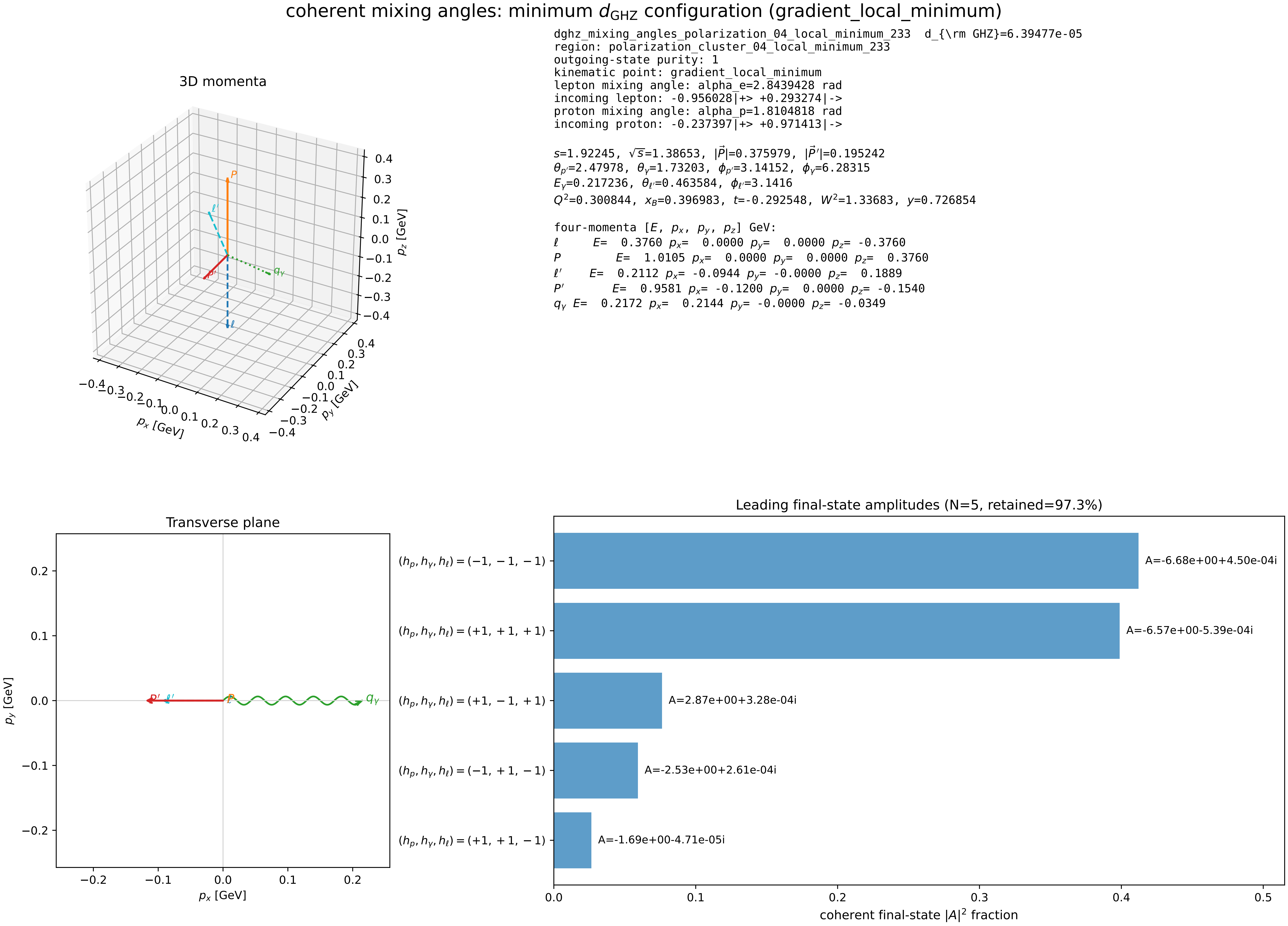


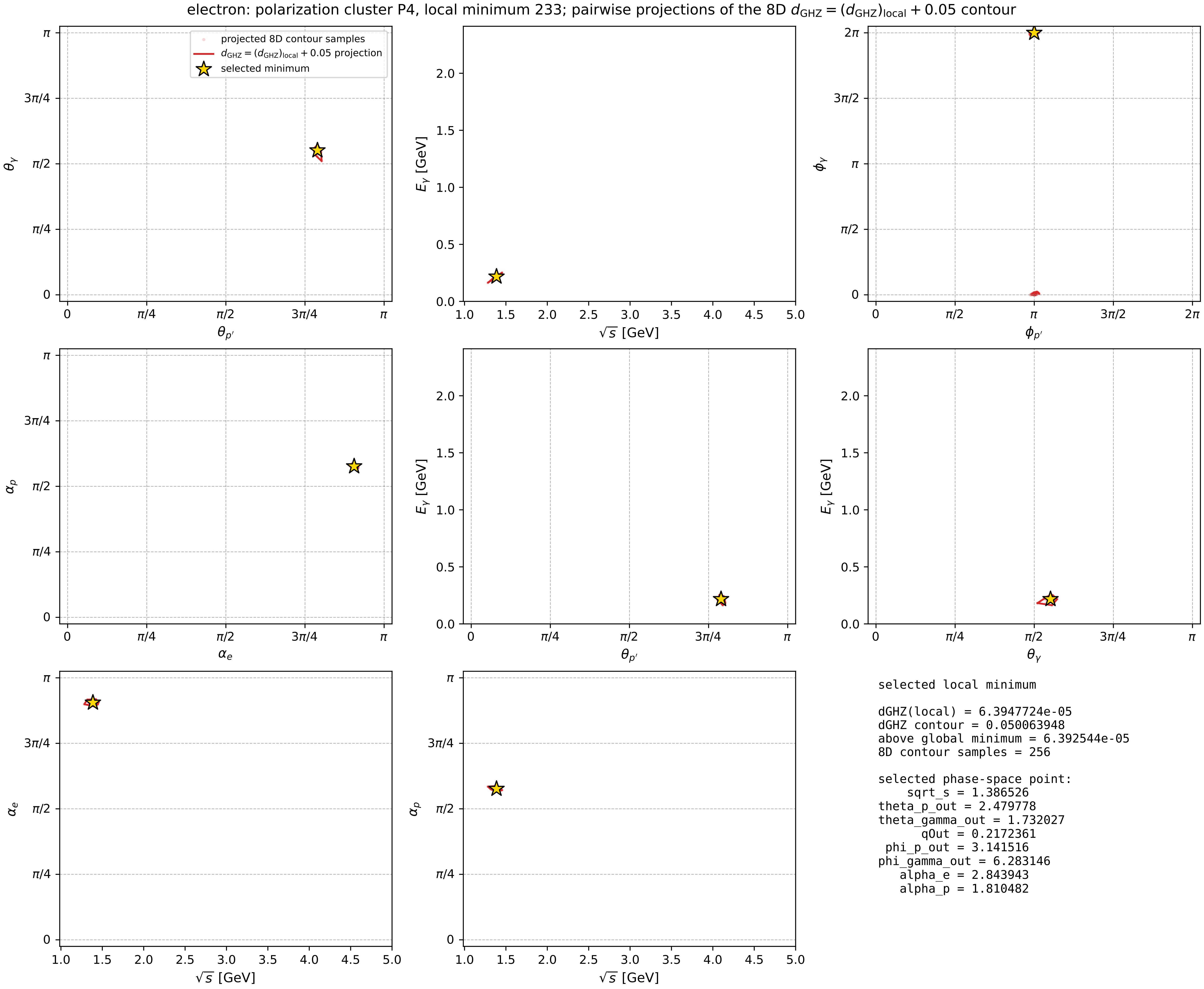

```
selected local minimum

dGHZ(local) = 5.7210653e-05
dGHZ contour = 0.050057211
above global minimum = 5.718837e-05
8D contour samples = 256

selected phase-space point:
    sqrt_s = 1.699516
theta_p_out = 0.2772847
theta_gamma_out = 2.563715
      qOut = 0.4738646
 phi_p_out = 4.575982e-05
phi_gamma_out = 3.14163
   alpha_e = 2.266852
   alpha_p = 2.553001
```


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
3D momenta
P
ℓ'
q_γ
P'
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_232 d_{\rm GHZ}=5.94071e-05
region: polarization_cluster_04_local_minimum_232
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.8727031 rad
incoming lepton: -0.297341|+> +0.954771|->
proton mixing angle: alpha_p=2.8988967 rad
incoming proton: -0.970694|+> +0.24032|->
s=1.87378, $\sqrt{s}$=1.36886, $|\vec{P}|$=0.363052, $|\vec{P}'|$=0.205658
$\theta_{p'}$=2.51609, θ_γ=1.61144, $\phi_{p'}$=6.28309, ϕ_γ=3.14153
E_γ=0.211229, $\theta_{\ell'}$=0.47719, $\phi_{\ell'}$=6.28316
Q^2=0.270587, x_B=0.373621, t=-0.293087, W^2=1.33349, y=0.728648
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3631 p_x= 0.0000 p_y= 0.0000 p_z= -0.3631
P E= 1.0058 p_x= 0.0000 p_y= 0.0000 p_z= 0.3631
ℓ' E= 0.1974 p_x= 0.0906 p_y= -0.0000 p_z= 0.1753
P' E= 0.9603 p_x= 0.1204 p_y= -0.0000 p_z= -0.1667
q_γ E= 0.2112 p_x= -0.2111 p_y= 0.0000 p_z= -0.0086
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=5, retained=97.5%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=6.72e+00+6.05e-04i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=6.60e+00-7.21e-04i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-2.83e+00+5.22e-04i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=2.55e+00+3.29e-04i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=1.65e+00-4.24e-05i
coherent final-state $|A|^2$ fraction

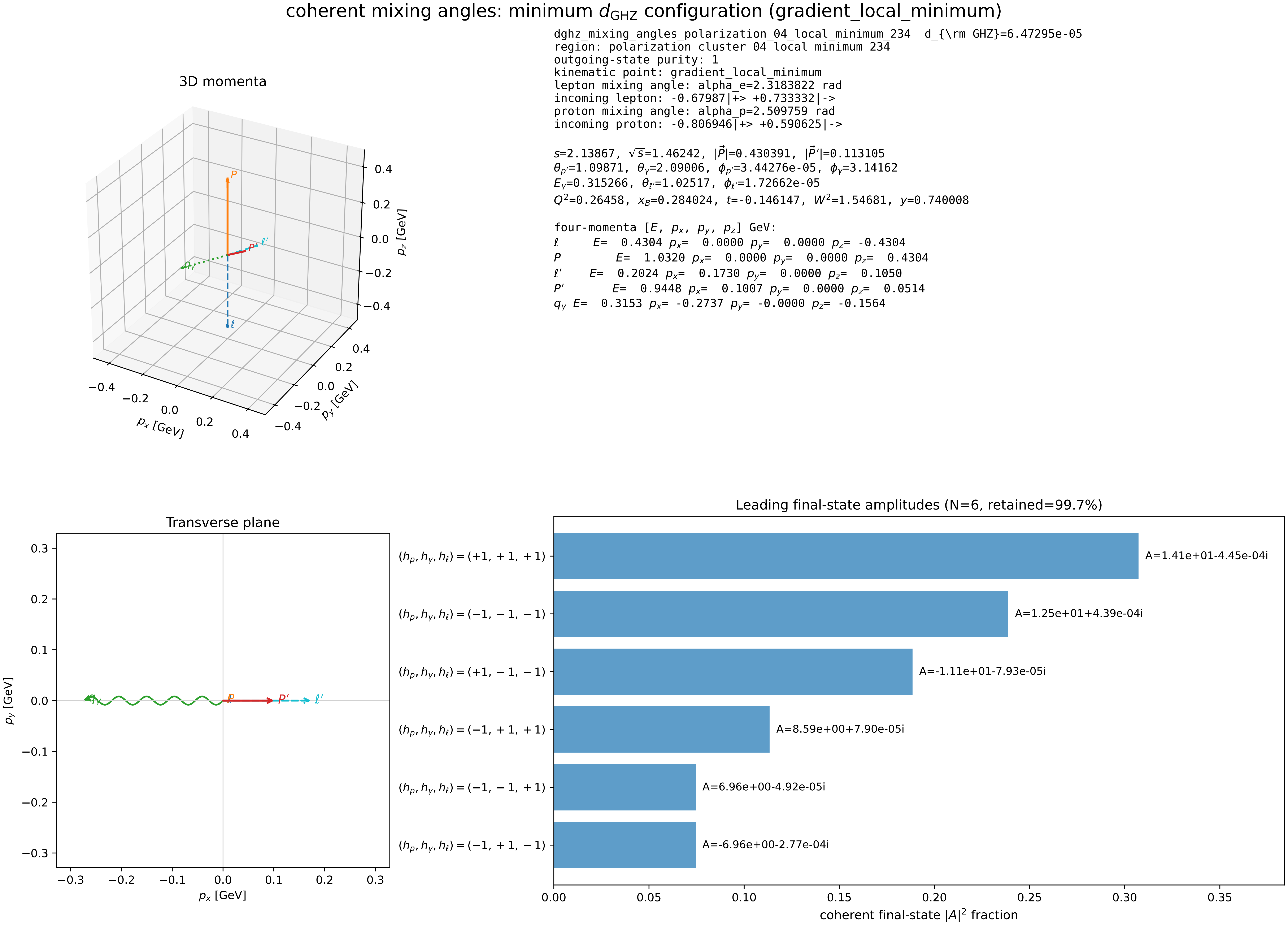
electron: polarization cluster P4, local minimum 232; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



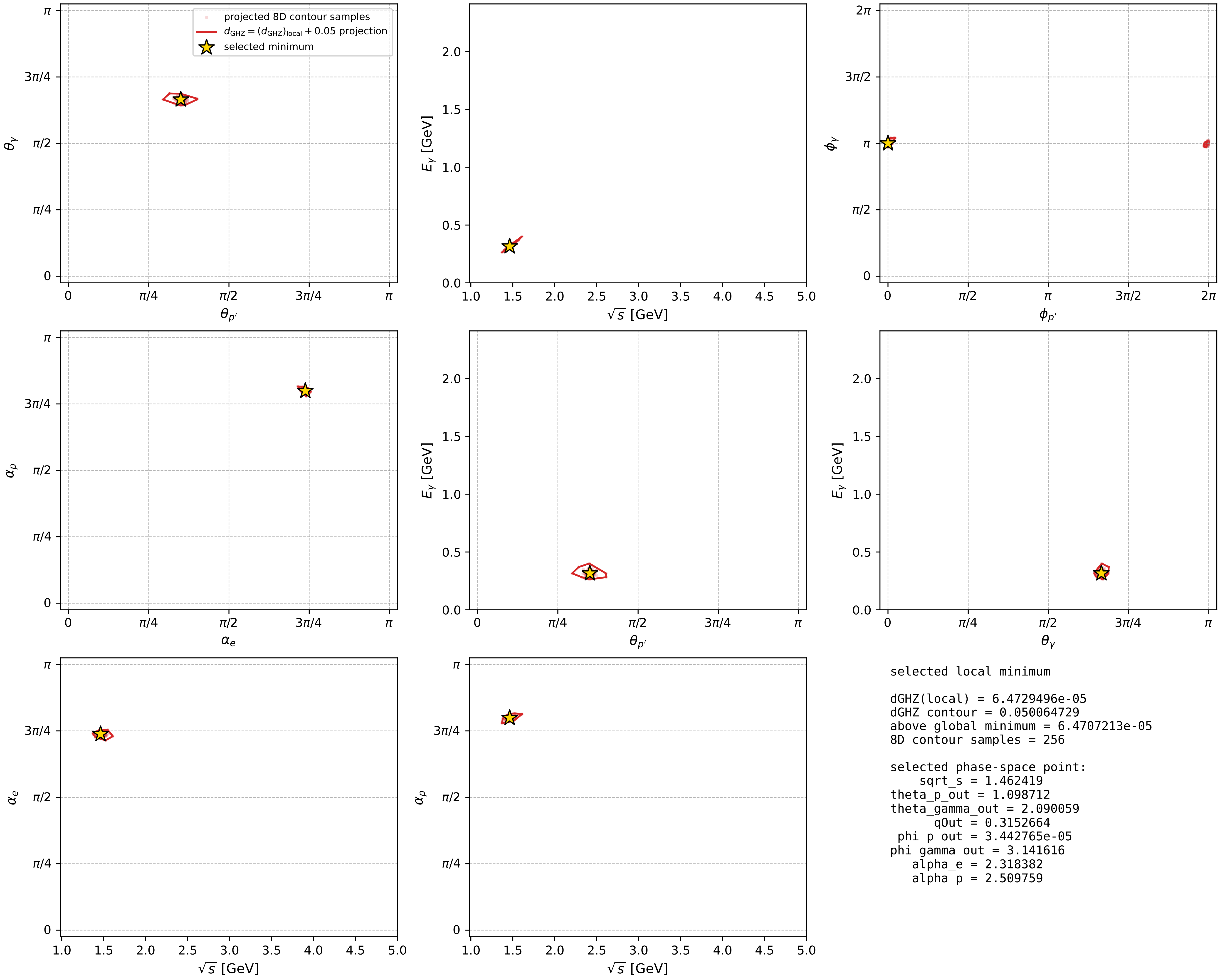

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_233 d_{\rm GHZ}=6.39477e-05
region: polarization_cluster_04_local_minimum_233
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.8439428 rad
incoming lepton: -0.956028|+> +0.293274|->
proton mixing angle: alpha_p=1.8104818 rad
incoming proton: -0.237397|+> +0.971413|->
s=1.92245, $\sqrt{s}$=1.38653, $|\vec{P}|$=0.375979, $|\vec{P}'|$=0.195242
$\theta_{p'}$=2.47978, θ_γ=1.73203, $\phi_{p'}$=3.14152, ϕ_γ=6.28315
E_γ=0.217236, $\theta_{\ell'}$=0.463584, $\phi_{\ell'}$=3.1416
Q^2=0.300844, x_B=0.396983, t=-0.292548, W^2=1.33683, y=0.726854
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3760 p_x= 0.0000 p_y= 0.0000 p_z= -0.3760
P E= 1.0105 p_x= 0.0000 p_y= 0.0000 p_z= 0.3760
ℓ' E= 0.2112 p_x= -0.0944 p_y= -0.0000 p_z= 0.1889
P' E= 0.9581 p_x= -0.1200 p_y= 0.0000 p_z= -0.1540
q_γ E= 0.2172 p_x= 0.2144 p_y= -0.0000 p_z= -0.0349
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=5, retained=97.3%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-6.68e+00+4.50e-04i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-6.57e+00-5.39e-04i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=2.87e+00+3.28e-04i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-2.53e+00+2.61e-04i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-1.69e+00-4.71e-05i
coherent final-state $|A|^2$ fraction



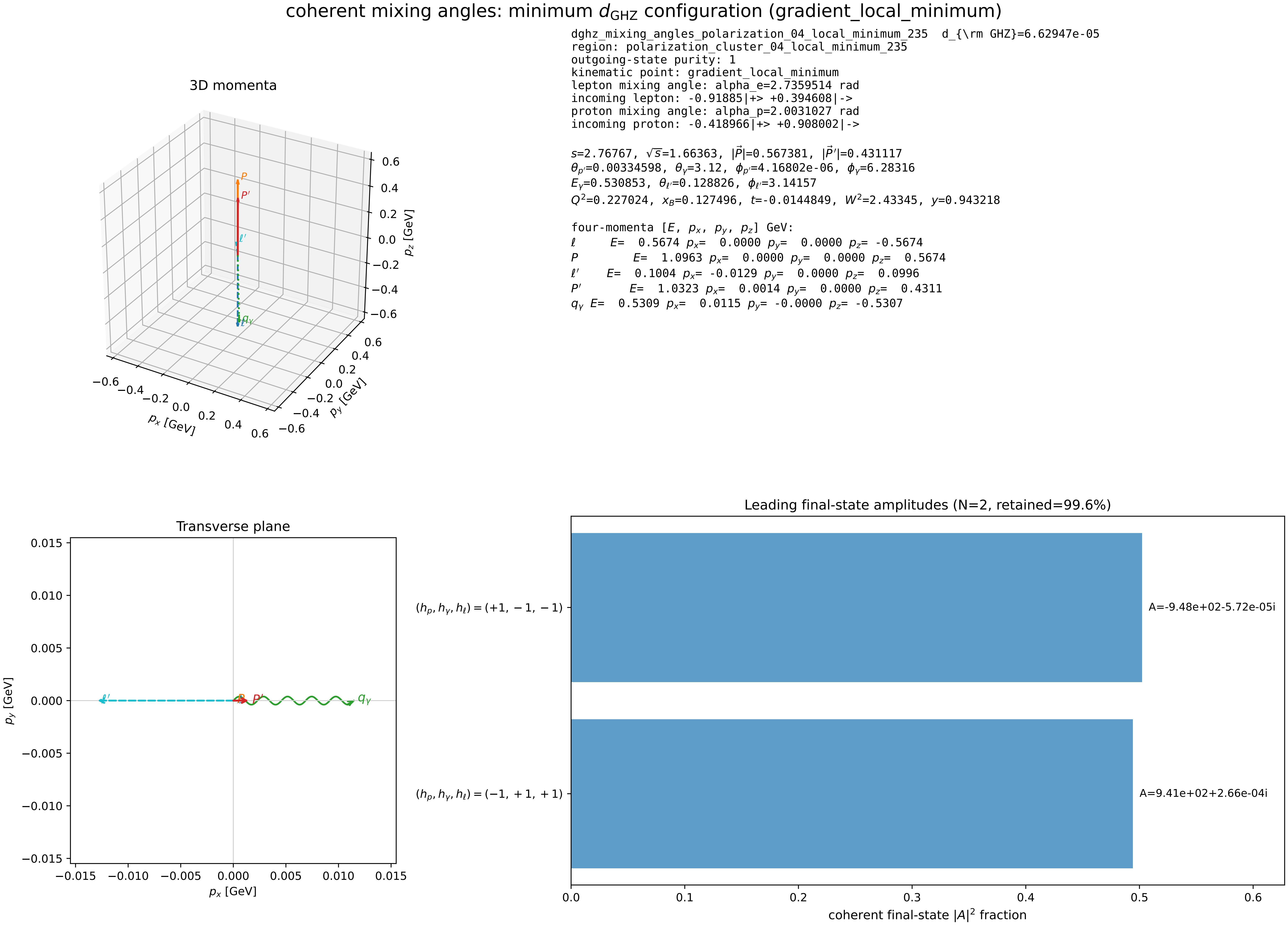
electron: polarization cluster P4, local minimum 233; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_{γ}
$\theta_{p'}$
E_{γ} [GeV]
$\sqrt{s}$ [GeV]
ϕ_{γ}
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 6.3947724e-05
dGHZ contour = 0.050063948
above global minimum = 6.392544e-05
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.386526
theta_p_out = 2.479778
theta_gamma_out = 1.732027
qOut = 0.2172361
phi_p_out = 3.141516
phi_gamma_out = 6.283146
alpha_e = 2.843943
alpha_p = 1.810482


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
dghz_mixing_angles_polarization_04_local_minimum_234 d_{\rm GHZ}=6.47295e-05
region: polarization_cluster_04_local_minimum_234
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3183822 rad
incoming lepton: -0.67987|+> +0.733332|->
proton mixing angle: alpha_p=2.509759 rad
incoming proton: -0.806946|+> +0.590625|->
s=2.13867, $\sqrt{s}$=1.46242, $|\vec{P}|$=0.430391, $|\vec{P}'|$=0.113105
$\theta_{p'}$=1.09871, θ_γ=2.09006, $\phi_{p'}$=3.44276e-05, ϕ_γ=3.14162
E_γ=0.315266, $\theta_{\ell'}$=1.02517, $\phi_{\ell'}$=1.72662e-05
Q^2=0.26458, x_B=0.284024, t=-0.146147, W^2=1.54681, y=0.740008
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4304 p_x= 0.0000 p_y= 0.0000 p_z= -0.4304
P E= 1.0320 p_x= 0.0000 p_y= 0.0000 p_z= 0.4304
ℓ' E= 0.2024 p_x= 0.1730 p_y= 0.0000 p_z= 0.1050
P' E= 0.9448 p_x= 0.1007 p_y= 0.0000 p_z= 0.0514
q_γ E= 0.3153 p_x= -0.2737 p_y= -0.0000 p_z= -0.1564
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading final-state amplitudes (N=6, retained=99.7%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=1.41e+01-4.45e-04i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=1.25e+01+4.39e-04i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-1.11e+01-7.93e-05i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=8.59e+00+7.90e-05i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=6.96e+00-4.92e-05i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-6.96e+00-2.77e-04i
coherent final-state $|A|^2$ fraction

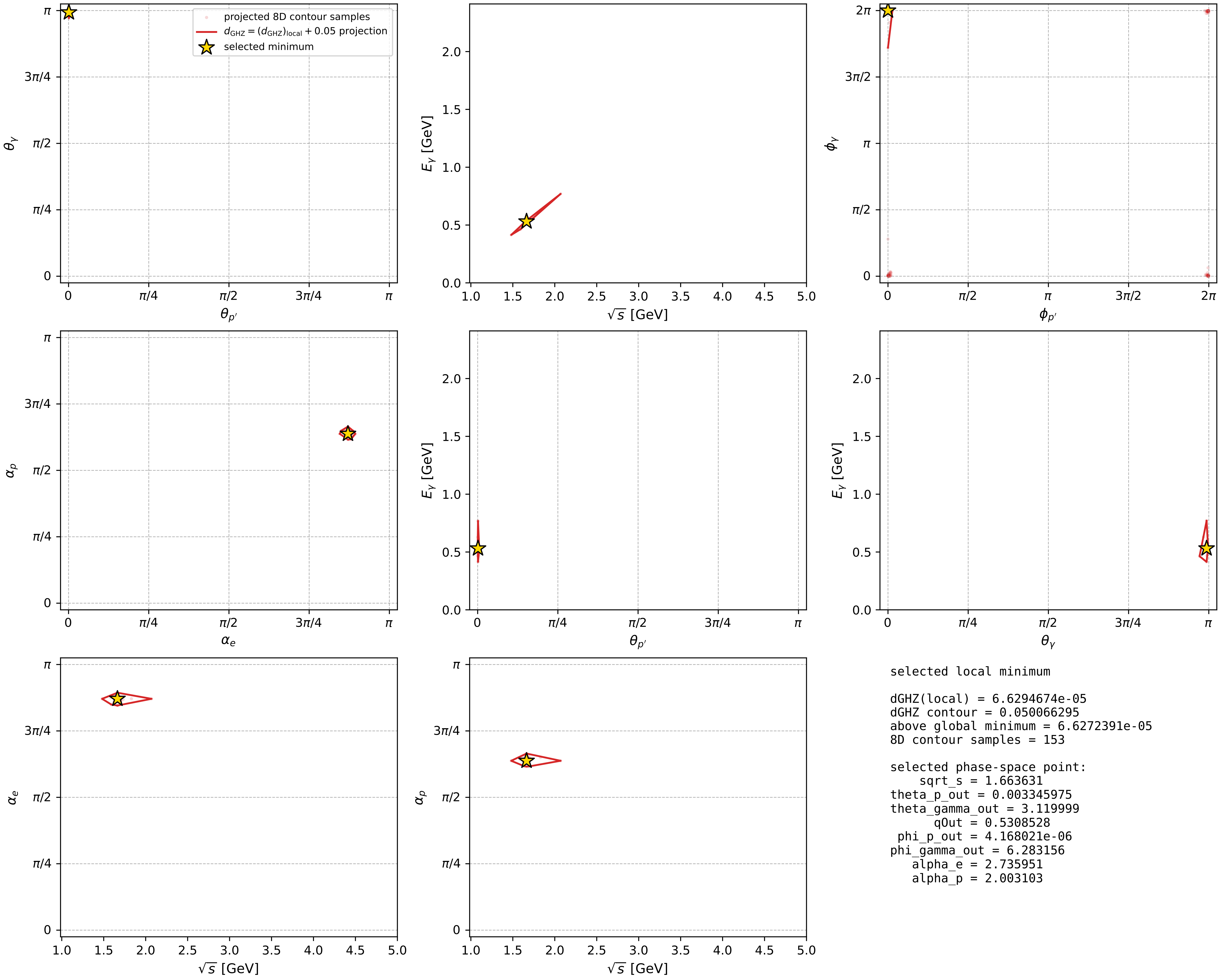
electron: polarization cluster P4, local minimum 234; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



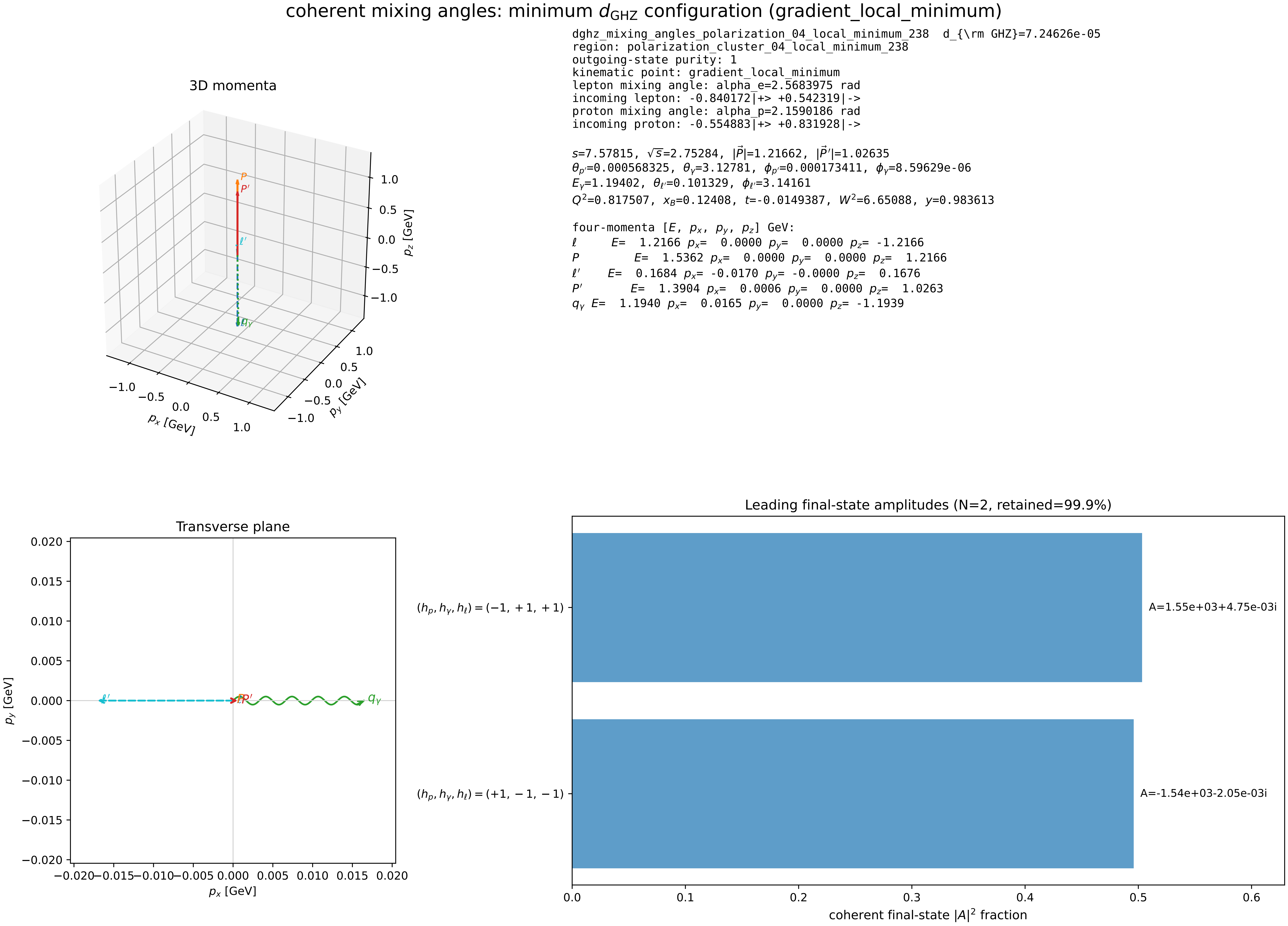

```
selected local minimum

dGHZ(local) = 6.4729496e-05
dGHZ contour = 0.050064729
above global minimum = 6.4707213e-05
8D contour samples = 256

selected phase-space point:
    sqrt_s = 1.462419
theta_p_out = 1.098712
theta_gamma_out = 2.090059
       qOut = 0.3152664
 phi_p_out = 3.442765e-05
phi_gamma_out = 3.141616
   alpha_e = 2.318382
   alpha_p = 2.509759
```

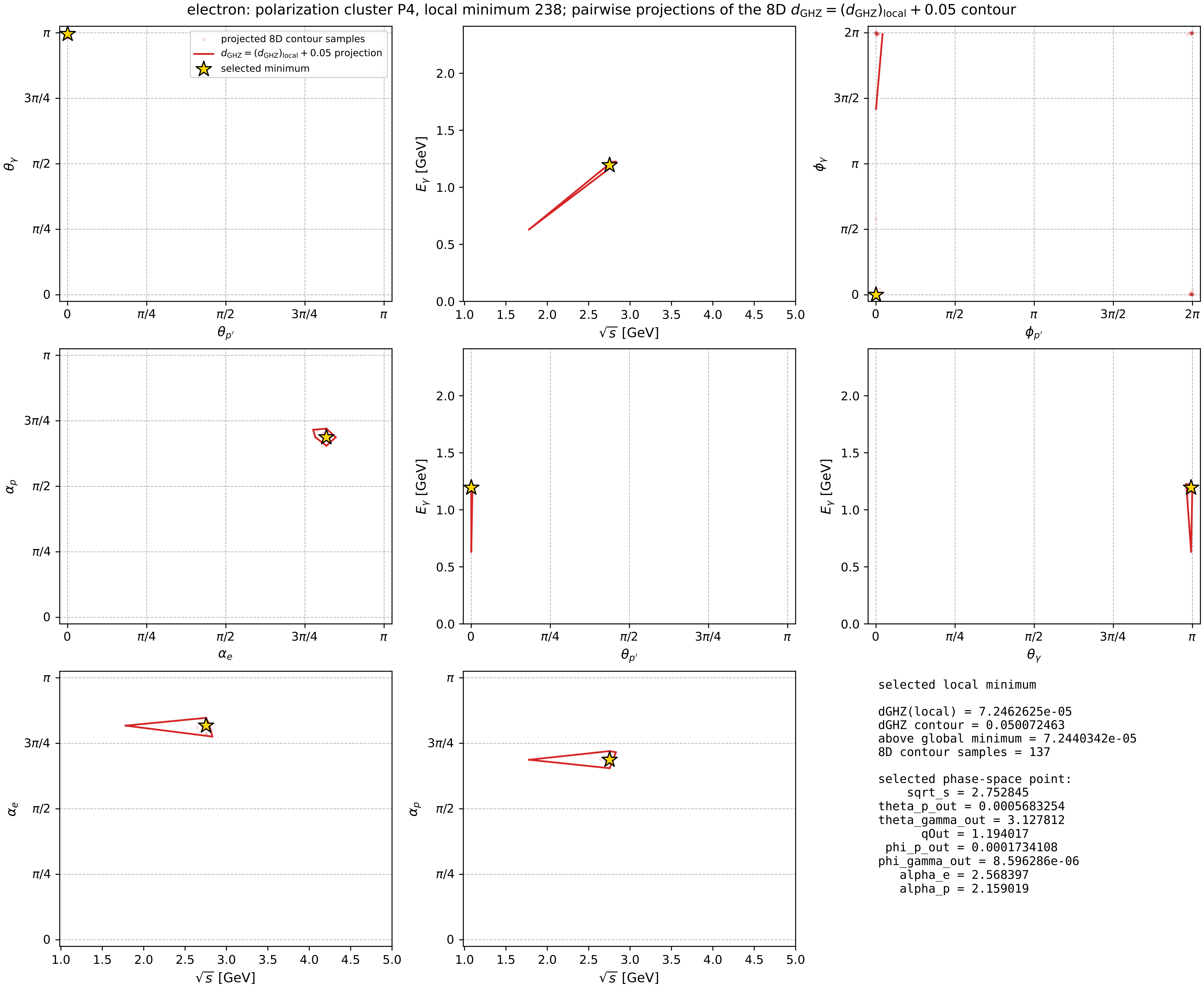

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_235 d_{\rm GHZ}=6.62947e-05
region: polarization_cluster_04_local_minimum_235
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.7359514 rad
incoming lepton: -0.91885|+> +0.394608|->
proton mixing angle: alpha_p=2.0031027 rad
incoming proton: -0.418966|+> +0.908002|->
s=2.76767, $\sqrt{s}$=1.66363, $|\vec{P}|$=0.567381, $|\vec{P}'|$=0.431117
$\theta_{p'}$=0.00334598, θ_γ=3.12, $\phi_{p'}$=4.16802e-06, ϕ_γ=6.28316
E_γ=0.530853, $\theta_{\ell'}$=0.128826, $\phi_{\ell'}$=3.14157
Q^2=0.227024, x_B=0.127496, t=-0.0144849, W^2=2.43345, y=0.943218
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5674 p_x= 0.0000 p_y= 0.0000 p_z= -0.5674
P E= 1.0963 p_x= 0.0000 p_y= 0.0000 p_z= 0.5674
ℓ' E= 0.1004 p_x= -0.0129 p_y= 0.0000 p_z= 0.0996
P' E= 1.0323 p_x= 0.0014 p_y= 0.0000 p_z= 0.4311
q_γ E= 0.5309 p_x= 0.0115 p_y= -0.0000 p_z= -0.5307
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=99.6%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-9.48e+02-5.72e-05i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=9.41e+02+2.66e-04i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P4, local minimum 235; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



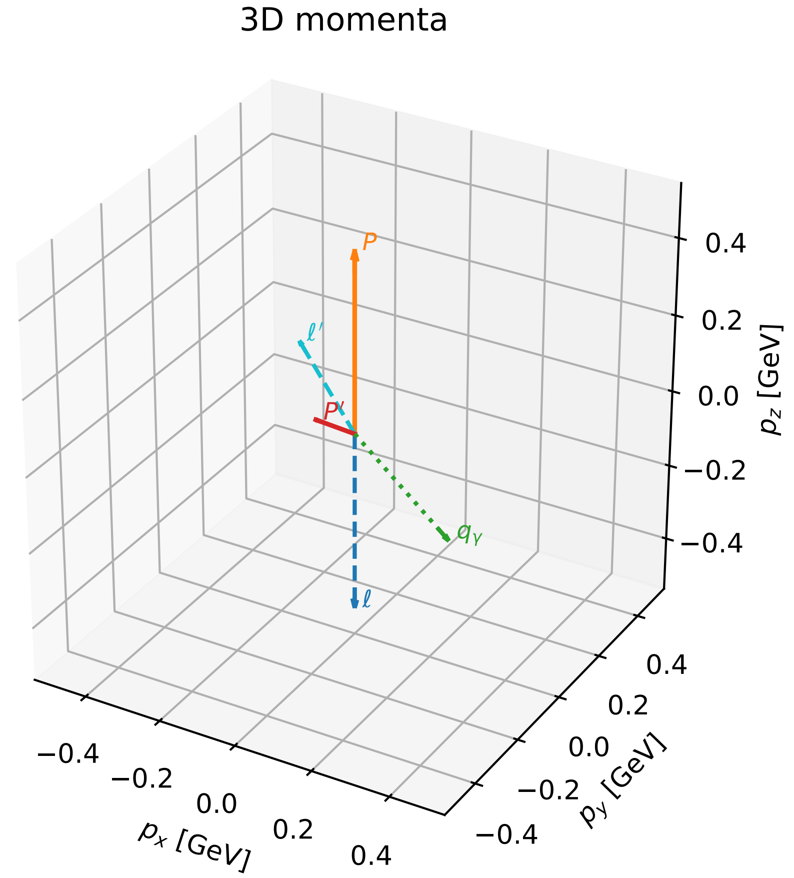

```
selected local minimum

dGHZ(local) = 6.6294674e-05
dGHZ contour = 0.050066295
above global minimum = 6.6272391e-05
8D contour samples = 153

selected phase-space point:
    sqrt_s = 1.663631
theta_p_out = 0.003345975
theta_gamma_out = 3.119999
       qOut = 0.5308528
 phi_p_out = 4.168021e-06
phi_gamma_out = 6.283156
   alpha_e = 2.735951
   alpha_p = 2.003103
```


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_238 d_{\rm GHZ}=7.24626e-05
region: polarization_cluster_04_local_minimum_238
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.5683975 rad
incoming lepton: -0.840172|+> +0.542319|->
proton mixing angle: alpha_p=2.1590186 rad
incoming proton: -0.554883|+> +0.831928|->
s=7.57815, $\sqrt{s}$=2.75284, $|\vec{P}|$=1.21662, $|\vec{P}'|$=1.02635
$\theta_{p'}$=0.000568325, θ_γ=3.12781, $\phi_{p'}$=0.000173411, ϕ_γ=8.59629e-06
E_γ=1.19402, $\theta_{\ell'}$=0.101329, $\phi_{\ell'}$=3.14161
Q^2=0.817507, x_B=0.12408, t=-0.0149387, W^2=6.65088, y=0.983613
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.2166 p_x= 0.0000 p_y= 0.0000 p_z= -1.2166
P E= 1.5362 p_x= 0.0000 p_y= 0.0000 p_z= 1.2166
ℓ' E= 0.1684 p_x= -0.0170 p_y= -0.0000 p_z= 0.1676
P' E= 1.3904 p_x= 0.0006 p_y= 0.0000 p_z= 1.0263
q_γ E= 1.1940 p_x= 0.0165 p_y= 0.0000 p_z= -1.1939
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=99.9%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=1.55e+03+4.75e-03i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-1.54e+03-2.05e-03i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P4, local minimum 238; pairwise projections of the 8D $d_{GHZ} = (d_{GHZ})_{local} + 0.05$ contour
projected 8D contour samples
$d_{GHZ} = (d_{GHZ})_{local} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 7.2462625e-05
dGHZ contour = 0.050072463
above global minimum = 7.2440342e-05
8D contour samples = 137
selected phase-space point:
sqrt_s = 2.752845
theta_p_out = 0.0005683254
theta_gamma_out = 3.127812
qOut = 1.194017
phi_p_out = 0.0001734108
phi_gamma_out = 8.596286e-06
alpha_e = 2.568397
alpha_p = 2.159019

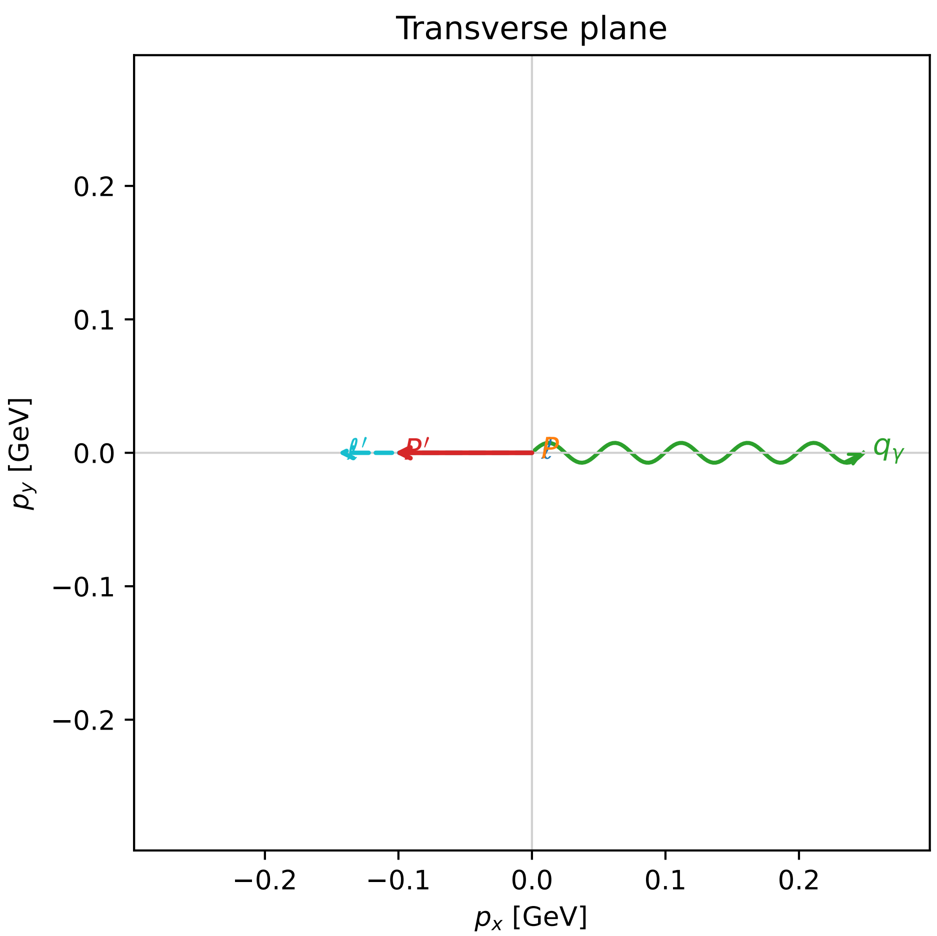
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

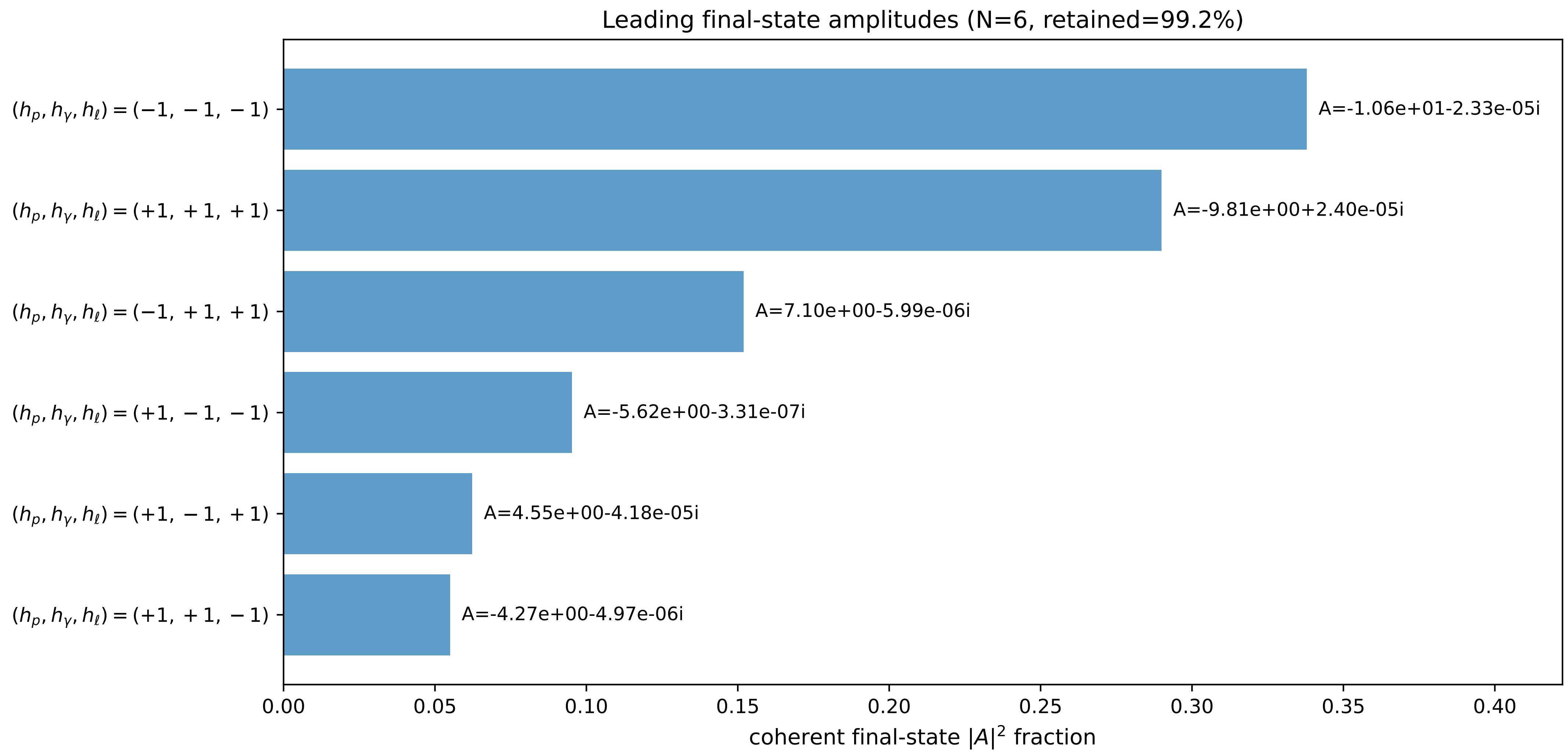


dghz_mixing_angles_polarization_04_local_minimum_243 d_{\rm GHZ}=0.000102715
region: polarization_cluster_04_local_minimum_243
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.6883821 rad
incoming lepton: -0.899046|+> +0.437854|->
proton mixing angle: alpha_p=1.9477293 rad
incoming proton: -0.36807|+> +0.929798|->

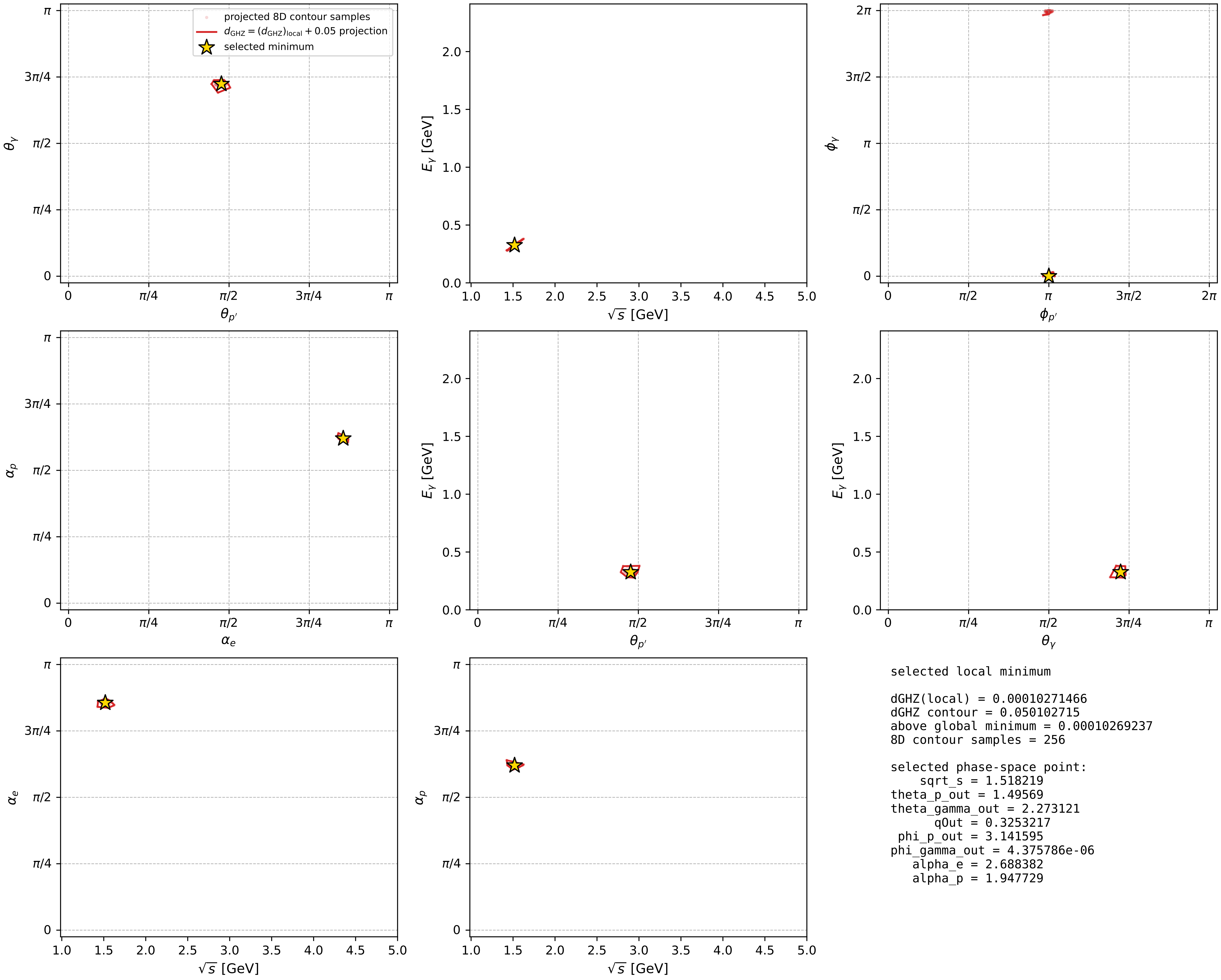
s=2.30499, $\sqrt{s}$=1.51822, $|\vec{P}|$=0.469347, $|\vec{P}'|$=0.103198
$\theta_{p'}$=1.49569, θ_γ=2.27312, $\phi_{p'}$=3.14159, ϕ_γ=4.37579e-06
E_γ=0.325322, $\theta_{\ell'}$=0.623006, $\phi_{\ell'}$=3.1416
Q^2=0.423961, x_B=0.388132, t=-0.212599, W^2=1.5482, y=0.766457

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4693 p_x= 0.0000 p_y= 0.0000 p_z= -0.4693
P E= 1.0489 p_x= 0.0000 p_y= 0.0000 p_z= 0.4693
ℓ' E= 0.2492 p_x= -0.1454 p_y= -0.0000 p_z= 0.2024
P' E= 0.9437 p_x= -0.1029 p_y= -0.0000 p_z= 0.0077
q_γ E= 0.3253 p_x= 0.2483 p_y= 0.0000 p_z= -0.2102

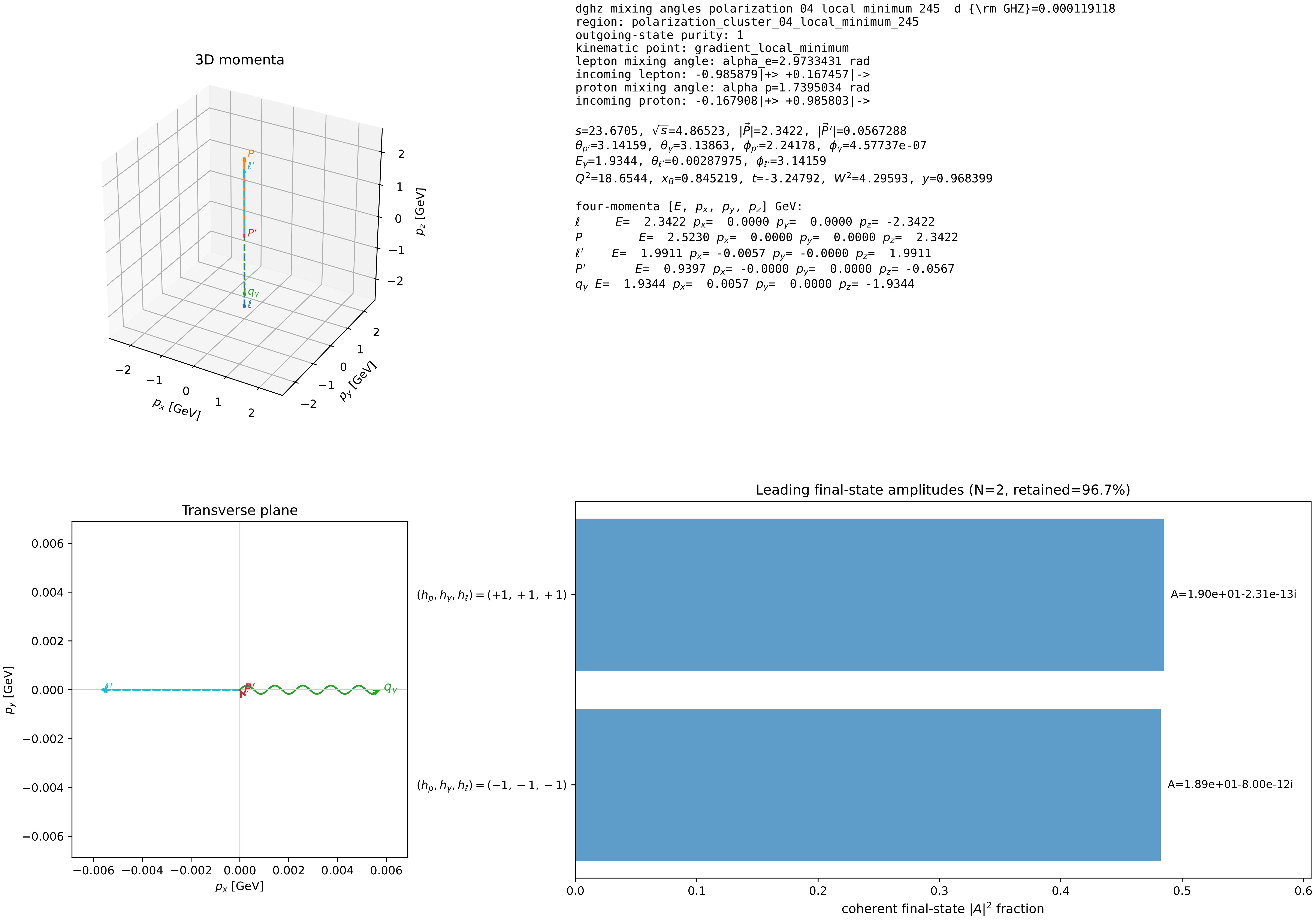




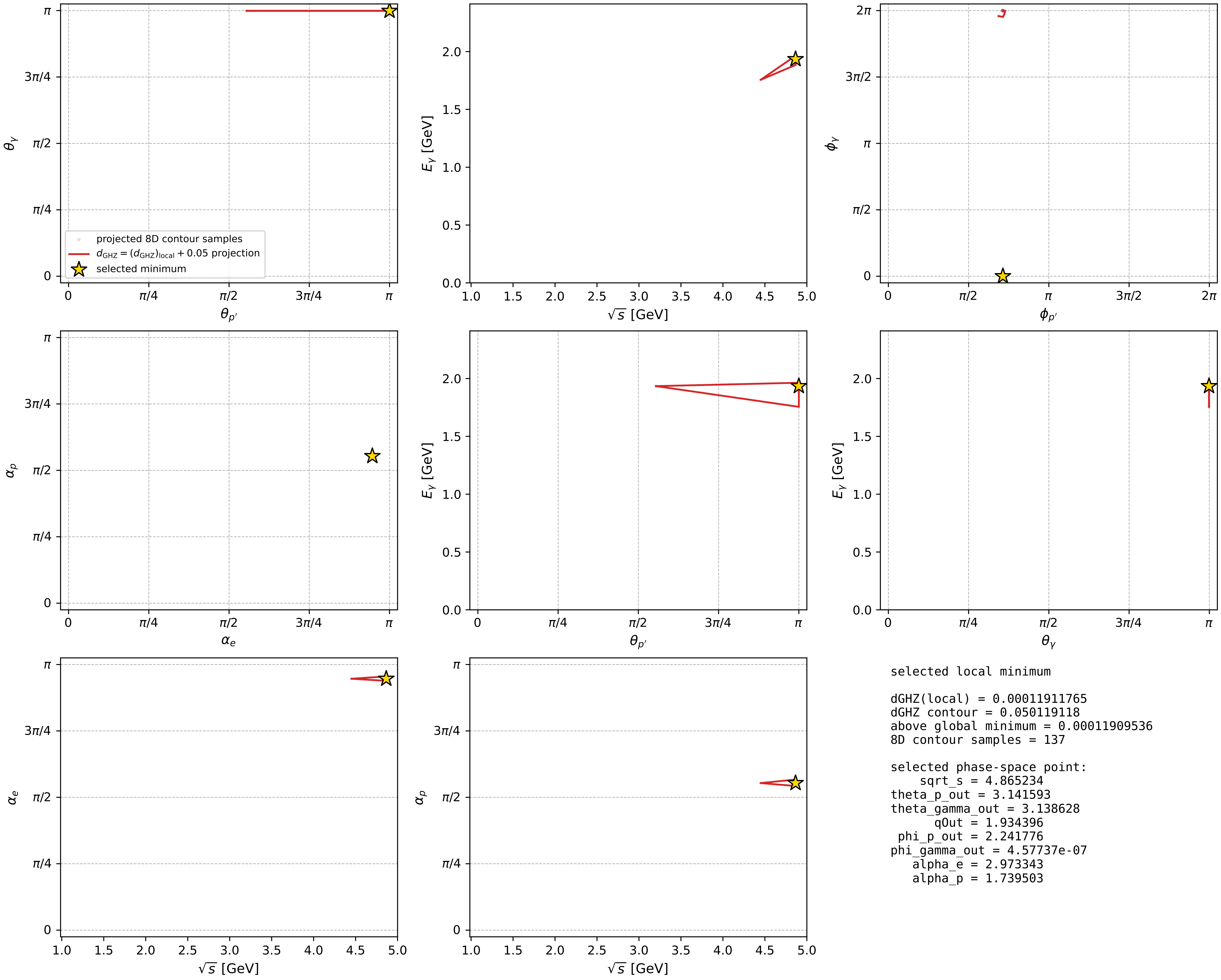
electron: polarization cluster P4, local minimum 243; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



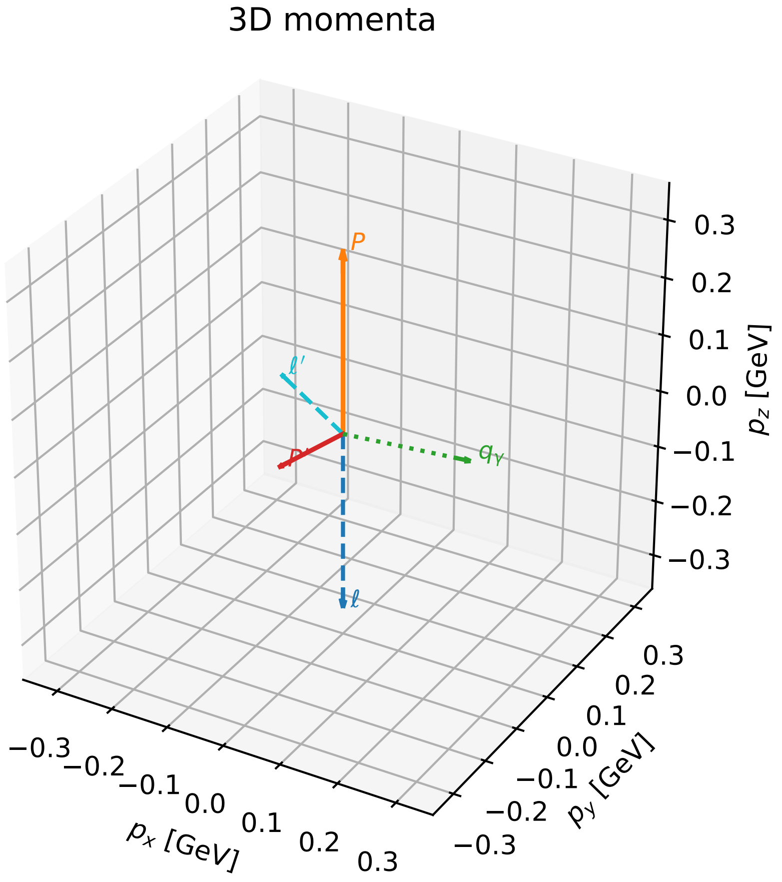
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)



electron: polarization cluster P4, local minimum 245; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



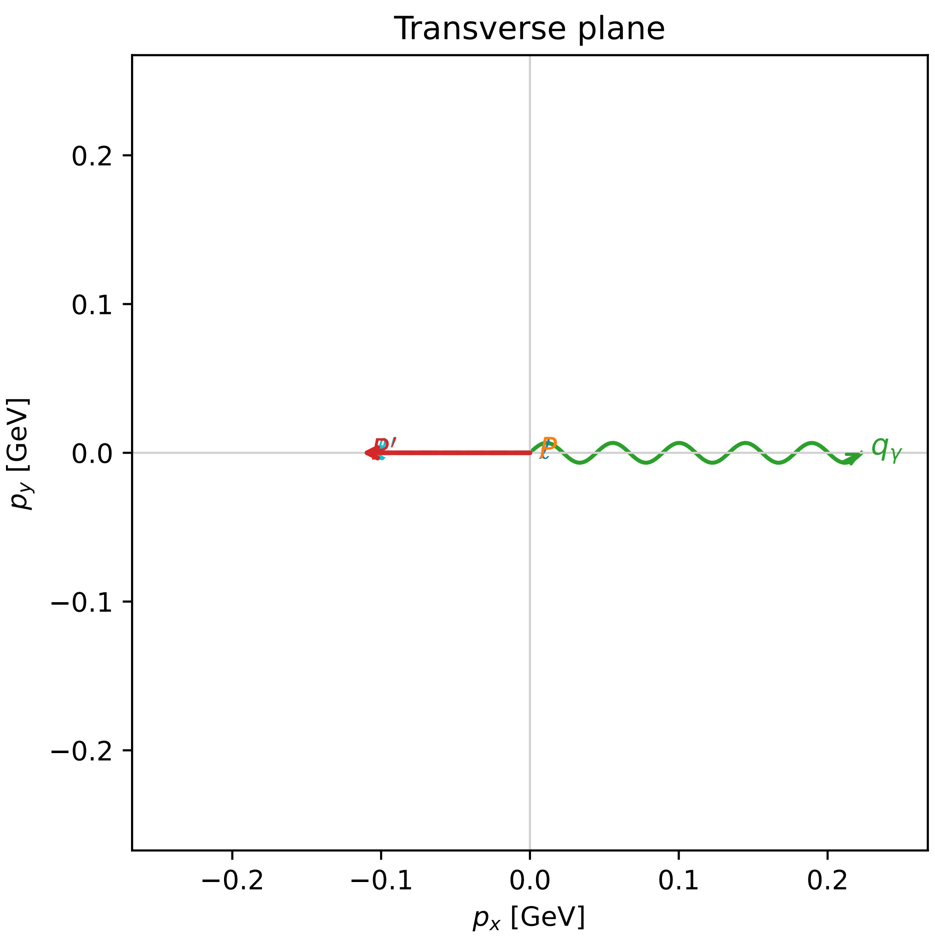
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

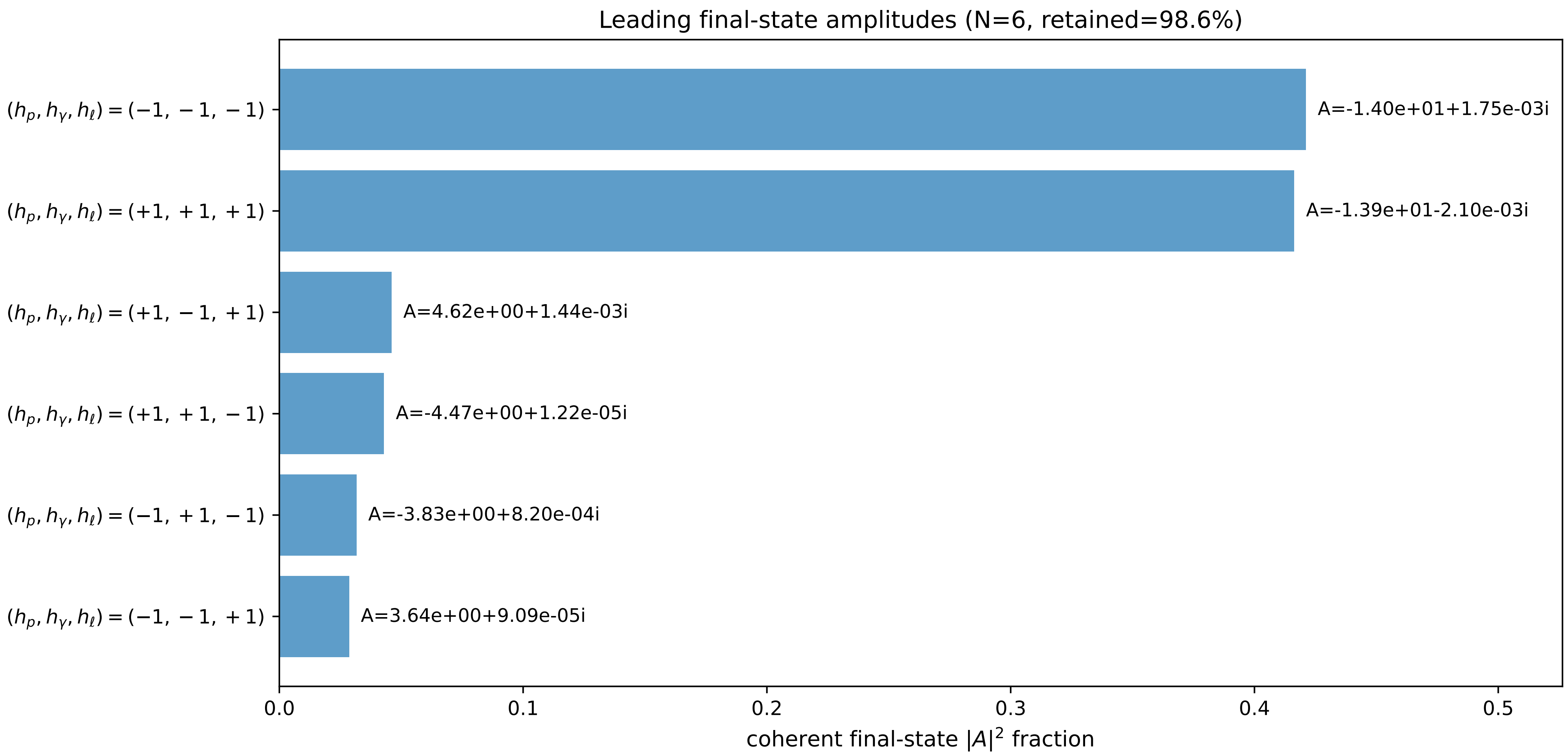


dghz_mixing_angles_polarization_04_local_minimum_247 d_{\rm GHZ}=0.000124428
region: polarization_cluster_04_local_minimum_247
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4927847 rad
incoming lepton: -0.796805|+> +0.604237|->
proton mixing angle: alpha_p=2.1188154 rad
incoming proton: -0.520997|+> +0.853558|->

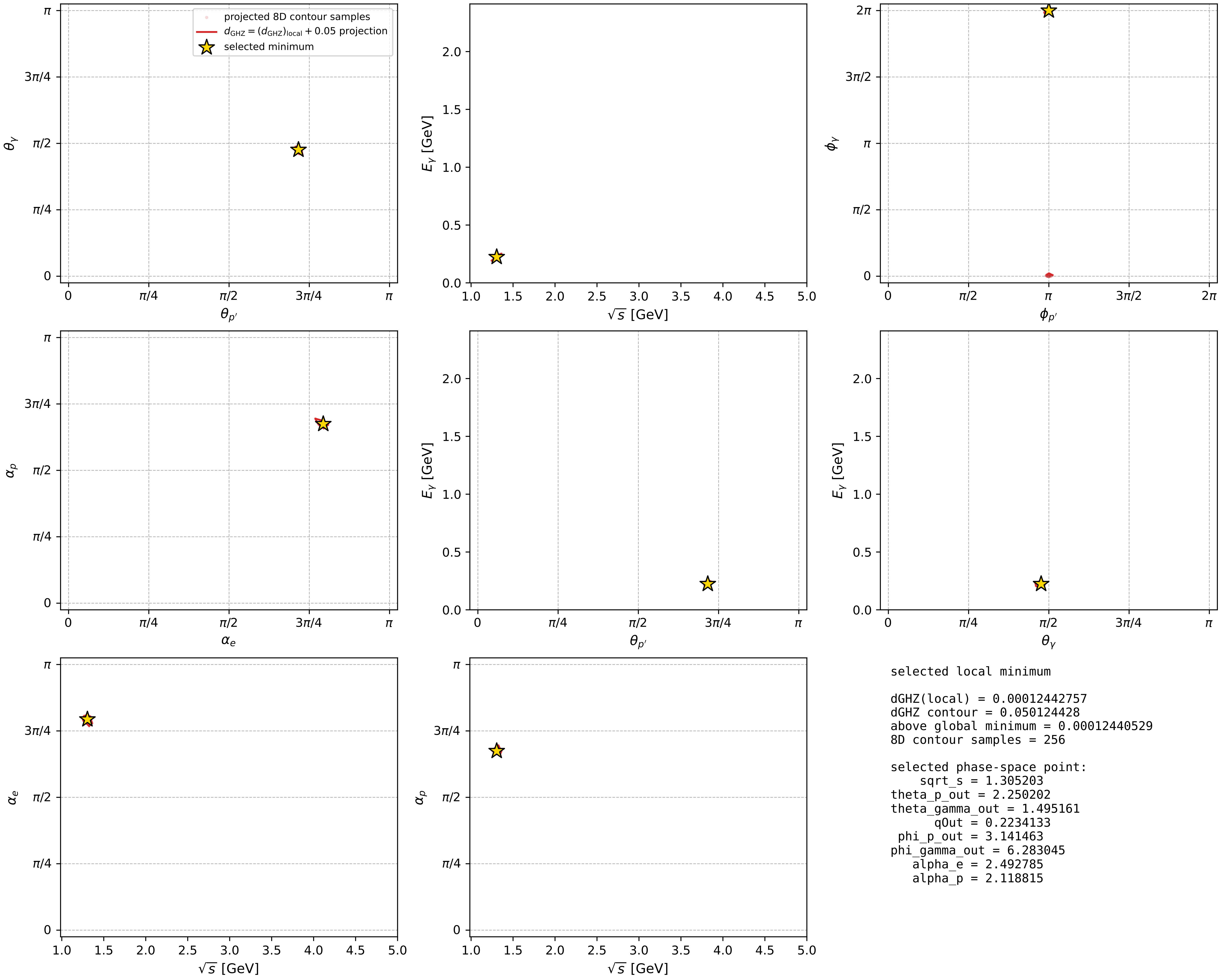
s=1.70355, $\sqrt{s}$=1.3052, $|\vec{P}|$=0.315549, $|\vec{P}'|$=0.145218
$\theta_{p'}$=2.2502, θ_γ=1.49516, $\phi_{p'}$=3.14146, ϕ_γ=6.28305
E_γ=0.223413, $\theta_{\ell'}$=0.975506, $\phi_{\ell'}$=3.14144
Q^2=0.130623, x_B=0.214786, t=-0.176605, W^2=1.35738, y=0.738311

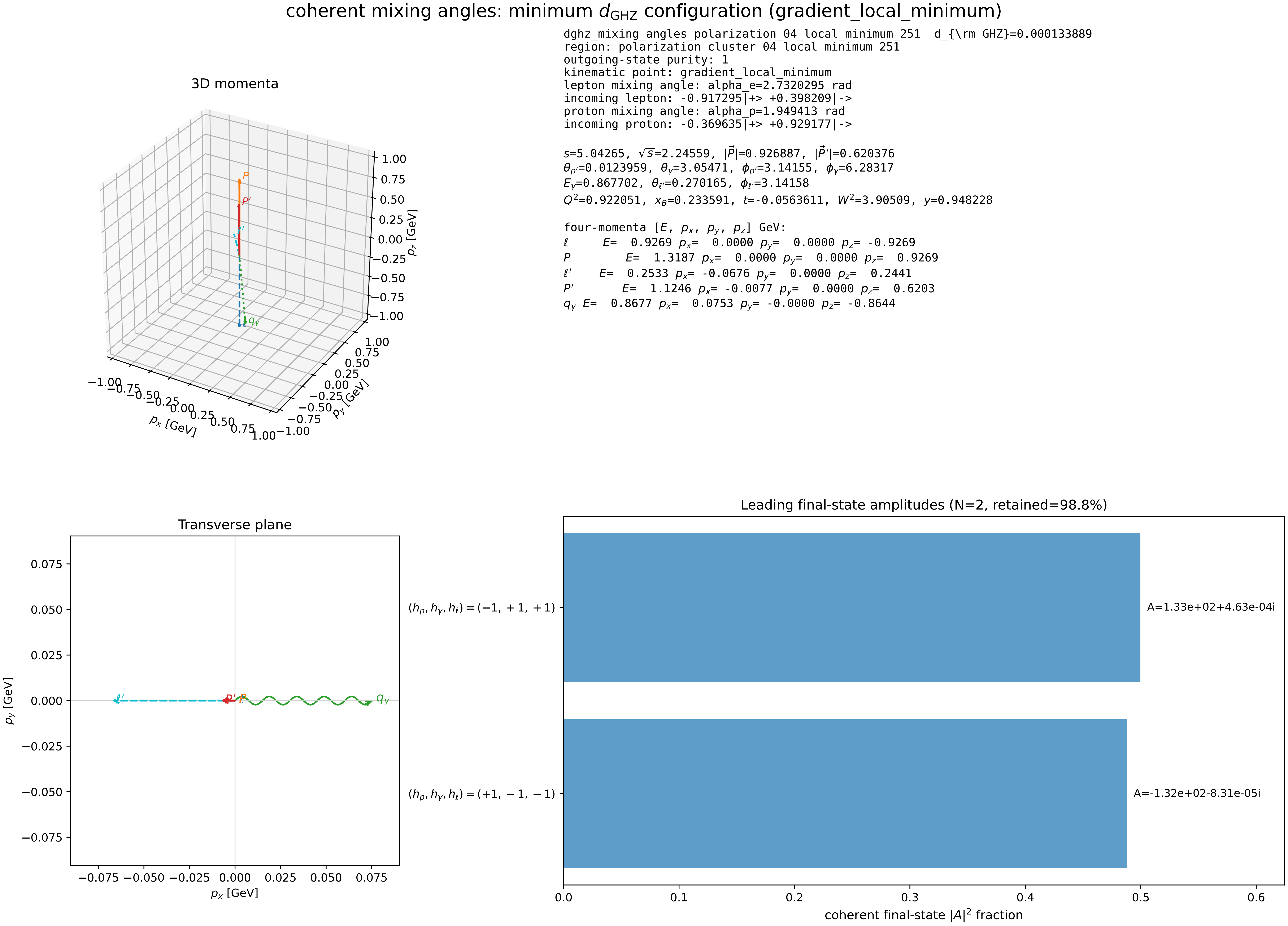
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3155 p_x= 0.0000 p_y= 0.0000 p_z= -0.3155
P E= 0.9897 p_x= 0.0000 p_y= 0.0000 p_z= 0.3155
ℓ' E= 0.1326 p_x= -0.1098 p_y= 0.0000 p_z= 0.0744
P' E= 0.9492 p_x= -0.1130 p_y= 0.0000 p_z= -0.0912
q_γ E= 0.2234 p_x= 0.2228 p_y= -0.0000 p_z= 0.0169

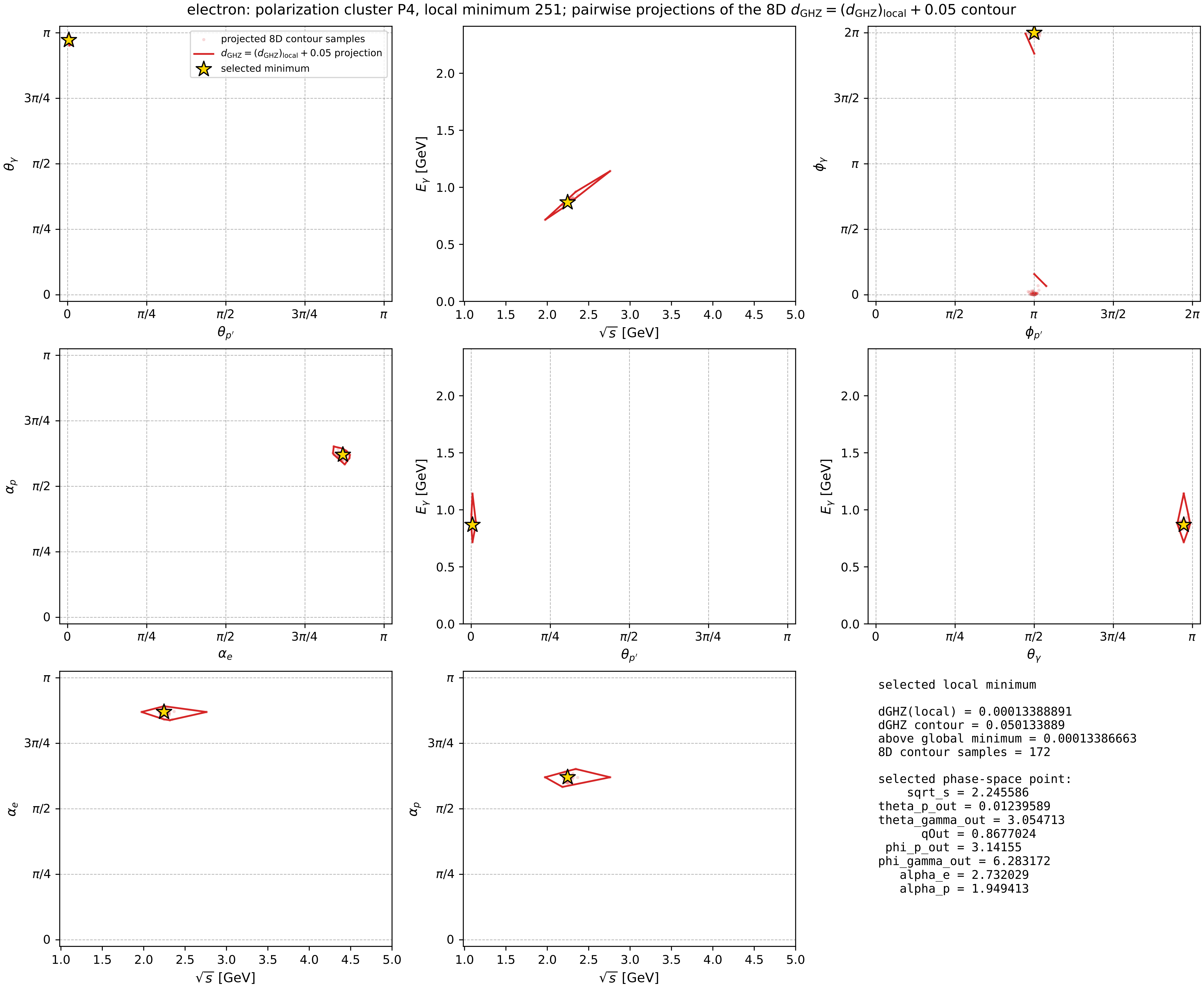




electron: polarization cluster P4, local minimum 247; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour

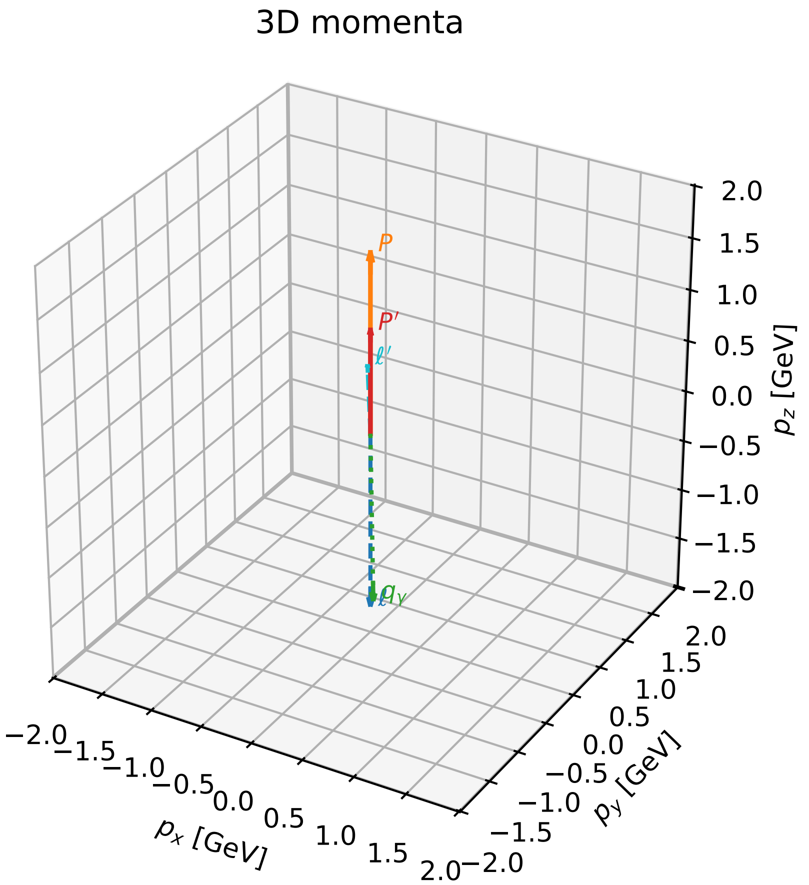



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dghz_mixing_angles_polarization_04_local_minimum_251 d_{\rm GHZ}=0.000133889
region: polarization_cluster_04_local_minimum_251
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.7320295 rad
incoming lepton: -0.917295|+> +0.398209|->
proton mixing angle: alpha_p=1.949413 rad
incoming proton: -0.369635|+> +0.929177|->
s=5.04265, $\sqrt{s}$=2.24559, $|\vec{P}|$=0.926887, $|\vec{P}'|$=0.620376
$\theta_{p'}$=0.0123959, θ_γ=3.05471, $\phi_{p'}$=3.14155, ϕ_γ=6.28317
E_γ=0.867702, $\theta_{\ell'}$=0.270165, $\phi_{\ell'}$=3.14158
Q^2=0.922051, x_B=0.233591, t=-0.0563611, W^2=3.90509, y=0.948228
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.9269 p_x= 0.0000 p_y= 0.0000 p_z= -0.9269
P E= 1.3187 p_x= 0.0000 p_y= 0.0000 p_z= 0.9269
ℓ' E= 0.2533 p_x= -0.0676 p_y= 0.0000 p_z= 0.2441
P' E= 1.1246 p_x= -0.0077 p_y= 0.0000 p_z= 0.6203
q_γ E= 0.8677 p_x= 0.0753 p_y= -0.0000 p_z= -0.8644
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=98.8%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=1.33e+02+4.63e-04i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-1.32e+02-8.31e-05i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P4, local minimum 251; pairwise projections of the 8D $d_{GHZ} = (d_{GHZ})_{local} + 0.05$ contour
projected 8D contour samples
$d_{GHZ} = (d_{GHZ})_{local} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 0.00013388891
dGHZ contour = 0.050133889
above global minimum = 0.00013386663
8D contour samples = 172
selected phase-space point:
sqrt_s = 2.245586
theta_p_out = 0.01239589
theta_gamma_out = 3.054713
qOut = 0.8677024
phi_p_out = 3.14155
phi_gamma_out = 6.283172
alpha_e = 2.732029
alpha_p = 1.949413

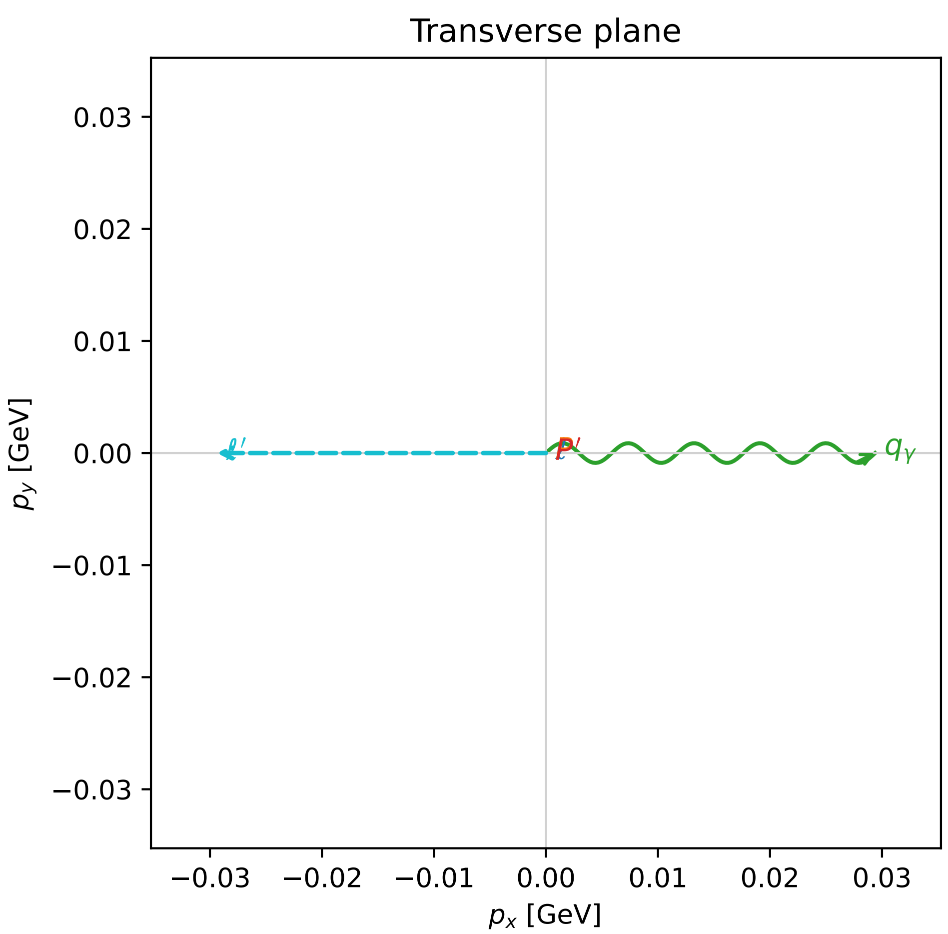
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

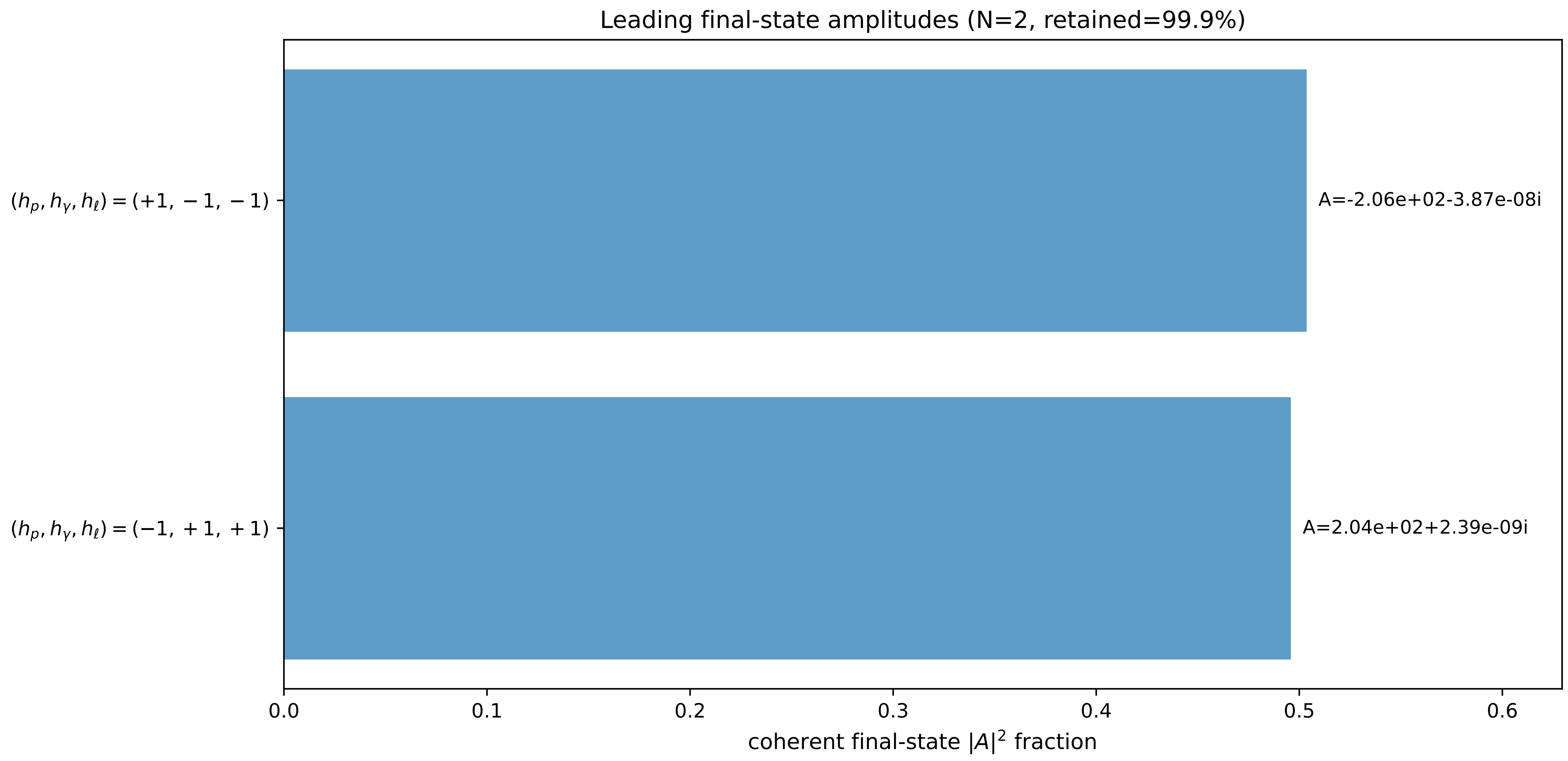


dghz_mixing_angles_polarization_04_local_minimum_252 d_{\rm GHZ}=0.000137569
region: polarization_cluster_04_local_minimum_252
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.8965137 rad
incoming lepton: -0.970118|+> +0.242633|->
proton mixing angle: alpha_p=1.8140654 rad
incoming proton: -0.240877|+> +0.970556|->

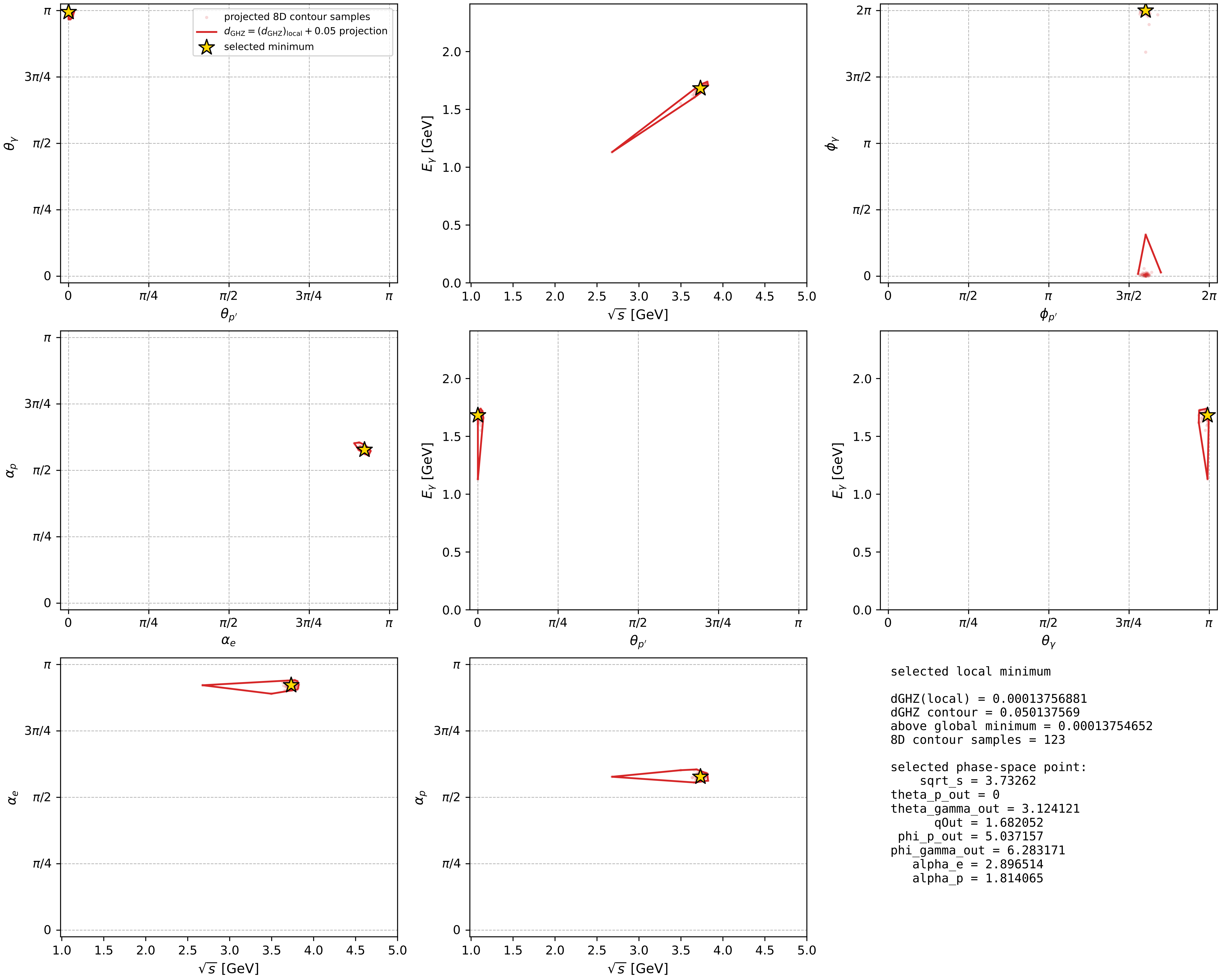
s=13.9325, $\sqrt{s}$=3.73262, $|\vec{P}|$=1.74845, $|\vec{P}'|$=1.01096
$\theta_{p'}$=0, θ_γ=3.12412, $\phi_{p'}$=5.03716, ϕ_γ=6.28317
E_γ=1.68205, $\theta_{\ell'}$=0.0437779, $\phi_{\ell'}$=3.14158
Q^2=4.69394, x_B=0.368621, t=-0.177771, W^2=8.9197, y=0.975575

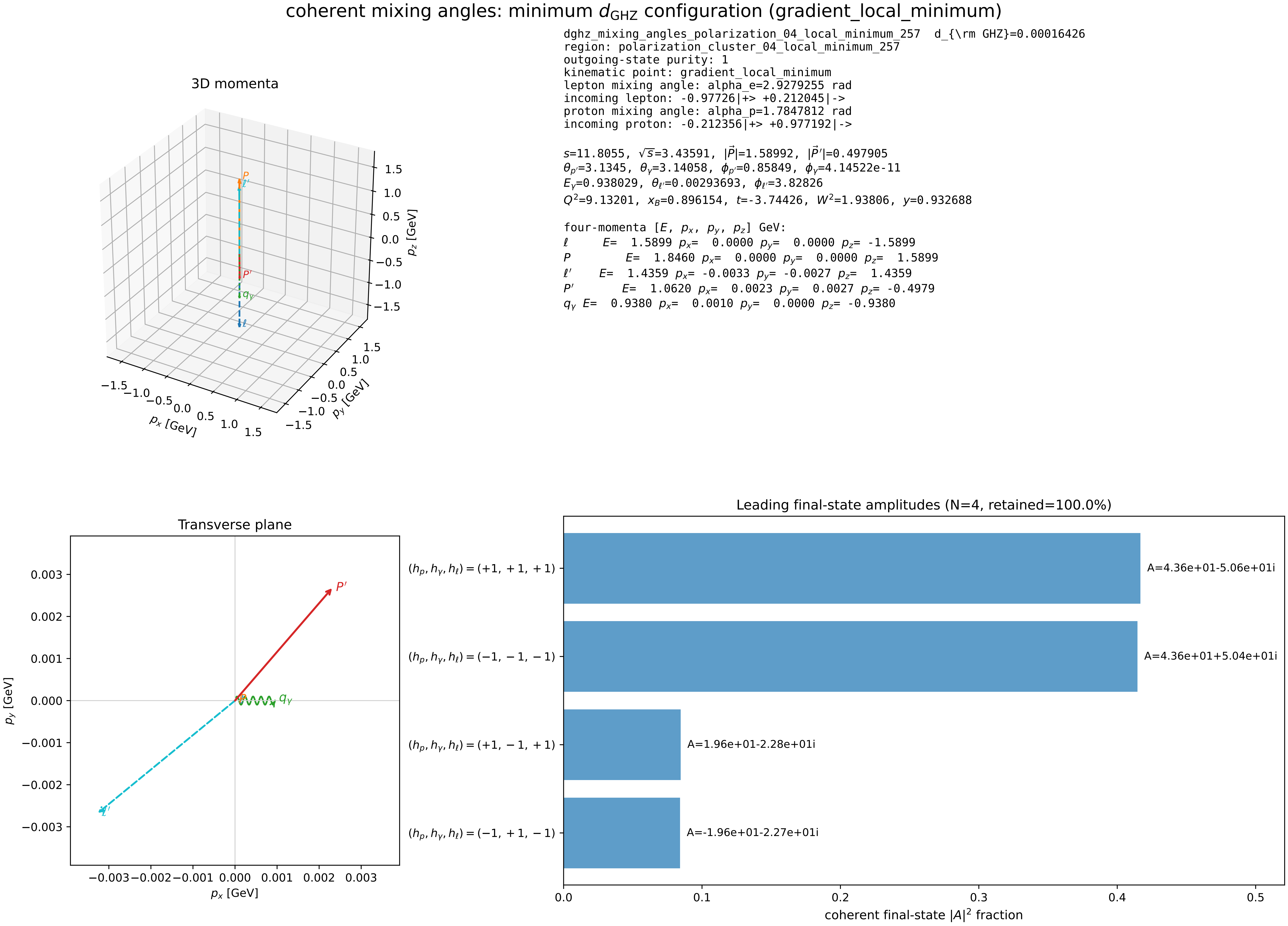
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.7485 p_x= 0.0000 p_y= 0.0000 p_z= -1.7485
P E= 1.9842 p_x= 0.0000 p_y= 0.0000 p_z= 1.7485
ℓ' E= 0.6715 p_x= -0.0294 p_y= 0.0000 p_z= 0.6708
P' E= 1.3791 p_x= 0.0000 p_y= -0.0000 p_z= 1.0110
q_γ E= 1.6821 p_x= 0.0294 p_y= -0.0000 p_z= -1.6818

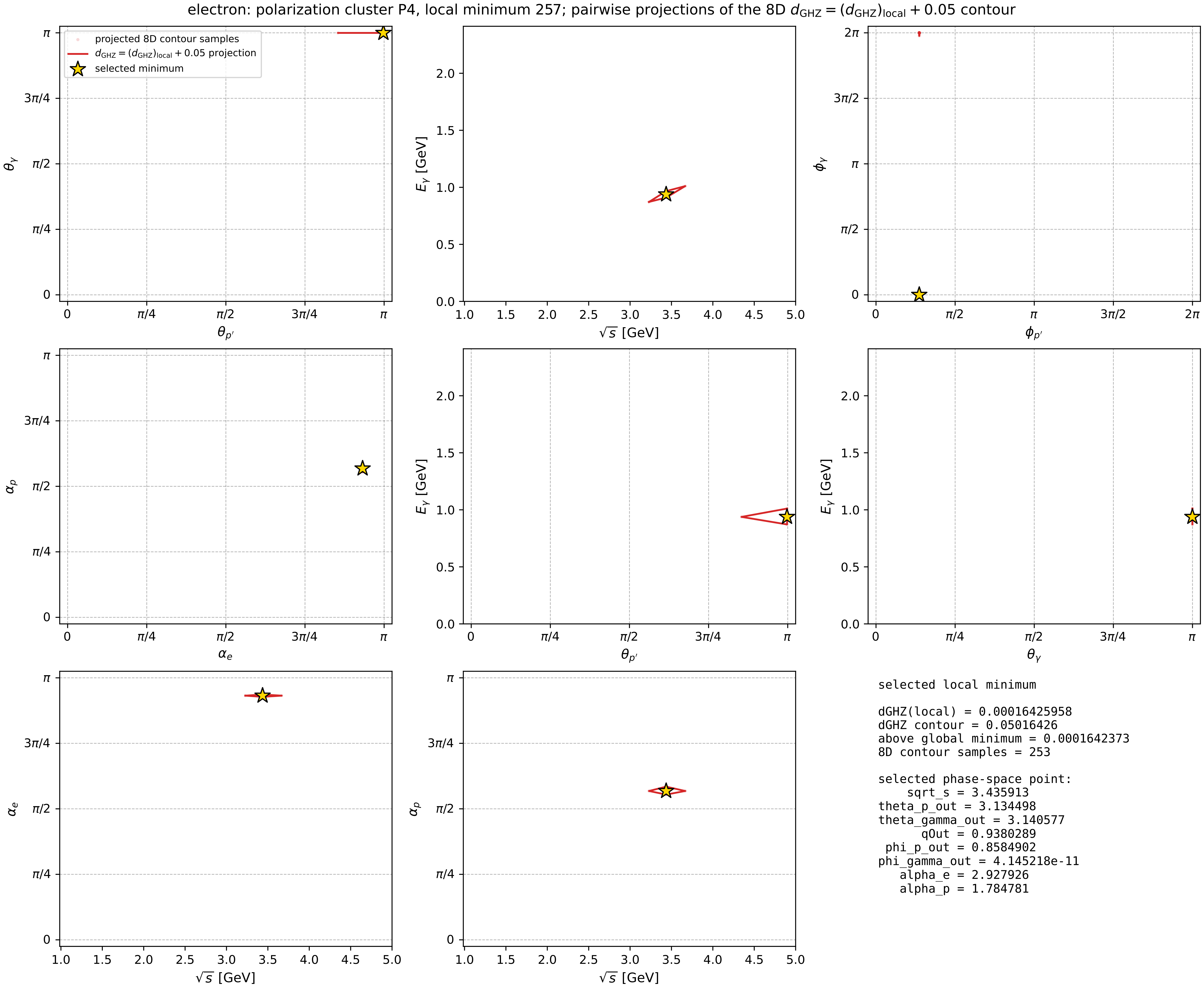




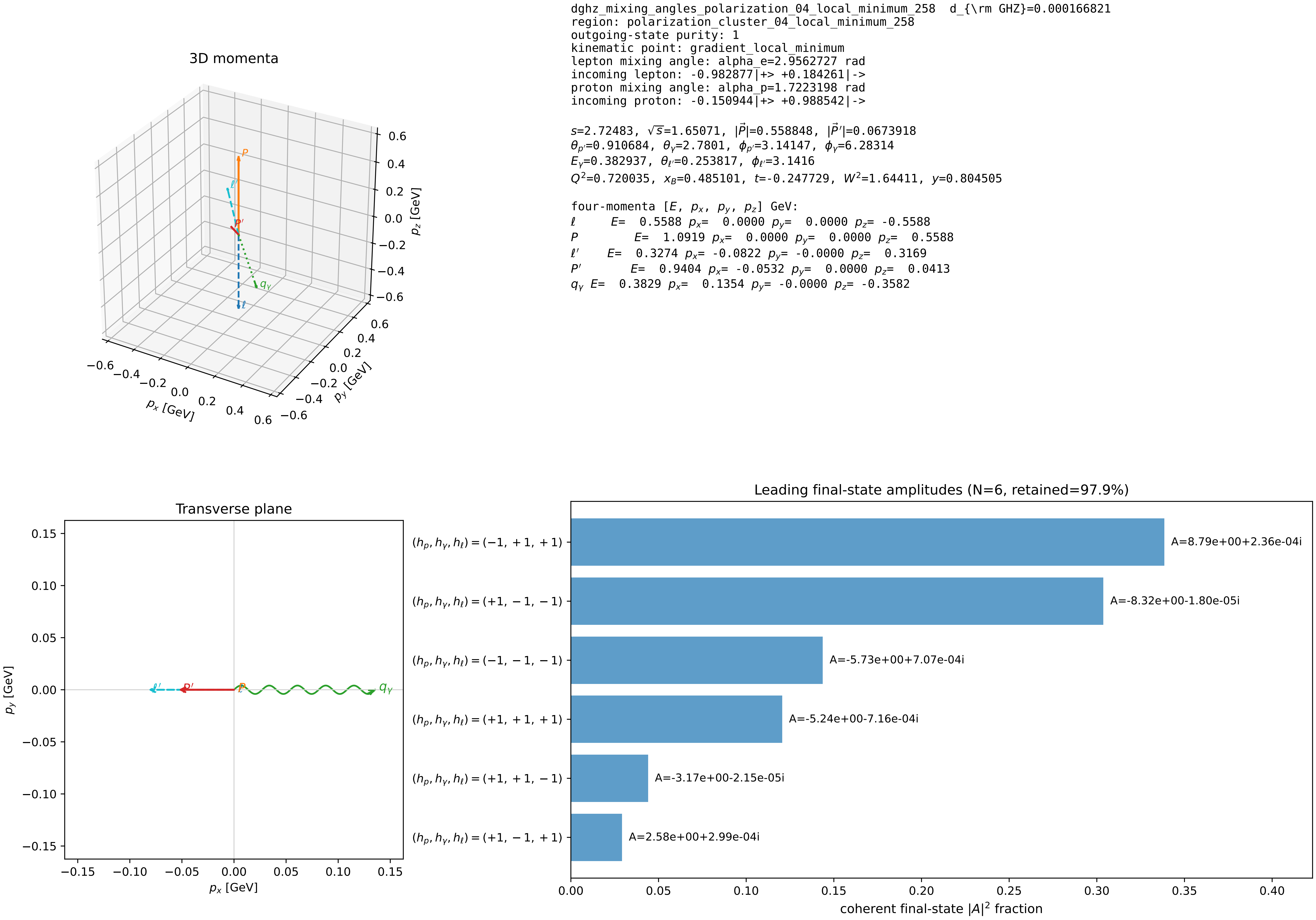
electron: polarization cluster P4, local minimum 252; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour

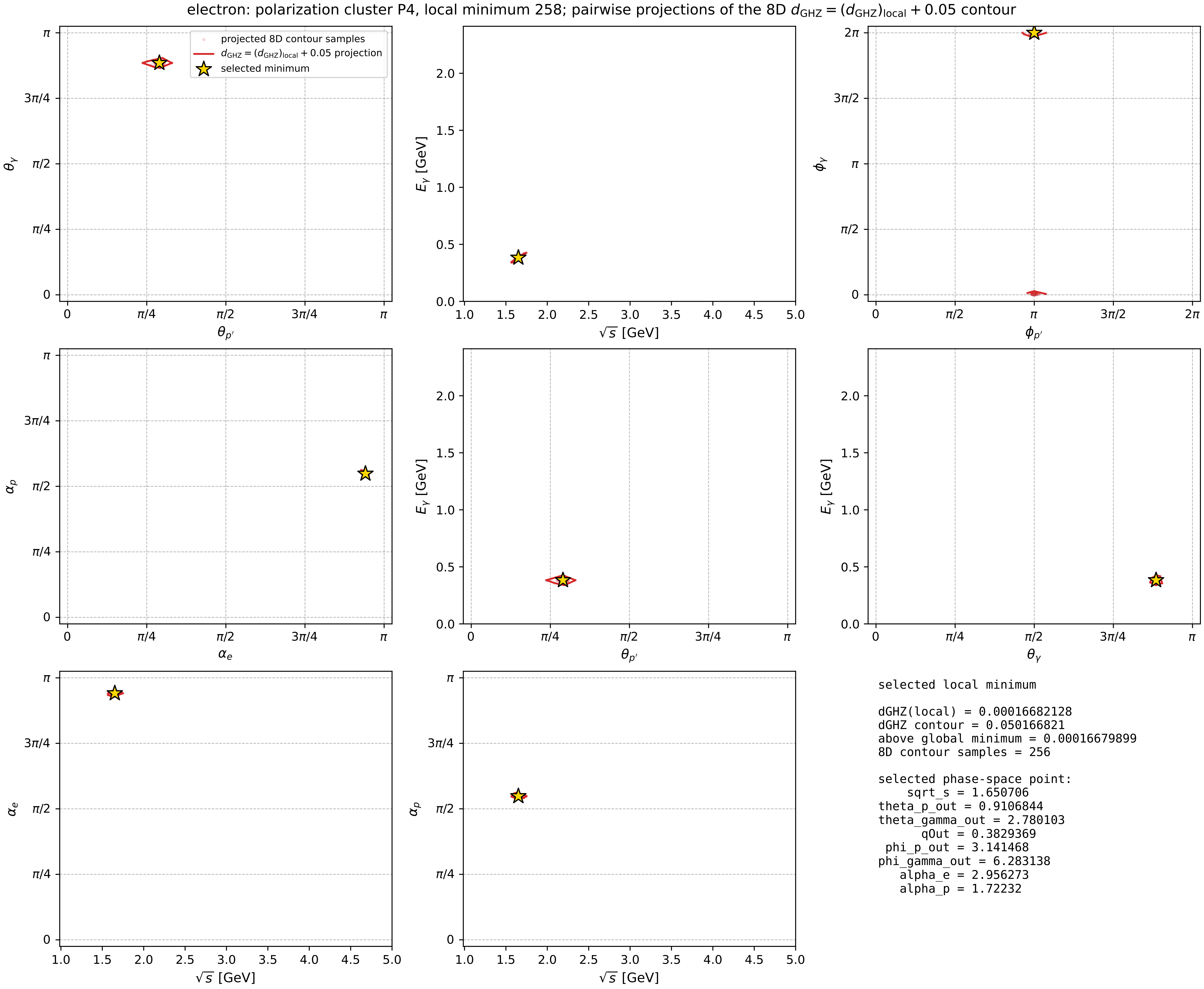



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_257 d_{\rm GHZ}=0.00016426
region: polarization_cluster_04_local_minimum_257
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.9279255 rad
incoming lepton: -0.97726|+> +0.212045|->
proton mixing angle: alpha_p=1.7847812 rad
incoming proton: -0.212356|+> +0.977192|->
s=11.8055, $\sqrt{s}$=3.43591, $|\vec{P}|$=1.58992, $|\vec{P}'|$=0.497905
$\theta_{p'}$=3.1345, θ_γ=3.14058, $\phi_{p'}$=0.85849, ϕ_γ=4.14522e-11
E_γ=0.938029, $\theta_{\ell'}$=0.00293693, $\phi_{\ell'}$=3.82826
Q^2=9.13201, x_B=0.896154, t=-3.74426, W^2=1.93806, y=0.932688
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.5899 p_x= 0.0000 p_y= 0.0000 p_z= -1.5899
P E= 1.8460 p_x= 0.0000 p_y= 0.0000 p_z= 1.5899
ℓ' E= 1.4359 p_x= -0.0033 p_y= -0.0027 p_z= 1.4359
P' E= 1.0620 p_x= 0.0023 p_y= 0.0027 p_z= -0.4979
q_γ E= 0.9380 p_x= 0.0010 p_y= 0.0000 p_z= -0.9380
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=4.36e+01-5.06e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=4.36e+01+5.04e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=1.96e+01-2.28e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.96e+01-2.27e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 257; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 0.00016425958
dGHZ contour = 0.05016426
above global minimum = 0.0001642373
8D contour samples = 253
selected phase-space point:
sqrt_s = 3.435913
theta_p_out = 3.134498
theta_gamma_out = 3.140577
qOut = 0.9380289
phi_p_out = 0.8584902
phi_gamma_out = 4.145218e-11
alpha_e = 2.927926
alpha_p = 1.784781

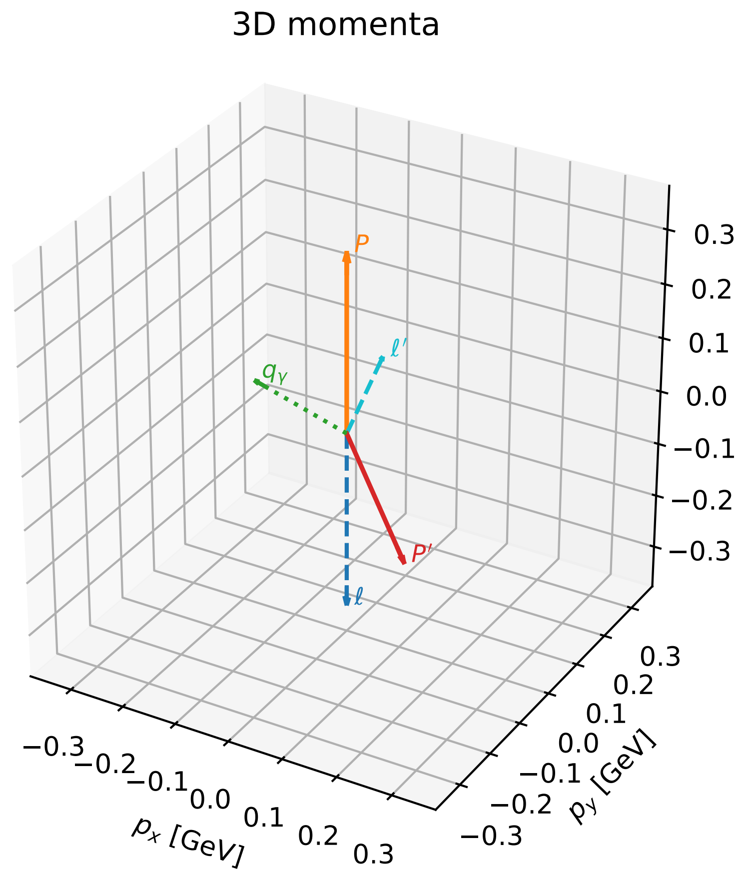
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)





electron: polarization cluster P4, local minimum 258; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 0.00016682128
dGHZ contour = 0.050166821
above global minimum = 0.00016679899
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.650706
theta_p_out = 0.9106844
theta_gamma_out = 2.780103
qOut = 0.3829369
phi_p_out = 3.141468
phi_gamma_out = 6.283138
alpha_e = 2.956273
alpha_p = 1.72232

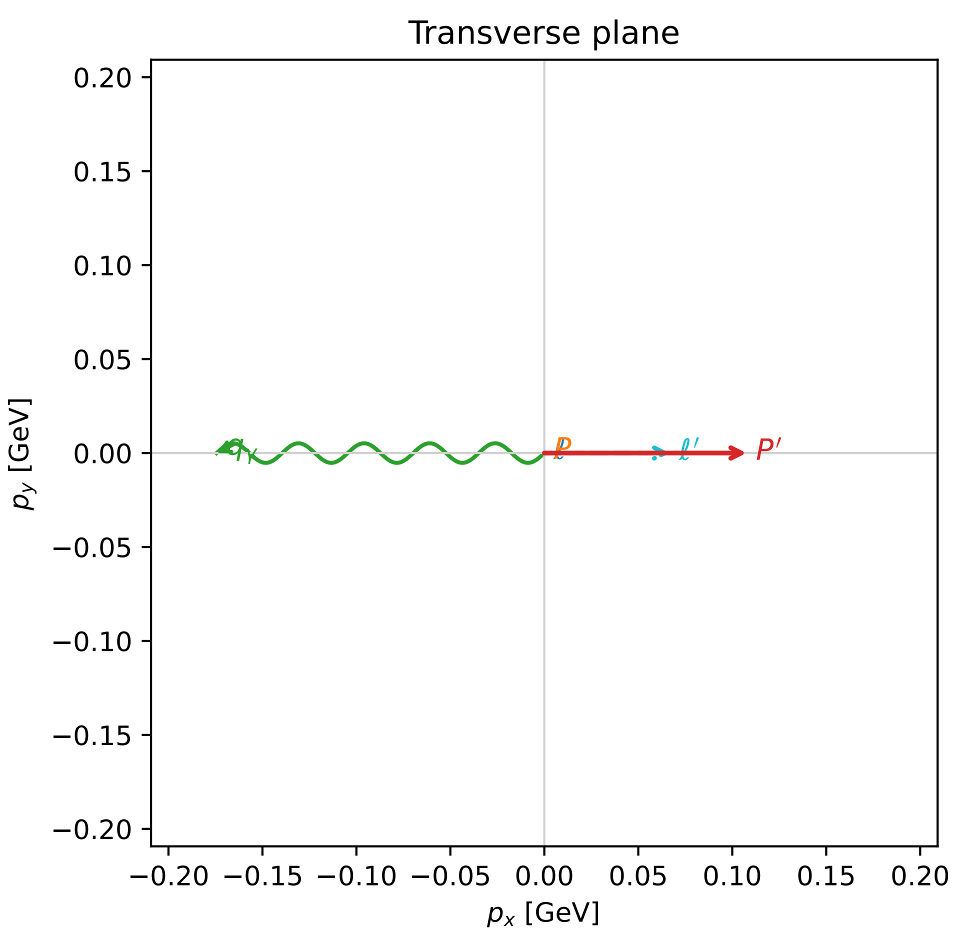
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

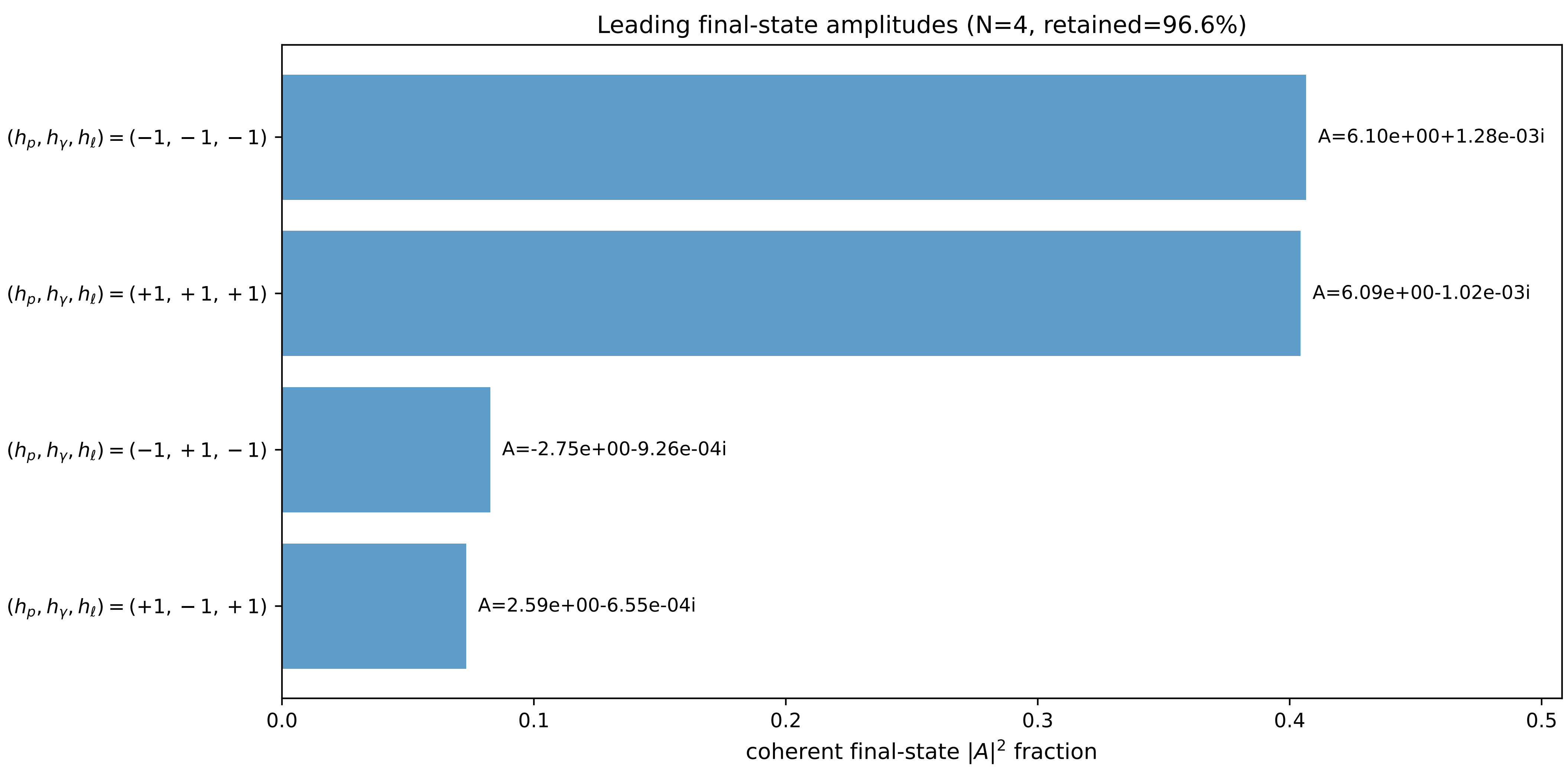


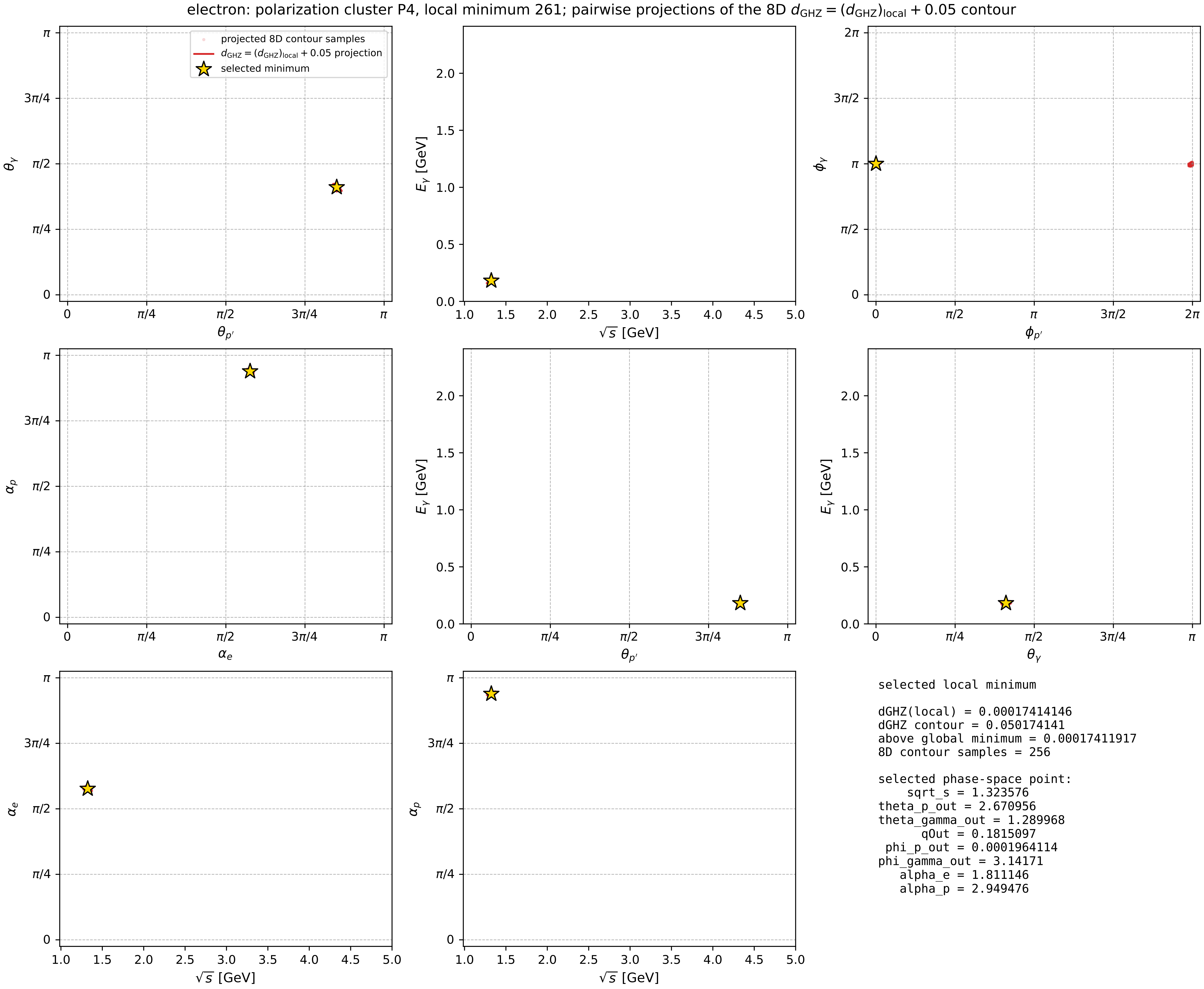
dghz_mixing_angles_polarization_04_local_minimum_261 d_{\rm GHZ}=0.000174141
region: polarization_cluster_04_local_minimum_261
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.8111462 rad
incoming lepton: -0.238042|+> +0.971255|->
proton mixing angle: alpha_p=2.9494759 rad
incoming proton: -0.981602|+> +0.190937|->

s=1.75185, $\sqrt{s}$=1.32358, $|\vec{P}|$=0.329414, $|\vec{P}'|$=0.237332
$\theta_{p'}$=2.67096, θ_γ=1.28997, $\phi_{p'}$=0.000196411, ϕ_γ=3.14171
E_γ=0.18151, $\theta_{\ell'}$=0.392696, $\phi_{\ell'}$=6.28317
Q^2=0.221188, x_B=0.350397, t=-0.303493, W^2=1.28991, y=0.723904

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3294 p_x= 0.0000 p_y= 0.0000 p_z= -0.3294
P E= 0.9942 p_x= 0.0000 p_y= 0.0000 p_z= 0.3294
ℓ' E= 0.1745 p_x= 0.0668 p_y= -0.0000 p_z= 0.1612
P' E= 0.9676 p_x= 0.1076 p_y= 0.0000 p_z= -0.2115
q_γ E= 0.1815 p_x= -0.1744 p_y= -0.0000 p_z= 0.0503

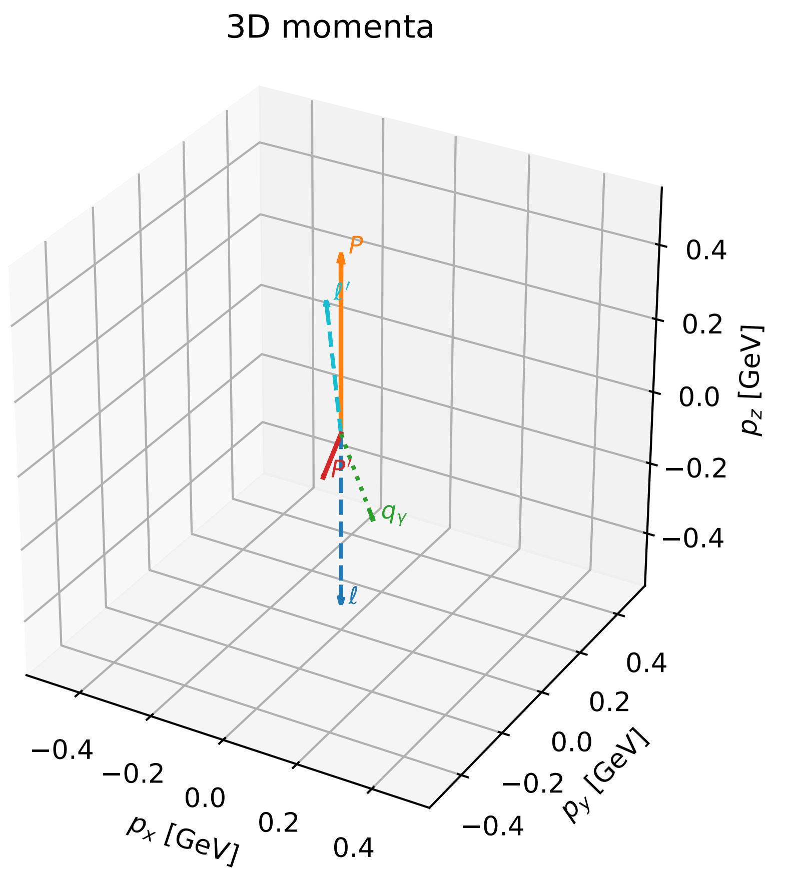






electron: polarization cluster P4, local minimum 261; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 0.00017414146
dGHZ contour = 0.050174141
above global minimum = 0.00017411917
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.323576
theta_p_out = 2.670956
theta_gamma_out = 1.289968
qOut = 0.1815097
phi_p_out = 0.0001964114
phi_gamma_out = 3.14171
alpha_e = 1.811146
alpha_p = 2.949476

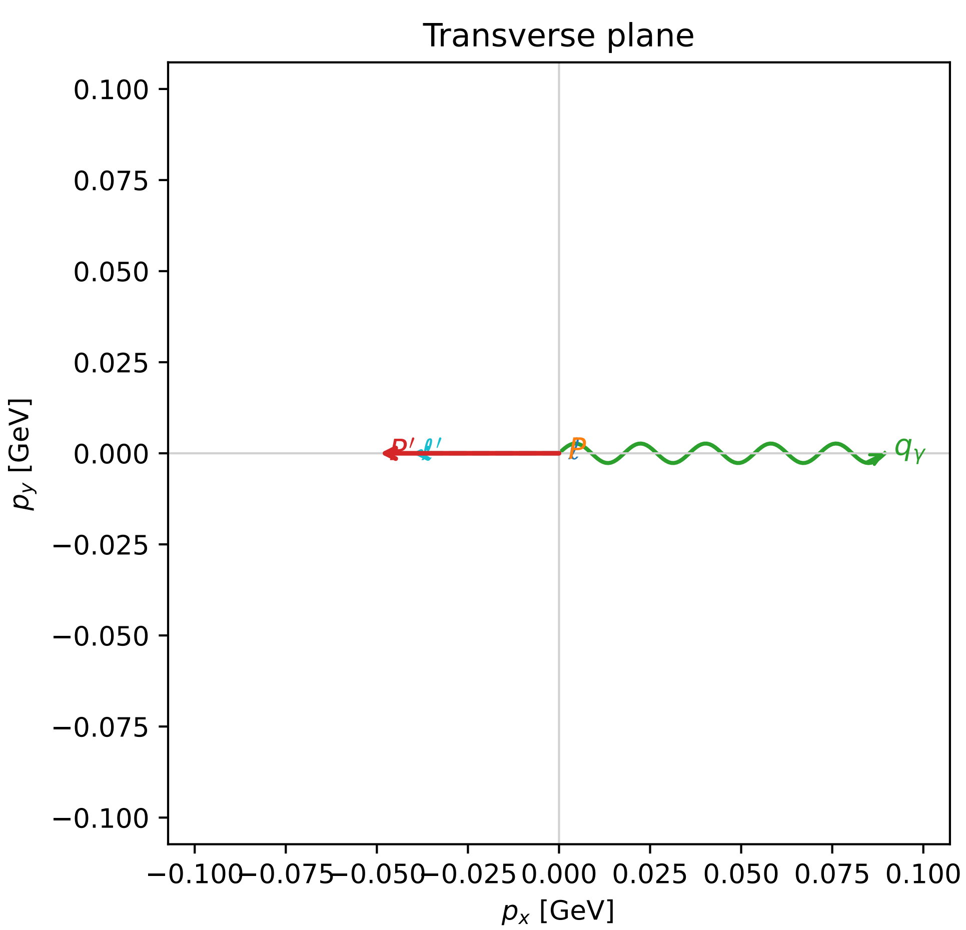
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

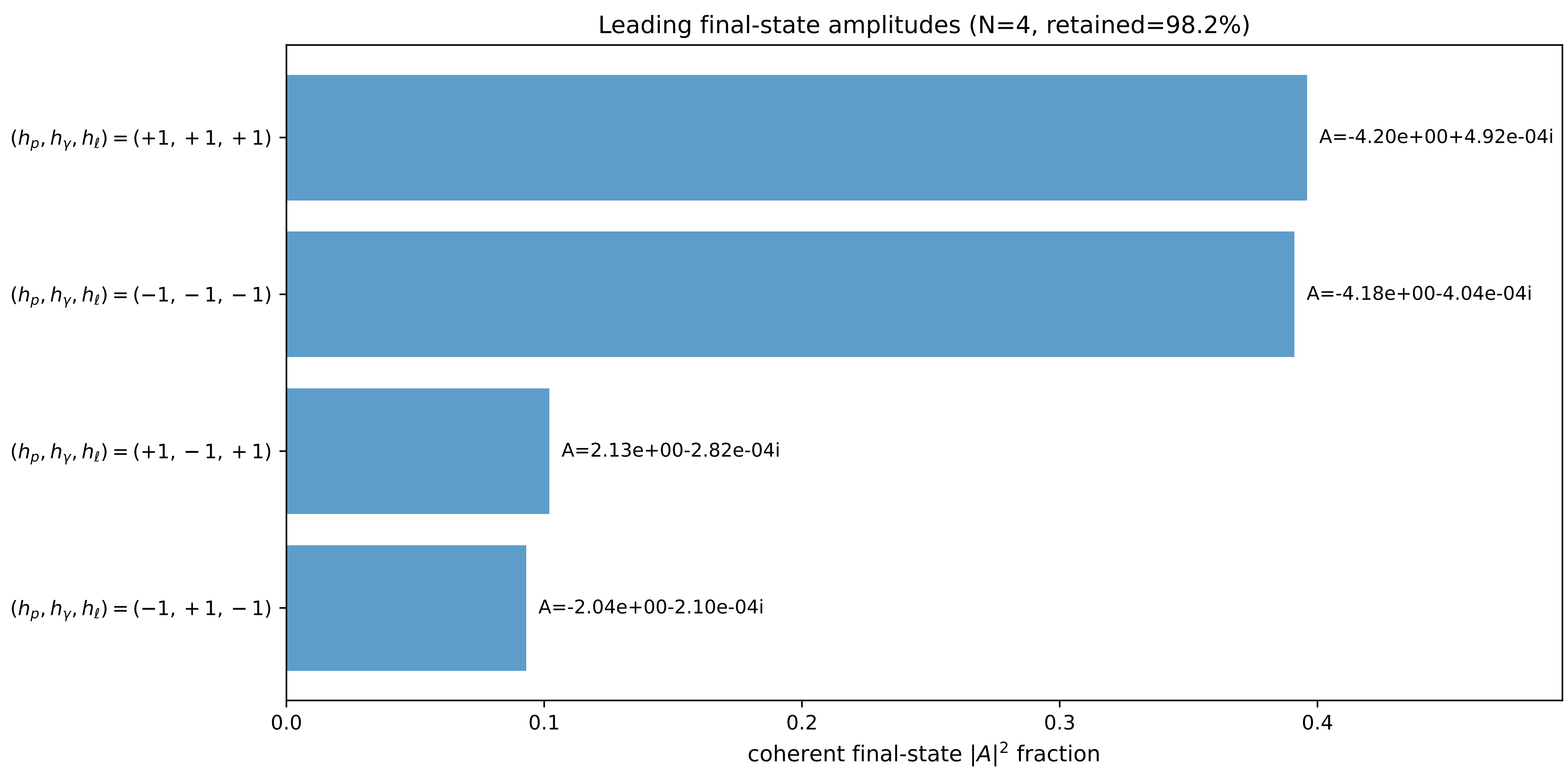


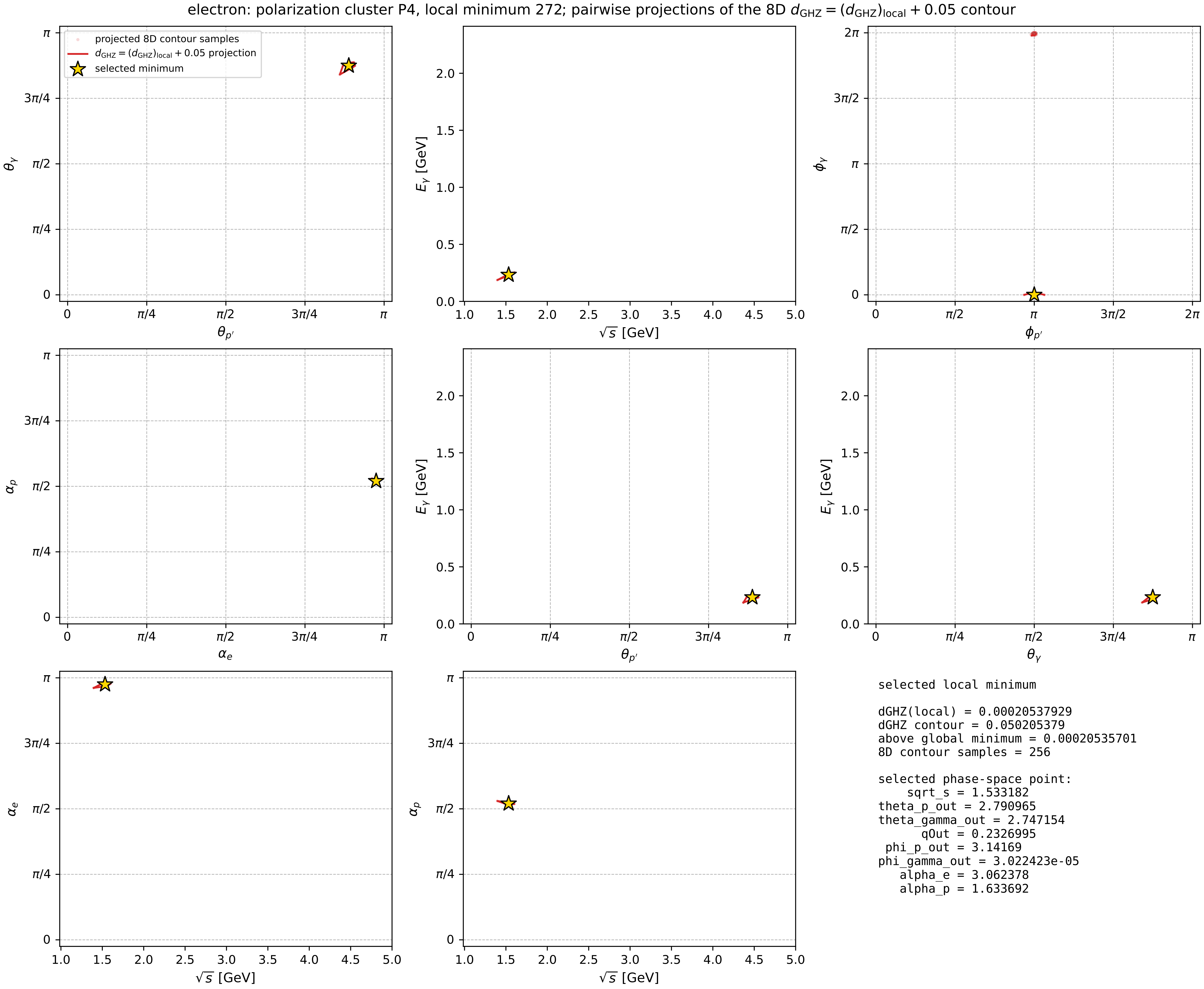
dghz_mixing_angles_polarization_04_local_minimum_272 d_{\rm GHZ}=0.000205379
region: polarization_cluster_04_local_minimum_272
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=3.062378 rad
incoming lepton: -0.996864|+> +0.0791318|->
proton mixing angle: alpha_p=1.6336923 rad
incoming proton: -0.0628545|+> +0.998023|->

s=2.35065, $\sqrt{s}$=1.53318, $|\vec{P}|$=0.479657, $|\vec{P}'|$=0.143187
$\theta_{p'}$=2.79096, θ_γ=2.74715, $\phi_{p'}$=3.14169, ϕ_γ=3.02242e-05
E_γ=0.2327, $\theta_{\ell'}$=0.114697, $\phi_{\ell'}$=3.14154
Q^2=0.672405, x_B=0.631352, t=-0.368624, W^2=1.27246, y=0.72411

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4797 p_x= 0.0000 p_y= 0.0000 p_z= -0.4797
P E= 1.0535 p_x= 0.0000 p_y= 0.0000 p_z= 0.4797
ℓ' E= 0.3516 p_x= -0.0402 p_y= 0.0000 p_z= 0.3493
P' E= 0.9489 p_x= -0.0492 p_y= -0.0000 p_z= -0.1345
q_γ E= 0.2327 p_x= 0.0894 p_y= 0.0000 p_z= -0.2148

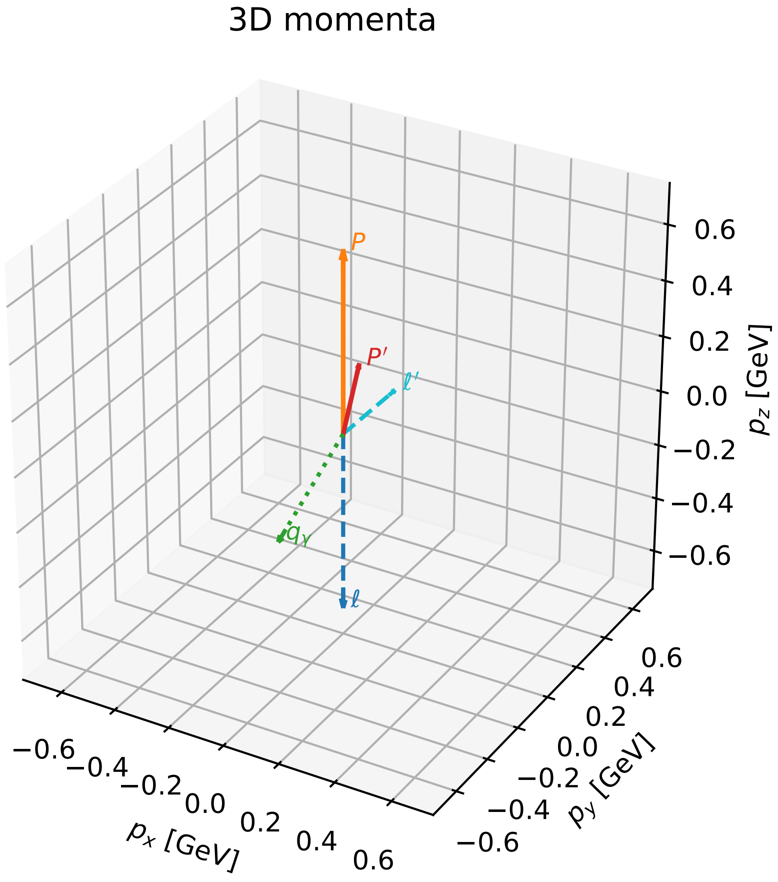







```
selected local minimum

dGHZ(local) = 0.00020537929
dGHZ contour = 0.050205379
above global minimum = 0.00020535701
8D contour samples = 256

selected phase-space point:
    sqrt_s = 1.533182
theta_p_out = 2.790965
theta_gamma_out = 2.747154
      qOut = 0.2326995
 phi_p_out = 3.14169
phi_gamma_out = 3.022423e-05
   alpha_e = 3.062378
   alpha_p = 1.633692
```

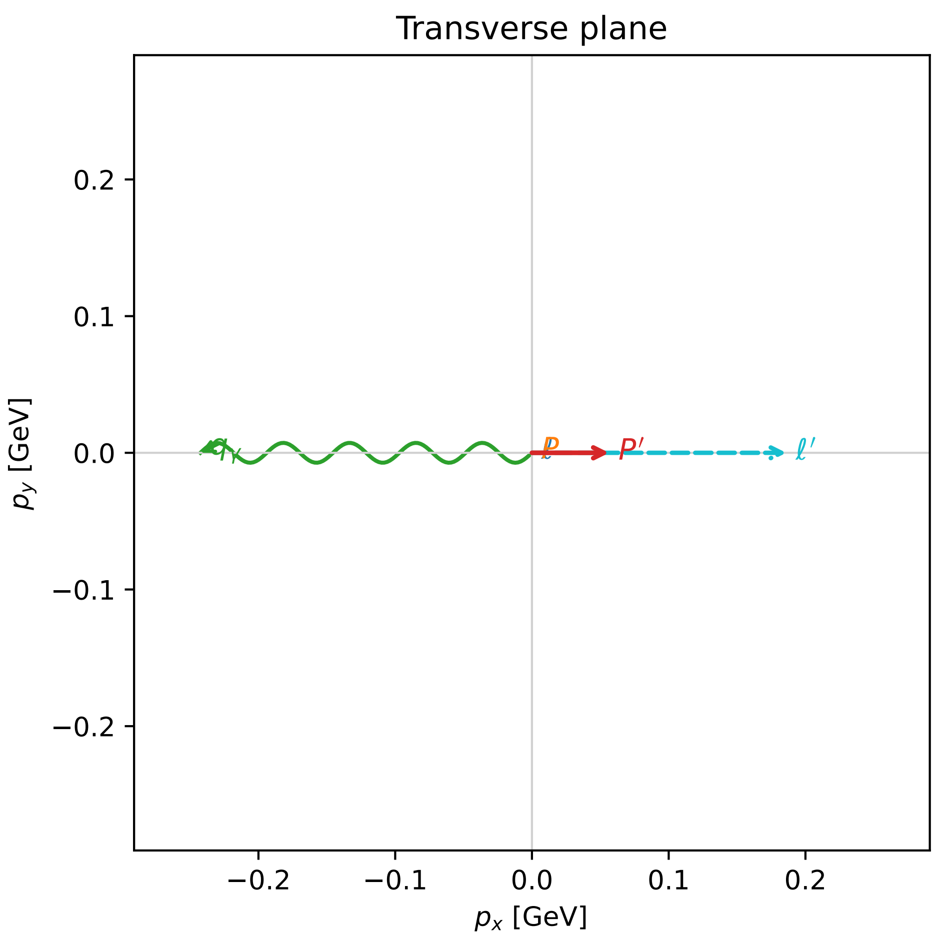
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

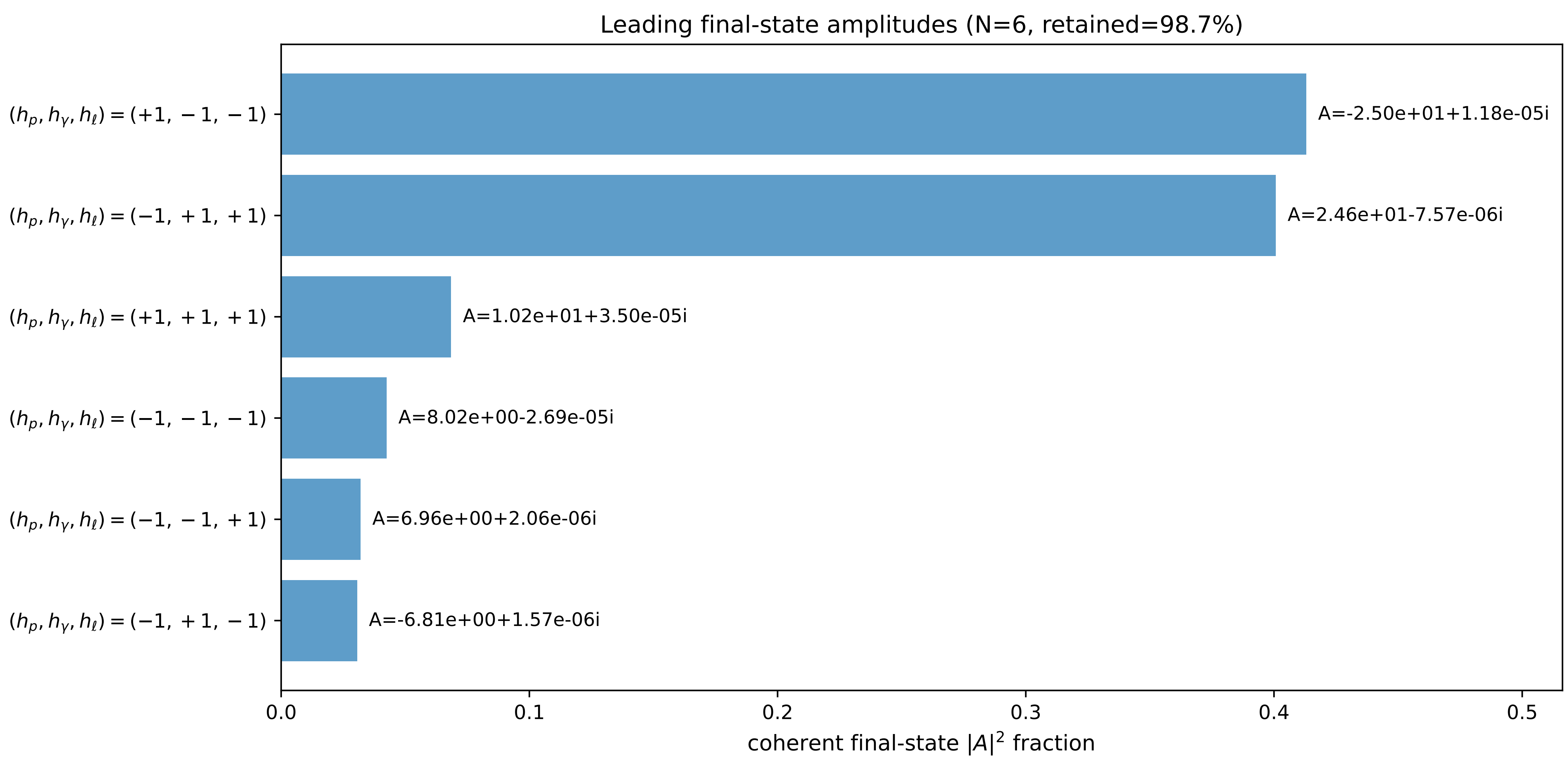


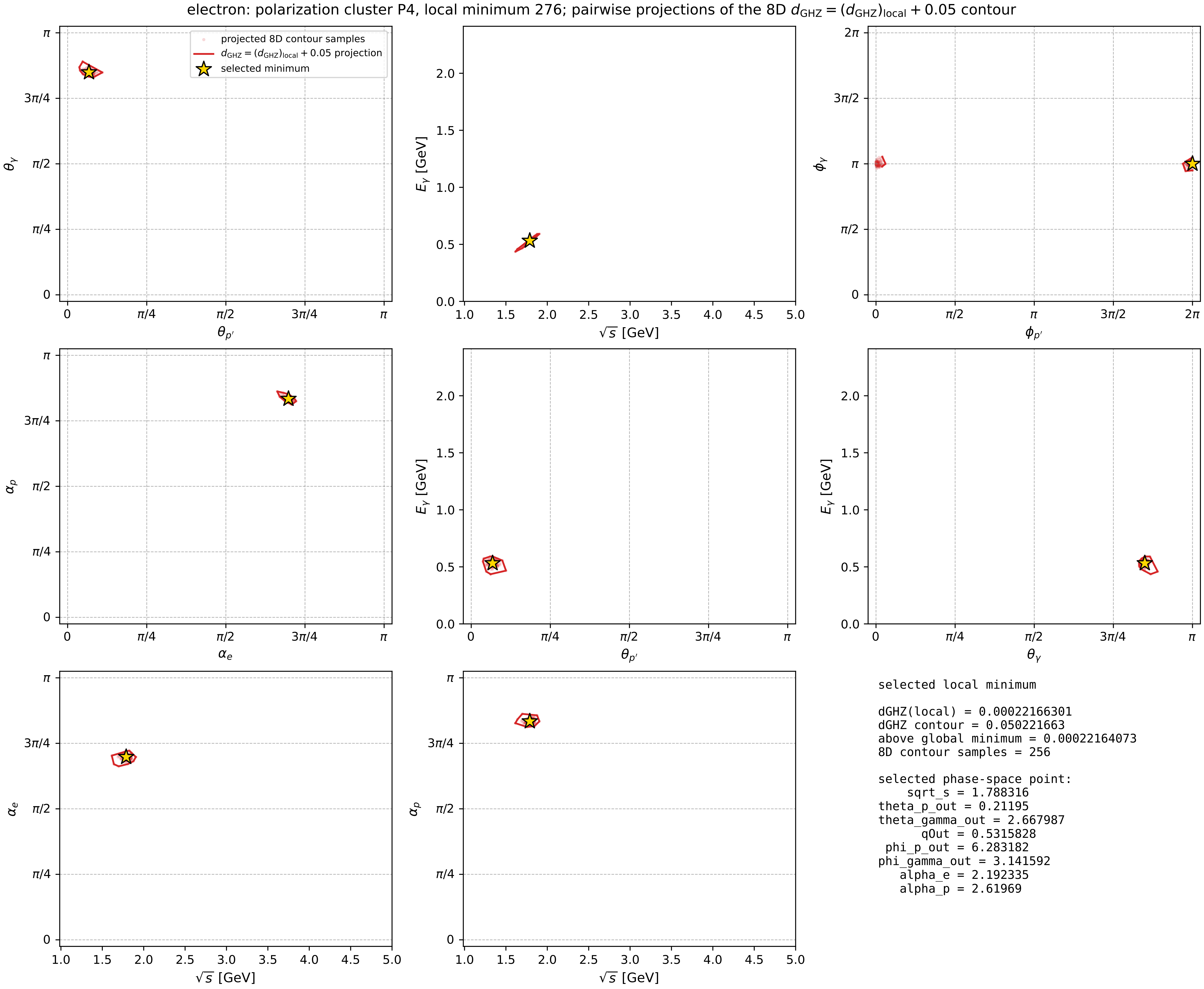
dghz_mixing_angles_polarization_04_local_minimum_276 d_{\rm GHZ}=0.000221663
region: polarization_cluster_04_local_minimum_276
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.1923354 rad
incoming lepton: -0.582287|+> +0.812983|->
proton mixing angle: alpha_p=2.6196902 rad
incoming proton: -0.866872|+> +0.49853|->

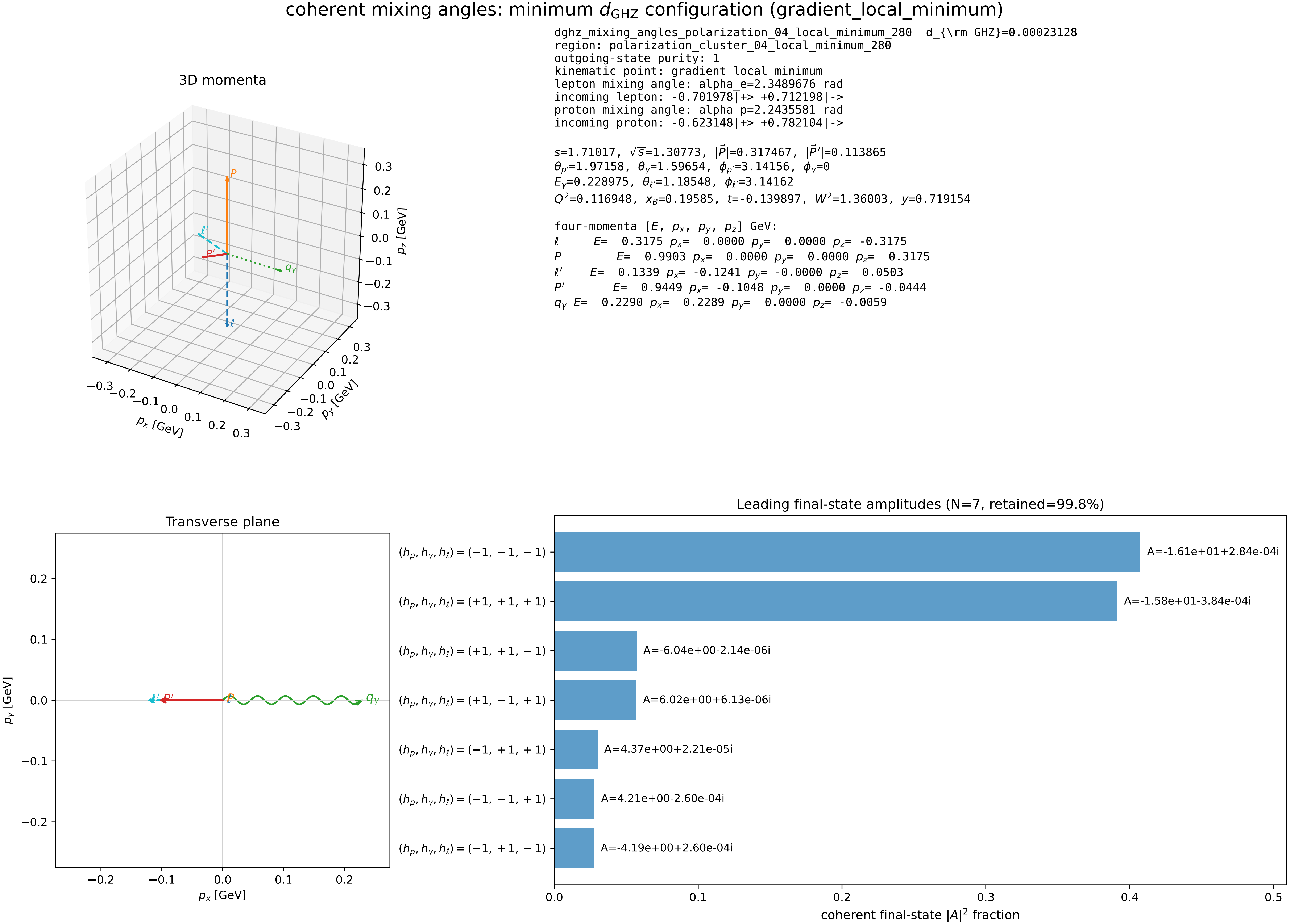
s=3.19807, $\sqrt{s}$=1.78832, $|\vec{P}|$=0.64816, $|\vec{P}'|$=0.268302
$\theta_{p'}$=0.21195, θ_γ=2.66799, $\phi_{p'}$=6.28318, ϕ_γ=3.14159
E_γ=0.531583, $\theta_{\ell'}$=0.723074, $\phi_{\ell'}$=4.63436e-07
Q^2=0.637645, x_B=0.326926, t=-0.125002, W^2=2.19263, y=0.841344

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.6482 p_x= 0.0000 p_y= 0.0000 p_z= -0.6482
P E= 1.1402 p_x= 0.0000 p_y= 0.0000 p_z= 0.6482
ℓ' E= 0.2811 p_x= 0.1860 p_y= 0.0000 p_z= 0.2108
P' E= 0.9756 p_x= 0.0564 p_y= -0.0000 p_z= 0.2623
q_γ E= 0.5316 p_x= -0.2425 p_y= 0.0000 p_z= -0.4731

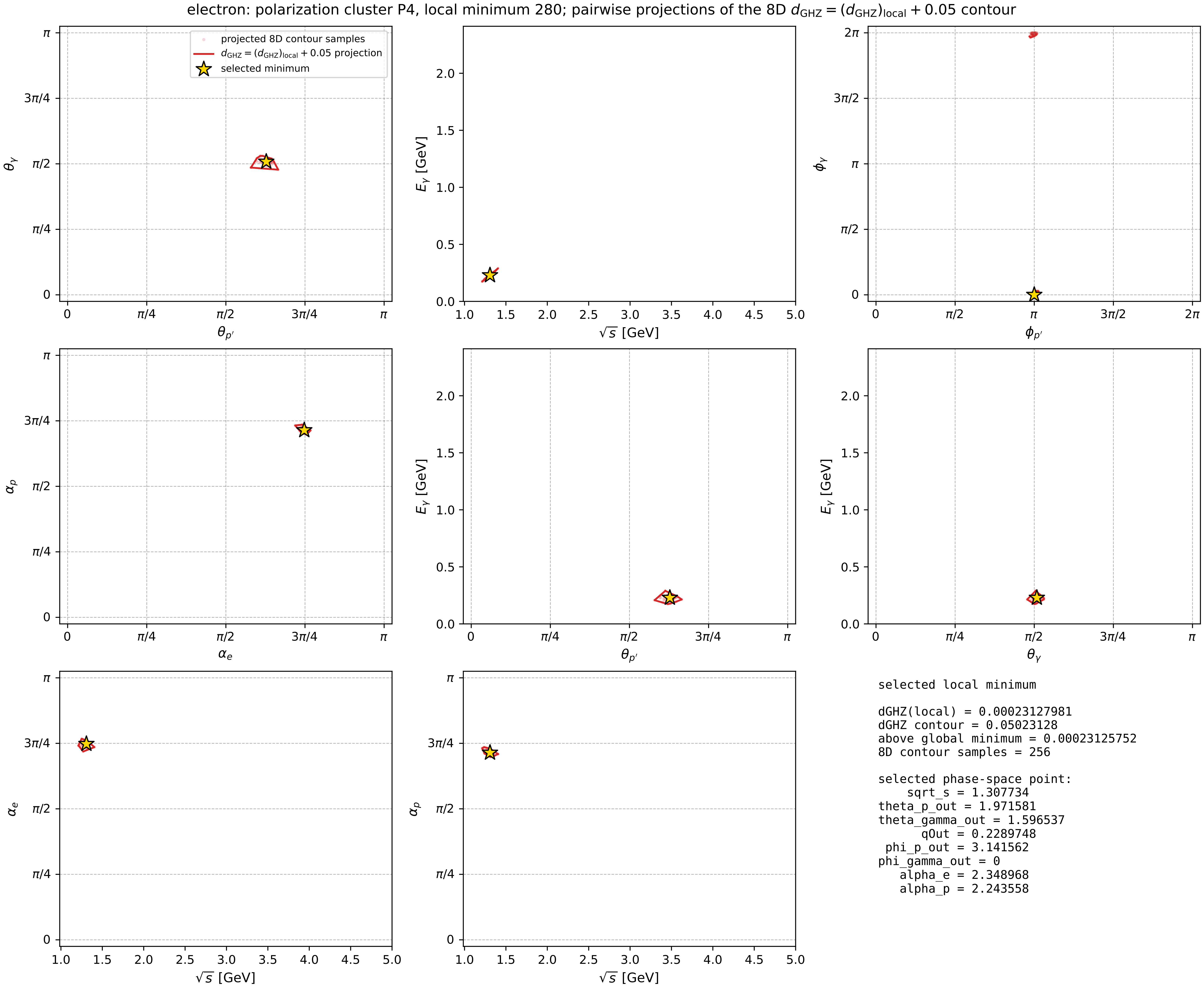


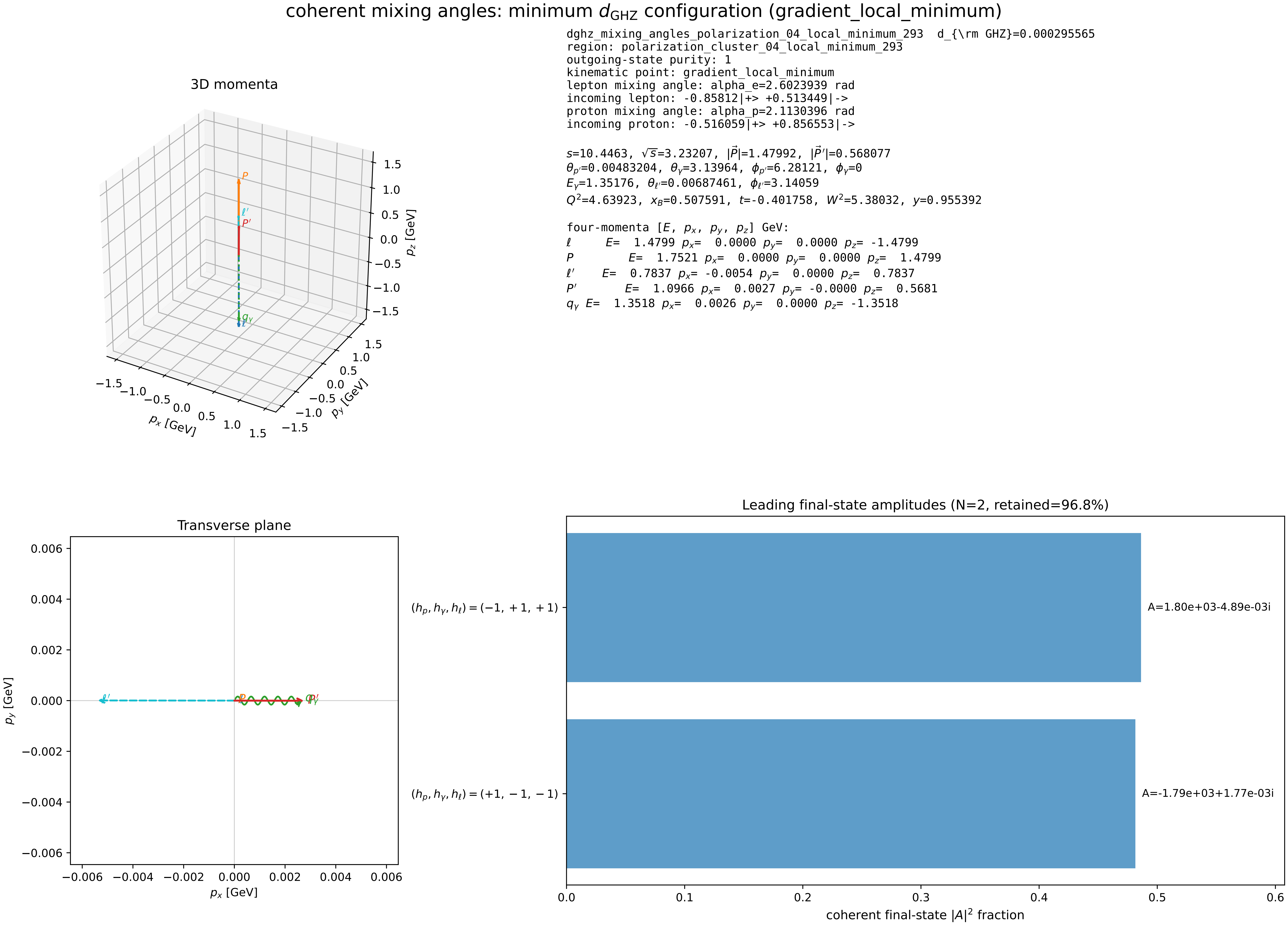



electron: polarization cluster P4, local minimum 276; pairwise projections of the 8D $d_{GHZ} = (d_{GHZ})_{local} + 0.05$ contour
projected 8D contour samples
$d_{GHZ} = (d_{GHZ})_{local} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 0.00022166301
dGHZ contour = 0.050221663
above global minimum = 0.00022164073
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.788316
theta_p_out = 0.21195
theta_gamma_out = 2.667987
qOut = 0.5315828
phi_p_out = 6.283182
phi_gamma_out = 3.141592
alpha_e = 2.192335
alpha_p = 2.61969

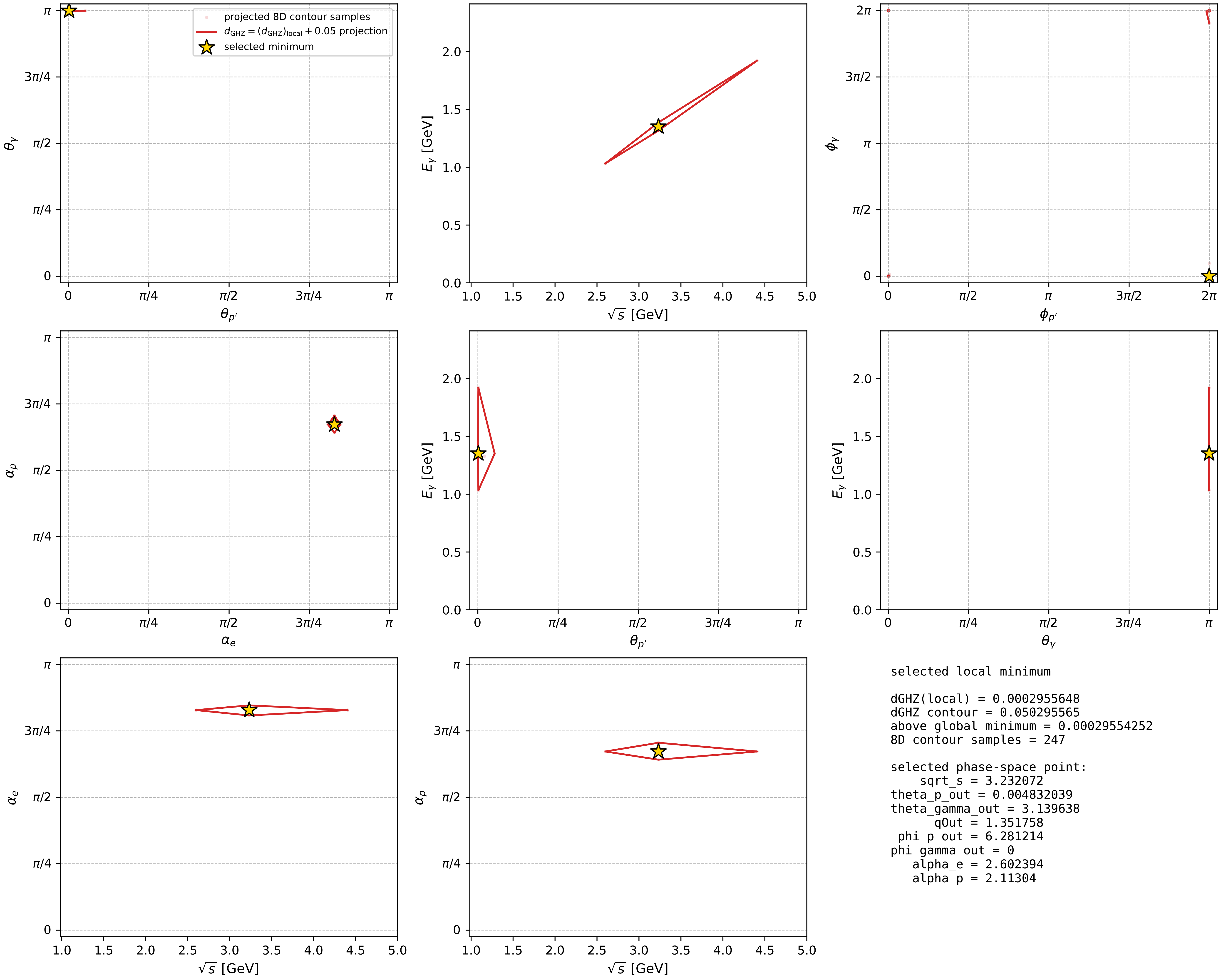


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_280 d_{\rm GHZ}=0.00023128
region: polarization_cluster_04_local_minimum_280
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3489676 rad
incoming lepton: -0.701978|+> +0.712198|->
proton mixing angle: alpha_p=2.2435581 rad
incoming proton: -0.623148|+> +0.782104|->
s=1.71017, $\sqrt{s}$=1.30773, $|\vec{P}|$=0.317467, $|\vec{P}'|$=0.113865
$\theta_{p'}$=1.97158, θ_γ=1.59654, $\phi_{p'}$=3.14156, ϕ_γ=0
E_γ=0.228975, $\theta_{\ell'}$=1.18548, $\phi_{\ell'}$=3.14162
Q^2=0.116948, x_B=0.19585, t=-0.139897, W^2=1.36003, y=0.719154
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3175 p_x= 0.0000 p_y= 0.0000 p_z= -0.3175
P E= 0.9903 p_x= 0.0000 p_y= 0.0000 p_z= 0.3175
ℓ' E= 0.1339 p_x= -0.1241 p_y= -0.0000 p_z= 0.0503
P' E= 0.9449 p_x= -0.1048 p_y= 0.0000 p_z= -0.0444
q_γ E= 0.2290 p_x= 0.2289 p_y= 0.0000 p_z= -0.0059
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=7, retained=99.8%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-1.61e+01+2.84e-04i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-1.58e+01-3.84e-04i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-6.04e+00-2.14e-06i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=6.02e+00+6.13e-06i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=4.37e+00+2.21e-05i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=4.21e+00-2.60e-04i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-4.19e+00+2.60e-04i
coherent final-state $|A|^2$ fraction

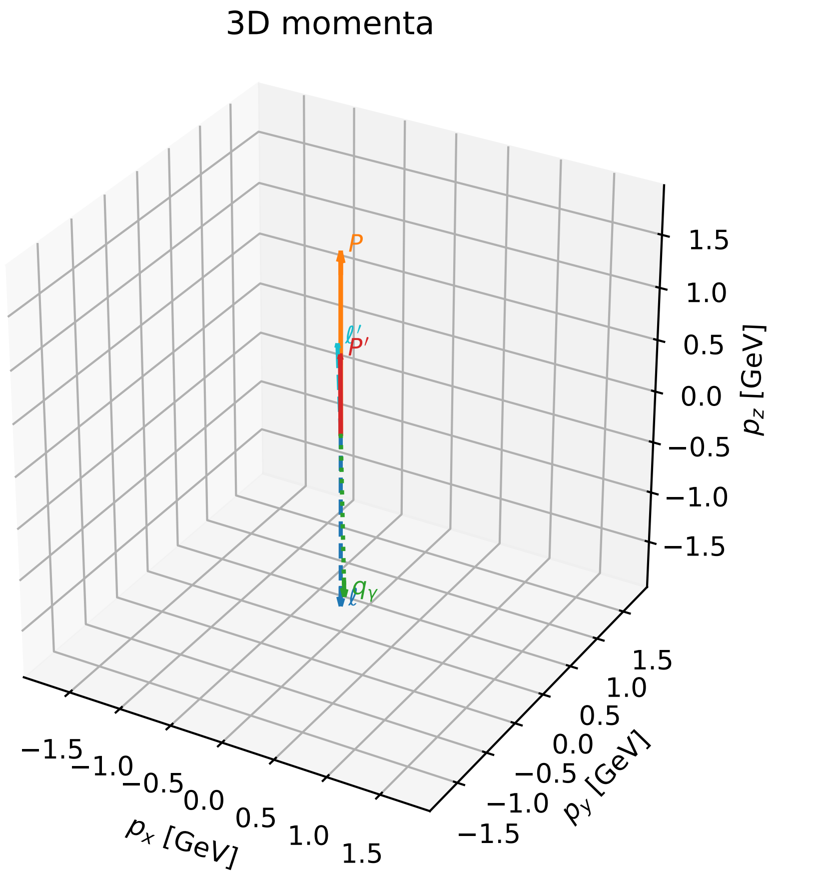

electron: polarization cluster P4, local minimum 280; pairwise projections of the 8D $d_{GHZ} = (d_{GHZ})_{local} + 0.05$ contour
projected 8D contour samples
$d_{GHZ} = (d_{GHZ})_{local} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 0.00023127981
dGHZ contour = 0.05023128
above global minimum = 0.00023125752
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.307734
theta_p_out = 1.971581
theta_gamma_out = 1.596537
qOut = 0.2289748
phi_p_out = 3.141562
phi_gamma_out = 0
alpha_e = 2.348968
alpha_p = 2.243558


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_293 d_{\rm GHZ}=0.000295565
region: polarization_cluster_04_local_minimum_293
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.6023939 rad
incoming lepton: -0.85812|+> +0.513449|->
proton mixing angle: alpha_p=2.1130396 rad
incoming proton: -0.516059|+> +0.856553|->
s=10.4463, $\sqrt{s}$=3.23207, $|\vec{P}|$=1.47992, $|\vec{P}'|$=0.568077
$\theta_{p'}$=0.00483204, θ_γ=3.13964, $\phi_{p'}$=6.28121, ϕ_γ=0
E_γ=1.35176, $\theta_{\ell'}$=0.00687461, $\phi_{\ell'}$=3.14059
Q^2=4.63923, x_B=0.507591, t=-0.401758, W^2=5.38032, y=0.955392
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.4799 p_x= 0.0000 p_y= 0.0000 p_z= -1.4799
P E= 1.7521 p_x= 0.0000 p_y= 0.0000 p_z= 1.4799
ℓ' E= 0.7837 p_x= -0.0054 p_y= 0.0000 p_z= 0.7837
P' E= 1.0966 p_x= 0.0027 p_y= -0.0000 p_z= 0.5681
q_γ E= 1.3518 p_x= 0.0026 p_y= 0.0000 p_z= -1.3518
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
ℓ'
P'
q_γ
ℓ
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=96.8%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=1.80e+03-4.89e-03i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-1.79e+03+1.77e-03i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P4, local minimum 293; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



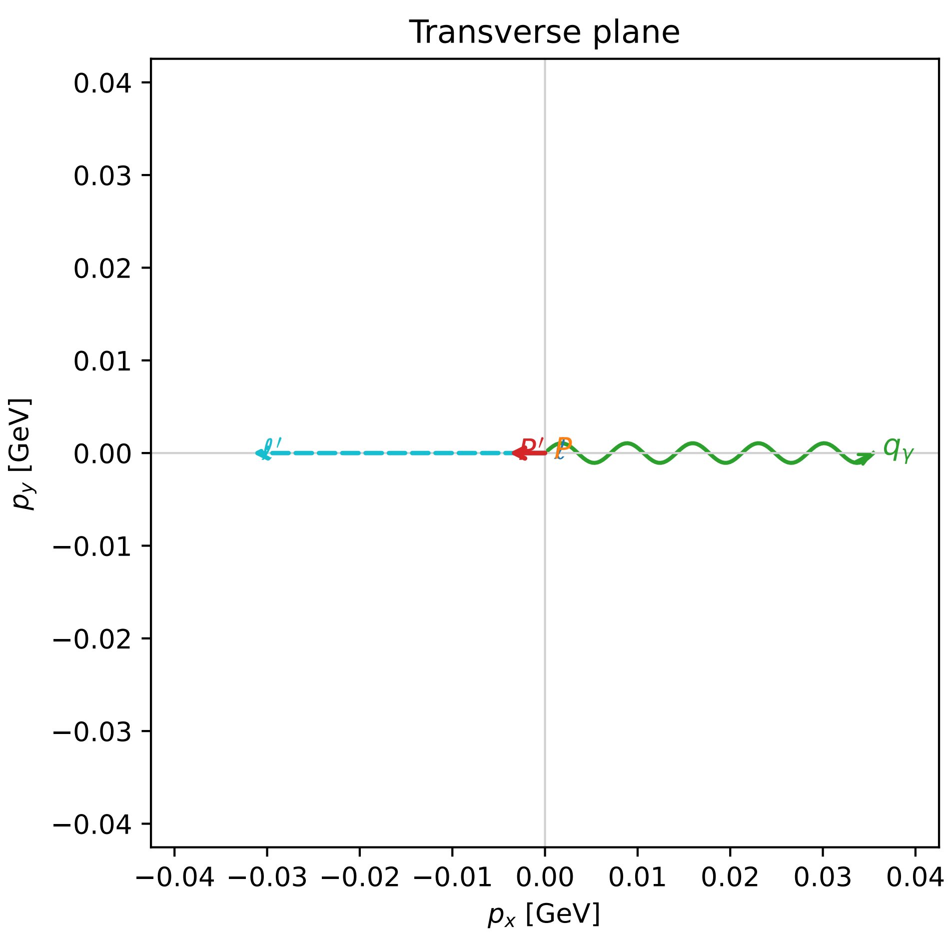
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

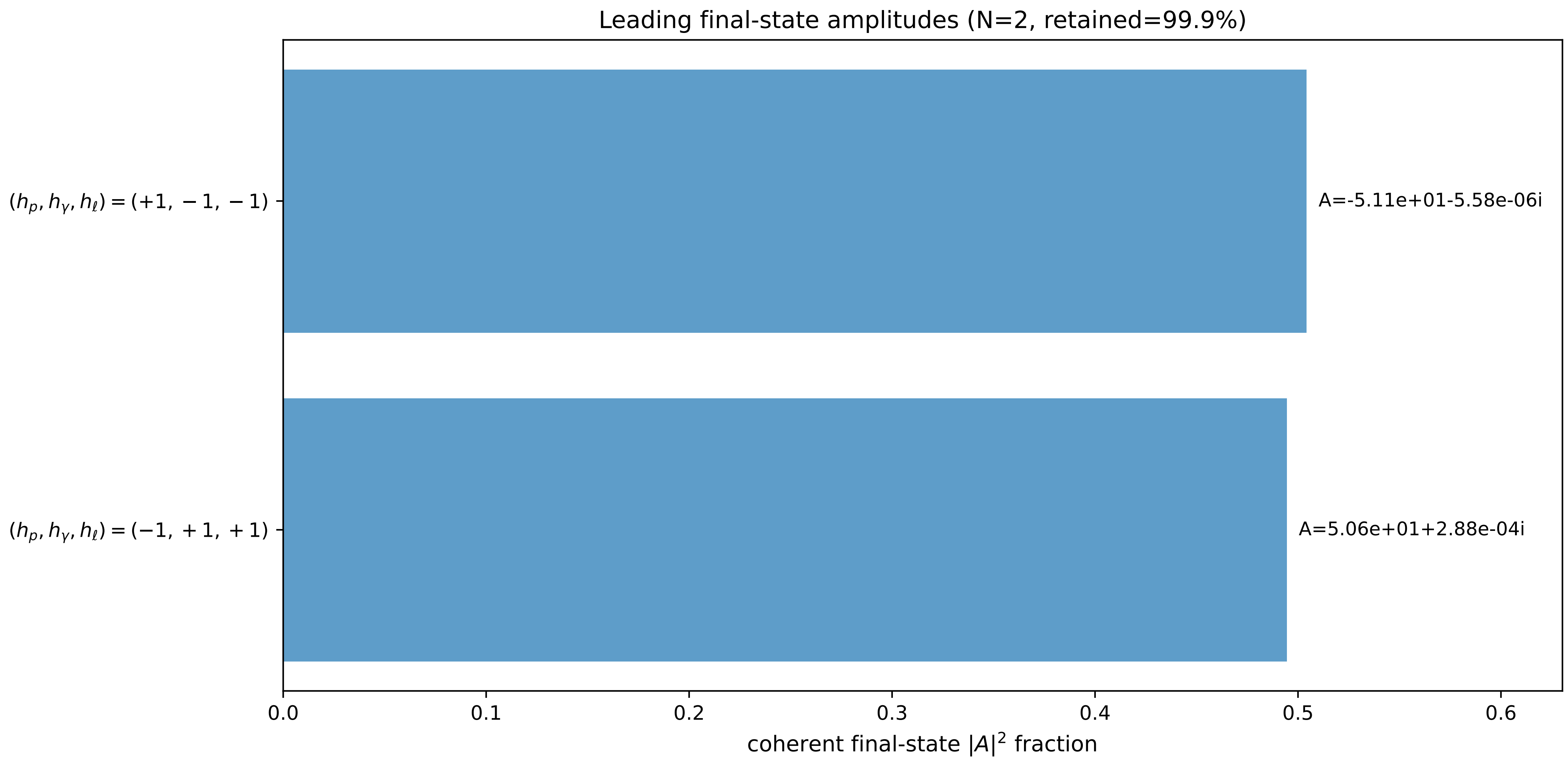


dghz_mixing_angles_polarization_04_local_minimum_297 d_{\rm GHZ}=0.000306617
region: polarization_cluster_04_local_minimum_297
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=3.0107596 rad
incoming lepton: -0.991454|+> +0.13046|->
proton mixing angle: alpha_p=1.6989407 rad
incoming proton: -0.127794|+> +0.991801|->

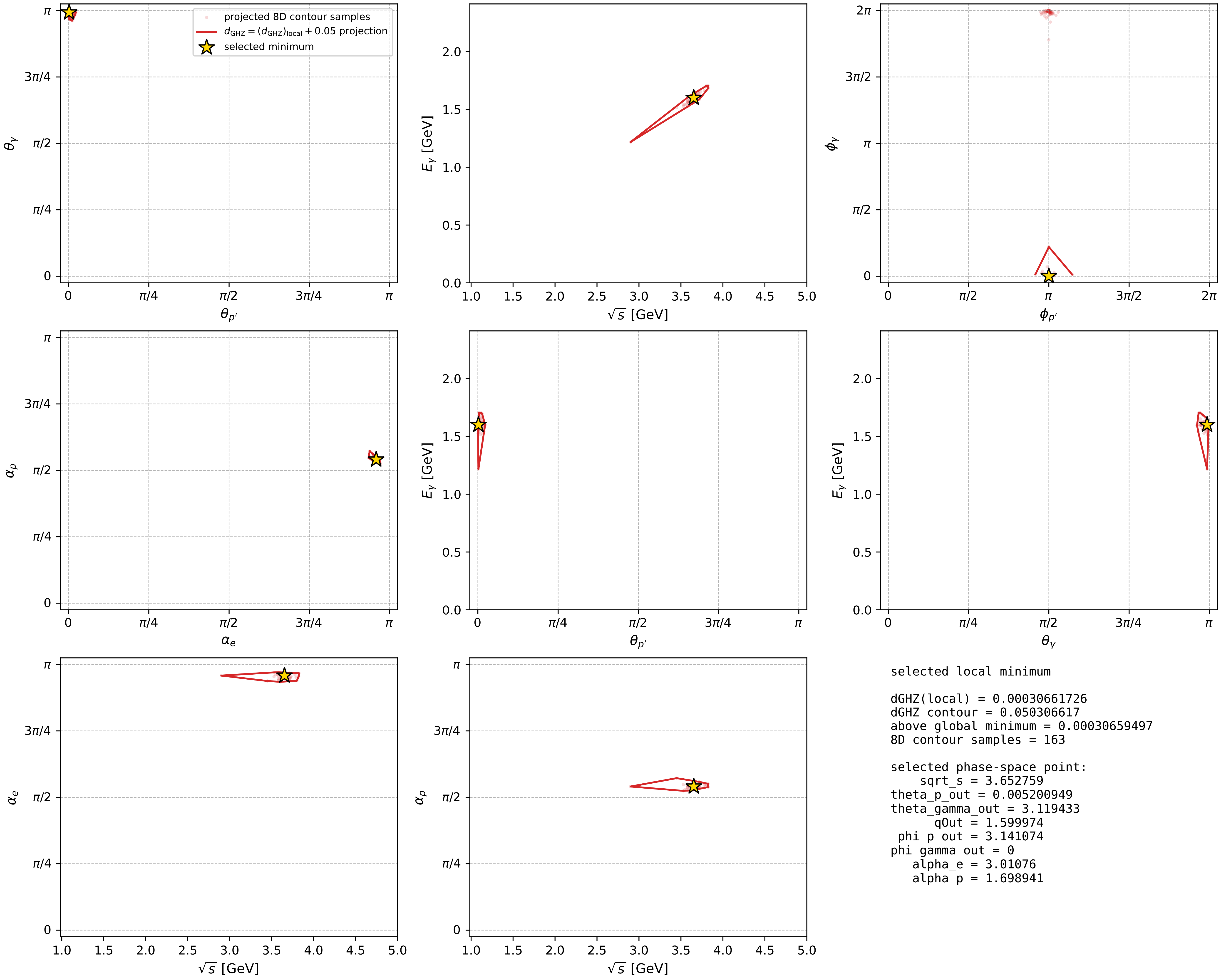
s=13.3426, $\sqrt{s}$=3.65276, $|\vec{P}|$=1.70594, $|\vec{P}'|$=0.745662
$\theta_{p'}$=0.00520095, θ_γ=3.11943, $\phi_{p'}$=3.14107, ϕ_γ=0
E_γ=1.59997, $\theta_{\ell'}$=0.0369583, $\phi_{\ell'}$=3.14166
Q^2=5.82901, x_B=0.483769, t=-0.361859, W^2=7.09999, y=0.96681

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.7059 p_x= 0.0000 p_y= 0.0000 p_z= -1.7059
P E= 1.9468 p_x= 0.0000 p_y= 0.0000 p_z= 1.7059
ℓ' E= 0.8545 p_x= -0.0316 p_y= -0.0000 p_z= 0.8539
P' E= 1.1983 p_x= -0.0039 p_y= 0.0000 p_z= 0.7457
q_γ E= 1.6000 p_x= 0.0355 p_y= 0.0000 p_z= -1.5996

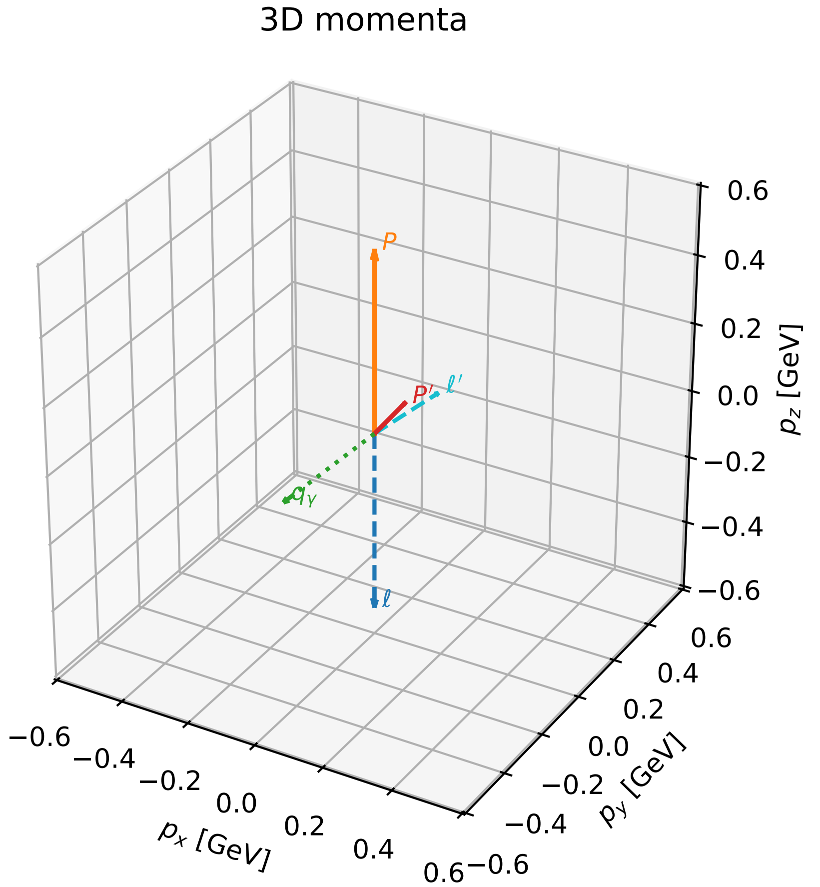




electron: polarization cluster P4, local minimum 297; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



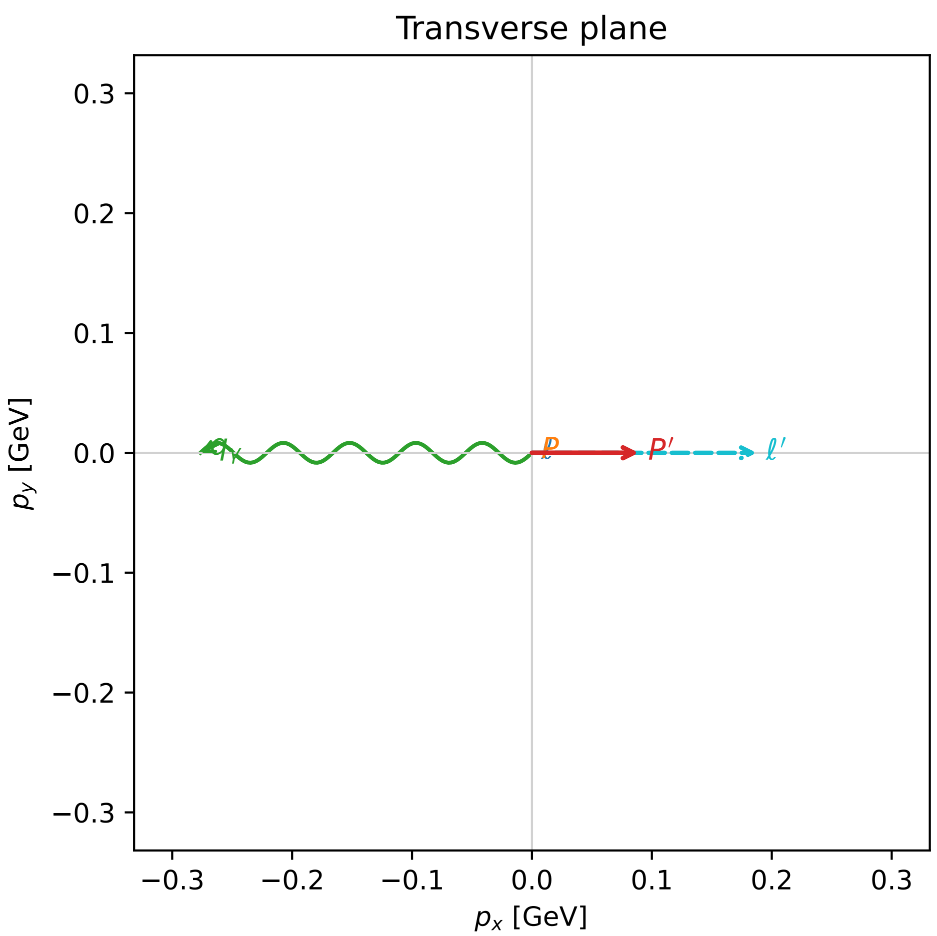
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

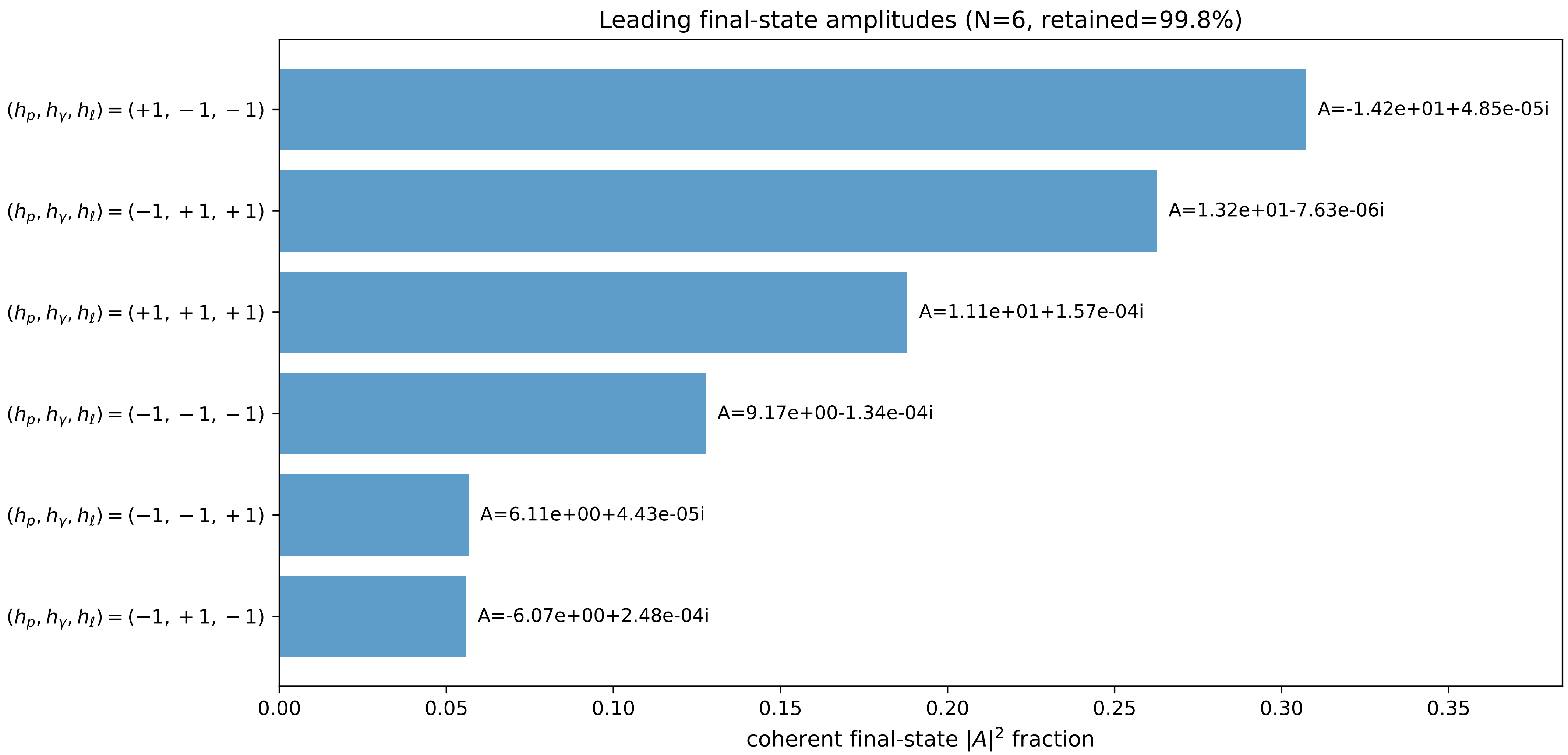


dghz_mixing_angles_polarization_04_local_minimum_300 d_{\rm GHZ}=0.000322149
region: polarization_cluster_04_local_minimum_300
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2107682 rad
incoming lepton: -0.597173|+> +0.802113|->
proton mixing angle: alpha_p=2.6090278 rad
incoming proton: -0.861508|+> +0.507745|->

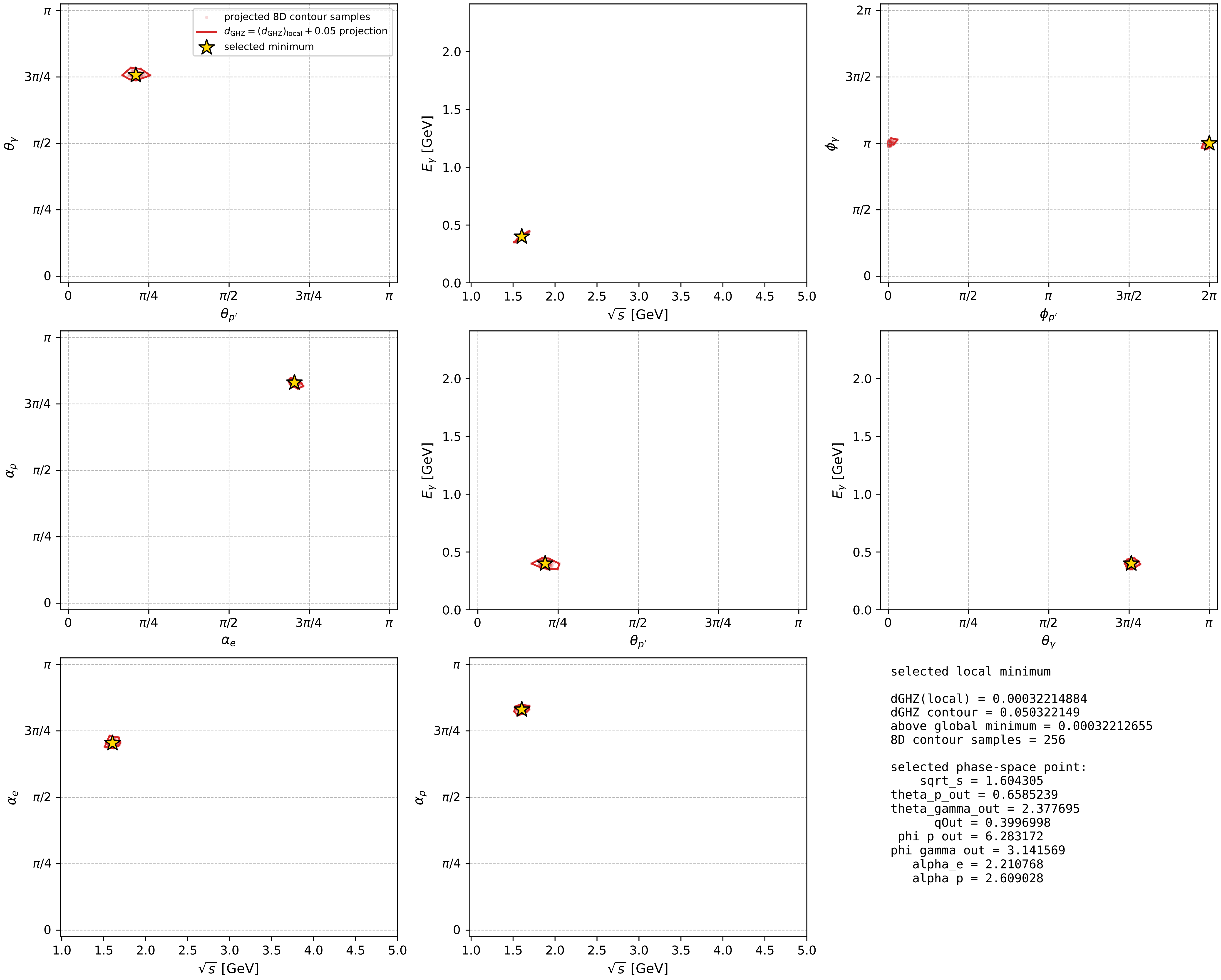
s=2.57379, $\sqrt{s}$=1.60431, $|\vec{P}|$=0.527939, $|\vec{P}'|$=0.145594
$\theta_{p'}$=0.658524, θ_γ=2.3777, $\phi_{p'}$=6.28317, ϕ_γ=3.14157
E_γ=0.3997, $\theta_{\ell'}$=0.823898, $\phi_{\ell'}$=6.28316
Q^2=0.452828, x_B=0.341143, t=-0.16217, W^2=1.7544, y=0.783604

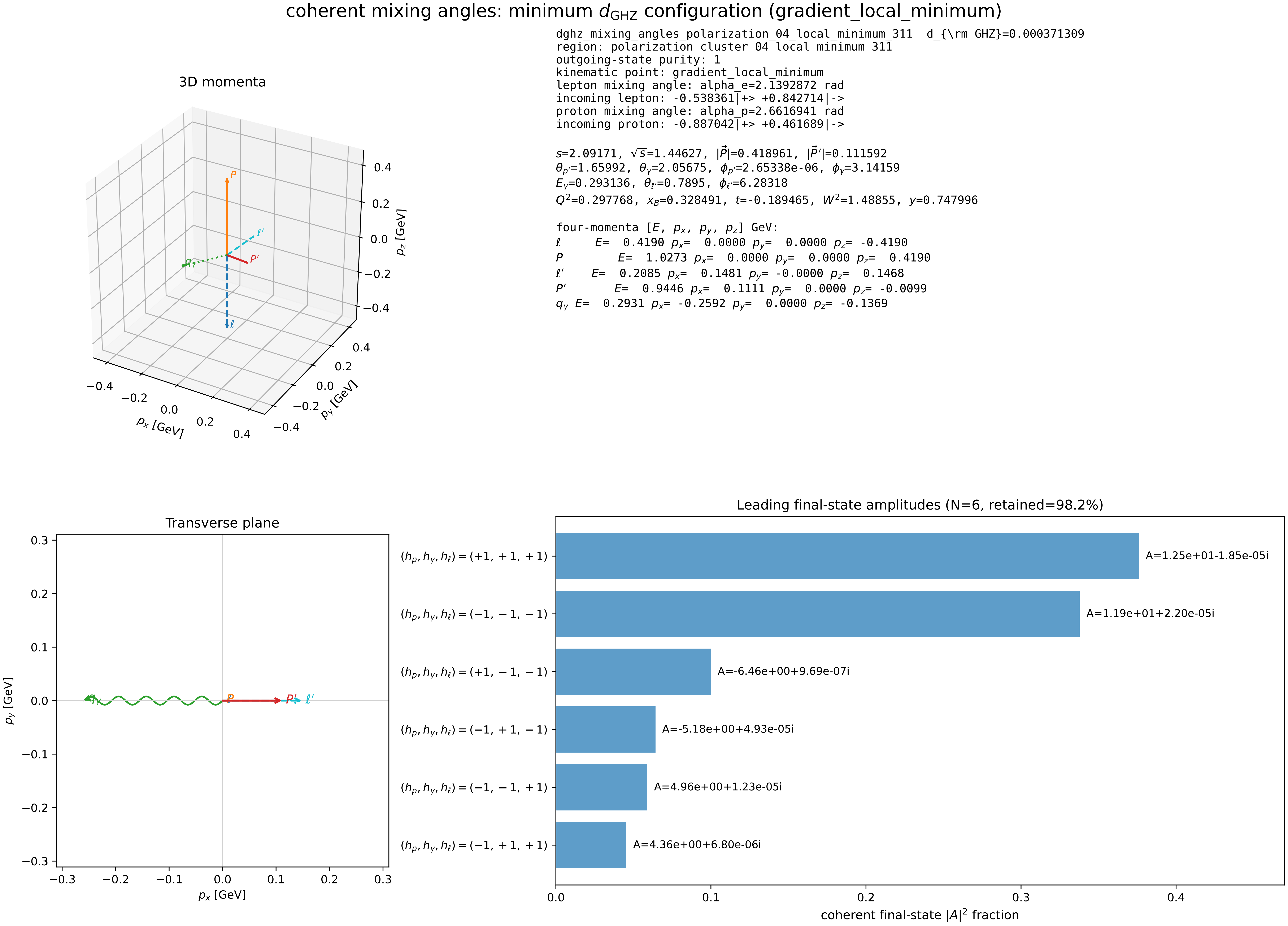
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5279 p_x= 0.0000 p_y= 0.0000 p_z= -0.5279
P E= 1.0764 p_x= 0.0000 p_y= 0.0000 p_z= 0.5279
ℓ' E= 0.2554 p_x= 0.1874 p_y= -0.0000 p_z= 0.1735
P' E= 0.9492 p_x= 0.0891 p_y= -0.0000 p_z= 0.1151
q_γ E= 0.3997 p_x= -0.2765 p_y= 0.0000 p_z= -0.2886

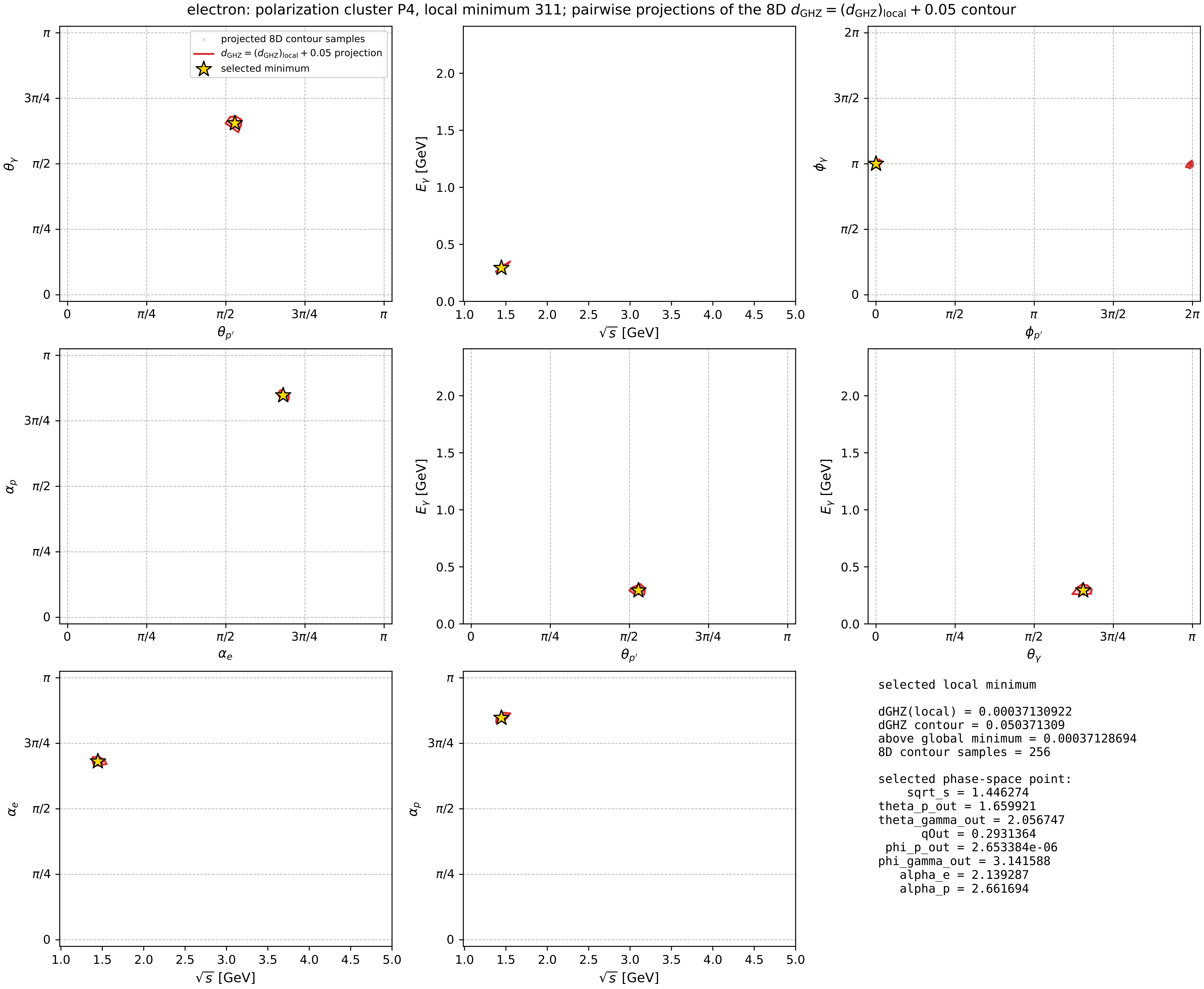




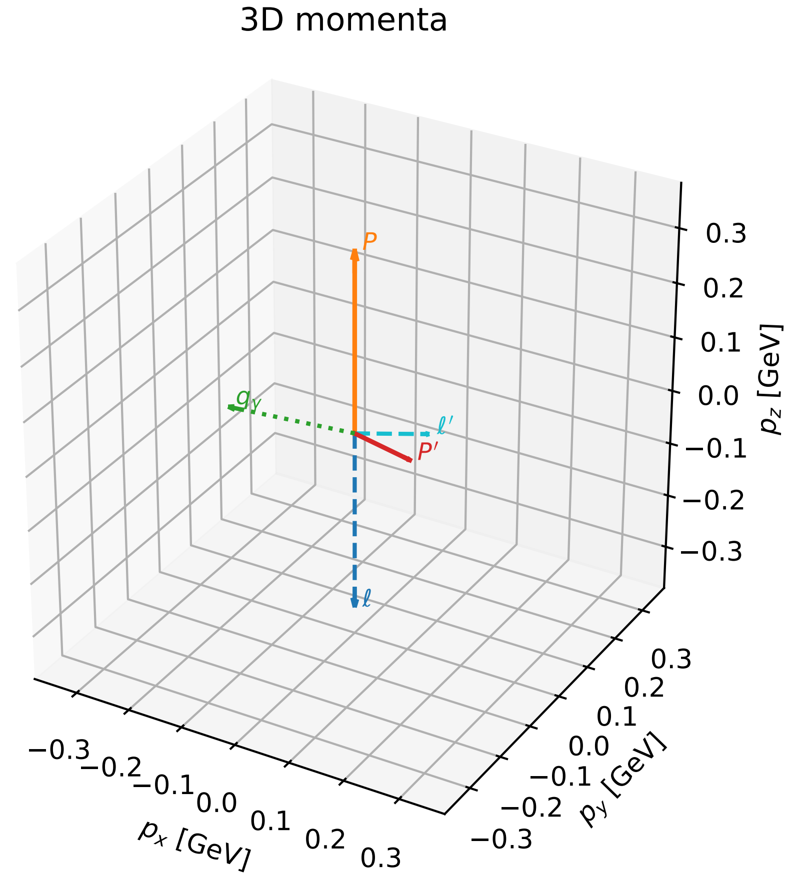
electron: polarization cluster P4, local minimum 300; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour




coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_311 d_{\rm GHZ}=0.000371309
region: polarization_cluster_04_local_minimum_311
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.1392872 rad
incoming lepton: -0.538361|+> +0.842714|->
proton mixing angle: alpha_p=2.6616941 rad
incoming proton: -0.887042|+> +0.461689|->
s=2.09171, $\sqrt{s}$=1.44627, $|\vec{P}|$=0.418961, $|\vec{P}'|$=0.111592
$\theta_{p'}$=1.65992, θ_γ=2.05675, $\phi_{p'}$=2.65338e-06, ϕ_γ=3.14159
E_γ=0.293136, $\theta_{\ell'}$=0.7895, $\phi_{\ell'}$=6.28318
Q^2=0.297768, x_B=0.328491, t=-0.189465, W^2=1.48855, y=0.747996
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4190 p_x= 0.0000 p_y= 0.0000 p_z= -0.4190
P E= 1.0273 p_x= 0.0000 p_y= 0.0000 p_z= 0.4190
ℓ' E= 0.2085 p_x= 0.1481 p_y= -0.0000 p_z= 0.1468
P' E= 0.9446 p_x= 0.1111 p_y= 0.0000 p_z= -0.0099
q_γ E= 0.2931 p_x= -0.2592 p_y= 0.0000 p_z= -0.1369
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=98.2%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=1.25e+01-1.85e-05i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=1.19e+01+2.20e-05i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-6.46e+00+9.69e-07i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-5.18e+00+4.93e-05i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=4.96e+00+1.23e-05i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=4.36e+00+6.80e-06i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 311; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 0.00037130922
dGHZ contour = 0.050371309
above global minimum = 0.00037128694
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.446274
theta_p_out = 1.659921
theta_gamma_out = 2.056747
qOut = 0.2931364
phi_p_out = 2.653384e-06
phi_gamma_out = 3.141588
alpha_e = 2.139287
alpha_p = 2.661694

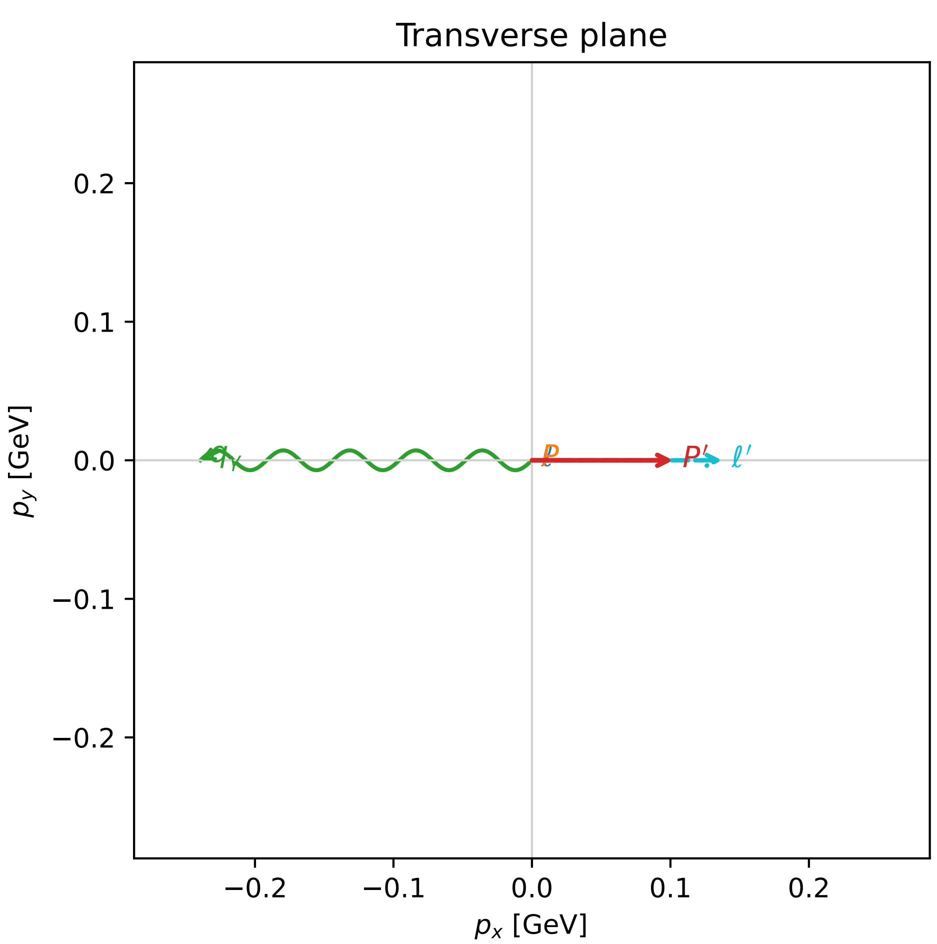
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

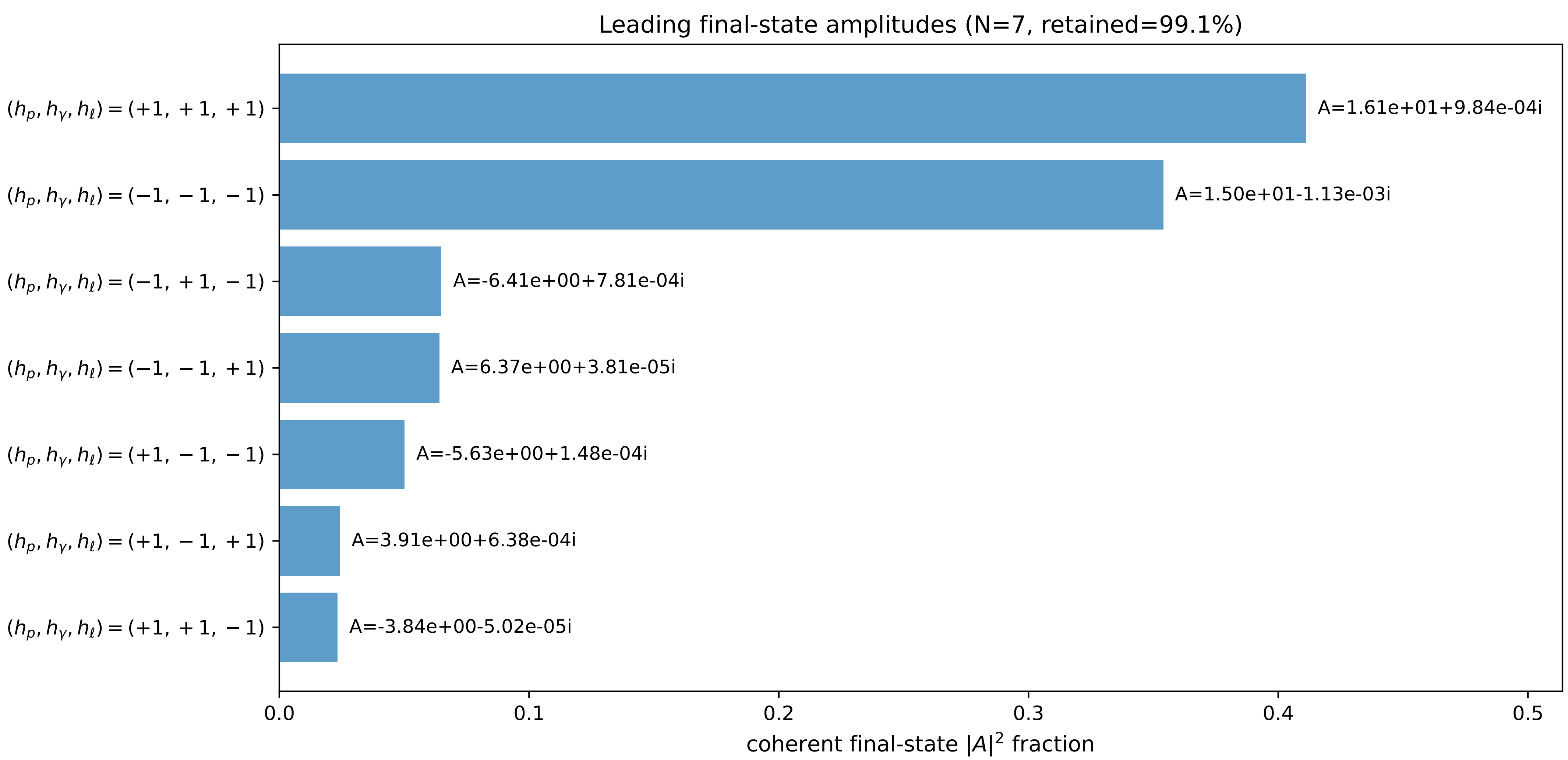


dghz_mixing_angles_polarization_04_local_minimum_312 d_{\rm GHZ}=0.000383267
region: polarization_cluster_04_local_minimum_312
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4329732 rad
incoming lepton: -0.759261|+> +0.650786|->
proton mixing angle: alpha_p=2.4210477 rad
incoming proton: -0.751446|+> +0.659794|->

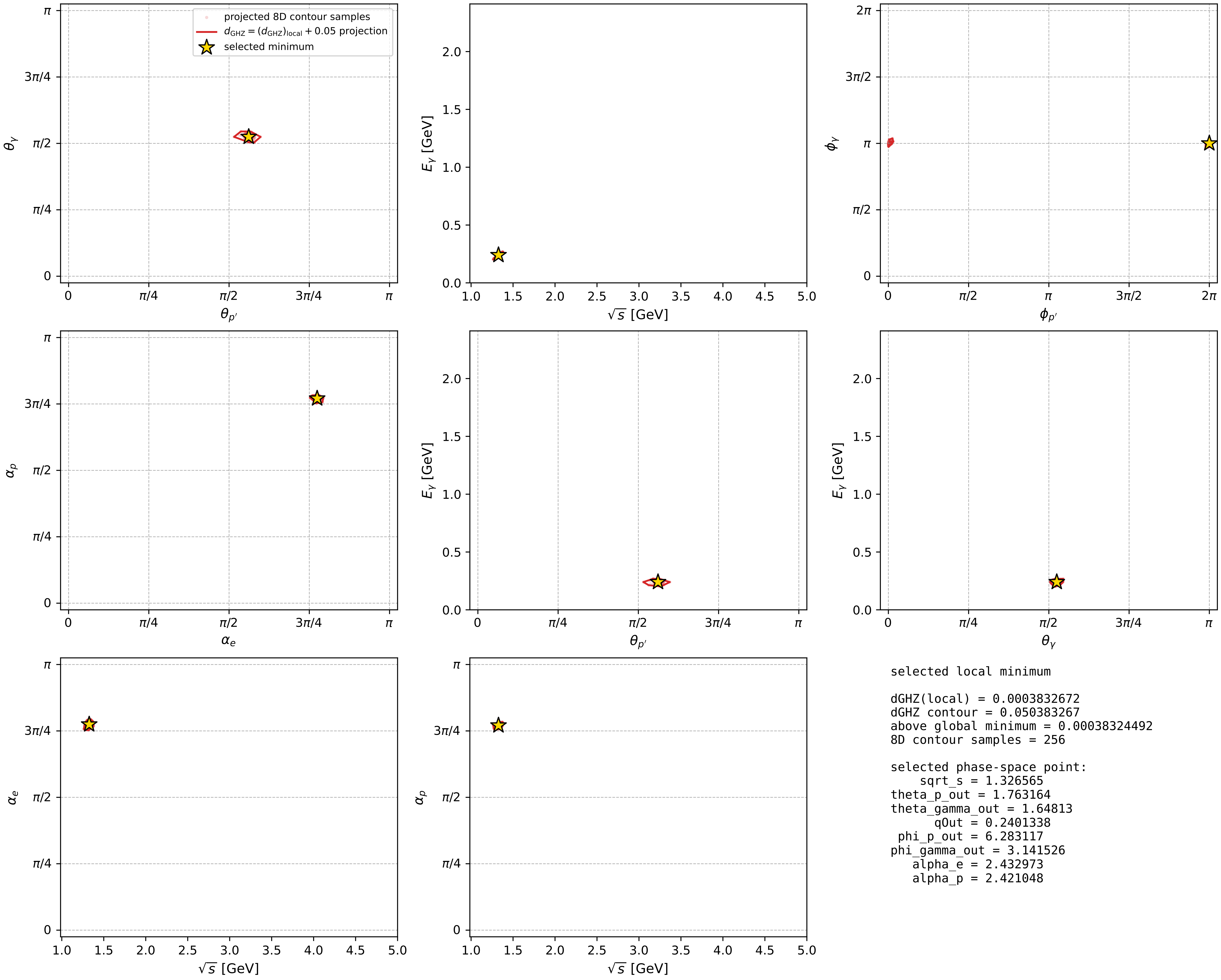
s=1.75977, $\sqrt{s}$=1.32656, $|\vec{P}|$=0.331657, $|\vec{P}'|$=0.10391
$\theta_{p'}$=1.76316, θ_γ=1.64813, $\phi_{p'}$=6.28312, ϕ_γ=3.14153
E_γ=0.240134, $\theta_{\ell'}$=1.2982, $\phi_{\ell'}$=6.28312
Q^2=0.120133, x_B=0.193301, t=-0.131353, W^2=1.38119, y=0.706284

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3317 p_x= 0.0000 p_y= 0.0000 p_z= -0.3317
P E= 0.9949 p_x= 0.0000 p_y= 0.0000 p_z= 0.3317
ℓ' E= 0.1427 p_x= 0.1374 p_y= -0.0000 p_z= 0.0384
P' E= 0.9437 p_x= 0.1020 p_y= -0.0000 p_z= -0.0199
q_γ E= 0.2401 p_x= -0.2394 p_y= 0.0000 p_z= -0.0186

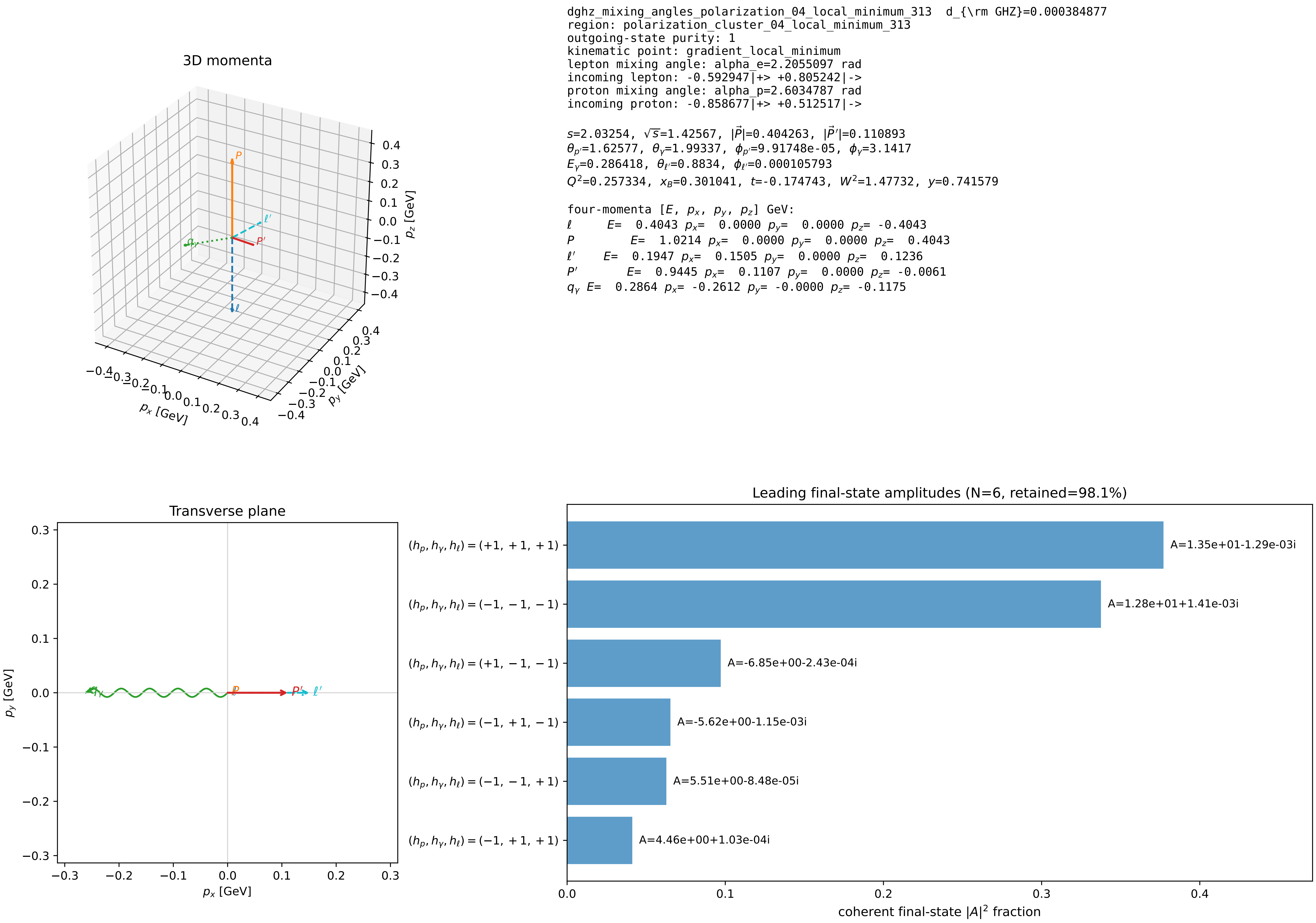




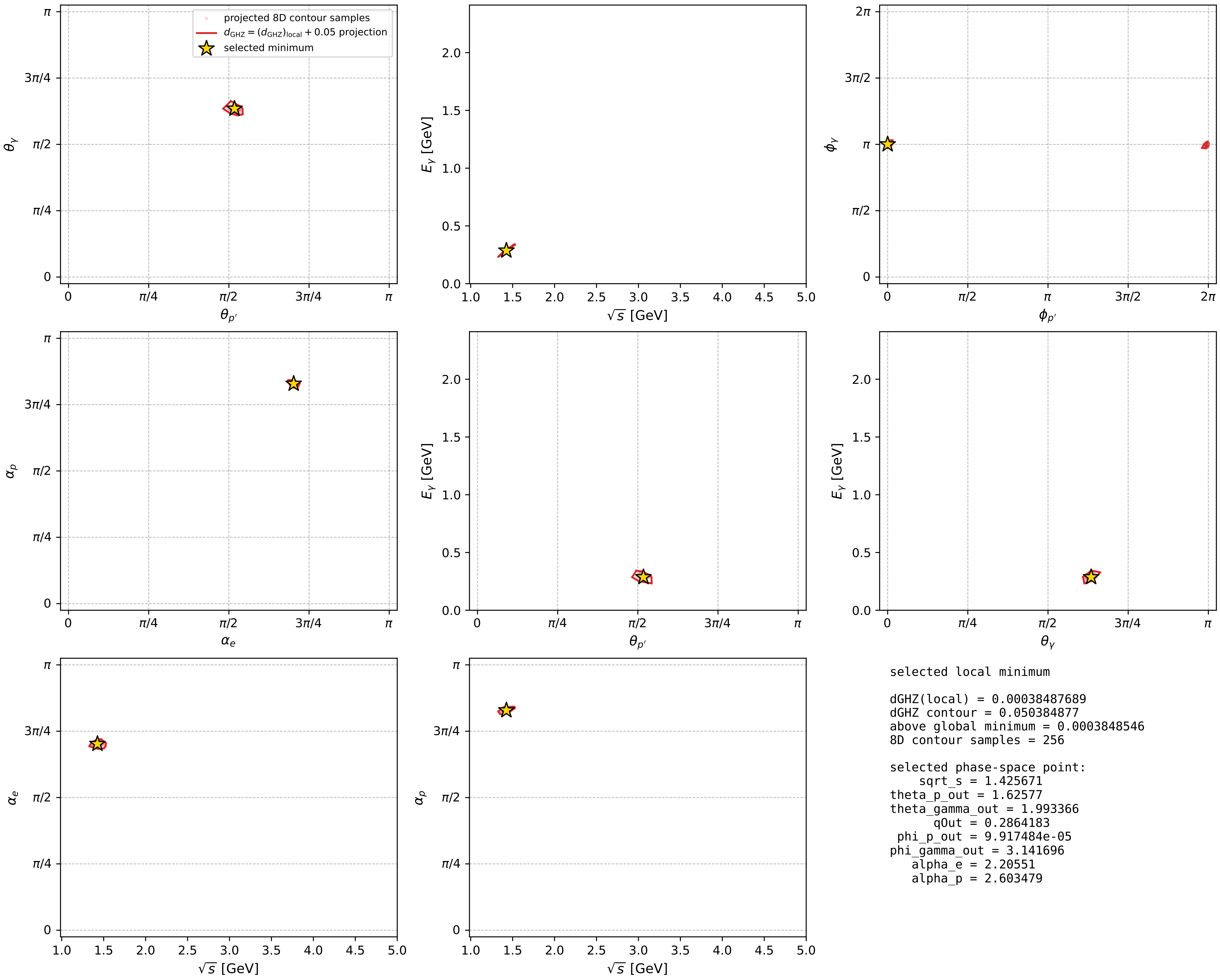
electron: polarization cluster P4, local minimum 312; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour

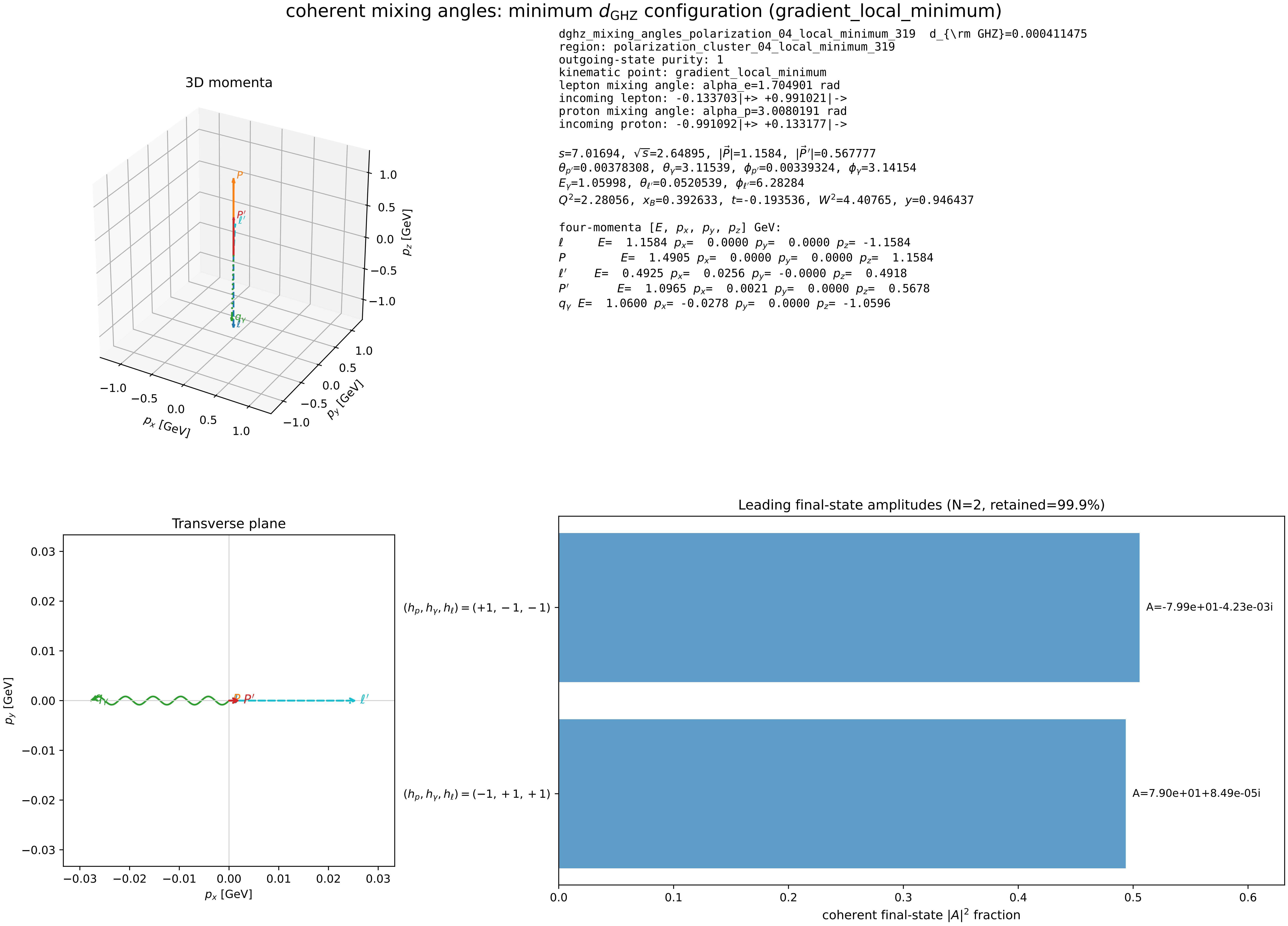


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



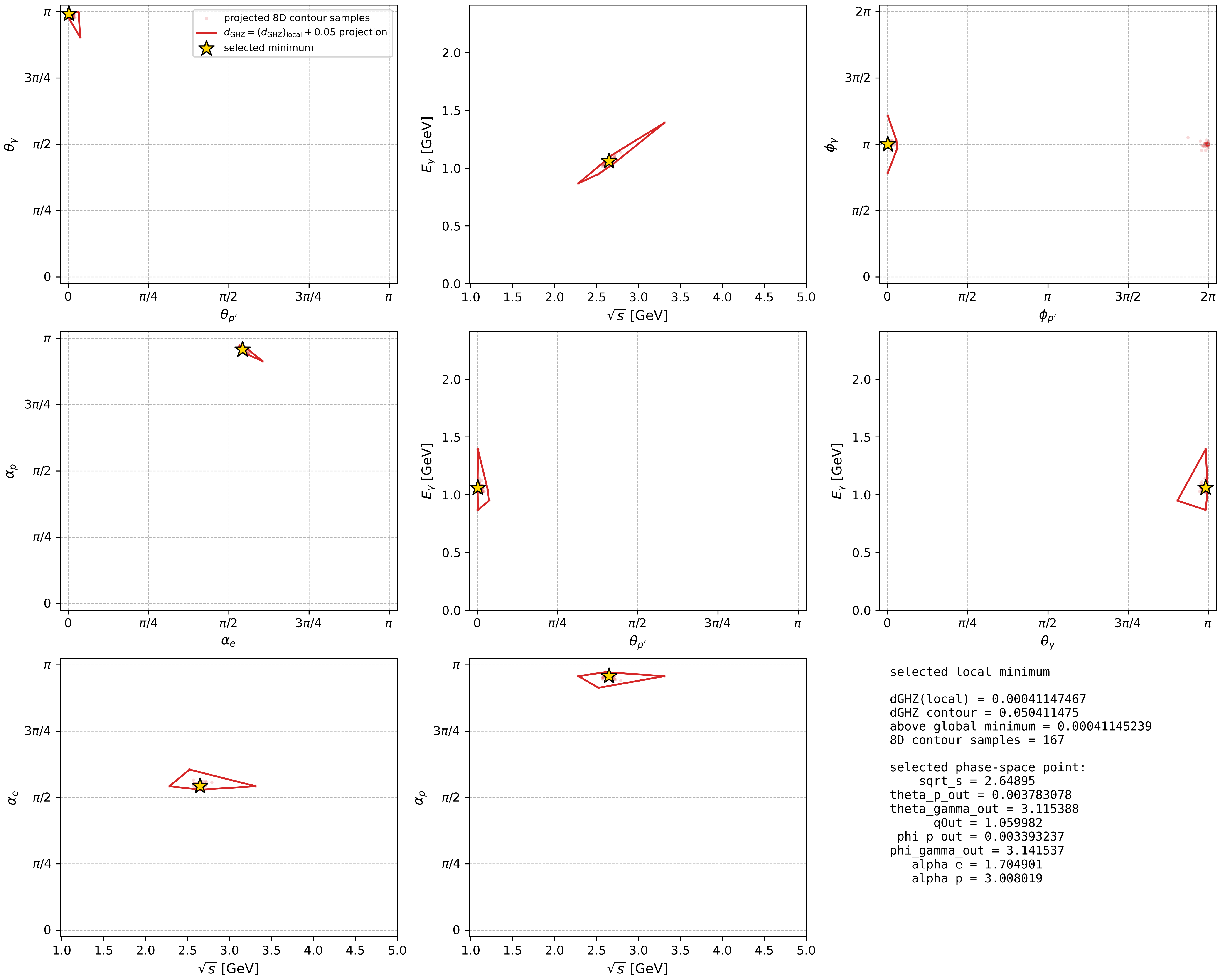
electron: polarization cluster P4, local minimum 313; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



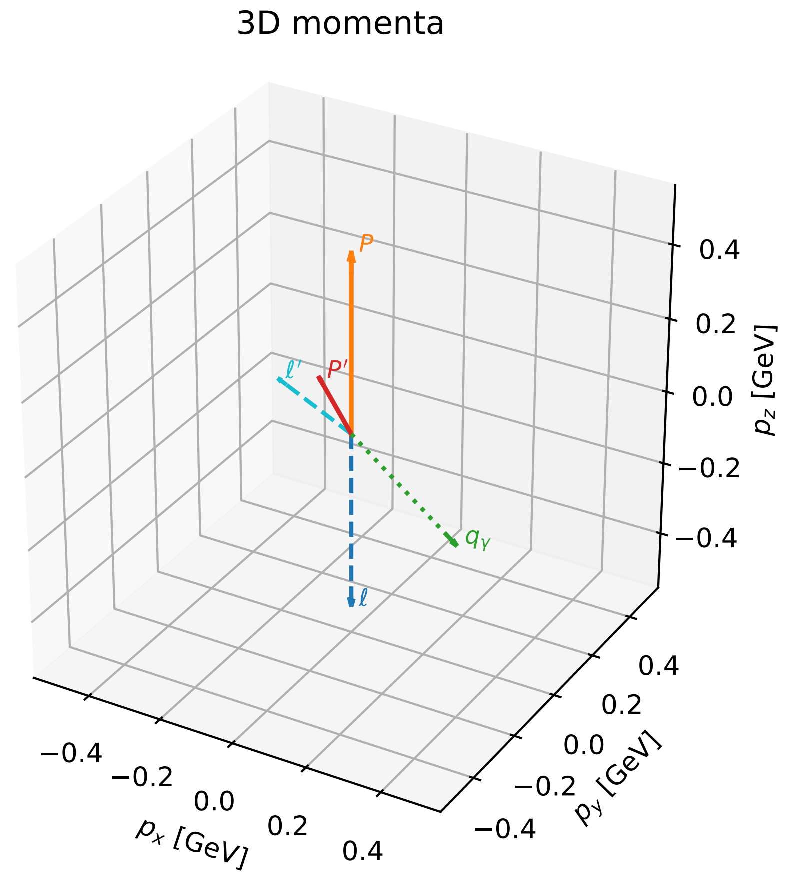


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_319 d_{\rm GHZ}=0.000411475
region: polarization_cluster_04_local_minimum_319
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.704901 rad
incoming lepton: -0.133703|+> +0.991021|->
proton mixing angle: alpha_p=3.0080191 rad
incoming proton: -0.991092|+> +0.133177|->
s=7.01694, $\sqrt{s}$=2.64895, $|\vec{P}|$=1.1584, $|\vec{P}'|$=0.567777
$\theta_{p'}$=0.00378308, θ_γ=3.11539, $\phi_{p'}$=0.00339324, ϕ_γ=3.14154
E_γ=1.05998, $\theta_{\ell'}$=0.0520539, $\phi_{\ell'}$=6.28284
Q^2=2.28056, x_B=0.392633, t=-0.193536, W^2=4.40765, y=0.946437
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.1584 p_x= 0.0000 p_y= 0.0000 p_z= -1.1584
P E= 1.4905 p_x= 0.0000 p_y= 0.0000 p_z= 1.1584
ℓ' E= 0.4925 p_x= 0.0256 p_y= -0.0000 p_z= 0.4918
P' E= 1.0965 p_x= 0.0021 p_y= 0.0000 p_z= 0.5678
q_γ E= 1.0600 p_x= -0.0278 p_y= 0.0000 p_z= -1.0596
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=2, retained=99.9%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-7.99e+01-4.23e-03i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=7.90e+01+8.49e-05i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P4, local minimum 319; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour



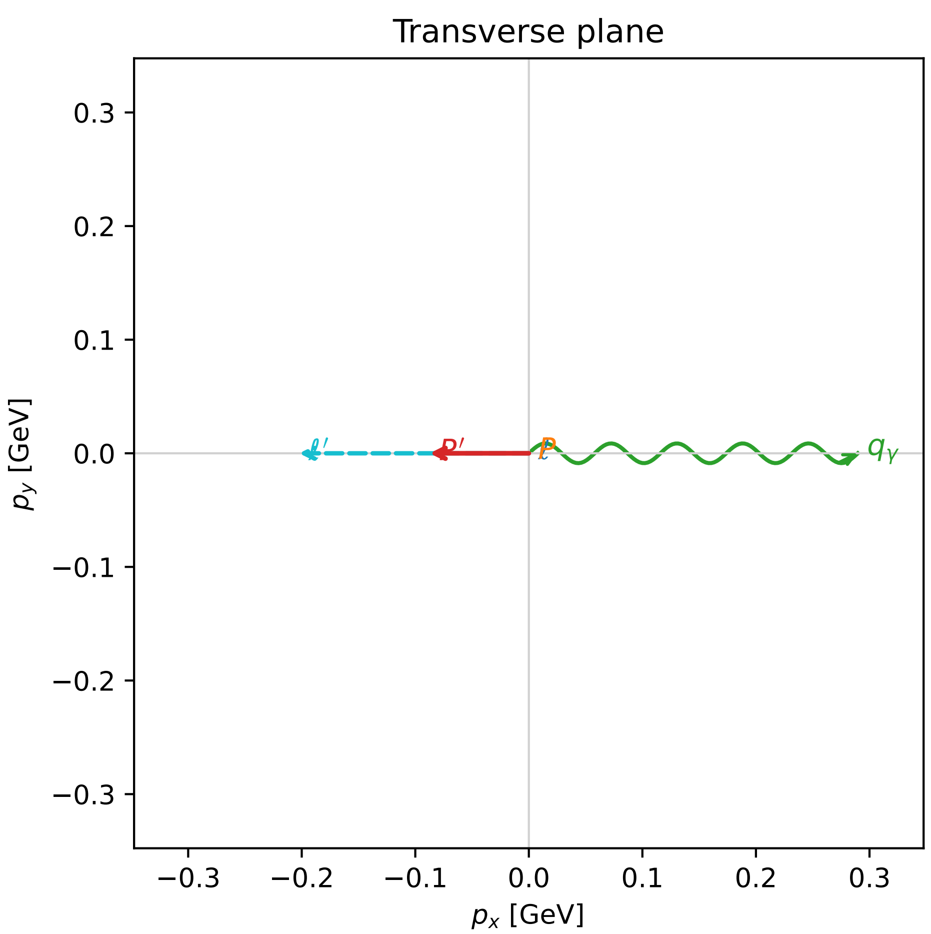
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

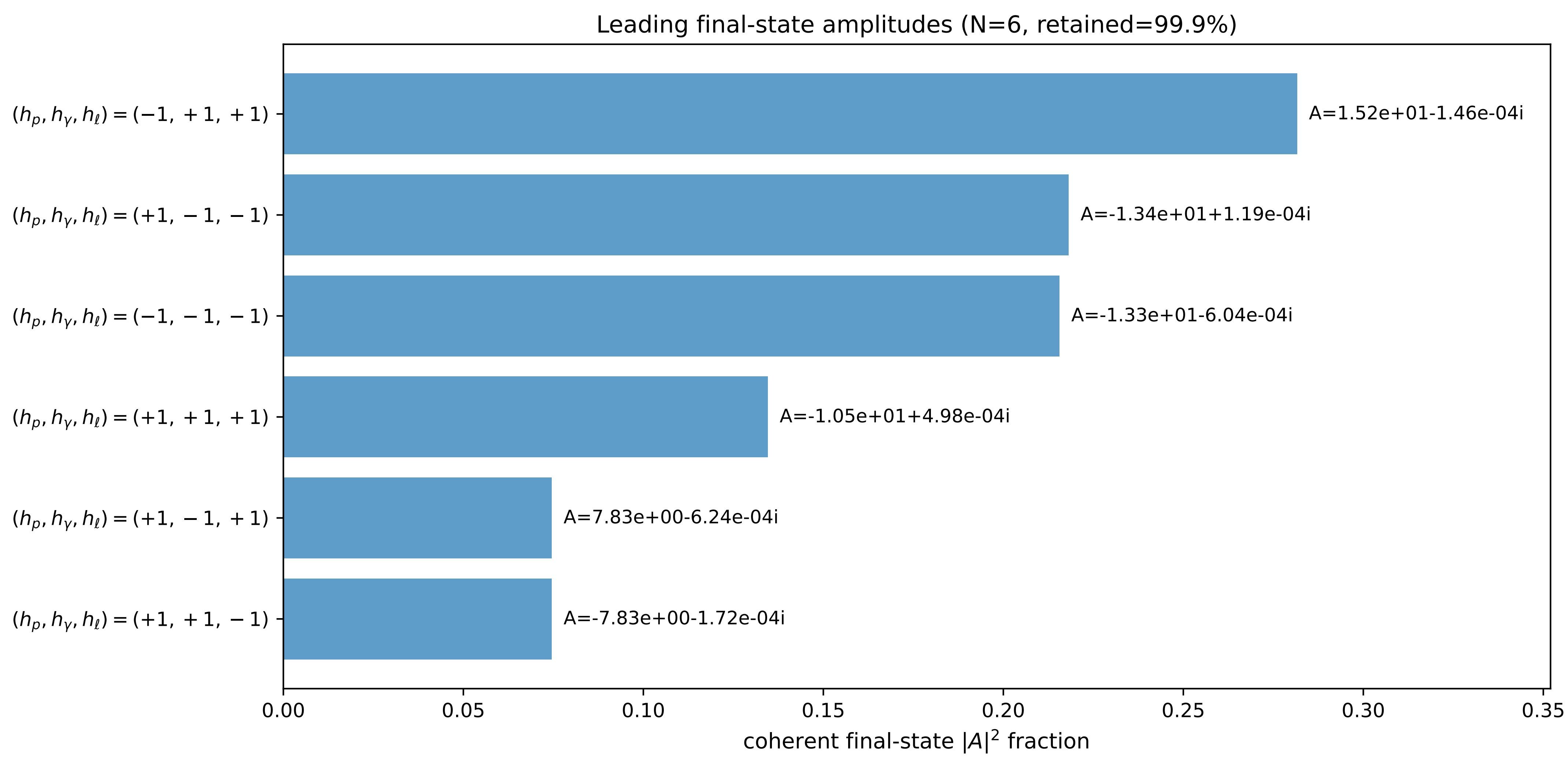



```
dghz_mixing_angles_polarization_04_local_minimum_325  d_{\rm GHZ}=0.000442859
region: polarization_cluster_04_local_minimum_325
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3094045 rad
incoming lepton: -0.673259|+> +0.739406|->
proton mixing angle: alpha_p=2.2635652 rad
incoming proton: -0.63867|+> +0.769481|->
```

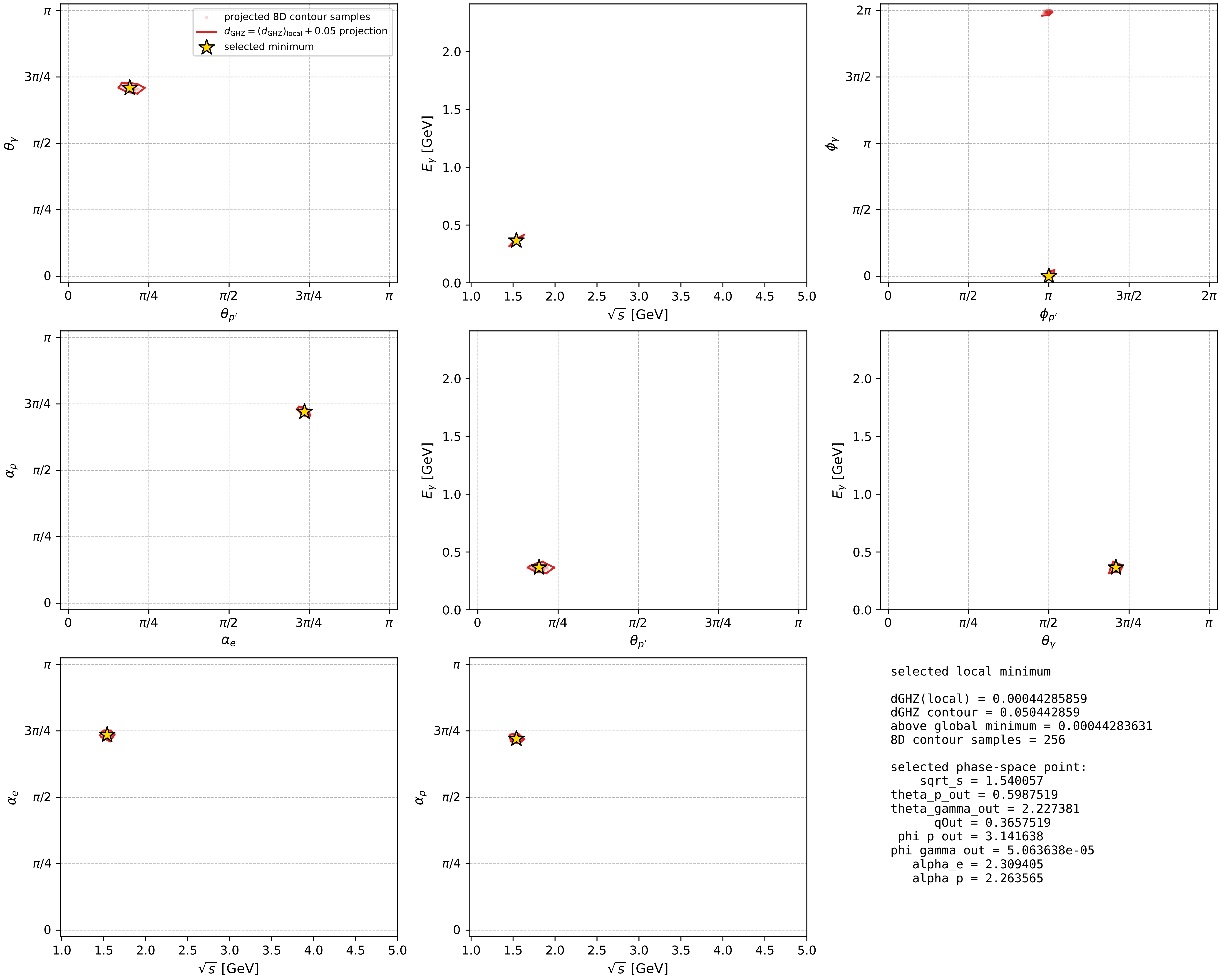
s=2.37177, $\sqrt{s}$=1.54006, $|\vec{P}|$=0.484375, $|\vec{P}'|$=0.155069
$\theta_{p'}$=0.598752, θ_γ=2.22738, $\phi_{p'}$=3.14164, ϕ_γ=5.06364e-05
E_γ=0.365752, $\theta_{\ell'}$=1.1311, $\phi_{\ell'}$=3.14165
Q^2=0.308779, x_B=0.277659, t=-0.123561, W^2=1.68314, y=0.745396

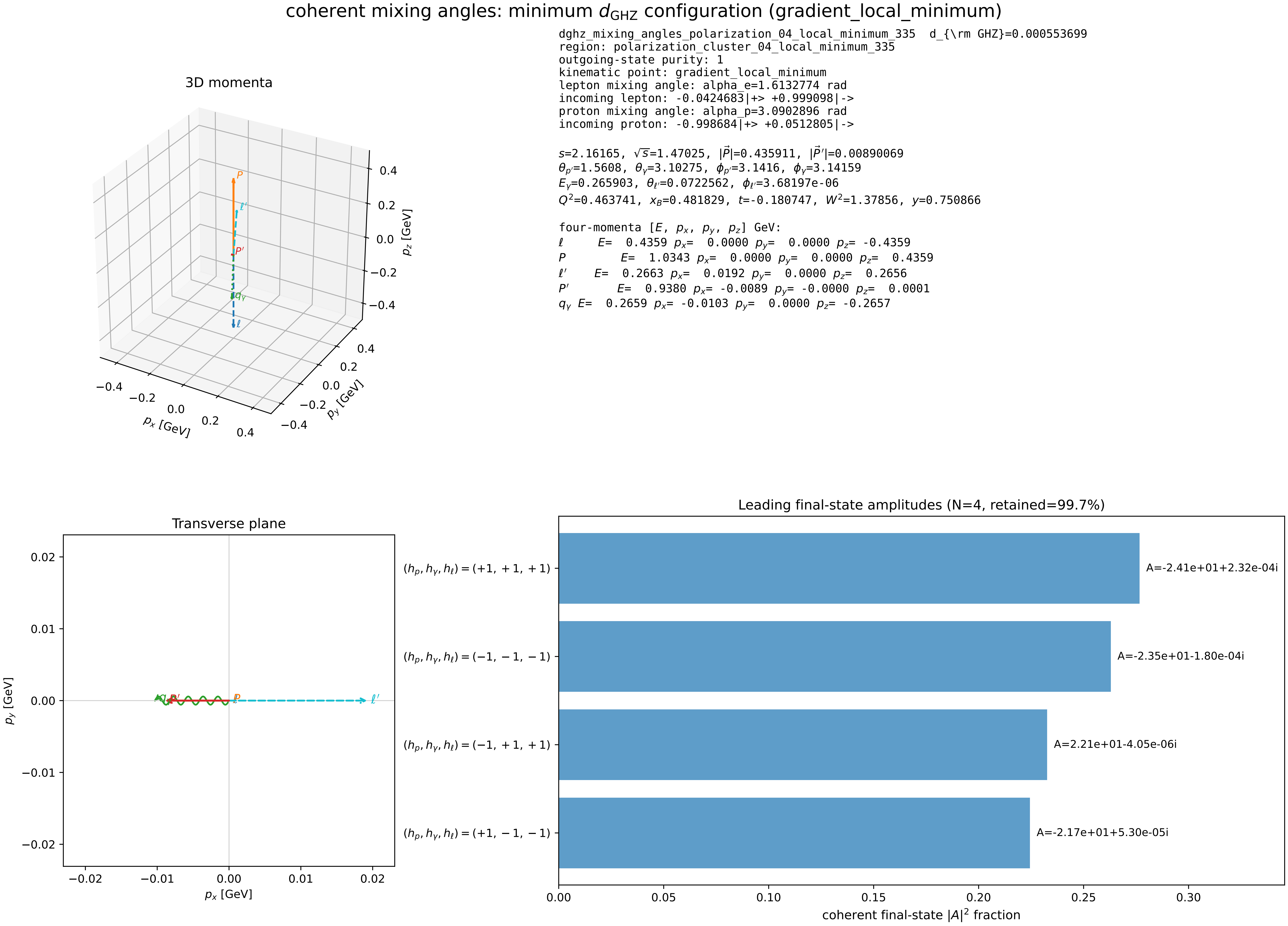
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4844 p_x= 0.0000 p_y= 0.0000 p_z= -0.4844
P E= 1.0557 p_x= 0.0000 p_y= 0.0000 p_z= 0.4844
ℓ' E= 0.2236 p_x= -0.2023 p_y= -0.0000 p_z= 0.0952
P' E= 0.9507 p_x= -0.0874 p_y= -0.0000 p_z= 0.1281
q_γ E= 0.3658 p_x= 0.2897 p_y= 0.0000 p_z= -0.2233

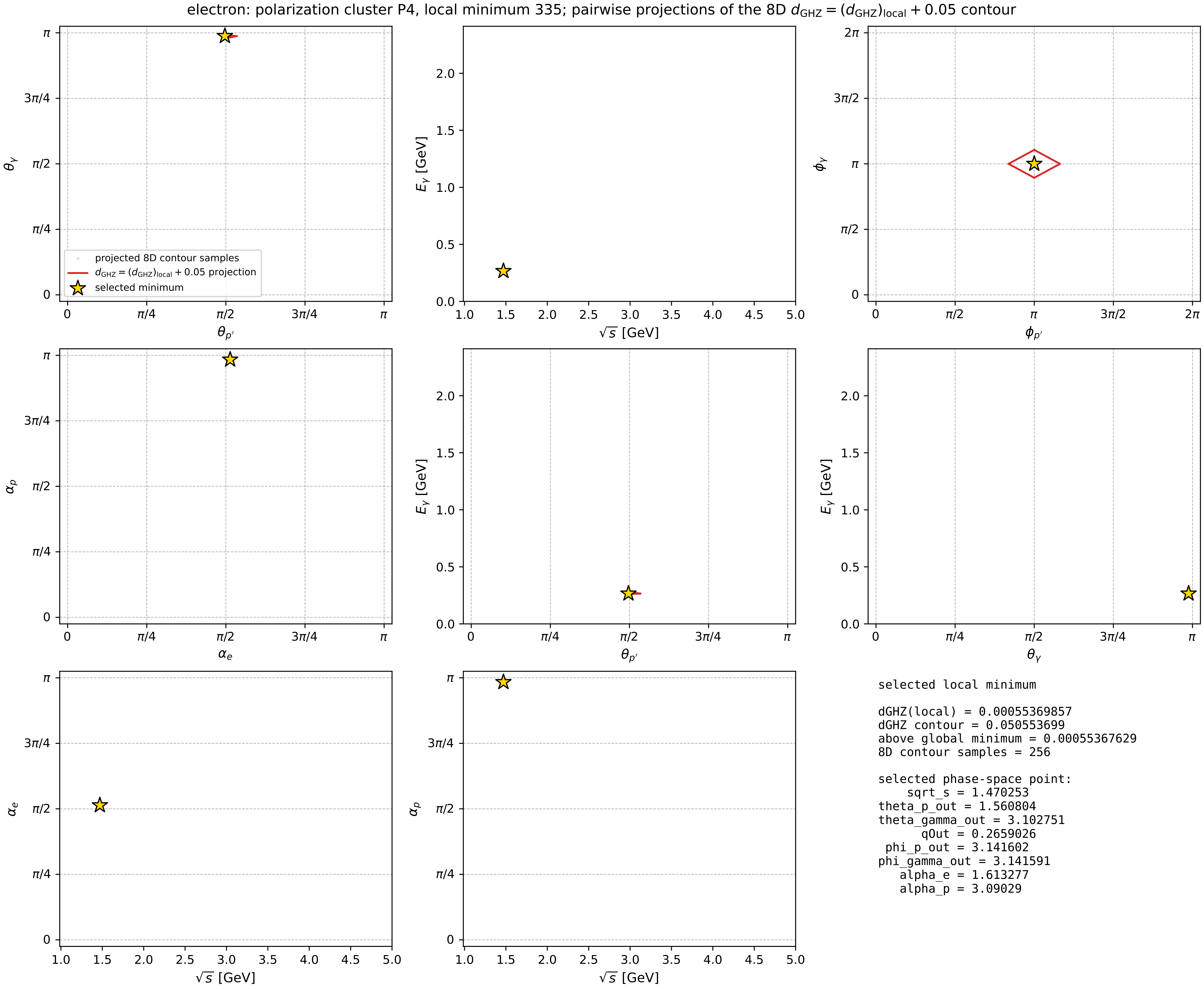




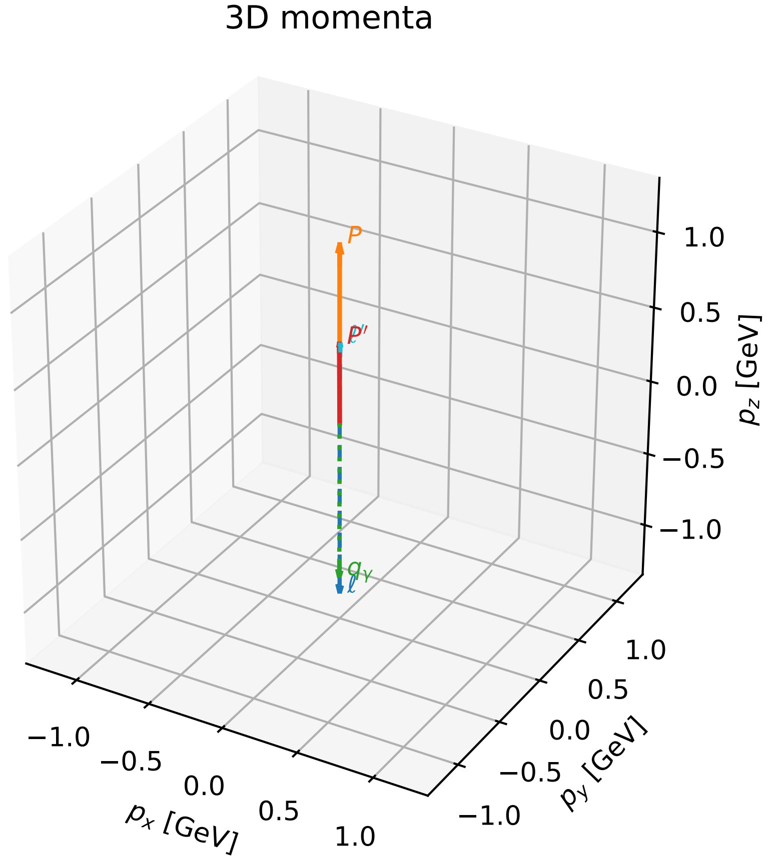
electron: polarization cluster P4, local minimum 325; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour




coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
dghz_mixing_angles_polarization_04_local_minimum_335 d_{\rm GHZ}=0.000553699
region: polarization_cluster_04_local_minimum_335
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.6132774 rad
incoming lepton: -0.0424683|+> +0.999098|->
proton mixing angle: alpha_p=3.0902896 rad
incoming proton: -0.998684|+> +0.0512805|->
s=2.16165, $\sqrt{s}$=1.47025, $|\vec{P}|$=0.435911, $|\vec{P}'|$=0.00890069
$\theta_{p'}$=1.5608, θ_γ=3.10275, $\phi_{p'}$=3.1416, ϕ_γ=3.14159
E_γ=0.265903, $\theta_{\ell'}$=0.0722562, $\phi_{\ell'}$=3.68197e-06
Q^2=0.463741, x_B=0.481829, t=-0.180747, W^2=1.37856, y=0.750866
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4359 p_x= 0.0000 p_y= 0.0000 p_z= -0.4359
P E= 1.0343 p_x= 0.0000 p_y= 0.0000 p_z= 0.4359
ℓ' E= 0.2663 p_x= 0.0192 p_y= 0.0000 p_z= 0.2656
P' E= 0.9380 p_x= -0.0089 p_y= -0.0000 p_z= 0.0001
q_γ E= 0.2659 p_x= -0.0103 p_y= 0.0000 p_z= -0.2657
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=99.7%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-2.41e+01+2.32e-04i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-2.35e+01-1.80e-04i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=2.21e+01-4.05e-06i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-2.17e+01+5.30e-05i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P4, local minimum 335; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 0.00055369857
dGHZ contour = 0.050553699
above global minimum = 0.00055367629
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.470253
theta_p_out = 1.560804
theta_gamma_out = 3.102751
qOut = 0.2659026
phi_p_out = 3.141602
phi_gamma_out = 3.141591
alpha_e = 1.613277
alpha_p = 3.09029

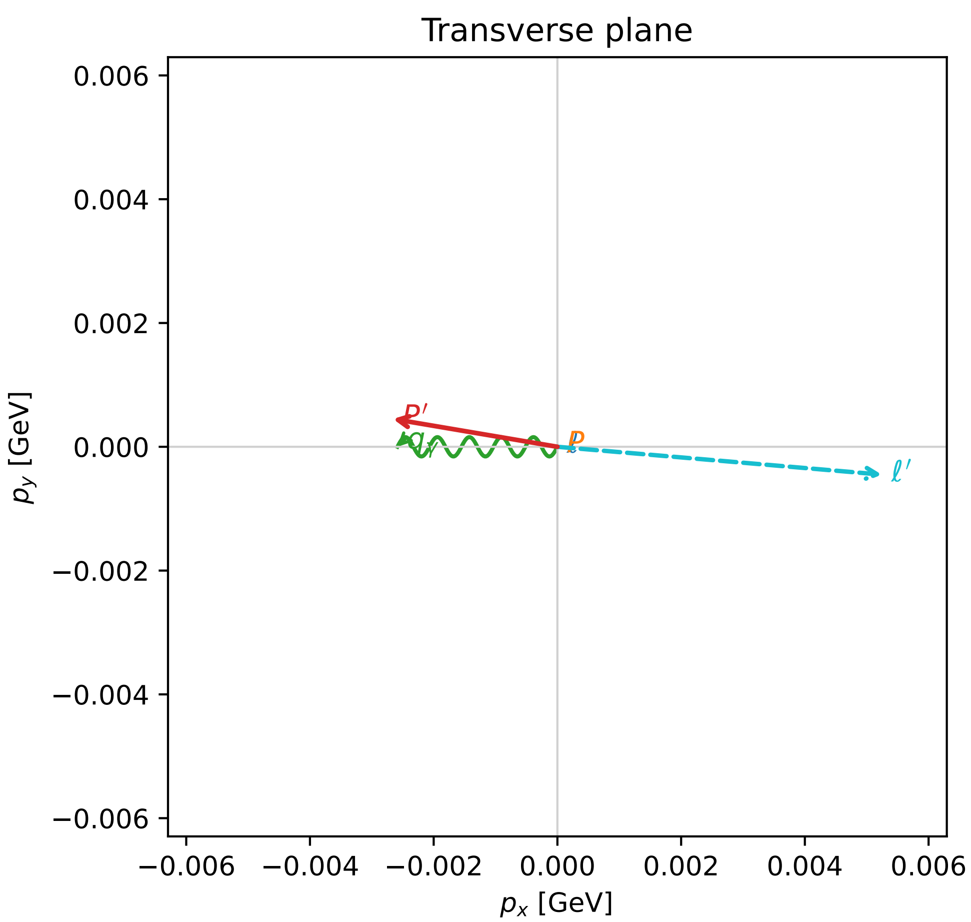
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

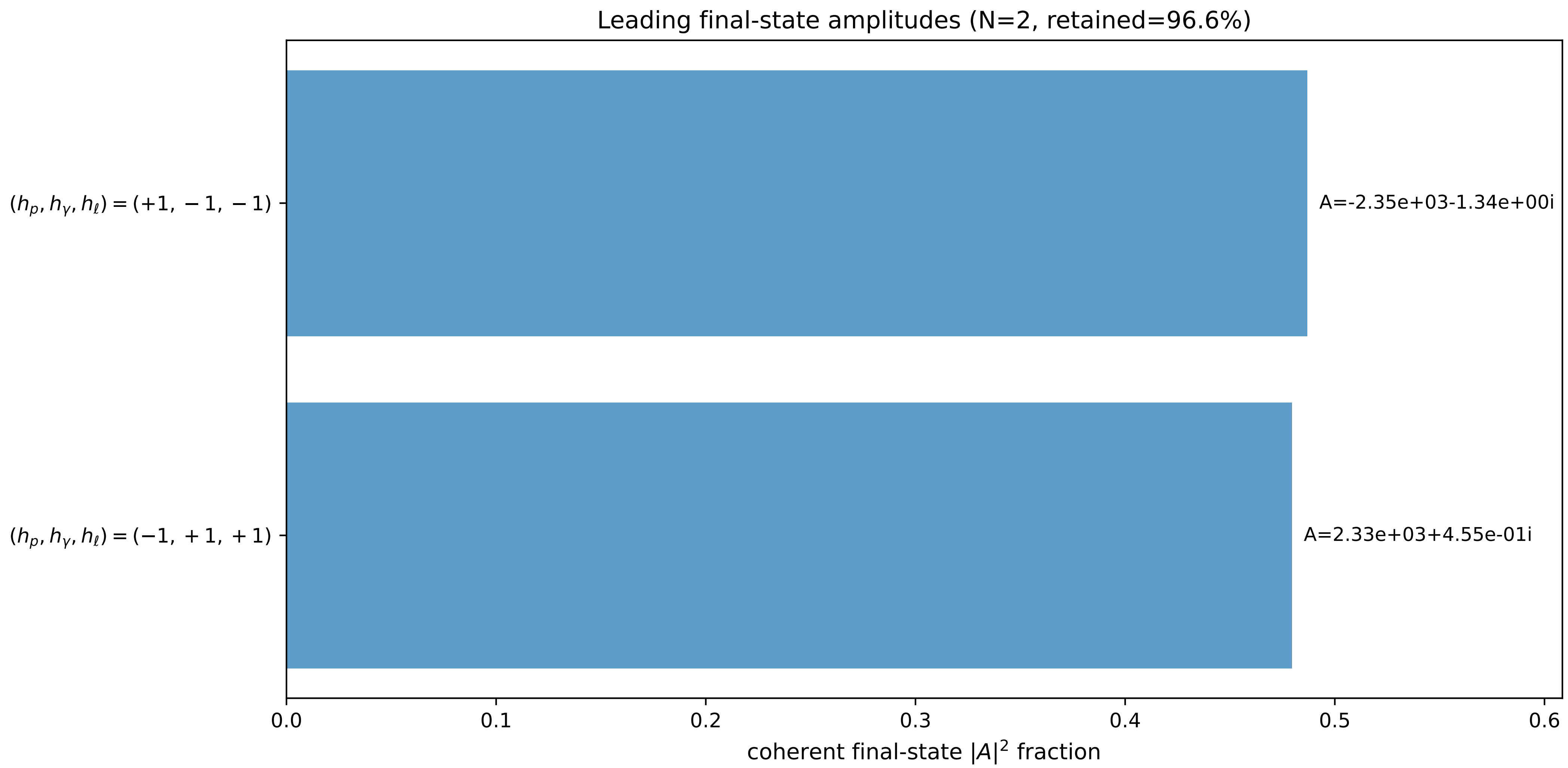


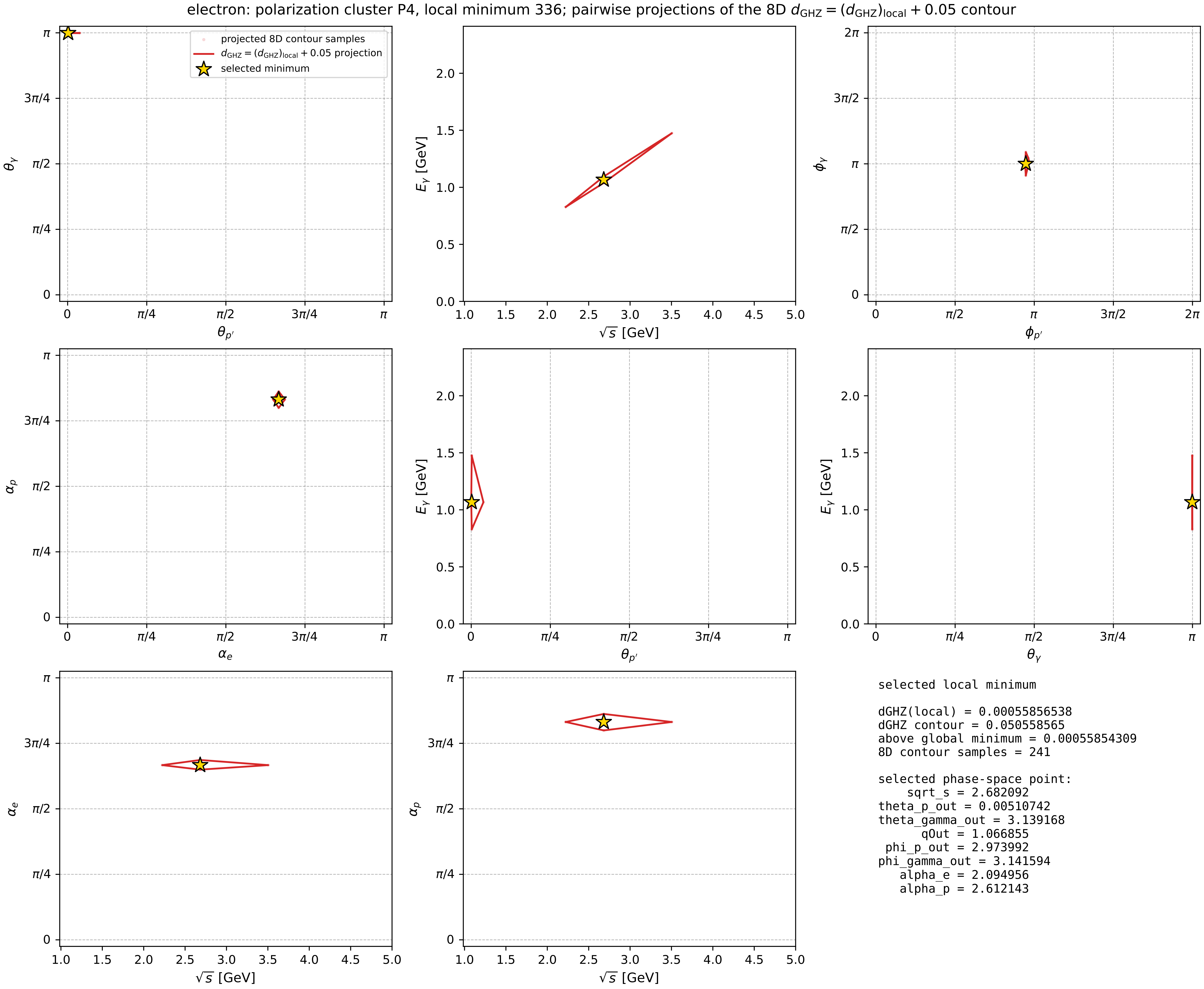
dghz_mixing_angles_polarization_04_local_minimum_336 d_{\rm GHZ}=0.000558565
region: polarization_cluster_04_local_minimum_336
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.0949563 rad
incoming lepton: -0.500486|+> +0.865745|->
proton mixing angle: alpha_p=2.6121434 rad
incoming proton: -0.863085|+> +0.505058|->

s=7.19362, $\sqrt{s}$=2.68209, $|\vec{P}|$=1.17702, $|\vec{P}'|$=0.528084
$\theta_{p'}$=0.00510742, θ_γ=3.13917, $\phi_{p'}$=2.97399, ϕ_γ=3.14159
E_γ=1.06685, $\theta_{\ell'}$=0.00977172, $\phi_{\ell'}$=6.19762
Q^2=2.53666, x_B=0.425599, t=-0.237414, W^2=4.30339, y=0.944002

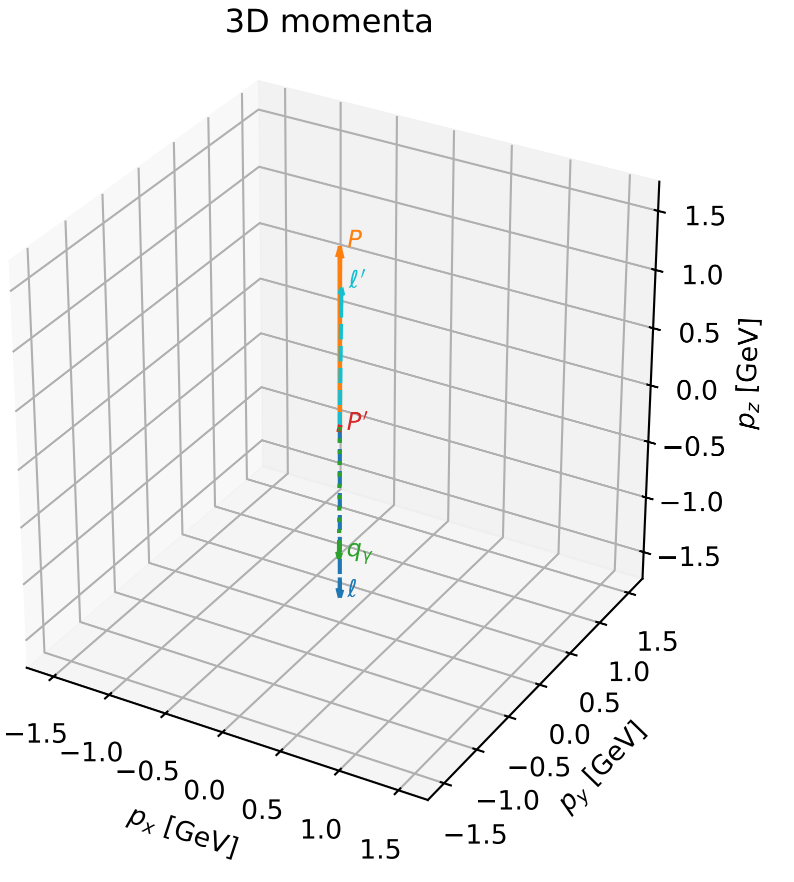
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.1770 p_x= 0.0000 p_y= 0.0000 p_z= -1.1770
P E= 1.5051 p_x= 0.0000 p_y= 0.0000 p_z= 1.1770
ℓ' E= 0.5388 p_x= 0.0052 p_y= -0.0004 p_z= 0.5388
P' E= 1.0764 p_x= -0.0027 p_y= 0.0004 p_z= 0.5281
q_γ E= 1.0669 p_x= -0.0026 p_y= -0.0000 p_z= -1.0669






electron: polarization cluster P4, local minimum 336; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
dGHZ(local) = 0.00055856538
dGHZ contour = 0.050558565
above global minimum = 0.00055854309
8D contour samples = 241
selected phase-space point:
sqrt_s = 2.682092
theta_p_out = 0.00510742
theta_gamma_out = 3.139168
qOut = 1.066855
phi_p_out = 2.973992
phi_gamma_out = 3.141594
alpha_e = 2.094956
alpha_p = 2.612143

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)




```
dghz_mixing_angles_polarization_04_local_minimum_343  d_{\rm GHZ}=0.000611825
region: polarization_cluster_04_local_minimum_343
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=1.6611867 rad
incoming lepton: -0.0902673|+> +0.995918|->
proton mixing angle: alpha_p=3.0508892 rad
incoming proton: -0.995889|+> +0.0905792|->
```

s=10.8933, $\sqrt{s}$=3.3005, $|\vec{P}|$=1.51696, $|\vec{P}'|$=0.0253145

$\theta_{p'}$=2.91168, θ_γ=3.13533, $\phi_{p'}$=3.15487, ϕ_γ=3.14156

E_γ=1.16873, $\theta_{\ell'}$=0.0109667, $\phi_{\ell'}$=0.00583323

Q^2=7.24134, x_B=0.772246, t=-1.66224, W^2=3.01549, y=0.936435

four-momenta [E, p_x, p_y, p_z] GeV:

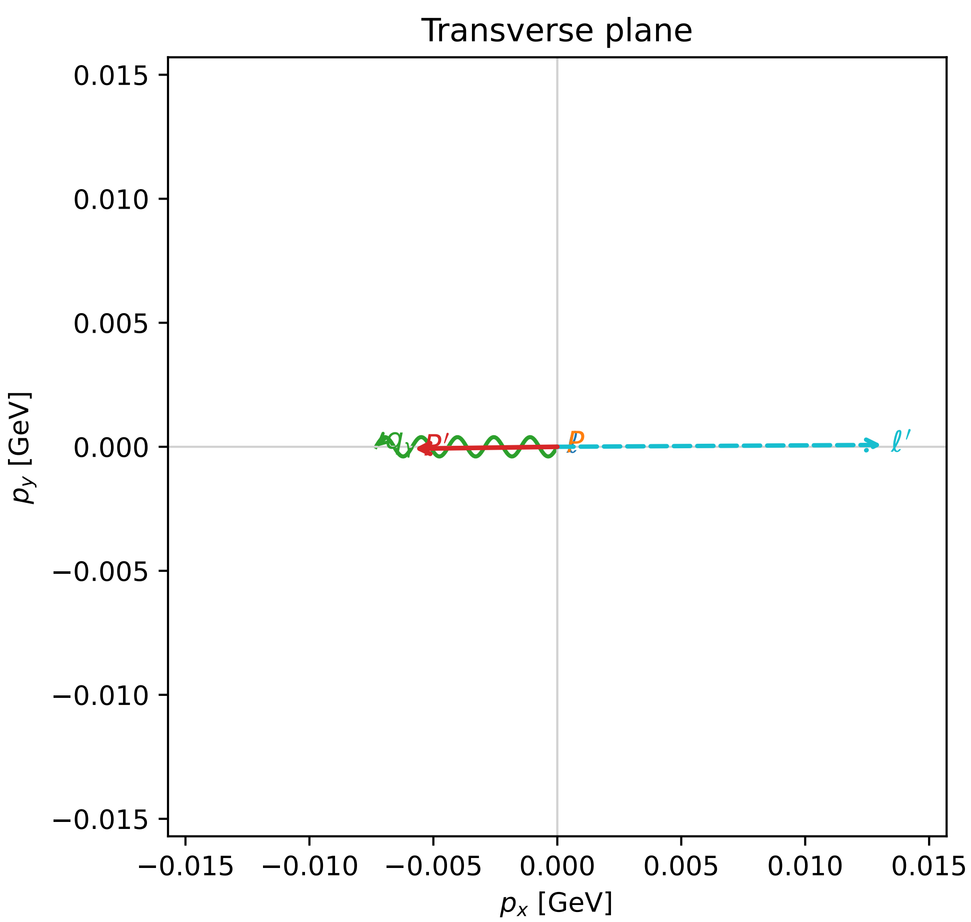
ℓ E= 1.5170 p_x= 0.0000 p_y= 0.0000 p_z= -1.5170

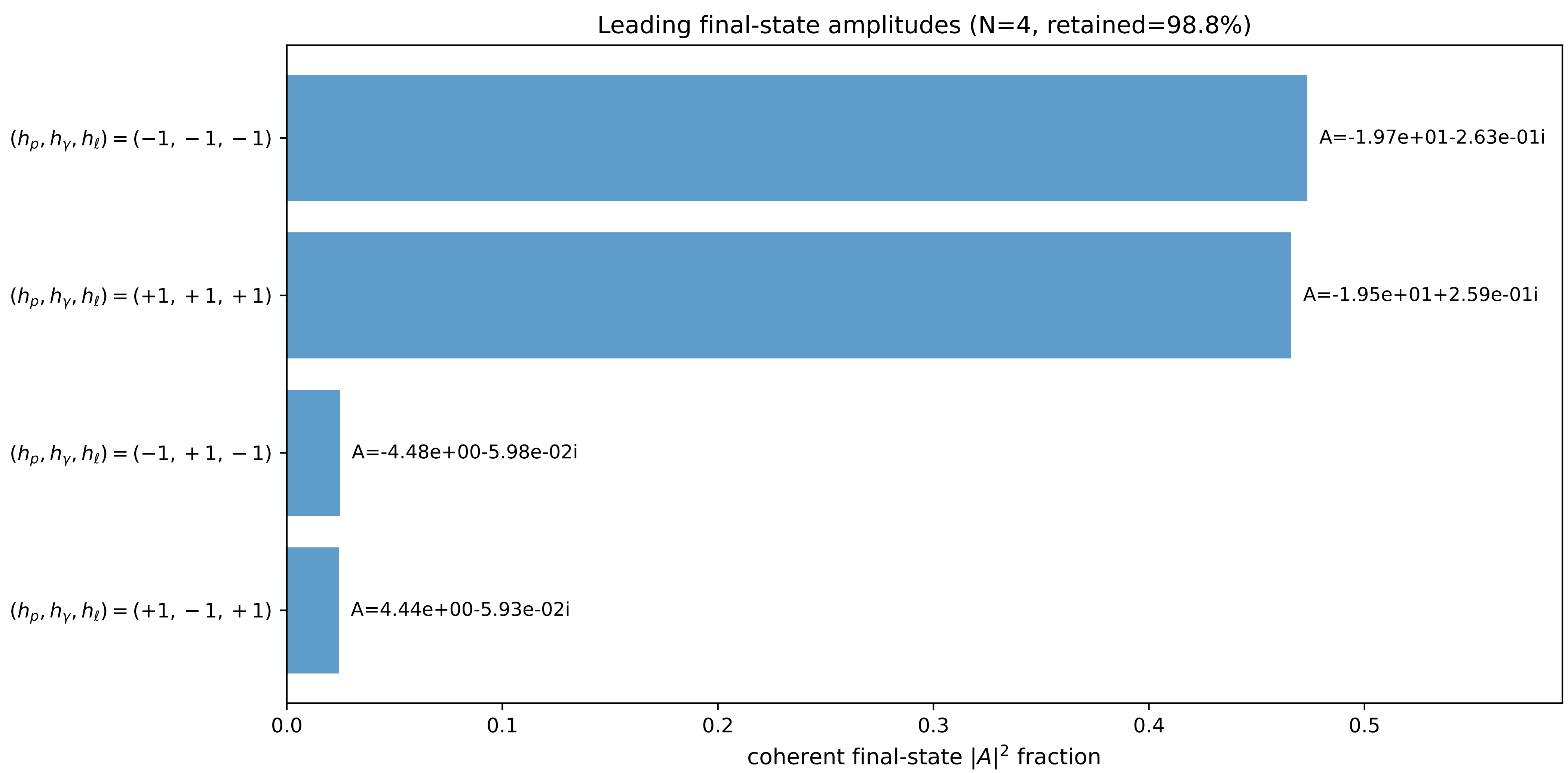
P E= 1.7835 p_x= 0.0000 p_y= 0.0000 p_z= 1.5170

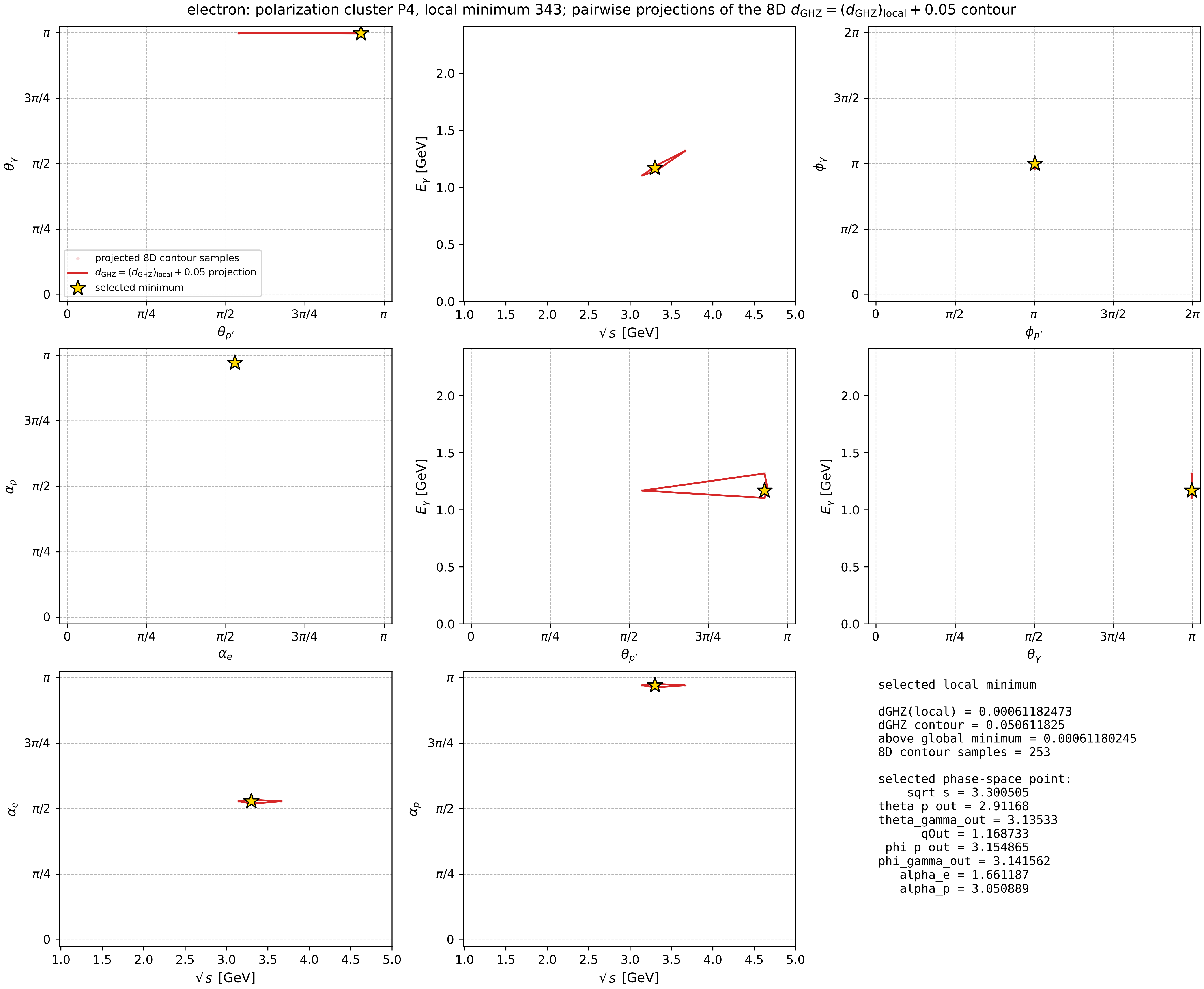
ℓ' E= 1.1934 p_x= 0.0131 p_y= 0.0001 p_z= 1.1934

P' E= 0.9383 p_x= -0.0058 p_y= -0.0001 p_z= -0.0246

q_γ E= 1.1687 p_x= -0.0073 p_y= 0.0000 p_z= -1.1687






electron: polarization cluster P4, local minimum 343; pairwise projections of the 8D $d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$d_{\mathrm{GHZ}} = (d_{\mathrm{GHZ}})_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_{γ}
$\theta_{p'}$
E_{γ} [GeV]
$\sqrt{s}$ [GeV]
ϕ_{γ}
$\phi_{p'}$
α_p
α_e
selected local minimum
dGHZ(local) = 0.00061182473
dGHZ contour = 0.050611825
above global minimum = 0.00061180245
8D contour samples = 253
selected phase-space point:
sqrt_s = 3.300505
theta_p_out = 2.91168
theta_gamma_out = 3.13533
qOut = 1.168733
phi_p_out = 3.154865
phi_gamma_out = 3.141562
alpha_e = 1.661187
alpha_p = 3.050889